

Lectures on Quantum Mechanics

Modernized Edition

Gordon Baym
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0.2 LECTURES ON QUANTUM MECHANICS

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0.5 PREFACE

These lecture notes comprise a three-semester graduate course in quantum mechanics given at the University of Illinois. There are a number of texts which present the basic topics very well; but since a fair quantity of the material discussed in my course was not available to the students in elementary quantum mechanics books, I was asked to prepare written notes. In retrospect these lecture notes seemed sufficiently interesting to warrant their publication in this format. The notes, presented here in slightly revised form, constitute a self-contained course in quantum mechanics from first principles to elementary relativistic one-particle mechanics. The student may want to look as well at one or more of the standard texts, such as K. Gottfried, *Quantum Mechanics* (W.A. Benjamin, New York, 1966); A. Messiah, *Quantum Mechanics* (North-Holland Publishing Company, Amsterdam, 1961), in two volumes; E. Merzbacher, *Quantum Mechanics* (John Wiley and Sons, New York, 1961); and for relativistic quantum mechanics, J. D. Bjorken and S. D. Drell, *Relativistic Quantum Mechanics* (McGraw-Hill Book Company, New York, 1964) and J. Sakurai, *Advanced Quantum Mechanics* (Addison-Wesley Publishing Company, Reading, Massachusetts, 1967). I occasionally refer in the notes to these books by author only. References to the Feynman Lectures are to R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics* (Addison-Wesley Publishing Company, Reading, Massachusetts, 1965), in three volumes.

Prerequisite to reading these notes is some familiarity with elementary quantum mechanics, at least at the undergraduate level; preferably the reader should already have met the uncertainty principle and the concept of a wave function. Mathematical prerequisites include sufficient acquaintance with complex variables to be able to do simple contour integrals and to understand words such as "poles" and "branch cuts." An elementary knowledge of Fourier transforms and series is necessary. I also assume an awareness of classical electrodynamics on the level of J. D. Jackson, *Classical Electrodynamics* (John Wiley and Sons, New York, 1962). Finally, I should mention that the figures in the notes are all sketches, not accurate graphs.

Among past and present colleagues I am particularly indebted, for their

patient criticisms, discussions, and comments, to R.W. Hellwarth, L.P. Kadanoff, and D.G. Ravenhall; and to J.D. Jackson, who read the notes through and made many helpful suggestions about revising for publication. I am grateful to Mrs. Marie Brosig who typed the original notes, to W. F. Saam for proofreading this volume, and to Nina Baym, for substantial editorial assistance.

Gordon B. Y. m.

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Chapter 1

Photon Polarization

1.1 Photon Polarization

To develop an intuitive understanding of the foundational concepts and mathematical formalism of quantum mechanics, we begin by studying one of the simplest and most accessible quantum mechanical systems: the polarization of light.

1.1.1 Classical Description of Polarized Light

Consider a classical monochromatic light wave propagating along the $+z$ -direction in free space. The wave is described by a transverse electric field vector:

$$\mathbf{E}(\mathbf{r}, t) = \begin{pmatrix} E_x(\mathbf{r}, t) \\ E_y(\mathbf{r}, t) \\ 0 \end{pmatrix}, \quad (1.1)$$

reflecting the transverse nature of electromagnetic waves in which both electric and magnetic fields are orthogonal to the propagation vector $\mathbf{k} = k\hat{\mathbf{z}}$. In Gaussian units, the associated magnetic field $\mathbf{H}$ satisfies $|\mathbf{H}| = |\mathbf{E}|$ and is given by

$$\mathbf{H} = \hat{\mathbf{z}} \times \mathbf{E}, \quad (1.2)$$

which implies the component relations $H_x = -E_y$ and $H_y = E_x$. Consequently, knowledge of $\mathbf{E}(\mathbf{r}, t)$ completely determines $\mathbf{H}(\mathbf{r}, t)$.

Since physical electric fields are real quantities, the components along the x - and y -axes can be expressed as

$$\begin{aligned} E_x(\mathbf{r}, t) &= E_x^0 \cos(kz - \omega t + \alpha_x), \\ E_y(\mathbf{r}, t) &= E_y^0 \cos(kz - \omega t + \alpha_y), \end{aligned} \quad (1.3)$$

where $k = 2\pi/\lambda$ is the wavenumber, ω is the angular frequency, E_x^0 and E_y^0 are real amplitudes, and α_x, α_y are the respective phase angles.

It is often mathematically convenient to employ complex representation. We define complex amplitudes $E_x = E_x^0 e^{i\alpha_x}$ and $E_y = E_y^0 e^{i\alpha_y}$, writing the physical fields as the real parts of complex wave functions:

$$E_x(\mathbf{r}, t) = \text{Re} \left[E_x e^{i(kz - \omega t)} \right], \quad E_y(\mathbf{r}, t) = \text{Re} \left[E_y e^{i(kz - \omega t)} \right]. \quad (1.4)$$

The polarization state of the wave is fully characterized by the relative amplitude and phase between E_x and E_y :

[(i)] **Linear x -polarization:** $E_y = 0$. The electric field oscillates exclusively along the x -axis. **Linear y -polarization:** $E_x = 0$. The electric field oscillates exclusively along the y -axis. **Linear polarization at 45° :** $E_x = E_y$. The field oscillates along the line $y = x$. **Right circular polarization (RCP):** $E_y = e^{i\pi/2} E_x = iE_x$. The y -component leads or lags the x -component by a quarter-cycle (90°). The field vector rotates clockwise when viewed looking into the oncoming wave:

$$\text{Re}[E_x(\mathbf{r}, t)] = E_0 \cos(kz - \omega t), \quad \text{Re}[E_y(\mathbf{r}, t)] = -E_0 \sin(kz - \omega t). \quad (1.5)$$

Left circular polarization (LCP): $E_y = -iE_x$. The field vector rotates counter-clockwise.

1.1.2 Field Energy and the Quantum Hypothesis

In Gaussian units, the instantaneous electromagnetic energy density is given by

$$\varepsilon(\mathbf{r}, t) = \frac{1}{8\pi} [(\text{Re } E_x)^2 + (\text{Re } E_y)^2 + (\text{Re } H_x)^2 + (\text{Re } H_y)^2]. \quad (1.6)$$

Using the relations $\text{Re } H_x = -\text{Re } E_y$ and $\text{Re } H_y = \text{Re } E_x$, this simplifies to

$$\begin{aligned} \varepsilon(\mathbf{r}, t) &= \frac{1}{4\pi} [(\text{Re } E_x)^2 + (\text{Re } E_y)^2] \\ &= \frac{1}{4\pi} [|E_x|^2 \cos^2(kz - \omega t + \alpha_x) + |E_y|^2 \cos^2(kz - \omega t + \alpha_y)]. \end{aligned} \quad (1.7)$$

Assuming the light wave occupies a spatial volume V spanning a large integer number of wavelengths along z , the spatial average of $\cos^2(kz - \omega t + \alpha)$ over the volume evaluates to $1/2$. The total electromagnetic energy contained within V is therefore

$$E_{\text{total}} = \int_V d^3r \varepsilon(\mathbf{r}, t) = \frac{V}{8\pi} (|E_x|^2 + |E_y|^2) = \frac{|\mathbf{E}|^2}{8\pi} V, \quad (1.8)$$

where $|\mathbf{E}|^2 \equiv |E_x|^2 + |E_y|^2$, and $|\mathbf{E}|^2/(8\pi)$ represents the time-averaged energy density.

Now consider what occurs when a light wave linearly polarized at 45° to the x -axis passes through a linear Polaroid filter whose transmission axis is aligned with the x -axis. Prior to entering the filter,

$$E_x = E_y = \frac{1}{\sqrt{2}} E. \quad (1.9)$$

After passing through the filter, the y -component is completely absorbed, leaving

$$E_x = \frac{1}{\sqrt{2}} E, \quad E_y = 0. \quad (1.10)$$

The transmitted beam is linearly polarized along the x -axis, and its total energy is reduced by a factor of two. A second identical filter placed downstream with its transmission axis also along x transmits the beam with no further reduction in intensity.

From a classical continuum viewpoint, the wave's amplitude is simply attenuated. However, quantum mechanics dictates that the energy of an electromagnetic field mode of frequency ω must be quantized in integer multiples of $\hbar\omega$:

$$E_{\text{total}} = N\hbar\omega, \quad (1.11)$$

where N is the number of photons present. Halving the total energy corresponds to halving the number of transmitted photons: on average, exactly half the incident photons emerge, while the other half are absorbed.

Since all incident photons are identical and encounter identical physical conditions at the filter, classical determinism fails. The only consistent physical interpretation is that each individual photon has an intrinsic *probability* of $1/2$ to pass through the filter. The probabilistic character of physical laws is thus directly mandated by the quantum postulate.

In an experiment with a finite number of photons, statistical fluctuations of order $\sqrt{N}$ occur about the mean transmitted count $N/2$. In the classical macroscopic limit ($N \rightarrow \infty$), the relative fluctuations vanish ($\sqrt{N}/N \rightarrow 0$), and deterministic behavior is recovered in accordance with the *correspondence principle*.

More generally, if a beam is linearly polarized at an arbitrary angle θ relative to the x -axis, the transmitted fraction of energy is

$$\frac{|E_x|^2}{|E_x|^2 + |E_y|^2} = \cos^2 \theta. \quad (1.12)$$

Quantum mechanically, each photon polarized along θ has a probability $P = \cos^2 \theta$ of passing through the x -polarizer. Furthermore, every photon that successfully passes emerges in a definite polarization state along the x -axis.

Similarly, an optical device (such as a birefringent prism or circular analyzer) that selectively transmits right circularly polarized light while rejecting left circularly polarized light decomposes an incident field into circular modes:

$$\mathbf{E} = \mathbf{E}_{\text{RCP}} + \mathbf{E}_{\text{LCP}}. \quad (1.13)$$

The fraction of energy transmitted is

$$\frac{|\mathbf{E}_{\text{RCP}}|^2}{|\mathbf{E}_{\text{RCP}}|^2 + |\mathbf{E}_{\text{LCP}}|^2}, \quad (1.14)$$

which represents the probability that an individual photon passes through the device and emerges with right circular polarization.

1.1.3 State Vectors and the Superposition Principle

Because any macroscopic light beam can be decomposed into an ensemble of single-photon states, we formulate the description directly for single photons. Setting $N = 1$ in Eq. (1.11), Eq. (1.8) gives

$$|\mathbf{E}|^2 V = 8\pi\hbar\omega. \quad (1.15)$$

We define the two-component *state vector* (or *ket*) $|\Psi\rangle$ for the polarization of a single photon as

$$|\Psi\rangle = \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix}, \quad (1.16)$$

with the dimensionless complex components

$$\psi_x = \sqrt{\frac{V}{8\pi\hbar\omega}} E_x, \quad \psi_y = \sqrt{\frac{V}{8\pi\hbar\omega}} E_y. \quad (1.17)$$

The vector $|\Psi\rangle$ belongs to a two-dimensional complex Hilbert space $\mathbb{C}^2$. By virtue of Eq. (1.15), the state vector satisfies the normalization condition:

$$|\psi_x|^2 + |\psi_y|^2 = 1. \quad (1.18)$$

Notice that the normalization and relative components are independent of the arbitrary quantization volume V . The state vector $|\Psi\rangle$ contains all physically accessible information regarding the polarization of the photon.

Canonical base states in this representation include:

$$|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{linear } x\text{-polarization}), \quad (1.19)$$

$$|y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{linear } y\text{-polarization}), \quad (1.20)$$

$$|R\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (\text{right circular polarization}), \quad (1.21)$$

$$|L\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (\text{left circular polarization}). \quad (1.22)$$

To each ket vector $|\Psi\rangle$, we associate a dual conjugate row vector (or *bra*) $\langle\Psi|$, defined by the Hermitian conjugate:

$$\langle\Psi| = |\Psi\rangle^\dagger = (\psi_x^* \quad \psi_y^*). \quad (1.23)$$

The inner product (scalar product) between a bra $\langle\Phi|$ and a ket $|\Psi\rangle$ is defined as

$$\langle\Phi|\Psi\rangle = \phi_x^* \psi_x + \phi_y^* \psi_y = \langle\Psi|\Phi\rangle^*. \quad (1.24)$$

In this notation, the normalization condition (1.18) becomes

$$\langle\Psi|\Psi\rangle = 1. \quad (1.25)$$

The base states are mutually orthonormal:

$$\langle x|x\rangle = \langle y|y\rangle = 1, \quad \langle x|y\rangle = 0, \quad (1.26)$$

and similarly,

$$\langle R|R\rangle = \langle L|L\rangle = 1, \quad \langle R|L\rangle = 0. \quad (1.27)$$

Because $\{|x\rangle, |y\rangle\}$ forms a complete orthonormal basis, any arbitrary polarization state can be expanded linearly as

$$|\Psi\rangle = \psi_x |x\rangle + \psi_y |y\rangle. \quad (1.28)$$

Taking the inner product of both sides with $\langle x|$ yields

$$\langle x|\Psi\rangle = \psi_x \langle x|x\rangle + \psi_y \langle x|y\rangle = \psi_x, \quad (1.29)$$

and likewise $\langle y|\Psi\rangle = \psi_y$. Thus, the state vector expansion can be written in the form of a completeness relation:

$$|\Psi\rangle = |x\rangle \langle x|\Psi\rangle + |y\rangle \langle y|\Psi\rangle = (|x\rangle \langle x| + |y\rangle \langle y|) |\Psi\rangle. \quad (1.30)$$

The circular polarization states $\{|R\rangle, |L\rangle\}$ form an alternative orthonormal basis:

$$|\Psi\rangle = |R\rangle \langle R|\Psi\rangle + |L\rangle \langle L|\Psi\rangle, \quad (1.31)$$

where the projection coefficients are

$$\langle R|\Psi\rangle = \frac{\psi_x - i\psi_y}{\sqrt{2}}, \quad \langle L|\Psi\rangle = \frac{\psi_x + i\psi_y}{\sqrt{2}}. \quad (1.32)$$

Equations (1.30) and (1.31) illustrate the quantum *superposition principle*: any state of polarization may be viewed as a coherent linear superposition of two orthogonal basis states.

1.1.4 Probability Amplitudes and Measurement

We can now formulate the measurement process in terms of the state vector formalism. In the classical analysis, the transmission probability through an x -polarizer is given by Eq. (1.12). In quantum terms, using the state components, this probability is

$$P(x) = \frac{|\psi_x|^2}{|\psi_x|^2 + |\psi_y|^2} = |\psi_x|^2 = |\langle x|\Psi\rangle|^2. \quad (1.33)$$

The complex number $\langle x|\Psi\rangle$ is called the *probability amplitude* for the photon in state $|\Psi\rangle$ to be detected in state $|x\rangle$, and its absolute square determines the detection probability.

Likewise, for a filter designed to transmit exclusively right circularly polarized photons, the probability of transmission is given by

$$P(R) = |\langle R|\Psi\rangle|^2. \quad (1.34)$$

More generally, if an ideal measurement apparatus selectively transmits photons in a state $|\Phi\rangle$ while rejecting all states orthogonal to $|\Phi\rangle$, the probability amplitude that an incident photon in state $|\Psi\rangle$ will pass through is given by the inner product:

$$\mathcal{A}(\Psi \rightarrow \Phi) = \langle \Phi|\Psi\rangle, \quad (1.35)$$

and the corresponding transition probability is

$$P(\Psi \rightarrow \Phi) = |\langle \Phi|\Psi\rangle|^2. \quad (1.36)$$

While the probability amplitude $\langle \Phi|\Psi\rangle$ depends on the overall global phases of $|\Psi\rangle$ and $|\Phi\rangle$, the physical probability $P(\Psi \rightarrow \Phi)$ is strictly invariant under global phase transformations $|\Psi\rangle \rightarrow e^{i\theta}|\Psi\rangle$.

1.2 Transformation of Bases

In the preceding discussion, we expressed the photon polarization state vector $|\Psi\rangle$ using the linear Cartesian basis $\{|x\rangle, |y\rangle\}$. However, the choice of basis is entirely arbitrary; physical predictions are invariant under unitary changes of basis. In this section, we develop the transformation laws connecting different representations of photon states and investigate the geometric and physical consequences of spatial rotations.

1.2.1 General Basis Transformations

Let $\{|x\rangle, |y\rangle\}$ be our original orthonormal basis, and let $\{|x'\rangle, |y'\rangle\}$ be an alternative orthonormal basis in the two-dimensional polarization Hilbert space. In the new basis, the state vector $|\Psi\rangle$ is represented by the column vector

$$|\Psi\rangle = \begin{pmatrix} \psi_{x'} \\ \psi_{y'} \end{pmatrix} = \begin{pmatrix} \langle x'|\Psi\rangle \\ \langle y'|\Psi\rangle \end{pmatrix}. \quad (1.37)$$

To relate the components in the new basis to those in the original basis, we insert the resolution of identity (completeness relation) $\mathbf{1} = |x\rangle\langle x| + |y\rangle\langle y|$:

$$\langle x'|\Psi\rangle = \langle x'|x\rangle\langle x|\Psi\rangle + \langle x'|y\rangle\langle y|\Psi\rangle, \quad (1.38)$$

$$\langle y'|\Psi\rangle = \langle y'|x\rangle\langle x|\Psi\rangle + \langle y'|y\rangle\langle y|\Psi\rangle. \quad (1.39)$$

In matrix form, this linear relation reads

$$\begin{pmatrix} \langle x'|\Psi\rangle \\ \langle y'|\Psi\rangle \end{pmatrix} = \begin{pmatrix} \langle x'|x\rangle & \langle x'|y\rangle \\ \langle y'|x\rangle & \langle y'|y\rangle \end{pmatrix} \begin{pmatrix} \langle x|\Psi\rangle \\ \langle y|\Psi\rangle \end{pmatrix}. \quad (1.40)$$

The matrix in Eq. (1.40) is the *transformation matrix* relating the two bases. Because both bases are orthonormal, this matrix is unitary ($U^\dagger U = \mathbf{1}$). Equation (1.40) is completely general and applies to any two orthonormal bases in $\mathbb{C}^2$.

1.2.2 Rotations in the Transverse Plane

As a concrete and fundamental example, consider a new Cartesian coordinate frame (x', y') obtained by rotating the original axes (x, y) about the z -axis through an angle θ in the counter-clockwise direction. The original basis vectors expressed in terms of the rotated basis vectors are

$$\begin{aligned} |x\rangle &= \cos \theta |x'\rangle - \sin \theta |y'\rangle, \\ |y\rangle &= \sin \theta |x'\rangle + \cos \theta |y'\rangle. \end{aligned} \quad (1.41)$$

The inner products between the new and old basis states are therefore

$$\begin{aligned} \langle x' | x \rangle &= \cos \theta, & \langle x' | y \rangle &= \sin \theta, \\ \langle y' | x \rangle &= -\sin \theta, & \langle y' | y \rangle &= \cos \theta. \end{aligned} \quad (1.42)$$

Hence, the transformation matrix, denoted by $\mathcal{R}(\theta)$, is given by

$$\mathcal{R}(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \quad (1.43)$$

and the transformation of components takes the form

$$\begin{pmatrix} \langle x' | \Psi \rangle \\ \langle y' | \Psi \rangle \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \langle x | \Psi \rangle \\ \langle y | \Psi \rangle \end{pmatrix}. \quad (1.44)$$

Equation (1.44) admits two complementary physical interpretations:

1. **Passive viewpoint:** The physical state $|\Psi\rangle$ remains unchanged, while the coordinate axes are rotated counter-clockwise by an angle θ . The vector on the left-hand side contains the components of $|\Psi\rangle$ relative to the new axes (x', y') .
2. **Active viewpoint:** The coordinate axes remain fixed, and the physical state vector itself is rotated clockwise by an angle θ to produce a new state $|\Psi'\rangle$:

$$|\Psi'\rangle = \mathcal{R}(\theta)|\Psi\rangle, \quad (1.45)$$

whose components in the original basis are identical to the components of $|\Psi\rangle$ in the rotated basis: $\langle x | \Psi' \rangle = \langle x' | \Psi \rangle$ and $\langle y | \Psi' \rangle = \langle y' | \Psi \rangle$.

1.2.3 Eigenvalues, Eigenvectors, and the Spin Operator

Under a general rotation $\mathcal{R}(\theta)$, the polarization state of a photon changes. We now seek the special states that remain form-invariant under rotation, changing at most by an overall phase factor:

$$\mathcal{R}(\theta)|\Psi\rangle = c|\Psi\rangle, \quad c \in \mathbb{C}. \quad (1.46)$$

Such states are the *eigenvectors* (or *eigenstates*) of $\mathcal{R}(\theta)$, and the scalars c are the corresponding *eigenvalues*.

To determine the eigensystem of $\mathcal{R}(\theta)$, we decompose the matrix in terms of the identity matrix $\mathbf{1}$ and the Pauli matrix σ_y :

$$\mathcal{R}(\theta) = \cos \theta \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + i \sin \theta \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \cos \theta \mathbf{1} + i \sin \theta \mathbf{S}, \quad (1.47)$$

where we have defined the Hermitian matrix $\mathbf{S}$ as

$$\mathbf{S} \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (1.48)$$

Because $\mathbf{S}^2 = \mathbf{1}$, Eq. (1.47) can be identified as the matrix exponential:

$$\mathcal{R}(\theta) = e^{i\theta\mathbf{S}}. \quad (1.49)$$

The eigenvalues λ of $\mathbf{S}$ satisfy

$$\mathbf{S}|\Psi\rangle = \lambda|\Psi\rangle \implies \mathbf{S}^2|\Psi\rangle = \lambda^2|\Psi\rangle. \quad (1.50)$$

Since $\mathbf{S}^2 = \mathbf{1}$, we have $\lambda^2 = 1$, which gives the two eigenvalues

$$\lambda = \pm 1. \quad (1.51)$$

The corresponding eigenvectors are obtained directly:

[(i)] For eigenvalue $\lambda = +1$:

$$|R\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \mathbf{S}|R\rangle = +|R\rangle. \quad (1.52)$$

This is precisely the state of right circular polarization (RCP). For eigenvalue $\lambda = -1$:

$$|L\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad \mathbf{S}|L\rangle = -|L\rangle. \quad (1.53)$$

This is the state of left circular polarization (LCP).

Applying $\mathcal{R}(\theta)$ to these states yields:

$$\mathcal{R}(\theta)|R\rangle = (\cos\theta + i\sin\theta)|R\rangle = e^{i\theta}|R\rangle, \quad (1.54)$$

$$\mathcal{R}(\theta)|L\rangle = (\cos\theta - i\sin\theta)|L\rangle = e^{-i\theta}|L\rangle. \quad (1.55)$$

Thus, circular polarization states are the eigenstates of spatial rotations around the propagation axis; upon rotation through an angle θ , their state vectors acquire only a geometric phase factor $e^{\pm i\theta}$.

Since $\{|R\rangle, |L\rangle\}$ forms a complete basis, any arbitrary state $|\Psi\rangle$ can be written as

$$|\Psi\rangle = |R\rangle\langle R|\Psi\rangle + |L\rangle\langle L|\Psi\rangle. \quad (1.56)$$

Under a rotation through θ , the state evolves according to

$$|\Psi\rangle \rightarrow \mathcal{R}(\theta)|\Psi\rangle = e^{i\theta}|R\rangle\langle R|\Psi\rangle + e^{-i\theta}|L\rangle\langle L|\Psi\rangle. \quad (1.57)$$

This confirms that $|R\rangle$ and $|L\rangle$ are the unique eigenstates of the rotation operator.

The operator $\mathbf{S}$ is the *spin operator* (or helicity operator in units of $\hbar$) for the photon along its propagation direction $\hat{\mathbf{z}}$. A right circularly polarized photon is in an eigenstate of spin with eigenvalue $+1$ (spin projection $+\hbar$), whereas a left circularly polarized photon possesses spin eigenvalue -1 (spin projection $-\hbar$). Any other polarization state represents a coherent quantum superposition of these two fundamental spin states.

1.3 Angular Momentum of the Photon

The spin operator introduced previously through rotational symmetry possesses a profound physical significance: $\hbar$ times the spin of a photon corresponds precisely to the component of its *angular momentum* along the direction of propagation. In this section, we derive this correspondence from classical electrodynamics and formalize the quantum description using Dirac notation.

1.3.1 Classical Electromagnetic Angular Momentum

In classical electrodynamics, the total angular momentum $\mathbf{J}$ stored in an electromagnetic field is given by the spatial integral of the moment of the Poynting vector:

$$\mathbf{J} = \frac{1}{4\pi c} \int d^3r \, \mathbf{r} \times [\mathbf{E}(\mathbf{r}, t) \times \mathbf{H}(\mathbf{r}, t)]. \quad (1.58)$$

For an idealized monochromatic plane wave of infinite extent propagating along the z -axis, the fields $\mathbf{E}$ and $\mathbf{H}$ lie strictly in the transverse xy -plane. The cross product $\mathbf{E} \times \mathbf{H}$ points purely along $\hat{\mathbf{z}}$, causing $\mathbf{r} \times (\mathbf{E} \times \mathbf{H})$ to lie entirely in the transverse plane, so that naively $J_z = 0$.

However, any physical light beam has a finite transverse cross-section. Near the lateral boundary of the beam, the condition $\nabla \cdot \mathbf{E} = 0$ requires the electric and magnetic fields to possess small longitudinal components. Consequently, $\mathbf{E} \times \mathbf{H}$ develops transverse components near the beam surface, which when crossed into $\mathbf{r}$ yield a nonzero contribution to J_z that scales with the beam radius times its surface area, yielding an extensive volume contribution.

To isolate this longitudinal angular momentum, we express the fields in the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$, $\phi = 0$ in the absence of free charges) in terms of the transverse vector potential $\mathbf{A}(\mathbf{r}, t)$:

$$\mathbf{H}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t), \quad \mathbf{E}(\mathbf{r}, t) = -\frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t}. \quad (1.59)$$

Substituting $\mathbf{H} = \nabla \times \mathbf{A}$ into Eq. (1.58), using vector identities, and integrating by parts under the assumption of localized fields (such that surface boundary terms at infinity vanish), the total angular momentum separates cleanly into two distinct contributions:

$$\mathbf{J} = \frac{1}{4\pi c} \int d^3r \, [\mathbf{E}(\mathbf{r}, t) \times \mathbf{A}(\mathbf{r}, t)] + \frac{1}{4\pi c} \int d^3r \sum_{i=x,y,z} E_i(\mathbf{r}, t) (\mathbf{r} \times \nabla) A_i(\mathbf{r}, t). \quad (1.60)$$

The first term represents the *intrinsic spin angular momentum* $\mathbf{S}_{\text{field}}$, which depends directly on polarization. The second term represents the *orbital angular momentum* $\mathbf{L}_{\text{field}}$. For a collimated beam traveling along z , the z -component of the orbital operator is $(\mathbf{r} \times \nabla)_z = \partial/\partial\varphi$ in cylindrical coordinates. For a wave with no azimuthal phase variation (such as a standard plane-wave-like beam), this orbital contribution vanishes or is localized solely to the beam edge without contributing an extensive volume term.

Hence, the dominant extensive longitudinal angular momentum is given by the spin term:

$$J_z = \frac{1}{4\pi c} \int d^3r \, [E_x(\mathbf{r}, t) A_y(\mathbf{r}, t) - E_y(\mathbf{r}, t) A_x(\mathbf{r}, t)]. \quad (1.61)$$

In terms of the real electric field amplitudes and phases, the vector potential components are

$$\begin{aligned} A_x(\mathbf{r}, t) &= \frac{c}{\omega} E_x^0 \sin(kz - \omega t + \alpha_x), \\ A_y(\mathbf{r}, t) &= \frac{c}{\omega} E_y^0 \sin(kz - \omega t + \alpha_y). \end{aligned} \quad (1.62)$$

Substituting these into the integral and averaging over oscillations yields

$$\begin{aligned} J_z &= \frac{1}{4\pi\omega} E_x^0 E_y^0 \int d^3r \sin(\alpha_y - \alpha_x) \\ &= \frac{V}{4\pi\omega} E_x^0 E_y^0 \sin(\alpha_y - \alpha_x). \end{aligned} \quad (1.63)$$

We rewrite Eq. (1.63) in terms of the complex field amplitudes $E_x = E_x^0 e^{i\alpha_x}$ and $E_y = E_y^0 e^{i\alpha_y}$:

$$\begin{aligned} J_z &= \frac{V}{8\pi i\omega} (E_x^* E_y - E_x E_y^*) \\ &= \frac{V}{8\pi\omega} \left(\left| \frac{E_x - iE_y}{\sqrt{2}} \right|^2 - \left| \frac{E_x + iE_y}{\sqrt{2}} \right|^2 \right). \end{aligned} \quad (1.64)$$

Recognizing the right- and left-circularly polarized field components:

$$E_{\text{RCP}} \equiv \frac{E_x - iE_y}{\sqrt{2}}, \quad E_{\text{LCP}} \equiv \frac{E_x + iE_y}{\sqrt{2}}, \quad (1.65)$$

we obtain the compact classical relation:

$$J_z = \frac{V}{8\pi\omega} (|E_{\text{RCP}}|^2 - |E_{\text{LCP}}|^2). \quad (1.66)$$

1.3.2 Quantum Interpretation and Expectation Values

To transition to the quantum mechanical description, we consider a single photon ($N = 1$). Expressing the electric fields in terms of the single-photon state vector components [Eq. (1.17)], Eq. (1.66) becomes

$$\begin{aligned} J_z &= \hbar \left(\left| \frac{\psi_x - i\psi_y}{\sqrt{2}} \right|^2 - \left| \frac{\psi_x + i\psi_y}{\sqrt{2}} \right|^2 \right) \\ &= \hbar |\langle R | \Psi \rangle|^2 - \hbar |\langle L | \Psi \rangle|^2. \end{aligned} \quad (1.67)$$

Experimentally, when a photon propagating along the z -direction is absorbed by an atom or macroscopic absorber, the z -component of angular momentum transferred to the matter is always observed to be quantized: it changes by either $+\hbar$ or $-\hbar$. It is never zero, and it never assumes any other value. For an arbitrary polarization state $|\Psi\rangle$, the outcome of an individual measurement cannot be predicted deterministically. Eq. (1.67) must therefore be interpreted as the statistical *expectation value* $\langle J_z \rangle$ over an ensemble of identically prepared single photons:

$$\langle J_z \rangle = (+\hbar)P(+1) + (-\hbar)P(-1), \quad (1.68)$$

where:

$$P(+1) = |\langle R | \Psi \rangle|^2 \quad (\text{probability of measuring } J_z = +\hbar), \quad (1.69)$$

$$P(-1) = |\langle L | \Psi \rangle|^2 \quad (\text{probability of measuring } J_z = -\hbar). \quad (1.70)$$

If the photon is in a pure right-circular state $|R\rangle$, then $P(+1) = 1$ and $J_z = +\hbar$ with certainty. If it is in a pure left-circular state $|L\rangle$, then $P(-1) = 1$ and $J_z = -\hbar$ with certainty. In general states, angular momentum exhibits quantum fluctuations.

In quantum mechanics, physical observables are represented by linear Hermitian operators acting on Hilbert space:

$$\hat{J}_z = \hbar \mathbf{S} = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (1.71)$$

The measurable outcomes of an observable correspond precisely to the eigenvalues of its associated operator ($\pm\hbar$). States with definite values of an observable are its eigenstates ($|R\rangle$ and $|L\rangle$). For an arbitrary state $|\Psi\rangle$, the expectation value is given by the sandwich product:

$$\langle J_z \rangle = \langle \Psi | \hat{J}_z | \Psi \rangle = \hbar \langle \Psi | \mathbf{S} | \Psi \rangle. \quad (1.72)$$

1.3.3 Dirac Notation: Outer Products and Spectral Decomposition

The manipulation of operators and states is rendered especially transparent through the use of Dirac bra-ket notation. We define the *outer product* $|\Psi\rangle\langle\Phi|$ of two vectors $|\Psi\rangle$ and $|\Phi\rangle$ as the linear operator whose matrix representation in the $\{|x\rangle, |y\rangle\}$ basis is

$$|\Psi\rangle\langle\Phi| \equiv \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix} \begin{pmatrix} \phi_x^* & \phi_y^* \end{pmatrix} = \begin{pmatrix} \psi_x\phi_x^* & \psi_x\phi_y^* \\ \psi_y\phi_x^* & \psi_y\phi_y^* \end{pmatrix}. \quad (1.73)$$

For example, the projection and transition operators in the Cartesian basis are

$$\begin{aligned} |x\rangle\langle x| &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & |y\rangle\langle y| &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \\ |x\rangle\langle y| &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, & |y\rangle\langle x| &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \end{aligned} \quad (1.74)$$

An outer product operator acts associatively on a ket $|\Theta\rangle$:

$$(|\Psi\rangle\langle\Phi|)|\Theta\rangle = |\Psi\rangle(\langle\Phi|\Theta\rangle), \quad (1.75)$$

yielding a state proportional to $|\Psi\rangle$ scaled by the scalar inner product $\langle\Phi|\Theta\rangle$.

Using outer products, the completeness of an orthonormal basis $\{|x\rangle, |y\rangle\}$ is expressed succinctly by the *resolution of identity*:

$$|x\rangle\langle x| + |y\rangle\langle y| = \mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (1.76)$$

Operating with Eq. (1.76) on an arbitrary ket $|\Psi\rangle$ immediately reproduces the basis expansion:

$$|\Psi\rangle = |x\rangle\langle x|\Psi\rangle + |y\rangle\langle y|\Psi\rangle. \quad (1.77)$$

Similarly, the circular basis $\{|R\rangle, |L\rangle\}$ satisfies

$$|R\rangle\langle R| + |L\rangle\langle L| = \mathbf{1}. \quad (1.78)$$

Multiplying Eq. (1.78) on the left by the spin operator $\mathbf{S}$ and using the eigenvalue equations $\mathbf{S}|R\rangle = +|R\rangle$ and $\mathbf{S}|L\rangle = -|L\rangle$, we obtain the *spectral decomposition* of $\mathbf{S}$:

$$\mathbf{S} = (+1)|R\rangle\langle R| + (-1)|L\rangle\langle L| = |R\rangle\langle R| - |L\rangle\langle L|. \quad (1.79)$$

Taking the expectation value of this operator decomposition directly yields Eq. (1.72):

$$\langle\Psi|\mathbf{S}|\Psi\rangle = \langle\Psi|R\rangle\langle R|\Psi\rangle - \langle\Psi|L\rangle\langle L|\Psi\rangle = |\langle R|\Psi\rangle|^2 - |\langle L|\Psi\rangle|^2. \quad (1.80)$$

Finally, for any linear operator M acting on this Hilbert space, its matrix elements in an arbitrary orthonormal basis $\{|i\rangle, |j\rangle\}$ are given by

$$M_{ij} = \langle i|M|j\rangle. \quad (1.81)$$

For instance, in the Cartesian basis:

$$M = \begin{pmatrix} \langle x|M|x\rangle & \langle x|M|y\rangle \\ \langle y|M|x\rangle & \langle y|M|y\rangle \end{pmatrix}, \quad (1.82)$$

and likewise in the circular basis, the off-diagonal matrix element M_{LR} is $\langle L|M|R\rangle$.

1.4 Amplitude Mechanics

In the quantum mechanical description of photon polarization, the inner product $\langle\Phi|\Psi\rangle$ plays a central role: its absolute square $|\langle\Phi|\Psi\rangle|^2$ represents the probability that a photon prepared in state $|\Psi\rangle$ will be found upon measurement to behave as if it were in state $|\Phi\rangle$. This probabilistic behavior arises directly from the superposition principle:

$$|\Psi\rangle = |R\rangle\langle R|\Psi\rangle + |L\rangle\langle L|\Psi\rangle. \quad (1.83)$$

In classical mechanics, a system residing in a definite state (e.g., at position x with velocity v) never exhibits the properties of a distinct physical state. In quantum mechanics, by contrast, a particle prepared in state $|\Psi\rangle$ possesses a nonvanishing probability $|\langle\Phi|\Psi\rangle|^2$ of exhibiting the physical characteristics of state $|\Phi\rangle$, vanishing only if the states are strictly orthogonal ($\langle\Phi|\Psi\rangle = 0$).

1.4.1 Quantum Superposition versus Classical Mixtures

Because a measurement of the longitudinal angular momentum J_z always yields either $+\hbar$ or $-\hbar$, one might naively suppose that light is simply an incoherent statistical mixture: that each individual photon is either in state $|R\rangle$ with probability w_R , or in state $|L\rangle$ with probability $w_L = 1 - w_R$.

This classical assumption leads to immediate contradictions with experiment. Consider a photon linearly polarized along the x -axis, $|x\rangle$. If this state were an incoherent 50/50 mixture of $|R\rangle$ and $|L\rangle$:

[(a)]The photon would have probability $|\langle R|x\rangle|^2 = 1/2$ of being right-circularly polarized, and an $|R\rangle$ photon has probability $|\langle y|R\rangle|^2 = 1/2$ of passing through a y -polarizer. The photon would have probability $|\langle L|x\rangle|^2 = 1/2$ of being left-circularly polarized, and an $|L\rangle$ photon has probability $|\langle y|L\rangle|^2 = 1/2$ of passing through a y -polarizer.

Summing these mutually exclusive classical probabilities would yield a transmission probability through a y -polarizer of

$$P_{\text{classical}}(y) = |\langle R|x\rangle|^2 |\langle y|R\rangle|^2 + |\langle L|x\rangle|^2 |\langle y|L\rangle|^2 = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}. \quad (1.84)$$

Yet experimentally, an x -polarized beam *never* passes through an ideal y -polarizer: $P(y) = 0$.

The classical reasoning fails because it omits the *quantum interference* between probability amplitudes. The quantum calculation proceeds by expanding the state vector:

$$|x\rangle = |R\rangle\langle R|x\rangle + |L\rangle\langle L|x\rangle, \quad (1.85)$$

so that the total transition amplitude for passing through the y -polarizer is

$$\langle y|x\rangle = \langle y|R\rangle\langle R|x\rangle + \langle y|L\rangle\langle L|x\rangle. \quad (1.86)$$

The true quantum probability is the absolute square of the total amplitude:

$$\begin{aligned} P(y) &= |\langle y|R\rangle\langle R|x\rangle + \langle y|L\rangle\langle L|x\rangle|^2 \\ &= |\langle y|R\rangle|^2 |\langle R|x\rangle|^2 + |\langle y|L\rangle|^2 |\langle L|x\rangle|^2 \\ &\quad + 2 \operatorname{Re} [\langle y|R\rangle\langle R|x\rangle\langle y|L\rangle^* \langle L|x\rangle^*]. \end{aligned} \quad (1.87)$$

Evaluating the inner products with $|x\rangle = \frac{1}{\sqrt{2}}(|R\rangle + |L\rangle)$ and $|y\rangle = -\frac{i}{\sqrt{2}}(|R\rangle - |L\rangle)$:

$$\langle R|x\rangle = \frac{1}{\sqrt{2}}, \quad \langle L|x\rangle = \frac{1}{\sqrt{2}}, \quad \langle y|R\rangle = \frac{i}{\sqrt{2}}, \quad \langle y|L\rangle = -\frac{i}{\sqrt{2}}. \quad (1.88)$$

The individual path amplitudes are therefore

$$\langle y|R\rangle\langle R|x\rangle = \frac{i}{2}, \quad \langle y|L\rangle\langle L|x\rangle = -\frac{i}{2}. \quad (1.89)$$

The sum of the amplitudes vanishes identically:

$$\langle y|x\rangle = \frac{i}{2} - \frac{i}{2} = 0 \implies P(y) = 0. \quad (1.90)$$

In Eq. (1.87), the classical sum of probabilities evaluates to $+1/2$, while the quantum cross-term evaluates to $-1/2$, producing destructive interference that suppresses the transition entirely.

1.4.2 The Fundamental Rules of Amplitude Mechanics

This illustrates the general principles of quantum probability (the Feynman amplitude calculus):

2. **Sequential Processes (Multiplication Rule):** The probability amplitude for a sequence of successive events is the product of the individual transition amplitudes. For example, the amplitude to start in $|x\rangle$, intermediate through $|R\rangle$, and end in $|y\rangle$ is $\langle y|R\rangle\langle R|x\rangle$.
2. **Indistinguishable Alternatives (Addition Rule):** When an event can occur via several indistinguishable, mutually coherent quantum pathways, the total probability amplitude is the linear sum of the amplitudes for each path:

$$\mathcal{A}_{\text{total}} = \sum_j \mathcal{A}_j. \quad (1.91)$$

3. **Probability Law (Born Rule):** The physical probability of the event is the absolute square of the total amplitude:

$$P = |\mathcal{A}_{\text{total}}|^2 = \left| \sum_j \mathcal{A}_j \right|^2. \quad (1.92)$$

Probabilities themselves are never added unless the alternative paths are physically distinguishable in principle.

1.4.3 Distinguishability, Complementarity, and the Uncertainty Principle

The coherence condition in Rule 2 hinges on the paths being strictly *indistinguishable*. If an apparatus detects which pathway a photon traverses, the interference pattern is destroyed.

Consider an idealized thought experiment (Fig. 1-4): a birefringent prism A splits an incident beam into right- and left-circularly polarized components. Each component is directed along separate spatial paths via mirrors M_1, M_2, M_3 (for $|R\rangle$) and M_4, M_5, M_6 (for $|L\rangle$), then recombined by an identical prism B before impinging on a linear analyzer.

To obtain which-path information, suppose mirror M_2 (or M_4) is mounted on a delicate spring to detect recoil. A photon of wavelength λ has linear momentum

$$p = \frac{2\pi\hbar}{\lambda} = \frac{h}{\lambda}. \quad (1.93)$$

Upon specular reflection at normal incidence, the photon reverses direction, imparting a momentum transfer $\Delta p_{\text{transfer}} = 2p$ to the mirror. To detect whether the photon traversed the $|R\rangle$ path without ambiguity, the initial vertical momentum uncertainty of the mirror Δp_y must satisfy

$$\Delta p_y < 2p = \frac{4\pi\hbar}{\lambda}. \quad (1.94)$$

According to the Heisenberg uncertainty principle,

$$\Delta y \Delta p_y \geq \frac{\hbar}{2} \quad (\text{or by Bohr's order-of-magnitude estimate, } \Delta y \Delta p_y \gtrsim h = 2\pi\hbar). \quad (1.95)$$

Substituting the requirement $\Delta p_y < 2p$ yields an irreducible uncertainty in the mirror's spatial coordinate:

$$\Delta y > \frac{2\pi\hbar}{2p} = \frac{\lambda}{2}. \quad (1.96)$$

Because the photon travels to and from the mirror, an uncertainty Δy in mirror position produces an uncertainty in the optical path length of

$$\Delta\ell = 2\Delta y > \lambda. \quad (1.97)$$

This path length fluctuation introduces a random optical phase shift $\Delta\alpha \sim k\Delta\ell > 2\pi$.

Mathematically, let the phase acquired along the right and left paths be α_R and α_L , respectively. The state emerging from prism B is

$$|\Phi'\rangle = e^{i\alpha_R}|R\rangle\langle R|\Phi\rangle + e^{i\alpha_L}|L\rangle\langle L|\Phi\rangle. \quad (1.98)$$

The probability of detecting this state through an x -polarizer is

$$P(x) = |\langle x|R\rangle|^2|\langle R|\Phi\rangle|^2 + |\langle x|L\rangle|^2|\langle L|\Phi\rangle|^2 + 2\operatorname{Re}\left[\langle x|R\rangle\langle R|\Phi\rangle\langle x|L\rangle^*\langle L|\Phi\rangle^*e^{i(\alpha_R-\alpha_L)}\right]. \quad (1.99)$$

Because $\Delta(\alpha_R - \alpha_L) \geq 2\pi$, the phase difference varies uncontrollably from shot to shot. Averaging over all relative phases uniformly distributed on $[0, 2\pi)$:

$$\int_0^{2\pi} \frac{d\theta}{2\pi} e^{i\theta} = 0, \quad (1.100)$$

the interference cross-term vanishes identically. The observed probability reduces to the incoherent classical sum:

$$\langle P(x) \rangle = |\langle x|R\rangle|^2|\langle R|\Phi\rangle|^2 + |\langle x|L\rangle|^2|\langle L|\Phi\rangle|^2. \quad (1.101)$$

The destruction of quantum interference does not require a conscious observer; it is an inevitable consequence of preparing a measurement apparatus whose quantum uncertainty is small enough to distinguish the alternative paths. Bohr's principle of complementarity is thus strictly enforced by the quantum mechanical uncertainty relations.

1.5 Unpolarized Light and Mixed States

Thus far, our discussion has focused exclusively on photons occupying definite, well-defined polarization states. However, typical natural and laboratory sources (such as thermal incandescent lamps or gas discharge tubes) produce *unpolarized* light. To describe unpolarized and partially polarized beams, we must expand our theoretical framework from pure states to statistical ensembles, introducing the fundamental distinction between pure states and mixed states.

1.5.1 Pure States versus Mixed States

Consider a monochromatic light beam generated by two independent, uncorrelated sources emitting photons randomly without phase coherence. Suppose Source 1 emits photons in state $|\Psi_1\rangle$ with statistical probability p_1 , while Source 2 emits photons in state $|\Psi_2\rangle$ with probability p_2 , where $p_1 + p_2 = 1$. For any individual photon, we cannot know a priori which source emitted it.

To compute the ensemble average of a physical observable, such as the longitudinal angular momentum $J_z = \hbar\mathbf{S}$, we first determine the probabilities of measuring outcomes $\pm\hbar$:

$$P(+\hbar) = p_1|\langle R|\Psi_1\rangle|^2 + p_2|\langle R|\Psi_2\rangle|^2, \quad (1.102)$$

$$P(-\hbar) = p_1|\langle L|\Psi_1\rangle|^2 + p_2|\langle L|\Psi_2\rangle|^2. \quad (1.103)$$

The ensemble average value $\langle J_z \rangle_{\text{ensemble}}$ is then

$$\begin{aligned} \langle J_z \rangle_{\text{ensemble}} &= (+\hbar)P(+\hbar) + (-\hbar)P(-\hbar) \\ &= p_1 \hbar (|\langle R | \Psi_1 \rangle|^2 - |\langle L | \Psi_1 \rangle|^2) + p_2 \hbar (|\langle R | \Psi_2 \rangle|^2 - |\langle L | \Psi_2 \rangle|^2) \\ &= p_1 \langle J_z \rangle_1 + p_2 \langle J_z \rangle_2 \\ &= p_1 \langle \Psi_1 | \hbar \mathbf{S} | \Psi_1 \rangle + p_2 \langle \Psi_2 | \hbar \mathbf{S} | \Psi_2 \rangle. \end{aligned} \quad (1.104)$$

The expectation value for the beam is the weighted statistical average of the quantum expectation values corresponding to the constituent states.

It is crucial to recognize that the statement “a photon is either in state $|\Psi_1\rangle$ with probability p_1 or in state $|\Psi_2\rangle$ with probability p_2 ” (a **mixed state**) is physically distinct from the statement “a photon is in a coherent linear superposition of $|\Psi_1\rangle$ and $|\Psi_2\rangle$ ” (a **pure state**):

$$|\Phi\rangle = \alpha|\Psi_1\rangle + \beta|\Psi_2\rangle. \quad (1.105)$$

In a pure coherent superposition, the relative phase between α and β is deterministic and physically meaningful. For example, if $|\Psi_1\rangle = |x\rangle$ and $|\Psi_2\rangle = |y\rangle$, choosing $\alpha = 1/\sqrt{2}$ and $\beta = i/\sqrt{2}$ produces a pure right-circularly polarized state $|R\rangle$, whereas choosing $\beta = -i/\sqrt{2}$ produces a pure left-circularly polarized state $|L\rangle$, which is orthogonal to $|R\rangle$. By contrast, in a statistical mixture, there is no phase relationship whatsoever between the states, completely precluding quantum interference.

Quantum mechanics therefore encompasses two distinct layers of probability:

1. **Quantum Probability (Pure States):** Governed by complex probability amplitudes and the superposition principle, exhibiting quantum interference.
2. **Classical Statistical Probability (Mixed States):** Arising from incomplete classical knowledge about the state of preparation, governed by standard probability weighting over quantum expectation values.

1.5.2 Phase Randomization and Decoherence

The connection between coherent superpositions and statistical mixtures becomes evident when considering phase fluctuations. For the pure superposition state $|\Phi\rangle$ in Eq. (1.105), the expectation value of J_z is

$$\begin{aligned} \langle J_z \rangle &= \langle \Phi | \hbar \mathbf{S} | \Phi \rangle \\ &= |\alpha|^2 \langle \Psi_1 | \hbar \mathbf{S} | \Psi_1 \rangle + |\beta|^2 \langle \Psi_2 | \hbar \mathbf{S} | \Psi_2 \rangle \\ &\quad + \alpha^* \beta \langle \Psi_1 | \hbar \mathbf{S} | \Psi_2 \rangle + \beta^* \alpha \langle \Psi_2 | \hbar \mathbf{S} | \Psi_1 \rangle. \end{aligned} \quad (1.106)$$

The first two terms represent the classical weighted average corresponding to probabilities $p_1 = |\alpha|^2$ and $p_2 = |\beta|^2$. The remaining two terms are quantum interference terms that depend explicitly on the relative phase:

$$\alpha = |\alpha|e^{i\varphi_1}, \quad \beta = |\beta|e^{i\varphi_2} \implies \alpha^* \beta = |\alpha||\beta|e^{-i(\varphi_1 - \varphi_2)}. \quad (1.107)$$

If the relative phase $\Delta\varphi = \varphi_1 - \varphi_2$ fluctuates randomly and uniformly over $[0, 2\pi)$ across the ensemble of photons, its statistical average vanishes:

$$\langle \alpha^* \beta \rangle_{\text{phase}} = |\alpha||\beta| \int_0^{2\pi} \frac{d(\Delta\varphi)}{2\pi} e^{-i\Delta\varphi} = 0. \quad (1.108)$$

The cross-terms cancel identically, and the pure-state expectation value (1.106) reduces precisely to the statistical mixture result (1.104). In modern terminology, phase randomization induces *decoherence*, transforming an entangled or coherent superposition into an incoherent statistical mixture.

1.5.3 Quantum Mechanical Definition of Unpolarized Light

We can now provide an exact operational definition: *Unpolarized light is light that has an equal probability of being found in any polarization state.* Its state is an isotropic mixed state.

Remarkably, this description is entirely equivalent to a much simpler specification: *An incoherent 50/50 statistical mixture of photons in any two mutually orthogonal states $\{|\Psi_1\rangle, |\Psi_2\rangle\}$ is indistinguishable from unpolarized light.*

To demonstrate this equivalence, consider averaging the expectation value of an arbitrary observable $\hat{A}$ over all possible polarization states:

$$|\Phi\rangle = \alpha|\Psi_1\rangle + \beta|\Psi_2\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (1.109)$$

Averaging over the uniform distribution of states involves:

1. Averaging over relative phases, which eliminates all off-diagonal cross-terms: $\langle\alpha^*\beta\rangle = 0$.
2. Averaging over the amplitudes $|\alpha|^2$ and $|\beta|^2$. By rotational symmetry in state space, $\langle|\alpha|^2\rangle = \langle|\beta|^2\rangle$. Because $|\alpha|^2 + |\beta|^2 = 1$, we must have

$$\langle|\alpha|^2\rangle = \langle|\beta|^2\rangle = \frac{1}{2}. \quad (1.110)$$

Thus, for any observable $\hat{A}$, the isotropic ensemble average evaluates to

$$\langle A \rangle_{\text{unpolarized}} = \frac{1}{2}\langle\Psi_1|\hat{A}|\Psi_1\rangle + \frac{1}{2}\langle\Psi_2|\hat{A}|\Psi_2\rangle. \quad (1.111)$$

This is identically the expectation value of an incoherent equal mixture of the two orthonormal states $|\Psi_1\rangle$ and $|\Psi_2\rangle$.

1.5.4 Basis Independence

The ensemble average in Eq. (1.111) is completely independent of the choice of orthonormal basis. Let $\{|\Phi_1\rangle, |\Phi_2\rangle\}$ be any other orthonormal basis. Using the resolution of identity $\mathbf{1} = |\Phi_1\rangle\langle\Phi_1| + |\Phi_2\rangle\langle\Phi_2|$, we obtain:

$$\begin{aligned} \langle A \rangle &= \frac{1}{2} \sum_{i=1,2} \langle\Psi_i|\hat{A}|\Psi_i\rangle \\ &= \frac{1}{2} \sum_{i=1,2} \sum_{j=1,2} \langle\Psi_i|\hat{A}|\Phi_j\rangle\langle\Phi_j|\Psi_i\rangle \\ &= \frac{1}{2} \sum_{j=1,2} \langle\Phi_j|\hat{A} \left(\sum_{i=1,2} |\Psi_i\rangle\langle\Psi_i| \right) |\Phi_j\rangle \\ &= \frac{1}{2} \sum_{j=1,2} \langle\Phi_j|\hat{A}|\Phi_j\rangle = \frac{1}{2} \text{Tr}(\hat{A}). \end{aligned} \quad (1.112)$$

In terms of the density operator formalism, unpolarized light is described by the maximally mixed density matrix proportional to the identity operator:

$$\rho_{\text{unpolarized}} = \frac{1}{2}|x\rangle\langle x| + \frac{1}{2}|y\rangle\langle y| = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}\mathbf{1}. \quad (1.113)$$

Consequently, an unpolarized beam cannot be distinguished experimentally from an equal mixture of horizontal and vertical linear polarizations, an equal mixture of right- and left-circular polarizations, or an equal mixture along any pair of orthogonal axes.

1.6 Behavior of Photon Polarization in Matter

In preceding sections, we examined optical elements such as polarizers and circular analyzers that selectively transmit photons in specific polarization states while absorbing or deflecting orthogonal states. We now consider transparent anisotropic media—such as birefringent crystals—which transmit essentially all incident photons without absorption, but continuously transform their polarization state as they propagate.

1.6.1 Birefringence and the Evolution Operator

A classic example of an anisotropic medium is a uniaxial birefringent crystal, such as calcite (CaCO_3). Calcite possesses an optical symmetry axis known as the *optic axis*. The crystal exhibits distinct refractive indices for light polarized parallel and perpendicular to the optic axis:

- The **ordinary ray** (*o*-ray), polarized perpendicular to the optic axis, experiences refractive index n_o .
- The **extraordinary ray** (*e*-ray), polarized parallel to the optic axis, experiences refractive index n_e .

In calcite at optical wavelengths, $n_o > n_e$ (by approximately 10%). The phase velocity of the extraordinary ray, $v_e = c/n_e$, exceeds that of the ordinary ray, $v_o = c/n_o$. Consequently, as a photon traverses the crystal along the z -axis (with the optic axis oriented in the transverse xy -plane), a progressive phase difference accumulates between the ordinary and extraordinary polarization components.

Let $|o\rangle$ and $|e\rangle$ denote the orthonormal polarization base states corresponding to the ordinary and extraordinary rays. For a monochromatic beam of angular frequency ω , the respective wavenumbers are

$$k_e = \frac{\omega n_e}{c}, \quad k_o = \frac{\omega n_o}{c}. \quad (1.114)$$

Over a crystal of propagation length ℓ , the phase accumulated by an extraordinary wave is $k_e \ell$, while an ordinary wave accumulates $k_o \ell$. Thus, the crystal maps the basis states according to

$$|e\rangle \rightarrow e^{ik_e \ell} |e\rangle, \quad |o\rangle \rightarrow e^{ik_o \ell} |o\rangle. \quad (1.115)$$

An arbitrary incident photon state $|\Psi_{\text{in}}\rangle$ expanded in the $\{|e\rangle, |o\rangle\}$ basis:

$$|\Psi_{\text{in}}\rangle = |e\rangle \langle e | \Psi_{\text{in}} \rangle + |o\rangle \langle o | \Psi_{\text{in}} \rangle, \quad (1.116)$$

emerges from the crystal in the transformed state

$$\begin{aligned} |\Psi_{\text{out}}\rangle &= e^{ik_e \ell} |e\rangle \langle e | \Psi_{\text{in}} \rangle + e^{ik_o \ell} |o\rangle \langle o | \Psi_{\text{in}} \rangle \\ &= U_\ell |\Psi_{\text{in}}\rangle, \end{aligned} \quad (1.117)$$

where the *evolution operator* U_ℓ is given by

$$U_\ell = e^{ik_e \ell} |e\rangle \langle e| + e^{ik_o \ell} |o\rangle \langle o|. \quad (1.118)$$

1.6.2 Transition Amplitudes and Probabilities

The complex amplitude for a photon entering the medium in state $|\Psi\rangle$ to emerge in state $|\Phi\rangle$ is the *transition amplitude*:

$$\mathcal{A}(\Psi \rightarrow \Phi) = \langle \Phi | U_\ell | \Psi \rangle. \quad (1.119)$$

The corresponding physical probability that the transmitted photon behaves upon measurement as state $|\Phi\rangle$ is the *transition probability*:

$$P(\Psi \rightarrow \Phi) = |\langle \Phi | U_\ell | \Psi \rangle|^2. \quad (1.120)$$

This concept of an evolution operator mediating transitions between initial and final quantum states is a cornerstone of quantum dynamics.

1.6.3 Spatial Evolution: The Wavenumber Operator

To understand the continuous evolution of the state vector as a function of propagation distance z , we write

$$|\Psi(z)\rangle = U_z |\Psi_{\text{in}}\rangle. \quad (1.121)$$

Because spatial translation is continuous and homogeneous, the evolution operator satisfies the group composition property:

$$U_{z+a} = U_a U_z = U_z U_a. \quad (1.122)$$

Hence, the state at $z + a$ is related to the state at z by

$$|\Psi(z + a)\rangle = U_a |\Psi(z)\rangle. \quad (1.123)$$

For an infinitesimal step $a \ll \lambda$ (such that $k_o a \ll 1$ and $k_e a \ll 1$), we expand U_a to first order in a :

$$\begin{aligned} U_a &= (1 + ik_e a) |e\rangle\langle e| + (1 + ik_o a) |o\rangle\langle o| + \mathcal{O}(a^2) \\ &= (|e\rangle\langle e| + |o\rangle\langle o|) + ia(k_e |e\rangle\langle e| + k_o |o\rangle\langle o|). \end{aligned} \quad (1.124)$$

Using the completeness relation $|e\rangle\langle e| + |o\rangle\langle o| = \mathbf{1}$ and defining the *wavenumber operator* (or propagation matrix) K as

$$K \equiv k_e |e\rangle\langle e| + k_o |o\rangle\langle o|, \quad (1.125)$$

the infinitesimal propagator becomes

$$U_a = \mathbf{1} + iaK + \mathcal{O}(a^2). \quad (1.126)$$

The eigenvalues of K are the characteristic wavenumbers $\{k_e, k_o\}$, and its eigenstates are the principal polarization modes $\{|e\rangle, |o\rangle\}$.

Applying U_a to $|\Psi(z)\rangle$:

$$|\Psi(z + a)\rangle - |\Psi(z)\rangle = iaK |\Psi(z)\rangle. \quad (1.127)$$

Dividing by a and taking the limit $a \rightarrow 0$ yields the spatial differential equation:

$$\frac{d}{dz} |\Psi(z)\rangle = iK |\Psi(z)\rangle. \quad (1.128)$$

Equation (1.128) is the optical spatial analog of the time-dependent Schrödinger equation: spatial propagation along z is generated by the Hermitian operator K , just as time evolution is generated by the Hamiltonian operator H . The formal solution is the matrix exponential:

$$U_z = e^{iKz}. \quad (1.129)$$

1.6.4 Conservation of Probability, Hermiticity, and Unitarity

In a lossless crystal, photons are neither absorbed nor scattered; the total photon flux is strictly conserved. Therefore, the norm of the polarization state must remain invariant under propagation:

$$\langle \Psi(z+a) | \Psi(z+a) \rangle = \langle \Psi(z) | \Psi(z) \rangle = 1. \quad (1.130)$$

To determine the algebraic constraints imposed by probability conservation, we express Eq. (1.127) in an arbitrary basis, such as the Cartesian $\{|x\rangle, |y\rangle\}$ basis:

$$\langle x | \Psi(z+a) \rangle - \langle x | \Psi(z) \rangle = ia (\langle x | K | x \rangle \langle x | \Psi(z) \rangle + \langle x | K | y \rangle \langle y | \Psi(z) \rangle), \quad (1.131)$$

$$\langle y | \Psi(z+a) \rangle - \langle y | \Psi(z) \rangle = ia (\langle y | K | x \rangle \langle x | \Psi(z) \rangle + \langle y | K | y \rangle \langle y | \Psi(z) \rangle). \quad (1.132)$$

Evaluating the squared norm $\langle \Psi(z+a) | \Psi(z+a) \rangle = |\langle x | \Psi(z+a) \rangle|^2 + |\langle y | \Psi(z+a) \rangle|^2$ to first order in a :

$$\begin{aligned} \langle \Psi(z+a) | \Psi(z+a) \rangle &= |\langle x | \Psi(z) \rangle|^2 + |\langle y | \Psi(z) \rangle|^2 \\ &\quad + ia (\langle x | K | x \rangle - \langle x | K | x \rangle^*) |\langle x | \Psi(z) \rangle|^2 \\ &\quad + ia (\langle y | K | y \rangle - \langle y | K | y \rangle^*) |\langle y | \Psi(z) \rangle|^2 \\ &\quad + ia (\langle x | K | y \rangle - \langle y | K | x \rangle^*) \langle y | \Psi(z) \rangle \langle x | \Psi(z) \rangle^* \\ &\quad + ia (\langle y | K | x \rangle - \langle x | K | y \rangle^*) \langle x | \Psi(z) \rangle \langle y | \Psi(z) \rangle^* + \mathcal{O}(a^2). \end{aligned} \quad (1.133)$$

Because this relation must hold identically for every arbitrary state $|\Psi(z)\rangle$, all first-order coefficients must vanish independently:

$$\begin{aligned} \langle x | K | x \rangle &= \langle x | K | x \rangle^*, & \langle y | K | y \rangle &= \langle y | K | y \rangle^*, \\ \langle x | K | y \rangle &= \langle y | K | x \rangle^*, & \langle y | K | x \rangle &= \langle x | K | y \rangle^*. \end{aligned} \quad (1.134)$$

In terms of the Hermitian adjoint $K^\dagger \equiv (K^*)^T$, these relations state that

$$K^\dagger = K. \quad (1.135)$$

The generator of spatial evolution, K , must be a *Hermitian operator*.

Consequently, the finite evolution operator $U_z = e^{iKz}$ satisfies

$$U_z^\dagger U_z = e^{-iK^\dagger z} e^{iKz} = e^{-iKz} e^{iKz} = \mathbf{1}. \quad (1.136)$$

The operator U_z is *unitary*. The unitarity of the evolution operator is mathematically equivalent to the Hermiticity of its generator K , guaranteeing the conservation of total probability throughout the propagation through matter.

1.7 Problems

[label=0.]Energy Fluctuations and the Correspondence Principle.

A beam of red light propagating along the z -axis passes through an ideal linear polarizer whose transmission axis is aligned along the x -axis. The beam is initially linearly polarized at an angle of 30° relative to the x -axis, and its total incident energy is measured to be $E_{\text{in}} = 10 \text{ J}$. [label=()]

1. (a) Estimate the statistical fluctuations (standard deviation) ΔE in the energy of the beam emerging from the polarizer.
- (b) Determine how these fluctuations scale with Planck's constant $\hbar$, and discuss the recovery of classical determinism in the limit $\hbar \rightarrow 0$.

2. Polarization Bases.

[label=()]Construct an orthonormal basis $\{|45^\circ\rangle, |135^\circ\rangle\}$ corresponding to linear polarizations oriented at 45° and 135° relative to the x -axis, expressing each state vector in terms of the Cartesian basis $\{|x\rangle, |y\rangle\}$. Construct an orthonormal basis $\{|\psi_1\rangle, |\psi_2\rangle\}$ consisting of general elliptical polarization states (neither purely linear nor purely circular).

(3) Transformation Matrices and Group Composition.

[label=()]Compute the 2×2 unitary transformation matrix relating the Cartesian basis $\{|x\rangle, |y\rangle\}$ to the circular basis $\{|R\rangle, |L\rangle\}$. Compute the transformation matrix from the circular basis $\{|R\rangle, |L\rangle\}$ to the elliptical basis constructed in Problem 2(b). Compute the transformation matrix directly from $\{|x\rangle, |y\rangle\}$ to the elliptical basis of Problem 2(b), and verify that it equals the matrix product of the two previous transformation matrices. What is the required order of matrix multiplication? Prove generally that if $U_{1 \rightarrow 2}$ transforms basis 1 to basis 2, and $U_{2 \rightarrow 3}$ transforms basis 2 to basis 3, the combined transformation matrix is

$$U_{1 \rightarrow 3} = U_{2 \rightarrow 3} U_{1 \rightarrow 2}. \quad (1.137)$$

(Hint: Insert the resolution of identity $\sum_k |k_{(2)}\rangle \langle k_{(2)}| = \mathbf{1}$ for basis 2.)

(4) Adjoint Operators and Eigensystems.

[label=()]Prove that for any linear operator M and arbitrary states $|\Phi\rangle$ and $|\Psi\rangle$,

$$\langle \Phi | M | \Psi \rangle^* = \langle \Psi | M^\dagger | \Phi \rangle. \quad (1.138)$$

Show that if $|\Psi\rangle$ is an eigenvector of M with eigenvalue λ ($M|\Psi\rangle = \lambda|\Psi\rangle$), the dual bra vector $\langle \Psi |$ satisfies

$$\langle \Psi | M^\dagger = \lambda^* \langle \Psi |. \quad (1.139)$$

(5) Unitarity of Basis Transformations.

Let $\{|u_i\rangle\}$ and $\{|v_j\rangle\}$ ($i, j = 1, 2$) be two complete orthonormal bases in $\mathbb{C}^2$. Define the transformation matrix elements by $U_{ij} = \langle v_i | u_j \rangle$. Prove from first principles that U is unitary, i.e., $U^\dagger U = U U^\dagger = \mathbf{1}$.

6. Hermiticity of Physical Observables.

[label=()]Prove that the projection operator $P_\Phi \equiv |\Phi\rangle \langle \Phi|$ onto a normalized state $|\Phi\rangle$ is Hermitian ($P_\Phi^\dagger = P_\Phi$) and idempotent ($P_\Phi^2 = P_\Phi$). Prove that the photon spin operator $\mathbf{S} = |R\rangle \langle R| - |L\rangle \langle L|$ is Hermitian, and verify its eigenvalues and eigenvectors.

(6) Differential Equations for Basis Rotations.

Let $\{|x'(\theta)\rangle, |y'(\theta)\rangle\}$ denote the Cartesian basis rotated counter-clockwise through an angle θ in the transverse plane.

[label=()]Show that for an arbitrary fixed state $|\Psi\rangle$, the rotated projection components satisfy the coupled differential equations:

$$\begin{aligned} -i \frac{\partial}{\partial \theta} \langle x'(\theta) | \Psi \rangle &= \langle x'(\theta) | \mathbf{S} | \Psi \rangle, \\ -i \frac{\partial}{\partial \theta} \langle y'(\theta) | \Psi \rangle &= \langle y'(\theta) | \mathbf{S} | \Psi \rangle. \end{aligned} \quad (1.140)$$

Solve these equations explicitly for the eigenstates $|\Psi\rangle = |R\rangle$ and $|\Psi\rangle = |L\rangle$.

(8) Projection Operators as Observables.

The probability that a photon in state $|\Psi\rangle$ passes through an ideal x -polarizer can be viewed as the expectation value of a projection observable P_x (the “ x -polarization filter” observable).

[label=()]Construct the operator P_x in Dirac outer-product notation and in matrix form in the Cartesian basis. Verify that P_x is Hermitian, and determine its eigenvalues and corresponding eigenstates. Write down the spectral decomposition of P_x . Confirm explicitly that $\langle\Psi|P_x|\Psi\rangle$ gives the transition probability $|\langle x|\Psi\rangle|^2$.

(9) Quantum Interference versus Which-Path Information.

Photons linearly polarized at 30° relative to the x -axis are directed toward a y -polarizer. An experimenter attempts to measure the frequency with which transmitted photons behave as right- or left-circularly polarized by inserting circular filters:

[label=()]Experiment A: An ideal right-circular polarizer is placed between the source and the y -polarizer. Calculate the probability P_R that a photon reaches the detector. Experiment B: An ideal left-circular polarizer replaces the right-circular polarizer. Calculate the corresponding probability P_L . Show that $P_R + P_L \neq P_{\text{direct}}$, where P_{direct} is the probability of transmission when only the y -polarizer is present with no intermediate filter. Explain the physical origin of this discrepancy in terms of quantum interference and relate it to Young’s double-slit experiment with which-path detectors.

(10) Photon Number–Phase Uncertainty Relation.

A monochromatic pulse of light containing N photons is prepared by reflecting a laser beam of frequency ω from a movable mirror. The photon number N is measured by detecting the mechanical momentum p_y transferred to the mirror upon specular reflection.

[label=()]Using the Heisenberg uncertainty relation for the mirror’s position and momentum, prove that the uncertainty in photon number ΔN and the uncertainty in optical phase $\Delta\varphi$ satisfy

$$\Delta N \Delta\varphi \gtrsim 2\pi. \quad (1.141)$$

Show that this uncertainty relation is strictly independent of the angle of incidence θ of the light beam on the mirror.

(11) Transition Probability in a Calcite Slab.

A photon linearly polarized at an angle θ to the optic axis is incident normally along the z -direction on a slab of calcite of thickness $\ell = 10^{-2}$ cm = 100 μ m. The optic axis lies in the transverse xy -plane. The refractive indices are $n_e = 1.50$ and $n_o = 1.65$, and the vacuum wavelength is $\lambda = 2\pi c/\omega = 5000$ Å.

[label=()]Calculate the total relative phase retardance $\Delta\phi = (k_o - k_e)\ell$ between the ordinary and extraordinary waves. Calculate the transition probability $P(\theta \rightarrow L)$ for the photon to emerge from the slab with left circular polarization, as a function of the input angle θ . Sketch $P(\theta \rightarrow L)$ over the interval $\theta \in [0, \pi]$.

(12) Circular Polarization Filter Design.

Using a slab of calcite with indices $n_e = 1.50$ and $n_o = 1.65$ together with a linear Polaroid sheet, design an optical device that selectively transmits incident photons of vacuum wavelength $\lambda = 5000$ Å if and only if they are right-circularly polarized ($|R\rangle$). Specify the required crystal thickness, orientation of the optic axis, and alignment of the Polaroid transmission axis.

13. Full-Wave Plate Condition.

Determine the condition on the propagation length ℓ of a birefringent crystal of indices

n_e and n_o such that every incident state $|\Psi_{\text{in}}\rangle$ of frequency ω emerges in identically the same physical state (up to an unobservable overall global phase factor):

$$|\Psi_{\text{out}}\rangle = e^{i\phi_0} |\Psi_{\text{in}}\rangle. \quad (1.142)$$

14. Optical Activity in Turpentine.

Turpentine is an optically active liquid that rotates the plane of linear polarization by approximately $\theta/\ell \approx 5^\circ$ per centimeter of path length traversed in the counter-clockwise direction (a dextrorotatory/levorotatory medium).

[label=()] Write down the 2×2 evolution operator $U(\ell)$ describing the transmission of a polarization state through a length ℓ of turpentine in the Cartesian basis. Prove that $U(\ell)$ is unitary. Find the eigenvalues and eigenstates of $U(\ell)$ as a function of propagation distance ℓ . Which polarization states propagate through turpentine without changing their polarization state?

(5) Evolution of Unpolarized Light in Matter.

[label=()] An unpolarized light beam propagating along the z -axis passes through a slab of calcite whose optic axis lies in the transverse plane. Describe the polarization properties of the transmitted beam. What is the effect of an optically active medium (such as turpentine) on an unpolarized beam? Prove your conclusions using the density matrix formalism.

(6) Algebraic Properties of the Trace.

[label=()] For an $n \times n$ matrix A , define the trace $\text{Tr}(A) = \sum_{i=1}^n A_{ii}$. Prove the cyclic invariance of the trace:

$$\text{Tr}(AB) = \text{Tr}(BA). \quad (1.143)$$

Show that the trace is basis-independent: for any unitary transformation U ,

$$\text{Tr}(U^\dagger A U) = \text{Tr}(A). \quad (1.144)$$

(7) The Density Operator Formalism.

[label=()] **Pure States:** For a system prepared in a normalized pure state $|\Psi\rangle$, define the density operator $\rho \equiv |\Psi\rangle\langle\Psi|$. Show that the expectation value of any observable represented by the Hermitian operator Q is given by

$$\langle Q \rangle = \text{Tr}(\rho Q). \quad (1.145)$$

Mixed States: For an ensemble prepared as a statistical mixture of states $\{|\Psi_i\rangle\}$ with classical probabilities p_i ($\sum_i p_i = 1$), define the mixed-state density operator

$$\rho = \sum_i p_i |\Psi_i\rangle\langle\Psi_i|. \quad (1.146)$$

Prove that the ensemble expectation value of Q satisfies $\langle Q \rangle = \text{Tr}(\rho Q)$. Prove that for any physical density operator, $\text{Tr}(\rho) = 1$ and $\text{Tr}(\rho^2) \leq 1$, with equality holding if and only if the state is pure.

(8) Quantum State Tomography.

[label=()] An experimentalist receives a beam of photons whose polarization state is completely unknown (which may be pure, partially polarized, or unpolarized). What is the minimum number of independent optical measurements required to completely determine the density matrix ρ , and what set of measurement

analyzers can be used? What is the explicit density matrix ρ for completely unpolarized light? How can one experimentally verify that a beam is unpolarized? Consider an ensemble consisting of a 50/50 statistical mixture of horizontally polarized photons ($|x\rangle$) and right-circularly polarized photons ($|R\rangle$).[label=()]

- (a)**
- i. Compute the density matrix ρ in the Cartesian basis.
 - ii. Diagonalize ρ to find its eigenvalues and mutually orthogonal eigenstates.
 - iii. Express ρ as an incoherent mixture of these two orthogonal states, proving that different physical preparation ensembles can yield identical quantum density operators.

Chapter 2

Neutral K Mesons

2.1 Neutral K Mesons

The mathematical formalism developed in Chapter 1 to describe the two-state polarization of the photon provides a universal framework for two-level quantum mechanical systems. One of the most fascinating and historically significant physical realizations of such a system is found in the physics of neutral K mesons (kaons).

2.1.1 Strangeness and Kaon Production

Neutral kaons are produced abundantly via strong interactions, for example in inelastic pion-proton collisions:

$$\pi^- + p \rightarrow \Lambda^0 + K^0. \quad (2.1)$$

In this reaction, electric charge is manifestly conserved: the initial state has total charge $Q = (-1) + (+1) = 0$, and the final state has $Q = 0 + 0 = 0$.

In addition to electric charge, hadrons carry another additive quantum number known as *strangeness* (S), which is strictly conserved in all strong and electromagnetic interactions. By convention, nucleons (protons p and neutrons n) and pions ($\pi^\pm, \pi^0$) have strangeness $S = 0$, whereas the lambda hyperon Λ^0 is assigned strangeness $S = -1$. Because the initial state in reaction (2.1) has total strangeness $S = 0$, conservation of strangeness demands that the produced neutral kaon, denoted by K^0 , must carry strangeness $S = +1$:

$$S(\pi^-) + S(p) = 0 + 0 = 0 \quad \implies \quad S(\Lambda^0) + S(K^0) = -1 + (+1) = 0. \quad (2.2)$$

To each particle in quantum field theory there corresponds an antiparticle carrying opposite additive internal quantum numbers. The antiparticle of the K^0 is the antikaon, denoted by $\bar{K}^0$. Since the K^0 is electrically neutral, $\bar{K}^0$ is likewise neutral, but carries strangeness $S = -1$ and baryon number $B = 0$. While K^0 cannot produce a Λ^0 upon interacting with a nucleon at low energies (as that would violate strangeness conservation), the $\bar{K}^0$ readily participates in strong absorption reactions such as

$$\bar{K}^0 + p \rightarrow \Lambda^0 + \pi^+. \quad (2.3)$$

Here, the total initial strangeness is $S = -1 + 0 = -1$, and the total final strangeness is $S = -1 + 0 = -1$, confirming that strangeness is indeed conserved in this strong interaction.

2.1.2 The Two-State Kaon System and the Strangeness Operator

Remarkably, neutral kaons can exist in arbitrary coherent linear superpositions of K^0 and $\bar{K}^0$ states, in direct analogy to the way a photon can exist in an arbitrary superposition of right- and left-circularly polarized states $|R\rangle$ and $|L\rangle$.

We represent the state of a neutral kaon by a state vector in a two-dimensional complex Hilbert space $\mathbb{C}^2$. Let $|K^0\rangle$ denote the state vector of a pure K^0 meson ($S = +1$), and let $|\bar{K}^0\rangle$ denote the state vector of a pure $\bar{K}^0$ meson ($S = -1$). Because they correspond to distinct eigenvalues of an observable (strangeness), these states are mutually orthogonal and can be chosen to be normalized:

$$\langle K^0|K^0\rangle = 1, \quad \langle \bar{K}^0|\bar{K}^0\rangle = 1, \quad \langle \bar{K}^0|K^0\rangle = 0. \quad (2.4)$$

The states $\{|K^0\rangle, |\bar{K}^0\rangle\}$ thus constitute a complete orthonormal basis:

$$\mathbf{1} = |K^0\rangle\langle K^0| + |\bar{K}^0\rangle\langle \bar{K}^0|. \quad (2.5)$$

The general state $|\Psi\rangle$ of a neutral kaon is a linear combination of these basis states. Choosing the basis states to be time-independent, any time evolution is carried entirely by the complex expansion coefficients.

We define a strangeness operator $\hat{S}$ acting on kaon states by

$$\hat{S}|K^0\rangle = +|K^0\rangle, \quad \hat{S}|\bar{K}^0\rangle = -|\bar{K}^0\rangle. \quad (2.6)$$

In the orthonormal basis $\{|K^0\rangle, |\bar{K}^0\rangle\}$, represented by the column vectors

$$|K^0\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\bar{K}^0\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (2.7)$$

the strangeness operator is represented by the Pauli matrix σ_z :

$$\hat{S} \doteq \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.8)$$

The eigenstates of $\hat{S}$ are $|K^0\rangle$ and $|\bar{K}^0\rangle$, with eigenvalues $+1$ and -1 , respectively. In coordinate-free Dirac notation, using the completeness relation, the operator $\hat{S}$ is expressed as

$$\hat{S} = |K^0\rangle\langle K^0| - |\bar{K}^0\rangle\langle \bar{K}^0|. \quad (2.9)$$

2.1.3 Charge Conjugation Parity and CP Eigenstates

Under the combined operation of charge conjugation ($\hat{C}$) and spatial parity ($\hat{P}$), a neutral pseudo-scalar kaon is converted into its antiparticle. Adopting standard phase conventions, the action of the unitary operator $\hat{C}\hat{P}$ on the strangeness eigenstates is defined by

$$\hat{C}\hat{P}|K^0\rangle = |\bar{K}^0\rangle, \quad \hat{C}\hat{P}|\bar{K}^0\rangle = |K^0\rangle. \quad (2.10)$$

In the $\{|K^0\rangle, |\bar{K}^0\rangle\}$ basis, $\hat{C}\hat{P}$ is represented by the off-diagonal Pauli matrix σ_x :

$$\hat{C}\hat{P} \doteq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (2.11)$$

Because $(\hat{C}\hat{P})^2 = \mathbf{1}$, the eigenvalues of $\hat{C}\hat{P}$ must be ± 1 . The corresponding normalized eigenstates are readily found to be

$$|K_S\rangle = \frac{1}{\sqrt{2}} (|K^0\rangle + |\bar{K}^0\rangle), \quad |K_L\rangle = \frac{1}{\sqrt{2}} (|K^0\rangle - |\bar{K}^0\rangle), \quad (2.12)$$

which satisfy

$$\hat{C}\hat{P}|K_S\rangle = +|K_S\rangle, \quad \hat{C}\hat{P}|K_L\rangle = -|K_L\rangle. \quad (2.13)$$

The quantum number associated with $\hat{C}P$ is called *charge conjugation parity* (or CP parity). Thus, $|K_S\rangle$ has positive CP parity ($CP = +1$), while $|K_L\rangle$ has negative CP parity ($CP = -1$).

Notice that the strangeness operator $\hat{S}$ and the $\hat{C}P$ operator do not commute:

$$[\hat{S}, \hat{C}P] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = 2i\sigma_y \neq 0. \quad (2.14)$$

Consequently, a neutral kaon cannot simultaneously have a well-defined strangeness and a well-defined CP parity. The CP eigenstates $|K_S\rangle$ and $|K_L\rangle$ are equal coherent mixtures of states with strangeness $+1$ and -1 .

The states $\{|K_S\rangle, |K_L\rangle\}$ form an alternative orthonormal basis for the neutral kaon space:

$$\langle K_S | K_S \rangle = 1, \quad \langle K_L | K_L \rangle = 1, \quad \langle K_S | K_L \rangle = 0. \quad (2.15)$$

Inverting Eq. (2.12), the strangeness eigenstates can be expressed in terms of the CP basis as

$$|K^0\rangle = \frac{1}{\sqrt{2}}(|K_S\rangle + |K_L\rangle), \quad |\bar{K}^0\rangle = \frac{1}{\sqrt{2}}(|K_S\rangle - |K_L\rangle). \quad (2.16)$$

Just as the CP eigenstates carry indefinite strangeness, the strangeness eigenstates $|K^0\rangle$ and $|\bar{K}^0\rangle$ carry indefinite CP parity, each being an equal superposition of $CP = +1$ and $CP = -1$ components.

2.1.4 Weak Decays and Lifetimes

Neutral kaons are unstable and decay spontaneously via the weak interaction. While strangeness is strictly conserved in strong interactions, it is violated by the weak interaction with $\Delta S = \pm 1$. To an excellent approximation, the weak interaction preserves CP symmetry.¹

Because CP is conserved in the decay process, the initial kaon state must possess the same CP eigenvalue as the final decay products:

[(i)] Two-pion decay ($CP = +1$): A state of two neutral or charged pions in an s -wave state ($\pi^+\pi^-$ or $\pi^0\pi^0$) has intrinsic parity $P = (-1)^2 = +1$ and charge conjugation eigenvalue $C = +1$, so its combined CP parity is $+1$. Therefore, only the $CP = +1$ eigenstate $|K_S\rangle$ can decay into two pions:

$$K_S \rightarrow \pi^+ + \pi^-, \quad K_S \rightarrow \pi^0 + \pi^0. \quad (2.17)$$

Because the available phase space is substantial and there is no angular momentum barrier, this decay occurs rapidly, with a mean lifetime

$$\tau_S \approx 0.895 \times 10^{-10} \text{ s}. \quad (2.18)$$

The subscript S stands for *short-lived*. **Three-pion and semileptonic decays ($CP = -1$):** A three-pion state ($\pi\pi\pi$) can form a state with $CP = -1$. Under exact CP invariance, the $CP = -1$ eigenstate $|K_L\rangle$ is forbidden from decaying into two pions and must instead decay into three pions or via semileptonic modes:

$$K_L \rightarrow \pi^0\pi^0\pi^0, \quad K_L \rightarrow \pi^+\pi^-\pi^0, \quad K_L \rightarrow \pi^\mp e^\pm \nu_e, \quad K_L \rightarrow \pi^\mp \mu^\pm \nu_\mu. \quad (2.19)$$

Due to the severe phase-space suppression in three-body decays and the weak coupling in semileptonic transitions, the lifetime of $|K_L\rangle$ is vastly longer:

$$\tau_L \approx 5.11 \times 10^{-8} \text{ s} \approx 518 \times 10^{-10} \text{ s}. \quad (2.20)$$

The subscript L stands for *long-lived*.

The ratio of lifetimes is $\tau_L/\tau_S \approx 580$, representing nearly three orders of magnitude difference in longevity.

¹Small violations of CP symmetry do occur in nature at the level of $|\varepsilon| \sim 2 \times 10^{-3}$, as discovered experimentally by Christenson, Cronin, Fitch, and Turlay in 1964; this effect is explored in Problem 2. For the present discussion, we treat CP conservation as exact.

2.1.5 Time Evolution and Decaying States

We now examine the time evolution of neutral kaon states. Recall from the discussion of birefringence in Chapter 1 that an ordinary wave $|o\rangle$ propagates with a definite wavenumber $k_o = \omega n_o/c$, and an extraordinary wave $|e\rangle$ propagates with $k_e = \omega n_e/c$, whereas an arbitrary superposition has no single wavenumber.

In an identical manner, the stationary states in quantum mechanics have well-defined energies and frequencies. A short-lived kaon $|K_S\rangle$ of momentum $\mathbf{p}$ possesses a definite rest mass m_S and energy

$$E_S = \sqrt{p^2 c^2 + m_S^2 c^4}, \quad (2.21)$$

and its quantum phase oscillates with angular frequency

$$\omega_S = \frac{E_S}{\hbar}. \quad (2.22)$$

Likewise, a long-lived kaon $|K_L\rangle$ possesses rest mass m_L and energy

$$E_L = \sqrt{p^2 c^2 + m_L^2 c^4}, \quad (2.23)$$

oscillating with angular frequency

$$\omega_L = \frac{E_L}{\hbar}. \quad (2.24)$$

A general neutral kaon state, being a superposition of $|K_S\rangle$ and $|K_L\rangle$, does not evolve with a single harmonic frequency; rather, its evolution is a superposition of the two distinct frequencies ω_S and ω_L .

To incorporate the radioactive decay of the kaons into the state vector formalism, suppose that at $t = 0$ a meson is prepared in a pure $|K_S\rangle$ state:

$$|\Psi(0)\rangle = |K_S\rangle. \quad (2.25)$$

The probability of finding the meson in the undecayed state at a later proper time t obeys the standard exponential decay law:

$$P(t) = |\langle K_S | \Psi(t) \rangle|^2 = e^{-t/\tau_S}. \quad (2.26)$$

Consequently, the probability amplitude $\langle K_S | \Psi(t) \rangle$ must decrease by the factor $e^{-t/(2\tau_S)}$. Combining this exponential damping with the harmonic phase factor $e^{-i\omega_S t}$ dictated by the Schrödinger equation, the full time-dependent state vector becomes

$$|\Psi(t)\rangle = e^{-i\omega_S t - t/(2\tau_S)} |K_S\rangle. \quad (2.27)$$

Similarly, if a neutral kaon is prepared at $t = 0$ in a pure long-lived state $|K_L\rangle$, its subsequent state vector at time t is given by

$$|\Psi(t)\rangle = e^{-i\omega_L t - t/(2\tau_L)} |K_L\rangle. \quad (2.28)$$

Equations (2.27) and (2.28) demonstrate that the time evolution of decaying quantum systems can be elegantly captured by assigning an effective complex energy to each decaying eigenstate:

$$\mathcal{E}_S = E_S - \frac{i\hbar}{2\tau_S}, \quad \mathcal{E}_L = E_L - \frac{i\hbar}{2\tau_L}, \quad (2.29)$$

so that the dynamical phase factor takes the unified form $e^{-i\mathcal{E}t/\hbar}$.

2.2 Quantum Interference Effects

The neutral kaon system exhibits some of the most striking and counter-intuitive manifestations of quantum superposition and interference in fundamental physics. In this section, we analyze two remarkable phenomena: strangeness oscillations—in which a particle spontaneously transforms into its own antiparticle—and the regeneration of short-lived kaons upon passing through matter.

2.2.1 Strangeness Oscillations ($K^0 \leftrightarrow \bar{K}^0$)

Consider a neutral kaon produced at time $t = 0$ via a strong interaction process such as $\pi^- + p \rightarrow \Lambda^0 + K^0$. Because strangeness is strictly conserved in strong interactions, the kaon is prepared in a pure strangeness state with $S = +1$:

$$|\Psi(0)\rangle = |K^0\rangle. \quad (2.30)$$

Using the transformation relations derived in Section 2.0, this initial state can be expanded in terms of the CP eigenstates $|K_S\rangle$ and $|K_L\rangle$:

$$|\Psi(0)\rangle = \frac{1}{\sqrt{2}} (|K_S\rangle + |K_L\rangle). \quad (2.31)$$

The dynamical time evolution of the state vector is determined by the evolution of its stationary components $|K_S\rangle$ and $|K_L\rangle$, each having a definite oscillation frequency and exponential decay rate:

$$|\Psi(t)\rangle = \frac{1}{\sqrt{2}} \left[e^{-i\omega_S t - t/(2\tau_S)} |K_S\rangle + e^{-i\omega_L t - t/(2\tau_L)} |K_L\rangle \right]. \quad (2.32)$$

To determine the probability that the meson behaves as an antiparticle $\bar{K}^0$ at proper time t , we compute the transition amplitude $\langle \bar{K}^0 | \Psi(t) \rangle$. From Eq. (2.12), the projections of the CP eigenstates onto the antikaon state are

$$\langle \bar{K}^0 | K_S \rangle = \frac{1}{\sqrt{2}}, \quad \langle \bar{K}^0 | K_L \rangle = -\frac{1}{\sqrt{2}}. \quad (2.33)$$

Taking the inner product of $\langle \bar{K}^0 |$ with Eq. (2.32), we obtain the time-dependent amplitude:

$$\langle \bar{K}^0 | \Psi(t) \rangle = \frac{1}{2} \left(e^{-i\omega_S t - t/(2\tau_S)} - e^{-i\omega_L t - t/(2\tau_L)} \right). \quad (2.34)$$

The corresponding probability $P_{\bar{K}^0}(t)$ of finding an antikaon at time t is the absolute square of this amplitude:

$$\begin{aligned} P_{\bar{K}^0}(t) &= |\langle \bar{K}^0 | \Psi(t) \rangle|^2 \\ &= \frac{1}{4} \left| e^{-i\omega_S t - t/(2\tau_S)} - e^{-i\omega_L t - t/(2\tau_L)} \right|^2 \\ &= \frac{1}{4} \left[e^{-t/\tau_S} + e^{-t/\tau_L} - 2e^{-\frac{t}{2}(\tau_S^{-1} + \tau_L^{-1})} \cos((\omega_S - \omega_L)t) \right]. \end{aligned} \quad (2.35)$$

Equation (2.35) reveals the profound quantum nature of the neutral kaon system:

[(i)]Initial state: At $t = 0$, the probability vanishes identically, $P_{\bar{K}^0}(0) = 0$, consistent with the orthogonality of the strangeness eigenstates, $\langle \bar{K}^0 | K^0 \rangle = 0$. **Antiparticle appearance:** As proper time elapses, $P_{\bar{K}^0}(t)$ increases from zero. A particle originally created with strangeness $S = +1$ spontaneously develops a component with strangeness $S = -1$, transforming into its own antiparticle. **Quantum interference:** The oscillatory

cross-term proportional to $\cos((\omega_S - \omega_L)t)$ represents direct quantum mechanical interference between the K_S and K_L amplitudes. The frequency difference is related to the mass difference $\Delta m = m_L - m_S$ by

$$\omega_S - \omega_L \approx \frac{(m_S - m_L)c^2}{\hbar}. \quad (2.36)$$

Experimentally, the mass difference satisfies

$$|m_L - m_S| \approx \frac{\hbar}{2c^2\tau_S} \approx 3.5 \times 10^{-12} \text{ MeV}/c^2, \quad (2.37)$$

which corresponds to a fractional mass difference $\Delta m/m_K \sim 10^{-14}$. Despite being exceedingly small, this tiny energy split is measured with extraordinary precision via the oscillation term in Eq. (2.35). **Long-time behavior:** In the laboratory frame, the proper time translates to a distance of travel $l = vt = (p/m)t$. Over the first several centimeters ($t \sim$ a few τ_S), the short-lived component e^{-t/τ_S} decays away. For times $\tau_S \ll t \ll \tau_L$, only the long-lived component remains:

$$P_{\bar{K}^0}(t) \approx \frac{1}{4}e^{-t/\tau_L}. \quad (2.38)$$

Since the surviving beam is in the state $|K_L\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle)$, half of the surviving undecayed kaons will interact as $\bar{K}^0$ and half as K^0 .

Experimentally, the presence of the $\bar{K}^0$ component is detected by placing a target in the beam and observing strangeness-conserving absorption reactions such as $\bar{K}^0 + p \rightarrow \Lambda^0 + \pi^+$ (Eq. (2.3)), which can be uniquely triggered by $\bar{K}^0$ and cannot be induced by K^0 at low energies.

An instructive optical analogue to this phenomenon is a dichroic birefringent medium that strongly absorbs light polarized along the x -axis while transmitting light polarized along the y -axis with little attenuation. If a right-circularly polarized photon $|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle)$ enters the medium, its $|x\rangle$ component is quickly absorbed, leaving predominantly the $|y\rangle$ state. Since $|y\rangle = \frac{1}{i\sqrt{2}}(|R\rangle - |L\rangle)$, the beam develops a substantial probability amplitude for emerging with left circular polarization $|L\rangle$.

2.2.2 Regeneration of Short-Lived Kaons

A second remarkable quantum interference effect is the *regeneration* of short-lived kaons (K_S) from a beam consisting purely of long-lived kaons (K_L).

Suppose a neutral kaon beam is produced and allowed to propagate in vacuum over a distance of several meters ($l \gg v\tau_S$). Because the lifetime τ_S is nearly 600 times shorter than τ_L , the K_S component completely decays away via two-pion modes ($K_S \rightarrow 2\pi$), leaving a beam composed exclusively of $|K_L\rangle$ mesons. Any subsequent decays observed in vacuum are three-body or semileptonic decays characteristic of K_L .

Now let this pure $|K_L\rangle$ beam pass through a thin plate of ordinary matter. The kaons interact strongly with the atomic nuclei (protons and neutrons) in the plate. Because ordinary matter contains nucleons but no antinucleons, it is fundamentally asymmetric with respect to strangeness:

- The K^0 component ($S = +1$) can only undergo elastic or quasi-elastic scattering with nucleons, since there are no $S = +1$ baryons into which the system could transition.
- The $\bar{K}^0$ component ($S = -1$) can undergo scattering and, in addition, readily undergoes exothermic inelastic absorption to form hyperons: $\bar{K}^0 + p \rightarrow \Lambda^0 + \pi^+$ or $\bar{K}^0 + n \rightarrow \Sigma^- + \pi^+$.

As a result of this physical asymmetry, the matter affects the two strangeness components differently. Mathematically, the transmission through the plate multiplies the $|K^0\rangle$ amplitude by a complex transmission coefficient $\alpha e^{i\varphi}$ and the $|\bar{K}^0\rangle$ amplitude by $\beta e^{i\varphi'}$:

$$|K^0\rangle \longrightarrow \alpha e^{i\varphi} |K^0\rangle, \quad |\bar{K}^0\rangle \longrightarrow \beta e^{i\varphi'} |\bar{K}^0\rangle, \quad (2.39)$$

where the attenuation parameters satisfy $0 < \beta < \alpha \leq 1$ due to the stronger absorption of $\bar{K}^0$, and the real phase shifts φ and φ' differ because of different forward scattering lengths (refractive indices).

Immediately before entering the absorber, the state of the kaon is pure K_L :

$$|\Psi_{\text{before}}\rangle = |K_L\rangle = \frac{1}{\sqrt{2}} (|K^0\rangle - |\bar{K}^0\rangle). \quad (2.40)$$

Upon emerging from the absorber, the state vector is

$$|\Psi_{\text{after}}\rangle = \frac{1}{\sqrt{2}} (\alpha e^{i\varphi} |K^0\rangle - \beta e^{i\varphi'} |\bar{K}^0\rangle). \quad (2.41)$$

Projecting this transmitted state onto the short-lived state $|K_S\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle)$, the amplitude for finding a K_S meson is

$$\langle K_S | \Psi_{\text{after}} \rangle = \frac{1}{2} (\alpha e^{i\varphi} - \beta e^{i\varphi'}). \quad (2.42)$$

So long as $\alpha \neq \beta$ or $\varphi \neq \varphi'$, this projection amplitude is non-zero.

The slab of matter acts as a coherent quantum mechanical filter: by differentially attenuating and phase-shifting the $|K^0\rangle$ and $|\bar{K}^0\rangle$ amplitudes, it destroys the precise cancellation between them that defined the pure $|K_L\rangle$ state. Consequently, a K_S component is *regenerated* in the beam. Immediately downstream of the plate, two-pion decays ($K_S \rightarrow \pi^+\pi^-$ and $K_S \rightarrow \pi^0\pi^0$) are observed once again, confirming this extraordinary prediction of quantum mechanics.

2.3 Problems

[label=0.]Coherent Regeneration of K_S Mesons by Phase Retardation.

A beam of purely long-lived neutral kaons (K_L) is directed through a thin non-absorbing plate whose only physical effect is to introduce a relative phase shift of $\delta = 10^\circ$ between the K^0 and $\bar{K}^0$ components of the wave function:

$$|K^0\rangle \longrightarrow e^{i\delta/2} |K^0\rangle, \quad |\bar{K}^0\rangle \longrightarrow e^{-i\delta/2} |\bar{K}^0\rangle. \quad (2.43)$$

[label=()]

1. (a) Compute the state vector $|\Psi_{\text{after}}\rangle$ immediately after emerging from the plate, and determine the probability $P(K_S) = |\langle K_S | \Psi_{\text{after}} \rangle|^2$ that an incident kaon is converted into the short-lived state $|K_S\rangle$.
- (b) Assume for simplicity that all kaons in the beam have relativistic momentum $p = m_K c$. Taking the lifetime of the short-lived kaon to be $\tau_S \approx 0.895 \times 10^{-10}$ s (with $c\tau_S \approx 2.68$ cm), calculate the fraction of incident particles that decay via the $K_S \rightarrow 2\pi$ channel in the first $d = 5$ cm of travel in the laboratory frame immediately following the absorber.

2. CP Violation in the Neutral Kaon System.

In 1964, Christenson, Cronin, Fitch, and Turlay discovered that the long-lived neutral kaon occasionally decays into two charged pions ($K_L \rightarrow \pi^+\pi^-$ with a branching ratio of

$\approx 2 \times 10^{-3}$), demonstrating that CP invariance is not an exact symmetry of the weak interaction. Because of CP violation, the stationary states of the effective Hamiltonian (the states with well-defined mass and lifetime) are not the pure CP eigenstates $|K_S\rangle$ and $|K_L\rangle$, but rather slightly perturbed linear combinations denoted by $|K'_S\rangle$ and $|K'_L\rangle$.

To first order in the CP -violation parameter ε , these physical states can be written as

$$|K'_S\rangle \propto (1 + \varepsilon)|K^0\rangle + (1 - \varepsilon)|\bar{K}^0\rangle, \quad (2.44)$$

$$|K'_L\rangle \propto (1 + \varepsilon)|K^0\rangle - (1 - \varepsilon)|\bar{K}^0\rangle, \quad (2.45)$$

where ε is a small complex parameter with $|\varepsilon| \approx 2.23 \times 10^{-3}$. In proper time t , these states evolve independently as

$$|K'_S(t)\rangle = e^{-i\omega_S t - t/(2\tau_S)} |K'_S\rangle, \quad |K'_L(t)\rangle = e^{-i\omega_L t - t/(2\tau_L)} |K'_L\rangle, \quad (2.46)$$

where $\hbar\omega_S = \sqrt{p^2 c^2 + m_S^2 c^4}$ and $\hbar\omega_L = \sqrt{p^2 c^2 + m_L^2 c^4}$, with $\tau_L \gg \tau_S$.

[label=()]Express the normalized states $|K'_S\rangle$ and $|K'_L\rangle$ in terms of the strangeness basis $\{|K^0\rangle, |\bar{K}^0\rangle\}$ and in terms of the pure CP basis $\{|K_S\rangle, |K_L\rangle\}$, retaining terms up to first order in ε . Let $\langle 2\pi|T|K\rangle$ denote the transition amplitude for a neutral kaon state $|K\rangle$ to decay into a two-pion state $|2\pi\rangle$ ($CP = +1$). Assuming that CP violation arises solely from state mixing in the effective Hamiltonian (so that the decay interaction itself conserves CP , meaning $\langle 2\pi|T|K_L\rangle = 0$ and $\langle 2\pi|T|K_S\rangle \neq 0$), calculate the amplitude ratio

$$\eta_{+-} \equiv \frac{\langle 2\pi|T|K'_L\rangle}{\langle 2\pi|T|K'_S\rangle}. \quad (2.47)$$

Show that $\eta_{+-} = \varepsilon$, and relate the measured branching ratio $\Gamma(K_L \rightarrow 2\pi)/\Gamma(K_S \rightarrow 2\pi)$ to $|\varepsilon|$. Suppose a beam of purely long-lived kaons $|K'_L\rangle$ traverses a thin regenerator plate that introduces a relative phase shift δ between the K^0 and $\bar{K}^0$ components.[label=()]

- (b)
- i. Determine the state vector of the beam emerging from the regenerator, and show that it consists of a coherent superposition of $|K'_L\rangle$ and a regenerated short-lived component $|K'_S\rangle$.
 - ii. Derive an expression for the rate of two-pion decay events $I_{2\pi}(t)$ observed as a function of proper time t downstream from the regenerator.
 - iii. Explain how the quantum interference term between the regenerated amplitude and the intrinsic CP -violating amplitude in $I_{2\pi}(t)$ allows an experimental determination of both the phase of ε ($\arg \varepsilon \approx 43.5^\circ$) and the kaon mass difference $\Delta m = m_L - m_S$.

Chapter 3

The Motion Of Particles In Quantum Mechanics

3.1 The Motion of Particles in Quantum Mechanics

The state of a particle in non-relativistic quantum mechanics is completely described by a complex-valued wave function, $\psi(\mathbf{r}, t)$, representing the probability amplitude for locating the particle at position $\mathbf{r}$ at time t . Specifically, the quantity

$$P(\mathbf{r}, t) d^3r = |\psi(\mathbf{r}, t)|^2 d^3r \quad (3.1)$$

gives the probability of finding the particle within an infinitesimal volume element d^3r centered at $\mathbf{r}$ at time t .

Because $\psi(\mathbf{r}, t)$ represents a probability amplitude rather than a deterministic trajectory, quantum mechanics does not predict the precise outcome of an individual measurement on a single particle. Instead, it predicts the statistical distribution of outcomes observed across an ensemble of identically prepared systems. This probabilistic character is identical to the behavior observed for photon polarization states, where one cannot predict with certainty whether an individual photon will pass through a polarizer, but only the transmission probability for a large statistical ensemble.

3.2 The Schrödinger Equation

Suppose the wave function of a particle is known at an initial time t . How does it evolve dynamically over time? To address this question, we first consider a particle constrained to move in one spatial dimension. It is instructive to discretize the spatial coordinate by dividing the real line into small intervals, each of width λ .¹

Labeling the intervals by indices $i, j, \dots$, let $\psi_i(t)$ denote the probability amplitude for finding the particle in interval i at time t . The corresponding probability is $|\psi_i(t)|^2$. Because the particle must be found somewhere along the line, the total probability summed over all intervals must equal unity:

$$\sum_i |\psi_i(t)|^2 = 1. \quad (3.2)$$

¹This tight-binding formulation, similar to the presentation in *The Feynman Lectures on Physics*, Vol. III, is not a rigorous first-principles derivation of the Schrödinger equation, but rather an intuitive physical model clarifying its underlying structure.

The amplitudes $\psi_i(t)$ can be regarded as components of an infinite-dimensional state vector:

$$|\Psi(t)\rangle = \begin{pmatrix} \vdots \\ \psi_{i-1}(t) \\ \psi_i(t) \\ \psi_{i+1}(t) \\ \vdots \end{pmatrix}. \quad (3.3)$$

This state vector directly generalizes the finite-dimensional state vectors encountered in two-state systems (such as photon polarization). Unlike the two-dimensional photon polarization space spanned by orthogonal states, the spatial state vector $|\Psi(t)\rangle$ possesses countably infinite dimensions corresponding to the discrete spatial cells.

3.2.1 Amplitude Hopping and Time Evolution

As time progresses, the particle is free to move, causing probability amplitude initially localized in interval i to disperse or “leak” into adjacent intervals. In a small time increment Δt , the probability amplitude transferred from interval i into interval $i+1$ is proportional to Δt and to the initial amplitude $\psi_i(t)$. Parametrizing this transition amplitude per unit time as $-iw_{i+1,i}$, the net amplitude transferred from i to $i+1$ during Δt is

$$-i\Delta t w_{i+1,i} \psi_i(t). \quad (3.4)$$

The amplitude in interval i at time $t + \Delta t$ receives contributions from incoming flow from the nearest neighbors $i-1$ and $i+1$, as well as an outgoing flow from cell i itself. Retaining terms up to first order in Δt :

$$\psi_i(t + \Delta t) = \psi_i(t) - i\Delta t w_{i,i-1} \psi_{i-1}(t) - i\Delta t w_{i,i+1} \psi_{i+1}(t) - i\Delta t w_{ii} \psi_i(t). \quad (3.5)$$

For sufficiently small Δt , transitions between non-adjacent cells can be neglected. The diagonal coefficient w_{ii} accounts for amplitude depletion due to outward hopping as well as any local phase rotation accumulated in interval i .

Conservation of total probability requires that the norm of the state vector be preserved under time evolution:

$$\sum_i |\psi_i(t + \Delta t)|^2 = \sum_i |\psi_i(t)|^2 = 1. \quad (3.6)$$

Substituting Eq. (3.5) into Eq. (3.6) and expanding to order $\mathcal{O}(\Delta t)$, we obtain

$$\sum_i |\psi_i(t)|^2 (w_{ii} - w_{ii}^*) + \sum_i \psi_i^*(t) \psi_{i+1}(t) (w_{i,i+1} - w_{i+1,i}^*) + \sum_i \psi_i^*(t) \psi_{i-1}(t) (w_{i,i-1} - w_{i-1,i}^*) = 0. \quad (3.7)$$

Because this condition must hold for arbitrary state vectors $|\Psi(t)\rangle$, the coefficients must vanish identically:

$$w_{ii} = w_{ii}^*, \quad w_{i,i+1} = w_{i+1,i}^*. \quad (3.8)$$

Thus, the transition amplitude between adjacent sites satisfies Hermitian symmetry: the amplitude for hopping in one direction is the complex conjugate of the amplitude for the reverse hop. In matrix representation, the operator W is tridiagonal:

$$W = \begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \\ \dots & w_{i-1,i-1} & w_{i-1,i} & 0 & \dots \\ \dots & w_{i-1,i}^* & w_{ii} & w_{i,i+1} & \dots \\ \dots & 0 & w_{i,i+1}^* & w_{i+1,i+1} & \dots \\ & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (3.9)$$

which is Hermitian ($W^\dagger = W$). In operator notation, Eq. (3.5) reads:

$$|\Psi(t + \Delta t)\rangle = (\mathbb{I} - i\Delta t W)|\Psi(t)\rangle. \quad (3.10)$$

Taking the infinitesimal limit $\Delta t \rightarrow 0$ yields the differential equation of motion:

$$i\frac{\partial\psi_i(t)}{\partial t} = w_{i,i-1}\psi_{i-1}(t) + w_{i,i+1}\psi_{i+1}(t) + w_{ii}\psi_i(t), \quad (3.11)$$

or equivalently,

$$i\frac{\partial}{\partial t}|\Psi(t)\rangle = W|\Psi(t)\rangle. \quad (3.12)$$

3.2.2 Potential Energy and the Continuum Limit

We may reparametrize the diagonal term w_{ii} by defining a local potential parameter v_i :

$$w_{ii} = -w_{i,i+1} - w_{i,i-1} + \frac{v_i}{\hbar}. \quad (3.13)$$

Equation (3.11) then assumes the form:

$$i\frac{\partial\psi_i(t)}{\partial t} = w_{i,i-1}[\psi_{i-1}(t) - \psi_i(t)] + w_{i,i+1}[\psi_{i+1}(t) - \psi_i(t)] + \frac{v_i}{\hbar}\psi_i(t). \quad (3.14)$$

The first two terms resemble a discrete diffusion process: the net rate of amplitude change at site i depends on the difference in amplitude between site i and its neighboring sites. However, the prefactor of i on the time derivative makes this a diffusion equation with an *imaginary* diffusion coefficient, leading to oscillatory, wave-like propagation rather than dissipative relaxation.

To interpret v_i , suppose inter-site hopping is suppressed ($w_{i,i\pm 1} = 0$), corresponding to a completely stationary particle possessing only potential energy. Equation (3.14) then simplifies to

$$\psi_i(t) = \psi_i(0)e^{-iv_it/\hbar}. \quad (3.15)$$

Since the oscillation frequency ω is related to energy by $E = \hbar\omega$, the quantity v_i corresponds directly to the local potential energy at interval i .

We now take the continuum limit as the lattice spacing $\lambda \rightarrow 0$. In this limit, the discrete probability amplitude relates to the continuous wave function $\psi(x)$ via

$$\psi_i(t) \rightarrow \sqrt{\lambda}\psi(x_i, t), \quad (3.16)$$

where x_i denotes the center of interval i . The dimensional factor $\sqrt{\lambda}$ ensures that the probability $\lambda|\psi(x_i, t)|^2$ within interval i matches $|\psi_i(t)|^2$, reflecting the fact that $\psi(x)$ has physical dimensions of $[\text{length}]^{-1/2}$.

Substituting this relation into Eq. (3.14) and canceling $\sqrt{\lambda}$ yields:

$$i\frac{\partial\psi(x_i, t)}{\partial t} = w_{i,i-1}[\psi(x_{i-1}, t) - \psi(x_i, t)] + w_{i,i+1}[\psi(x_{i+1}, t) - \psi(x_i, t)] + \frac{v(x_i, t)}{\hbar}\psi(x_i, t). \quad (3.17)$$

With $x_{i\pm 1} = x_i \pm \lambda$, we perform a Taylor expansion for small λ :

$$\psi(x_i \pm \lambda, t) = \psi(x_i, t) \pm \lambda \frac{\partial\psi(x_i, t)}{\partial x_i} + \frac{\lambda^2}{2} \frac{\partial^2\psi(x_i, t)}{\partial x_i^2} + \mathcal{O}(\lambda^3). \quad (3.18)$$

Defining directional hopping rates $w_L(x_i) \equiv w_{i,i+1}$ (motion to the left into site i) and $w_R(x_i) \equiv w_{i,i-1}$ (motion to the right into site i), Eq. (3.8) gives $w_R(x) = w_L(x)^*$. Dropping the site index i , the continuum evolution equation becomes

$$i\frac{\partial\psi(x, t)}{\partial t} = \frac{\lambda^2}{2}[w_L(x) + w_R(x)]\frac{\partial^2\psi(x, t)}{\partial x^2} + \lambda[w_L(x) - w_R(x)]\frac{\partial\psi(x, t)}{\partial x} + \frac{v(x, t)}{\hbar}\psi(x, t). \quad (3.19)$$

In the absence of velocity-dependent forces (such as external magnetic fields),² spatial reflection symmetry requires $w_R(x) = w_L(x) \equiv w$, where $w \in \mathbb{R}$ is independent of position. In order for the second-derivative kinetic term to remain finite and non-vanishing as $\lambda \rightarrow 0$, the hopping parameter must scale as

$$\lim_{\lambda \rightarrow 0} \lambda^2 w = -\frac{\hbar}{2m}, \quad (3.20)$$

where m is the particle mass. Multiplying Eq. (3.19) by $\hbar$ then recovers the celebrated one-dimensional time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + v(x, t) \psi(x, t). \quad (3.21)$$

Generalizing this derivation to three dimensions gives:

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + v(\mathbf{r}, t) \right] \psi(\mathbf{r}, t). \quad (3.22)$$

The scaling behavior in Eq. (3.20) has a clear physical interpretation. The characteristic time τ required for amplitude to hop between adjacent cells is of order $\tau \sim |1/w| \sim 2m\lambda^2/\hbar$. Heavier particles diffuse more slowly ($\tau \propto m$), quantum diffusion slows down in the classical limit ($\hbar \rightarrow 0$), and the quadratic dependence $\lambda^2 \propto \tau$ mirrors the fundamental scaling of random walk processes.

3.2.3 Free Particle Wave Solutions

For a free particle ($v(\mathbf{r}, t) \equiv 0$), the Schrödinger equation (3.22) admits plane wave solutions of the form

$$\psi(\mathbf{r}, t) = C \exp \left[\frac{i}{\hbar} (\mathbf{p} \cdot \mathbf{r} - Et) \right]. \quad (3.23)$$

Substituting this ansatz into Eq. (3.22) yields the energy dispersion relation:

$$E = \frac{p^2}{2m}. \quad (3.24)$$

Identifying $\omega = E/\hbar$ as the angular frequency and $\mathbf{k} = \mathbf{p}/\hbar$ as the wave vector, Eq. (3.24) reproduces the classical kinetic energy of a free particle.

The probability density for a single plane wave state is spatially uniform:

$$P(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2 = |C|^2 = \text{const.} \quad (3.25)$$

Hence, a state with a precisely defined momentum $\mathbf{p}$ is completely delocalized throughout all space, illustrating Heisenberg's uncertainty principle in its extreme limit.

3.3 State Vectors in Continuous Space

To establish a unified mathematical framework connecting discrete quantum systems (such as photon polarization) with spatial wave mechanics, we formulate wave functions in terms of abstract state vectors residing in an infinite-dimensional Hilbert space.

Let a particle at a given instant be described by the spatial wave function $\psi(\mathbf{r})$. In analogy with the discrete amplitude column vector introduced earlier, we associate with this physical state an abstract ket vector $|\Psi\rangle$, which can be visualized as a continuous column vector:

$$|\Psi\rangle = \begin{pmatrix} \vdots \\ \psi(\mathbf{r}) \\ \vdots \end{pmatrix}, \quad (3.26)$$

²The inclusion of magnetic vector potentials will be addressed in Section 3.8.

where the continuous spatial coordinate $\mathbf{r} \in \mathbb{R}^3$ replaces the discrete index i .

3.3.1 Inner Product and Normalization

The scalar product between two state vectors $|\Phi\rangle$ and $|\Psi\rangle$, associated with wave functions $\varphi(\mathbf{r})$ and $\psi(\mathbf{r})$ respectively, is defined as the spatial overlap integral:

$$\langle\Phi|\Psi\rangle = \int d^3r \varphi^*(\mathbf{r})\psi(\mathbf{r}). \quad (3.27)$$

The inner product satisfies the standard conjugate symmetry relation:

$$\langle\Phi|\Psi\rangle = \langle\Psi|\Phi\rangle^*. \quad (3.28)$$

The requirement that the total probability of finding the particle somewhere in space equals unity imposes the normalization condition:

$$\int d^3r |\psi(\mathbf{r})|^2 = \langle\Psi|\Psi\rangle = 1. \quad (3.29)$$

3.3.2 Position Basis and the Completeness Relation

We introduce an idealized position eigenstate $|\mathbf{r}\rangle$, corresponding to a state strictly localized at the point $\mathbf{r}$. In this representation, the wave function $\psi(\mathbf{r})$ is simply the projection of the state vector $|\Psi\rangle$ onto the position basis ket $|\mathbf{r}\rangle$:

$$\psi(\mathbf{r}) = \langle\mathbf{r}|\Psi\rangle. \quad (3.30)$$

Because the collection of values $\psi(\mathbf{r})$ for all $\mathbf{r}$ uniquely and completely specifies the state, the set $\{|\mathbf{r}\rangle\}$ forms a continuous basis for the Hilbert space.

Expressing the inner product (3.27) in terms of projections, we have:

$$\langle\Phi|\Psi\rangle = \int d^3r \langle\Phi|\mathbf{r}\rangle\langle\mathbf{r}|\Psi\rangle = \langle\Phi|\left(\int d^3r |\mathbf{r}\rangle\langle\mathbf{r}|\right)|\Psi\rangle. \quad (3.31)$$

Since Eq. (3.31) holds for arbitrary states $|\Phi\rangle$ and $|\Psi\rangle$, the operator in parentheses must be the identity operator $\mathbb{I}$:

$$\int d^3r |\mathbf{r}\rangle\langle\mathbf{r}| = \mathbb{I}. \quad (3.32)$$

This is the continuous resolution of the identity (or completeness relation) for the position basis. Applying this identity to any ket $|\Psi\rangle$ yields its continuous expansion in position space:

$$|\Psi\rangle = \int d^3r' |\mathbf{r}'\rangle\langle\mathbf{r}'|\Psi\rangle = \int d^3r' \psi(\mathbf{r}')|\mathbf{r}'\rangle. \quad (3.33)$$

3.3.3 Orthogonality and the Dirac Delta Function

Projecting Eq. (3.33) onto a bra $\langle\mathbf{r}|$, we obtain:

$$\langle\mathbf{r}|\Psi\rangle = \psi(\mathbf{r}) = \int d^3r' \langle\mathbf{r}|\mathbf{r}'\rangle\psi(\mathbf{r}'). \quad (3.34)$$

Since this identity must hold for any arbitrary, well-behaved wave function $\psi(\mathbf{r})$, the overlap between two position eigenstates must satisfy the defining property of the three-dimensional Dirac delta function:

$$\langle\mathbf{r}|\mathbf{r}'\rangle = \delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (3.35)$$

Thus, position eigenstates are generalized continuous orthogonal states normalized to the Dirac delta distribution.

3.4 The Dirac Delta Function and Momentum Representation

The formulation of quantum mechanics in continuous bases relies heavily on the theory of distributions, specifically the Dirac delta function. In this section, we review the fundamental properties of the delta function and develop the momentum representation via Fourier analysis.

3.4.1 Properties of the Dirac Delta Function

In one dimension, the Dirac delta function $\delta(x - x')$ may be characterized heuristically as vanishing everywhere except at $x = x'$, where it diverges such that its integral over the real line is unity:

$$\int_{-\infty}^{\infty} dx' \delta(x - x') = 1. \quad (3.36)$$

Its defining property is the sifting property, which extracts the value of any continuous test function $f(x)$ evaluated at x' :

$$\int_{-\infty}^{\infty} dx' \delta(x - x') f(x') = f(x). \quad (3.37)$$

Elementary properties include symmetry under reflection:

$$\delta(x - x') = \delta(x' - x), \quad (3.38)$$

and scaling under coordinate dilation:

$$\delta(ax) = \frac{1}{|a|} \delta(x) \quad (a \neq 0). \quad (3.39)$$

The delta distribution can be rigorously represented as the weak limit of a sequence of normalized functions of narrowing width. For instance, the normalized Gaussian distribution

$$f_{\lambda}(x' - x) = \frac{1}{\lambda\sqrt{2\pi}} \exp\left[-\frac{(x' - x)^2}{2\lambda^2}\right], \quad (3.40)$$

satisfies $\int_{-\infty}^{\infty} dx' f_{\lambda}(x' - x) = 1$ for all $\lambda > 0$, and in the limit of vanishing width converges to the delta function:

$$\lim_{\lambda \rightarrow 0} f_{\lambda}(x' - x) = \delta(x - x'). \quad (3.41)$$

In three spatial dimensions, the delta function factorizes in Cartesian coordinates:

$$\delta^{(3)}(\mathbf{r} - \mathbf{r}') = \delta(x - x') \delta(y - y') \delta(z - z'), \quad (3.42)$$

satisfying the corresponding volume sifting identity:

$$\int d^3r' \delta^{(3)}(\mathbf{r} - \mathbf{r}') f(\mathbf{r}') = f(\mathbf{r}). \quad (3.43)$$

An idealized position eigenstate $|\mathbf{r}_0\rangle$ possesses the position-space wave function:

$$\psi(\mathbf{r}) = \langle \mathbf{r} | \mathbf{r}_0 \rangle = \delta^{(3)}(\mathbf{r} - \mathbf{r}_0). \quad (3.44)$$

While not square-integrable in the strict Hilbert space $L^2(\mathbb{R}^3)$ sense (a perfectly localized particle would possess infinite kinetic energy and momentum uncertainty), these generalized states form an indispensable continuous basis.

3.4.2 Fourier Transform and the Momentum Basis

Given a spatial wave function $\psi(\mathbf{r})$, we define its Fourier transform by

$$\tilde{\psi}(\mathbf{p}) \equiv \int d^3r e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi(\mathbf{r}). \quad (3.45)$$

The inclusion of Planck's constant $\hbar$ in the exponent ensures that $\mathbf{p}$ has the dimensions of physical momentum. The inverse Fourier transform is then given by

$$\psi(\mathbf{r}) = \int \frac{d^3p}{(2\pi\hbar)^3} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \tilde{\psi}(\mathbf{p}). \quad (3.46)$$

Equation (3.46) expresses the arbitrary state $\psi(\mathbf{r})$ as a continuous superposition of monochromatic plane waves $e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$ corresponding to definite momentum $\mathbf{p}$.

In Dirac notation, we associate each plane wave with a momentum eigenstate $|\mathbf{p}\rangle$, defined such that

$$\langle \mathbf{r} | \mathbf{p} \rangle = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \quad (3.47)$$

Using the position completeness relation $\int d^3r |\mathbf{r}\rangle\langle \mathbf{r}| = \mathbb{I}$, the Fourier amplitude can be identified as the projection of the state $|\Psi\rangle$ onto $|\mathbf{p}\rangle$:

$$\tilde{\psi}(\mathbf{p}) = \int d^3r \langle \mathbf{p} | \mathbf{r} \rangle \langle \mathbf{r} | \Psi \rangle = \langle \mathbf{p} | \Psi \rangle. \quad (3.48)$$

Rewriting Eq. (3.46) in bra-ket form:

$$\langle \mathbf{r} | \Psi \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} \langle \mathbf{r} | \mathbf{p} \rangle \langle \mathbf{p} | \Psi \rangle, \quad (3.49)$$

which implies the completeness relation (resolution of the identity) for momentum states:

$$\int \frac{d^3p}{(2\pi\hbar)^3} |\mathbf{p}\rangle\langle \mathbf{p}| = \mathbb{I}. \quad (3.50)$$

Setting $|\Psi\rangle = |\mathbf{r}\rangle$ in this expansion yields

$$|\mathbf{r}\rangle = \int \frac{d^3p}{(2\pi\hbar)^3} |\mathbf{p}\rangle \langle \mathbf{p} | \mathbf{r} \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} |\mathbf{p}\rangle. \quad (3.51)$$

Projecting onto $\langle \mathbf{r}' |$, we obtain the Fourier representation of the three-dimensional delta function:

$$\int \frac{d^3p}{(2\pi\hbar)^3} e^{i\mathbf{p}\cdot(\mathbf{r}'-\mathbf{r})/\hbar} = \delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (3.52)$$

3.4.3 Physical Interpretation of the Momentum Wave Function

The kernel $\langle \mathbf{r} | \mathbf{p} \rangle = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$ functions as the unitary transformation matrix connecting the position representation and the momentum representation.

Using the completeness of the momentum basis, the normalization condition $\langle \Psi | \Psi \rangle = 1$ becomes

$$\langle \Psi | \Psi \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} \langle \Psi | \mathbf{p} \rangle \langle \mathbf{p} | \Psi \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} |\langle \mathbf{p} | \Psi \rangle|^2 = 1. \quad (3.53)$$

By Born's statistical postulate, the quantity

$$dP(\mathbf{p}) = \frac{d^3p}{(2\pi\hbar)^3} |\langle \mathbf{p} | \Psi \rangle|^2 \quad (3.54)$$

represents the probability of finding the particle with momentum in the infinitesimal volume d^3p centered at $\mathbf{p}$. Thus, $\psi(\mathbf{r}) = \langle \mathbf{r} | \Psi \rangle$ is the probability amplitude in position space, while $\tilde{\psi}(\mathbf{p}) = \langle \mathbf{p} | \Psi \rangle$ is the probability amplitude in momentum space.

3.5 Quantum Operators and Observables

In the operator formalism of quantum mechanics, every physical observable is associated with a linear operator acting on the state space. When a system resides in an eigenstate of an observable's operator, a measurement of that quantity yields with certainty the corresponding eigenvalue.

3.5.1 The Momentum Operator

Consider a particle in a plane wave state $|\mathbf{p}\rangle$. This state possesses a well-defined momentum $\mathbf{p}$ and is therefore an eigenstate of the momentum operator $\hat{\mathbf{p}}$:

$$\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle. \quad (3.55)$$

Using the resolution of the identity in momentum space, $\int \frac{d^3p}{(2\pi\hbar)^3} |\mathbf{p}\rangle\langle\mathbf{p}| = \mathbb{I}$, the action of $\hat{\mathbf{p}}$ on an arbitrary state $|\Psi\rangle$ is

$$\hat{\mathbf{p}}|\Psi\rangle = \int \frac{d^3p}{(2\pi\hbar)^3} \mathbf{p}|\mathbf{p}\rangle\langle\mathbf{p}|\Psi\rangle. \quad (3.56)$$

Hence, the spectral resolution of the momentum operator is given by

$$\hat{\mathbf{p}} = \int \frac{d^3p}{(2\pi\hbar)^3} \mathbf{p}|\mathbf{p}\rangle\langle\mathbf{p}|. \quad (3.57)$$

To determine the action of $\hat{\mathbf{p}}$ in the position representation, consider its projection onto the spatial basis $\langle\mathbf{r}|$. Taking the gradient of the plane wave $\langle\mathbf{r}|\mathbf{p}\rangle = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$:

$$\langle\mathbf{r}|\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} = -i\hbar\nabla\langle\mathbf{r}|\mathbf{p}\rangle. \quad (3.58)$$

Expanding an arbitrary ket $|\Psi\rangle$ over the momentum basis immediately yields

$$\langle\mathbf{r}|\hat{\mathbf{p}}|\Psi\rangle = -i\hbar\nabla\langle\mathbf{r}|\Psi\rangle = -i\hbar\nabla\psi(\mathbf{r}). \quad (3.59)$$

Thus, in position space, the momentum observable is realized as the differential operator:

$$\hat{\mathbf{p}} \longleftrightarrow -i\hbar\nabla. \quad (3.60)$$

For a free particle, the kinetic energy operator is constructed as

$$\hat{T} = \frac{\hat{\mathbf{p}}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2. \quad (3.61)$$

The free-particle Schrödinger equation can therefore be written in abstract vector form as

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = \frac{\hat{\mathbf{p}}^2}{2m}|\Psi(t)\rangle, \quad (3.62)$$

which reproduces Eq. (3.22) when projected onto the position bra $\langle\mathbf{r}|$:

$$i\hbar\frac{\partial}{\partial t}\langle\mathbf{r}|\Psi(t)\rangle = \langle\mathbf{r}|\frac{\hat{\mathbf{p}}^2}{2m}|\Psi(t)\rangle = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r},t). \quad (3.63)$$

3.5.2 The Position Operator and Potential Energy

Similarly, the position operator $\hat{\mathbf{r}}$ acts diagonally on the continuous localized basis $\{|\mathbf{r}'\rangle\}$:

$$\hat{\mathbf{r}}|\mathbf{r}'\rangle = \mathbf{r}'|\mathbf{r}'\rangle, \quad (3.64)$$

yielding the spectral decomposition:

$$\hat{\mathbf{r}} = \int d^3r' \mathbf{r}' |\mathbf{r}'\rangle \langle \mathbf{r}'|. \quad (3.65)$$

Taking matrix elements between $\langle \mathbf{r}|$ and an arbitrary state $|\Psi\rangle$:

$$\langle \mathbf{r}|\hat{\mathbf{r}}|\Psi\rangle = \int d^3r' \mathbf{r}' \delta^{(3)}(\mathbf{r} - \mathbf{r}') \langle \mathbf{r}'|\Psi\rangle = \mathbf{r} \langle \mathbf{r}|\Psi\rangle = \mathbf{r} \psi(\mathbf{r}). \quad (3.66)$$

In the position representation, the position operator acts simply as multiplication by the coordinate $\mathbf{r}$. More generally, for any function $v(\mathbf{r})$, the operator $v(\hat{\mathbf{r}})$ acts as

$$\langle \mathbf{r}|v(\hat{\mathbf{r}})|\Psi\rangle = v(\mathbf{r})\psi(\mathbf{r}). \quad (3.67)$$

3.5.3 The Hamiltonian Operator and Canonical Quantization

Combining the kinetic and potential energy operators gives the total quantum Hamiltonian operator:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + v(\hat{\mathbf{r}}, t). \quad (3.68)$$

The general time-dependent Schrödinger equation is expressed coordinate-independently as

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle. \quad (3.69)$$

Projecting Eq. (3.69) onto $\langle \mathbf{r}|$ recovers the wave equation in position space:

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = H(\mathbf{r}, -i\hbar \nabla, t) \psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + v(\mathbf{r}, t) \right] \psi(\mathbf{r}, t). \quad (3.70)$$

This correspondence principle—substituting classical canonical variables by their corresponding operators $\mathbf{r} \rightarrow \hat{\mathbf{r}}$ and $\mathbf{p} \rightarrow \hat{\mathbf{p}}$ —ensures that the classical equations of motion emerge naturally in the macroscopic limit.

3.6 The Free Particle and Wave Packet Dynamics

We now examine the dynamics of a free non-relativistic particle ($v(\mathbf{r}, t) \equiv 0$). If at $t = 0$ the particle resides in a momentum eigenstate $\psi(\mathbf{r}, 0) = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$, its subsequent time evolution is governed by the plane wave solution:

$$\psi(\mathbf{r}, t) = \exp \left[\frac{i}{\hbar} (\mathbf{p} \cdot \mathbf{r} - Et) \right], \quad E = \frac{p^2}{2m}. \quad (3.71)$$

An arbitrary initial wave packet can be decomposed into momentum eigenstates via Fourier analysis. The time-dependent state then evolves as

$$\psi(\mathbf{r}, t) = \int \frac{d^3p}{(2\pi\hbar)^3} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} e^{-iEt/\hbar} \tilde{\psi}(\mathbf{p}, 0). \quad (3.72)$$

In Dirac notation, using $\tilde{\psi}(\mathbf{p}, 0) = \langle \mathbf{p} | \Psi(0) \rangle$, this is expressed as

$$\langle \mathbf{r} | \Psi(t) \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} \langle \mathbf{r} | \mathbf{p} \rangle e^{-iEt/\hbar} \langle \mathbf{p} | \Psi(0) \rangle. \quad (3.73)$$

Because momentum eigenstates are stationary energy eigenstates for a free particle, the momentum-space amplitude evolves purely by a dynamical phase:

$$\langle \mathbf{p} | \Psi(t) \rangle = e^{-iEt/\hbar} \langle \mathbf{p} | \Psi(0) \rangle. \quad (3.74)$$

The formal time-evolution operator acting on the state vector is given by

$$|\Psi(t)\rangle = \left[\int \frac{d^3p}{(2\pi\hbar)^3} |\mathbf{p}\rangle e^{-iEt/\hbar} \langle \mathbf{p}| \right] |\Psi(0)\rangle = e^{-i\hat{H}_0 t/\hbar} |\Psi(0)\rangle. \quad (3.75)$$

3.6.1 Wave Packets and Group Velocity

While a pure plane wave is completely delocalized in space, a physical particle can be modeled by a localized *wave packet* constructed from a distribution of momentum components peaked about a central value $\mathbf{p}_0$ with spread Δp .

Writing the initial momentum amplitude in polar form, $\langle \mathbf{p} | \Psi(0) \rangle = g(\mathbf{p}) e^{i\alpha(\mathbf{p})}$, where $g(\mathbf{p}) \geq 0$ is localized around $\mathbf{p}_0$ and $\alpha(\mathbf{p})$ is a real phase, the position-space wave function is

$$\psi(\mathbf{r}, t) = \int \frac{d^3p}{(2\pi\hbar)^3} g(\mathbf{p}) \exp \left\{ \frac{i}{\hbar} [\mathbf{p} \cdot \mathbf{r} - E(\mathbf{p})t + \hbar\alpha(\mathbf{p})] \right\}. \quad (3.76)$$

For general spatial coordinates $\mathbf{r}$, rapid oscillations of the complex exponential lead to destructive interference, causing the integral to vanish. Constructive interference occurs at points $\mathbf{r}_t$ where the phase is stationary with respect to $\mathbf{p}$ near the peak $\mathbf{p}_0$:

$$\nabla_{\mathbf{p}} [\mathbf{p} \cdot \mathbf{r}_t - E(\mathbf{p})t + \hbar\alpha(\mathbf{p})] \big|_{\mathbf{p}=\mathbf{p}_0} = 0. \quad (3.77)$$

Solving for the center of the wave packet $\mathbf{r}_t$ yields

$$\mathbf{r}_t = \nabla_{\mathbf{p}} E(\mathbf{p}) \big|_{\mathbf{p}=\mathbf{p}_0} t - \hbar \nabla_{\mathbf{p}} \alpha(\mathbf{p}_0) = \frac{\mathbf{p}_0}{m} t + \mathbf{r}_0, \quad (3.78)$$

where $\mathbf{r}_0 = -\hbar \nabla_{\mathbf{p}} \alpha(\mathbf{p}_0)$ represents the initial centroid position. Thus, the center of the wave packet moves with a constant velocity matching the classical particle velocity:

$$\mathbf{v}_g = \nabla_{\mathbf{p}} E = \frac{\mathbf{p}_0}{m}. \quad (3.79)$$

This is the *group velocity* of the wave packet, whereas individual Fourier components propagate at the *phase velocity*:

$$v_p = \frac{\omega}{|\mathbf{k}|} = \frac{E}{p} = \frac{p}{2m} = \frac{1}{2} v_g. \quad (3.80)$$

3.6.2 The Heisenberg Uncertainty Principle

The spatial width Δx of the wave packet can be estimated from the condition for constructive interference. Expanding the phase along the x -direction about p_{0x} :

$$\Delta\Phi \approx \frac{1}{\hbar} \Delta p_x \frac{\partial}{\partial p_x} [\mathbf{p} \cdot \mathbf{r} - Et + \hbar\alpha] \bigg|_{\mathbf{p}=\mathbf{p}_0} = \frac{1}{\hbar} \Delta p_x (x - x_t). \quad (3.81)$$

Significant constructive interference requires $|\Delta\Phi| \lesssim 2\pi$, establishing that the spatial confinement $|x - x_t| \lesssim \Delta x$ must satisfy

$$\Delta x \Delta p_x \gtrsim 2\pi\hbar. \quad (3.82)$$

Identical considerations apply independently to each Cartesian dimension:

$$\Delta y \Delta p_y \gtrsim 2\pi\hbar, \quad \Delta z \Delta p_z \gtrsim 2\pi\hbar. \quad (3.83)$$

By contrast, orthogonal coordinates and momenta (e.g., x and p_y) are mutually unconstrained.

To establish the exact quantitative lower bound, consider the canonical commutation relation between the position and momentum operators:

$$[\hat{x}, \hat{p}_x] \equiv \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = i\hbar\mathbb{I}. \quad (3.84)$$

This follows directly from the operator action on a test wave function:

$$(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\psi(x) = x \left(-i\hbar \frac{\partial \psi}{\partial x} \right) - \left(-i\hbar \frac{\partial}{\partial x} \right) (x\psi) = i\hbar\psi(x). \quad (3.85)$$

For any two physical observables represented by Hermitian operators $\hat{A}$ and $\hat{B}$, their uncertainties in a normalized state $|\Psi\rangle$ are defined by $\Delta A = \sqrt{\langle(\hat{A} - \langle\hat{A}\rangle)^2\rangle}$ and $\Delta B = \sqrt{\langle(\hat{B} - \langle\hat{B}\rangle)^2\rangle}$. Defining the state vectors

$$|\Theta\rangle = (\hat{A} - \langle\hat{A}\rangle)|\Psi\rangle, \quad |\Phi\rangle = (\hat{B} - \langle\hat{B}\rangle)|\Psi\rangle, \quad (3.86)$$

the Cauchy-Schwarz inequality $\langle\Theta|\Theta\rangle\langle\Phi|\Phi\rangle \geq |\langle\Theta|\Phi\rangle|^2$ yields

$$(\Delta A)^2(\Delta B)^2 \geq |\langle\Theta|\Phi\rangle|^2 \geq [\text{Im}\langle\Theta|\Phi\rangle]^2 = \left| \frac{1}{2i} \langle\Psi|[\hat{A}, \hat{B}]|\Psi\rangle \right|^2. \quad (3.87)$$

Applying this Robertson-Schrödinger relation to $\hat{x}$ and $\hat{p}_x$ yields the exact Heisenberg uncertainty relation:

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}. \quad (3.88)$$

The equality $\Delta x \Delta p_x = \hbar/2$ is attained uniquely when $|\Theta\rangle$ is proportional to $|\Phi\rangle$:

$$\left(\frac{\hat{x} - \langle x \rangle}{\Delta x} + i \frac{\hat{p}_x - \langle p_x \rangle}{\Delta p_x} \right) |\Psi\rangle = 0. \quad (3.89)$$

In position representation, this operator equation reduces to the first-order differential equation:

$$\left[\frac{x - \langle x \rangle}{\Delta x} + \frac{\hbar}{\Delta p_x} \frac{\partial}{\partial x} - \frac{i\langle p_x \rangle}{\Delta p_x} \right] \psi(x) = 0, \quad (3.90)$$

whose normalized solution is the Gaussian wave packet:

$$\psi(x) = \left(\frac{1}{2\pi(\Delta x)^2} \right)^{1/4} \exp \left[\frac{i\langle p_x \rangle x}{\hbar} - \frac{(x - \langle x \rangle)^2}{4(\Delta x)^2} \right]. \quad (3.91)$$

Gaussian states are thus the minimal-uncertainty wave packets of quantum mechanics.

3.6.3 The Free Particle Propagator and Classical Action

The spatial evolution of a wave function from an initial time t' to a later time t can be expressed in integral form:

$$\psi(\mathbf{r}, t) = \int d^3r' K(\mathbf{r}, t; \mathbf{r}', t') \psi(\mathbf{r}', t'), \quad (3.92)$$

where the integral kernel $K(\mathbf{r}, t; \mathbf{r}', t')$, termed the *quantum propagator* or Green's function, denotes the transition amplitude:

$$K(\mathbf{r}, t; \mathbf{r}', t') \equiv \langle \mathbf{r} | e^{-i\hat{H}_0(t-t')/\hbar} | \mathbf{r}' \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} e^{i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')/\hbar} e^{-ip^2(t-t')/(2m\hbar)}. \quad (3.93)$$

This three-dimensional Gaussian integral can be evaluated analytically:

$$\int_{-\infty}^{\infty} \frac{dp_x}{2\pi\hbar} \exp \left[\frac{i}{\hbar} p_x (x - x') - \frac{i(t-t')}{2m\hbar} p_x^2 \right] = \left[\frac{m}{2\pi i\hbar(t-t')} \right]^{1/2} \exp \left[\frac{im(x-x')^2}{2\hbar(t-t')} \right]. \quad (3.94)$$

Taking the product over all three dimensions, the free propagator becomes

$$K(\mathbf{r}, t; \mathbf{r}', t') = \left[\frac{m}{2\pi i\hbar(t-t')} \right]^{3/2} \exp \left[\frac{im(\mathbf{r} - \mathbf{r}')^2}{2\hbar(t-t')} \right]. \quad (3.95)$$

In the limit $t \rightarrow t'$, the propagator reproduces the identity kernel: $\lim_{t \rightarrow t'} K(\mathbf{r}, t; \mathbf{r}', t') = \delta^{(3)}(\mathbf{r} - \mathbf{r}')$.

Remarkably, the phase of the propagator directly encodes the classical action along the extremal trajectory connecting $(\mathbf{r}', t')$ and $(\mathbf{r}, t)$. For a free particle with Lagrangian $\mathcal{L} = \frac{1}{2}m\mathbf{v}^2$, the classical path has constant velocity $\mathbf{v}_{cl} = (\mathbf{r} - \mathbf{r}')/(t - t')$, giving the classical action

$$S_{cl}(\mathbf{r}, t; \mathbf{r}', t') = \int_{t'}^t \mathcal{L} dt'' = \frac{1}{2}m \frac{(\mathbf{r} - \mathbf{r}')^2}{t - t'}. \quad (3.96)$$

Therefore, the free quantum propagator satisfies the fundamental semiclassical relation:

$$K(\mathbf{r}, t; \mathbf{r}', t') = \left[\frac{m}{2\pi i\hbar(t-t')} \right]^{3/2} \exp \left[\frac{i}{\hbar} S_{cl}(\mathbf{r}, t; \mathbf{r}', t') \right]. \quad (3.97)$$

This connection provides the stepping stone to the path integral formulation of quantum mechanics.

3.7 Quantum Mechanical Motion as a Sum Over Paths

A central feature of the time evolution kernel $K(\mathbf{r}, t; \mathbf{r}', t')$ is that it satisfies the Chapman-Kolmogorov composition law:

$$K(\mathbf{r}_1, t_1; \mathbf{r}_3, t_3) = \int d^3r_2 K(\mathbf{r}_1, t_1; \mathbf{r}_2, t_2) K(\mathbf{r}_2, t_2; \mathbf{r}_3, t_3) \quad (t_1 > t_2 > t_3). \quad (3.98)$$

This relation follows directly from the completeness of the spatial position basis at intermediate time t_2 :

$$\langle \mathbf{r}_1 | e^{-i\hat{H}(t_1-t_3)/\hbar} | \mathbf{r}_3 \rangle = \int d^3r_2 \langle \mathbf{r}_1 | e^{-i\hat{H}(t_1-t_2)/\hbar} | \mathbf{r}_2 \rangle \langle \mathbf{r}_2 | e^{-i\hat{H}(t_2-t_3)/\hbar} | \mathbf{r}_3 \rangle. \quad (3.99)$$

3.7.1 Path Discretization and the Action

Equation (3.98) provides a deep physical insight into the transition amplitude. It states that the total amplitude to propagate from $(\mathbf{r}_3, t_3)$ to $(\mathbf{r}_1, t_1)$ is obtained by summing the product of amplitudes along piecewise-linear two-legged paths passing through arbitrary intermediate coordinates $\mathbf{r}_2$ at time t_2 . Substituting the free-particle propagator into Eq. (3.98):

$$K(\mathbf{r}_1, t_1; \mathbf{r}_2, t_2) K(\mathbf{r}_2, t_2; \mathbf{r}_3, t_3) = \left[\frac{m}{2\pi i\hbar(t_1-t_2)} \right]^{3/2} \left[\frac{m}{2\pi i\hbar(t_2-t_3)} \right]^{3/2} \times \exp \left\{ \frac{i}{\hbar} [S_{cl}(\mathbf{r}_1, t_1; \mathbf{r}_2, t_2) + S_{cl}(\mathbf{r}_2, t_2; \mathbf{r}_3, t_3)] \right\}. \quad (3.100)$$

The exponent is precisely $i/\hbar$ times the classical action evaluated along the two-segment broken trajectory:

$$S_{path} = S_{cl}(\mathbf{r}_1, t_1; \mathbf{r}_2, t_2) + S_{cl}(\mathbf{r}_2, t_2; \mathbf{r}_3, t_3). \quad (3.101)$$

Dividing the total time interval into N equal subintervals $\Delta t = (t_1 - t_3)/N$ with intermediate times $t_k = t_3 + k\Delta t$, we repeatedly apply the composition rule across all time slices:

$$K(\mathbf{r}_1, t_1; \mathbf{r}_3, t_3) = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \Delta t} \right)^{3N/2} \int \prod_{k=1}^{N-1} d^3 r_k \exp \left[\frac{i}{\hbar} \sum_{k=1}^N S_{\text{cl}}(\mathbf{r}_k, t_k; \mathbf{r}_{k-1}, t_{k-1}) \right]. \quad (3.102)$$

In the continuum limit $N \rightarrow \infty$, the sum over discrete coordinates becomes a functional integration over all possible continuous trajectories $\mathbf{r}(t)$ satisfying the boundary conditions $\mathbf{r}(t_3) = \mathbf{r}_3$ and $\mathbf{r}(t_1) = \mathbf{r}_1$:³

$$K(\mathbf{r}_1, t_1; \mathbf{r}_3, t_3) = \int \mathcal{D}[\mathbf{r}(t)] \exp \left(\frac{i}{\hbar} S[\mathbf{r}(t)] \right), \quad (3.103)$$

where the action functional is the time integral of the classical Lagrangian $\mathcal{L}(\mathbf{r}, \dot{\mathbf{r}}, t)$:

$$S[\mathbf{r}(t)] = \int_{t_3}^{t_1} \mathcal{L}(\mathbf{r}(t), \dot{\mathbf{r}}(t), t) dt. \quad (3.104)$$

Although derived here for free particles, Eq. (3.103) remains valid in the presence of arbitrary scalar and vector potentials. The quantum mechanical particle simultaneously explores all conceivable space-time paths connecting the initial and final configurations, each path weighted by the phase factor $e^{iS/\hbar}$.

3.7.2 Diffraction as Path Truncation

This sum-over-paths formulation provides an intuitive geometric understanding of wave phenomena such as diffraction and interference.

- When an impenetrable barrier is placed between the source and the detector, all trajectories intersecting the barrier are eliminated from the functional domain in Eq. (3.103). The removal of these paths alters the interference pattern on the detection screen, producing the familiar diffraction pattern.
- In the classic double-slit experiment, opening a single slit restricts the accessible trajectories to those threading that slit. Opening both slits introduces an entirely new class of paths passing through the second slit; their mutual quantum interference yields the characteristic two-slit fringe pattern.

3.7.3 The Classical Limit and Hamilton's Principle

In the macroscopic or classical limit, the characteristic action S of the system is enormously larger than Planck's constant ($\hbar \rightarrow 0$). For two arbitrary neighboring paths $\mathbf{r}(t)$ and $\mathbf{r}(t) + \delta\mathbf{r}(t)$, the difference in action δS typically satisfies $\delta S \gg \hbar$. Consequently, the phase difference $\delta S/\hbar$ fluctuates wildly, leading to complete destructive interference across generic path bundles.

However, near an extremal trajectory $\mathbf{r}_{\text{cl}}(t)$ that renders the action stationary:

$$\delta S[\mathbf{r}_{\text{cl}}(t)] = 0, \quad (3.105)$$

the first-order phase variation vanishes. Neighboring paths within a corridor satisfying $|\delta S| \lesssim \hbar$ interfere constructively. As $\hbar \rightarrow 0$, this coherent bundle contracts to an infinitesimally narrow tube surrounding the classical trajectory. Thus, Hamilton's classical principle of stationary action emerges as the short-wavelength stationary-phase limit of the underlying quantum superposition of paths.

³This path-integral formulation of quantum mechanics was developed by R. P. Feynman, *Rev. Mod. Phys.* **20**, 367 (1948); see also R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGraw-Hill, New York, 1965).

3.8 Particle in a Magnetic Field and the Peierls Phase

In classical electrodynamics, the Hamiltonian for a non-relativistic particle of mass m and electric charge e in the presence of an external magnetic field $\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t)$ and scalar potential $V(\mathbf{r}, t)$ is given by minimal coupling:

$$H = \frac{1}{2m} \left[\mathbf{p} - \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + V(\mathbf{r}, t), \quad (3.106)$$

written here in Gaussian cgs units. Under canonical quantization, the physical position and kinematic momentum are promoted to operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$. The state vector evolves according to the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \left\{ \frac{1}{2m} \left[\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A}(\hat{\mathbf{r}}, t) \right]^2 + V(\hat{\mathbf{r}}, t) \right\} |\Psi(t)\rangle. \quad (3.107)$$

Because $\hat{\mathbf{p}} = -i\hbar \nabla$ and $\mathbf{A}(\mathbf{r}, t)$ do not generally commute ($[\hat{p}_j, A_k] = -i\hbar \partial_j A_k$), the squared kinetic term expands symmetrically:

$$\left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right)^2 = \hat{\mathbf{p}}^2 - \frac{e}{c} (\hat{\mathbf{p}} \cdot \mathbf{A} + \mathbf{A} \cdot \hat{\mathbf{p}}) + \frac{e^2}{c^2} A^2. \quad (3.108)$$

Projecting into the position representation yields

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + \frac{ie\hbar}{2mc} (\mathbf{A} \cdot \nabla + \nabla \cdot \mathbf{A}) + \frac{e^2}{2mc^2} A^2 + V(\mathbf{r}, t) \right] \psi(\mathbf{r}, t), \quad (3.109)$$

where $(\nabla \cdot \mathbf{A})\psi = (\nabla \cdot \mathbf{A})\psi + \mathbf{A} \cdot \nabla \psi$. In the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$), this simplifies to $\mathbf{A} \cdot \nabla + \nabla \cdot \mathbf{A} = 2\mathbf{A} \cdot \nabla$.

3.8.1 Discrete Lattice Perspective and the Peierls Substitution

It is instructive to revisit the discrete lattice hopping model introduced in Section 3.8. In one dimension with lattice spacing λ , matching the continuum expansion (??) to the magnetic Schrödinger equation (3.109) requires complex-valued hopping parameters:

$$w_{i+1,i} = -\frac{\hbar}{2m\lambda^2} \left[1 + \frac{ie\lambda}{\hbar c} A_x(x_i, t) \right], \quad (3.110)$$

with the diagonal term given by

$$w_{ii} = \frac{\hbar}{m\lambda^2} + \frac{e^2}{2mc^2\hbar} A_x(x_i, t)^2 + \frac{1}{\hbar} V(x_i, t). \quad (3.111)$$

In the small- λ limit, the linear bracket in Eq. (3.110) is the first-order Taylor expansion of a pure phase factor:

$$w_{i+1,i} = -\frac{\hbar}{2m\lambda^2} \exp \left[\frac{ie}{\hbar c} \int_{x_i}^{x_{i+1}} A_x(x, t) dx \right] \approx -\frac{\hbar}{2m\lambda^2} \exp \left[\frac{ie\lambda}{\hbar c} A_x(x_i, t) \right]. \quad (3.112)$$

This phase modification is known as the *Peierls substitution*: an external magnetic field dresses each hopping amplitude with a gauge phase proportional to the line integral of the vector potential along the hop.

3.8.2 Magnetic Vector Potential in the Path Integral

Consider a discretized trajectory consisting of sequential hops along sites x_j at time slices $t^{(j)}$. The cumulative phase acquired along the path is the product of the individual hopping phases:

$$\prod_j \left(1 + \frac{ie\lambda}{\hbar c} A_x(x_j, t^{(j)}) \right) \approx \exp \left[\frac{ie}{\hbar c} \sum_j \lambda A_x(x_j, t^{(j)}) \right]. \quad (3.113)$$

Taking the continuum limit $\lambda \rightarrow 0$ and generalizing to three dimensions,⁴ this phase factor becomes the path integral of the 1-form $\mathbf{A} \cdot d\mathbf{l}$:

$$\Phi_{\text{mag}}[\text{path}] = \exp \left[\frac{ie}{\hbar c} \int_{\text{path}} \mathbf{A}(\mathbf{r}, t) \cdot d\mathbf{l} \right]. \quad (3.114)$$

Consequently, the full quantum propagator in a magnetic field, $K_{\mathbf{A}}(\mathbf{r}, t; \mathbf{r}', t')$, decomposes into the zero-field amplitude weighted by this geometric phase:

$$K_{\mathbf{A}}(\mathbf{r}, t; \mathbf{r}', t') = \int \mathcal{D}[\mathbf{r}(t)] \exp \left\{ \frac{i}{\hbar} \left[S_0[\mathbf{r}(t)] + \frac{e}{c} \int_{\text{path}} \mathbf{A}(\mathbf{r}, t) \cdot d\mathbf{l} \right] \right\}, \quad (3.115)$$

where S_0 is the classical action in the absence of the magnetic field. This structure is in complete agreement with classical mechanics, where the interaction Lagrangian between a charged particle and the electromagnetic field is

$$\mathcal{L}_{\mathbf{A}} = \mathcal{L}_0 + \frac{e}{c} \mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t), \quad (3.116)$$

so that $\frac{e}{c} \int \mathbf{v} \cdot \mathbf{A} dt = \frac{e}{c} \int_{\text{path}} \mathbf{A} \cdot d\mathbf{l}$.

3.9 The Aharonov-Bohm Effect

The path-dependent phase factor established in Eq. (3.114) leads to profound quantum phenomena that have no counterpart in classical mechanics. Consider two paths, denoted Path 1 and Path 2, originating at an initial point $\mathbf{r}'$ and terminating at a detection point $\mathbf{r}$, traversing a static vector potential $\mathbf{A}(\mathbf{r})$.

The amplitudes associated with the two trajectories acquire individual phase shifts:

$$\phi_1 = \frac{e}{\hbar c} \int_1 \mathbf{A} \cdot d\mathbf{l}, \quad \phi_2 = \frac{e}{\hbar c} \int_2 \mathbf{A} \cdot d\mathbf{l}. \quad (3.117)$$

The relative phase difference $\Delta\phi$ between these two interfering amplitudes is therefore

$$\Delta\phi = \phi_1 - \phi_2 = \frac{e}{\hbar c} \left(\int_1 \mathbf{A} \cdot d\mathbf{l} - \int_2 \mathbf{A} \cdot d\mathbf{l} \right) = \frac{e}{\hbar c} \oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{l}, \quad (3.118)$$

where $\mathcal{C}$ denotes the closed oriented loop formed by traversing Path 1 in the forward direction and Path 2 in the reverse direction. By Stokes' theorem, the closed line integral equals the magnetic flux Φ threading the interior of the loop:

$$\oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{l} = \iint_{\mathcal{S}} (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \iint_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{S} = \Phi. \quad (3.119)$$

Thus, the relative phase difference is directly proportional to the enclosed magnetic flux:

$$\Delta\phi = \frac{e}{\hbar c} \Phi. \quad (3.120)$$

⁴Corrections of order $\mathcal{O}(\lambda^2)$ vanish upon summation in the continuum limit, ensuring that Eq. (3.114) is exact.

When the enclosed magnetic flux is an integer multiple of the fundamental magnetic flux quantum:

$$\Phi = n\Phi_0, \quad \Phi_0 \equiv \frac{2\pi\hbar c}{e} = \frac{hc}{e} \approx 4.135 \times 10^{-7} \text{ G} \cdot \text{cm}^2 \quad (4.135 \times 10^{-15} \text{ Wb}), \quad (3.121)$$

the phase difference $\Delta\phi = 2\pi n$ leaves the quantum interference pattern completely invariant.

3.9.1 Physical Implications and Gauge Invariance

This result underpins the famous *Aharonov-Bohm effect*.⁵ Consider a double-slit electron interference experiment in which an infinitely long, perfectly shielded microscopic solenoid is positioned behind the slit barrier. The magnetic field $\mathbf{B}$ is strictly confined to the interior of the solenoid, so that the magnetic field vanishes everywhere along the accessible electron trajectories ($\mathbf{B} = 0$).

According to classical mechanics, charged particles experiencing no Lorentz force ($\mathbf{F} = e\mathbf{v} \times \mathbf{B}/c = 0$) should undergo no deflection, leaving the distribution on the detector screen unaltered. In quantum mechanics, however, the vector potential $\mathbf{A}$ outside the solenoid cannot vanish identically because $\oint \mathbf{A} \cdot d\mathbf{l} = \Phi \neq 0$. As the enclosed magnetic flux Φ is varied, the phase difference shifts according to Eq. (3.118), producing an observable lateral translation of the interference fringes.

The Aharonov-Bohm effect demonstrates that in quantum mechanics, the electromagnetic potentials $\mathbf{A}$ and V are not mere computational conveniences (as in classical mechanics), but direct physical determinants of quantum phase and interference. Although the vector potential itself is gauge-dependent under $\mathbf{A} \rightarrow \mathbf{A} + \nabla\Lambda$, the enclosed flux $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$ and the observable phase shift $\Delta\phi$ are strictly gauge-invariant.

3.10 Problems

1. **Tight-Binding Lattice with Periodic Boundary Conditions.** Consider the discrete Schrödinger equation for a free particle ($v_i = 0$ and $w_{i,i-1} = w_{i,i+1} \equiv w$):

$$i\hbar \frac{\partial \psi_j(t)}{\partial t} = -\hbar w [\psi_{j+1}(t) + \psi_{j-1}(t) - 2\psi_j(t)]. \quad (3.122)$$

Impose periodic boundary conditions on a chain of N sites, such that $\psi_{j+N}(t) = \psi_j(t)$.

- (a) Show that there are N linearly independent stationary solutions of this equation.
- (b) Determine the N orthonormal energy eigenstates $\psi_j^{(n)}(t) = u_j^{(n)} e^{-iE_n t/\hbar}$ and their associated energy eigenvalues E_n . Demonstrate that in the continuum limit ($\lambda \rightarrow 0$ with total circumference $L = N\lambda$ held fixed), the eigenvalues approach the continuous free-particle dispersion $E = p^2/2m$.
- (c) Define the discrete propagator matrix $K_{jk}(t, t')$ via $\psi_j(t) = \sum_{k=1}^N K_{jk}(t, t') \psi_k(t')$. Derive an explicit expression for $K_{jk}(t, t')$.

2. **Dirac Delta Function Representations.**

- (a) Express the three-dimensional delta function $\delta^{(3)}(\mathbf{r} - \mathbf{r}')$ as a product of three one-dimensional delta functions in spherical polar coordinates (r, θ, ϕ) .
- (b) Calculate the Fourier transform of $\delta^{(3)}(\mathbf{r} - \mathbf{r}')$, and express $\delta^{(3)}(\mathbf{r} - \mathbf{r}')$ in terms of its Fourier inversion integral.

⁵Y. Aharonov and D. Bohm, *Phys. Rev.* **115**, 485 (1959); experimental verification was first achieved by R. G. Chambers, *Phys. Rev. Lett.* **5**, 3 (1960).

- (c) Using the plane wave normalization $\langle \mathbf{r} | \mathbf{p} \rangle = e^{i\mathbf{p} \cdot \mathbf{r} / \hbar}$, prove that momentum eigenstates satisfy the orthogonality relation:

$$\langle \mathbf{p} | \mathbf{p}' \rangle = (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}'). \quad (3.123)$$

3. **Position Operator in Momentum Space.** Derive the momentum-space representation of the position operator $\hat{\mathbf{r}}$, that is, determine the functional relationship between $\langle \mathbf{p} | \hat{\mathbf{r}} | \Psi \rangle$ and the momentum wave function $\tilde{\psi}(\mathbf{p}) = \langle \mathbf{p} | \Psi \rangle$.
4. **Harmonic Oscillator in Momentum Space.** For an isotropic harmonic potential $v(\mathbf{r}) = \frac{1}{2}kr^2$, write down the time-dependent Schrödinger equation satisfied by the momentum-space amplitude $\tilde{\psi}(\mathbf{p}, t) = \langle \mathbf{p} | \Psi(t) \rangle$.
5. **Commutation Relations and Poisson Brackets.** Evaluate the commutator $[\hat{y}, \hat{p}_x]$. Compare the canonical quantum commutation relations $[\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij}$ with the classical Poisson brackets $\{r_i, p_j\}_{\text{PB}} = \delta_{ij}$.
6. **Expectation Value of Kinetic Energy.** Derive an integral expression for the expectation value of the kinetic energy, $\langle \hat{T} \rangle = \langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m} | \Psi \rangle$, in terms of the spatial wave function $\psi(\mathbf{r})$ and its gradient $\nabla\psi(\mathbf{r})$, assuming suitable decay at spatial infinity.
7. **Minimal Uncertainty Wave Packet.** By integrating the first-order differential equation

$$\left(\frac{x - \langle x \rangle}{\Delta x} + \frac{\hbar}{\Delta p_x} \frac{\partial}{\partial x} - \frac{i\langle p_x \rangle}{\Delta p_x} \right) \psi(x) = 0, \quad (3.124)$$

derive the Gaussian wave packet that saturates the Heisenberg uncertainty lower bound $\Delta x \Delta p_x = \hbar/2$.

8. **Propagator Convolution.** Explicitly evaluate the spatial convolution integral

$$\int d^3r_2 K(\mathbf{r}_1, t_1; \mathbf{r}_2, t_2) K(\mathbf{r}_2, t_2; \mathbf{r}_3, t_3) \quad (3.125)$$

using the explicit Gaussian form of the free-particle propagator, verifying that it reproduces $K(\mathbf{r}_1, t_1; \mathbf{r}_3, t_3)$.

9. **Propagator Differential Equation and Dyson Equation.**

- (a) Show that for $t > t'$, the free propagator satisfies the free-particle Schrödinger equation:

$$\left(i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \nabla^2 \right) K(\mathbf{r}, t; \mathbf{r}', t') = 0. \quad (3.126)$$

- (b) Suppose a particle moves under a time-dependent potential $v(\mathbf{r}, t)$ with initial state $\psi(\mathbf{r}, t_0) = \psi_0(\mathbf{r})$. Show that the time-dependent wave function satisfies the integral equation (Lippmann-Schwinger / Dyson form):

$$\psi(\mathbf{r}, t) = \int d^3r' K(\mathbf{r}, t; \mathbf{r}', t_0) \psi_0(\mathbf{r}') + \frac{1}{i\hbar} \int_{t_0}^t dt' \int d^3r' K(\mathbf{r}, t; \mathbf{r}', t') v(\mathbf{r}', t') \psi(\mathbf{r}', t'). \quad (3.127)$$

10. **Probability Current Density and Continuity.**

- (a) The probability current density for a particle in a scalar potential $V(\mathbf{r}, t)$ is defined by

$$\mathbf{j}(\mathbf{r}, t) = \frac{\hbar}{2mi} [\psi^*(\mathbf{r}, t) \nabla \psi(\mathbf{r}, t) - \psi(\mathbf{r}, t) \nabla \psi^*(\mathbf{r}, t)]. \quad (3.128)$$

Using the time-dependent Schrödinger equation, prove directly that the local probability density $P(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ obeys the continuity equation:

$$\frac{\partial P}{\partial t} + \nabla \cdot \mathbf{j} = 0. \quad (3.129)$$

- (b) Generalize the definition of $\mathbf{j}(\mathbf{r}, t)$ in the presence of an external magnetic vector potential $\mathbf{A}(\mathbf{r}, t)$ such that probability conservation remains satisfied.

11. **Gauge Invariance in Quantum Mechanics.** Consider a charged particle in the electromagnetic potentials generated by a scalar function $\lambda(\mathbf{r}, t)$:

$$\mathbf{A}(\mathbf{r}, t) = -\nabla\lambda(\mathbf{r}, t), \quad \phi(\mathbf{r}, t) = \frac{1}{c} \frac{\partial\lambda(\mathbf{r}, t)}{\partial t}. \quad (3.130)$$

- (a) Determine the physical electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ corresponding to these potentials.
- (b) Show that if $\psi^{(0)}(\mathbf{r}, t)$ solves the Schrödinger equation with $\lambda \equiv 0$, then

$$\psi(\mathbf{r}, t) = \exp \left[-\frac{ie}{\hbar c} \lambda(\mathbf{r}, t) \right] \psi^{(0)}(\mathbf{r}, t) \quad (3.131)$$

is an exact solution to the Schrödinger equation with potentials $\mathbf{A}$ and ϕ .

- (c) For a spatially uniform, time-dependent electric potential $v(\mathbf{r}, t) = e\phi(t)$, show that

$$\psi(\mathbf{r}, t) = \exp \left[-\frac{ie}{\hbar} \int_{-\infty}^t \phi(t') dt' \right] \psi^{(0)}(\mathbf{r}, t), \quad (3.132)$$

and explain why the lower integration limit is taken at $-\infty$.

12. **Scalar Aharonov-Bohm Effect with Conducting Drift Tubes.** A beam of charged particles (e) is emitted from a coherent source in well-defined wave packets. The beam is split into two paths, each directed through a hollow cylindrical conducting tube. For a time interval $t \in [t_0, t_1]$ during which the wave packet is entirely contained within the interior of the tubes, constant voltages V_A and V_B are applied to tubes A and B , respectively. Because the tubes are Faraday cages, no electric fields exist inside the tubes ($\mathbf{E} = 0$, $\mathbf{B} = 0$). At all other times, $V_A = V_B = 0$.

- (a) Determine how the interference fringe pattern at the downstream screen depends on V_A and V_B .
- (b) If the spatial fringe phase difference at the detector is parameterized as $\Delta\phi = Kz$ with $K = 10^4 \text{ cm}^{-1}$, calculate the physical shift of the fringe pattern (in cm) produced by $V_A = 0$, $V_B = 10^{-6} \text{ V}$, and an interaction duration $t_1 - t_0 = 10^{-9} \text{ s}$ for an electron ($e = 4.80 \times 10^{-10} \text{ esu} = 1.602 \times 10^{-19} \text{ C}$).

Chapter 4

Potential Problems Mostly In One Dimension

4.1 Potential Problems, Mostly in One Dimension

Having established the foundational principles of wave mechanics and the Schrödinger equation in Chapter 3, we now investigate the behavior of quantum particles subject to external spatial potentials $V(x)$. One-dimensional systems serve as an indispensable paradigm in quantum mechanics: they allow exact mathematical treatment of quintessential non-classical phenomena—such as quantum tunneling, bound states, parity symmetry, transmission resonances, and band structure in periodic potentials—while providing essential physical intuition that carries directly over to three-dimensional problems.

In this chapter, we first develop the general spectral properties of Hermitian operators and stationary energy eigenstates. We then systematically explore scattering and bound-state phenomena across standard one-dimensional potential landscapes, culminating in the operator solution of the simple harmonic oscillator.

4.2 Properties of Hermitian Operators

Physical observables in quantum mechanics are represented by linear operators whose spectra define the possible measurement outcomes. To guarantee that measurement results are real numbers and that states can be decomposed into complete orthonormal bases, these observables must be represented by *Hermitian* (self-adjoint) operators.

4.2.1 Definition and Adjoint Algebra

Given a linear operator $\hat{B}$ acting on a Hilbert space, its Hermitian adjoint $\hat{B}^\dagger$ is defined via the inner product by

$$\langle \Psi | \hat{B}^\dagger | \Phi \rangle \equiv \langle \Phi | \hat{B} | \Psi \rangle^* \quad (4.1)$$

for all states $|\Psi\rangle$ and $|\Phi\rangle$. In matrix representations, the matrix of $\hat{B}^\dagger$ is the conjugate transpose of that of $\hat{B}$. For any complex scalar λ , the adjoint satisfies

$$(\lambda \hat{B})^\dagger = \lambda^* \hat{B}^\dagger. \quad (4.2)$$

For the product of two linear operators $\hat{B}\hat{C}$, inserting a complete basis $\mathbb{I} = \sum_\alpha |\alpha\rangle\langle\alpha|$ yields

$$\begin{aligned} \langle \Psi | (\hat{B}\hat{C})^\dagger | \Phi \rangle &= \left\langle \Phi \left| \hat{B} \left(\sum_\alpha |\alpha\rangle\langle\alpha| \right) \hat{C} \right| \Psi \right\rangle^* = \sum_\alpha \langle \alpha | \hat{C} | \Psi \rangle^* \langle \Phi | \hat{B} | \alpha \rangle^* \\ &= \sum_\alpha \langle \Psi | \hat{C}^\dagger | \alpha \rangle \langle \alpha | \hat{B}^\dagger | \Phi \rangle = \langle \Psi | \hat{C}^\dagger \hat{B}^\dagger | \Phi \rangle. \end{aligned} \quad (4.3)$$

Hence, the adjoint of an operator product reverses the order of operations:

$$(\hat{B}\hat{C})^\dagger = \hat{C}^\dagger\hat{B}^\dagger. \quad (4.4)$$

An operator is defined to be *Hermitian* if it coincides with its adjoint:

$$\hat{B}^\dagger = \hat{B}. \quad (4.5)$$

4.2.2 Hermiticity of Position and Momentum

The position operator $\hat{\mathbf{r}}$ is manifestly Hermitian:

$$\langle \Phi | \hat{\mathbf{r}} | \Psi \rangle^* = \left[\int d^3r \phi^*(\mathbf{r}) \mathbf{r} \psi(\mathbf{r}) \right]^* = \int d^3r \psi^*(\mathbf{r}) \mathbf{r} \phi(\mathbf{r}) = \langle \Psi | \hat{\mathbf{r}} | \Phi \rangle. \quad (4.6)$$

For the momentum operator $\hat{\mathbf{p}} = -i\hbar\nabla$, we compute the matrix element and integrate by parts:

$$\begin{aligned} \langle \Phi | \hat{\mathbf{p}} | \Psi \rangle &= \int d^3r \phi^*(\mathbf{r}) \left(\frac{\hbar}{i} \nabla \psi(\mathbf{r}) \right) \\ &= \int d^3r \left[-\frac{\hbar}{i} \nabla \phi^*(\mathbf{r}) \right] \psi(\mathbf{r}) = \left[\int d^3r \left(\frac{\hbar}{i} \nabla \phi(\mathbf{r}) \right) \psi^*(\mathbf{r}) \right]^* \\ &= \langle \Psi | \hat{\mathbf{p}} | \Phi \rangle^*, \end{aligned} \quad (4.7)$$

provided the wave functions vanish sufficiently rapidly as $|\mathbf{r}| \rightarrow \infty$ so that surface boundary terms vanish.¹ Thus, $\hat{\mathbf{p}}^\dagger = \hat{\mathbf{p}}$.

4.2.3 Spectral Theorems: Reality of Eigenvalues and Orthogonality

Hermitian operators possess fundamental spectral properties:

1. **Reality of Eigenvalues.** Let $|b\rangle$ be an eigenstate of a Hermitian operator $\hat{B}$ with eigenvalue b : $\hat{B}|b\rangle = b|b\rangle$. Then

$$b = \langle b | \hat{B} | b \rangle = \langle b | \hat{B}^\dagger | b \rangle = \langle b | \hat{B} | b \rangle^* = b^*, \quad (4.8)$$

proving that $b \in \mathbb{R}$.

2. **Orthogonality of Eigenstates.** Let $|b'\rangle$ and $|b''\rangle$ be two eigenstates of $\hat{B}$ corresponding to distinct eigenvalues $b' \neq b''$. Then

$$\langle b'' | \hat{B} | b' \rangle = b' \langle b'' | b' \rangle. \quad (4.9)$$

Alternatively, acting to the left with $\hat{B}^\dagger = \hat{B}$:

$$\langle b'' | \hat{B} | b' \rangle = \langle b' | \hat{B} | b'' \rangle^* = (b'' \langle b' | b'' \rangle)^* = b'' \langle b'' | b' \rangle. \quad (4.10)$$

Subtracting Eq. (4.10) from Eq. (4.9) gives

$$(b' - b'') \langle b'' | b' \rangle = 0. \quad (4.11)$$

Since $b' \neq b''$, the eigenstates are strictly orthogonal:

$$\langle b'' | b' \rangle = 0. \quad (4.12)$$

¹The requirement that boundary terms vanish at spatial infinity or satisfy self-adjoint boundary conditions is essential for differential operators to be truly Hermitian on their domain.

4.2.4 Completeness and Spectral Resolution

By the spectral theorem for self-adjoint operators, the set of orthonormal eigenstates $\{|b'\rangle\}$ forms a complete basis spanning the Hilbert space (with degenerate subspaces orthonormalized via Gram-Schmidt orthogonalization). Any arbitrary state $|\Psi\rangle$ can be expanded as

$$|\Psi\rangle = \sum_{b'} |b'\rangle \langle b'|\Psi\rangle, \quad (4.13)$$

where the sum includes integration over continuous parts of the spectrum where appropriate. Correspondingly, the resolution of the identity is

$$\sum_{b'} |b'\rangle \langle b'| = \mathbb{I}. \quad (4.14)$$

Applying $\hat{B}$ to this decomposition establishes the spectral representation of the operator:

$$\hat{B} = \sum_{b'} b' |b'\rangle \langle b'|. \quad (4.15)$$

Every Hermitian operator can be represented in diagonal form in its eigenbasis.

4.3 Energy Eigenstates and Stationary States

When the potential energy $V(\mathbf{r})$ is independent of time, the Hamiltonian operator $\hat{H}$ is time-independent. Under these conditions, the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right] \psi(\mathbf{r}, t), \quad (4.16)$$

admits separable solutions of the form

$$\psi(\mathbf{r}, t) = e^{-iEt/\hbar} \psi_E(\mathbf{r}). \quad (4.17)$$

Substituting this ansatz into Eq. (4.16) cancels the harmonic time factor $e^{-iEt/\hbar}$, yielding the *time-independent Schrödinger equation*:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right] \psi_E(\mathbf{r}) = E \psi_E(\mathbf{r}). \quad (4.18)$$

In abstract Dirac notation, this is the eigenvalue problem for the Hamiltonian operator:

$$\hat{H}|E\rangle = E|E\rangle, \quad (4.19)$$

where $\psi_E(\mathbf{r}) = \langle \mathbf{r}|E\rangle$. The allowed energy eigenvalues E constitute the energy spectrum of the system. Eigenstates sharing the same energy eigenvalue are termed degenerate.

4.3.1 Stationary Properties of Energy Eigenstates

A system prepared in a single energy eigenstate $|E\rangle$ is stationary in the sense that all physical observable distributions are time-independent. At any later time t , the state vector has evolved to

$$|E(t)\rangle = e^{-iEt/\hbar} |E\rangle, \quad \langle E(t)| = e^{iEt/\hbar} \langle E|. \quad (4.20)$$

For any time-independent observable $\hat{Q}$, its expectation value at time t satisfies

$$\langle \hat{Q} \rangle_t = \langle E(t)|\hat{Q}|E(t)\rangle = \langle E|e^{iEt/\hbar} \hat{Q} e^{-iEt/\hbar} |E\rangle = \langle E|\hat{Q}|E\rangle = \text{const.} \quad (4.21)$$

Similarly, the spatial probability density is strictly static:

$$P(\mathbf{r}, t) = |\psi_E(\mathbf{r}, t)|^2 = |\psi_E(\mathbf{r})|^2. \quad (4.22)$$

4.3.2 General Time Evolution via Spectral Expansion

Because $\hat{H}$ is Hermitian, its orthonormal energy eigenstates form a complete resolution of the identity:

$$\sum_E |E\rangle\langle E| = \mathbb{I}, \quad (4.23)$$

where the summation symbol represents a discrete sum over bound states and an integral over the continuum spectrum. Any arbitrary initial state $|\Psi(0)\rangle$ can be expanded in this basis:

$$|\Psi(0)\rangle = \sum_E c_E |E\rangle, \quad c_E = \langle E|\Psi(0)\rangle. \quad (4.24)$$

Applying the time-evolution operator $\hat{U}(t, 0) = e^{-i\hat{H}t/\hbar}$, the state at time t is given by

$$|\Psi(t)\rangle = \sum_E c_E e^{-iEt/\hbar} |E\rangle. \quad (4.25)$$

Projecting Eq. (4.25) into position space:

$$\begin{aligned} \psi(\mathbf{r}, t) &= \langle \mathbf{r}|\Psi(t)\rangle = \sum_E e^{-iEt/\hbar} \langle \mathbf{r}|E\rangle \langle E|\Psi(0)\rangle \\ &= \sum_E e^{-iEt/\hbar} \psi_E(\mathbf{r}) \int d^3r' \psi_E^*(\mathbf{r}') \psi(\mathbf{r}', 0). \end{aligned} \quad (4.26)$$

Interchanging the order of integration and summation, this relation can be written in terms of the general quantum propagator:

$$\psi(\mathbf{r}, t) = \int d^3r' K(\mathbf{r}, t; \mathbf{r}', t') \psi(\mathbf{r}', t'), \quad (4.27)$$

where the spectral representation of the propagator is given by the bilinear expansion:

$$K(\mathbf{r}, t; \mathbf{r}', t') = \sum_E e^{-iE(t-t')/\hbar} \psi_E(\mathbf{r}) \psi_E^*(\mathbf{r}'). \quad (4.28)$$

Equation (4.28) generalizes the free-particle propagator derived in Chapter 3 to arbitrary time-independent potentials.

4.4 One-Dimensional Potential Step and Scattering

We begin our analysis of one-dimensional problems governed by the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_E(x)}{dx^2} + V(x)\psi_E(x) = E\psi_E(x). \quad (4.29)$$

For any finite potential $V(x)$, the second derivative $\psi''(x)$ remains finite everywhere. Consequently, the wave function $\psi_E(x)$ and its first spatial derivative $\psi'_E(x)$ must be continuous across all points.

In a region where the potential is spatially uniform ($V(x) = V_0 = \text{const}$), Eq. (4.29) reduces to the Helmholtz equation with general solution

$$\psi_E(x) = Ae^{i\bar{p}x/\hbar} + Be^{-i\bar{p}x/\hbar}, \quad \bar{p} \equiv \sqrt{2m(E - V_0)}. \quad (4.30)$$

4.4.1 Scattering Above the Barrier ($E > V_0$)

Consider a sharp potential step of height $V_0 > 0$:

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & x > 0. \end{cases} \quad (4.31)$$

For particles incident from the left with energy $E > V_0$, the asymptotic wave function in each region is given by

$$\psi_E(x) = \begin{cases} Ae^{ipx/\hbar} + Be^{-ipx/\hbar}, & x < 0, \\ Ce^{i\bar{p}x/\hbar}, & x > 0, \end{cases} \quad (4.32)$$

where $p = \sqrt{2mE}$ is the incident and reflected momentum, $\bar{p} = \sqrt{2m(E - V_0)}$ is the transmitted momentum, and the absence of a left-traveling wave in $x > 0$ reflects the boundary condition of an incident beam arriving solely from $x = -\infty$.

Matching the boundary conditions of continuity for $\psi_E(x)$ and $\psi'_E(x)$ at the junction $x = 0$:

$$A + B = C, \quad p(A - B) = \bar{p}C, \quad (4.33)$$

yields the reflection and transmission amplitudes:

$$\frac{B}{A} = \frac{p - \bar{p}}{p + \bar{p}}, \quad \frac{C}{A} = \frac{2p}{p + \bar{p}}. \quad (4.34)$$

To verify conservation of probability, we compute the probability current density $j(x) = \frac{\hbar}{m} \text{Im}[\psi^* \psi']$. In the left region ($x < 0$), interference between the incident and reflected waves gives

$$j_L = \frac{p}{m} |A|^2 - \frac{p}{m} |B|^2 = j_{\text{inc}} - j_{\text{refl}}. \quad (4.35)$$

In the right region ($x > 0$), the transmitted current is

$$j_R = \frac{\bar{p}}{m} |C|^2 = j_{\text{trans}}. \quad (4.36)$$

Using the explicit solutions in Eq. (4.34):

$$j_{\text{inc}} - j_{\text{refl}} = \frac{p}{m} |A|^2 \left[1 - \left(\frac{p - \bar{p}}{p + \bar{p}} \right)^2 \right] = \frac{p}{m} |A|^2 \frac{4p\bar{p}}{(p + \bar{p})^2} = \frac{\bar{p}}{m} |A|^2 \left(\frac{2p}{p + \bar{p}} \right)^2 = j_{\text{trans}}. \quad (4.37)$$

Thus, current conservation $j_L = j_R$ is strictly satisfied. The reflection coefficient R and transmission coefficient T are

$$R \equiv \frac{j_{\text{refl}}}{j_{\text{inc}}} = \left(\frac{p - \bar{p}}{p + \bar{p}} \right)^2, \quad T \equiv \frac{j_{\text{trans}}}{j_{\text{inc}}} = \frac{\bar{p}}{p} \left(\frac{2p}{p + \bar{p}} \right)^2 = \frac{4p\bar{p}}{(p + \bar{p})^2}, \quad (4.38)$$

satisfying $R + T = 1$. Quantum mechanical reflection occurs even though the particle's energy strictly exceeds the step height ($E > V_0$), directly mirroring the partial reflection of electromagnetic waves across an interface with mismatched wave impedances.

4.4.2 Wave Packet Dynamics

To elucidate the time-dependent scattering process, consider an incident wave packet composed of positive-momentum eigenstates:

$$\psi(x, t) = \int_0^\infty \frac{dp}{2\pi\hbar} f(p) \left[e^{ipx/\hbar} + \left(\frac{p - \bar{p}}{p + \bar{p}} \right) e^{-ipx/\hbar} \right] e^{-iEt/\hbar} \quad (x < 0), \quad (4.39)$$

$$\psi(x, t) = \int_0^\infty \frac{dp}{2\pi\hbar} f(p) \left(\frac{2p}{p + \bar{p}} \right) e^{i\bar{p}x/\hbar} e^{-iEt/\hbar} \quad (x > 0), \quad (4.40)$$

where $f(p)$ is a spectral envelope peaked at $p_0 > \sqrt{2mV_0}$.

Applying the stationary phase approximation:

- For $t < 0$, the incident packet $\psi_{\text{inc}}(x, t)$ is localized around $x_{\text{inc}}(t) = (p_0/m)t < 0$, propagating toward the step. The reflected and transmitted components are negligible.
- At $t = 0$, the wave packet collides with the step at $x = 0$.
- For $t > 0$, the probability amplitude splits into a reflected wave packet localized at $x_{\text{refl}}(t) = -(p_0/m)t < 0$ and a transmitted wave packet propagating at reduced group velocity $x_{\text{trans}}(t) = (\bar{p}_0/m)t > 0$.

The particle does not fragment; rather, its probability amplitude bifurcates into distinct spatial channels.

4.4.3 Total Reflection and Evanescent Waves ($E < V_0$)

When the incident energy is below the step barrier ($E < V_0$), the momentum in the right-hand region becomes purely imaginary:

$$\bar{p} = i\hbar\kappa, \quad \kappa \equiv \frac{\sqrt{2m(V_0 - E)}}{\hbar} > 0. \quad (4.41)$$

The boundary condition requiring square-integrability as $x \rightarrow \infty$ eliminates the exponentially growing branch, yielding an evanescent wave:

$$\psi_E(x) = \begin{cases} Ae^{ipx/\hbar} + Be^{-ipx/\hbar}, & x < 0, \\ Ce^{-\kappa x}, & x > 0. \end{cases} \quad (4.42)$$

Continuity of ψ and ψ' at $x = 0$ gives

$$\frac{B}{A} = \frac{p - i\hbar\kappa}{p + i\hbar\kappa} = -e^{2i\delta(E)}, \quad (4.43)$$

where the scattering phase shift $\delta(E)$ is determined by

$$\hbar\kappa = p \cot \delta(E). \quad (4.44)$$

Since $|B/A| = 1$, the reflection coefficient is unity ($R = 1$); all incident probability current is completely reflected, and the net transmitted current vanishes identically ($j_R = 0$). However, the wave function does not vanish abruptly at the barrier: it penetrates into the classically forbidden region $x > 0$ with a characteristic penetration depth

$$d \sim \frac{1}{\kappa} = \frac{\hbar}{\sqrt{2m(V_0 - E)}}. \quad (4.45)$$

This non-vanishing probability in the forbidden region does not violate energy conservation. Determining experimentally that the particle is located in $x > 0$ confines it within a spatial interval $\Delta x \lesssim 1/\kappa$. By Heisenberg's uncertainty principle, this measurement introduces a momentum uncertainty $\Delta p \gtrsim \hbar/\Delta x \sim \hbar\kappa$, imparting an unavoidable kinetic energy uncertainty:

$$\Delta E_k \sim \frac{(\Delta p)^2}{2m} \gtrsim \frac{\hbar^2 \kappa^2}{2m} = V_0 - E. \quad (4.46)$$

Thus, any measurement capable of localizing the particle inside the barrier necessarily supplies the requisite energy to place it above V_0 .

4.5 Quantum Tunneling

The existence of evanescent wave penetration into classically forbidden regions leads to one of the most celebrated and consequential phenomena in quantum mechanics: *quantum tunneling*. When a classically forbidden potential barrier has finite spatial width, the probability amplitude does not decay to zero, allowing particles with energy below the barrier height to penetrate and emerge on the opposite side.

4.5.1 Transmission Amplitude and Tunneling Probability

Consider a rectangular potential barrier of height $V_0 > 0$ and width a :

$$V(x) = \begin{cases} V_0, & 0 < x < a, \\ 0, & x < 0 \quad \text{and} \quad x > a. \end{cases} \quad (4.47)$$

For an incident particle with energy below the barrier summit ($E < V_0$), the stationary wave function across the three regions is given by

$$\psi(x) = \begin{cases} Ae^{ipx/\hbar} + Be^{-ipx/\hbar}, & x < 0, \\ Ce^{-\kappa x} + De^{\kappa x}, & 0 < x < a, \\ AS(E)e^{ip(x-a)/\hbar}, & x > a, \end{cases} \quad (4.48)$$

where $p = \sqrt{2mE}$ and $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Because the barrier has finite thickness a , the exponentially growing solution $e^{\kappa x}$ remains bounded and is physically admissible.

Matching the wave function and its derivative continuously across the interfaces at $x = 0$ and $x = a$ yields the transmission amplitude $S(E)$:

$$S(E) = \frac{2i\kappa p \hbar}{2i\kappa p \hbar \cosh \kappa a + (p^2 - \hbar^2 \kappa^2) \sinh \kappa a}. \quad (4.49)$$

The transmission coefficient (or *transmissivity*) $T(E) \equiv |S(E)|^2$ defines the probability that an incident particle tunnels through the barrier:

$$T(E) = \left[1 + \frac{\sinh^2 \kappa a}{4 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right)} \right]^{-1}. \quad (4.50)$$

For wide or high barriers ($\kappa a \gg 1$), $\sinh \kappa a \approx \frac{1}{2}e^{\kappa a}$, and the transmission probability decays exponentially with barrier thickness:

$$T(E) \approx 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) \exp(-2\kappa a) = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) \exp\left(-\frac{2a}{\hbar} \sqrt{2m(V_0 - E)}\right). \quad (4.51)$$

4.5.2 Energy-Time Uncertainty and Transit Time

Does a tunneling particle violate energy conservation while traversing the interior of the barrier where $E < V_0$? The resolution lies in the quantum energy-time uncertainty relation:

$$\Delta E \Delta t \gtrsim \hbar. \quad (4.52)$$

To demonstrate an empirical violation of energy conservation, one would have to measure the particle's energy while simultaneously verifying its presence inside the barrier. However, the traversal time Δt between the arrival of the incident wave packet and the emergence of the transmitted packet satisfies $\Delta t \lesssim \hbar/(V_0 - E)$. Consequently, any measurement confirming that

the particle is within the barrier induces an energy uncertainty $\Delta E \gtrsim V_0 - E$, precluding an experimental determination that $E < V_0$ while inside.²

Here, Δt reflects the intrinsic indeterminacy in event timing rather than the duration of the measurement itself. An arbitrarily rapid momentum measurement can yield the energy with high precision, but doing so completely delocalizes the particle, rendering the time of arrival at any spatial boundary uncertain by $\Delta t \sim \hbar/\Delta E$.

4.5.3 Physical Applications

Nuclear α -Decay. The Gamow theory of radioactive α -decay models the decaying nucleus as an α -particle trapped in a quasi-bound state inside a potential well formed by the attractive strong nuclear force and bounded by a Coulomb repulsive barrier $V(r) = 2(Z - 2)e^2/r$. With positive kinetic energy $E > 0$ below the Coulomb barrier peak, the α -particle impinges on the barrier with an attempt frequency $\nu \sim v/(2R) \sim E/\hbar$. At each collision, it possesses a finite tunneling probability T . The radioactive decay constant λ and half-life τ are given by

$$\lambda = \frac{1}{\tau} \sim \frac{E}{\hbar} T \propto \exp\left(-\frac{2}{\hbar} \int_R^{r_b} \sqrt{2m[V(r) - E]} dr\right). \quad (4.53)$$

Because the exponent depends inversely on $\sqrt{E}$, small variations in decay energy E produce immense variations (spanning over twenty orders of magnitude) in half-life, providing a first-principles derivation of the empirical Geiger-Nuttall law.

Solid-State Tunnel Junctions. In condensed matter systems, two metallic conductors separated by an ultrathin insulating oxide layer ($\sim 10\text{--}30 \text{ \AA}$) form a tunnel junction. Electrons tunnel quantum mechanically through the insulating potential barrier under an applied bias voltage V_{bias} , producing a net tunneling current:

$$I \propto \int dE D_1(E) D_2(E + eV_{\text{bias}}) T(E) [f(E) - f(E + eV_{\text{bias}})]. \quad (4.54)$$

This mechanism forms the operational basis of scanning tunneling microscopy (STM), semiconductor tunnel diodes, and Josephson junction superconducting circuits.

4.6 Bound States and the Infinite Potential Well

Unlike scattering states that form a continuous spectrum for $E > 0$, attractive potential wells admit spatially localized *bound states*. These states decay exponentially as $|x| \rightarrow \infty$, belong to the Hilbert space $L^2(\mathbb{R})$, and possess a discrete energy spectrum:

$$E_1 < E_2 < E_3 < \dots \quad (4.55)$$

Such states characterize electrons bound in atomic orbitals, nucleons within a nucleus, and molecular valence bonds.

4.6.1 The One-Dimensional Infinite Square Well

The prototypical model of quantum confinement is the one-dimensional infinite potential well (particle in a box) of width a , symmetrically positioned on the interval $[-a/2, a/2]$:

$$V(x) = \begin{cases} 0, & |x| < \frac{a}{2}, \\ \infty, & |x| \geq \frac{a}{2}. \end{cases} \quad (4.56)$$

²For comprehensive analyses of tunneling transit times and the Hartman effect, see T. Hartman, *J. Appl. Phys.* **33**, 3422 (1962).

Within the well ($|x| < a/2$), the potential vanishes and the general solution is sinusoidal:

$$\psi(x) = A \sin\left(\frac{px}{\hbar} + \varphi\right), \quad p = \sqrt{2mE}. \quad (4.57)$$

Outside the well, an infinite potential barrier requires the wave function to vanish identically, as verified by considering a finite barrier of height V_0 where the wave function decays as $e^{-\kappa|x|}$ with $\kappa = \sqrt{2m(V_0 - E)}/\hbar \rightarrow \infty$. Requiring continuity of $\psi(x)$ at the hard walls implies the Dirichlet boundary conditions:

$$\psi\left(-\frac{a}{2}\right) = \psi\left(\frac{a}{2}\right) = 0. \quad (4.58)$$

These boundary conditions restrict the allowable stationary states to two alternating symmetry classes:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \begin{cases} \cos\left(\frac{n\pi x}{a}\right), & n = 1, 3, 5, \dots \quad (\text{even parity}), \\ \sin\left(\frac{n\pi x}{a}\right), & n = 2, 4, 6, \dots \quad (\text{odd parity}), \end{cases} \quad (4.59)$$

with quantized kinematic momenta:

$$p_n = \frac{n\pi\hbar}{a}, \quad n = 1, 2, 3, \dots \quad (4.60)$$

The corresponding quantized energy levels are

$$E_n = \frac{p_n^2}{2m} = n^2 \frac{\pi^2 \hbar^2}{2ma^2}, \quad n \in \mathbb{N}^+. \quad (4.61)$$

The state $n = 0$ is trivial ($\psi \equiv 0$) and physically inadmissible because the probability of finding the particle would be zero.

4.6.2 Zero-Point Energy and Quantum Confinement

The lowest allowable energy level—the *ground state* ($n = 1$)—possesses strictly positive kinetic energy:

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2} > 0. \quad (4.62)$$

This non-vanishing minimum energy is the *zero-point energy*. Its origin is an inevitable consequence of the Heisenberg uncertainty principle. Confining a particle within a box of size $\Delta x \approx a$ requires an intrinsic momentum uncertainty:

$$\Delta p \gtrsim \frac{\hbar}{\Delta x} \sim \frac{\hbar}{a}. \quad (4.63)$$

Consequently, the particle cannot be at rest; its mean kinetic energy must satisfy

$$\langle E_k \rangle = \frac{\langle p^2 \rangle}{2m} \geq \frac{(\Delta p)^2}{2m} \sim \frac{\hbar^2}{2ma^2}. \quad (4.64)$$

This can be seen directly by decomposing the ground-state cosine into counter-propagating plane waves, $\cos(\pi x/a) = \frac{1}{2}(e^{i\pi x/a} + e^{-i\pi x/a})$, representing a superposition of states with momentum peaked at $\pm\pi\hbar/a$.

4.6.3 Macroscopic Quantum Effects: Liquid Helium

Zero-point energy plays a decisive macroscopic role in low-temperature physics. For the vast majority of substances, thermal cooling reduces kinetic energy until attractive van der Waals intermolecular forces freeze the atoms into a periodic crystalline lattice.

For liquid helium (^4He and ^3He), however, the small atomic mass m combined with weak interatomic Lennard-Jones potentials produces a zero-point energy so large that the atoms cannot be localized onto lattice sites without expanding the crystal. Consequently, helium remains a liquid under saturated vapor pressure all the way down to absolute zero ($T = 0\text{ K}$), exhibiting macroscopic superfluidity. Solidification into a crystal requires an external pressure exceeding approximately 25 atm to overcome this zero-point motion.

4.7 Parity Symmetry and Commuting Observables

Symmetry principles play a profound role in simplifying quantum mechanical eigenvalue problems and classifying physical states. One of the most fundamental discrete spatial symmetries is *space inversion*, or *parity*.

4.7.1 The Parity Operator

The parity operator $\hat{\Pi}$ (or $\hat{P}$) acts on spatial state vectors by reflecting position coordinates through the origin:

$$\langle \mathbf{r} | \hat{P} | \Psi \rangle \equiv \langle -\mathbf{r} | \Psi \rangle = \psi(-\mathbf{r}). \quad (4.65)$$

Applying the parity operator twice returns the coordinate to its original state:

$$\hat{P}^2 = \mathbb{I}. \quad (4.66)$$

Thus, $\hat{P}$ is unitary and involutory ($\hat{P}^{-1} = \hat{P}^\dagger = \hat{P}$), which guarantees that $\hat{P}$ is Hermitian. Its allowed eigenvalues π satisfy $\pi^2 = 1$, yielding the two discrete eigenvalues:

$$\pi = \pm 1. \quad (4.67)$$

States satisfying $\hat{P}|\Psi_+\rangle = +|\Psi_+\rangle$ ($\psi(-\mathbf{r}) = \psi(\mathbf{r})$) possess *even parity*, while states satisfying $\hat{P}|\Psi_-\rangle = -|\Psi_-\rangle$ ($\psi(-\mathbf{r}) = -\psi(\mathbf{r})$) possess *odd parity*.

If the potential energy function is symmetric under spatial inversion,

$$V(-\mathbf{r}) = V(\mathbf{r}), \quad (4.68)$$

the Hamiltonian satisfies

$$\hat{P}^\dagger \hat{H} \hat{P} = \hat{H} \iff [\hat{H}, \hat{P}] = 0. \quad (4.69)$$

When the Hamiltonian commutes with the parity operator, the stationary energy eigenstates can always be chosen to have definite parity.

4.7.2 General Theory of Commuting Observables

The connection between parity and energy eigenstates illustrates a fundamental theorem of quantum mechanics:

Two Hermitian operators $\hat{A}$ and $\hat{B}$ commute ($[\hat{A}, \hat{B}] = 0$) if and only if there exists a complete orthonormal basis of simultaneous eigenstates of both operators.

Proof. Case 1: Nondegenerate Spectrum. Suppose $|a\rangle$ is an eigenstate of $\hat{A}$ with nondegenerate eigenvalue a : $\hat{A}|a\rangle = a|a\rangle$. Since $[\hat{A}, \hat{B}] = 0$,

$$\hat{A}(\hat{B}|a\rangle) = \hat{B}(\hat{A}|a\rangle) = a(\hat{B}|a\rangle). \quad (4.70)$$

Thus $\hat{B}|a\rangle$ is also an eigenvector of $\hat{A}$ with eigenvalue a . Since the eigenspace is one-dimensional, $\hat{B}|a\rangle$ must be proportional to $|a\rangle$:

$$\hat{B}|a\rangle = b|a\rangle. \quad (4.71)$$

Hence, $|a\rangle$ is a simultaneous eigenstate of both $\hat{A}$ and $\hat{B}$.

Case 2: Degenerate Spectrum. Let the eigenvalue a be n -fold degenerate, spanned by an orthonormal basis $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ such that $\hat{A}|i\rangle = a|i\rangle$ for all $i = 1, \dots, n$. Since $\hat{A}(\hat{B}|i\rangle) = a(\hat{B}|i\rangle)$, the state $\hat{B}|i\rangle$ lies entirely within this n -dimensional degenerate subspace:

$$\hat{B}|i\rangle = \sum_{j=1}^n B_{ji}|j\rangle, \quad B_{ji} = \langle j|\hat{B}|i\rangle. \quad (4.72)$$

The matrix B_{ji} is an $n \times n$ Hermitian matrix and can therefore be diagonalized by a unitary transformation. Solving the finite-dimensional eigenvalue problem

$$\sum_{i=1}^n B_{ji}c_i^{(\alpha)} = b_\alpha c_j^{(\alpha)}, \quad \alpha = 1, \dots, n, \quad (4.73)$$

the resulting linear combinations $|\alpha\rangle = \sum_i c_i^{(\alpha)}|i\rangle$ satisfy

$$\hat{A}|\alpha\rangle = a|\alpha\rangle \quad \text{and} \quad \hat{B}|\alpha\rangle = b_\alpha|\alpha\rangle. \quad (4.74)$$

Thus, a complete set of mutual eigenstates can always be constructed. $\square$

Physically, two observables can be measured simultaneously to arbitrary precision without mutual quantum backaction if and only if their operators commute ($[\hat{A}, \hat{B}] = 0$).

4.7.3 Application: The Finite Symmetric Square Well

We utilize parity symmetry to determine the bound-state spectrum of an attractive finite square well of depth $V_0 > 0$ and width a :

$$V(x) = \begin{cases} -V_0, & |x| < \frac{a}{2}, \\ 0, & |x| \geq \frac{a}{2}. \end{cases} \quad (4.75)$$

Because $V(-x) = V(x)$, bound states ($-V_0 < E < 0$) separate rigorously into decoupled even and odd parity sectors. Defining internal and external wave numbers:

$$p = \sqrt{2m(E + V_0)}, \quad \hbar\kappa = \sqrt{-2mE}, \quad (4.76)$$

the solutions take the form:

- **Even Parity States ($\pi = +1$):**

$$\psi(x) = \begin{cases} A \cos\left(\frac{px}{\hbar}\right), & |x| < \frac{a}{2}, \\ C e^{-\kappa|x|}, & |x| \geq \frac{a}{2}. \end{cases} \quad (4.77)$$

Matching $\psi(x)$ and $\psi'(x)$ continuously at $x = a/2$ gives

$$A \cos\left(\frac{pa}{2\hbar}\right) = C e^{-\kappa a/2}, \quad -\frac{p}{\hbar} A \sin\left(\frac{pa}{2\hbar}\right) = -\kappa C e^{-\kappa a/2}. \quad (4.78)$$

Dividing these equations yields the transcendental quantization condition:

$$p \tan\left(\frac{pa}{2\hbar}\right) = \hbar\kappa. \quad (4.79)$$

• **Odd Parity States** ($\pi = -1$):

$$\psi(x) = \begin{cases} A \sin\left(\frac{px}{\hbar}\right), & |x| < \frac{a}{2}, \\ \text{sgn}(x) C e^{-\kappa|x|}, & |x| \geq \frac{a}{2}. \end{cases} \quad (4.80)$$

Matching boundary conditions at $x = a/2$ yields:

$$p \cot\left(\frac{pa}{2\hbar}\right) = -\hbar\kappa. \quad (4.81)$$

Graphical analysis reveals that the even- and odd-parity bound states alternate in energy. A fundamental feature of one-dimensional quantum mechanics is that *at least one bound state* (of even parity) always exists, no matter how shallow or narrow the attractive well ($V_0 a^2 \rightarrow 0$). In the shallow well limit ($V_0 \rightarrow 0$), the ground-state binding energy scales as

$$E \approx -\frac{mV_0^2 a^2}{2\hbar^2}, \quad (4.82)$$

with the particle's wave function extending far beyond the well into the asymptotic region ($1/\kappa \propto 1/V_0$).

4.8 Transmission Resonances and Complex Energy Poles

For energies $E > 0$, the finite square well of width a and depth $V_0 > 0$ admits a continuum of scattering states. Particles incident from $x = -\infty$ undergo partial reflection and partial transmission.

Writing the incident and transmitted asymptotic wave functions as

$$\psi_{\text{inc}}(x) = A e^{ipx/\hbar} \quad \left(x < -\frac{a}{2}\right), \quad (4.83)$$

$$\psi_{\text{trans}}(x) = A S(E) e^{ip(x-a)/\hbar} \quad \left(x > \frac{a}{2}\right), \quad (4.84)$$

the transmission amplitude $S(E)$ is determined by matching boundary conditions at $x = \pm a/2$:

$$S(E) = \left[\cos\left(\frac{\bar{p}a}{\hbar}\right) - \frac{i}{2} \left(\frac{p}{\bar{p}} + \frac{\bar{p}}{p} \right) \sin\left(\frac{\bar{p}a}{\hbar}\right) \right]^{-1}, \quad (4.85)$$

where $p = \sqrt{2mE}$ and $\bar{p} = \sqrt{2m(E + V_0)}$. The corresponding transmission coefficient (transmissivity) is

$$T(E) = |S(E)|^2 = \left[1 + \frac{\sin^2(\bar{p}a/\hbar)}{4 \frac{E}{V_0} \left(1 + \frac{E}{V_0}\right)} \right]^{-1}. \quad (4.86)$$

4.8.1 Resonance Condition and Perfect Transmission

The transmission probability $T(E)$ reaches unity ($T = 1$) whenever the interference inside the well is fully constructive:

$$\sin\left(\frac{\bar{p}a}{\hbar}\right) = 0 \implies \frac{\bar{p}a}{\hbar} = n\pi, \quad n \in \mathbb{N}^+. \quad (4.87)$$

This yields transmission resonance energies at

$$E_n = n^2 \frac{\pi^2 \hbar^2}{2ma^2} - V_0 > 0. \quad (4.88)$$

At these resonant energies, the potential well becomes completely transparent to the incoming particle—a quantum phenomenon observed experimentally in electron-atom scattering as the *Ramsauer-Townsend effect*.

4.8.2 Analytic Structure of the S-Matrix

When viewed as an analytic function of the complex energy E , the transmission amplitude $S(E)$ reveals the deep unity between bound states and scattering resonances:

- **Bound States as Real Poles.** Bound states correspond to solutions possessing outgoing exponential decay without an incident wave, which formally requires $S(E) \rightarrow \infty$. The denominator of Eq. (4.85) vanishes when

$$\cos\left(\frac{\bar{p}a}{\hbar}\right) = \frac{i}{2} \left(\frac{p}{\bar{p}} + \frac{\bar{p}}{p} \right) \sin\left(\frac{\bar{p}a}{\hbar}\right). \quad (4.89)$$

Analytically continuing to negative energies $E < 0$, the physical branch of momentum is imaginary: $p = i\hbar\kappa = i\sqrt{2m|E|}$. Using trigonometric double-angle identities, Eq. (4.89) factorizes precisely into the two bound-state conditions:

$$\bar{p} \tan\left(\frac{\bar{p}a}{2\hbar}\right) = \hbar\kappa \quad (\text{even parity}), \quad (4.90)$$

and

$$\bar{p} \cot\left(\frac{\bar{p}a}{2\hbar}\right) = -\hbar\kappa \quad (\text{odd parity}). \quad (4.91)$$

Thus, every bound state of the potential corresponds to a simple pole of $S(E)$ located on the negative real axis of the physical Riemann sheet.

- **Resonances as Second-Sheet Complex Poles.** Near a transmission resonance $E_0 > 0$, expanding the denominator of $S(E)$ to first order in $(E - E_0)$ yields the classic *Breit-Wigner* resonance formula:

$$S(E) \approx \frac{\pm i\Gamma/2}{E - E_0 + i\Gamma/2}, \quad (4.92)$$

where the resonance width $\Gamma > 0$ is defined by

$$\frac{4}{\Gamma} \equiv \left(\frac{p}{\bar{p}} + \frac{\bar{p}}{p} \right) \frac{1}{\hbar} \left. \frac{d(\bar{p}a)}{dE} \right|_{E=E_0}. \quad (4.93)$$

Across the branch cut running along the positive real energy axis, $S(E)$ possesses a pole located at

$$E_{\text{pole}} = E_0 - i\frac{\Gamma}{2} \quad (4.94)$$

in the lower half of the unphysical (second) Riemann sheet.

Writing $S(E) = \sqrt{T(E)} e^{i\delta(E)}$, the scattering phase shift $\delta(E)$ near resonance satisfies

$$\delta(E) \approx \arctan \left[\frac{2(E - E_0)}{\Gamma} \right]. \quad (4.95)$$

As the energy traverses the resonance window of width Γ , the phase shift undergoes a rapid increase of π , and its derivative $d\delta/dE$ attains a sharp maximum.

4.8.3 Wave Packet Dynamics and Exponential Decay

A resonance represents a quasi-bound, metastable state: an incident particle enters the well, is temporarily trapped for a characteristic lifetime $\tau = \hbar/\Gamma$, and eventually escapes.

To demonstrate this dynamically, consider an incident wave packet $\psi_{\text{inc}}(x, t)$ with spectral distribution $f(p)$ centered at the resonant momentum $p_0 = \sqrt{2mE_0}$. The transmitted wave packet is

$$\psi_{\text{trans}}(x, t) \approx f(p_0) \int \frac{dp}{2\pi\hbar} e^{i[p(x-a/2)-Et]/\hbar} S(E). \quad (4.96)$$

Linearizing the dispersion relation around E_0 , $p \approx p_0 + \frac{m}{p_0}(E - E_0)$, and substituting the Breit-Wigner amplitude (4.92), the transmitted packet takes the form:

$$\begin{aligned} \psi_{\text{trans}}(x, t) \approx & \pm \frac{i\Gamma m f(p_0)}{2p_0} \exp \left\{ \frac{i}{\hbar} \left[p_0 \left(x - \frac{a}{2} \right) - E_0 t \right] \right\} \\ & \times \int_{-\infty}^{\infty} \frac{dE}{2\pi\hbar} \frac{1}{E - E_0 + i\Gamma/2} \exp \left\{ -\frac{i(E - E_0)}{\hbar} \left[t - \frac{m}{p_0} \left(x - \frac{a}{2} \right) \right] \right\}. \end{aligned} \quad (4.97)$$

Evaluating this integral via Cauchy contour integration:

- For $x > \frac{a}{2} + \frac{p_0}{m}t$, the contour is closed in the upper half of the complex E -plane where the integrand is analytic, giving identically zero.
- For $x < \frac{a}{2} + \frac{p_0}{m}t$, closing the contour in the lower half-plane encloses the Breit-Wigner pole at $E = E_0 - i\Gamma/2$.

By the residue theorem, the transmitted wave function is

$$\psi_{\text{trans}}(x, t) \approx \pm \frac{m\Gamma f(p_0)}{2p_0\hbar} \Theta \left(\frac{a}{2} + \frac{p_0}{m}t - x \right) e^{i[p_0(x-a/2)-E_0t]/\hbar} \exp \left\{ -\frac{\Gamma}{2\hbar} \left[t - \frac{m}{p_0} \left(x - \frac{a}{2} \right) \right] \right\}. \quad (4.98)$$

Equation (4.98) depicts a causal wave front propagating at the group velocity $v_0 = p_0/m$, trailing an exponential tail. The local probability density decays as

$$P(x, t) \propto e^{-\Gamma t/\hbar} = e^{-t/\tau}, \quad \tau = \frac{\hbar}{\Gamma}. \quad (4.99)$$

This exponential decay law governs all metastable quantum systems—from radioactive α -emitters to unstable short-lived hadronic resonances in high-energy particle physics.³

4.9 The One-Dimensional Delta-Function Potential

The one-dimensional Dirac delta-function potential provides a transparent and exactly solvable model illustrating both scattering theory and bound-state formation:

$$V(x) = v_0 \delta(x), \quad (4.100)$$

where the coupling parameter v_0 has physical dimensions of [energy $\times$ length].

4.9.1 Green's Function Formulation and Scattering States

The stationary Schrödinger equation reads

$$\left(E + \frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \psi(x) = v_0 \delta(x) \psi(0). \quad (4.101)$$

This inhomogeneous differential equation can be recast into the Lippmann-Schwinger integral form:

$$\psi(x) = \psi_{\text{inc}}(x) + \int_{-\infty}^{\infty} \frac{dp'}{2\pi\hbar} \frac{e^{ip'x/\hbar}}{E - \frac{p'^2}{2m} + i\varepsilon} v_0 \psi(0), \quad (4.102)$$

where $\psi_{\text{inc}}(x) = e^{ipx/\hbar}$ with $p = \sqrt{2mE} > 0$ represents the incident plane wave arriving from $x = -\infty$. The infinitesimal $+i\varepsilon$ shifts the poles at $p' = \pm p \mp i\varepsilon$ into the complex plane, selecting the causal outgoing radiation boundary condition.

³At asymptotically long times ($t \gg \tau$), subtle branch-cut contributions replace the exponential decay with a Khalfin power-law falloff $\sim t^{-3/2}$; see L. A. Khalfin, *Sov. Phys. JETP* **6**, 1053 (1958).

Evaluating the Green's function integral via contour integration:

$$\int_{-\infty}^{\infty} \frac{dp'}{2\pi\hbar} \frac{e^{ip'x/\hbar}}{E - \frac{p'^2}{2m} + i\varepsilon} = \frac{m}{i\hbar p} e^{ip|x|/\hbar}. \quad (4.103)$$

Substituting this result into Eq. (4.102) gives

$$\psi(x) = \psi_{\text{inc}}(x) + \frac{mv_0}{i\hbar p} e^{ip|x|/\hbar} \psi(0). \quad (4.104)$$

Evaluating at $x = 0$ yields a self-consistent algebraic equation for $\psi(0)$:

$$\psi(0) = \frac{ip\hbar}{ip\hbar - mv_0} \psi_{\text{inc}}(0). \quad (4.105)$$

For an incident wave of unit amplitude ($\psi_{\text{inc}}(0) = 1$), the full spatial wave function is

$$\psi(x) = e^{ipx/\hbar} + \frac{mv_0}{ip\hbar - mv_0} e^{ip|x|/\hbar}. \quad (4.106)$$

For $x > 0$, combining the incident and scattered waves gives the transmitted wave:

$$\psi_{\text{trans}}(x) = \left(1 + \frac{mv_0}{ip\hbar - mv_0}\right) e^{ipx/\hbar} = \frac{ip\hbar}{ip\hbar - mv_0} e^{ipx/\hbar}. \quad (4.107)$$

The transmission amplitude is therefore

$$S(E) = \frac{ip\hbar}{ip\hbar - mv_0}, \quad (4.108)$$

and the corresponding transmission coefficient is

$$T(E) = |S(E)|^2 = \frac{p^2\hbar^2}{p^2\hbar^2 + m^2v_0^2} = \frac{E}{E + \frac{mv_0^2}{2\hbar^2}}. \quad (4.109)$$

The reflection coefficient is $R(E) = 1 - T(E) = \frac{mv_0^2/(2\hbar^2)}{E + mv_0^2/(2\hbar^2)}$. Notably, $T(E)$ is symmetric under $v_0 \rightarrow -v_0$; an attractive delta potential and a repulsive delta potential produce identical scattering probabilities.

4.9.2 Bound State for Attractive Potential ($v_0 < 0$)

The transmission amplitude $S(E)$ possesses a simple pole when its denominator vanishes:

$$ip\hbar - mv_0 = 0 \implies p = -i \frac{mv_0}{\hbar}. \quad (4.110)$$

For a physical bound state with $E < 0$, the momentum must lie on the positive imaginary axis, $p = i\sqrt{2m|E|}$, to ensure exponential decay at spatial infinity. This condition can only be satisfied if the potential is attractive:

$$v_0 < 0 \implies p = i \frac{m|v_0|}{\hbar}. \quad (4.111)$$

Squaring this relation gives the unique discrete bound-state energy:

$$E = -\frac{mv_0^2}{2\hbar^2}. \quad (4.112)$$

In the absence of an incident wave ($\psi_{\text{inc}} = 0$), the normalized bound-state wave function is

$$\psi_0(x) = \frac{\sqrt{m|v_0|}}{\hbar} \exp\left(-\frac{m|v_0|}{\hbar^2}|x|\right). \quad (4.113)$$

This exact result recovers the shallow-well asymptotic bound state obtained in Eq. (4.82) by identifying $v_0 = -V_0a$.

4.10 Periodic Potentials and Band Theory

A particle moving through a crystalline lattice experiences an external potential with discrete spatial translation symmetry:

$$V(x + a) = V(x), \quad (4.114)$$

where a is the primitive lattice constant. The study of periodic potentials explains the electronic properties of solids, establishing why materials classify into electrical conductors, semiconductors, and insulators.

4.10.1 Translation Symmetry and Bloch's Theorem

We define the discrete spatial translation operator $\hat{T}_a$ acting on wave functions by

$$\langle x | \hat{T}_a | \Psi \rangle = \psi(x + a). \quad (4.115)$$

Because the Hamiltonian $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$ is periodic, $\hat{T}_a$ commutes with $\hat{H}$:

$$[\hat{H}, \hat{T}_a] = 0. \quad (4.116)$$

Consequently, stationary energy eigenstates can be chosen to be simultaneous eigenstates of $\hat{T}_a$.

Let $\psi(x)$ be an eigenstate of $\hat{T}_a$ with eigenvalue λ :

$$\hat{T}_a \psi(x) = \psi(x + a) = \lambda \psi(x). \quad (4.117)$$

To ensure that $\psi(x)$ remains bounded over the entire real line ($x \in \mathbb{R}$), the eigenvalue must lie on the unit circle: $\lambda = e^{ika}$ with crystal wave vector $k \in \mathbb{R}$.⁴ Defining $u_k(x) \equiv e^{-ikx} \psi(x)$, we find

$$u_k(x + a) = e^{-ik(x+a)} \psi(x + a) = e^{-ika} e^{-ikx} [e^{ika} \psi(x)] = u_k(x). \quad (4.118)$$

This proves *Bloch's Theorem*: stationary eigenstates in a periodic potential take the form of a plane wave modulated by a cell-periodic amplitude:

$$\psi_k(x) = e^{ikx} u_k(x), \quad u_k(x + a) = u_k(x). \quad (4.119)$$

Because $e^{i(k+2\pi n/a)a} = e^{ika}$ for any integer $n \in \mathbb{Z}$, the crystal wave vector k is uniquely defined modulo reciprocal lattice vectors $G_n = \frac{2\pi n}{a}$. All inequivalent eigenvalues are contained within the *first Brillouin zone*:

$$-\frac{\pi}{a} < k \leq \frac{\pi}{a}. \quad (4.120)$$

4.10.2 The Kronig-Penney Model with Delta-Function Barriers

To obtain an exact dispersion relation, consider the Dirac comb (Kronig-Penney model):

$$V(x) = \sum_{n=-\infty}^{\infty} v_0 \delta(x - na), \quad v_0 > 0. \quad (4.121)$$

In the inter-site region $0 < x < a$, the potential vanishes, and the wave function solves the free Helmholtz equation:

$$\psi(x) = Ae^{iqx} + Be^{-iqx}, \quad q \equiv \frac{\sqrt{2mE}}{\hbar}. \quad (4.122)$$

The cell-periodic Bloch function $u_k(x) = e^{-ikx} \psi(x)$ is therefore

$$u_k(x) = Ae^{i(q-k)x} + Be^{-i(q+k)x}. \quad (4.123)$$

We impose two boundary conditions across the delta barrier at $x = 0$:

⁴Complex values of k with $\text{Im}(k) \neq 0$ describe evanescent states localized at surfaces or interfaces (Tamm/Shockley surface states).

1. **Continuity of the wave function:** $u_k(0^+) = u_k(a^-)$, which gives

$$A + B = Ae^{i(q-k)a} + Be^{-i(q+k)a}. \quad (4.124)$$

2. **Derivative discontinuity condition:** Integrating the Schrödinger equation across $x = 0$ from $-\varepsilon$ to $+\varepsilon$ yields

$$\left. \frac{d\psi}{dx} \right|_{0^+} - \left. \frac{d\psi}{dx} \right|_{0^-} = \frac{2mv_0}{\hbar^2} \psi(0). \quad (4.125)$$

Using the Bloch condition $\psi'(0^-) = e^{-ika}\psi'(a^-)$, this becomes

$$iq(A - B) - iqe^{-ika}(Ae^{iqa} - Be^{-iqa}) = \frac{2mv_0}{\hbar^2}(A + B). \quad (4.126)$$

Eliminating A and B from Eqs. (4.124) and (4.126) yields the dispersion relation:

$$\cos(ka) = \cos(qa) + \frac{mv_0a}{\hbar^2} \frac{\sin(qa)}{qa}. \quad (4.127)$$

Defining the dimensionless coupling strength $P \equiv \frac{mv_0a}{\hbar^2}$, the condition for physical solutions is

$$\left| \cos(qa) + P \frac{\sin(qa)}{qa} \right| \leq 1. \quad (4.128)$$

For values of q where the magnitude of the right-hand side exceeds unity, no real wave vector k exists. Consequently, the energy spectrum separates into continuous *allowed energy bands* separated by forbidden *band gaps*.

4.10.3 Born-von Kármán Boundary Conditions and Band Filling

For a finite macroscopic crystal containing N primitive cells ($L = Na$), we impose Born-von Kármán periodic boundary conditions:

$$\psi(x + L) = \psi(x) \implies e^{ikL} = 1. \quad (4.129)$$

This quantizes the allowed crystal wave vectors into N discrete values within the first Brillouin zone:

$$k_n = \frac{2\pi n}{Na}, \quad n = -\frac{N}{2} + 1, \dots, \frac{N}{2}. \quad (4.130)$$

Thus, each energy band contains precisely N spatial orbital states.

Accounting for electron spin ($s = 1/2$, with spin degeneracy $g_s = 2$), each spatial state can accommodate two electrons with opposite spin projections. By the Pauli exclusion principle:

$$N_{\text{electrons/band}} = 2N. \quad (4.131)$$

4.10.4 Insulators versus Conductors

The ground state of the solid is constructed by filling the single-particle energy states up to the Fermi energy E_F :

- **Insulators and Semiconductors:** If the total number of valence electrons is an even integer multiple of N , the electrons can completely fill an integer number of bands. The highest occupied band (valence band) is fully filled, and the lowest unoccupied band (conduction band) is separated by a finite energy gap E_g . A weak external electric field cannot supply the finite energy $\Delta E \geq E_g$ required to promote electrons across the gap into unoccupied states. Hence, the net electric current remains zero.
- **Metals (Conductors):** If the number of valence electrons per atom is odd, or if bands overlap in energy, the topmost band is only partially occupied. Infinitesimally close unoccupied states are available directly above E_F . An applied electric field accelerates electrons into adjacent momentum states, generating a net conduction current.

4.11 The Quantum Harmonic Oscillator

The simple harmonic oscillator represents one of the most fundamental models in quantum physics, describing small-amplitude vibrations across mechanical systems, molecular bonds, lattice phonons, and quantized electromagnetic radiation fields.

The one-dimensional Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \quad (4.132)$$

where m is the particle mass and ω is the classical oscillation frequency.

4.11.1 Dirac Ladder Operators and the Number Operator

Following Dirac's algebraic operator formulation, we introduce non-Hermitian dimensionless ladder operators:

$$\hat{a} \equiv \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i\hat{p}}{\sqrt{2m\hbar\omega}}, \quad (4.133)$$

$$\hat{a}^\dagger \equiv \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - \frac{i\hat{p}}{\sqrt{2m\hbar\omega}}. \quad (4.134)$$

Using the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar\mathbb{I}$, the commutator of the ladder operators is

$$[\hat{a}, \hat{a}^\dagger] = \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} = \mathbb{I}. \quad (4.135)$$

Defining the Hermitian *number operator*:

$$\hat{N} \equiv \hat{a}^\dagger\hat{a}, \quad (4.136)$$

the Hamiltonian is expressed linearly in terms of $\hat{N}$:

$$\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right). \quad (4.137)$$

Thus, the stationary energy eigenstates of $\hat{H}$ are the eigenstates of $\hat{N}$.

Let $|n\rangle$ denote an eigenstate of $\hat{N}$ with eigenvalue n :

$$\hat{N}|n\rangle = n|n\rangle \implies \hat{H}|n\rangle = \hbar\omega \left(n + \frac{1}{2} \right) |n\rangle. \quad (4.138)$$

Since $\hat{N} = \hat{a}^\dagger\hat{a}$, the norm of any physical state vector guarantees that eigenvalues are non-negative:

$$n = \langle n|\hat{N}|n\rangle = \langle n|\hat{a}^\dagger\hat{a}|n\rangle = \|\hat{a}|n\rangle\|^2 \geq 0. \quad (4.139)$$

4.11.2 Energy Spectrum and Ladder Algebra

The commutation relation $[\hat{a}, \hat{a}^\dagger] = \mathbb{I}$ leads directly to the operator identities:

$$[\hat{N}, \hat{a}] = [\hat{a}^\dagger\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}]\hat{a} = -\hat{a}, \quad (4.140)$$

$$[\hat{N}, \hat{a}^\dagger] = [\hat{a}^\dagger\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger[\hat{a}, \hat{a}^\dagger] = +\hat{a}^\dagger. \quad (4.141)$$

Acting with these relations on an eigenstate $|n\rangle$:

$$\hat{N}(\hat{a}|n\rangle) = (\hat{a}\hat{N} - \hat{a})|n\rangle = (n-1)(\hat{a}|n\rangle), \quad (4.142)$$

$$\hat{N}(\hat{a}^\dagger|n\rangle) = (\hat{a}^\dagger\hat{N} + \hat{a}^\dagger)|n\rangle = (n+1)(\hat{a}^\dagger|n\rangle). \quad (4.143)$$

Thus, $\hat{a}$ acts as an *annihilation* (lowering) operator that decreases the eigenvalue n by one unit, while $\hat{a}^\dagger$ acts as a *creation* (raising) operator that increases n by one unit.

Computing the norms of these shifted states:

$$\|\hat{a}|n\rangle\|^2 = \langle n|\hat{a}^\dagger\hat{a}|n\rangle = n \implies \hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \quad (4.144)$$

$$\|\hat{a}^\dagger|n\rangle\|^2 = \langle n|\hat{a}\hat{a}^\dagger|n\rangle = \langle n|(\hat{N}+1)|n\rangle = n+1 \implies \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (4.145)$$

Repeated application of $\hat{a}$ generates descending states $|n-1\rangle, |n-2\rangle, \dots$. To avoid producing states with forbidden negative eigenvalues ($n < 0$), the descending sequence must terminate at a ground state $|0\rangle$ that is annihilated by $\hat{a}$:

$$\hat{a}|0\rangle = 0. \quad (4.146)$$

Consequently, the eigenvalues of $\hat{N}$ must be non-negative integers:

$$n = 0, 1, 2, 3, \dots \quad (4.147)$$

The quantized energy levels of the harmonic oscillator are therefore equally spaced:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n \in \mathbb{N}_0. \quad (4.148)$$

The ground state energy $E_0 = \frac{1}{2}\hbar\omega$ is the zero-point energy of the oscillator.

All normalized excited states are generated algebraically from the vacuum $|0\rangle$:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle. \quad (4.149)$$

4.11.3 Wave Functions in the Position Representation

The coordinate wave function of the ground state is determined by the annihilation condition (4.146). Projecting onto the position bra $\langle x|$:

$$\langle x|\hat{a}|0\rangle = \left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx} \right) \psi_0(x) = 0. \quad (4.150)$$

Integrating this first-order differential equation yields the normalized Gaussian wave function:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \exp \left(-\frac{m\omega x^2}{2\hbar} \right). \quad (4.151)$$

Evaluating the spatial and momentum uncertainties:

$$\Delta x = \sqrt{\frac{\hbar}{2m\omega}}, \quad \Delta p = \sqrt{\frac{\hbar m\omega}{2}}, \quad (4.152)$$

confirming that the ground state saturates the Heisenberg lower bound:

$$\Delta x \Delta p = \frac{\hbar}{2}. \quad (4.153)$$

The excited stationary wave functions are obtained by repeatedly applying the differential creation operator $\hat{a}^\dagger$:

$$\psi_n(x) = \langle x|n\rangle = \frac{1}{\sqrt{n!}} \left[\frac{1}{\sqrt{2}} \left(\xi - \frac{d}{d\xi} \right) \right]^n \psi_0(x), \quad \xi \equiv \sqrt{\frac{m\omega}{\hbar}}x. \quad (4.154)$$

This yields the closed-form expression:

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) \exp\left(-\frac{m\omega x^2}{2\hbar}\right), \quad (4.155)$$

where $H_n(\xi)$ denotes the n -th Hermite polynomial:

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2, \quad H_3(\xi) = 8\xi^3 - 12\xi. \quad (4.156)$$

Because $H_n(-\xi) = (-1)^n H_n(\xi)$, the eigenstates have definite parity matching $(-1)^n$. The wave functions satisfy the orthonormality condition:

$$\int_{-\infty}^{\infty} dx \psi_n^*(x) \psi_m(x) = \langle n|m \rangle = \delta_{nm}. \quad (4.157)$$

4.12 Problems

1. **Absence of Bound States for a Repulsive Step.** Explain why there exist no physically acceptable solutions with energy $E < 0$ for the potential step

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0 > 0, & x > 0. \end{cases} \quad (4.158)$$

2. **Wigner Time Delay at a Potential Step.** Consider the potential step of Problem 1. Assume that a wave packet with mean energy $E_0 = p_0^2/(2m) < V_0$ is incident from the left.

- (a) Using the energy dependence of the reflection phase shift $\delta(E)$, calculate the time delay $\tau_{\text{delay}} = 2\hbar \frac{d\delta}{dE}$ between the arrival of the incident wave packet at $x = 0$ and the emergence of the reflected packet.
- (b) Express this time delay as an effective penetration depth $d_{\text{eff}} = v_0 \tau_{\text{delay}}/2$ into the classically forbidden barrier.

3. **Rectangular Potential Barrier Scattering.**

- (a) For a rectangular barrier of height V_0 and width a , derive the reflection amplitude $R(E)$ and transmission amplitude $S(E)$ for both $E < V_0$ (tunneling) and $E > V_0$ (transmission over the barrier).
- (b) Calculate the transit time of a wave packet traversing the barrier, and examine its limiting behavior at transmission resonances.

4. **Probability Current Density for Step Scattering.** Using the general definition of the probability current density $\mathbf{j}(x) = \frac{\hbar}{m} \text{Im}[\psi^*(x)\psi'(x)]$, explicitly verify the current conservation relations:

$$j_L = \frac{p}{m} (|A|^2 - |B|^2), \quad j_R = \frac{\bar{p}}{m} |C|^2, \quad (4.159)$$

and demonstrate that $j_L = j_R$ for the potential step with $E > V_0$.

5. **Reciprocity and S-Matrix Symmetries.** Consider an arbitrary localized potential $V(x)$ vanishing outside the interval $[-a, a]$. Let $S_{L \rightarrow R}(E)$ and $N_L(E)$ be the transmission and reflection amplitudes for a wave incident from the left ($x = -\infty$).

- (a) Using time-reversal symmetry (the fact that the stationary Schrödinger equation is real), prove reciprocity: the transmission amplitude for a wave incident from the right ($x = +\infty$) satisfies $S_{R \rightarrow L}(E) = S_{L \rightarrow R}(E) \equiv S(E)$.

- (b) Express the reflection amplitude from the right, $N_R(E)$, in terms of $N_L(E)$ and $S(E)$, and verify the unitarity relations of the scattering matrix.

6. **Phase Shift Perturbation and Energy Level Shifts.** Consider a potential step bounded by an infinite wall:

$$V_I(x) = \begin{cases} 0, & x < a, \\ V(x), & a < x < b, \\ \infty, & x > b, \end{cases} \quad (4.160)$$

where $V(x)$ is an unknown barrier potential and $a > 0$. The reflection amplitude at $x = a$ is parameterized by $N(E) = -e^{2i\delta(E)}$.

- (a) Prove that $\delta(E)$ must be real.
 (b) Consider a particle confined in the well $V_{II}(x)$ with an infinite wall at $x = 0$ ($V(x) = \infty$ for $x < 0$, and $V(x) = V_I(x)$ for $x > 0$). Show that to first order in the phase shift $\delta(E)$, the bound-state energy levels are shifted relative to the unperturbed infinite well $V_{III}(x)$ (of width a) by

$$E = E_0 - \frac{p_0 \hbar}{ma} \delta(E_0), \quad (4.161)$$

where $p_0 = \sqrt{2mE_0}$ and $E_0 = n^2 \pi^2 \hbar^2 / (2ma^2)$.

- (c) Provide a physical argument showing that $\delta(E) > 0$ for penetrable repulsive barriers.

7. **Tunneling Splitting and the One-Electron Chemical Bond.**

- (a) Show that the bound-state energy levels of a decoupled double square well,

$$V_S(x) = \begin{cases} \infty, & |x| > b, \\ 0, & a < |x| < b, \\ \infty, & |x| < a, \end{cases} \quad (4.162)$$

are two-fold degenerate.

- (b) Replace the central hard wall with a high, symmetric barrier of width $2a$. Parametrize the reflection amplitude at the barrier face as $-e^{2i\delta(E)}$ ($\delta > 0$) and the transmission amplitude as $S(E) = iJ(E)$ with $J \ll 1$. Show that tunneling lifts the degeneracy, splitting each level into an even-parity (symmetric bonding) state and an odd-parity (antisymmetric antibonding) state:

$$E_{\pm} = E_0 - \frac{p_0 \hbar}{m(b-a)} \left[\delta(E_0) \mp \frac{J(E_0)}{2} \right]. \quad (4.163)$$

- (c) Discuss how this energy splitting explains the formation of the molecular orbital and covalent bonding in the ionized hydrogen molecule H_2^+ , and estimate the binding energy in terms of the tunneling rate.

8. **Transmission Resonances in a Well with High Shoulders.** Calculate the transmission amplitude $S(E)$ for a symmetric potential well flanked by two finite barrier shoulders:

$$V(x) = \begin{cases} 0, & |x| < \frac{b}{2}, \\ V_0 > 0, & \frac{b}{2} \leq |x| < \frac{a}{2}, \\ 0, & |x| \geq \frac{a}{2}. \end{cases} \quad (4.164)$$

Determine the resonance energies and widths Γ in the limit of high barrier shoulders ($V_0 \gg E$).

9. **Spatial Density Profile of Emitted α -Particles.** An α -emitter containing N_0 radioactive nuclei, each decaying exponentially with decay constant γ , is placed at the origin at $t = 0$. Assuming particles are emitted unidirectionally in the $+x$ direction with speed v :
- Derive the spatial particle density $n(x, t)$ along $x > 0$.
 - Sketch $n(x, t)$ at successive times, including $t > \gamma^{-1} \ln N_0$, and connect this distribution to the wave packet envelope derived in Eq. (4.98).
10. **Delta-Function Limit of a Finite Square Well.**
- Show that taking the limit of an attractive rectangular well of depth V_0 and width a such that $V_0 \rightarrow \infty$ and $a \rightarrow 0$ with $V_0 a = |v_0| = \text{const}$ rigorously reproduces the bound-state energy $E = -\frac{mv_0^2}{2\hbar^2}$ and the exponential wave function of the Dirac delta potential.
 - Show that the transmission amplitude $S(E)$ of the finite square well converges to Eq. (4.108) in the same scaling limit.
11. **Double Delta-Function Potential.** Consider a particle of mass m subject to two attractive delta-function wells:

$$V(x) = -v_0\delta(x-a) - v_0\delta(x+a), \quad v_0 > 0. \quad (4.165)$$

- Determine the stationary wave functions for bound states of even parity ($\psi(-x) = \psi(x)$).
 - Derive the transcendental eigenvalue equation for even-parity bound-state energies, and prove graphically that at least one even bound state exists for all values of $v_0 > 0$ and $a > 0$.
 - Solve analytically for the even-parity binding energy in the weak-coupling limit $mv_0a/\hbar^2 \ll 1$.
 - Repeat the derivation for odd-parity bound states ($\psi(-x) = -\psi(x)$), and determine the threshold condition on v_0 required for an odd bound state to exist.
 - In the asymptotic limit of large separation ($a \rightarrow \infty$), show that both bound-state energies approach the single-well value $E \rightarrow -\frac{mv_0^2}{2\hbar^2}$, and explain the physical origin of this asymptotic degeneracy.
12. **Effective Mass in the Kronig-Penney Lattice.** In solid-state physics, the effective mass m^* of an electron in a crystal band $E(k)$ is defined by

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2}. \quad (4.166)$$

For the delta-comb Kronig-Penney dispersion relation, calculate m^* analytically at the band bottom ($k = 0$) and at the band edge (Brillouin zone boundary $k = \pi/a$) of the lowest energy band. Discuss the physical interpretation of a negative effective mass.

13. **Completeness and Expansion of the Delta Function.** Using the completeness of the orthonormal Hermite eigenbasis $\psi_n(x)$ of the harmonic oscillator, express the localized Dirac delta distribution $\delta(x - x_0)$ as a bilinear series in Hermite functions.
14. **Harmonic Oscillator Ladder Matrix Elements and the Virial Theorem.**
- Express the coordinate $\hat{x}$ and momentum $\hat{p}$ operators in terms of ladder creation and annihilation operators $\hat{a}$ and $\hat{a}^\dagger$.

- (b) Calculate the matrix elements $\langle n|\hat{x}|m\rangle$, $\langle n|\hat{p}|m\rangle$, $\langle n|\hat{x}^2|m\rangle$, $\langle n|\hat{p}^2|m\rangle$, $\langle n|\hat{x}\hat{p}|m\rangle$, $\langle n|\hat{p}\hat{x}|m\rangle$, and $\langle n|\hat{H}|m\rangle$.
- (c) Using these matrix elements, prove the quantum virial theorem for any stationary energy eigenstate $|n\rangle$:

$$\langle n|\hat{T}|n\rangle = \langle n|\hat{V}|n\rangle = \frac{1}{2}E_n. \quad (4.167)$$

15. **Zero-Point Energy Estimates via the Uncertainty Principle.** Use the Heisenberg uncertainty relation $\Delta x \Delta p \approx \hbar/2$ to estimate the ground-state zero-point energy and spatial size for:

- (a) An anharmonic oscillator potential $V(x) = \lambda x^4$ ($\lambda > 0$) in one dimension.
- (b) The attractive Coulomb potential $V(r) = -e^2/r$ in three dimensions (hydrogen atom), expressing the resulting ground-state energy estimate in eV.
- (c) A screened Coulomb (Yukawa) potential $V(r) = -\frac{q^2}{r}e^{-\kappa r}$ with screening parameter $\kappa = 1.3 \times 10^6 \text{ cm}^{-1}$ and effective charge $q^2 = e^2/14$ (modeling a donor impurity in germanium).

16. **Two-State Tunneling Dynamics in a Double-Well Box.** A particle is confined in a box divided into two symmetric compartments by a thin central barrier. Let $\psi_1(t)$ and $\psi_2(t)$ denote the probability amplitudes for the particle to occupy the left and right compartments, respectively. The inter-well tunneling dynamics are modeled by the coupled Hamiltonian system:

$$i\hbar \frac{d\psi_1}{dt} = K\psi_2, \quad (4.168)$$

$$i\hbar \frac{d\psi_2}{dt} = K\psi_1, \quad (4.169)$$

where K is a real tunneling coupling constant.

- (a) Determine the normalized stationary energy eigenstates (eigenvectors in $\mathbb{C}^2$) and their eigenvalues. Determine the spatial parity of each eigenstate.
- (b) Suppose the system is prepared at $t = 0$ in the state $\psi_1(0) = A$ and $\psi_2(0) = Ae^{i\delta}$. Calculate the time-dependent probability $P_1(t) = |\psi_1(t)|^2$ of locating the particle in the left compartment, and find its rate of change $\frac{dP_1}{dt}$ as a function of time.

Chapter 5

Equations Of Motion For Operators

5.1 Equations of Motion for Operators

In the preceding chapters, quantum dynamics was formulated primarily within the Schrödinger picture, where physical states $|\Psi(t)\rangle$ evolve continuously in time according to the Schrödinger equation, while the operators corresponding to physical observables remain stationary (unless they possess explicit time dependence).

In this chapter, we develop the alternative, fundamentally equivalent Heisenberg picture of quantum mechanics. In this formulation, the state vectors remain fixed at their initial values, and the entire dynamical evolution of the physical system is carried by the time-dependent operators representing observables.

We begin by establishing the algebraic foundation of functions of operators and their spectral representations. We then introduce the unitary time-evolution operator, formulate the Heisenberg equations of motion, and demonstrate their deep correspondence with classical Hamilton–Jacobi mechanics and Poisson brackets, culminating in Ehrenfest’s theorem. Finally, we investigate formal solutions for the time-development operator in both time-independent and time-dependent situations, introducing the time-ordering operator and Dyson series expansion.

5.2 Functions of Operators

5.2.1 Power Series and Spectral Definitions

Suppose a scalar function $f(x)$ admits a convergent Taylor or power series expansion about the origin:

$$f(x) = f_0 + f_1x + f_2x^2 + f_3x^3 + \cdots = \sum_{n=0}^{\infty} f_n x^n. \quad (5.1)$$

If $\hat{A}$ is a linear operator acting on a Hilbert space, we can naturally define the operator function $f(\hat{A})$ by the corresponding operator series:

$$f(\hat{A}) = f_0\mathbb{I} + f_1\hat{A} + f_2\hat{A}^2 + f_3\hat{A}^3 + \cdots = \sum_{n=0}^{\infty} f_n \hat{A}^n. \quad (5.2)$$

A quintessential example is the operator exponential:

$$e^{\lambda\hat{A}} = \mathbb{I} + \lambda\hat{A} + \frac{\lambda^2}{2!}\hat{A}^2 + \frac{\lambda^3}{3!}\hat{A}^3 + \cdots = \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} \hat{A}^n, \quad (5.3)$$

where λ is a complex parameter.

If $|\alpha\rangle$ is an eigenstate of $\hat{A}$ associated with the eigenvalue α , such that $\hat{A}|\alpha\rangle = \alpha|\alpha\rangle$, repeated application yields $\hat{A}^n|\alpha\rangle = \alpha^n|\alpha\rangle$. Consequently, acting on $|\alpha\rangle$ with $f(\hat{A})$ gives

$$\begin{aligned} f(\hat{A})|\alpha\rangle &= (f_0\mathbb{I} + f_1\hat{A} + f_2\hat{A}^2 + \cdots)|\alpha\rangle \\ &= (f_0 + f_1\alpha + f_2\alpha^2 + \cdots)|\alpha\rangle = f(\alpha)|\alpha\rangle. \end{aligned} \quad (5.4)$$

Hence, $|\alpha\rangle$ is an eigenstate of $f(\hat{A})$ with eigenvalue $f(\alpha)$.

Assuming the eigenstates of $\hat{A}$ form a complete orthonormal basis (which is always guaranteed for self-adjoint operators by the spectral theorem), the resolution of the identity is

$$\mathbb{I} = \sum_{\alpha} |\alpha\rangle\langle\alpha|, \quad (5.5)$$

where the summation symbol represents a sum over discrete eigenvalues and an integral over any continuous spectrum. Applying $f(\hat{A})$ to both sides and using Eq. (5.4), we obtain the spectral representation:

$$f(\hat{A}) = \sum_{\alpha} f(\alpha)|\alpha\rangle\langle\alpha|. \quad (5.6)$$

Indeed, Eq. (5.6) serves as an elegant, general definition of $f(\hat{A})$ that does not require $f(x)$ to be analytic or possess a power series expansion. For instance, one can define the Dirac delta function of an operator, $\delta(\hat{A} - x)$, where $x \in \mathbb{R}$:

$$\delta(\hat{A} - x) = \sum_{\alpha} \delta(\alpha - x)|\alpha\rangle\langle\alpha|. \quad (5.7)$$

5.2.2 Non-Commutativity and Weyl's Formula

Care must be exercised when manipulating functions of non-commuting operators. In general, if $[\hat{A}, \hat{B}] \neq 0$, the exponential of their sum does not factorize:

$$e^{\hat{A}+\hat{B}} \neq e^{\hat{A}}e^{\hat{B}} \neq e^{\hat{B}}e^{\hat{A}}. \quad (5.8)$$

In the special case where the commutator $[\hat{A}, \hat{B}]$ commutes with both $\hat{A}$ and $\hat{B}$ —that is, when $[[\hat{A}, \hat{B}], \hat{A}] = [[\hat{A}, \hat{B}], \hat{B}] = 0$, so that $[\hat{A}, \hat{B}]$ behaves as a *c*-number (scalar multiple of the identity), as with $[\hat{x}, \hat{p}] = i\hbar\mathbb{I}$ —the Baker–Campbell–Hausdorff formula reduces to *Weyl's formula*:

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{-\frac{1}{2}[\hat{A}, \hat{B}]}. \quad (5.9)$$

5.2.3 Time-Evolution Operator in the Schrödinger Picture

As an immediate physical application, consider the temporal evolution of a quantum state. If $|\Psi(0)\rangle$ is the state of the system at $t = 0$, its state at any subsequent time t can be expressed in terms of the complete set of energy eigenstates $\{|E\rangle\}$ of the Hamiltonian $\hat{H}$ as

$$|\Psi(t)\rangle = \sum_E e^{-iEt/\hbar} |E\rangle\langle E|\Psi(0)\rangle. \quad (5.10)$$

Recognizing the spectral expansion in Eq. (5.6), Eq. (5.10) is identically written as

$$|\Psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\Psi(0)\rangle. \quad (5.11)$$

Differentiating Eq. (5.11) with respect to t confirms that it satisfies the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle. \quad (5.12)$$

We emphasize that Eq. (5.11) is valid whenever $\hat{H}$ has no explicit time dependence. The corresponding dual state vector $\langle\Psi(t)|$ is given by taking the Hermitian conjugate:

$$\langle\Psi(t)| = \langle\Psi(0)| \left(e^{-i\hat{H}t/\hbar}\right)^\dagger = \langle\Psi(0)|e^{i\hat{H}t/\hbar}, \quad (5.13)$$

where we have used the Hermiticity of the Hamiltonian, $\hat{H}^\dagger = \hat{H}$.

5.3 The Heisenberg Representation

5.3.1 State Evolution versus Operator Evolution

In physical experiments, observational predictions are formulated in terms of expectation values of operators. In the Schrödinger picture, the expectation value of an observable $\hat{A}$ at time t in a state $|\Psi(t)\rangle$ is given by

$$\langle\hat{A}\rangle_t = \langle\Psi(t)|\hat{A}|\Psi(t)\rangle. \quad (5.14)$$

Assuming for now that $\hat{A}$ does not possess explicit time dependence, and using the formal time-evolution relation $|\Psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\Psi(0)\rangle$ together with its Hermitian adjoint $\langle\Psi(t)| = \langle\Psi(0)|e^{i\hat{H}t/\hbar}$, Eq. (5.14) can be recast as

$$\langle\hat{A}\rangle_t = \langle\Psi(0)|e^{i\hat{H}t/\hbar}\hat{A}e^{-i\hat{H}t/\hbar}|\Psi(0)\rangle. \quad (5.15)$$

This algebraic rewriting motivates the definition of the *Heisenberg representation operator*:

$$\hat{A}(t) \equiv e^{i\hat{H}t/\hbar}\hat{A}e^{-i\hat{H}t/\hbar}. \quad (5.16)$$

The expectation value is therefore equivalently expressed as

$$\langle\hat{A}\rangle_t = \langle\Psi(0)|\hat{A}(t)|\Psi(0)\rangle. \quad (5.17)$$

Equation (5.17) reflects a profound conceptual shift: rather than regarding the state vector as rotating through Hilbert space while operators remain static (the *Schrödinger picture*), one can regard the state vector as permanently frozen at its initial condition $|\Psi(0)\rangle$, with all temporal dynamics carried by the operators $\hat{A}(t)$ (the *Heisenberg picture*). At $t = 0$, both representations coincide: $\hat{A}(0) = \hat{A}$ and $|\Psi(0)\rangle_H = |\Psi(0)\rangle_S$. Because matrix elements and expectation values are strictly identical in both pictures, the two formulations are physically equivalent.

5.3.2 The Heisenberg Equation of Motion

To determine the dynamic equation governing $\hat{A}(t)$, we differentiate Eq. (5.16) with respect to time (assuming $\partial\hat{A}/\partial t = 0$):

$$\begin{aligned} i\hbar \frac{d\hat{A}(t)}{dt} &= i\hbar \left[\frac{d}{dt} e^{i\hat{H}t/\hbar} \right] \hat{A} e^{-i\hat{H}t/\hbar} + i\hbar e^{i\hat{H}t/\hbar} \hat{A} \left[\frac{d}{dt} e^{-i\hat{H}t/\hbar} \right] \\ &= e^{i\hat{H}t/\hbar} \left(-\hat{H}\hat{A} + \hat{A}\hat{H} \right) e^{-i\hat{H}t/\hbar} = e^{i\hat{H}t/\hbar} [\hat{A}, \hat{H}] e^{-i\hat{H}t/\hbar}. \end{aligned} \quad (5.18)$$

Since $\hat{H}$ commutes with $e^{\pm i\hat{H}t/\hbar}$, this simplifies directly to the *Heisenberg equation of motion*:

$$i\hbar \frac{d\hat{A}(t)}{dt} = [\hat{A}(t), \hat{H}]. \quad (5.19)$$

A fundamental consequence of Eq. (5.19) is that any operator that commutes with the Hamiltonian is a constant of the motion. In particular, the Hamiltonian itself satisfies

$$\hat{H}(t) = e^{i\hat{H}t/\hbar}\hat{H}e^{-i\hat{H}t/\hbar} = \hat{H}, \quad (5.20)$$

so that $[\hat{H}(t), \hat{H}] = 0$ and $d\hat{H}/dt = 0$, expressing quantum energy conservation.

5.3.3 Particle Dynamics and the Classical Correspondence

Consider a single particle of mass m moving in three dimensions governed by the standard Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{r}}). \quad (5.21)$$

Because $\hat{H}$ is time-independent, we have

$$\hat{H} = \hat{H}(t) = \frac{\hat{\mathbf{p}}(t)^2}{2m} + V(\hat{\mathbf{r}}(t)). \quad (5.22)$$

The Heisenberg position operator satisfies

$$i\hbar \frac{d\hat{\mathbf{r}}(t)}{dt} = [\hat{\mathbf{r}}(t), \hat{H}]. \quad (5.23)$$

Since an operator always commutes with any function of itself, $[\hat{\mathbf{r}}(t), V(\hat{\mathbf{r}}(t))] = 0$. At equal times, the canonical commutation relations are preserved under unitary transformation:

$$\begin{aligned} [\hat{r}_i(t), \hat{p}_j(t)] &= e^{i\hat{H}t/\hbar} [\hat{r}_i, \hat{p}_j] e^{-i\hat{H}t/\hbar} \\ &= e^{i\hat{H}t/\hbar} (i\hbar \delta_{ij} \mathbb{I}) e^{-i\hat{H}t/\hbar} = i\hbar \delta_{ij} \mathbb{I}. \end{aligned} \quad (5.24)$$

Caution: The canonical commutation relations hold only at *equal* times; for unequal times $t \neq t'$, $[\hat{r}_i(t), \hat{p}_j(t')]$ does not vanish for $i \neq j$ nor does it reduce to $i\hbar \delta_{ij}$, due to intervening factors of $e^{\pm i\hat{H}(t-t')/\hbar}$.

Evaluating the commutator with the kinetic energy yields

$$\left[\hat{\mathbf{r}}(t), \frac{\hat{\mathbf{p}}(t)^2}{2m} \right] = \frac{i\hbar \hat{\mathbf{p}}(t)}{m}, \quad (5.25)$$

from which we recover the quantum velocity equation:

$$\frac{d\hat{\mathbf{r}}(t)}{dt} = \frac{\hat{\mathbf{p}}(t)}{m}. \quad (5.26)$$

For the momentum operator, we evaluate $[\hat{\mathbf{p}}(t), V(\hat{\mathbf{r}}(t))] = e^{i\hat{H}t/\hbar} [\hat{\mathbf{p}}, V(\hat{\mathbf{r}})] e^{-i\hat{H}t/\hbar}$. In coordinate space, for any test state $|\Psi\rangle$:

$$\begin{aligned} \langle \mathbf{r} | [\hat{\mathbf{p}}, V(\hat{\mathbf{r}})] | \Psi \rangle &= \frac{\hbar}{i} \nabla [V(\mathbf{r}) \langle \mathbf{r} | \Psi \rangle] - V(\mathbf{r}) \frac{\hbar}{i} \nabla \langle \mathbf{r} | \Psi \rangle \\ &= -i\hbar \nabla V(\mathbf{r}) \langle \mathbf{r} | \Psi \rangle, \end{aligned} \quad (5.27)$$

establishing that

$$[\hat{\mathbf{p}}, V(\hat{\mathbf{r}})] = -i\hbar \nabla V(\hat{\mathbf{r}}). \quad (5.28)$$

Transforming to the Heisenberg picture, we find

$$[\hat{\mathbf{p}}(t), V(\hat{\mathbf{r}}(t))] = -i\hbar \nabla V(\hat{\mathbf{r}}(t)), \quad (5.29)$$

yielding the Heisenberg equation of motion for momentum:

$$\frac{d\hat{\mathbf{p}}(t)}{dt} = -\nabla V(\hat{\mathbf{r}}(t)). \quad (5.30)$$

Equations (5.26) and (5.30) are in exact formal correspondence with Hamilton's classical equations of motion:

$$\dot{\mathbf{r}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{r}}. \quad (5.31)$$

This striking structural isomorphism stems from Dirac's quantization rule, wherein the quantum commutator divided by $i\hbar$, namely $\frac{1}{i\hbar} [\hat{A}, \hat{B}]$, plays the exact role of the classical Poisson bracket $\{A, B\}_{\text{PB}}$.

5.3.4 Exactly Solvable Cases

The Free Particle

For a free particle ($V(\mathbf{r}) = 0$), the momentum operator satisfies

$$\frac{d\hat{\mathbf{p}}(t)}{dt} = 0 \quad \implies \quad \hat{\mathbf{p}}(t) = \hat{\mathbf{p}}(0), \quad (5.32)$$

so momentum is conserved. Integrating Eq. (5.26) then immediately yields

$$\hat{\mathbf{r}}(t) = \hat{\mathbf{r}}(0) + \frac{\hat{\mathbf{p}}(0)}{m}t. \quad (5.33)$$

Taking the expectation value of Eq. (5.33) in an arbitrary initial wave packet $|\Psi(0)\rangle$, we obtain

$$\langle \hat{\mathbf{r}} \rangle_t = \langle \hat{\mathbf{r}} \rangle_0 + \frac{\langle \hat{\mathbf{p}} \rangle_0}{m}t, \quad (5.34)$$

confirming that the center of the wave packet moves uniformly along a rectilinear classical trajectory (Ehrenfest's theorem).

The Simple Harmonic Oscillator

For a one-dimensional harmonic oscillator with $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$, the Heisenberg equations of motion are linear:

$$\frac{d\hat{x}(t)}{dt} = \frac{\hat{p}(t)}{m}, \quad \frac{d\hat{p}(t)}{dt} = -m\omega^2\hat{x}(t). \quad (5.35)$$

Because these coupled operator equations are linear with constant scalar coefficients, their operator solutions have the same form as the classical solutions:

$$\begin{aligned} \hat{x}(t) &= \hat{x}(0) \cos(\omega t) + \frac{\hat{p}(0)}{m\omega} \sin(\omega t), \\ \hat{p}(t) &= \hat{p}(0) \cos(\omega t) - m\omega \hat{x}(0) \sin(\omega t). \end{aligned} \quad (5.36)$$

Consequently, the mean position $\langle \hat{x} \rangle_t = \langle \hat{x} \rangle_0 \cos(\omega t) + \frac{\langle \hat{p} \rangle_0}{m\omega} \sin(\omega t)$ oscillates purely sinusoidally at the classical natural frequency ω .

5.3.5 Explicitly Time-Dependent Hamiltonians

In many realistic physical situations, the Hamiltonian contains explicit time dependence due to interaction with external fields, such as

$$\hat{H}(t) = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{r}}) - \mathbf{F}(t) \cdot \hat{\mathbf{r}}, \quad (5.37)$$

where $\mathbf{F}(t)$ represents an externally imposed time-varying force (e.g., an electric field).

When $\hat{H}$ depends on time, the simple solution $|\Psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\Psi(0)\rangle$ is no longer valid. Instead, we introduce a unitary time-evolution operator $\hat{U}(t)$ such that

$$|\Psi(t)\rangle = \hat{U}(t)|\Psi(0)\rangle, \quad (5.38)$$

with $\hat{U}(0) = \mathbb{I}$ and unitarity $\hat{U}^\dagger(t)\hat{U}(t) = \hat{U}(t)\hat{U}^\dagger(t) = \mathbb{I}$ to preserve probability conservation. Substituting into the Schrödinger equation yields

$$i\hbar \frac{d\hat{U}(t)}{dt} = \hat{H}_S(t)\hat{U}(t), \quad (5.39)$$

where the subscript S denotes the Schrödinger representation.

The expectation value of an observable $\hat{A}$ at time t is

$$\langle \hat{A} \rangle_t = \langle \Psi(t) | \hat{A} | \Psi(t) \rangle = \langle \Psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \Psi(0) \rangle. \quad (5.40)$$

We define the Heisenberg operator in the presence of explicit time dependence by

$$\hat{A}_H(t) \equiv \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t). \quad (5.41)$$

Differentiating Eq. (5.41) with respect to t , and noting that $-i\hbar \frac{d\hat{U}^\dagger(t)}{dt} = \hat{U}^\dagger(t) \hat{H}_S(t)$, we find

$$\begin{aligned} i\hbar \frac{d\hat{A}_H(t)}{dt} &= i\hbar \frac{d\hat{U}^\dagger(t)}{dt} \hat{A}_S \hat{U}(t) + \hat{U}^\dagger(t) \hat{A}_S \left(i\hbar \frac{d\hat{U}(t)}{dt} \right) + i\hbar \hat{U}^\dagger(t) \frac{\partial \hat{A}_S}{\partial t} \hat{U}(t) \\ &= \hat{U}^\dagger(t) \left[-\hat{H}_S(t) \hat{A}_S + \hat{A}_S \hat{H}_S(t) \right] \hat{U}(t) + i\hbar \left(\frac{\partial \hat{A}}{\partial t} \right)_H \\ &= [\hat{A}_H(t), \hat{H}_H(t)] + i\hbar \left(\frac{\partial \hat{A}}{\partial t} \right)_H, \end{aligned} \quad (5.42)$$

where $\hat{H}_H(t) \equiv \hat{U}^\dagger(t) \hat{H}_S(t) \hat{U}(t)$ is the Hamiltonian in the Heisenberg picture.

For the Heisenberg Hamiltonian itself, $[\hat{H}_H(t), \hat{H}_H(t)] = 0$, which yields

$$\frac{d\hat{H}_H(t)}{dt} = \hat{U}^\dagger(t) \frac{\partial \hat{H}_S(t)}{\partial t} \hat{U}(t) \equiv \frac{\partial \hat{H}_H(t)}{\partial t}, \quad (5.43)$$

in direct agreement with classical mechanics.

For the particle governed by Eq. (5.37), the Heisenberg equations of motion become

$$\frac{d\hat{\mathbf{r}}(t)}{dt} = \frac{\hat{\mathbf{p}}(t)}{m}, \quad (5.44)$$

and

$$\frac{d\hat{\mathbf{p}}(t)}{dt} = -\nabla V(\hat{\mathbf{r}}(t)) + \mathbf{F}(t)\mathbb{I}. \quad (5.45)$$

In the case of a driven (forced) harmonic oscillator in one dimension, combining Eqs. (5.44) and (5.45) gives

$$m \left[\frac{d^2}{dt^2} + \omega^2 \right] \hat{x}(t) = F(t)\mathbb{I}, \quad (5.46)$$

which is the quantum mechanical operator analog of the driven harmonic oscillator.

5.4 Formal Solution for the Time-Development Operator

5.4.1 Perturbative Expansion and the Dyson Series

We now construct a formal solution to the operator differential equation

$$i\hbar \frac{d\hat{U}(t)}{dt} = \hat{H}_S(t) \hat{U}(t), \quad \text{with } \hat{U}(0) = \mathbb{I}. \quad (5.47)$$

Were the Hamiltonian a scalar function or an operator that commuted with itself at all different times ($[\hat{H}_S(t'), \hat{H}_S(t'')] = 0$), the solution would simply be the ordinary exponential $\exp\left(-\frac{i}{\hbar} \int_0^t \hat{H}_S(t') dt'\right)$. However, in generic quantum systems, the Hamiltonian at time t' does not commute with itself at a different time t'' . Consequently, the ordinary exponential does not satisfy Eq. (5.47).

To resolve this ordering complication, we expand $\hat{U}(t)$ in powers of the interaction Hamiltonian:

$$\hat{U}(t) = \hat{U}_0(t) + \hat{U}_1(t) + \hat{U}_2(t) + \cdots = \sum_{n=0}^{\infty} \hat{U}_n(t), \quad (5.48)$$

where $\hat{U}_n(t)$ contains n factors of $\hat{H}_S$. Substituting this expansion into Eq. (5.47) and matching terms of equal order yields the hierarchy of coupled equations:

$$i\hbar \frac{d\hat{U}_0(t)}{dt} = 0, \quad (5.49)$$

$$i\hbar \frac{d\hat{U}_n(t)}{dt} = \hat{H}_S(t)\hat{U}_{n-1}(t) \quad (n \geq 1). \quad (5.50)$$

Applying the initial condition $\hat{U}(0) = \mathbb{I}$ requires $\hat{U}_0(0) = \mathbb{I}$ and $\hat{U}_n(0) = 0$ for all $n \geq 1$. Therefore, $\hat{U}_0(t) = \mathbb{I}$, and successive integration yields

$$\begin{aligned} \hat{U}_1(t) &= \frac{1}{i\hbar} \int_0^t dt' \hat{H}_S(t'), \\ \hat{U}_2(t) &= \frac{1}{i\hbar} \int_0^t dt' \hat{H}_S(t') \hat{U}_1(t') = \frac{1}{(i\hbar)^2} \int_0^t dt' \int_0^{t'} dt'' \hat{H}_S(t') \hat{H}_S(t''), \end{aligned}$$

and in general,

$$\hat{U}_n(t) = \frac{1}{(i\hbar)^n} \int_0^t dt' \int_0^{t'} dt'' \cdots \int_0^{t^{(n-1)}} dt^{(n)} \hat{H}_S(t') \hat{H}_S(t'') \cdots \hat{H}_S(t^{(n)}). \quad (5.51)$$

In this nested integral, the chronological sequence of time arguments is strictly ordered:

$$t \geq t' \geq t'' \geq \cdots \geq t^{(n)} \geq 0. \quad (5.52)$$

The operator evaluated at the latest time $\hat{H}_S(t')$ stands furthest to the left, followed by the operator at the next latest time, down to the earliest time $\hat{H}_S(t^{(n)})$ on the far right. The total evolution operator is therefore represented by the *Dyson series*:

$$\hat{U}(t) = \mathbb{I} + \sum_{n=1}^{\infty} \frac{1}{(i\hbar)^n} \int_0^t dt' \int_0^{t'} dt'' \cdots \int_0^{t^{(n-1)}} dt^{(n)} \hat{H}_S(t') \hat{H}_S(t'') \cdots \hat{H}_S(t^{(n)}). \quad (5.53)$$

5.4.2 The Time-Ordering Operator

The Dyson series can be cast into an exceptionally compact form through the introduction of the *time-ordering operator* $\mathcal{T}$ (also denoted by $(\dots)_+$). For any set of time-dependent operators, the time-ordered product rearranges them so that operators evaluated at later times appear to the left of operators evaluated at earlier times:

$$\mathcal{T} [\hat{A}(t') \hat{B}(t'')] \equiv \begin{cases} \hat{A}(t') \hat{B}(t''), & t' \geq t'', \\ \hat{B}(t'') \hat{A}(t'), & t'' > t'. \end{cases} \quad (5.54)$$

Consider the square of the integrated Hamiltonian under the time-ordering operation:

$$\begin{aligned} \mathcal{T} \left[\left(\int_0^t dt' \hat{H}_S(t') \right)^2 \right] &= \int_0^t dt' \int_0^t dt'' \mathcal{T} [\hat{H}_S(t') \hat{H}_S(t'')] \\ &= \int_0^t dt' \int_0^{t'} dt'' \hat{H}_S(t') \hat{H}_S(t'') + \int_0^t dt'' \int_0^{t''} dt' \hat{H}_S(t'') \hat{H}_S(t') \\ &= 2! \int_0^t dt' \int_0^{t'} dt'' \hat{H}_S(t') \hat{H}_S(t''). \end{aligned} \quad (5.55)$$

By extension to arbitrary order n , the n -dimensional hypercube $[0, t]^n$ can be partitioned into $n!$ congruent simplexes corresponding to each permutation of the time orderings. Because the time-ordered integrand is symmetric under permutations of its integration variables, the integral over the full hypercube is simply $n!$ times the integral over the ordered simplex:

$$\mathcal{T} \left[\left(\int_0^t dt' \hat{H}_S(t') \right)^n \right] = n! \int_0^t dt' \int_0^{t'} dt'' \cdots \int_0^{t^{(n-1)}} dt^{(n)} \hat{H}_S(t') \cdots \hat{H}_S(t^{(n)}). \quad (5.56)$$

Dividing by $n!$, the n -th order contribution in Eq. (5.51) becomes

$$\hat{U}_n(t) = \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \mathcal{T} \left[\left(\int_0^t dt' \hat{H}_S(t') \right)^n \right]. \quad (5.57)$$

Summing over all $n \geq 0$, the series inside the time-ordered product is recognized as the power series expansion of the exponential function:

$$\begin{aligned} \hat{U}(t) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \mathcal{T} \left[\left(\int_0^t dt' \hat{H}_S(t') \right)^n \right] \\ &= \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_0^t dt' \hat{H}_S(t') \right). \end{aligned} \quad (5.58)$$

Equation (5.58) is the celebrated *Dyson chronological exponential*.

5.4.3 Direct Verification

We can verify directly that Eq. (5.58) satisfies the evolution equation (5.47). Differentiating with respect to t :

$$i\hbar \frac{d\hat{U}(t)}{dt} = \mathcal{T} \left[\hat{H}_S(t) \exp \left(-\frac{i}{\hbar} \int_0^t dt' \hat{H}_S(t') \right) \right]. \quad (5.59)$$

Inside the time-ordering bracket, the operator order is automatically assigned by the time argument. Since the evaluation time t is the upper limit of integration, it is greater than or equal to any intermediate time $t' \in [0, t]$. Consequently, the time-ordering operator automatically places $\hat{H}_S(t)$ to the extreme left of all other factors:

$$i\hbar \frac{d\hat{U}(t)}{dt} = \hat{H}_S(t) \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_0^t dt' \hat{H}_S(t') \right) = \hat{H}_S(t) \hat{U}(t). \quad (5.60)$$

This confirms that the time-ordered exponential identically solves the Schrödinger evolution equation for any time-dependent Hamiltonian.

5.5 Problems

1. Harmonic oscillator ladder operators in the Heisenberg picture.

- Derive and solve the Heisenberg equations of motion for the annihilation and creation operators $\hat{a}(t)$ and $\hat{a}^\dagger(t)$ of a simple harmonic oscillator.
- Calculate the unequal-time commutator $[\hat{a}(t), \hat{a}^\dagger(t')]$.

2. Spatial translations and momentum as the generator of translations.

- Show that $[\hat{\mathbf{r}}, f(\hat{\mathbf{p}})] = i\hbar \nabla_{\mathbf{p}} f(\hat{\mathbf{p}})$ for any differentiable function $f(\hat{\mathbf{p}})$ of the momentum operator.

- (b) Using this result, prove the spatial translation identity:

$$e^{i\hat{\mathbf{p}}\cdot\boldsymbol{\lambda}/\hbar}\hat{\mathbf{r}}e^{-i\hat{\mathbf{p}}\cdot\boldsymbol{\lambda}/\hbar} = \hat{\mathbf{r}} + \boldsymbol{\lambda}\mathbb{I}, \quad (5.61)$$

where $\boldsymbol{\lambda}$ is a constant real displacement vector.

- (c) Show that the position-space wave function of the translated state

$$|\Phi\rangle \equiv e^{-i\hat{\mathbf{p}}\cdot\boldsymbol{\lambda}/\hbar}|\Psi\rangle \quad (5.62)$$

is given by $\langle\mathbf{r}|\Phi\rangle = \psi(\mathbf{r} - \boldsymbol{\lambda})$, where $\psi(\mathbf{r}) = \langle\mathbf{r}|\Psi\rangle$. Write out $\langle x|\Phi\rangle$ explicitly for the ground state of a one-dimensional harmonic oscillator.

- (d) Show that in the Schrödinger representation, the state $|\Phi(t)\rangle$ evolves according to

$$|\Phi(t)\rangle = e^{-i\hat{\mathbf{p}}(-t)\cdot\boldsymbol{\lambda}/\hbar}|\Psi(t)\rangle, \quad (5.63)$$

where $\hat{\mathbf{p}}(-t)$ is the Heisenberg momentum operator evaluated at time $-t$.

3. **Coherent-like displaced states of the harmonic oscillator.** Let $|\Psi\rangle$ be an energy eigenstate of a one-dimensional harmonic oscillator, and define the displaced state

$$|\Phi\rangle = e^{i\hat{p}\lambda/\hbar}|\Psi\rangle. \quad (5.64)$$

Show that in the Schrödinger representation, $|\Phi(t)\rangle$ is a wave packet whose expectation value $\langle\hat{x}\rangle_t$ oscillates sinusoidally between $-\lambda$ and $+\lambda$ with angular frequency ω , while maintaining a constant shape (no wave-packet dispersion). In particular, verify that if $|\Psi\rangle = |0\rangle$ is the ground state, the state remains a minimum-uncertainty wave packet with $\Delta x \cdot \Delta p = \hbar/2$ at all times. *Hint:* Apply Weyl's formula, Eq. (5.9).

4. **Correlation functions and the retarded Green's function.**

- (a) Calculate the two-time correlation function $\langle 0|\hat{x}(t)\hat{x}(t')|0\rangle$, where $|0\rangle$ is the ground state of the one-dimensional harmonic oscillator and $\hat{x}(t)$ is the Heisenberg position operator.
- (b) Suppose the oscillator is driven by a time-dependent classical force $F(t)$. Show that $\hat{x}(t)$ obeys the equation of motion

$$m\left(\frac{d^2}{dt^2} + \omega^2\right)\hat{x}(t) = F(t)\mathbb{I}. \quad (5.65)$$

- (c) Let $\theta(t - t')$ denote the Heaviside step function:

$$\theta(t - t') = \begin{cases} 1, & t \geq t', \\ 0, & t < t'. \end{cases} \quad (5.66)$$

Show that the retarded commutator expectation value

$$D(t - t') \equiv -\frac{i}{\hbar}\langle 0|[\hat{x}(t), \hat{x}(t')]|0\rangle\theta(t - t') \quad (5.67)$$

is the retarded Green's function for the differential equation in part (b), satisfying $D(t - t') = 0$ for $t < t'$.

- (d) Compute $D(t - t')$ explicitly and write the general solution for $\hat{x}(t)$ in terms of $F(t)$.

5. **Work done by a pulsed external force.** Consider a one-dimensional harmonic oscillator initially in an energy eigenstate $|n\rangle$ at $t < t_0$. A constant force F acts on the particle during the interval $t_0 \leq t \leq t_0 + \tau$, after which the force vanishes.

- (a) Determine the expectation value $\langle \hat{x} \rangle_t$ as a function of time for $t > t_0 + \tau$.
- (b) Show that the average energy transferred to the oscillator (the work done by the force) is

$$\Delta E = \frac{2F^2}{m\omega^2} \sin^2\left(\frac{\omega\tau}{2}\right). \quad (5.68)$$

Analyze this result in the impulsive limit $\omega\tau \ll 1$. Why does the transferred energy vanish identically whenever τ is an integer multiple of the oscillator period $T = 2\pi/\omega$? In classical mechanics, can the work done on an arbitrary initial trajectory be negative, and how does this contrast with the quantum expectation value starting from an energy eigenstate?

6. Free particle subject to a uniform time-dependent force.

- (a) Solve the Heisenberg equations of motion for the position and momentum operators of a free particle driven by a spatially uniform force $\mathbf{F}(t)$. Find the net momentum and kinetic energy transferred to the particle after time t .
- (b) If at $t = 0$ the particle's wave function has the form $\psi(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\varphi(\mathbf{r})$, where $\varphi(\mathbf{r})$ is real-valued, show that the spatial uncertainty $(\Delta r)_t \equiv \sqrt{\langle (\hat{\mathbf{r}} - \langle \hat{\mathbf{r}} \rangle)^2 \rangle_t}$ at a later time t is given by

$$(\Delta r)_t = \left[(\Delta r)_0^2 + \frac{2t^2}{m} \left(\langle E \rangle_0 - \frac{\hbar^2 k^2}{2m} \right) \right]^{1/2}, \quad (5.69)$$

independent of the applied force $\mathbf{F}(t)$, where $\langle E \rangle_0$ is the expectation value of the energy at $t = 0$.

- (c) Determine an explicit closed-form expression for the time-evolution operator $\hat{U}(t)$ in this system without employing time-ordered products. *Hint:* Consider an ansatz of the form $\hat{U}(t) = e^{i\alpha(t)\hat{\mathbf{r}}/\hbar}\bar{U}(t)$.

7. Gaussian expectation values and operator delta functions.

- (a) For the ground state $|0\rangle$ of a one-dimensional harmonic oscillator, prove that

$$\langle 0 | e^{ik\hat{x}} | 0 \rangle = \exp\left(-\frac{k^2}{2} \langle 0 | \hat{x}^2 | 0 \rangle\right). \quad (5.70)$$

- (b) Using this result, compute the expectation value $\langle 0 | \delta(\hat{x} - a) | 0 \rangle$, where $a \in \mathbb{R}$, and interpret the result physically in terms of the probability density.

- 8. Position representation of the inverse momentum operator.** Determine the integral kernel (matrix elements) $\langle \mathbf{r} | \frac{1}{\hat{p}} | \mathbf{r}' \rangle$ in the coordinate representation, where $\hat{p} = \sqrt{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}$. Given a state $|\Psi\rangle$ with wave function $\psi(\mathbf{r}) = \langle \mathbf{r} | \Psi \rangle$, express the wave function of $(1/\hat{p})|\Psi\rangle$ in terms of an integral over $\psi(\mathbf{r}')$.

- 9. Heisenberg dynamics in a two-well system.** In the symmetric double-well box problem (Chapter 4, Problem 16), let $\hat{Q}$ be the projection operator whose expectation value represents the probability of finding the particle in the left half of the box. Write down the effective two-state Hamiltonian and formulate the Heisenberg equation of motion for $\hat{Q}(t)$. Integrate this equation to rederive the oscillatory tunneling probability found previously in the Schrödinger picture.

Chapter 6

Orbital Angular Momentum And Central Potentials

6.1 Orbital Angular Momentum and Central Potentials

In three-dimensional quantum mechanics, rotational symmetry and the conservation of angular momentum play a central role in the classification of states and the solution of quantum mechanical problems.

6.1.1 Definition and Hermiticity of Orbital Angular Momentum

The orbital angular momentum of a particle is defined in direct correspondence with classical mechanics: if $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are the quantum mechanical position and momentum operators, the orbital angular momentum operator $\hat{\mathbf{L}}$ about the origin is given by the vector cross product:

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}. \quad (6.1)$$

In Cartesian components, this definition reads

$$\begin{aligned} \hat{L}_x &= \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \\ \hat{L}_y &= \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \\ \hat{L}_z &= \hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \end{aligned} \quad (6.2)$$

or in compact index notation using the Levi-Civita alternating symbol ϵ_{ijk} :

$$\hat{L}_i = \sum_{j,k=1}^3 \epsilon_{ijk} \hat{r}_j \hat{p}_k. \quad (6.3)$$

Because orthogonal components of position and momentum commute (e.g., $[\hat{y}, \hat{p}_z] = 0$), there is no operator ordering ambiguity in products of the form $\hat{r}_j \hat{p}_k$ for $j \neq k$. Furthermore, since $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are Hermitian operators, the components of $\hat{\mathbf{L}}$ are manifestly Hermitian. For example, for the z -component:

$$\begin{aligned} \hat{L}_z^\dagger &= (\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)^\dagger = \hat{p}_y^\dagger \hat{x}^\dagger - \hat{p}_x^\dagger \hat{y}^\dagger \\ &= \hat{p}_y \hat{x} - \hat{p}_x \hat{y} = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = \hat{L}_z. \end{aligned} \quad (6.4)$$

Identical calculations for the x - and y -components confirm that

$$\hat{L}_x^\dagger = \hat{L}_x, \quad \hat{L}_y^\dagger = \hat{L}_y, \quad \hat{L}_z^\dagger = \hat{L}_z, \quad (6.5)$$

so that $\hat{\mathbf{L}}^\dagger = \hat{\mathbf{L}}$ is a valid physical vector observable.

6.2 Commutation Relations

To understand the algebraic structure of angular momentum in quantum mechanics, we systematically determine the commutation relations among the components of $\hat{\mathbf{L}}$, and their commutators with the position and momentum operators.

6.2.1 Commutators with Position and Momentum

Evaluating the commutator of $\hat{L}_x$ with the Cartesian coordinates gives

$$\begin{aligned} [\hat{L}_x, \hat{x}] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{x}] = 0, \\ [\hat{L}_x, \hat{y}] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{y}] = -\hat{z}[\hat{p}_y, \hat{y}] = i\hbar\hat{z}, \\ [\hat{L}_x, \hat{z}] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}] = \hat{y}[\hat{p}_z, \hat{z}] = -i\hbar\hat{y}. \end{aligned} \quad (6.6)$$

By cyclic permutation of (x, y, z) , the commutators of $\hat{L}_y$ and $\hat{L}_z$ are

$$[\hat{L}_y, \hat{x}] = -i\hbar\hat{z}, \quad [\hat{L}_y, \hat{y}] = 0, \quad [\hat{L}_y, \hat{z}] = i\hbar\hat{x}, \quad (6.7)$$

$$[\hat{L}_z, \hat{x}] = i\hbar\hat{y}, \quad [\hat{L}_z, \hat{y}] = -i\hbar\hat{x}, \quad [\hat{L}_z, \hat{z}] = 0. \quad (6.8)$$

Denoting the Cartesian axes by indices 1, 2, 3 (corresponding to x, y, z), these relations are unified using the completely antisymmetric Levi-Civita tensor ϵ_{ijk} :

$$[\hat{L}_i, \hat{r}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{r}_k. \quad (6.9)$$

Multiplying Eq. (6.9) by the components of an arbitrary numerical unit vector $\hat{\mathbf{n}}$ yields the coordinate-free vector identity:

$$[\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{r}}] = i\hbar(\hat{\mathbf{r}} \times \hat{\mathbf{n}}). \quad (6.10)$$

Equation (6.10) demonstrates that the commutator of the angular momentum along any direction $\hat{\mathbf{n}}$ with the position vector $\hat{\mathbf{r}}$ produces an infinitesimal displacement perpendicular to both $\hat{\mathbf{r}}$ and $\hat{\mathbf{n}}$.

An identical calculation for the momentum operator $\hat{\mathbf{p}}$ gives

$$[\hat{L}_i, \hat{p}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{p}_k, \quad (6.11)$$

or equivalently in vector form,

$$[\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{p}}] = i\hbar(\hat{\mathbf{p}} \times \hat{\mathbf{n}}). \quad (6.12)$$

6.2.2 Angular Momentum Commutation Relations

Using the operator product rule $[\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C}$, we evaluate the commutator of $\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}$ with $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$:

$$\begin{aligned} [\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{L}}] &= \hat{\mathbf{r}} \times [\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{p}}] + [\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{r}}] \times \hat{\mathbf{p}} \\ &= i\hbar[\hat{\mathbf{r}} \times (\hat{\mathbf{p}} \times \hat{\mathbf{n}}) + (\hat{\mathbf{r}} \times \hat{\mathbf{n}}) \times \hat{\mathbf{p}}]. \end{aligned} \quad (6.13)$$

Applying the Jacobi vector identity $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{0}$, we obtain

$$[\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{L}}] = i\hbar(\hat{\mathbf{r}} \times \hat{\mathbf{p}}) \times \hat{\mathbf{n}} = i\hbar\hat{\mathbf{L}} \times \hat{\mathbf{n}}. \quad (6.14)$$

In index notation, this fundamental Lie algebra reads

$$[\hat{L}_i, \hat{L}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{L}_k, \quad (6.15)$$

which is compactly written in operator vector notation as

$$\hat{\mathbf{L}} \times \hat{\mathbf{L}} = i\hbar \hat{\mathbf{L}}. \quad (6.16)$$

Because the individual components do not commute ($[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z \neq 0$), the three components of angular momentum cannot be simultaneously diagonalized.

6.3 Rotations and Angular Momentum

The angular momentum operator is inextricably connected to the geometric symmetry of spatial rotations. Just as linear momentum generates spatial translations, angular momentum acts as the generator of rotations in quantum mechanics.

6.3.1 Infinitesimal and Finite Operator Transformations

Consider a coordinate rotation specified by an infinitesimal rotation vector $\boldsymbol{\alpha}$, whose direction $\hat{\mathbf{a}} = \boldsymbol{\alpha}/|\boldsymbol{\alpha}|$ defines the rotation axis and whose magnitude $|\boldsymbol{\alpha}|$ denotes the right-handed angle of rotation. A classical position vector $\mathbf{r}_0$ is mapped under this rotation to

$$\mathbf{r}'_0 = \mathbf{r}_0 + \boldsymbol{\alpha} \times \mathbf{r}_0. \quad (6.17)$$

In quantum mechanics, consider the transformed position operator

$$\hat{\mathbf{r}}' = \hat{\mathbf{r}} + \boldsymbol{\alpha} \times \hat{\mathbf{r}}. \quad (6.18)$$

If $|\mathbf{r}_0\rangle$ is an eigenstate of the position operator with eigenvalue $\mathbf{r}_0$, then

$$\hat{\mathbf{r}}'|\mathbf{r}_0\rangle = (\mathbf{r}_0 + \boldsymbol{\alpha} \times \mathbf{r}_0)|\mathbf{r}_0\rangle, \quad (6.19)$$

demonstrating that the eigenvalues of $\hat{\mathbf{r}}'$ are the rotated position vectors.

Using the commutation relation $[\hat{\boldsymbol{\alpha}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{r}}] = i\hbar(\hat{\mathbf{r}} \times \hat{\boldsymbol{\alpha}}) = -i\hbar(\boldsymbol{\alpha} \times \hat{\mathbf{r}})$, we can express the infinitesimal rotation as

$$\hat{\mathbf{r}}' = \hat{\mathbf{r}} + \frac{i}{\hbar}[\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}, \hat{\mathbf{r}}]. \quad (6.20)$$

To first order in $\boldsymbol{\alpha}$, Eq. (6.20) is equivalent to the unitary similarity transformation:

$$\hat{\mathbf{r}}' = \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) \hat{\mathbf{r}} \exp\left(-\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right). \quad (6.21)$$

Remarkably, Eq. (6.21) holds for finite rotations through arbitrary angles $\theta = |\boldsymbol{\alpha}|$. Let $\hat{\mathbf{r}}'(\theta)$ denote the position operator rotated by an angle θ about the axis $\hat{\mathbf{a}}$. An incremental rotation $\delta\theta$ satisfies the differential equation

$$\frac{d\hat{\mathbf{r}}'(\theta)}{d\theta} = \hat{\mathbf{a}} \times \hat{\mathbf{r}}'(\theta), \quad \text{with initial condition } \hat{\mathbf{r}}'(0) = \hat{\mathbf{r}}. \quad (6.22)$$

Differentiating the right-hand side of Eq. (6.21) with respect to θ yields

$$\begin{aligned} \frac{d\hat{\mathbf{r}}'}{d\theta} &= \exp\left(\frac{i}{\hbar}\theta \hat{\mathbf{a}} \cdot \hat{\mathbf{L}}\right) \left(\frac{i}{\hbar}[\hat{\mathbf{a}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{r}}]\right) \exp\left(-\frac{i}{\hbar}\theta \hat{\mathbf{a}} \cdot \hat{\mathbf{L}}\right) \\ &= \exp\left(\frac{i}{\hbar}\theta \hat{\mathbf{a}} \cdot \hat{\mathbf{L}}\right) (\hat{\mathbf{a}} \times \hat{\mathbf{r}}) \exp\left(-\frac{i}{\hbar}\theta \hat{\mathbf{a}} \cdot \hat{\mathbf{L}}\right) = \hat{\mathbf{a}} \times \hat{\mathbf{r}}'(\theta). \end{aligned} \quad (6.23)$$

Because Eq. (6.21) satisfies both the differential equation (6.22) and the initial condition, it provides the exact finite rotation of the position operator. Furthermore, because this derivation relied solely on the vector commutation relations $[\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{V}}] = i\hbar(\hat{\mathbf{V}} \times \hat{\mathbf{n}})$, the same transformation applies to any vector operator $\hat{\mathbf{V}}$ (such as linear momentum $\hat{\mathbf{p}}$):

$$\hat{\mathbf{V}}' = \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) \hat{\mathbf{V}} \exp\left(-\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right). \quad (6.24)$$

6.3.2 Transformation of State Vectors and Wave Functions

To determine how the unitary rotation operator $\hat{\mathcal{R}}(\boldsymbol{\alpha}) \equiv \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right)$ affects physical states, consider the position eigenstate bra $\langle \mathbf{r}_0 |$:

$$\langle \mathbf{r}_0 | \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) \hat{\mathbf{r}} \exp\left(-\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) = \langle \mathbf{r}_0 | \hat{\mathbf{r}}' = \mathbf{r}'_0 \langle \mathbf{r}_0 |. \quad (6.25)$$

Multiplying from the right by $\exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right)$ gives

$$\left[\langle \mathbf{r}_0 | \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) \right] \hat{\mathbf{r}} = \mathbf{r}'_0 \left[\langle \mathbf{r}_0 | \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) \right], \quad (6.26)$$

which proves that $\langle \mathbf{r}_0 | \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right)$ is a position eigenstate with eigenvalue $\mathbf{r}'_0$:

$$\langle \mathbf{r}_0 | \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) = \langle \mathbf{r}'_0 |. \quad (6.27)$$

For an infinitesimal rotation, expanding the exponential gives

$$\langle \mathbf{r}_0 | \left(\mathbb{I} + \frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}} \right) = \langle \mathbf{r}_0 + \boldsymbol{\alpha} \times \mathbf{r}_0 |, \quad (6.28)$$

identifying $\hat{\mathbf{L}}$ as the generator of infinitesimal rotations.

Under an active rotation of a physical state, $|\Psi'\rangle = \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) |\Psi\rangle$, the transformed wave function $\psi'(\mathbf{r}_0) = \langle \mathbf{r}_0 | \Psi' \rangle$ is

$$\psi'(\mathbf{r}_0) = \langle \mathbf{r}_0 | \exp\left(\frac{i}{\hbar}\boldsymbol{\alpha} \cdot \hat{\mathbf{L}}\right) |\Psi\rangle = \langle \mathbf{r}'_0 | \Psi \rangle = \psi(\mathbf{r}'_0). \quad (6.29)$$

The wave function evaluated at $\mathbf{r}_0$ after the rotation equals the original wave function evaluated at the rotated coordinate $\mathbf{r}'_0$.

6.3.3 Comparison with Spatial Translations

This structure is completely analogous to spatial translations generated by linear momentum $\hat{\mathbf{p}}$. For any numerical displacement vector $\boldsymbol{\lambda}$,

$$\exp\left(\frac{i}{\hbar}\boldsymbol{\lambda} \cdot \hat{\mathbf{p}}\right) \hat{\mathbf{r}} \exp\left(-\frac{i}{\hbar}\boldsymbol{\lambda} \cdot \hat{\mathbf{p}}\right) = \hat{\mathbf{r}} + \boldsymbol{\lambda} \mathbb{I}, \quad (6.30)$$

and correspondingly,

$$\langle \mathbf{r} | \exp\left(\frac{i}{\hbar}\boldsymbol{\lambda} \cdot \hat{\mathbf{p}}\right) = \langle \mathbf{r} + \boldsymbol{\lambda} |. \quad (6.31)$$

Thus, while linear momentum $\hat{\mathbf{p}}$ generates translations along coordinate axes, angular momentum $\hat{\mathbf{L}}$ generates rotations about those axes.

6.4 Differential Operator Representation

Using the geometric action of infinitesimal rotations on position eigenstates, Eq. (6.28), we can construct explicit differential operator representations for the orbital angular momentum components, in direct analogy with the momentum representation $\hat{\mathbf{p}} \doteq -i\hbar\nabla$.

6.4.1 Spherical Coordinates and the z -Component

In standard spherical polar coordinates (r, θ, φ) , defined by

$$\begin{aligned} x &= r \sin \theta \cos \varphi, \\ y &= r \sin \theta \sin \varphi, \\ z &= r \cos \theta, \end{aligned} \tag{6.32}$$

a position eigenstate bra is denoted by $\langle \mathbf{r} | \equiv \langle r, \theta, \varphi |$. Because rotations preserve the radial distance r , we consider rotations acting purely on the angular coordinates (θ, φ) .

An infinitesimal rotation through angle α about the z -axis shifts the azimuthal angle:

$$\theta \rightarrow \theta, \quad \varphi \rightarrow \varphi + \alpha. \tag{6.33}$$

Applying Eq. (6.28) to $\langle r, \theta, \varphi |$ yields

$$\begin{aligned} \langle r, \theta, \varphi | \left(\mathbb{I} + \frac{i\alpha}{\hbar} \hat{L}_z \right) &= \langle r, \theta, \varphi + \alpha | \\ &= \langle r, \theta, \varphi | + \alpha \frac{\partial}{\partial \varphi} \langle r, \theta, \varphi | + \mathcal{O}(\alpha^2). \end{aligned} \tag{6.34}$$

Comparing first-order terms in α , we find

$$\langle r, \theta, \varphi | \hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \varphi} \langle r, \theta, \varphi | = -i\hbar \frac{\partial}{\partial \varphi} \langle r, \theta, \varphi |. \tag{6.35}$$

Acting to the right on a wave function $\psi(r, \theta, \varphi) = \langle r, \theta, \varphi | \Psi \rangle$, the operator $\hat{L}_z$ has the familiar differential representation:

$$\hat{L}_z \doteq -i\hbar \frac{\partial}{\partial \varphi}. \tag{6.36}$$

6.4.2 Transverse Components and Ladder Operators

Under an infinitesimal rotation by α about the x -axis, the Cartesian coordinates transform as

$$x \rightarrow x, \quad y \rightarrow y - \alpha z, \quad z \rightarrow z + \alpha y. \tag{6.37}$$

From Eq. (6.32), the variation of z gives

$$\delta z = \alpha y = \alpha r \sin \theta \sin \varphi = \delta(r \cos \theta) = -r \sin \theta \delta \theta, \tag{6.38}$$

which implies

$$\delta \theta = -\alpha \sin \varphi \implies \theta \rightarrow \theta - \alpha \sin \varphi. \tag{6.39}$$

Since x remains invariant, $\delta(\sin \theta \cos \varphi) = 0$, leading to

$$\cos \theta \cos \varphi \delta \theta - \sin \theta \sin \varphi \delta \varphi = 0, \tag{6.40}$$

whence

$$\delta \varphi = \cot \theta \cot \varphi \delta \theta = -\alpha \cot \theta \cos \varphi. \tag{6.41}$$

Therefore,

$$\langle r, \theta, \varphi | \left(\mathbb{I} + \frac{i\alpha}{\hbar} \hat{L}_x \right) = \langle r, \theta - \alpha \sin \varphi, \varphi - \alpha \cot \theta \cos \varphi |, \quad (6.42)$$

which yields the differential representation of $\hat{L}_x$:

$$\hat{L}_x \doteq i\hbar \left(\sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi} \right). \quad (6.43)$$

Repeating the analysis for an infinitesimal rotation about the y -axis yields

$$\hat{L}_y \doteq i\hbar \left(-\cos \varphi \frac{\partial}{\partial \theta} + \cot \theta \sin \varphi \frac{\partial}{\partial \varphi} \right). \quad (6.44)$$

It is extraordinarily convenient to define the angular momentum raising and lowering (ladder) operators:

$$\hat{L}_{\pm} \equiv \hat{L}_x \pm i\hat{L}_y. \quad (6.45)$$

Combining Eqs. (6.43) and (6.44), their differential representations are

$$\hat{L}_{\pm} \doteq \hbar e^{\pm i\varphi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right). \quad (6.46)$$

6.4.3 Total Angular Momentum and the Laplacian

The Casimir operator corresponding to the total squared angular momentum is

$$\hat{\mathbf{L}}^2 \equiv \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \frac{1}{2}(\hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+) + \hat{L}_z^2. \quad (6.47)$$

A straightforward computation using Eq. (6.46) yields

$$\hat{\mathbf{L}}^2 \doteq -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right]. \quad (6.48)$$

Remarkably, the differential operator appearing in brackets in Eq. (6.48) is precisely the angular portion of the three-dimensional Laplacian. The full Laplacian in spherical coordinates can therefore be written in the elegantly partitioned form:

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{\mathbf{L}}^2}{\hbar^2 r^2} = \frac{1}{r} \frac{\partial^2}{\partial r^2} r - \frac{\hat{\mathbf{L}}^2}{\hbar^2 r^2}. \quad (6.49)$$

This decomposition directly separates the radial kinetics from the rotational centrifugal barrier.

6.5 Eigenfunctions and Eigenvalues of Angular Momentum

We now systematically determine the eigenvalue spectrum and eigenstates of angular momentum operators using purely algebraic methods. The derivation relies solely on the Hermiticity of $\hat{\mathbf{L}}$ and the commutation relations $[\hat{L}_i, \hat{L}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{L}_k$. Consequently, all conclusions derived here apply equally to orbital angular momentum, intrinsic spin, and total angular momentum.

6.5.1 Commuting Observables and the Choice of Basis

We first observe that the total angular momentum squared commutes with every Cartesian component:

$$[\hat{\mathbf{L}}^2, \hat{\mathbf{L}}] = 0. \quad (6.50)$$

Because $\hat{\mathbf{L}}^2$ commutes with $\hat{\mathbf{L}}$, one can construct a complete orthonormal basis of simultaneous eigenstates of $\hat{\mathbf{L}}^2$ and *one* selected component of $\hat{\mathbf{L}}$. By convention, this quantization axis is chosen along the z -direction ($\hat{L}_z$).

Simultaneous eigenstates of two or three distinct components cannot exist because non-parallel components fail to commute. For instance, if $|l_x, l_z\rangle$ were a simultaneous eigenstate of $\hat{L}_x$ and $\hat{L}_z$ with respective eigenvalues l_x and l_z , then

$$-i\hbar\hat{L}_y|l_x, l_z\rangle = [\hat{L}_x, \hat{L}_z]|l_x, l_z\rangle = (l_x l_z - l_z l_x)|l_x, l_z\rangle = 0. \quad (6.51)$$

If a complete basis of such states existed, $\hat{L}_y$ would annihilate every state in the Hilbert space and thus vanish identically, which is false. Hence, a complete commuting set of observables consists of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$.

Because $\hat{\mathbf{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$ is the sum of squares of Hermitian operators, its expectation value in any normalized state $|\Psi\rangle$ is positive semi-definite:

$$\langle\Psi|\hat{\mathbf{L}}^2|\Psi\rangle = \sum_{i=x,y,z} \|\hat{L}_i|\Psi\rangle\|^2 \geq 0. \quad (6.52)$$

Accordingly, all eigenvalues of $\hat{\mathbf{L}}^2$ are non-negative. We parametrize these eigenvalues without loss of generality as $\hbar^2 l(l+1)$ with $l \geq 0$. The eigenvalues of $\hat{L}_z$ are denoted by $\hbar m$, and the joint eigenstates are designated by $|l, m\rangle$:

$$\begin{aligned} \hat{\mathbf{L}}^2|l, m\rangle &= \hbar^2 l(l+1)|l, m\rangle, \\ \hat{L}_z|l, m\rangle &= \hbar m|l, m\rangle. \end{aligned} \quad (6.53)$$

Here, m is termed the *magnetic* or *azimuthal quantum number*.

6.5.2 Ladder Operators and the Spectrum

The spectral properties are uncovered using the ladder operators $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$. Their algebraic properties include:

$$[\hat{L}_z, \hat{L}_\pm] = \pm\hbar\hat{L}_\pm, \quad (6.54)$$

$$\hat{L}_+\hat{L}_- = \hat{\mathbf{L}}^2 - \hat{L}_z^2 + \hbar\hat{L}_z, \quad (6.55)$$

$$\hat{L}_-\hat{L}_+ = \hat{\mathbf{L}}^2 - \hat{L}_z^2 - \hbar\hat{L}_z, \quad (6.56)$$

$$[\hat{L}_+, \hat{L}_-] = 2\hbar\hat{L}_z. \quad (6.57)$$

Since $(\hat{L}_\pm)^\dagger = \hat{L}_\mp$, the expectation values of $\hat{L}_+\hat{L}_-$ and $\hat{L}_-\hat{L}_+$ are strictly non-negative:

$$\langle l, m|\hat{L}_+\hat{L}_-|l, m\rangle = \hbar^2 [l(l+1) - m(m-1)] \geq 0, \quad (6.58)$$

$$\langle l, m|\hat{L}_-\hat{L}_+|l, m\rangle = \hbar^2 [l(l+1) - m(m+1)] \geq 0. \quad (6.59)$$

These inequalities can be rewritten as

$$\left(l + \frac{1}{2}\right)^2 \geq \left(m \mp \frac{1}{2}\right)^2, \quad (6.60)$$

which immediately bounds m :

$$-l \leq m \leq l. \quad (6.61)$$

From the commutation relation $[\hat{L}_z, \hat{L}_\pm] = \pm \hbar \hat{L}_\pm$, we see that

$$\hat{L}_z (\hat{L}_\pm |l, m\rangle) = \hbar(m \pm 1) (\hat{L}_\pm |l, m\rangle). \quad (6.62)$$

Thus, $\hat{L}_+$ acts as a raising operator and $\hat{L}_-$ acts as a lowering operator for the azimuthal quantum number m . Adopting the standard Condon–Shortley phase convention (where the matrix elements are chosen positive real), we have

$$\hat{L}_+ |l, m\rangle = \hbar \sqrt{l(l+1) - m(m+1)} |l, m+1\rangle, \quad (6.63)$$

$$\hat{L}_- |l, m\rangle = \hbar \sqrt{l(l+1) - m(m-1)} |l, m-1\rangle. \quad (6.64)$$

Because Eq. (6.61) imposes strict bounds on m , the ascending sequence must terminate at a maximum value $m_{\max} = l$, where $\hat{L}_+ |l, l\rangle = 0$. Similarly, the descending sequence must terminate at a minimum value $m_{\min} = -l$, where $\hat{L}_- |l, -l\rangle = 0$. Because states are stepped in integer increments from $-l$ to $+l$, the span $l - (-l) = 2l$ must be a non-negative integer. Therefore:

$$l \in \left\{ 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots \right\},$$

$$m \in \{-l, -l+1, \dots, l-1, l\}. \quad (6.65)$$

For each value of l , there are precisely $2l+1$ distinct orthogonal states $|l, m\rangle$, satisfying

$$\langle l, m | l', m' \rangle = \delta_{ll'} \delta_{mm'}. \quad (6.66)$$

6.5.3 Physical Picture and Single-Valuedness for Orbital States

In a state $|l, m\rangle$, the total magnitude of angular momentum is $\hbar \sqrt{l(l+1)}$, while its projection on the quantization axis is sharply defined as $\hbar m$. The transverse component has magnitude $\sqrt{\langle \hat{L}_x^2 + \hat{L}_y^2 \rangle} = \hbar \sqrt{l(l+1) - m^2} > 0$. Because $\hat{L}_x$ and $\hat{L}_y$ do not commute, the transverse vector precesses with complete azimuthal uncertainty.

While the general Lie algebra of rotations allows both integer and half-integer values of l (as realized by intrinsic spin), *orbital* angular momentum operators $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ acting on spatial wave functions require l to be an integer. Under a rotation of 2π about the z -axis, a physical spatial state rotates back onto itself:

$$\langle r, \theta, \varphi + 2\pi | l, m \rangle = \langle r, \theta, \varphi | e^{i2\pi \hat{L}_z / \hbar} | l, m \rangle = e^{i2\pi m} \langle r, \theta, \varphi | l, m \rangle. \quad (6.67)$$

For the single-particle spatial wave function $\psi(\mathbf{r})$ to remain single-valued in coordinate space, we must have $e^{i2\pi m} = 1$, which strictly requires

$$m \in \mathbb{Z} \quad \implies \quad l \in \{0, 1, 2, 3, \dots\}. \quad (6.68)$$

Double-valued spatial wave functions are excluded because a sign change upon 360° rotation would necessitate singular branch lines in Euclidean three-space, which would violate fundamental rotational invariance.

6.6 Spherical Harmonics

Having established the algebraic structure of angular momentum, we now construct the coordinate-space wave functions corresponding to the eigenstates $|l, m\rangle$.

6.6.1 Construction via Ladder Operators

In spherical polar coordinates, the eigenvalue equation $\hat{L}_z|l, m\rangle = \hbar m|l, m\rangle$ takes the coordinate representation:

$$\langle r, \theta, \varphi | \hat{L}_z | l, m \rangle = -i\hbar \frac{\partial}{\partial \varphi} \langle r, \theta, \varphi | l, m \rangle = \hbar m \langle r, \theta, \varphi | l, m \rangle. \quad (6.69)$$

Integrating Eq. (6.69) isolates the azimuthal dependence:

$$\langle r, \theta, \varphi | l, m \rangle = e^{im\varphi} \langle r, \theta, 0 | l, m \rangle. \quad (6.70)$$

To determine the polar angle (θ) dependence, rather than solving the second-order Legendre differential equation directly, we exploit the ladder property: the state of maximal projection $|l, l\rangle$ must be annihilated by the raising operator $\hat{L}_+$:

$$\langle r, \theta, \varphi | \hat{L}_+ | l, l \rangle = \hbar e^{i\varphi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right) \langle r, \theta, \varphi | l, l \rangle = 0. \quad (6.71)$$

Substituting $\partial/\partial\varphi \rightarrow il$, we obtain the first-order ordinary differential equation:

$$\left(\frac{\partial}{\partial \theta} - l \cot \theta \right) \langle r, \theta, \varphi | l, l \rangle = 0. \quad (6.72)$$

Integrating Eq. (6.72) yields

$$\langle r, \theta, \varphi | l, l \rangle = R(r) e^{il\varphi} (\sin \theta)^l, \quad (6.73)$$

where $R(r)$ is an arbitrary radial function. Because the ladder operators $\hat{L}_\pm$ act strictly on the angular coordinates, application of $\hat{L}_-$ leaves the radial function $R(r)$ untouched. We therefore factorize the spatial wave function into separate radial and angular components:

$$\langle r, \theta, \varphi | l, m \rangle = R_{lm}(r) Y_{lm}(\theta, \varphi), \quad (6.74)$$

where the angular functions $Y_{lm}(\theta, \varphi) \equiv \langle \theta, \varphi | l, m \rangle$ are the *spherical harmonics*.

6.6.2 Normalization and General Formula

Assuming standard independent normalization for the radial factor $\int_0^\infty r^2 |R(r)|^2 dr = 1$, the spherical harmonics satisfy the orthonormality condition over the unit sphere:

$$\int d\Omega Y_{lm}^*(\theta, \varphi) Y_{l'm'}(\theta, \varphi) \equiv \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta Y_{lm}^*(\theta, \varphi) Y_{l'm'}(\theta, \varphi) = \delta_{ll'} \delta_{mm'}. \quad (6.75)$$

For $m = l$, writing $Y_{ll}(\theta, \varphi) = c_l e^{il\varphi} (\sin \theta)^l$, the normalization constant c_l is determined by

$$2\pi |c_l|^2 \int_0^\pi (\sin \theta)^{2l+1} d\theta = 2\pi |c_l|^2 \int_{-1}^1 (1 - u^2)^l du = 1. \quad (6.76)$$

Adopting the standard Condon–Shortley phase $(-1)^l$, this integration gives

$$c_l = \frac{(-1)^l}{2^l l!} \sqrt{\frac{2l+1}{4\pi}} (2l)!. \quad (6.77)$$

Repeatedly applying the lowering operator $\hat{L}_-$ via Eq. (6.64), the general normalized spherical harmonic for any $m \in \{-l, \dots, l\}$ is given by Rodrigues' formula:

$$Y_{lm}(\theta, \varphi) = \frac{(-1)^l}{2^l l!} \sqrt{\frac{2l+1}{4\pi} \frac{(l+m)!}{(l-m)!}} \frac{e^{im\varphi}}{(\sin \theta)^m} \left(\frac{d}{d \cos \theta} \right)^{l-m} (\sin \theta)^{2l}. \quad (6.78)$$

Equivalently, in terms of the associated Legendre polynomials $P_l^m(\cos \theta)$:

$$Y_{lm}(\theta, \varphi) = (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\varphi} \quad (m \geq 0). \quad (6.79)$$

6.6.3 Low-Order Harmonics and Symmetry Properties

For the lowest orbital angular momentum states ($l = 0, 1, 2$), the spherical harmonics are:

$$\begin{aligned} Y_{00} &= \frac{1}{\sqrt{4\pi}}, \\ Y_{10} &= \sqrt{\frac{3}{4\pi}} \cos \theta, & Y_{1,\pm 1} &= \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\varphi}, \\ Y_{20} &= \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1), & Y_{2,\pm 1} &= \mp \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\varphi}, \\ Y_{2,\pm 2} &= \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\varphi}. \end{aligned} \quad (6.80)$$

The spherical harmonics exhibit the complex conjugation relation:

$$Y_{l,-m}(\theta, \varphi) = (-1)^m Y_{lm}^*(\theta, \varphi). \quad (6.81)$$

Under the spatial inversion (parity) operator $\hat{\mathcal{P}} : \mathbf{r} \rightarrow -\mathbf{r}$, the spherical coordinates transform as

$$r \rightarrow r, \quad \theta \rightarrow \pi - \theta, \quad \varphi \rightarrow \varphi + \pi. \quad (6.82)$$

Under this reflection:

$$e^{im\varphi} \rightarrow e^{im(\varphi+\pi)} = (-1)^m e^{im\varphi}, \quad \cos \theta \rightarrow -\cos \theta, \quad \sin \theta \rightarrow \sin \theta. \quad (6.83)$$

Evaluating Eq. (6.78) under these transformations reveals that the spherical harmonics are parity eigenstates with eigenvalue $(-1)^l$:

$$\hat{\mathcal{P}} Y_{lm}(\theta, \varphi) = Y_{lm}(\pi - \theta, \varphi + \pi) = (-1)^l Y_{lm}(\theta, \varphi). \quad (6.84)$$

Even- l states possess even (positive) intrinsic parity, while odd- l states possess odd (negative) parity.

6.6.4 Completeness on the Sphere

Because the angular momentum eigenstates $|l, m\rangle$ span the angular Hilbert space $\mathcal{L}^2(S^2)$, the spherical harmonics form a complete orthonormal basis:

$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}(\theta, \varphi) Y_{lm}^*(\theta', \varphi') = \frac{\delta(\theta - \theta') \delta(\varphi - \varphi')}{\sin \theta}. \quad (6.85)$$

Any square-integrable angular function $f(\theta, \varphi)$ can be uniquely expanded in spherical harmonics:

$$f(\theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_{lm} Y_{lm}(\theta, \varphi), \quad (6.86)$$

with expansion coefficients determined by projection:

$$f_{lm} = \int d\Omega Y_{lm}^*(\theta, \varphi) f(\theta, \varphi). \quad (6.87)$$

6.7 Central Forces and the Radial Equation

We now apply the theory of orbital angular momentum to a particle moving in a spherically symmetric central potential $V(r)$, where $r = |\mathbf{r}|$.

6.7.1 Rotational Invariance and Multiplet Degeneracy

The Hamiltonian for a particle of mass m in a central potential is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(r). \quad (6.88)$$

Because both $\hat{\mathbf{p}}^2$ and $V(r)$ are scalar operators invariant under rotations about the origin, $\hat{H}$ commutes with all three components of orbital angular momentum:

$$[\hat{H}, \hat{\mathbf{L}}] = 0 \quad \implies \quad [\hat{H}, \hat{\mathbf{L}}^2] = 0. \quad (6.89)$$

Therefore, $\hat{H}$, $\hat{\mathbf{L}}^2$, and $\hat{L}_z$ form a complete set of mutually commuting observables, admitting a joint eigenbasis denoted by $|E, l, m\rangle$.

Since $[\hat{H}, \hat{L}_\pm] = 0$, the ladder operators preserve energy:

$$\hat{H} \left(\hat{L}_\pm |E, l, m\rangle \right) = \hat{L}_\pm \hat{H} |E, l, m\rangle = E \left(\hat{L}_\pm |E, l, m\rangle \right). \quad (6.90)$$

Hence, states $|E, l, m\rangle$ for all $2l+1$ values of $m \in \{-l, \dots, l\}$ share the identical energy eigenvalue E . Rotational symmetry thus dictates that every energy level with orbital angular momentum l exhibits at least a $(2l+1)$ -fold *essential degeneracy*, corresponding to the spatial isotropy of the potential.

6.7.2 Separation of Variables and the Radial Equation

Using the spherical Laplacian decomposition, Eq. (6.49), the stationary Schrödinger equation becomes

$$\left[-\frac{\hbar^2}{2m} \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{\hat{\mathbf{L}}^2}{2mr^2} + V(r) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}). \quad (6.91)$$

The first term can be understood as $\hat{p}_r^2/2m$, where the Hermitian radial momentum operator is

$$\hat{p}_r = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right). \quad (6.92)$$

Factoring the wave function into radial and angular parts,

$$\psi(\mathbf{r}) = R_l(r) Y_{lm}(\theta, \varphi), \quad (6.93)$$

and substituting into Eq. (6.91), the angular operator acts as $\hat{\mathbf{L}}^2 Y_{lm} = \hbar^2 l(l+1) Y_{lm}$, yielding the radial differential equation:

$$\left[-\frac{\hbar^2}{2m} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{\hbar^2 l(l+1)}{2mr^2} + V(r) \right] R_l(r) = E R_l(r). \quad (6.94)$$

Defining the reduced radial wave function $u_l(r) \equiv r R_l(r)$, Eq. (6.94) simplifies to an effective one-dimensional Schrödinger equation on the half-line $r \in [0, \infty)$:

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + V_{\text{eff}}(r) \right] u_l(r) = E u_l(r), \quad (6.95)$$

governed by the effective potential

$$V_{\text{eff}}(r) \equiv V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}. \quad (6.96)$$

The repulsive term $\hbar^2 l(l+1)/(2mr^2)$ represents the *centrifugal barrier*, which penalizes states of non-zero angular momentum near the origin.

6.7.3 Boundary Conditions at the Origin and Asymptotics

Assuming that $r^2 V(r) \rightarrow 0$ as $r \rightarrow 0$, the centrifugal term dominates as $r \rightarrow 0$, giving

$$\left[\frac{d^2}{dr^2} - \frac{l(l+1)}{r^2} \right] u_l(r) \approx 0. \quad (6.97)$$

The linearly independent solutions are $u_l(r) \sim r^{l+1}$ and $u_l(r) \sim r^{-l}$. The irregular solution r^{-l} diverges too severely for $l \geq 1$ to remain square-integrable. Even for $l = 0$, where $u_0(r) \rightarrow \text{const}$ would lead to $R_0(r) \sim 1/r$, this is unphysical because $\nabla^2(1/r) = -4\pi\delta(\mathbf{r})$, which would introduce an extraneous delta-function source at the origin. Hence, physical acceptability demands the Dirichlet boundary condition:

$$u_l(0) = 0 \quad \implies \quad R_l(r) \propto r^l \quad \text{as } r \rightarrow 0. \quad (6.98)$$

For potentials satisfying $V(r) \rightarrow 0$ as $r \rightarrow \infty$:

- **Scattering states** ($E > 0$): With $k = \sqrt{2mE}/\hbar$, $u_l(r) \sim \sin(kr - l\pi/2 + \delta_l)$ as $r \rightarrow \infty$.
- **Bound states** ($E < 0$): With $\kappa = \sqrt{-2mE}/\hbar$, $u_l(r) \sim e^{-\kappa r}$ as $r \rightarrow \infty$.

Spectroscopic states are traditionally classified by letters according to l :

l	0	1	2	3	4	...
Symbol	s	p	d	f	g	...

6.7.4 Free Solutions and Spherical Bessel Functions

In regions where the potential is constant, $V(r) = V_0$, the radial equation for $E > V_0$ with $k = \sqrt{2m(E - V_0)}/\hbar$ reduces to the spherical Bessel differential equation:

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} + k^2 \right) R_l(r) = 0. \quad (6.99)$$

The two linearly independent real solutions are:

1. The *spherical Bessel functions* $j_l(kr)$, regular at the origin:

$$j_l(x) \equiv \left(\frac{\pi}{2x} \right)^{1/2} J_{l+1/2}(x) = (-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \left(\frac{\sin x}{x} \right). \quad (6.100)$$

2. The *spherical Neumann functions* $n_l(kr)$ (or $y_l(kr)$), irregular at the origin:

$$n_l(x) \equiv (-1)^{l+1} \left(\frac{\pi}{2x} \right)^{1/2} J_{-l-1/2}(x) = -(-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \left(\frac{\cos x}{x} \right). \quad (6.101)$$

The explicit lowest-order functions are:

$$\begin{aligned} j_0(x) &= \frac{\sin x}{x}, & n_0(x) &= -\frac{\cos x}{x}, \\ j_1(x) &= \frac{\sin x}{x^2} - \frac{\cos x}{x}, & n_1(x) &= -\frac{\cos x}{x^2} - \frac{\sin x}{x}. \end{aligned} \quad (6.102)$$

Their asymptotic limits are:

$$x \rightarrow 0: \quad j_l(x) \rightarrow \frac{x^l}{(2l+1)!!}, \quad n_l(x) \rightarrow -\frac{(2l-1)!!}{x^{l+1}}, \quad (6.103)$$

$$x \rightarrow \infty: \quad j_l(x) \rightarrow \frac{1}{x} \sin \left(x - \frac{l\pi}{2} \right), \quad n_l(x) \rightarrow -\frac{1}{x} \cos \left(x - \frac{l\pi}{2} \right). \quad (6.104)$$

The *spherical Hankel functions* represent traveling spherical waves:

$$h_l^{(1)}(x) \equiv j_l(x) + in_l(x) \xrightarrow{x \rightarrow \infty} \frac{(-i)^{l+1}}{x} e^{ix}, \quad h_l^{(2)}(x) = [h_l^{(1)}(x)]^*, \quad (6.105)$$

corresponding to outgoing and incoming spherical waves, respectively. For bound-state problems where $E < V_0$, writing $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, the solution decaying at infinity is proportional to $h_l^{(1)}(i\kappa r) \sim e^{-\kappa r}/r$.

6.7.5 Partial-Wave Expansion of a Plane Wave

A free particle plane wave $e^{i\mathbf{k} \cdot \mathbf{r}}$ with energy $E = \hbar^2 k^2/2m$ can be decomposed into an infinite sum of angular momentum eigenstates (Rayleigh's formula):

$$e^{i\mathbf{k} \cdot \mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l Y_{lm}^*(\hat{\mathbf{k}}) j_l(kr) Y_{lm}(\hat{\mathbf{r}}), \quad (6.106)$$

where $\hat{\mathbf{k}} = (\theta_k, \varphi_k)$ and $\hat{\mathbf{r}} = (\theta, \varphi)$. Using the spherical harmonic addition theorem, $\sum_m Y_{lm}^*(\hat{\mathbf{k}}) Y_{lm}(\hat{\mathbf{r}}) = \frac{2l+1}{4\pi} P_l(\hat{\mathbf{k}} \cdot \hat{\mathbf{r}})$, this takes the form

$$e^{i\mathbf{k} \cdot \mathbf{r}} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos \Theta), \quad (6.107)$$

where $\cos \Theta = \hat{\mathbf{k}} \cdot \hat{\mathbf{r}}$.

6.8 Problems

1. Rotational transformation of scalar, vector, and tensor operators.

- (a) Prove that any scalar operator $S(\hat{\mathbf{r}}, \hat{\mathbf{p}})$ constructed from the position and momentum operators (such as $\hat{\mathbf{r}} \cdot \hat{\mathbf{p}}$ or $\hat{\mathbf{p}}^2$) commutes with the angular momentum operator $\hat{\mathbf{L}}$:

$$[S(\hat{\mathbf{r}}, \hat{\mathbf{p}}), \hat{\mathbf{L}}] = 0. \quad (6.108)$$

- (b) Show that any vector operator $\hat{\mathbf{V}}(\hat{\mathbf{r}}, \hat{\mathbf{p}})$ satisfies

$$[\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{V}}] = i\hbar(\hat{\mathbf{V}} \times \hat{\mathbf{n}}). \quad (6.109)$$

- (c) If $\hat{T}_{ij}(\hat{\mathbf{r}}, \hat{\mathbf{p}})$ transforms under rotations as a second-rank Cartesian tensor, derive the commutation relation $[\hat{L}_k, \hat{T}_{ij}]$ in terms of the Levi-Civita symbol.

2. Matrix representation for $l = 1$.

- (a) In the standard ordered basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$, construct the explicit 3×3 matrix representations for $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$:

$$(L_k)_{m,m'} = \langle 1, m | \hat{L}_k | 1, m' \rangle = \int d\Omega Y_{1m}^*(\theta, \varphi) \hat{L}_k Y_{1m'}(\theta, \varphi). \quad (6.110)$$

- (b) Verify by explicit matrix multiplication that these matrices satisfy the angular momentum Lie algebra:

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y. \quad (6.111)$$

- (c) Determine the 3×3 matrix representations of the ladder operators $\hat{L}_+$, $\hat{L}_-$, and the Casimir operator $\hat{\mathbf{L}}^2$.

3. **Hermiticity of the radial momentum operator.** Consider the radial momentum operator

$$\hat{p}_r = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right). \quad (6.112)$$

Demonstrate that $\hat{p}_r$ is Hermitian on the Hilbert space of physical radial wave functions with inner product $\langle \phi | \psi \rangle = \int_0^\infty r^2 \phi^*(r) \psi(r) dr$, assuming the wave functions remain finite at $r = 0$ and vanish sufficiently rapidly as $r \rightarrow \infty$.

4. **Two-dimensional circular box.** Determine the energy eigenvalues and normalized stationary eigenfunctions of a particle of mass m confined within a two-dimensional circular billiard of radius R with infinitely rigid walls ($V(r) = 0$ for $r \leq R$ and $V(r) = \infty$ for $r > R$). Express the eigenenergies in terms of the zeros of ordinary Bessel functions.
5. **Two-dimensional isotropic harmonic oscillator.** Consider a particle of mass m moving in the two-dimensional isotropic potential $V(r) = \frac{1}{2}m\omega_0^2(x^2 + y^2)$.
- Solve the Schrödinger equation in Cartesian coordinates. Find the energy eigenvalues E_n , identify the degeneracy of each level, and write the normalized wave functions for the ground state and each of the degenerate first excited states.
 - Formulate the problem in polar coordinates (r, φ) . Construct the explicit wave functions of the first excited states that are simultaneous eigenfunctions of $\hat{L}_z$ with eigenvalues $+\hbar$ and $-\hbar$, and show how they are expressed as linear combinations of the Cartesian eigenstates.
6. **Landau levels in a uniform magnetic field.** A charged particle of mass m and charge e moves in a spatially uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$. Using the Landau gauge $\mathbf{A} = (0, Bx, 0)$:
- Determine the complete set of energy eigenvalues (Landau levels) and stationary wave functions.
 - Do these eigenstates simultaneously diagonalize $\hat{L}_z$?
 - Contrast the quantum probability distribution of these states with the classical cyclotron motion in the xy -plane, explaining how the choice of gauge influences the manifest symmetries.

Chapter 7

The Hydrogen Atom

7.1 The Hydrogen Atom

The hydrogen atom—composed of an electron and a proton interacting via an attractive central Coulomb potential—occupies a central role in quantum mechanics. As an exactly solvable three-dimensional system, it provides the fundamental benchmark for atomic physics, quantum spectroscopy, and perturbation theory.

In this chapter, we begin by formulating the quantum mechanics of two-particle systems, demonstrating how translation invariance enables the exact separation of the center-of-mass motion from the internal relative coordinates. We then solve the radial Schrödinger equation for the attractive $1/r$ Coulomb potential, deriving the discrete bound-state energy spectrum, the characteristic Bohr radius, and the complete orthonormal set of hydrogenic wave functions in terms of associated Laguerre polynomials. Finally, we investigate the conserved Laplace–Runge–Lenz vector, which explains the “accidental” l -degeneracy of the Coulomb spectrum through its underlying four-dimensional rotational symmetry $SO(4)$.

7.2 Structure of the Hydrogen Atom

The simplest realistic atomic system is the hydrogen atom, consisting of a negatively charged electron of mass m_e and charge $-e$, and a positively charged nucleus (a proton) of mass m_p and charge $+e$. In the non-relativistic approximation, their mutual interaction is governed by the attractive central Coulomb potential:

$$V(\mathbf{r}_e - \mathbf{r}_p) = -\frac{e^2}{|\mathbf{r}_e - \mathbf{r}_p|}, \quad (7.1)$$

in Gaussian units (where e^2 is replaced by $e^2/4\pi\epsilon_0$ in SI units).

Because the physical system involves two distinct moving particles, a rigorous quantum mechanical description requires formulating the dynamics within the composite two-particle Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$. Before solving for the discrete bound-state spectra and electronic wave functions of hydrogen, we must first establish the canonical transformation that separates the external center-of-mass kinematics from the internal relative motion.

7.3 Two-Particle Systems and Center-of-Mass Separation

A quantum mechanical system comprising two distinct particles cannot be described by a wave function living in ordinary three-dimensional space. Instead, it is described by a state vector $|\Psi(t)\rangle$ inhabiting the tensor product Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$, represented in the coordinate basis

by a wave function on the six-dimensional configuration space:

$$\psi(\mathbf{r}_1, \mathbf{r}_2, t) = \langle \mathbf{r}_1, \mathbf{r}_2 | \Psi(t) \rangle. \quad (7.2)$$

The quantity $|\psi(\mathbf{r}_1, \mathbf{r}_2, t)|^2 d^3r_1 d^3r_2$ represents the joint probability of simultaneously finding particle 1 in volume element d^3r_1 about $\mathbf{r}_1$ and particle 2 in d^3r_2 about $\mathbf{r}_2$ at time t . The normalization condition is

$$\int d^3r_1 \int d^3r_2 |\psi(\mathbf{r}_1, \mathbf{r}_2, t)|^2 = 1. \quad (7.3)$$

7.3.1 Commutation Relations and Observables

The dynamical variables are represented by Hermitian operators acting on the respective particle subspaces:

$$\begin{aligned} \hat{\mathbf{r}}_1 &= \mathbf{r}_1 \otimes \mathbb{I}_2, & \hat{\mathbf{p}}_1 &= -i\hbar \nabla_1 \otimes \mathbb{I}_2, \\ \hat{\mathbf{r}}_2 &= \mathbb{I}_1 \otimes \mathbf{r}_2, & \hat{\mathbf{p}}_2 &= \mathbb{I}_1 \otimes (-i\hbar \nabla_2). \end{aligned} \quad (7.4)$$

Operators acting on different particles commute identically:

$$[(\hat{\mathbf{r}}_1)_i, (\hat{\mathbf{r}}_2)_j] = 0, \quad [(\hat{\mathbf{p}}_1)_i, (\hat{\mathbf{p}}_2)_j] = 0, \quad [(\hat{\mathbf{r}}_1)_i, (\hat{\mathbf{p}}_2)_j] = 0, \quad (7.5)$$

while the canonical commutation relations hold for each individual particle:

$$[(\hat{r}_1)_i, (\hat{p}_1)_j] = i\hbar \delta_{ij} \mathbb{I}, \quad [(\hat{r}_2)_i, (\hat{p}_2)_j] = i\hbar \delta_{ij} \mathbb{I}. \quad (7.6)$$

The individual orbital angular momenta are $\hat{\mathbf{L}}_1 = \hat{\mathbf{r}}_1 \times \hat{\mathbf{p}}_1$ and $\hat{\mathbf{L}}_2 = \hat{\mathbf{r}}_2 \times \hat{\mathbf{p}}_2$.

7.3.2 Transformation to Center-of-Mass and Relative Coordinates

Consider two particles with masses m_1 and m_2 interacting via a mutual translationally invariant potential $V(\mathbf{r}_1 - \mathbf{r}_2)$. The two-particle Hamiltonian is

$$\hat{H} = \frac{\hat{\mathbf{p}}_1^2}{2m_1} + \frac{\hat{\mathbf{p}}_2^2}{2m_2} + V(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2). \quad (7.7)$$

The time-dependent Schrödinger equation in coordinate representation is

$$\left[-\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{\hbar^2}{2m_2} \nabla_2^2 + V(\mathbf{r}_1 - \mathbf{r}_2) \right] \psi(\mathbf{r}_1, \mathbf{r}_2, t) = i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}_1, \mathbf{r}_2, t). \quad (7.8)$$

To decouple the external kinematics from the internal dynamics, we define the canonical center-of-mass and relative operators:

$$\begin{aligned} \hat{\mathbf{R}} &\equiv \frac{m_1 \hat{\mathbf{r}}_1 + m_2 \hat{\mathbf{r}}_2}{M}, & \hat{\mathbf{P}} &\equiv \hat{\mathbf{p}}_1 + \hat{\mathbf{p}}_2, \\ \hat{\mathbf{r}} &\equiv \hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2, & \hat{\mathbf{p}} &\equiv \frac{m_2 \hat{\mathbf{p}}_1 - m_1 \hat{\mathbf{p}}_2}{M}, \end{aligned} \quad (7.9)$$

where

$$M \equiv m_1 + m_2, \quad \mu \equiv \frac{m_1 m_2}{m_1 + m_2} \quad (7.10)$$

are the total mass and reduced mass, respectively.

One directly verifies that the new coordinates and conjugate momenta satisfy canonical commutation relations:

$$[\hat{R}_i, \hat{P}_j] = i\hbar \delta_{ij} \mathbb{I}, \quad [\hat{r}_i, \hat{p}_j] = i\hbar \delta_{ij} \mathbb{I}, \quad (7.11)$$

with all cross-commutators vanishing:

$$[\hat{\mathbf{R}}, \hat{\mathbf{r}}] = 0, \quad [\hat{\mathbf{P}}, \hat{\mathbf{p}}] = 0, \quad [\hat{\mathbf{R}}, \hat{\mathbf{p}}] = 0, \quad [\hat{\mathbf{r}}, \hat{\mathbf{P}}] = 0. \quad (7.12)$$

Furthermore, the kinetic energy decomposes cleanly into center-of-mass and relative terms:

$$\frac{\hat{\mathbf{p}}_1^2}{2m_1} + \frac{\hat{\mathbf{p}}_2^2}{2m_2} = \frac{\hat{\mathbf{P}}^2}{2M} + \frac{\hat{\mathbf{p}}^2}{2\mu}. \quad (7.13)$$

7.3.3 Separation of Variables

In terms of the differential operators $\hat{\mathbf{P}} \doteq -i\hbar\nabla_{\mathbf{R}}$ and $\hat{\mathbf{p}} \doteq -i\hbar\nabla_{\mathbf{r}}$, the Hamiltonian operator becomes

$$\hat{H} = -\frac{\hbar^2}{2M}\nabla_{\mathbf{R}}^2 - \frac{\hbar^2}{2\mu}\nabla_{\mathbf{r}}^2 + V(\mathbf{r}). \quad (7.14)$$

Because the potential $V(\mathbf{r})$ depends only on the relative separation vector $\mathbf{r}$ and is entirely independent of the center-of-mass coordinate $\mathbf{R}$, the Hamiltonian separates:

$$\hat{H} = \hat{H}_{\text{CM}} + \hat{H}_{\text{rel}}, \quad [\hat{H}_{\text{CM}}, \hat{H}_{\text{rel}}] = 0. \quad (7.15)$$

This allows an exact product separation of the wave function:

$$\psi(\mathbf{r}_1, \mathbf{r}_2, t) = \Phi_{\text{CM}}(\mathbf{R}, t) \psi_{\text{rel}}(\mathbf{r}, t). \quad (7.16)$$

Substituting this ansatz into Eq. (7.14) decomposes the six-dimensional Schrödinger equation into two uncoupled three-dimensional equations:

$$-\frac{\hbar^2}{2M}\nabla_{\mathbf{R}}^2 \Phi_{\text{CM}}(\mathbf{R}, t) = i\hbar \frac{\partial}{\partial t} \Phi_{\text{CM}}(\mathbf{R}, t), \quad (7.17)$$

$$\left[-\frac{\hbar^2}{2\mu}\nabla_{\mathbf{r}}^2 + V(\mathbf{r}) \right] \psi_{\text{rel}}(\mathbf{r}, t) = i\hbar \frac{\partial}{\partial t} \psi_{\text{rel}}(\mathbf{r}, t). \quad (7.18)$$

Equation (7.17) describes the center-of-mass moving as a free particle of total mass M , with plane-wave momentum eigenstates:

$$\Phi_{\mathbf{K}}(\mathbf{R}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \exp \left[\frac{i}{\hbar} \left(\mathbf{P} \cdot \mathbf{R} - \frac{P^2}{2M} t \right) \right]. \quad (7.19)$$

Meanwhile, Eq. (7.18) demonstrates that the internal relative motion is formally identical to the problem of a single fictitious particle of reduced mass μ subject to the stationary central potential $V(\mathbf{r})$.

Stationary relative states $\psi_{\text{rel}}(\mathbf{r}, t) = e^{-iEt/\hbar} \psi(\mathbf{r})$ satisfy

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 + V(\mathbf{r}) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (7.20)$$

and the complete two-particle stationary state is given by

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, t) = \exp \left[-\frac{i}{\hbar} \left(E + \frac{P^2}{2M} \right) t \right] e^{i\mathbf{P} \cdot \mathbf{R}/\hbar} \psi(\mathbf{r}), \quad (7.21)$$

with total system energy $E_{\text{total}} = E + \frac{P^2}{2M}$.

7.4 Hydrogen Atom Wave Functions

The mutual electrostatic attraction between the electron and proton is described by the central Coulomb potential:

$$V(r) = -\frac{e^2}{r}. \quad (7.22)$$

Because the potential is strictly spherically symmetric, the relative orbital angular momentum $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ is a conserved constant of the motion: $[\hat{H}, \hat{\mathbf{L}}] = 0$. Consequently, stationary energy eigenstates can be chosen to simultaneously diagonalize $\hat{\mathbf{L}}^2$ and $\hat{L}_z$:

$$\psi_{nlm}(\mathbf{r}) = R_{nl}(r) Y_{lm}(\theta, \varphi). \quad (7.23)$$

7.4.1 The Radial Coulomb Equation and Energy Spectrum

Setting $u(r) = rR(r)$, the radial Schrödinger equation, Eq. (6.95), becomes

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{e^2}{r} \right] u(r) = Eu(r), \quad (7.24)$$

where $\mu = m_e m_p / (m_e + m_p) \approx m_e$ is the reduced mass of the electron-proton system.

For bound states ($E < 0$), introducing the dimensionless coordinate $\rho = 2\kappa r$ with $\kappa = \sqrt{-2\mu E}/\hbar$, the square-integrable solutions are labeled by the *principal quantum number* $n \in \{1, 2, 3, \dots\}$. For a given n , the orbital angular momentum takes integer values:

$$l \in \{0, 1, 2, \dots, n-1\}. \quad (7.25)$$

The bound-state energy eigenvalues are given by the Bohr–Balmer formula:

$$E_n = -\frac{e^2}{2a_0} \frac{1}{n^2} = -\frac{\text{Ry}}{n^2}, \quad n = 1, 2, 3, \dots, \quad (7.26)$$

where a_0 is the *Bohr radius*:

$$a_0 \equiv \frac{\hbar^2}{\mu e^2} \approx 0.529 \times 10^{-10} \text{ m} = 0.529 \text{ \AA}, \quad (7.27)$$

and $\text{Ry} \equiv e^2/(2a_0) \approx 13.606 \text{ eV}$ is the Rydberg energy unit.

7.4.2 Normalized Radial Wave Functions

The normalized radial eigenfunctions $R_{nl}(r)$ can be expressed analytically in terms of associated Laguerre polynomials:¹

$$R_{nl}(r) = - \left[\left(\frac{2}{na_0} \right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3} \right]^{1/2} e^{-r/(na_0)} \left(\frac{2r}{na_0} \right)^l L_{n+l}^{2l+1} \left(\frac{2r}{na_0} \right), \quad (7.28)$$

where the associated Laguerre polynomial is defined by

$$L_{n+l}^{2l+1}(\rho) = \sum_{k=0}^{n-l-1} \frac{(-1)^{k+1} [(n+l)!]^2 \rho^k}{k! (n-l-1-k)! (2l+1+k)!}. \quad (7.29)$$

For the lowest three electronic energy levels, the explicit radial wave functions are:

$$\begin{aligned} R_{10}(r) &= \frac{2}{a_0^{3/2}} e^{-r/a_0}, \\ R_{20}(r) &= \frac{1}{(2a_0)^{3/2}} \left(2 - \frac{r}{a_0} \right) e^{-r/(2a_0)}, \\ R_{21}(r) &= \frac{1}{\sqrt{3}(2a_0)^{3/2}} \left(\frac{r}{a_0} \right) e^{-r/(2a_0)}. \end{aligned} \quad (7.30)$$

These radial functions satisfy the mutual orthogonality condition:

$$\int_0^\infty r^2 R_{nl}(r) R_{n'l}(r) dr = \delta_{nn'}. \quad (7.31)$$

¹We follow the historical convention commonly used in quantum mechanics texts (e.g., Bethe–Salpeter, Messiah): $L_p^q(\rho) = \frac{d^q}{d\rho^q} L_p(\rho)$. In modern mathematical literature, the generalized Laguerre polynomial is often denoted by $L_k^{(\alpha)}(\rho)$, related to this definition by $L_{n+l}^{2l+1}(\rho) = (-1)^{2l+1} (n+l)! L_{n-l-1}^{(2l+1)}(\rho)$.

7.4.3 Energy Level Multiplets and Accidental Degeneracy

For a specified principal quantum number n , the orbital quantum number l ranges from 0 to $n - 1$. Because each orbital state has $(2l + 1)$ magnetic sublevels labeled by $m \in \{-l, \dots, l\}$, the total spatial degeneracy of the n -th energy level is

$$d_n = \sum_{l=0}^{n-1} (2l + 1) = n^2. \quad (7.32)$$

(Including the two intrinsic spin projections of the electron, $s = \pm 1/2$, the full degeneracy is $2n^2$.)

The structural hierarchy of the low-lying hydrogen bound states is summarized in Table 7.1.

Table 7.1: Structure and spatial degeneracies of low-lying hydrogen bound states.

Level	$l = 0$ (s)	$l = 1$ (p)	$l = 2$ (d)	$l = 3$ (f)	Total Degeneracy (n^2)
$n = 4$	$4s$ (1)	$4p$ (3)	$4d$ (5)	$4f$ (7)	16
$n = 3$	$3s$ (1)	$3p$ (3)	$3d$ (5)	—	9
$n = 2$	$2s$ (1)	$2p$ (3)	—	—	4
$n = 1$	$1s$ (1)	—	—	—	1

The fact that states with different values of l (e.g., $2s$ and $2p$) possess identical energy is unique to the pure $1/r$ Coulomb potential. In a generic central potential $V(r)$, rotational symmetry guarantees only a $(2l + 1)$ -fold degeneracy for a given l . The additional n^2 degeneracy in hydrogen is known as an *accidental degeneracy*, and arises from a hidden dynamical symmetry.

7.4.4 Asymptotic Behavior and Spatial Characteristics

At very large distances $r \gg n^2 a_0$, the attractive Coulomb potential $V(r) = -e^2/r$ dominates the centrifugal barrier. The radial equation for $u_{nl}(r) = rR_{nl}(r)$ asymptotically satisfies

$$\left(-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} - \frac{e^2}{r} \right) u_{nl}(r) \approx E_n u_{nl}(r). \quad (7.33)$$

The asymptotic solution takes the characteristic Coulomb-modified form:

$$u_{nl}(r) \sim r^n e^{-r/(na_0)}. \quad (7.34)$$

The long-range tail of the $1/r$ potential produces the pre-exponential power factor r^n , contrasting sharply with the purely exponential decay $\sim e^{-\kappa r}$ of short-range Yukawa potentials.

In the ground state ($n = 1, l = 0$), the radial probability distribution for observing the electron in a spherical shell between r and $r + dr$ is

$$P(r) dr = r^2 |R_{10}(r)|^2 dr = \frac{4}{a_0^3} r^2 e^{-2r/a_0} dr. \quad (7.35)$$

Setting $dP/dr = 0$ reveals that the most probable radius is precisely the Bohr radius $r = a_0$. However, due to the asymmetric radial tail, the expectation value of the radial distance is somewhat larger:

$$\langle r \rangle_{1s} = \int_0^\infty r P(r) dr = \frac{3}{2} a_0. \quad (7.36)$$

7.5 The Dynamical Symmetry of the Hydrogen Atom

The n^2 -fold degeneracy of the hydrogen atom energy levels is an exceptional phenomenon unique to the $1/r$ Coulomb potential. In classical celestial mechanics, Keplerian bound orbits do not precess; the major axis of the elliptical trajectory remains permanently fixed in space. This orbital stability is governed by the conservation of the classical Laplace–Runge–Lenz vector:

$$\mathbf{R}_{\text{classical}} = \frac{1}{m} \mathbf{p} \times \mathbf{L} - \frac{e^2}{r} \mathbf{r}, \quad (7.37)$$

which points along the Laplace perihelion vector from the focus to the pericenter.

7.5.1 The Quantum Runge–Lenz Operator

To quantize the Runge–Lenz vector, we must construct a Hermitian operator by symmetrizing the non-commuting product $\hat{\mathbf{p}} \times \hat{\mathbf{L}}$:

$$\hat{\mathbf{R}} = \frac{1}{2\mu} \left(\hat{\mathbf{p}} \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times \hat{\mathbf{p}} \right) - \frac{e^2}{r} \hat{\mathbf{r}} = \frac{1}{\mu} \hat{\mathbf{p}} \times \hat{\mathbf{L}} - \frac{i\hbar}{\mu} \hat{\mathbf{p}} - \frac{e^2}{r} \hat{\mathbf{r}}. \quad (7.38)$$

A straightforward operator calculation verifies that $\hat{\mathbf{R}}$ is an exact constant of the motion:

$$[\hat{\mathbf{R}}, \hat{H}] = 0. \quad (7.39)$$

Furthermore, the Runge–Lenz vector is orthogonal to the angular momentum:

$$\hat{\mathbf{R}} \cdot \hat{\mathbf{L}} = \hat{\mathbf{L}} \cdot \hat{\mathbf{R}} = 0. \quad (7.40)$$

The operator square of $\hat{\mathbf{R}}$ is directly related to the Hamiltonian and $\hat{\mathbf{L}}^2$:

$$\hat{\mathbf{R}}^2 = e^4 \mathbb{I} + \frac{2\hat{H}}{\mu} (\hat{\mathbf{L}}^2 + \hbar^2 \mathbb{I}). \quad (7.41)$$

7.5.2 The SO(4) Symmetry Lie Algebra

Because $\hat{\mathbf{R}}$ is a vector operator, its commutator with angular momentum satisfies

$$[\hat{R}_i, \hat{L}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{R}_k. \quad (7.42)$$

Meanwhile, the commutator of $\hat{\mathbf{R}}$ with itself yields

$$[\hat{R}_i, \hat{R}_j] = i\hbar \left(-\frac{2\hat{H}}{\mu} \right) \sum_{k=1}^3 \epsilon_{ijk} \hat{L}_k. \quad (7.43)$$

For the bound-state subspace where the energy is strictly negative ($E < 0$), the operator $(-2\hat{H}/\mu)$ is positive-definite. We can therefore rescale $\hat{\mathbf{R}}$ to define a normalized vector operator:

$$\hat{\mathbf{K}} \equiv \sqrt{\frac{-\mu}{2\hat{H}}} \hat{\mathbf{R}} = \hat{\mathbf{R}} \sqrt{\frac{-\mu}{2\hat{H}}}. \quad (7.44)$$

Because $[\hat{H}, \hat{\mathbf{R}}] = [\hat{H}, \hat{\mathbf{L}}] = 0$, this scaling factor commutes with all relevant generators. The operators $\hat{\mathbf{L}}$ and $\hat{\mathbf{K}}$ satisfy the closed Lie algebra:

$$[\hat{L}_i, \hat{L}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{L}_k, \quad (7.45)$$

$$[\hat{K}_i, \hat{L}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{K}_k, \quad (7.46)$$

$$[\hat{K}_i, \hat{K}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{L}_k. \quad (7.47)$$

Equations (7.45)–(7.47) are the structure relations of the four-dimensional rotation group $\text{SO}(4)$.

Substituting $\hat{\mathbf{K}}$ into Eq. (7.41), the Hamiltonian can be inverted and expressed purely in terms of the Casimir invariants:

$$\hat{H} = -\frac{\mu e^4}{2(\hat{\mathbf{K}}^2 + \hat{\mathbf{L}}^2 + \hbar^2 \mathbb{I})}. \quad (7.48)$$

7.5.3 Decoupling into Independent $\text{SU}(2)$ Algebras

Because the Lie algebra $\mathfrak{so}(4)$ is isomorphic to the direct sum $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, we define two decoupled vector operators:

$$\hat{\mathbf{M}} \equiv \frac{\hat{\mathbf{L}} + \hat{\mathbf{K}}}{2}, \quad \hat{\mathbf{N}} \equiv \frac{\hat{\mathbf{L}} - \hat{\mathbf{K}}}{2}. \quad (7.49)$$

Direct evaluation of their commutators confirms:

$$\begin{aligned} [\hat{M}_i, \hat{M}_j] &= i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{M}_k, \\ [\hat{N}_i, \hat{N}_j] &= i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{N}_k, \\ [\hat{M}_i, \hat{N}_j] &= 0. \end{aligned} \quad (7.50)$$

Thus, $\hat{\mathbf{M}}$ and $\hat{\mathbf{N}}$ form two mutually commuting, independent angular momentum algebras.

In terms of $\hat{\mathbf{M}}$ and $\hat{\mathbf{N}}$, we have

$$\hat{\mathbf{L}} = \hat{\mathbf{M}} + \hat{\mathbf{N}}, \quad \hat{\mathbf{K}} = \hat{\mathbf{M}} - \hat{\mathbf{N}}, \quad (7.51)$$

so that

$$\hat{\mathbf{K}}^2 + \hat{\mathbf{L}}^2 = 2(\hat{\mathbf{M}}^2 + \hat{\mathbf{N}}^2). \quad (7.52)$$

Furthermore, the orthogonality relation $\hat{\mathbf{K}} \cdot \hat{\mathbf{L}} = 0$ imposes a crucial constraint:

$$0 = \hat{\mathbf{K}} \cdot \hat{\mathbf{L}} = (\hat{\mathbf{M}} - \hat{\mathbf{N}}) \cdot (\hat{\mathbf{M}} + \hat{\mathbf{N}}) = \hat{\mathbf{M}}^2 - \hat{\mathbf{N}}^2 \implies \hat{\mathbf{M}}^2 = \hat{\mathbf{N}}^2. \quad (7.53)$$

7.5.4 Spectrum and Degeneracy from Group Representations

The representation theory of $\mathfrak{su}(2)$ dictates that the eigenvalues of $\hat{\mathbf{M}}^2$ and $\hat{\mathbf{N}}^2$ are parametrized by $j_1(j_1 + 1)\hbar^2$ and $j_2(j_2 + 1)\hbar^2$, where $j_1, j_2 \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$. Because $\hat{\mathbf{M}}^2 = \hat{\mathbf{N}}^2$, physical bound states must have identical representation labels:

$$j_1 = j_2 \equiv j \in \left\{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\right\}. \quad (7.54)$$

Acting on simultaneous eigenstates $|j, j, \mu, \nu\rangle$ of $\hat{\mathbf{M}}^2, \hat{\mathbf{N}}^2, \hat{M}_z, \hat{N}_z$, the denominator operator of Eq. (7.48) evaluates to

$$2(\hat{\mathbf{M}}^2 + \hat{\mathbf{N}}^2) + \hbar^2 \mathbb{I} \rightarrow 2\hbar^2[2j(j+1)] + \hbar^2 = \hbar^2(4j^2 + 4j + 1) = \hbar^2(2j+1)^2. \quad (7.55)$$

Defining the principal quantum number $n \equiv 2j+1$, which takes values $n \in \{1, 2, 3, \dots\}$, we obtain the energy spectrum purely from symmetry:

$$E_n = -\frac{\mu e^4}{2\hbar^2 n^2}, \quad n = 1, 2, 3, \dots \quad (7.56)$$

This is precisely the Bohr formula.

For a given j , the magnetic quantum numbers $\mu \in \{-j, \dots, j\}$ and $\nu \in \{-j, \dots, j\}$ independently assume $(2j+1)$ values. The total number of degenerate quantum states for energy level E_n is therefore

$$(2j+1)(2j+1) = n \times n = n^2. \quad (7.57)$$

The hidden four-dimensional rotational symmetry $\text{SO}(4)$ generated by $\{\hat{\mathbf{L}}, \hat{\mathbf{K}}\}$ thus fully explains the n^2 degeneracy of the Coulomb potential.

The states $|j, j, \mu, \nu\rangle$ diagonalize $\hat{L}_z = \hat{M}_z + \hat{N}_z$ with integer eigenvalue $m = \mu + \nu \in \{-(n-1), \dots, n-1\}$, but they do not diagonalize $\hat{\mathbf{L}}^2$. Rather, they correspond to parabolic eigenstates (related to separation of variables in parabolic coordinates), which are linear superpositions of spherical states $|n, l, m\rangle$ with Clebsch–Gordan coefficients $\langle l, m | j, j; \mu, \nu \rangle$.

7.6 Problems

1. Ground state spatial properties and uncertainty relations.

- In classical mechanics, an electron with total energy $E_1 = -e^2/(2a_0)$ in the Coulomb potential $V(r) = -e^2/r$ cannot exceed the classical turning radius $r_{\text{cl}} = 2a_0$ because its kinetic energy would be negative. Compute the quantum probability of finding the electron in the classically forbidden region $r > 2a_0$ in the hydrogen ground state $|1, 0, 0\rangle$.
- Show that the Heisenberg uncertainty relation $\Delta x \cdot \Delta p_x \geq \hbar/2$ is satisfied by an electron in the ground state.
- Describe and sketch the contours of constant probability density in the xz -plane for the $1s$, $2s$, $2p$ ($m=0$), and $2p$ ($m=1$) states.

2. Electric dipole transition matrix elements.

Calculate all non-vanishing coordinate matrix elements of $\hat{x}$ between the first excited states ($n=2$) and the ground state ($n=1$):

$$\langle 1, 0, 0 | \hat{x} | 2, l, m \rangle. \quad (7.58)$$

Confirm that the electric dipole selection rules $\Delta l = \pm 1$ and $\Delta m = \pm 1$ are respected.

3. Schwinger's transformation to a 2D isotropic harmonic oscillator.

Schwinger developed an elegant method to solve the Coulomb bound-state problem by transforming the radial Schrödinger equation into that of a two-dimensional isotropic harmonic oscillator. Let $r = \frac{1}{2}\lambda\rho^2$, where $\lambda > 0$ is a scaling constant, and substitute $R_{nl}(r) = F(\rho)/\rho$.

- Show that $F(\rho)$ satisfies the radial differential equation of a two-dimensional isotropic harmonic oscillator with angular frequency $\omega = \sqrt{-2\lambda^2 E/\mu}$, effective angular momentum quantum number $M = 2l+1$, and fictitious oscillator energy $\mathcal{E} = 2e^2\lambda$.

- (b) Using the known energy eigenvalues and degeneracies of the 2D oscillator (Chapter 6, Problem 5), deduce the discrete energy levels $E_n = -\mu e^4/(2\hbar^2 n^2)$ and their n^2 degeneracy.
- (c) Using this mapping, construct the normalized hydrogen ground-state radial wave function $R_{10}(r)$.

4. **Algebra of the Laplace–Runge–Lenz vector.** Consider the Hermitian quantum Runge–Lenz operator:

$$\hat{\mathbf{R}} = \frac{1}{2\mu}(\hat{\mathbf{p}} \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times \hat{\mathbf{p}}) - \frac{e^2}{r}\hat{\mathbf{r}}. \quad (7.59)$$

- (a) Prove by explicit commutator evaluation that $[\hat{\mathbf{R}}, \hat{H}] = 0$, confirming that $\hat{\mathbf{R}}$ is an exact constant of the motion.
- (b) Verify the operator identities:

$$\hat{\mathbf{R}}^2 = e^4 \mathbb{I} + \frac{2\hat{H}}{\mu}(\hat{\mathbf{L}}^2 + \hbar^2 \mathbb{I}), \quad (7.60)$$

$$[\hat{R}_i, \hat{L}_j] = i\hbar \sum_{k=1}^3 \epsilon_{ijk} \hat{R}_k, \quad (7.61)$$

$$[\hat{R}_i, \hat{R}_j] = i\hbar \left(-\frac{2\hat{H}}{\mu} \right) \sum_{k=1}^3 \epsilon_{ijk} \hat{L}_k. \quad (7.62)$$

- (c) Verify that $\hat{\mathbf{L}} \cdot \hat{\mathbf{R}} = \hat{\mathbf{R}} \cdot \hat{\mathbf{L}} = 0$.

Chapter 8

Cooper Pairs

8.1 Cooper Pairs

Superconductivity arises from intricate correlations among electrons in a metal. To understand its physical mechanism, we investigate a simplified model due to Leon Cooper (1956), describing the interaction of a single pair of electrons just above a filled quiescent Fermi sea. Despite its simplicity, this model illustrates fundamental principles of quantum mechanics and forms the conceptual foundation for the Bardeen–Cooper–Schrieffer (BCS) theory of superconductivity.

8.1.1 The Noninteracting Fermi Sea

Neglecting crystal lattice periodicities, a metal of volume $V = L^3$ can be modeled as a three-dimensional box filled with electrons. In the absence of mutual interactions, the stationary spatial eigenfunctions under periodic boundary conditions are plane waves:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}/\hbar}, \quad (8.1)$$

with corresponding single-particle kinetic energies

$$\varepsilon_k = \frac{k^2}{2m}, \quad (8.2)$$

where $\mathbf{k}$ is the momentum vector. For a cubic box of side length L , periodic boundary conditions require

$$k_x = \frac{2\pi n_x \hbar}{L}, \quad k_y = \frac{2\pi n_y \hbar}{L}, \quad k_z = \frac{2\pi n_z \hbar}{L}, \quad (n_x, n_y, n_z \in \mathbb{Z}). \quad (8.3)$$

By the Pauli exclusion principle, each orbital state can accommodate at most two electrons with opposite spin orientations ($s_z = \pm\hbar/2$). In the ground state of N noninteracting electrons, the lowest single-particle momentum states are sequentially occupied up to the Fermi momentum k_f :

$$N = \sum_{|\mathbf{k}| \leq k_f} 2. \quad (8.4)$$

The occupied states constitute the *Fermi sea*, bounded by the spherical *Fermi surface* of radius k_f .

For macroscopic dimensions L and particle number N , the momentum level spacing $\Delta k = 2\pi\hbar/L$ becomes infinitesimally small, enabling the discrete summation to be replaced by a phase-space integral:

$$\sum_{\mathbf{k}} \longrightarrow V \int \frac{d^3k}{(2\pi\hbar)^3}. \quad (8.5)$$

The density of states in momentum space is $V/(2\pi\hbar)^3$. Evaluating Eq. (8.4) in the continuum limit yields

$$N = 2V \int_0^{k_f} \frac{4\pi k^2 dk}{(2\pi\hbar)^3} = \frac{V k_f^3}{3\pi^2 \hbar^3}, \quad (8.6)$$

so that the Fermi momentum is related to the particle density $n = N/V$ by

$$k_f = (3\pi^2 n)^{1/3} \hbar, \quad (8.7)$$

and the corresponding Fermi energy is $\varepsilon_f = k_f^2/(2m)$.

8.1.2 Formulation of the Two-Electron Problem

We now consider two extra electrons added to the system, occupying states immediately above the Fermi surface ($k > k_f$), and subject to a mutual attractive interaction. We assume that the electrons in the underlying Fermi sea remain frozen in their states, neglecting interactions between the Fermi sea electrons themselves as well as interactions between the pair and the sea, except through the Pauli principle: the pair is strictly forbidden from scattering into any state inside the Fermi sea ($k \leq k_f$). Furthermore, the two electrons are assumed to form a spin-singlet configuration (antiparallel spins), allowing their spatial wave function to be completely symmetric under particle exchange.

In the absence of interaction, the stationary state of the pair is given by the product

$$\psi(\mathbf{r}_1, \mathbf{r}_2, t) = \frac{1}{V} e^{i\mathbf{k}_1 \cdot \mathbf{r}_1 / \hbar} e^{i\mathbf{k}_2 \cdot \mathbf{r}_2 / \hbar} e^{-i(\varepsilon_{k_1} + \varepsilon_{k_2})t / \hbar}. \quad (8.8)$$

When an interaction is introduced, momentum is exchanged between the electrons via scattering, and the two-body state becomes a superposition of noninteracting product states:

$$\psi(\mathbf{r}_1, \mathbf{r}_2, t) = \sum_{\mathbf{k}_1, \mathbf{k}_2} a_{\mathbf{k}_1 \mathbf{k}_2}(t) \frac{1}{V} e^{i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2) / \hbar}. \quad (8.9)$$

For a stationary energy eigenstate of total energy E , the coefficients oscillate with a single frequency:

$$a_{\mathbf{k}_1 \mathbf{k}_2}(t) = a_{\mathbf{k}_1 \mathbf{k}_2} e^{-iEt / \hbar}. \quad (8.10)$$

Crucially, Pauli blocking mandates that

$$a_{\mathbf{k}_1 \mathbf{k}_2} = 0 \quad \text{whenever } |\mathbf{k}_1| < k_f \text{ or } |\mathbf{k}_2| < k_f. \quad (8.11)$$

The time evolution of the probability amplitudes $a_{\mathbf{k}_1 \mathbf{k}_2}(t)$ is governed by the two-particle Schrödinger equation. In the presence of an interaction potential operator $\hat{\mathcal{V}}$, transitions occur between different momentum configurations:

$$i\hbar \frac{\partial}{\partial t} a_{\mathbf{k}_1 \mathbf{k}_2}(t) = (\varepsilon_{k_1} + \varepsilon_{k_2}) a_{\mathbf{k}_1 \mathbf{k}_2}(t) + \sum_{\mathbf{k}'_1, \mathbf{k}'_2} \langle \mathbf{k}_1 \mathbf{k}_2 | \hat{\mathcal{V}} | \mathbf{k}'_1 \mathbf{k}'_2 \rangle a_{\mathbf{k}'_1 \mathbf{k}'_2}(t). \quad (8.12)$$

The matrix element $\langle \mathbf{k}_1 \mathbf{k}_2 | \hat{\mathcal{V}} | \mathbf{k}'_1 \mathbf{k}'_2 \rangle$ represents the amplitude per unit time for the pair to scatter from an initial state $|\mathbf{k}'_1 \mathbf{k}'_2\rangle$ into $|\mathbf{k}_1 \mathbf{k}_2\rangle$. Conservation of total probability requires $\hat{\mathcal{V}}$ to be Hermitian:

$$\langle \mathbf{k}_1 \mathbf{k}_2 | \hat{\mathcal{V}} | \mathbf{k}'_1 \mathbf{k}'_2 \rangle^* = \langle \mathbf{k}'_1 \mathbf{k}'_2 | \hat{\mathcal{V}} | \mathbf{k}_1 \mathbf{k}_2 \rangle. \quad (8.13)$$

8.1.3 Nonlocal Potentials and Momentum Conservation

In the coordinate representation, Eq. (8.12) takes the form of an integro-differential equation:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}_1, \mathbf{r}_2, t) = \left(-\frac{\hbar^2}{2m} \nabla_1^2 - \frac{\hbar^2}{2m} \nabla_2^2 \right) \psi(\mathbf{r}_1, \mathbf{r}_2, t) + \int d^3r'_1 d^3r'_2 \langle \mathbf{r}_1 \mathbf{r}_2 | \hat{\mathcal{V}} | \mathbf{r}'_1 \mathbf{r}'_2 \rangle \psi(\mathbf{r}'_1, \mathbf{r}'_2, t), \quad (8.14)$$

where the coordinate matrix elements are given by

$$\langle \mathbf{r}_1 \mathbf{r}_2 | \hat{\mathcal{V}} | \mathbf{r}'_1 \mathbf{r}'_2 \rangle = \frac{1}{V^2} \sum_{\substack{\mathbf{k}_1, \mathbf{k}_2 \\ \mathbf{k}'_1, \mathbf{k}'_2}} e^{i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2)/\hbar} \langle \mathbf{k}_1 \mathbf{k}_2 | \hat{\mathcal{V}} | \mathbf{k}'_1 \mathbf{k}'_2 \rangle e^{-i(\mathbf{k}'_1 \cdot \mathbf{r}'_1 + \mathbf{k}'_2 \cdot \mathbf{r}'_2)/\hbar}. \quad (8.15)$$

Equation (8.14) illustrates that the interaction is generally *nonlocal*: the rate of change of the wave amplitude at $(\mathbf{r}_1, \mathbf{r}_2)$ is influenced by the wave function values at separated points $(\mathbf{r}'_1, \mathbf{r}'_2)$. For a conventional local pairwise potential $v(\mathbf{r}_1 - \mathbf{r}_2)$, the kernel collapses to

$$\langle \mathbf{r}_1 \mathbf{r}_2 | \hat{\mathcal{V}} | \mathbf{r}'_1 \mathbf{r}'_2 \rangle = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) v(\mathbf{r}_1 - \mathbf{r}_2), \quad (8.16)$$

in which case the momentum matrix elements reduce to

$$\langle \mathbf{k}_1 \mathbf{k}_2 | \hat{\mathcal{V}} | \mathbf{k}'_1 \mathbf{k}'_2 \rangle = \frac{1}{V} \tilde{v}(\mathbf{k} - \mathbf{k}') \delta_{\mathbf{K}, \mathbf{K}'}, \quad (8.17)$$

where $\mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2$ is the total momentum, $\mathbf{k} = (\mathbf{k}_1 - \mathbf{k}_2)/2$ is the relative momentum, and $\tilde{v}(\mathbf{q}) = \int d^3r e^{-i\mathbf{q} \cdot \mathbf{r}/\hbar} v(\mathbf{r})$.

Whenever translation invariance holds, the total momentum $\mathbf{K}$ is strictly conserved, so the matrix elements take the general form

$$\langle \mathbf{k}_1 \mathbf{k}_2 | \hat{\mathcal{V}} | \mathbf{k}'_1 \mathbf{k}'_2 \rangle = \mathcal{V}_{\mathbf{k}, \mathbf{k}'}(\mathbf{K}) \delta_{\mathbf{K}, \mathbf{K}'}. \quad (8.18)$$

In center-of-mass and relative coordinates, $\mathbf{R} = (\mathbf{r}_1 + \mathbf{r}_2)/2$ and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, the interaction kernel can be expressed as

$$\langle \mathbf{r}_1 \mathbf{r}_2 | \hat{\mathcal{V}} | \mathbf{r}'_1 \mathbf{r}'_2 \rangle = \langle \mathbf{r} | \hat{\mathcal{V}}(\mathbf{R} - \mathbf{R}') | \mathbf{r}' \rangle. \quad (8.19)$$

For an energy eigenstate with conserved total momentum $\mathbf{K}$, we write

$$a_{\mathbf{k}_1 \mathbf{k}_2}(t) = a_{\mathbf{k}}(\mathbf{K}) e^{-iEt/\hbar}, \quad (8.20)$$

and Eq. (8.12) reduces to the stationary eigenvalue equation:

$$(E - \varepsilon_{\mathbf{k}_1} - \varepsilon_{\mathbf{k}_2}) a_{\mathbf{k}}(\mathbf{K}) = \sum_{\mathbf{k}'} \mathcal{V}_{\mathbf{k}, \mathbf{k}'}(\mathbf{K}) a_{\mathbf{k}'}(\mathbf{K}). \quad (8.21)$$

8.1.4 The Cooper Model and Bound State Solution

In a crystal, electrons interact repulsively through the screened Coulomb force and attractively through phonon-mediated lattice polarizations: a moving electron displaces the positively charged ionic lattice, creating a localized region of increased positive charge density that attracts a second electron. Over a narrow energy window around the Fermi level, this phonon-mediated attraction can overcome the screened Coulomb repulsion.

Cooper modeled this effective net attraction by an idealized separable potential:

$$\mathcal{V}_{\mathbf{k}, \mathbf{k}'}(\mathbf{K}) = \begin{cases} -\frac{v_0}{V}, & k_f < |\mathbf{k}_1|, |\mathbf{k}_2|, |\mathbf{k}'_1|, |\mathbf{k}'_2| < k_a, \\ 0, & \text{otherwise,} \end{cases} \quad (8.22)$$

where $v_0 > 0$ is a coupling constant and k_a represents a momentum cutoff slightly above k_f , corresponding to the maximum energy of acoustic phonons (the Debye energy $\hbar\omega_D$).

With this interaction, Eq. (8.21) simplifies to

$$(E - \varepsilon_{\mathbf{k}_1} - \varepsilon_{\mathbf{k}_2})a_{\mathbf{k}}(\mathbf{K}) = -\frac{v_0}{V} \sum'_{\mathbf{k}'} a_{\mathbf{k}'}(\mathbf{K}), \quad (8.23)$$

where the primed summation is restricted to relative momenta $\mathbf{k}'$ satisfying the Pauli and cutoff constraints: $k_f < |\mathbf{K}/2 \pm \mathbf{k}'| < k_a$. Since the right-hand side is independent of $\mathbf{k}$, the relative amplitude is directly proportional to the energy denominator:

$$a_{\mathbf{k}}(\mathbf{K}) = -\frac{v_0}{V} \frac{1}{E - \varepsilon_{\mathbf{k}_1} - \varepsilon_{\mathbf{k}_2}} \sum'_{\mathbf{k}'} a_{\mathbf{k}'}(\mathbf{K}). \quad (8.24)$$

Summing both sides over the allowed $\mathbf{k}$ and dividing by the common factor $\sum'_{\mathbf{k}} a_{\mathbf{k}}(\mathbf{K})$ (which is non-zero for a non-trivial bound state), we obtain the implicit eigenvalue equation:

$$1 = -\frac{v_0}{V} \sum'_{\mathbf{k}} \frac{1}{E - \varepsilon_{\mathbf{k}_1} - \varepsilon_{\mathbf{k}_2}}. \quad (8.25)$$

Defining the dispersion function

$$\Phi(E) \equiv \frac{1}{V} \sum'_{\mathbf{k}} \frac{1}{E - \varepsilon_{\mathbf{k}_1} - \varepsilon_{\mathbf{k}_2}}, \quad (8.26)$$

the eigenvalue condition is $\Phi(E) = -1/v_0$. The function $\Phi(E)$ exhibits simple poles at every unperturbed two-particle kinetic energy $\varepsilon_{\mathbf{k}_1} + \varepsilon_{\mathbf{k}_2} \geq 2\varepsilon_f$. For $E < 2\varepsilon_f$, all denominators in the sum are strictly negative, and $\Phi(E)$ is a monotonically increasing, negative, smooth function that tends to 0^- as $E \rightarrow -\infty$ and diverges toward $-\infty$ as $E \rightarrow (2\varepsilon_f)^-$.

Consequently, for any attractive interaction strength $v_0 > 0$, the horizontal line $-1/v_0$ must intersect the curve $\Phi(E)$ at precisely one energy $E_b < 2\varepsilon_f$. This demonstrates the remarkable result that an attractive interaction, no matter how weak, always creates a bound pair state below the continuum threshold of the Fermi sea.

8.1.5 Properties of the Bound State and Supercurrents

We now calculate the binding energy explicitly for a pair with zero center-of-mass momentum ($\mathbf{K} = \mathbf{0}$). In this case, $\mathbf{k}_1 = \mathbf{k}$, $\mathbf{k}_2 = -\mathbf{k}$, so that $\varepsilon_{\mathbf{k}_1} + \varepsilon_{\mathbf{k}_2} = 2\varepsilon_k = k^2/m$. Replacing the sum by an integral,

$$\Phi(E) = \int_{k_f}^{k_a} \frac{d^3k}{(2\pi\hbar)^3} \frac{1}{E - 2\varepsilon_k} = \frac{1}{2\pi^2\hbar^3} \int_{k_f}^{k_a} k^2 dk \frac{1}{E - 2\varepsilon_k}. \quad (8.27)$$

Changing variables from k to single-particle energy $\varepsilon = k^2/(2m)$, with $d\varepsilon = (k/m) dk$, gives

$$\Phi(E) = \frac{m}{2\pi^2\hbar^3} \int_{\varepsilon_f}^{\varepsilon_a} k(\varepsilon) \frac{d\varepsilon}{E - 2\varepsilon}. \quad (8.28)$$

Since the active energy layer is narrow ($\varepsilon_a - \varepsilon_f \ll \varepsilon_f$), we approximate $k(\varepsilon) \approx k_f$. Defining the single-spin density of states at the Fermi energy,

$$N(0) \equiv \frac{mk_f}{2\pi^2\hbar^3} = \int \frac{d^3k}{(2\pi\hbar)^3} \delta(\varepsilon_k - \varepsilon_f), \quad (8.29)$$

the integral evaluates to

$$\Phi(E) = -\frac{N(0)}{2} \int_{\varepsilon_f}^{\varepsilon_a} \frac{d\varepsilon}{2\varepsilon - E} = -\frac{N(0)}{2} \ln \left(\frac{2\varepsilon_a - E}{2\varepsilon_f - E} \right). \quad (8.30)$$

Setting $\Phi(E) = -1/v_0$ and parameterizing the bound state energy as $E_b = 2\varepsilon_f - 2\Delta$, where Δ is the binding energy per electron, we obtain

$$\ln \left(\frac{2(\varepsilon_a - \varepsilon_f) + 2\Delta}{2\Delta} \right) = \frac{2}{v_0 N(0)}. \quad (8.31)$$

Solving for Δ yields

$$\Delta = \frac{\varepsilon_a - \varepsilon_f}{e^{2/[v_0 N(0)]} - 1}. \quad (8.32)$$

The cutoff interval $\varepsilon_a - \varepsilon_f$ is set by the Debye energy $\hbar\omega_D$, which represents the highest frequency of acoustic lattice vibrations. In typical superconductors, $\hbar\omega_D/\varepsilon_f \sim 10^{-2}$ and the dimensionless coupling constant satisfies $v_0 N(0) \ll 1$ (weak coupling). In this regime, $e^{2/[v_0 N(0)]} \gg 1$, giving

$$\Delta \approx \hbar\omega_D \exp \left(-\frac{2}{v_0 N(0)} \right). \quad (8.33)$$

Equation (8.33) exhibits an essential singularity at $v_0 = 0$; the binding energy cannot be expanded in powers of the coupling constant v_0 , which explains why ordinary perturbation theory fails to capture superconductivity.

When the total momentum $\mathbf{K}$ is nonzero, the kinematically accessible phase space satisfying $k_f < |\mathbf{K}/2 \pm \mathbf{k}'| < k_a$ is substantially reduced due to the geometric overlap of the two shifted Fermi spheres. As a result, $|\Phi(E)|$ decreases, and the bound state energy shifts closer to the continuum threshold $2\varepsilon_f$. Maximum binding is achieved uniquely when $\mathbf{K} = \mathbf{0}$, where the two electrons have exactly opposite momenta ($\mathbf{k}$ and $-\mathbf{k}$).

The spatial structure of the Cooper pair is obtained from the Fourier transform of the amplitude:

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{V} \sum_{\mathbf{k}}' a_{\mathbf{k}}(\mathbf{0}) e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)/\hbar} \propto \int_{k_f}^{k_a} \frac{d^3k}{(2\pi\hbar)^3} \frac{e^{i\mathbf{k} \cdot \mathbf{r}/\hbar}}{2\varepsilon_k - E_b}, \quad (8.34)$$

where $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. Performing the angular integration reveals that the relative wave function is spherically symmetric (an s -wave state):

$$\varphi(r) \sim \frac{1}{r} \sin \left(\frac{k_f r}{\hbar} \right) f(r), \quad (8.35)$$

where $f(r)$ is an envelope function decaying on the characteristic length scale

$$\xi = \frac{2\hbar v_f}{\Delta} = \frac{2\hbar k_f}{m\Delta}. \quad (8.36)$$

This coherence length ξ represents the physical spatial extent of a Cooper pair, typically of the order of 10^3 to 10^4 Å, vastly exceeding the average interelectronic spacing in a metal.

In a macroscopic superconductor, the pairing instability affects all electrons within $\hbar\omega_D$ of the Fermi surface simultaneously. In a current-carrying state, every pair condenses into the same nonzero center-of-mass momentum $\mathbf{K}$. Dissipation requires scattering pairs out of this collective condensate; because all pairs are bound and phase-locked together, a single pair cannot be scattered independently without paying the macroscopic energy cost of breaking the superconducting gap. As a consequence, electric supercurrents persist without resistance.

8.1.6 Angular Momentum Decomposition of Nonlocal Potentials

If the interaction potential preserves relative orbital angular momentum, the two-body wave function $\psi(\mathbf{r}_1, \mathbf{r}_2, t)$ must remain an angular momentum eigenstate for all times. This requires the nonlocal interaction kernel to admit a partial-wave expansion:

$$\langle \mathbf{r}_1, \mathbf{r}_2 | \hat{\mathcal{V}} | \mathbf{r}'_1, \mathbf{r}'_2 \rangle = \sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}(\hat{\mathbf{r}}) Y_{lm}^*(\hat{\mathbf{r}}') \mathcal{V}_l(\mathbf{R} - \mathbf{R}'; r, r'), \quad (8.37)$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ denotes unit angular directions and $r = |\mathbf{r}|$. Taking the double Fourier transform yields the momentum-space decomposition:

$$\mathcal{V}_{\mathbf{k}, \mathbf{k}'}(\mathbf{K}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}(\hat{\mathbf{k}}) Y_{lm}^*(\hat{\mathbf{k}}') \mathcal{V}_{l, k, k'}(\mathbf{K}). \quad (8.38)$$

Here, $\mathcal{V}_{l, k, k'}(\mathbf{K})$ governs scattering between relative states of magnitude k' and k within the orbital angular momentum channel l . The isotropic Cooper interaction in Eq. (8.22) corresponds to a pure s -wave channel ($l = 0$), which directly explains why the Cooper pair forms an orbital singlet state. In other condensed matter systems, such as superfluid ^3He below 2.7 mK, pairing occurs in the p -wave channel ($l = 1$). Nucleons in atomic nuclei and neutron matter in neutron stars also experience pairing instabilities in higher angular momentum channels.

8.2 Problems

1. **Mean-square radius of a Cooper pair.** Using the relative wave function $\varphi(\mathbf{r})$ obtained in Eq. (8.34) for a zero center-of-mass momentum bound pair ($\mathbf{K} = \mathbf{0}$), calculate the mean-square radius of the pair:

$$\langle r^2 \rangle = \frac{\int d^3r r^2 |\varphi(\mathbf{r})|^2}{\int d^3r |\varphi(\mathbf{r})|^2}. \quad (8.39)$$

Express your result in terms of the Fermi momentum k_f and the pairing energy gap Δ .

2. **Role of the Fermi sea in pair binding.** To elucidate the essential role played by the filled Fermi sea in Cooper pairing, consider the two-body problem in the absence of a Fermi sea ($k_f = 0$). Assuming the attractive interaction $\mathcal{V}_{\mathbf{k}, \mathbf{k}'}(\mathbf{0}) = -v_0/V$ acts only up to an ultraviolet cutoff momentum k_a (i.e., for $0 < k, k' < k_a$), determine the exact condition on the interaction strength v_0 for a bound state ($E < 0$) to exist. Contrast this threshold condition with the Cooper instability at finite k_f .
3. **Phase space volume for moving pairs ($\mathbf{K} \neq \mathbf{0}$).** For pairs with nonzero center-of-mass momentum $\mathbf{K}$, the summation in Eq. (8.26) is restricted to relative wavevectors $\mathbf{k}'$ satisfying $k_f < |\mathbf{K}/2 \pm \mathbf{k}'| < k_a$. As a function of $K = |\mathbf{K}|$, estimate the volume of accessible $\mathbf{k}$ -space entering this sum. Show that this phase-space volume decreases monotonically as K increases, thereby reducing the pair binding energy.
4. **d -wave Cooper pairing.** Consider an attractive interaction that is active exclusively in the d -wave relative angular momentum channel ($l = 2$):

$$\mathcal{V}_{\mathbf{k}, \mathbf{k}'}(\mathbf{0}) = -\frac{v_2}{V} \sum_{m=-2}^2 Y_{2m}(\hat{\mathbf{k}}) Y_{2m}^*(\hat{\mathbf{k}}'), \quad (v_2 > 0), \quad (8.40)$$

for electrons in a thin momentum shell $k_f < k, k' < k_a$ around the Fermi surface, and zero otherwise. For zero center-of-mass momentum ($\mathbf{K} = \mathbf{0}$), find the possible bound state energies and the corresponding two-particle wave functions.

Chapter 9

Potential Scattering

9.1 Potential Scattering

Scattering processes provide the primary experimental probe for deciphering the microscopic laws of physics, the internal structure of matter, and the nature of fundamental forces. From Rutherford's discovery of the atomic nucleus to modern subatomic collision experiments, the analysis of how incoming beams of particles deflect upon encountering an interaction potential yields direct quantitative information about the underlying quantum dynamics.

In this chapter, we develop the quantum mechanical theory of nonrelativistic potential scattering. We begin with a time-dependent description using wave packets, demonstrating how localized wave trains evolve, interact with a target, and produce outgoing probability currents. We then define the fundamental observable quantities—the differential and total cross sections—and show how the stationary asymptotic form of the wave function determines the scattering amplitude.

Next, we establish the powerful method of partial-wave analysis for spherically symmetric potentials, introducing phase shifts and deriving the optical theorem from probability conservation. We develop systematic approximation methods, including the Born series, and analyze the general analytical properties of the scattering amplitude, such as its threshold behavior, Levinson's theorem, and low-energy Breit–Wigner resonances.

9.2 Wave Packets in Potential Scattering

To establish the physical foundation of scattering theory, we examine the collision of a particle with a localized, short-range potential $v(\mathbf{r})$. (Long-range interactions, such as the Coulomb potential, exhibit distinct asymptotic behavior and will be treated separately in Chapter 10.) Although stationary scattering states provide an indispensable computational framework, actual scattering experiments involve localized particles prepared in the distant past and detected in the far field. We therefore begin by describing the scattering process using time-dependent wave packets.

9.2.1 Initial Wave Packet and Stationary State Formulation

Consider a source located far to the left of the scattering target (Fig. 9-1). At an early time t_0 , long before the particle enters the interaction region, the particle is prepared in a localized wave packet state:

$$\psi(\mathbf{r}, t_0) = \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}, \quad (9.1)$$

where the amplitude distribution $a(\mathbf{k})$ is sharply peaked around a mean wavevector $\mathbf{k}_0$. In free space, the center of this packet propagates toward the target with group velocity $\mathbf{v}_0 = \hbar\mathbf{k}_0/m$.

Our objective is to determine the state $\psi(\mathbf{r}, t)$ at a time $t > t_0$ after the particle has interacted with the potential.

The time evolution is determined by expanding the state in terms of exact stationary energy eigenfunctions $\psi_{\mathbf{k}}(\mathbf{r})$ of the full Hamiltonian:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + v(\mathbf{r})\right)\psi_{\mathbf{k}}(\mathbf{r}) = E_k\psi_{\mathbf{k}}(\mathbf{r}), \quad E_k = \frac{\hbar^2 k^2}{2m} > 0. \quad (9.2)$$

Because the initial wave packet is localized far away in the asymptotic region where bound states are exponentially negligible, only the continuous spectrum ($E_k > 0$) enters the expansion:

$$\psi(\mathbf{r}, t_0) = \int \frac{d^3k}{(2\pi)^3} A(\mathbf{k}) \psi_{\mathbf{k}}(\mathbf{r}). \quad (9.3)$$

Once the expansion coefficients $A(\mathbf{k})$ are known, the state at any subsequent time t follows by applying the time-evolution operator:

$$\psi(\mathbf{r}, t) = \int \frac{d^3k}{(2\pi)^3} A(\mathbf{k}) \psi_{\mathbf{k}}(\mathbf{r}) e^{-iE_k(t-t_0)/\hbar}. \quad (9.4)$$

9.2.2 The Free-Particle Green's Function and Lippmann–Schwinger Equation

To construct the appropriate stationary states $\psi_{\mathbf{k}}(\mathbf{r})$, we rewrite the Schrödinger equation (9.2) as an inhomogeneous Helmholtz equation:

$$\left(\frac{\hbar^2}{2m}\nabla^2 + E_k\right)\psi_{\mathbf{k}}(\mathbf{r}) = v(\mathbf{r})\psi_{\mathbf{k}}(\mathbf{r}). \quad (9.5)$$

We define the outgoing Green's function $G(\mathbf{r}, k)$ as the response to a point source:

$$\left(\frac{\hbar^2}{2m}\nabla^2 + E_k\right)G(\mathbf{r}, k) = \delta(\mathbf{r}). \quad (9.6)$$

The stationary Schrödinger equation is then formally equivalent to the Lippmann–Schwinger integral equation:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \varphi_0(\mathbf{r}) + \int d^3r' G(\mathbf{r} - \mathbf{r}', k) v(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}'), \quad (9.7)$$

where $\varphi_0(\mathbf{r})$ is an arbitrary solution to the homogeneous free-particle equation $(\frac{\hbar^2}{2m}\nabla^2 + E_k)\varphi_0(\mathbf{r}) = 0$. For a particle incident along $\mathbf{k}$, we set $\varphi_0(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}$.

In momentum representation, the Green's function is given by the Fourier integral

$$G(\mathbf{r}, k) = \int \frac{d^3p}{(2\pi)^3} \frac{e^{i\mathbf{p}\cdot\mathbf{r}}}{E_k - \frac{\hbar^2 p^2}{2m} + i\epsilon} = -\frac{m}{2\pi^2 i r \hbar^2} \int_{-\infty}^{\infty} \frac{p dp e^{ipr}}{p^2 - (k + i\epsilon)^2}, \quad (9.8)$$

where the $+i\epsilon$ prescription ($\epsilon \rightarrow 0^+$) shifts the pole at $p = k$ into the upper half-plane and $p = -k$ into the lower half-plane, thereby selecting purely outgoing spherical waves. Closing the contour in the upper complex half-plane ($r > 0$) around the pole at $p = k + i\epsilon$ yields

$$G(\mathbf{r}, k) = -\frac{m}{2\pi\hbar^2} \frac{e^{ikr}}{r}. \quad (9.9)$$

Substituting Eq. (9.9) into Eq. (9.7), the stationary scattering state satisfies the integral equation

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} - \frac{m}{2\pi\hbar^2} \int d^3r' \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} v(\mathbf{r}')\psi_{\mathbf{k}}(\mathbf{r}'). \quad (9.10)$$

9.2.3 Asymptotic Scattering Amplitude

In an experimental configuration, detectors are placed far from the scattering center ($r \gg r'$, where r' is within the interaction range of v). In this far-field limit, we expand the distance:

$$|\mathbf{r} - \mathbf{r}'| = r - \hat{\mathbf{r}} \cdot \mathbf{r}' + \mathcal{O}\left(\frac{r'^2}{r}\right), \quad (9.11)$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$ is the unit vector pointing in the observation direction. Defining the scattered wavevector $\mathbf{k}' \equiv k\hat{\mathbf{r}}$ (which has the same magnitude k as the incident wavevector by energy conservation), the asymptotic form of the wave function becomes

$$\psi_{\mathbf{k}}(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{r}} + f(\mathbf{k} \rightarrow \mathbf{k}') \frac{e^{ikr}}{r}, \quad (9.12)$$

where the *scattering amplitude* $f(\mathbf{k} \rightarrow \mathbf{k}') \equiv f_{\mathbf{k}}(\hat{\mathbf{r}})$ is given by

$$f_{\mathbf{k}}(\hat{\mathbf{r}}) = -\frac{m}{2\pi\hbar^2} \int d^3r' e^{-i\mathbf{k}' \cdot \mathbf{r}'} v(\mathbf{r}') \psi_{\mathbf{k}}(\mathbf{r}'). \quad (9.13)$$

The scattering amplitude has dimensions of length and governs the angular distribution of scattered flux.

9.2.4 Time Evolution of the Scattering Wave Packet

We now connect the expansion in exact eigenstates [Eq. (9.3)] with the initial free wave packet [Eq. (9.1)]. Substituting the integral equation (9.10) into Eq. (9.1), we write

$$\psi(\mathbf{r}, t_0) = \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) \left[\psi_{\mathbf{k}}(\mathbf{r}) + \frac{m}{2\pi\hbar^2} \int d^3r' \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} v(\mathbf{r}') \psi_{\mathbf{k}}(\mathbf{r}') \right]. \quad (9.14)$$

The second term contains the integral

$$I_{\text{scatt}}(\mathbf{r}) \equiv \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) e^{ik|\mathbf{r}-\mathbf{r}'|} \psi_{\mathbf{k}}(\mathbf{r}'). \quad (9.15)$$

Since $a(\mathbf{k})$ is sharply localized near $\mathbf{k}_0$, we can approximate $\psi_{\mathbf{k}}(\mathbf{r}') \approx \psi_{\mathbf{k}_0}(\mathbf{r}')$ (away from sharp resonances) and linearize $k \approx \mathbf{k} \cdot \hat{\mathbf{k}}_0$. The integral then becomes

$$I_{\text{scatt}}(\mathbf{r}) \approx \psi_{\mathbf{k}_0}(\mathbf{r}') \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) e^{i\mathbf{k} \cdot \hat{\mathbf{k}}_0 |\mathbf{r}-\mathbf{r}'|} = \psi_{\mathbf{k}_0}(\mathbf{r}') \psi_0(\hat{\mathbf{k}}_0 |\mathbf{r}-\mathbf{r}'|, t_0). \quad (9.16)$$

The spatial argument $\hat{\mathbf{k}}_0 |\mathbf{r}-\mathbf{r}'|$ lies strictly downfield (to the right) of the scattering center, whereas the incident wave packet at t_0 is localized strictly upfield (far to the left). Consequently, $\psi_0(\hat{\mathbf{k}}_0 |\mathbf{r}-\mathbf{r}'|, t_0) = 0$, which implies that the second term in Eq. (9.14) vanishes identically:

$$A(\mathbf{k}) = a(\mathbf{k}). \quad (9.17)$$

The expansion coefficients for an incoming wave packet prepared in the distant past are identical whether expanded in free plane waves or in exact outgoing scattering eigenstates.

At a later time $t > t_0$, when the packet has collided with the potential, its state is given by Eq. (9.4):

$$\psi(\mathbf{r}, t) = \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) \psi_{\mathbf{k}}(\mathbf{r}) e^{-iE_k(t-t_0)/\hbar}. \quad (9.18)$$

Using the asymptotic form (9.12) in the observation region, we find

$$\psi(\mathbf{r}, t) = \psi_0(\mathbf{r}, t) + \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) \frac{e^{i[kr - E_k(t-t_0)/\hbar]}}{r} f_{\mathbf{k}}(\hat{\mathbf{r}}), \quad (9.19)$$

where

$$\psi_0(\mathbf{r}, t) = \int \frac{d^3k}{(2\pi)^3} a(\mathbf{k}) e^{i[\mathbf{k} \cdot \mathbf{r} - E_k(t-t_0)/\hbar]} \quad (9.20)$$

represents the unscattered wave packet that would have propagated freely in the absence of the potential. Assuming $f_{\mathbf{k}}(\hat{\mathbf{r}})$ varies slowly over the spectral width of $a(\mathbf{k})$, we evaluate it at the central wavevector $\mathbf{k}_0$ and approximate $k \approx \mathbf{k} \cdot \hat{\mathbf{k}}_0$, arriving at the transparent physical result:

$$\psi(\mathbf{r}, t) = \psi_0(\mathbf{r}, t) + \frac{f_{\mathbf{k}_0}(\hat{\mathbf{r}})}{r} \psi_0(\hat{\mathbf{k}}_0 r, t). \quad (9.21)$$

Equation (9.21) provides a clear heuristic picture of quantum scattering: the total wave function consists of the unperturbed incoming wave packet $\psi_0(\mathbf{r}, t)$ continuing along its original trajectory, accompanied by a spherical outgoing shell whose radial profile matches that of the incoming packet at distance r , weighted by the angular scattering amplitude $f_{\mathbf{k}_0}(\hat{\mathbf{r}})$ and attenuated by $1/r$ to conserve probability flux in three dimensions.

This simple asymptotic superposition breaks down in two important circumstances:

1. Near a narrow resonance ($E_0 \approx \hbar^2 k_0^2/2m$), where the scattering amplitude $f_{\mathbf{k}}(\hat{\mathbf{r}})$ varies rapidly with energy, introducing substantial phase delay and packet distortion (resulting in an exponentially decaying spatial tail).
2. For long-range potentials, such as the Coulomb potential $v(r) \propto 1/r$, whose infinite range distorts the asymptotic states at all distances and requires modified parabolic coordinates.

9.3 Scattering Cross Sections

The quantitative results of scattering experiments are universally expressed in terms of cross sections, which measure the effective geometrical area that a target presents to an incoming particle beam for a specified reaction.

9.3.1 Definition of the Differential Cross Section

Consider an incident beam of particles directed toward a target centered at the origin. The *differential cross section*, denoted by $d\sigma/d\Omega$, is defined as the number of particles scattered into an element of solid angle $d\Omega = \sin\theta d\theta d\phi$ per unit time, normalized by the incident particle flux (the number of incident particles crossing unit area perpendicular to the beam per unit time):

$$\frac{d\sigma}{d\Omega}(\theta, \phi) \equiv \frac{1}{J_{\text{inc}}} \frac{dN_{\text{scatt}}(\Omega)}{d\Omega dt}. \quad (9.22)$$

By definition, $d\sigma/d\Omega$ has dimensions of area per steradian (cm^2/sr or barns/sr, where $1 \text{ b} = 10^{-24} \text{ cm}^2$).

In terms of an individual quantum wave packet, the differential cross section represents the ratio of the total probability dP_{scatt} that the particle scatters into the solid angle $d\Omega$ to the integrated incident probability fluence F_{inc} per unit area upstream of the target:

$$\frac{d\sigma}{d\Omega} d\Omega = \frac{dP_{\text{scatt}}(\Omega)}{F_{\text{inc}}}. \quad (9.23)$$

Using the asymptotic wave packet solution obtained in Eq. (9.21), the scattered probability entering the detector area $dA = r^2 d\Omega$ integrated over the entire collision history is

$$dP_{\text{scatt}}(\Omega) = \int_{-\infty}^{\infty} dt \left(\frac{\hbar k_0}{m} \right) \left| \frac{f_{\mathbf{k}_0}(\Omega)}{r} \psi_0(\hat{\mathbf{k}}_0 r, t) \right|^2 r^2 d\Omega = |f_{\mathbf{k}_0}(\Omega)|^2 d\Omega \left[\frac{\hbar k_0}{m} \int_{-\infty}^{\infty} dt |\psi_0(\hat{\mathbf{k}}_0 r, t)|^2 \right]. \quad (9.24)$$

Similarly, the total incident probability crossing a unit transverse area at a point $\mathbf{r}_0$ upstream of the target is

$$F_{\text{inc}} = \frac{\hbar k_0}{m} \int_{-\infty}^{\infty} dt |\psi_0(\mathbf{r}_0, t)|^2. \quad (9.25)$$

Neglecting the longitudinal dispersive spreading of the packet during the flight from the source to the detector, the time integrals in Eqs. (9.24) and (9.25) are identical. Taking their ratio yields the fundamental relation:

$$\frac{d\sigma}{d\Omega}(\Omega) = |f_{\mathbf{k}_0}(\Omega)|^2. \quad (9.26)$$

Remarkably, the differential cross section is completely independent of the shape, duration, or transverse profile of the incident wave packet, justifying the use of stationary scattering states.

9.3.2 Probability Current Formulation and Total Cross Section

The same conclusion follows directly from the quantum probability current density in stationary scattering states:

$$\mathbf{j}(\mathbf{r}) = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) = \frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi). \quad (9.27)$$

For the incident plane wave $\psi_{\text{inc}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}$, the incident flux is uniform:

$$\mathbf{j}_{\text{inc}} = \frac{\hbar \mathbf{k}}{m} = v \hat{\mathbf{k}}, \quad J_{\text{inc}} = |\mathbf{j}_{\text{inc}}| = \frac{\hbar k}{m}. \quad (9.28)$$

For the scattered spherical wave $\psi_{\text{scatt}}(\mathbf{r}) = f(\Omega) \frac{e^{ikr}}{r}$, the leading radial gradient as $r \rightarrow \infty$ gives

$$\nabla \psi_{\text{scatt}} = ik f(\Omega) \frac{e^{ikr}}{r} \hat{\mathbf{r}} + \mathcal{O}(r^{-2}), \quad (9.29)$$

which yields an exclusively radial outward current density:

$$\mathbf{j}_{\text{scatt}}(\mathbf{r}) = \frac{\hbar k}{mr^2} |f(\Omega)|^2 \hat{\mathbf{r}} + \mathcal{O}(r^{-3}). \quad (9.30)$$

The scattered particle rate passing through the surface element $dA = r^2 d\Omega \hat{\mathbf{r}}$ is

$$dI_{\text{scatt}} = \mathbf{j}_{\text{scatt}} \cdot d\mathbf{A} = \frac{\hbar k}{m} |f(\Omega)|^2 d\Omega. \quad (9.31)$$

Dividing by the incident flux $J_{\text{inc}} = \hbar k/m$ immediately reproduces Eq. (9.26):

$$\frac{d\sigma}{d\Omega} = \frac{1}{J_{\text{inc}}} \frac{dI_{\text{scatt}}}{d\Omega} = |f(\Omega)|^2. \quad (9.32)$$

Integrating the differential cross section over all 4π steradians gives the *total elastic cross section* σ :

$$\sigma \equiv \int \frac{d\sigma}{d\Omega} d\Omega = \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta |f(\theta, \phi)|^2. \quad (9.33)$$

The total cross section represents the total effective area removed from the incident beam by elastic scattering.

9.4 Partial Wave Analysis

When the interaction potential is spherically symmetric, $v(\mathbf{r}) = v(r)$, the Hamiltonian commutes with the orbital angular momentum operators $\hat{\mathbf{L}}^2$ and $\hat{L}_z$. Consequently, angular momentum is conserved throughout the collision, and the three-dimensional scattering problem decomposes into decoupled one-dimensional radial problems for each orbital angular momentum quantum number l .

9.4.1 Partial Wave Expansion of Plane and Scattering Waves

We choose the coordinate system such that the incident wavevector $\mathbf{k}$ defines the positive z -axis. The incident plane wave $e^{i\mathbf{k}\cdot\mathbf{r}} = e^{ikz} = e^{ikr\cos\theta}$ possesses complete azimuthal symmetry about $\mathbf{k}$ and can be expanded in terms of spherical Bessel functions $j_l(kr)$ and Legendre polynomials $P_l(\cos\theta)$:

$$\begin{aligned} e^{i\mathbf{k}\cdot\mathbf{r}} &= \sum_{l=0}^{\infty} i^l (2l+1) j_l(kr) P_l(\cos\theta) \\ &= \frac{1}{2} \sum_{l=0}^{\infty} i^l (2l+1) \left[h_l^{(2)}(kr) + h_l^{(1)}(kr) \right] P_l(\cos\theta), \end{aligned} \quad (9.34)$$

where $h_l^{(1)}(kr) = j_l(kr) + in_l(kr)$ and $h_l^{(2)}(kr) = [h_l^{(1)}(kr)]^* = j_l(kr) - in_l(kr)$ are spherical Hankel functions representing outgoing and incoming spherical waves, respectively. The Legendre polynomials are related to the spherical harmonics with $m = 0$ by

$$P_l(\cos\theta) = \sqrt{\frac{4\pi}{2l+1}} Y_{l0}(\theta, \phi). \quad (9.35)$$

Because the potential is central, azimuthal symmetry is strictly preserved: the scattering amplitude depends only on the polar scattering angle θ , $f_{\mathbf{k}}(\Omega) = f(\theta)$, and the scattered wave couples only within the same angular momentum channel ($l, m = 0$).

We expand the full stationary scattering wave function $\psi_{\mathbf{k}}(\mathbf{r})$ in the partial wave series:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{l=0}^{\infty} i^l (2l+1) R_l(r) P_l(\cos\theta). \quad (9.36)$$

Substituting Eq. (9.36) into the Schrödinger equation yields the decoupled radial equations for $u_l(r) \equiv rR_l(r)$:

$$\left[\frac{d^2}{dr^2} + k^2 - \frac{l(l+1)}{r^2} - \frac{2m}{\hbar^2} v(r) \right] u_l(r) = 0, \quad (9.37)$$

subject to the regular boundary condition at the origin: $u_l(0) = 0$, or equivalently $R_l(r) \propto r^l$ as $r \rightarrow 0$.

9.4.2 Phase Shifts and the S -Matrix

Beyond the range R_0 of a short-range potential (where $v(r) \approx 0$ for $r > R_0$), the radial equation reduces to the spherical Bessel differential equation. The general solution outside the interaction zone is a linear combination of incoming and outgoing spherical waves:

$$R_l(r) = B_l \left[h_l^{(2)}(kr) + S_l(E) h_l^{(1)}(kr) \right]. \quad (9.38)$$

In the absence of a potential ($v(r) = 0$), the physical regular solution is $R_l(r) = j_l(kr) = \frac{1}{2} [h_l^{(2)}(kr) + h_l^{(1)}(kr)]$, requiring $B_l = 1/2$ and $S_l = 1$.

Because the potential cannot alter the incoming wave packet approaching from infinity, the coefficient of the incoming wave $h_l^{(2)}(kr)$ must remain unperturbed: $B_l = 1/2$. All dynamical effects of the scattering potential are encoded entirely in the complex coefficient $S_l(E)$, known as the *scattering matrix element* (or *S-matrix*) in the channel l .

Conservation of probability requires that the net radial probability flux through any sphere enclosing the target vanishes in a stationary state:

$$j_r(r) = \frac{\hbar}{m} \operatorname{Im} \left(R_l^*(r) \frac{dR_l(r)}{dr} \right) = 0. \quad (9.39)$$

Using the asymptotic Wronskian relation for spherical Hankel functions, $W[h_l^{(1)}(\rho), h_l^{(2)}(\rho)] = -2i/\rho^2$, Eq. (9.39) directly implies that the S -matrix is unitary:

$$|S_l(E)| = 1 \quad (\text{for purely elastic scattering}). \quad (9.40)$$

We can therefore parameterize $S_l(E)$ in terms of a single real quantity $\delta_l(E)$, the *scattering phase shift*:

$$S_l(E) = e^{2i\delta_l(E)}, \quad (\delta_l \in \mathbb{R}). \quad (9.41)$$

The phase shift δ_l measures the asymptotic phase displacement of the outgoing partial wave relative to the unscattered free wave.

9.4.3 Cross Sections in Terms of Phase Shifts

Using the asymptotic behavior of the spherical Hankel functions as $kr \rightarrow \infty$,

$$h_l^{(1)}(kr) \sim \frac{(-i)^{l+1}}{kr} e^{ikr}, \quad h_l^{(2)}(kr) \sim \frac{i^{l+1}}{kr} e^{-ikr}, \quad (9.42)$$

the radial solution outside the potential assumes the asymptotic form

$$R_l(r) \xrightarrow{r \rightarrow \infty} \frac{e^{i\delta_l}}{kr} \sin\left(kr - \frac{l\pi}{2} + \delta_l\right). \quad (9.43)$$

Comparing the full expansion of $\psi_{\mathbf{k}}(\mathbf{r})$ with the asymptotic scattering boundary condition,

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} + \frac{1}{2} \sum_{l=0}^{\infty} i^l (2l+1) (e^{2i\delta_l} - 1) h_l^{(1)}(kr) P_l(\cos \theta) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{r}} + f(\theta) \frac{e^{ikr}}{r}, \quad (9.44)$$

we obtain the partial-wave expansion of the scattering amplitude:

$$\begin{aligned} f(\theta) &= \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) (e^{2i\delta_l} - 1) P_l(\cos \theta) \\ &= \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta) = \sum_{l=0}^{\infty} (2l+1) f_l(k) P_l(\cos \theta), \end{aligned} \quad (9.45)$$

where the *partial wave amplitude* is defined by

$$f_l(k) \equiv \frac{e^{2i\delta_l} - 1}{2ik} = \frac{e^{i\delta_l} \sin \delta_l}{k} = \frac{1}{k(\cot \delta_l - i)}. \quad (9.46)$$

To determine the total cross section σ , we integrate $|f(\theta)|^2$ over the unit sphere. Applying the orthogonality relation for Legendre polynomials,

$$\int_{-1}^1 P_l(\cos \theta) P_{l'}(\cos \theta) d(\cos \theta) = \frac{2}{2l+1} \delta_{ll'}, \quad (9.47)$$

all cross-interference terms between different partial waves integrate to zero, yielding

$$\sigma = \int |f(\theta)|^2 d\Omega = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l = \sum_{l=0}^{\infty} \sigma_l, \quad (9.48)$$

where the *partial cross section* for channel l is

$$\sigma_l = \frac{4\pi}{k^2} (2l+1) \sin^2 \delta_l. \quad (9.49)$$

Since $\sin^2 \delta_l \leq 1$, each partial cross section satisfies the strict *unitarity bound*:

$$\sigma_l \leq \frac{4\pi(2l+1)}{k^2}. \quad (9.50)$$

The maximum cross section in each channel occurs at a resonance where $\delta_l(E) = (n + 1/2)\pi$. Although different partial waves do not interfere in the total cross section σ , interference between different l values plays a crucial role in shaping the angular dependence of the differential cross section $d\sigma/d\Omega = |f(\theta)|^2$.

9.4.4 Physical Interpretation of the Phase Shift

The sign of the phase shift δ_l directly reveals whether the interaction is predominantly attractive or repulsive:

- **Attractive potential** ($v(r) < 0$): Inside the potential well, the local kinetic energy increases, shortening the de Broglie wavelength. Consequently, the radial wave function completes its oscillations more rapidly than a free wave, pulling the asymptotic wave inward toward the origin:

$$\delta_l > 0. \quad (9.51)$$

- **Repulsive potential** ($v(r) > 0$): The local kinetic energy decreases, lengthening the local wavelength and pushing the wave outward away from the origin:

$$\delta_l < 0. \quad (9.52)$$

Semiclassically, an incident particle with angular momentum $L \approx \hbar(l + 1/2)$ and momentum $p = \hbar k$ has an impact parameter $b \approx l/k$. If the potential has an effective range R_0 , particles with impact parameter $b \gg R_0$ (i.e., $l \gg kR_0$) do not penetrate the potential barrier and experience virtually no scattering:

$$\delta_l \approx 0 \quad \text{for } l \gg kR_0. \quad (9.53)$$

At low collision energies ($kR_0 \ll 1$), only the *s*-wave ($l = 0$) contributes significantly to scattering, causing the angular distribution to become purely isotropic.

9.5 The Optical Theorem

A profound consequence of the conservation of probability in quantum mechanics is the *optical theorem* (also known as the Bohr–Peierls–Placzek relation). It establishes a direct, universal connection between the total cross section σ and the imaginary part of the forward scattering amplitude $f(0) \equiv f(\theta = 0)$.

9.5.1 Derivation via Partial Wave Analysis

For a central potential, the optical theorem can be established algebraically from the partial wave expansion. Taking the imaginary part of the scattering amplitude in Eq. (9.45),

$$\text{Im } f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l P_l(\cos \theta). \quad (9.54)$$

In the exact forward direction ($\theta = 0$), the Legendre polynomials satisfy $P_l(1) = 1$ for all l . Hence,

$$\text{Im } f(0) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l. \quad (9.55)$$

Comparing Eq. (9.55) with the total cross section formula in Eq. (9.48),

$$\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l, \quad (9.56)$$

we immediately obtain the optical theorem:

$$\sigma = \frac{4\pi}{k} \text{Im } f(0). \quad (9.57)$$

9.5.2 Physical Origin: Conservation of Probability and the Forward Shadow

The optical theorem is by no means restricted to spherically symmetric potentials; it is an exact, general expression of probability conservation. Physically, every particle scattered out of the incident beam into non-forward angles ($\theta \neq 0$) must correspond to an identical loss of particle flux in the original beam direction. This attenuation is produced by destructive quantum interference between the unscattered incident wave and the forward scattered wave, generating a “shadow” downstream of the scatterer.

To demonstrate this mechanism explicitly for an arbitrary potential, consider the time-dependent probability current density:

$$\mathbf{j}(\mathbf{r}, t) = \frac{\hbar}{m} \text{Im} [\psi^*(\mathbf{r}, t) \nabla \psi(\mathbf{r}, t)]. \quad (9.58)$$

Substituting the asymptotic wave packet solution [Eq. (9.21)],

$$\psi(\mathbf{r}, t) = \psi_0(\mathbf{r}, t) + \frac{f_{\mathbf{k}_0}(\hat{\mathbf{r}})}{r} \psi_0(\hat{\mathbf{k}}_0 r, t), \quad (9.59)$$

the radial current in the far field ($r \rightarrow \infty$) decomposes into three distinct contributions:

$$j_r(\mathbf{r}, t) = j_{r,0}(\mathbf{r}, t) + j_{r,\text{scatt}}(\mathbf{r}, t) + j_{r,\text{int}}(\mathbf{r}, t), \quad (9.60)$$

where $j_{r,0}$ is the unperturbed incident current, $j_{r,\text{scatt}}$ is the purely outgoing scattered current, and $j_{r,\text{int}}$ is the interference term:

$$j_{r,\text{int}}(\mathbf{r}, t) = \frac{\hbar k_0}{mr} \text{Im} \left\{ i f_{\mathbf{k}_0}(\hat{\mathbf{r}}) \psi_0^*(\mathbf{r}, t) \psi_0(\hat{\mathbf{k}}_0 r, t) (1 + \hat{\mathbf{k}}_0 \cdot \hat{\mathbf{r}}) \right\}, \quad (9.61)$$

obtained using $\partial \psi_0(\hat{\mathbf{k}}_0 r, t) / \partial r \approx i k_0 \psi_0(\hat{\mathbf{k}}_0 r, t)$ and $\partial \psi_0(\mathbf{r}, t) / \partial r \approx i(\mathbf{k}_0 \cdot \hat{\mathbf{r}}) \psi_0(\mathbf{r}, t)$, retaining terms through order $\mathcal{O}(r^{-2})$ upon angular integration.

Because $\psi_0(\hat{\mathbf{k}}_0 r, t)$ is non-zero only after the collision, the product $\psi_0^*(\mathbf{r}, t) \psi_0(\hat{\mathbf{k}}_0 r, t)$ in the far field is non-vanishing strictly along the forward axis ($\hat{\mathbf{r}} = \hat{\mathbf{k}}_0$, where $\theta = 0$ and $\hat{\mathbf{k}}_0 \cdot \hat{\mathbf{r}} = 1$). Outside the forward beam, the unscattered packet and the scattered spherical shell do not overlap spatially.

To evaluate the net interference flux passing through a large sphere of radius r , we integrate over all solid angles:

$$I_{\text{int}}(t) = \int r^2 d\Omega j_{r,\text{int}}(\mathbf{r}, t) = \frac{2\hbar k_0 r}{m} \text{Im} \left\{ i f_{\mathbf{k}_0}(0) \psi_0(\hat{\mathbf{k}}_0 r, t) \int d\Omega \psi_0^*(\mathbf{r}, t) \right\}. \quad (9.62)$$

For any Fourier component $e^{i\mathbf{k} \cdot \mathbf{r}}$ of $\psi_0(\mathbf{r}, t)$, the solid angle integral is elementary:

$$\int d\Omega e^{i\mathbf{k} \cdot \mathbf{r}} = 2\pi \int_{-1}^1 e^{ikr \cos \theta} d(\cos \theta) = \frac{2\pi}{ikr} (e^{ikr} - e^{-ikr}). \quad (9.63)$$

Linearizing near $\mathbf{k}_0$ ($kr \approx \mathbf{k} \cdot \hat{\mathbf{k}}_0 r$), this angular average gives

$$\int d\Omega \psi_0(\mathbf{r}, t) = \frac{2\pi}{ik_0 r} \left[\psi_0(\hat{\mathbf{k}}_0 r, t) - \psi_0(-\hat{\mathbf{k}}_0 r, t) \right]. \quad (9.64)$$

Since the packet is moving forward, the backward component $\psi_0(-\hat{\mathbf{k}}_0 r, t)$ vanishes at positive times t . Substituting the complex conjugate of Eq. (9.64) into Eq. (9.62), the factors of r cancel:

$$\begin{aligned} I_{\text{int}}(t) &= \frac{2\hbar k_0 r}{m} \text{Im} \left\{ i f_{\mathbf{k}_0}(0) \psi_0(\hat{\mathbf{k}}_0 r, t) \left(-\frac{2\pi}{ik_0 r} \psi_0^*(\hat{\mathbf{k}}_0 r, t) \right) \right\} \\ &= -\frac{4\pi\hbar}{m} \text{Im} f_{\mathbf{k}_0}(0) |\psi_0(\hat{\mathbf{k}}_0 r, t)|^2. \end{aligned} \quad (9.65)$$

The negative sign signifies a net removal of flux from the forward direction.

Total conservation of probability demands that the net integrated probability flux exiting the sphere equals zero at all times:

$$\int_{-\infty}^{\infty} dt \int r^2 d\Omega j_{r,\text{scatt}}(\mathbf{r}, t) + \int_{-\infty}^{\infty} dt I_{\text{int}}(t) = 0. \quad (9.66)$$

Using the scattered flux from Eq. (9.24),

$$\int_{-\infty}^{\infty} dt \int r^2 d\Omega j_{r,\text{scatt}} = \sigma \frac{\hbar k_0}{m} \int_{-\infty}^{\infty} dt |\psi_0(\hat{\mathbf{k}}_0 r, t)|^2, \quad (9.67)$$

and balancing this with $-\int_{-\infty}^{\infty} dt I_{\text{int}}(t)$ yields

$$\sigma \frac{\hbar k_0}{m} \int_{-\infty}^{\infty} dt |\psi_0(\hat{\mathbf{k}}_0 r, t)|^2 = \frac{4\pi\hbar}{m} \text{Im} f_{\mathbf{k}_0}(0) \int_{-\infty}^{\infty} dt |\psi_0(\hat{\mathbf{k}}_0 r, t)|^2, \quad (9.68)$$

which immediately gives the general optical theorem:

$$\sigma = \frac{4\pi}{k_0} \text{Im} f(0). \quad (9.69)$$

This derivation makes it transparent that the imaginary part of the forward scattering amplitude is a direct measure of beam extinction caused by shadow formation.

9.6 The Born Approximation

When the scattering interaction is sufficiently weak, or when the incident particle energy is high compared to the potential depth, the wave function is only slightly perturbed from a free plane wave. Under these conditions, perturbation theory provides accurate and computationally tractable closed-form approximations for the scattering amplitude and phase shifts, known as the *Born approximation*.

9.6.1 Born Approximation for Partial Wave Phase Shifts

To derive an exact integral representation for the phase shift in a central potential $v(r)$, we substitute the partial-wave expansion of $\psi_{\mathbf{k}}(\mathbf{r})$ [Eq. (9.36)] into the general expression for the scattering amplitude [Eq. (9.13)]:

$$f(\theta) = -\frac{m}{2\pi\hbar^2} \int d^3r' e^{-i\mathbf{k}' \cdot \mathbf{r}'} v(r') \sum_{l=0}^{\infty} i^l (2l+1) R_l(r') P_l(\hat{\mathbf{k}} \cdot \hat{\mathbf{r}}'). \quad (9.70)$$

Expanding the plane wave $e^{-i\mathbf{k}'\cdot\mathbf{r}'}$ in spherical harmonics using the addition theorem,

$$e^{-i\mathbf{k}'\cdot\mathbf{r}'} = 4\pi \sum_{l'=0}^{\infty} (-i)^{l'} j_{l'}(kr') \sum_{m'=-l'}^{l'} Y_{l'm'}(\hat{\mathbf{k}}') Y_{l'm'}^*(\hat{\mathbf{r}}'), \quad (9.71)$$

and carrying out the angular integration over $\hat{\mathbf{r}}'$, the orthogonality of the spherical harmonics collapses the double sum to

$$f(\theta) = -\frac{2m}{\hbar^2} \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \int_0^{\infty} r^2 dr v(r) j_l(kr) R_l(r). \quad (9.72)$$

Comparing Eq. (9.72) term by term with the partial-wave expansion $f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta)$, we establish the exact relation:

$$e^{i\delta_l} \sin \delta_l = -\frac{2mk}{\hbar^2} \int_0^{\infty} r^2 dr v(r) j_l(kr) R_l(r). \quad (9.73)$$

If the potential produces only a modest perturbation on the radial wave function (which is typically true for large angular momentum $l \gg kR_0$ due to centrifugal barrier exclusion, or for high energies), the phase shift is small ($|\delta_l| \ll 1$). We may then approximate $e^{i\delta_l} \sin \delta_l \approx \delta_l$ on the left, and replace the true radial wave function $R_l(r)$ by its unperturbed free-particle counterpart $j_l(kr)$ in the integrand:

$$\delta_l \approx -\frac{2mk}{\hbar^2} \int_0^{\infty} r^2 dr v(r) [j_l(kr)]^2. \quad (9.74)$$

Equation (9.74) is the *first Born approximation for the phase shift* δ_l . We note that this approximation assumes $|v(r)|$ is bounded; it cannot be applied directly to singular or hard-core potentials ($v(r) \rightarrow \infty$ for $r < r_0$), where the physical wave function vanishes ($R_l \equiv 0$) inside the core while $j_l(kr)$ remains finite.

9.6.2 First Born Approximation for the Scattering Amplitude

When the potential is weak across all angular momentum channels, we approximate the full wave function $\psi_{\mathbf{k}}(\mathbf{r}')$ in the Lippmann–Schwinger integral equation (9.13) directly by the incident plane wave $\psi_{\mathbf{k}}^{(0)}(\mathbf{r}') = e^{i\mathbf{k}\cdot\mathbf{r}'}$. This yields the *first Born approximation*:

$$f^{(1)}(\mathbf{k} \rightarrow \mathbf{k}') = -\frac{m}{2\pi\hbar^2} \int d^3r' e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}'} v(\mathbf{r}') = -\frac{m}{2\pi\hbar^2} \tilde{v}(\mathbf{q}), \quad (9.75)$$

where $\tilde{v}(\mathbf{q}) = \int d^3r e^{-i\mathbf{q}\cdot\mathbf{r}} v(\mathbf{r})$ is the three-dimensional spatial Fourier transform of the interaction potential, evaluated at the momentum transfer wavevector

$$\mathbf{q} \equiv \mathbf{k}' - \mathbf{k}. \quad (9.76)$$

Since elastic scattering conserves energy, $|\mathbf{k}'| = |\mathbf{k}| = k$, the magnitude of the momentum transfer depends solely on the scattering angle θ :

$$q = |\mathbf{k}' - \mathbf{k}| = \sqrt{k^2 + k^2 - 2k^2 \cos \theta} = 2k \sin \left(\frac{\theta}{2} \right). \quad (9.77)$$

For a spherically symmetric potential $v(r)$, the angular integration can be performed explicitly:

$$\tilde{v}(q) = \int_0^{\infty} r^2 dr v(r) \int_{-1}^1 d(\cos \alpha) 2\pi e^{-iqr \cos \alpha} = \frac{4\pi}{q} \int_0^{\infty} r dr v(r) \sin(qr), \quad (9.78)$$

which implies that $f^{(1)}(\theta)$ is purely real.

9.6.3 Application to the Yukawa and Coulomb Potentials

As an illustrative and physically significant application, consider the *Yukawa potential*:

$$v(r) = a \frac{e^{-\kappa r}}{r}, \quad (9.79)$$

which describes screened electrostatic interactions in plasmas and electrolytes (Debye screening), as well as short-range meson-mediated nuclear forces. The Fourier transform is readily computed:

$$\tilde{v}(\mathbf{q}) = \frac{4\pi a}{q} \int_0^\infty dr e^{-\kappa r} \sin(qr) = \frac{4\pi a}{q^2 + \kappa^2}. \quad (9.80)$$

Substituting this into Eq. (9.75), the first Born scattering amplitude is

$$f^{(1)}(\theta) = -\frac{2ma}{\hbar^2} \frac{1}{q^2 + \kappa^2} = -\frac{2ma}{\hbar^2} \frac{1}{4k^2 \sin^2(\theta/2) + \kappa^2}. \quad (9.81)$$

The resulting differential cross section is

$$\frac{d\sigma}{d\Omega} = |f^{(1)}(\theta)|^2 = \frac{a^2}{\left[4E_k \sin^2(\theta/2) + \frac{\hbar^2 \kappa^2}{2m}\right]^2}, \quad (9.82)$$

where $E_k = \hbar^2 k^2 / (2m)$.

In the unscreened Coulomb limit ($\kappa \rightarrow 0$, $a = Z_1 Z_2 e^2$), Eq. (9.82) reproduces the celebrated classical Rutherford scattering formula:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Rutherford}} = \left(\frac{Z_1 Z_2 e^2}{4E_k \sin^2(\theta/2)}\right)^2. \quad (9.83)$$

Although Eq. (9.82) coincides with the exact quantum mechanical cross section for a pure Coulomb potential, this exact agreement in the Born approximation is an accidental mathematical coincidence. Because of the long-range tail of $1/r$, the standard plane-wave Born series does not rigorously converge for an unscreened Coulomb potential, a pathology that requires specialized treatment via parabolic coordinates.

9.7 Properties of the Scattering Amplitude

In this section, we analyze the analytic and threshold properties of the partial wave phase shifts $\delta_l(k)$, establishing how low-energy scattering parameters are intimately connected to the bound states and virtual states of the interaction potential.

9.7.1 Logarithmic Derivative and Phase Shifts

For clarity, consider a short-range central potential that vanishes identically beyond a finite radius b : $v(r) = 0$ for $r > b$ (the conclusions extend to any potential falling off faster than r^{-2} as $r \rightarrow \infty$). In the external region ($r \geq b$), the radial wave function is a superposition of free spherical Hankel functions:

$$R_l(r) = \frac{1}{2} \left[h_l^{(2)}(kr) + e^{2i\delta_l} h_l^{(1)}(kr) \right] = j_l(kr) + i f_l(k) k h_l^{(1)}(kr). \quad (9.84)$$

Within the interaction zone ($r \leq b$), $R_l(r)$ is obtained by integrating the radial Schrödinger equation (9.37) subject to regularity at the origin ($R_l(0) = 0$). Matching $R_l(r)$ and its radial derivative continuously at the boundary $r = b$ determines the *logarithmic derivative*:

$$\alpha_l(E) \equiv \frac{1}{R_l(r)} \frac{dR_l(r)}{dr} \Big|_{r=b}. \quad (9.85)$$

Equating α_l to the logarithmic derivative of the exterior solution gives

$$\alpha_l = \frac{\frac{dh_l^{(2)}(kr)}{dr} + e^{2i\delta_l} \frac{dh_l^{(1)}(kr)}{dr}}{h_l^{(2)}(kr) + e^{2i\delta_l} h_l^{(1)}(kr)} \Big|_{r=b}. \quad (9.86)$$

Solving Eq. (9.86) for the S -matrix element $S_l = e^{2i\delta_l}$ yields

$$e^{2i\delta_l} - 1 = \frac{2 \left(\frac{dj_l}{dr} - \alpha_l j_l \right)}{\alpha_l h_l^{(1)} - \frac{dh_l^{(1)}}{dr}} \Big|_{r=b}. \quad (9.87)$$

Expressing $h_l^{(1)} = j_l + in_l$ and utilizing the algebraic identity $e^{2i\delta_l} - 1 = 2i/(\cot \delta_l - i)$, we find

$$\cot \delta_l(k) = \frac{\frac{dn_l(kr)}{dr} - \alpha_l(E)n_l(kr)}{\frac{dj_l(kr)}{dr} - \alpha_l(E)j_l(kr)} \Big|_{r=b}. \quad (9.88)$$

Thus, the entire influence of the interior potential on the exterior phase shift is distilled into the boundary value $\alpha_l(E)$.

Example: Hard Sphere Scattering. For an impenetrable hard sphere of radius b , $v(r) = \infty$ for $r < b$, requiring the wave function to vanish at the contact surface: $R_l(b) = 0$. Consequently, $\alpha_l \rightarrow \infty$, and Eq. (9.88) reduces to

$$\cot \delta_l = \frac{n_l(kb)}{j_l(kb)}. \quad (9.89)$$

For s -waves ($l = 0$), where $j_0(\rho) = \sin \rho/\rho$ and $n_0(\rho) = -\cos \rho/\rho$, this gives $\cot \delta_0 = -\cot(kb)$, which implies

$$\delta_0 = -kb. \quad (9.90)$$

As expected for a purely repulsive core, the phase shift is negative, reflecting the outward displacement of the wave function by the sphere radius b .

9.7.2 Threshold Behavior and the Scattering Length

At low incident momenta ($kb \ll 1$), we use the small-argument asymptotic expansions of the spherical Bessel functions:

$$j_l(\rho) \approx \frac{\rho^l}{(2l+1)!!}, \quad n_l(\rho) \approx -\frac{(2l-1)!!}{\rho^{l+1}}, \quad (9.91)$$

where $(2l+1)!! = 1 \cdot 3 \cdot 5 \cdots (2l+1)$ and $(-1)!! \equiv 1$. Evaluating Eq. (9.88) in this limit yields

$$\cot \delta_l \approx \frac{(2l-1)!!(2l+1)!!}{(kb)^{2l+1}} \left[\frac{l+1+b\alpha_l(E)}{l-b\alpha_l(E)} \right]. \quad (9.92)$$

As $k \rightarrow 0$, the logarithmic derivative approaches a constant $\alpha_l(0)$. Therefore, the phase shifts scale generically as

$$\delta_l(k) \propto k^{2l+1}, \quad (k \rightarrow 0). \quad (9.93)$$

Because higher angular momentum phase shifts vanish rapidly with increasing l (e.g., $\delta_1 \propto k^3$, $\delta_2 \propto k^5$), only the s -wave channel ($l = 0$) survives in the zero-energy limit:

$$\frac{d\sigma}{d\Omega} \xrightarrow{k \rightarrow 0} \frac{\sin^2 \delta_0}{k^2}. \quad (9.94)$$

Low-energy scattering is universally isotropic. For instance, for the hard sphere with $\delta_0 = -kb$, the total cross section becomes $\sigma = 4\pi b^2$, which is exactly four times its classical geometric cross section πb^2 . (In quantum mechanics, diffraction around the sphere doubles the forward shadow, contributing an additional πb^2 equal to the classical reflection area).

For s -waves ($l = 0$), Eq. (9.92) motivates the definition of the *scattering length* a :

$$\lim_{k \rightarrow 0} k \cot \delta_0 = -\frac{1}{a}, \quad (9.95)$$

where

$$a = -\lim_{k \rightarrow 0} \frac{\tan \delta_0(k)}{k} = b \left[1 - \frac{1}{b\alpha_0(0)} \right]. \quad (9.96)$$

In terms of the scattering length, the zero-energy total cross section is

$$\sigma_0 = 4\pi a^2. \quad (9.97)$$

9.7.3 Analytic Continuation, Bound States, and Virtual States

The partial-wave amplitude $f_l(k) = (e^{2i\delta_l} - 1)/(2ik)$ can be analytically continued to complex values of k . Bound states correspond to normalizable solutions that decay exponentially as $r \rightarrow \infty$ in the absence of an incident beam:

$$R_l(r) \propto h_l^{(1)}(ikr) \xrightarrow{r \rightarrow \infty} \frac{e^{-\kappa r}}{r}, \quad (\kappa > 0), \quad (9.98)$$

with negative bound state energy $E_b = -\hbar^2 \kappa^2 / (2m)$. In Eq. (9.84), an unperturbed incoming wave $h_l^{(2)}(kr) \propto e^{+i\kappa r}$ would diverge exponentially; hence, a bound state requires the coefficient of the outgoing wave to diverge relative to the incoming wave. This condition corresponds precisely to a simple pole of the S -matrix:

$$S_l(E) \rightarrow \infty \iff \cot \delta_l(E) = i. \quad (9.99)$$

At an s -wave bound state close to threshold ($\kappa b \ll 1$), matching logarithmic derivatives requires

$$\alpha_0(E_b) = \frac{1}{h_0^{(1)}(i\kappa r)} \frac{dh_0^{(1)}(i\kappa r)}{dr} \bigg|_{r=b} = -\kappa. \quad (9.100)$$

Substituting this into Eq. (9.95) gives

$$-\frac{1}{a} \approx -\kappa \implies a \approx \frac{1}{\kappa} > 0. \quad (9.101)$$

The scattering amplitude near threshold becomes

$$f_0(k) = \frac{1}{k \cot \delta_0 - ik} \approx \frac{1}{-\frac{1}{a} - ik} = -\frac{a}{1 + ika}. \quad (9.102)$$

This amplitude exhibits a simple pole at $k = i/a = i\kappa$ in the upper half of the complex k -plane. When $a > 0$ and $a \gg b$, the potential possesses a weakly bound s -wave state with binding energy

$$E_b = -\frac{\hbar^2 \kappa^2}{2m} \approx -\frac{\hbar^2}{2ma^2}. \quad (9.103)$$

The low-energy elastic cross section is then determined entirely by the bound-state energy:

$$\sigma(E) = \frac{4\pi}{k^2 + \kappa^2} = \frac{4\pi \hbar^2 / (2m)}{\frac{\hbar^2 k^2}{2m} + \frac{\hbar^2 \kappa^2}{2m}} = \frac{2\pi \hbar^2 / m}{E - E_b}. \quad (9.104)$$

If the potential is slightly too shallow to form a bound state, the pole moves to the negative imaginary axis, $k = -i|\kappa|$ ($\kappa < 0$), corresponding to a negative scattering length $a < 0$. This pole resides on the non-physical second Riemann sheet and describes an antibound or *virtual state* with energy $E_v = -\hbar^2 / (2ma^2)$.

9.7.4 Application: Low-Energy Neutron–Proton Scattering

A classic experimental realization of these principles occurs in low-energy neutron–proton scattering. The deuteron is a weakly bound state of a neutron and proton with binding energy $|E_b| = 2.225$ MeV. Because both nucleons possess spin $1/2$, the total spin of the two-nucleon system can be either a spin-triplet ($S = 1$, parallel spins) or a spin-singlet ($S = 0$, antiparallel spins).

The strong nuclear force is spin-dependent:

- **Triplet channel (3S_1):** The interaction is sufficiently attractive to form the deuteron bound state, with positive scattering length $a_t \approx +5.42$ fm. The triplet cross section is well described by Eq. (9.104).
- **Singlet channel (1S_0):** The potential is slightly weaker and fails to form a stable bound state, producing instead a virtual state at $E_v \approx -67$ keV, corresponding to a large negative scattering length $a_s \approx -23.75$ fm.

For unpolarized thermal neutrons incident on unpolarized protons, the statistical average over the triplet (degeneracy 3) and singlet (degeneracy 1) channels yields

$$\sigma_{\text{tot}} = \frac{3}{4}\sigma_t + \frac{1}{4}\sigma_s = \frac{3}{4}(4\pi a_t^2) + \frac{1}{4}(4\pi a_s^2) \approx 20.4 \text{ b}, \quad (9.105)$$

in excellent agreement with low-energy thermal neutron measurements.

9.8 Low-Energy Resonances and the Breit–Wigner Formula

In the low-energy regime ($kb \ll 1$), we showed that the phase shift for non-zero angular momentum channels ($l \geq 1$) typically satisfies $\cot \delta_l \propto k^{-(2l+1)} \gg 1$. Consequently, the off-resonance partial cross section

$$\sigma_l = \frac{4\pi}{k^2}(2l+1)\frac{1}{\cot^2 \delta_l + 1} \propto k^{4l} \quad (9.106)$$

is suppressed by high powers of k . However, if the potential parameters and energy are tuned such that $\cot \delta_l$ passes through zero at an energy $E_r > 0$, the partial cross section rises sharply to saturate its maximum theoretical unitarity limit:

$$\sigma_l(E_r) = \frac{4\pi(2l+1)}{k^2}. \quad (9.107)$$

Such a pronounced peak in the scattering cross section is called a *scattering resonance*.

9.8.1 Derivation of the Breit–Wigner Resonance Formula

From Eq. (9.92), $\cot \delta_l$ vanishes at an energy E_r where the numerator of the bracketed expression vanishes:

$$l + 1 + b\alpha_l(E_r) = 0, \quad (l \geq 1), \quad (9.108)$$

which is equivalent to the phase shift passing through half-integer multiples of π :

$$\delta_l(E_r) = \left(n + \frac{1}{2}\right)\pi, \quad (n \in \mathbb{Z}). \quad (9.109)$$

Expanding the logarithmic derivative $\alpha_l(E)$ in a Taylor series about the resonance energy E_r ,

$$l + 1 + b\alpha_l(E) \approx (E - E_r)b \left(\frac{\partial \alpha_l}{\partial E} \right)_{E=E_r}, \quad (9.110)$$

$$l - b\alpha_l(E) \approx 2l + 1, \quad (9.111)$$

and substituting these into Eq. (9.92), we obtain

$$\cot \delta_l(E) \approx -\frac{2(E - E_r)}{\Gamma_k}, \quad (9.112)$$

where the energy width Γ_k is defined by

$$\Gamma_k \equiv -\frac{2k^{2l+1}b^{2l}}{[(2l-1)!!]^2 \left(\frac{\partial \alpha_l}{\partial E} \right)_{E_r}}. \quad (9.113)$$

Since the logarithmic derivative is a monotonically decreasing function of energy ($\partial \alpha_l / \partial E < 0$, see below), the width Γ_k is strictly positive.

Evaluating $\Gamma \equiv \Gamma_k(E_r)$, the partial cross section (9.106) takes the celebrated *Breit–Wigner form*:

$$\sigma_l(E) = \frac{4\pi(2l+1)}{k^2} \frac{(\Gamma/2)^2}{(E - E_r)^2 + (\Gamma/2)^2}. \quad (9.114)$$

When the collision energy deviates from E_r by $\pm \Gamma/2$, the cross section drops to exactly half of its peak value. Thus, Γ represents the full-width at half-maximum (FWHM) of the resonance peak.

From Eq. (9.112), the resonant phase shift varies as

$$\delta_l(E) = \delta_{bg} + \arctan \left(\frac{\Gamma/2}{E_r - E} \right) = \frac{\pi}{2} + \arctan \left(\frac{E - E_r}{\Gamma/2} \right). \quad (9.115)$$

As the collision energy sweeps through the resonance from $E \ll E_r$ to $E \gg E_r$, the phase shift rises sharply by π over an energy interval of width $\sim \Gamma$, with maximal slope $d\delta_l/dE = 2/\Gamma$ attained at $E = E_r$.

9.8.2 Physical Origin: Centrifugal Barrier and Quasi-Bound States

The physical mechanism underlying scattering resonances for $l \geq 1$ is the existence of a *quasi-bound state* (or metastably trapped state) behind the centrifugal barrier. The effective radial potential,

$$V_{\text{eff}}(r) = v(r) + \frac{\hbar^2 l(l+1)}{2mr^2}, \quad (9.116)$$

features an attractive core surrounded by a repulsive centrifugal barrier of height $V_{\text{barrier}} \sim \hbar^2 l(l+1)/(2mb^2)$. At a resonance energy $E_r > 0$ lying below the barrier peak, the interior well supports a standing wave state that would be a genuine bound state if the barrier were infinitely thick.

Because the barrier has finite width, an incident particle whose energy matches E_r quantum-mechanically tunnels through the centrifugal barrier into the interior well with resonant transmission probability. Once inside, the particle reflects back and forth numerous times before tunneling back out into the continuum. The mean lifetime τ of this trapped state is governed by the energy-time uncertainty relation:

$$\tau = \frac{\hbar}{\Gamma}. \quad (9.117)$$

Higher angular momentum l and smaller kb enhance the centrifugal barrier thickness, dramatically narrowing the resonance width $\Gamma \propto (kb)^{2l+1}$ and exponentially prolonging the lifetime of the quasi-bound state.

9.8.3 Complex Energy Poles of the S -Matrix

Near a Breit-Wigner resonance, the partial-wave S -matrix element $S_l(E) = e^{2i\delta_l(E)}$ can be expressed using Eq. (9.112):

$$S_l(E) = \frac{\cot \delta_l + i}{\cot \delta_l - i} \approx \frac{E - E_r - i\Gamma/2}{E - E_r + i\Gamma/2}. \quad (9.118)$$

Similarly, the partial-wave scattering amplitude is

$$f_l(E) = \frac{S_l(E) - 1}{2ik} = -\frac{1}{k} \frac{\Gamma/2}{E - E_r + i\Gamma/2}. \quad (9.119)$$

Analytically continuing the energy E into the complex plane, $f_l(E)$ has a simple pole on the unphysical second Riemann sheet at the complex energy:

$$E_{\text{pole}} = E_r - i\frac{\Gamma}{2}. \quad (9.120)$$

Under unitary time evolution, the temporal phase factor associated with this pole state is

$$e^{-iE_{\text{pole}}t/\hbar} = e^{-iE_r t/\hbar} e^{-\Gamma t/(2\hbar)}, \quad (9.121)$$

so that the probability density decays exponentially with characteristic decay rate $\Gamma/\hbar$:

$$P(t) \propto \left| e^{-iE_{\text{pole}}t/\hbar} \right|^2 = e^{-\Gamma t/\hbar} = e^{-t/\tau}. \quad (9.122)$$

Thus, stationary scattering theory naturally incorporates the finite lifetime and decay dynamics of unstable metastable particles and resonances.

9.8.4 Monotonicity of the Logarithmic Derivative

To verify that $\Gamma > 0$, we prove that $\partial\alpha_l/\partial E < 0$. Dividing the radial Schrödinger equation (9.37) by the reduced wave function $u_l(r) = rR_l(r)$ and differentiating with respect to $k^2 = 2mE/\hbar^2$, we obtain

$$\frac{\partial}{\partial k^2} \left(\frac{1}{u_l} \frac{d^2 u_l}{dr^2} \right) = -1. \quad (9.123)$$

Using the identity $\frac{d}{dr} \left[u_l^2 \frac{\partial}{\partial k^2} \left(\frac{1}{u_l} \frac{du_l}{dr} \right) \right] = u_l^2 \frac{\partial}{\partial k^2} \left(\frac{1}{u_l} \frac{d^2 u_l}{dr^2} \right) = -u_l^2$, and integrating from $r = 0$ to $r = b$ with the boundary condition $u_l(0) = 0$, we find

$$\frac{\partial}{\partial k^2} \left(\frac{1}{u_l(r)} \frac{du_l(r)}{dr} \right) \Big|_{r=b} = -\frac{1}{u_l(b)^2} \int_0^b u_l(r')^2 dr' < 0. \quad (9.124)$$

Since $\alpha_l(b) = \frac{1}{R_l} \frac{dR_l}{dr} \Big|_{r=b} = \frac{1}{u_l} \frac{du_l}{dr} \Big|_{r=b} - \frac{1}{b}$, this immediately confirms that

$$\frac{\partial\alpha_l}{\partial E} = \frac{2m}{\hbar^2} \frac{\partial\alpha_l}{\partial k^2} < 0, \quad (9.125)$$

guaranteeing that the physical resonance width Γ is strictly positive.

9.9 Problems

1. **Laboratory frame angular distribution with recoil.** Consider the elastic scattering of a projectile of mass m_1 and initial momentum $\hbar\mathbf{k}$ from an initially stationary target particle of mass m_2 . Suppose that only the s -wave and p -wave phase shifts, $\delta_0(E)$ and $\delta_1(E)$, are nonzero in the center-of-mass frame.
 - (a) Express the center-of-mass differential cross section $(\frac{d\sigma}{d\Omega})_{\text{cm}}$ in terms of δ_0 , δ_1 , and the center-of-mass scattering angle θ_{cm} .
 - (b) Transform the angular distribution to the laboratory frame where the target particle m_2 was initially at rest. Taking full account of target recoil, derive an explicit expression for the laboratory differential cross section $(\frac{d\sigma}{d\Omega})_{\text{lab}}$ as a function of the laboratory scattering angle θ_{lab} and the mass ratio $\gamma = m_1/m_2$.
2. **Neutron scattering from a fixed diatomic molecule.** Consider the elastic scattering of slow neutrons with wavevector $\mathbf{k} = k\hat{\mathbf{z}}$ by a stationary diatomic molecule. The two identical nuclei are held fixed along the y -axis at positions $\mathbf{r}_1 = (0, b, 0)$ and $\mathbf{r}_2 = (0, -b, 0)$. The interaction between the incident neutron and each nucleus is modeled by Fermi pseudopotentials (delta functions):

$$V(\mathbf{r}) = a \delta(x) \delta(z) [\delta(y - b) + \delta(y + b)], \quad (9.126)$$

where a characterizes the neutron-nucleus scattering strength.

- (a) Using the first Born approximation, calculate the scattering amplitude $f(\theta, \phi)$ and the differential cross section $\frac{d\sigma}{d\Omega}(\theta, \phi)$, where θ and ϕ denote the standard spherical polar and azimuthal angles of the scattered wavevector $\mathbf{k}'$.
- (b) Analyze the resulting interference pattern. In what specific qualitative and quantitative ways does the quantum mechanical diffraction result differ from classical expectation?

Chapter 10

Coulomb Scattering

10.1 Coulomb Scattering

All previous scattering analyses rested upon the assumption that the interaction potential is short-ranged, falling off faster than r^{-1} as $r \rightarrow \infty$, so that the scattered particle behaves asymptotically as a free particle. The Coulomb potential,

$$v(r) = -\frac{e^2}{r} \quad (\text{or in general } \frac{Z_1 Z_2 e^2}{r}), \quad (10.1)$$

is long-ranged and represents a singular exception: its force extends to infinity, permanently altering the asymptotic phase of the particle at arbitrarily large distances. Consequently, standard plane waves and ordinary partial-wave boundary conditions must be fundamentally generalized.

10.1.1 Asymptotic Behavior and the Logarithmic Phase Distortion

The breakdown of free-particle asymptotics is evident from the bound states of the hydrogen atom. At large distances, the discrete bound-state radial wave functions behave as

$$R_{nl}(r) \sim r^{n-1} e^{-\kappa r} = \frac{1}{r} e^{-\kappa r + n \ln r}, \quad (r \rightarrow \infty), \quad (10.2)$$

in sharp contrast to the localized exponential decay $R_l(r) \sim r^{-1} e^{-\kappa r}$ of a short-range potential. Expressing the principal quantum number n in terms of the bound-state energy $E = -\hbar^2 \kappa^2 / (2m) = -me^4 / (2\hbar^2 n^2)$, we have

$$n = \frac{me^2}{\hbar^2 \kappa} = \frac{1}{a_0 \kappa}, \quad (10.3)$$

where $a_0 = \hbar^2 / (me^2)$ is the Bohr radius. Thus, the bound state wave function exhibits a logarithmic coordinate dependence:

$$R_{nl}(r) \sim \frac{1}{r} \exp \left(-\kappa r + \frac{me^2}{\hbar^2 \kappa} \ln r \right). \quad (10.4)$$

Analytically continuing this behavior to positive continuum energies ($E = \hbar^2 k^2 / (2m) > 0$) via the substitution $\kappa \rightarrow \mp i k$, the continuum radial wave functions must asymptotically take the form

$$R(r) \sim \frac{1}{r} \exp [\pm i (kr - \gamma \ln(2kr))], \quad (10.5)$$

where the dimensionless *Sommerfeld parameter* γ is defined by

$$\gamma \equiv -\frac{me^2}{\hbar^2 k} = -\frac{1}{a_0 k}. \quad (10.6)$$

(For general charges q_1, q_2 , the parameter is $\eta = q_1 q_2 / (\hbar v)$). The long-range $1/r$ tail accumulates an unbounded phase distortion $-\gamma \ln(2kr)$ that grows logarithmically with distance.

10.1.2 Exact Solution in Parabolic Coordinates

The Schrödinger equation for an attractive Coulomb potential,

$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{r}\right)\psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (10.7)$$

admits an exact analytic solution by separating variables in parabolic coordinates $\xi = r + z$, $\eta = r - z$, $\phi = \arctan(y/x)$. Choosing the incident wavevector $\mathbf{k} = k\hat{\mathbf{z}}$ along the polar axis, the solution regular at the origin is

$$\psi(\mathbf{r}) = \Gamma(1 + i\gamma) e^{-\pi\gamma/2} e^{ikz} {}_1F_1(-i\gamma; 1; ik(r - z)), \quad (10.8)$$

where ${}_1F_1(a; b; \zeta)$ is Kummer's confluent hypergeometric function:

$${}_1F_1(a; b; \zeta) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n} \frac{\zeta^n}{n!}, \quad (a)_n \equiv a(a+1)\cdots(a+n-1). \quad (10.9)$$

Using the asymptotic expansion of ${}_1F_1(-i\gamma; 1; \zeta)$ for large $|\zeta| = k(r - z) \gg 1$:

$${}_1F_1(-i\gamma; 1; \zeta) \sim \frac{e^{\pi\gamma/2}}{\Gamma(1 + i\gamma)} \left\{ e^{i\gamma \ln(-i\zeta)} \left(1 - \frac{\gamma^2}{\zeta} + \cdots\right) + \frac{\Gamma(1 + i\gamma)}{\Gamma(-i\gamma)} \frac{e^{\zeta}}{\zeta} e^{-i\gamma \ln(-i\zeta)} \right\}, \quad (10.10)$$

the full wave function decomposes asymptotically as

$$\psi(\mathbf{r}) \sim \psi_{\text{inc}}(\mathbf{r}) + \psi_{\text{scatt}}(\mathbf{r}). \quad (10.11)$$

The distorted incident plane wave is

$$\psi_{\text{inc}}(\mathbf{r}) = e^{ikz + i\gamma \ln[k(r - z)]} \left[1 - \frac{\gamma^2}{ik(r - z)}\right], \quad (10.12)$$

which yields an asymptotic current density $\mathbf{j}_{\text{inc}} = (\hbar\mathbf{k}/m)[1 + \mathcal{O}((r - z)^{-1})]$. The outgoing scattered wave is

$$\psi_{\text{scatt}}(\mathbf{r}) = \frac{\Gamma(1 + i\gamma)}{\Gamma(-i\gamma)} \frac{e^{ikr - i\gamma \ln[k(r - z)]}}{ik(r - z)}. \quad (10.13)$$

10.1.3 The Coulomb Scattering Amplitude and Rutherford Cross Section

Noting that $r - z = r(1 - \cos\theta) = 2r \sin^2(\theta/2)$, we express the scattered wave in standard spherical form:

$$\psi_{\text{scatt}}(\mathbf{r}) = f_c(\theta) \frac{e^{i[kr - \gamma \ln(2kr)]}}{r}, \quad (10.14)$$

where the *Coulomb scattering amplitude* is

$$f_c(\theta) = \frac{\Gamma(1 + i\gamma)}{\Gamma(-i\gamma)} \frac{e^{-i\gamma \ln \sin^2(\theta/2)}}{2ik \sin^2(\theta/2)}. \quad (10.15)$$

Using the recurrence relation $\Gamma(1 + z) = z\Gamma(z)$ and complex conjugation $\Gamma(z^*) = [\Gamma(z)]^*$, we write

$$\frac{\Gamma(1 + i\gamma)}{\Gamma(-i\gamma)} = -i\gamma \frac{\Gamma(1 + i\gamma)}{\Gamma(1 - i\gamma)} = -i\gamma e^{2i\sigma_0}, \quad (10.16)$$

where the *s-wave Coulomb phase shift* σ_0 is defined by

$$e^{2i\sigma_0} \equiv \frac{\Gamma(1 + i\gamma)}{\Gamma(1 - i\gamma)}, \quad (\sigma_0 \in \mathbb{R}). \quad (10.17)$$

Substituting this relation into Eq. (10.15), the Coulomb amplitude takes the compact form:

$$f_c(\theta) = -\frac{\gamma}{2k} e^{2i\sigma_0} \left(\sin^2 \frac{\theta}{2} \right)^{-i\gamma-1}. \quad (10.18)$$

The scattered radial probability current is

$$j_{r,\text{scatt}} = \frac{\hbar k}{mr^2} |f_c(\theta)|^2, \quad (10.19)$$

which gives the quantum mechanical differential cross section:

$$\frac{d\sigma}{d\Omega} = |f_c(\theta)|^2 = \frac{\gamma^2}{4k^2 \sin^4(\theta/2)} = \frac{e^4}{16E^2 \sin^4(\theta/2)}. \quad (10.20)$$

Remarkably, Eq. (10.20) is mathematically identical to the classical *Rutherford scattering cross section*. In this unique system, quantum wave mechanics and classical Newtonian orbit mechanics predict precisely the same angular cross section, because $\hbar$ cancels out identically from the ratio $\gamma/k = -me^2/(\hbar^2 k^2) \cdot k = -e^2/(2E)$.

10.1.4 Forward Divergence and Screening

The differential cross section (10.20) diverges severely as $\theta \rightarrow 0$ ($d\sigma/d\Omega \propto \theta^{-4}$), causing the total cross section $\sigma = \int (d\sigma/d\Omega) d\Omega$ to diverge to infinity. Classically, this infinite cross section reflects the fact that every incident particle, regardless of how large its impact parameter b , experiences a non-vanishing transverse force and is deflected through an infinitesimally small angle.

In realistic physical systems, pure isolated Coulomb charges do not exist in isolation. In atomic matter and plasmas, positive nuclear charges are surrounded by electron clouds that effectively screen the long-range potential:

$$v_{\text{screened}}(r) = -\frac{Ze^2}{r} e^{-r/a_{\text{TF}}}, \quad (10.21)$$

where a_{TF} is the Thomas–Fermi atomic screening radius. For impact parameters $b \gg a_{\text{TF}}$, the projectile encounters a neutral atom with exponentially suppressed fields. This screening cuts off small-angle scattering for $\theta \lesssim (\hbar k a_{\text{TF}})^{-1}$, rendering the total physical cross section finite.

10.2 Analytic Properties of the Scattering Amplitude

In this section, we investigate the analytic continuation of the Coulomb scattering amplitude into the complex momentum and energy planes, demonstrating how the discrete hydrogen spectrum is naturally encoded within the analytic structure of the continuum scattering amplitude.

10.2.1 Complex Poles and the Hydrogen Bound States

The Coulomb scattering amplitude obtained in Eq. (10.15),

$$f_c(\theta) = \frac{\Gamma(1+i\gamma)}{\Gamma(-i\gamma)} \frac{e^{-i\gamma \ln \sin^2(\theta/2)}}{2ik \sin^2(\theta/2)}, \quad (10.22)$$

is analytic in the momentum k except at the singularities of the Euler gamma function $\Gamma(z)$. The gamma function has no zeros anywhere in the complex plane, but possesses isolated simple poles at zero and the negative integers:

$$z = 0, -1, -2, -3, \dots \quad (10.23)$$

This is readily seen from the reflection and recurrence identity:

$$\Gamma(z) = \frac{\Gamma(z+n)}{z(z+1)\cdots(z+n-1)}, \quad (10.24)$$

which exhibits simple poles at $z = -m$ ($m = 0, 1, 2, \dots$) with residues

$$\text{Res}[\Gamma(z), z = -m] = \frac{(-1)^m}{m!}. \quad (10.25)$$

Consequently, the numerator $\Gamma(1+i\gamma)$ in $f_c(\theta)$ develops simple poles whenever

$$1+i\gamma = 1-m \implies i\gamma = -n, \quad (n = 1, 2, 3, \dots). \quad (10.26)$$

Recalling the definition $\gamma = -1/(ka_0)$, this condition requires

$$-\frac{i}{ka_0} = -n \implies k = \frac{i}{na_0}, \quad (n = 1, 2, 3, \dots). \quad (10.27)$$

Evaluating the corresponding energy values,

$$E_n = \frac{\hbar^2 k^2}{2m} = -\frac{\hbar^2}{2m(na_0)^2} = -\frac{me^4}{2\hbar^2 n^2}, \quad (10.28)$$

we recover precisely the discrete Bohr energy levels of the hydrogen atom!

These poles correspond to genuine, physically normalizable bound states because $k = i\kappa$ lies on the positive imaginary axis ($\kappa = 1/(na_0) > 0$), producing exponential asymptotic damping $e^{ikr} = e^{-\kappa r}$. For a repulsive Coulomb interaction ($e^2 \rightarrow -e^2$), the parameter γ changes sign, and the poles migrate to the negative imaginary k -axis ($k = -i/(na_0)$, $\kappa < 0$). Such poles lie on the non-physical sheet, reflecting the physical fact that two like charges cannot form bound states.

10.2.2 Partial Wave Expansion in Spherical Coordinates

The Schrödinger equation with a Coulomb potential can also be solved directly by separation of variables in spherical coordinates (r, θ, ϕ) :

$$\psi_c(\mathbf{r}) = \frac{1}{kr} \sum_{l=0}^{\infty} (2l+1) i^l e^{i\sigma_l} F_l(\gamma, kr) P_l(\cos \theta), \quad (10.29)$$

where $F_l(\gamma, \rho)$ is the regular Coulomb wave function. Asymptotically ($kr \gg l$),

$$F_l(\gamma, kr) \sim \sin \left(kr - \gamma \ln(2kr) - \frac{l\pi}{2} + \sigma_l \right), \quad (10.30)$$

and the *Coulomb phase shifts* σ_l are given by

$$e^{2i\sigma_l} = \frac{\Gamma(l+1+i\gamma)}{\Gamma(l+1-i\gamma)}. \quad (10.31)$$

Here, the logarithmic term $-\gamma \ln(2kr)$ represents an l -independent distortion common to all partial waves, while σ_l acts as the channel-specific phase shift.

10.2.3 Partial Wave Projection and Analytic Continuation

To express the full Coulomb amplitude $f_c(\theta)$ in terms of partial-wave amplitudes,

$$f_c(\theta) = \sum_{l=0}^{\infty} (2l+1) f_l(k) P_l(\cos \theta), \quad (10.32)$$

we project onto the Legendre polynomials:

$$f_l(k) = \frac{1}{2} \int_{-1}^1 P_l(\cos \theta) f_c(\theta) d(\cos \theta) = -\frac{\gamma e^{2i\sigma_0}}{4k} \int_{-1}^1 P_l(x) \left(\frac{1-x}{2} \right)^{-i\gamma-1} dx, \quad (10.33)$$

where $x = \cos \theta$. Because $f_c(\theta)$ diverges as $(1-x)^{-1}$ in the forward direction, this integral diverges for real γ . However, it converges absolutely in the complex half-plane $\text{Re}(-i\gamma) > 0$. Using Rodrigues' formula,

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l, \quad (10.34)$$

and integrating by parts l times, the boundary terms vanish, yielding

$$\begin{aligned} f_l(k) &= \frac{1}{2ik} \frac{(l+i\gamma)(l-1+i\gamma)\cdots(1+i\gamma)}{(l-i\gamma)(l-1-i\gamma)\cdots(1-i\gamma)} e^{2i\sigma_0} \\ &= \frac{1}{2ik} \frac{\Gamma(l+1+i\gamma)}{\Gamma(l+1-i\gamma)} = \frac{e^{2i\sigma_l}}{2ik}. \end{aligned} \quad (10.35)$$

Analytically continuing back to real γ , the partial-wave expansion becomes

$$f_c(\theta) = \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) e^{2i\sigma_l} P_l(\cos \theta). \quad (10.36)$$

Subtracting the unscattered delta function distribution $(2ik)^{-1} \sum_l (2l+1) P_l(\cos \theta) = (i/k) \delta(1 - \cos \theta)$, which is active solely at $\theta = 0$, Eq. (10.36) can be written in the standard form:

$$f_c(\theta) = \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) (e^{2i\sigma_l} - 1) P_l(\cos \theta), \quad (\theta \neq 0). \quad (10.37)$$

10.2.4 Channel-by-Channel Bound State Poles

Examining the partial-wave amplitude $f_l(k)$ as an analytic function of k , its poles are dictated by the poles of $\Gamma(l+1+i\gamma)$:

$$l+1+i\gamma = -m, \quad (m = 0, 1, 2, \dots). \quad (10.38)$$

Defining the integer principal quantum number $n \equiv l+1+m \geq l+1$, this condition reads

$$i\gamma = -n \implies k = \frac{i}{na_0}, \quad (n = l+1, l+2, \dots). \quad (10.39)$$

Thus, for each orbital angular momentum channel l , the poles of the partial-wave scattering amplitude $f_l(k)$ reproduce the complete hydrogen bound-state spectrum with the correct orbital angular momentum constraint:

$$l \leq n-1. \quad (10.40)$$

This beautiful synthesis demonstrates that bound states and scattering states are unified as different aspects of a single analytic S -matrix function.

10.3 Regge Poles and Complex Angular Momentum

In traditional scattering theory, the orbital angular momentum l is restricted to non-negative integers ($l \in \mathbb{N}_0$), and one examines the analytic properties of the partial-wave scattering amplitude $f_l(E)$ as a function of the complex energy E or momentum k . Tullio Regge (1959) pioneered an alternative, deeply insightful perspective: fixing the physical energy $E \in \mathbb{R}$ and analytically continuing the partial-wave amplitude into the *complex angular momentum plane* $l \in \mathbb{C}$.

10.3.1 Definition of Regge Poles and Trajectories

In the complex l -plane, the poles of the partial-wave amplitude $f(l, E) = [S(l, E) - 1]/(2ik)$ are termed *Regge poles*. As the real energy E varies continuously, each pole traces out a continuous curve in the complex l -plane known as a *Regge trajectory*, denoted by $\alpha(E) \equiv l(E)$.

For the attractive Coulomb potential, the poles of $\Gamma(l + 1 + i\gamma)$ occur at

$$l + 1 + i\gamma = 1 - p \implies l(E) = -p - i\gamma = -p + \frac{ime^2}{\hbar^2 k}, \quad (p = 1, 2, 3, \dots). \quad (10.41)$$

For negative energies ($E < 0$), the momentum is purely imaginary: $k = i\kappa = i\sqrt{2m|E|}/\hbar$. The Coulomb Regge trajectory for $p = 1$ is then purely real:

$$\alpha(E) = -1 + \frac{me^2}{\hbar^2 \kappa} = -1 + \frac{1}{a_0 \kappa} = -1 + \sqrt{\frac{\text{Ry}}{|E|}}, \quad (10.42)$$

where $\text{Ry} = me^4/(2\hbar^2) = 13.6$ eV. As the energy increases from $E = -\infty$ toward $E = 0^-$, the trajectory $\alpha(E)$ travels along the positive real axis from -1 to $+\infty$. Whenever the trajectory passes through a non-negative integer $l \in \{0, 1, 2, \dots\}$, the energy satisfies

$$l = -1 + n \implies E_n = -\frac{\text{Ry}}{n^2}, \quad (n = l + 1, l + 2, \dots), \quad (10.43)$$

reproducing the infinite sequence of hydrogen bound states. Because the Coulomb potential possesses an infinite range, it supports bound states of arbitrarily large angular momentum l , allowing the trajectory to extend to $+\infty$ along the real axis.

10.3.2 Regge Trajectories for Short-Range Potentials

For a short-range potential of finite range b , the effective centrifugal barrier $\hbar^2 l(l + 1)/(2mr^2)$ overpowers the attractive well for sufficiently large l . Hence, only a finite number of bound states can exist below threshold ($E < 0$).

As established in Section 9.6, a bound state at energy $E < 0$ satisfies the S -matrix pole condition:

$$\cot \delta_l(E) = i. \quad (10.44)$$

Regarding Eq. (10.44) as an implicit equation for complex l as a function of real E , its solutions define the Regge trajectories $l = \alpha(E)$. For $E < 0$, the trajectory remains on the real axis ($\text{Im } l = 0$). Each time $\alpha(E)$ crosses a non-negative integer $l_b \in \{0, 1, 2, \dots\}$ at an energy $E_b < 0$, the system possesses a genuine physical bound state with orbital angular momentum l_b .

As E approaches the continuum threshold ($E \rightarrow 0^-$), the trajectory turns away from the real axis and enters the upper half-plane:

$$\text{Im } \alpha(E) > 0 \quad \text{for } E > 0. \quad (10.45)$$

This reflects the fundamental requirement of probability conservation: for continuum energies $E > 0$ and real physical l , the phase shift $\delta_l(E)$ is strictly real, so $\cot \delta_l(E)$ is real and cannot satisfy $\cot \delta_l = i$. Thus, for positive energies, the poles of the S -matrix can exist only at complex values of l .

10.3.3 Physical Resonances from Complex Angular Momentum Poles

When a Regge trajectory passes close to the real axis at positive energy, we parameterize the complex pole position as

$$l(E) = l_0(E) + il_1(E), \quad (l_0, l_1 \in \mathbb{R}, \quad 0 < l_1 \ll 1). \quad (10.46)$$

Expanding the pole condition (10.44) to first order in the small imaginary component l_1 ,

$$\cot \delta_{l_0}(E) + il_1 \left. \frac{\partial \cot \delta_l(E)}{\partial l} \right|_{l=l_0} = i. \quad (10.47)$$

Equating the real and imaginary parts separately yields two fundamental relations:

$$\cot \delta_{l_0}(E) = 0, \quad (10.48)$$

$$l_1(E) = \left[\left. \frac{\partial \cot \delta_l(E)}{\partial l} \right|_{l=l_0} \right]^{-1}. \quad (10.49)$$

Equation (10.48) is precisely the condition for a scattering resonance [Eq. (9.109)] in the partial-wave channel l_0 . Therefore:

Whenever the real part of a Regge trajectory, $\text{Re } \alpha(E) = l_0(E)$, passes through a positive integer at energy $E_r > 0$, the system exhibits a physical scattering resonance with orbital angular momentum l_0 .

The imaginary part $l_1(E_r)$ determines the physical energy width Γ of the resonance. Expanding $\cot \delta_{l_0}(E)$ about E_r ,

$$\cot \delta_{l_0}(E) \approx -\frac{2(E - E_r)}{\Gamma}, \quad (10.50)$$

and relating the energy and angular momentum derivatives via the chain rule along the trajectory,

$$\left. \frac{\partial \cot \delta_l}{\partial l} \right|_{l_0} = \left(\left. \frac{\partial \cot \delta_{l_0}}{\partial E} \right|_{E_r} \right) \left(\frac{dE}{dl_0} \right) = -\frac{2}{\Gamma} \left(\left. \frac{d \text{Re } \alpha}{dE} \right|_{E_r} \right)^{-1}, \quad (10.51)$$

Eq. (10.49) provides the direct relation:

$$l_1(E_r) = \frac{\Gamma}{2} \left. \frac{d \text{Re } \alpha(E)}{dE} \right|_{E_r}. \quad (10.52)$$

The closer the Regge pole is to the physical real axis (smaller l_1), the sharper and longer-lived the resonance.

10.3.4 Low-Energy Connection to the Breit–Wigner Width

In the low-energy limit ($kb \ll 1$), we evaluate l_1 explicitly using the threshold expansion (9.92) for $\cot \delta_l(E)$. At resonance, $l_0 + 1 + b\alpha_{l_0}(E_r) = 0$, and differentiating with respect to l gives

$$l_1 = \frac{(kb)^{2l_0+1}}{[(2l_0 - 1)!!]^2 \left[1 + b \left. \frac{\partial \alpha_{l_0}}{\partial l} \right|_{l_0} \right]}. \quad (10.53)$$

Using Eq. (9.113) for the Breit–Wigner width Γ_k , Eq. (10.53) can be expressed compactly as

$$l_1 = -\frac{\Gamma_k}{2} b \left(\left. \frac{\partial \alpha_{l_0}}{\partial E} \right|_{E_r} \right) \frac{1}{1 + b \left. \frac{\partial \alpha_{l_0}}{\partial l} \right|_{l_0}}. \quad (10.54)$$

Regge theory unifies bound states, virtual states, and resonances as manifestations of a single dynamical entity: a family of quantum states lying along the same Regge trajectory $\alpha(E)$. In modern hadron physics, this concept led directly to the dual resonance model and laid the foundational cornerstone of string theory.

10.4 Problems

1. **Scattering by a screened hydrogen atom.** To a first approximation, the electrostatic potential experienced by a particle of charge q and mass m colliding with a neutral hydrogen atom can be modeled as that of a positive point nucleus $+e$ at the origin enveloped by a uniform spherical electron cloud of total charge $-e$ and radius a_0 (the Bohr radius).
 - (a) Using electrostatics, determine the potential $V(r)$ inside ($r \leq a_0$) and outside ($r > a_0$) the atom. Calculate the differential cross section $\frac{d\sigma}{d\Omega}$ and total cross section σ in the first Born approximation, neglecting atomic recoil.
 - (b) Examine the low-energy limit ($ka_0 \ll 1$) of the differential cross section. Contrast its finite forward value with the forward divergence of the unscreened Coulomb cross section.
 - (c) Demonstrate that as the incident energy increases ($ka_0 \gg 1$), the differential cross section approaches the pure Coulomb Rutherford cross section for scattering by the bare nucleus. Provide a physical interpretation for this transition.
 - (d) In the first Born approximation, compute the phase shifts $\delta_0(E)$ and $\delta_1(E)$. Determine the scattering length a and compare the low-energy cross section obtained from partial waves with your result from part (b).
2. **Electron scattering from a spherical shell atom.** Consider an idealized model of a heavy neutral atom of atomic number $Z \gg 1$: a point nucleus of charge $+Ze$ is located at the origin, and the Z atomic electrons are modeled as a spherically symmetric surface charge distribution uniformly coating an infinitesimally thin shell of radius $R = a_0/Z^{1/3}$.
 - (a) Determine the electrostatic potential $V(r)$ throughout space. By matching boundary conditions at $r = R$, calculate *exactly* the low-energy scattering amplitude for an incident electron, including both s -wave and p -wave contributions. Find the electron scattering length a . Sketch the total cross section as a function of energy.
 - (b) Investigate whether this potential supports near-threshold bound states or resonances as the parameters Z and R vary.
3. **Finite nuclear charge radius corrections to Coulomb scattering.** Consider the scattering of a charged particle from an extended nucleus modeled as a uniformly charged sphere of total charge e and radius $a \ll a_0$. For low energies ($ka \ll 1$), the scattering closely resembles point-charge Coulomb scattering. Calculate, to lowest non-trivial order in ka , the deviation of the scattering amplitude $\Delta f(\theta) = f(\theta) - f_c(\theta)$ from the ideal Coulomb amplitude.
4. **Imaginary part of a Regge trajectory near resonance.** Provide the detailed step-by-step mathematical derivation of Eq. (10.53):

$$l_1(E) = \frac{(kb)^{2l_0+1}}{[(2l_0-1)!!]^2 \left[1 + b \frac{\partial \alpha_{l_0}}{\partial l} \Big|_{l_0} \right]} = -\frac{\Gamma_k}{2} b \left(\frac{\partial \alpha_{l_0}}{\partial E} \Big|_{E_r} \right) \frac{1}{1 + b \frac{\partial \alpha_{l_0}}{\partial l} \Big|_{l_0}}, \quad (10.55)$$

which directly connects the imaginary displacement of a Regge pole from the physical real axis to the Breit–Wigner energy width Γ_k .

Chapter 11

Stationary State Perturbation Theory

11.1 Stationary-State Perturbation Theory

In quantum mechanics, one frequently encounters problems that cannot be solved exactly, yet bear a close resemblance to solvable models. Consider, for example, the problem of determining the energy levels and stationary states of a system described by the Hamiltonian

$$H = H_0 + V, \quad (11.1)$$

where the unperturbed Hamiltonian H_0 possesses known orthonormal eigenstates $|n\rangle$ and corresponding discrete eigenvalues ε_n :

$$H_0|n\rangle = \varepsilon_n|n\rangle, \quad \langle m|n\rangle = \delta_{mn}. \quad (11.2)$$

When the interaction potential V constitutes a small correction to H_0 , we anticipate that the true eigenstates $|N\rangle$ and eigenenergies E_n of H deviate only slightly from $|n\rangle$ and ε_n . To systematize this approximation, we introduce a dimensionless tracking parameter $\lambda \in [0, 1]$ and consider the parametric Hamiltonian

$$H(\lambda) = H_0 + \lambda V. \quad (11.3)$$

As λ increases adiabatically from 0 to 1, each unperturbed state $|n\rangle$ smoothly deforms into the interacting eigenstate $|N\rangle$, while the eigenvalue ε_n evolves continuously into E_n .

We therefore seek power-series expansions in λ for both the state vectors and the energy eigenvalues:

$$|N\rangle = |n\rangle + \lambda|N^{(1)}\rangle + \lambda^2|N^{(2)}\rangle + \dots, \quad (11.4)$$

$$E_n = \varepsilon_n + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots. \quad (11.5)$$

Ultimately, the physical results correspond to setting $\lambda = 1$; the parameter λ serves bookkeeping purposes to organize terms order-by-order in powers of V .

11.1.1 Validity and Non-Perturbative Limitations

The assumption that the energy and state vectors admit an analytic Taylor expansion around $\lambda = 0$ is highly nontrivial and does not universally hold. For instance, bound states formed in an arbitrarily weak attractive potential cannot be captured at any finite order of perturbation theory starting from free-particle continuum states. An analytic series expansion in λ implies that reversing the sign of λ (turning an attractive potential into a repulsive one) merely changes

signs symmetrically. Yet, while an attractive potential in one or two dimensions always supports a bound state, a repulsive potential never does. A canonical example is the shallow square well in one dimension, whose ground-state binding energy is given by

$$E \approx -\frac{mV_0^2 a^2}{2\hbar^2}. \quad (11.6)$$

This non-analytic dependence on the potential depth V_0 underscores that bound states represent singular, non-perturbative phenomena. Similarly, the pairing energy of Cooper pairs in BCS superconductivity, $\Delta \propto \exp(-1/[N(0)V])$, exhibits an essential singularity at $V = 0$ and cannot be expanded in a Taylor series. Keeping these physical caveats in mind, we now proceed with Rayleigh–Schrödinger perturbation theory for regular, non-singular potentials.

11.1.2 Intermediate Normalization and Order-by-Order Expansion

It is mathematically convenient to adopt the *intermediate normalization* condition,

$$\langle n|N \rangle = 1. \quad (11.7)$$

Substituting the perturbative ansatz (11.4) into this condition, we obtain

$$\langle n|N \rangle = \langle n|n \rangle + \lambda \langle n|N^{(1)} \rangle + \lambda^2 \langle n|N^{(2)} \rangle + \dots = 1. \quad (11.8)$$

Since $\langle n|n \rangle = 1$, the coefficient of every non-zero power of λ must vanish independently:

$$\langle n|N^{(k)} \rangle = 0 \quad \text{for } k = 1, 2, 3, \dots \quad (11.9)$$

Thus, all higher-order state corrections $|N^{(k)} \rangle$ are strictly orthogonal to the unperturbed reference state $|n \rangle$.

We substitute the expansions (11.4) and (11.5) into the time-independent Schrödinger equation:

$$(H_0 + \lambda V)|N \rangle = E_n|N \rangle. \quad (11.10)$$

Equating coefficients of identical powers of λ on both sides yields a hierarchy of equations:

$$\lambda^0 : H_0|n \rangle = \varepsilon_n|n \rangle, \quad (11.11)$$

$$\lambda^1 : (H_0 - \varepsilon_n)|N^{(1)} \rangle + V|n \rangle = E_n^{(1)}|n \rangle, \quad (11.12)$$

$$\lambda^k : (H_0 - \varepsilon_n)|N^{(k)} \rangle + V|N^{(k-1)} \rangle = \sum_{j=1}^k E_n^{(j)}|N^{(k-j)} \rangle. \quad (11.13)$$

Taking the inner product of the first-order relation (11.12) with the unperturbed bra $\langle n|$, and using $\langle n|H_0 = \varepsilon_n \langle n|$ along with the orthogonality condition $\langle n|N^{(1)} \rangle = 0$, we immediately find

$$E_n^{(1)} = \langle n|V|n \rangle. \quad (11.14)$$

To first order in the perturbation, the energy shift is simply the diagonal expectation value of the perturbing operator in the unperturbed state:

$$\Delta E_n^{(1)} = \langle n|V|n \rangle. \quad (11.15)$$

More generally, taking the inner product of the k -th order equation (11.13) with $\langle n|$ and utilizing $\langle n|(H_0 - \varepsilon_n) = 0$ alongside $\langle n|N^{(k-j)} \rangle = \delta_{jk}$, we obtain the compact recurrence relation for the energy corrections:

$$E_n^{(k)} = \langle n|V|N^{(k-1)} \rangle. \quad (11.16)$$

Once the state correction is determined up to order $k - 1$, the k -th order energy correction follows directly by taking the matrix element of V . Multiplying Eq. (11.16) by λ^k and summing over all $k \geq 1$, we arrive at the exact closed-form relation:

$$E_n = \varepsilon_n + \lambda \langle n|V|N \rangle. \quad (11.17)$$

11.1.3 State Corrections and the Second-Order Energy Shift

To resolve the state corrections $|N^{(k)}\rangle$, we expand them in terms of the complete, unperturbed eigenbasis $\{|m\rangle\}$ of H_0 :

$$|N^{(k)}\rangle = \sum_{m \neq n} |m\rangle \langle m|N^{(k)}\rangle, \quad (k \geq 1), \quad (11.18)$$

where the prime on the summation (or explicit condition $m \neq n$) enforces the orthogonality condition $\langle n|N^{(k)}\rangle = 0$. Projecting the k -th order equation (11.13) onto an arbitrary unperturbed bra $\langle m|$ with $m \neq n$, we have

$$(\varepsilon_m - \varepsilon_n) \langle m|N^{(k)}\rangle + \langle m|V|N^{(k-1)}\rangle = \sum_{j=1}^{k-1} E_n^{(j)} \langle m|N^{(k-j)}\rangle. \quad (11.19)$$

Assuming that the unperturbed state $|n\rangle$ is strictly non-degenerate (so that $\varepsilon_n \neq \varepsilon_m$ for all $m \neq n$), we obtain

$$\langle m|N^{(k)}\rangle = \frac{1}{\varepsilon_n - \varepsilon_m} \left[\langle m|V|N^{(k-1)}\rangle - \sum_{j=1}^{k-1} E_n^{(j)} \langle m|N^{(k-j)}\rangle \right]. \quad (11.20)$$

For $k = 1$, this yields the standard first-order state projection:

$$\langle m|N^{(1)}\rangle = \frac{\langle m|V|n\rangle}{\varepsilon_n - \varepsilon_m}, \quad (m \neq n). \quad (11.21)$$

The first-order perturbed state vector is therefore

$$|N\rangle = |n\rangle + \sum_{m \neq n} \frac{\langle m|V|n\rangle}{\varepsilon_n - \varepsilon_m} |m\rangle + \mathcal{O}(V^2). \quad (11.22)$$

Substituting this first-order state correction into the energy relation (11.16) for $k = 2$ gives the famous Rayleigh–Schrödinger second-order energy formula:

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{\varepsilon_n - \varepsilon_m}. \quad (11.23)$$

An essential property of Eq. (11.23) is that when $|n\rangle = |0\rangle$ is the unperturbed *ground state*, every denominator satisfies $\varepsilon_0 - \varepsilon_m < 0$. Consequently,

$$E_0^{(2)} \leq 0. \quad (11.24)$$

The second-order correction always pushes the ground-state energy downward, reflecting the level repulsion characteristic of quantum states under mutual perturbation.

Evaluating Eq. (11.20) for $k = 2$ gives the second-order wave function correction:

$$|N^{(2)}\rangle = \sum_{m \neq n} \sum_{k \neq n} \frac{\langle m|V|k\rangle \langle k|V|n\rangle}{(\varepsilon_n - \varepsilon_m)(\varepsilon_n - \varepsilon_k)} |m\rangle - \sum_{m \neq n} \frac{\langle m|V|n\rangle \langle n|V|n\rangle}{(\varepsilon_n - \varepsilon_m)^2} |m\rangle. \quad (11.25)$$

11.1.4 Example: The Displaced Harmonic Oscillator

To illustrate the perturbative formalism, consider a one-dimensional harmonic oscillator subjected to a uniform external force F , described by the perturbation

$$V = -Fx. \quad (11.26)$$

In the unperturbed number basis $\{|n\rangle\}$, the position operator $x = \sqrt{\hbar/(2m\omega)}(a + a^\dagger)$ couples only to neighboring levels:

$$\langle n+1|V|n\rangle = -F\sqrt{\frac{\hbar(n+1)}{2m\omega}}, \quad (11.27)$$

$$\langle n-1|V|n\rangle = -F\sqrt{\frac{\hbar n}{2m\omega}}. \quad (11.28)$$

Because parity is odd under inversion $x \rightarrow -x$, the diagonal matrix element vanishes identically:

$$E_n^{(1)} = \langle n|V|n\rangle = 0. \quad (11.29)$$

The second-order energy correction receives contributions exclusively from $m = n \pm 1$:

$$E_n^{(2)} = \frac{|\langle n+1|V|n\rangle|^2}{\varepsilon_n - \varepsilon_{n+1}} + \frac{|\langle n-1|V|n\rangle|^2}{\varepsilon_n - \varepsilon_{n-1}} = \frac{F^2 \frac{\hbar(n+1)}{2m\omega}}{-\hbar\omega} + \frac{F^2 \frac{\hbar n}{2m\omega}}{\hbar\omega} = -\frac{F^2}{2m\omega^2}. \quad (11.30)$$

All energy levels are uniformly shifted downward by the state-independent constant $-F^2/(2m\omega^2)$.

This exact result can be understood directly by completing the square in the full Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 - Fx = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 \left(x - \frac{F}{m\omega^2}\right)^2 - \frac{F^2}{2m\omega^2}. \quad (11.31)$$

This is simply a harmonic oscillator of identical frequency ω centered at the displaced equilibrium position $x_0 = F/(m\omega^2)$. The exact, normalized eigenstates $|\bar{N}\rangle$ are obtained by translating the unperturbed states via the unitary spatial displacement operator:

$$|\bar{N}\rangle = \exp\left(-\frac{iFp}{m\omega^2\hbar}\right)|n\rangle. \quad (11.32)$$

Expanding the exponential displacement operator in powers of F , one finds

$$|\bar{N}^{(1)}\rangle = -\frac{iF}{m\omega^2\hbar}p|n\rangle, \quad (11.33)$$

$$|\bar{N}^{(2)}\rangle = -\frac{1}{2}\left(\frac{F}{m\omega^2\hbar}\right)^2 p^2|n\rangle, \quad (11.34)$$

in complete agreement with the order-by-order results derived from Rayleigh–Schrödinger perturbation theory.

11.2 The Wave Function Renormalization Constant

In the preceding development, the perturbed state vector $|N\rangle$ was defined under the intermediate normalization convention $\langle n|N\rangle = 1$. To restore standard unit normalization,

$$\langle \bar{N}|\bar{N}\rangle = 1, \quad (11.35)$$

we define the physically normalized state $|\bar{N}\rangle$ by rescaling $|N\rangle$:

$$|\bar{N}\rangle = Z^{1/2}|N\rangle, \quad Z \equiv \frac{1}{\langle N|N\rangle}. \quad (11.36)$$

The positive real scalar Z is termed the *wave function renormalization constant* (or quasiparticle weight). Taking the inner product of Eq. (11.36) with the unperturbed state bra $\langle n|$, we see that $Z^{1/2}$ measures the probability amplitude of finding the interacting system in the unperturbed state:

$$Z^{1/2} = \langle n|\bar{N}\rangle, \quad Z = |\langle n|\bar{N}\rangle|^2 \leq 1. \quad (11.37)$$

To evaluate Z perturbatively, we expand the norm $\langle N|N\rangle$ using the order-by-order decomposition:

$$\begin{aligned} \langle N|N\rangle &= \left(\langle n| + \lambda \langle N^{(1)}| + \lambda^2 \langle N^{(2)}| + \dots \right) \left(|n\rangle + \lambda |N^{(1)}\rangle + \lambda^2 |N^{(2)}\rangle + \dots \right) \\ &= \langle n|n\rangle + \lambda \left(\langle n|N^{(1)}\rangle + \langle N^{(1)}|n\rangle \right) + \lambda^2 \left(\langle N^{(1)}|N^{(1)}\rangle + \langle n|N^{(2)}\rangle + \langle N^{(2)}|n\rangle \right) + \mathcal{O}(\lambda^3). \end{aligned} \quad (11.38)$$

Because intermediate normalization enforces $\langle n|N^{(k)}\rangle = 0$ for all $k \geq 1$, the first-order cross terms vanish identically. The first non-trivial correction to the norm appears at second order:

$$\langle N|N\rangle = 1 + \lambda^2 \langle N^{(1)}|N^{(1)}\rangle + \mathcal{O}(\lambda^3). \quad (11.39)$$

Substituting the first-order state expansion $|N^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m|V|n\rangle}{\varepsilon_n - \varepsilon_m} |m\rangle$, we find

$$\langle N|N\rangle = 1 + \lambda^2 \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{(\varepsilon_n - \varepsilon_m)^2} + \mathcal{O}(\lambda^3). \quad (11.40)$$

Inverting this expression to order λ^2 , the renormalization constant becomes

$$Z = 1 - \lambda^2 \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{(\varepsilon_n - \varepsilon_m)^2} + \mathcal{O}(\lambda^3). \quad (11.41)$$

Comparing Eq. (11.41) with the second-order Rayleigh–Schrödinger energy formula,

$$E_n = \varepsilon_n + \lambda \langle n|V|n\rangle + \lambda^2 \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{\varepsilon_n - \varepsilon_m} + \mathcal{O}(\lambda^3), \quad (11.42)$$

we observe that differentiating E_n with respect to the unperturbed energy eigenvalue ε_n (holding the matrix elements $\langle m|V|k\rangle$ and background energies ε_m fixed) gives

$$\frac{\partial E_n}{\partial \varepsilon_n} = 1 - \lambda^2 \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{(\varepsilon_n - \varepsilon_m)^2} + \mathcal{O}(\lambda^3) = Z. \quad (11.43)$$

This profound identity holds to all orders in perturbation theory: the overlap probability $Z = |\langle n|\bar{N}\rangle|^2$ of the fully interacting state with the bare unperturbed state equals the parametric derivative of the perturbed energy eigenvalue with respect to the unperturbed level,

$$Z = \frac{\partial E_n}{\partial \varepsilon_n}. \quad (11.44)$$

In many-body theory and quantum field theory, this quantity represents the quasiparticle pole residue of the single-particle Green's function.

11.3 Degenerate Perturbation Theory

The non-degenerate perturbation expansions developed in the preceding sections rely fundamentally on energy denominators of the form $\varepsilon_n - \varepsilon_m$. The convergence of the series is governed by the dimensionless ratios

$$\frac{\langle m|V|n\rangle}{\varepsilon_n - \varepsilon_m}. \quad (11.45)$$

Whenever the unperturbed system possesses degenerate energy levels ($\varepsilon_n = \varepsilon_m$) that are coupled by the interaction ($\langle m|V|n\rangle \neq 0$), these denominators vanish and ordinary perturbation theory breaks down completely. In such cases, one must reformulate the expansion by appropriately choosing the zeroth-order basis within the degenerate subspace.

11.3.1 Diagonalization in the Degenerate Subspace

Suppose the unperturbed Hamiltonian H_0 possesses a k -fold degenerate energy level ε_n spanned by the orthonormal basis $\{|n_a\rangle, |n_b\rangle, \dots, |n_k\rangle\}$:

$$H_0|n_i\rangle = \varepsilon_n|n_i\rangle, \quad \langle n_i|n_j\rangle = \delta_{ij}, \quad (i, j = a, b, \dots, k). \quad (11.46)$$

Any arbitrary linear combination within this subspace is likewise an exact eigenstate of H_0 with eigenvalue ε_n . To avoid singular zero denominators, we seek an adapted orthonormal basis $\{|n_\alpha\rangle\}$,

$$|n_\alpha\rangle = \sum_{i=a}^k C_{\alpha i}|n_i\rangle, \quad (11.47)$$

which diagonalizes the perturbation operator V within the degenerate manifold:

$$\langle n_\beta|V|n_\alpha\rangle = E_{n_\alpha}^{(1)}\delta_{\alpha\beta}. \quad (11.48)$$

In this adapted basis, any transition matrix element between distinct states possessing identical unperturbed energies vanishes identically ($\langle n_\beta|V|n_\alpha\rangle = 0$ for $\alpha \neq \beta$), eliminating the dangerous 0/0 indeterminate forms.

The determination of the coefficients $C_{\alpha i}$ is equivalent to the eigenvalue problem for the $k \times k$ secular matrix:

$$V_{ji} \equiv \langle n_j|V|n_i\rangle. \quad (11.49)$$

The secular equation reads

$$\sum_{i=a}^k \langle n_j|V|n_i\rangle C_{\alpha i} = E_{n_\alpha}^{(1)} C_{\alpha j}, \quad (11.50)$$

or in matrix form, $\mathbf{V}\mathbf{C}_\alpha = E_{n_\alpha}^{(1)}\mathbf{C}_\alpha$. Because V is Hermitian, the $k \times k$ matrix is Hermitian, guaranteeing real eigenvalues $E_{n_\alpha}^{(1)}$ and mutually orthogonal eigenvectors $\mathbf{C}_\alpha$ normalized to unity:

$$\sum_{j=a}^k C_{\beta j}^* C_{\alpha j} = \delta_{\alpha\beta}. \quad (11.51)$$

Computing the matrix element of V between the linear combinations (11.47) yields

$$\langle n_\beta|V|n_\alpha\rangle = \sum_{j,i} C_{\beta j}^* C_{\alpha i} \langle n_j|V|n_i\rangle = E_{n_\alpha}^{(1)} \sum_j C_{\beta j}^* C_{\alpha j} = E_{n_\alpha}^{(1)} \delta_{\alpha\beta}, \quad (11.52)$$

confirming Eq. (11.48). The roots $E_{n_\alpha}^{(1)}$ of the secular determinant,

$$\det \left(\langle n_j|V|n_i\rangle - E^{(1)}\delta_{ji} \right) = 0, \quad (11.53)$$

represent the first-order energy shifts that resolve or partially lift the degeneracy.

11.3.2 Perturbation Expansions in the Adapted Basis

Once the degenerate manifold has been diagonalized, higher-order corrections are calculated by applying the standard Rayleigh–Schrödinger formulas, excluding the degenerate subspace from the intermediate state summations. The perturbed state corresponding to $|n_\alpha\rangle$ is given to first order by

$$|N_\alpha\rangle = |n_\alpha\rangle + \sum_{m \notin D} \frac{\langle m|V|n_\alpha\rangle}{\varepsilon_n - \varepsilon_m} |m\rangle + \mathcal{O}(V^2), \quad (11.54)$$

where D denotes the set of states degenerate with ε_n . The corresponding energy eigenvalue to second order is

$$E_{n\alpha} = \varepsilon_n + \langle n_\alpha|V|n_\alpha\rangle + \sum_{m \notin D} \frac{|\langle m|V|n_\alpha\rangle|^2}{\varepsilon_n - \varepsilon_m} + \mathcal{O}(V^3). \quad (11.55)$$

If other energy levels in the spectrum of H_0 are degenerate among themselves, they do not cause divergences in Eq. (11.55) unless their unperturbed energy coincides with ε_n .

11.3.3 Symmetries and Selection Rules

The task of diagonalizing the secular matrix is greatly simplified when the perturbation V respects symmetries of the unperturbed system. Let A be a Hermitian operator representing an unperturbed constant of motion ($[H_0, A] = 0$). If V also commutes with A ,

$$[V, A] = 0, \quad (11.56)$$

then the matrix elements of V between eigenstates of A with different eigenvalues must vanish identically. Indeed,

$$0 = \langle a'|[V, A]|a\rangle = (a - a')\langle a'|V|a\rangle \implies \langle a'|V|a\rangle = 0 \quad \text{for } a \neq a'. \quad (11.57)$$

Such *selection rules* block-diagonalize the secular matrix into independent submatrices corresponding to each symmetry sector, dramatically reducing the dimensionality of the secular problem.

11.3.4 Example: The Linear Stark Effect in Hydrogen ($n = 2$)

As a classic application, consider the splitting of the $n = 2$ manifold of atomic hydrogen in a uniform external electric field $\mathcal{E} = \mathcal{E}\hat{z}$ (the *linear Stark effect*). Neglecting electron spin and fine-structure corrections, the unperturbed $n = 2$ shell is fourfold degenerate, consisting of the orbital states

$$|2s, m = 0\rangle, \quad |2p, m = 1\rangle, \quad |2p, m = 0\rangle, \quad |2p, m = -1\rangle. \quad (11.58)$$

The perturbation Hamiltonian is the dipole coupling

$$V = e\mathcal{E}z. \quad (11.59)$$

Because the electric field is oriented along the z -axis, the interaction possesses axial symmetry: $[z, L_z] = 0$. Consequently, the magnetic quantum number m is strictly conserved, and matrix elements connecting different m values vanish identically:

$$\langle 2p, m'|e\mathcal{E}z|2p, m\rangle = 0 \quad (m \neq m'), \quad \langle 2s, 0|e\mathcal{E}z|2p, \pm 1\rangle = 0. \quad (11.60)$$

Furthermore, the parity operator Π satisfies $\Pi^\dagger z \Pi = -z$, whereas unperturbed hydrogen wave functions have definite parity $(-1)^l$. Diagonal matrix elements between states of identical parity must vanish by inversion symmetry:

$$\langle 2s, 0|z|2s, 0\rangle = 0, \quad \langle 2p, m|z|2p, m\rangle = 0. \quad (11.61)$$

Thus, the states $|2p, m = 1\rangle$ and $|2p, m = -1\rangle$ experience no first-order Stark shift ($E^{(1)} = 0$). The only non-zero off-diagonal coupling connects the states $|2s, 0\rangle$ and $|2p, 0\rangle$ of opposite parity:

$$\langle 2p, 0 | z | 2s, 0 \rangle = \int d^3r \psi_{210}^*(\mathbf{r}) z \psi_{200}(\mathbf{r}) = -3a_0, \quad (11.62)$$

where $a_0 = \hbar^2/(m_e e^2)$ is the Bohr radius. Within the $m = 0$ subspace, the secular matrix becomes

$$\mathbf{V}_{m=0} = e\mathcal{E} \begin{pmatrix} 0 & -3a_0 \\ -3a_0 & 0 \end{pmatrix}. \quad (11.63)$$

Diagonalizing this symmetric matrix yields the first-order energy shifts:

$$E_{\pm}^{(1)} = \mp 3e\mathcal{E}a_0, \quad (11.64)$$

with corresponding zeroth-order eigenstates

$$|\psi_{\pm}\rangle = \frac{1}{\sqrt{2}} (|2s, 0\rangle \pm |2p, 0\rangle). \quad (11.65)$$

The external field mixes the spherical s and p orbitals into hybridized states with permanent electric dipole moments $\mp 3ea_0$ aligned parallel or antiparallel to the external field.

The characteristic atomic electric field scale is

$$\mathcal{E}_{\text{at}} = \frac{e}{a_0^2} \approx 5.14 \times 10^9 \text{ V/cm}. \quad (11.66)$$

For laboratory electric fields ($\mathcal{E} \ll \mathcal{E}_{\text{at}}$), the perturbation parameter $\mathcal{E}/\mathcal{E}_{\text{at}} \ll 1$ ensures exceptional convergence of the perturbation series. If an atom is initially prepared in the spherically symmetric $|2s, 0\rangle = \frac{1}{\sqrt{2}}(|\psi_+\rangle + |\psi_-\rangle)$ state and the field is switched on, an energy measurement yields the split values

$$E = -\frac{e^2}{8a_0} \mp 3e\mathcal{E}a_0 \quad (11.67)$$

with equal probability $1/2$. This linear Stark splitting is a distinct hallmark of degenerate states with opposite parity; in non-degenerate systems without accidental parity degeneracy, the leading Stark shift is strictly quadratic in $\mathcal{E}$.

11.4 Van der Waals Interaction

A foundational application of second-order perturbation theory is the derivation of the long-range dispersion force between two neutral, spherically symmetric atoms. Consider two neutral hydrogen atoms separated by a large internuclear displacement $\mathbf{R} = R\hat{\mathbf{z}}$, where the proton positions are treated as fixed classical centers in accordance with the Born–Oppenheimer approximation. Let $\mathbf{r}_1$ and $\mathbf{r}_2$ denote the electronic displacement vectors relative to proton 1 and proton 2, respectively.

11.4.1 The Dipole–Dipole Interaction Hamiltonian

The total electrostatic interaction between the charge distributions of the two neutral atoms is given by the sum of four Coulomb terms:

$$V = e^2 \left[\frac{1}{R} + \frac{1}{|\mathbf{R} + \mathbf{r}_2 - \mathbf{r}_1|} - \frac{1}{|\mathbf{R} + \mathbf{r}_2|} - \frac{1}{|\mathbf{R} - \mathbf{r}_1|} \right]. \quad (11.68)$$

At internuclear separations large compared to the atomic size ($R \gg a_0$), we expand the Coulomb potentials in powers of the small ratios r_1/R and r_2/R . Because each atom is electrically neutral,

the monopole–monopole, monopole–dipole, and dipole–monopole interactions vanish identically. The leading non-vanishing term in the multipole expansion is the dipole–dipole coupling:

$$V = -e^2(\mathbf{r}_1 \cdot \nabla_{\mathbf{R}})(\mathbf{r}_2 \cdot \nabla_{\mathbf{R}}) \frac{1}{R} = \frac{e^2}{R^3} \left[\mathbf{r}_1 \cdot \mathbf{r}_2 - \frac{3(\mathbf{r}_1 \cdot \mathbf{R})(\mathbf{r}_2 \cdot \mathbf{R})}{R^2} \right]. \quad (11.69)$$

Choosing the coordinate z -axis along the internuclear axis $\mathbf{R}$, the interaction Hamiltonian simplifies to

$$V = \frac{e^2}{R^3} (x_1 x_2 + y_1 y_2 - 2z_1 z_2). \quad (11.70)$$

11.4.2 First- and Second-Order Perturbation Energies

In the non-interacting limit ($R \rightarrow \infty$), the unperturbed product states are $|n_1, n_2\rangle = |n_1\rangle \otimes |n_2\rangle$ with unperturbed energy $E_0 = \varepsilon_{n_1} + \varepsilon_{n_2}$. The first-order energy correction is given by the diagonal expectation value:

$$\begin{aligned} E^{(1)}(R) &= \langle n_1, n_2 | V(R) | n_1, n_2 \rangle \\ &= \frac{e^2}{R^3} \left[\langle n_1 | x_1 | n_1 \rangle \langle n_2 | x_2 | n_2 \rangle + \langle n_1 | y_1 | n_1 \rangle \langle n_2 | y_2 | n_2 \rangle - 2 \langle n_1 | z_1 | n_1 \rangle \langle n_2 | z_2 | n_2 \rangle \right]. \end{aligned} \quad (11.71)$$

When either atom resides in a spherically symmetric state of definite parity (such as the ground state $|1s\rangle$), the parity selection rule requires $\langle n | \mathbf{r} | n \rangle = 0$. Consequently, the first-order energy shift vanishes identically:

$$E^{(1)}(R) = 0. \quad (11.72)$$

(If both atoms occupy degenerate manifolds of opposite parities, non-zero first-order matrix elements yield an interaction energy scaling as R^{-3} , describing resonant dipole exchange.)

For two non-degenerate ground-state atoms, the leading interaction arises at second order in Rayleigh–Schrödinger perturbation theory:

$$E^{(2)}(R) = \frac{e^4}{R^6} \sum_{m_1, m_2 \neq 1s, 1s} \frac{|\langle 1s, 1s | (x_1 x_2 + y_1 y_2 - 2z_1 z_2) | m_1, m_2 \rangle|^2}{2\varepsilon_{1s} - \varepsilon_{m_1} - \varepsilon_{m_2}}. \quad (11.73)$$

Because both atoms are initially in their lowest possible energy levels, every intermediate state satisfies $\varepsilon_{m_1} + \varepsilon_{m_2} > 2\varepsilon_{1s}$. The energy denominators are strictly negative, which guarantees that the second-order dispersion energy is strictly attractive:

$$E^{(2)}(R) < 0. \quad (11.74)$$

Factoring out dimensional scales, the effective interaction potential between two ground-state hydrogen atoms takes the universal van der Waals form:

$$E(R) = 2\varepsilon_{1s} - \frac{C_6}{R^6} = 2\varepsilon_{1s} - \frac{e^2}{a_0} \left(\frac{a_0}{R} \right)^6 \zeta, \quad (11.75)$$

where ζ is a dimensionless numerical coefficient defined by

$$\zeta = \frac{e^2}{a_0^5} \sum_{m_1, m_2 \neq 1s, 1s} \frac{|\langle 1s, 1s | (x_1 x_2 + y_1 y_2 - 2z_1 z_2) | m_1, m_2 \rangle|^2}{\varepsilon_{m_1} + \varepsilon_{m_2} - 2\varepsilon_{1s}}. \quad (11.76)$$

Detailed evaluation summing over discrete bound levels and integrating across continuum channels gives $\zeta \approx 6.499 \approx 6.5$.

11.4.3 Physical Origin and Short-Range Core Repulsion

The physical mechanism underlying the van der Waals attraction lies in correlated quantum vacuum fluctuations. Although isolated ground-state atoms possess zero permanent dipole moments, quantum zero-point motion generates instantaneous fluctuating dipoles $\mathbf{d}_1(t) = e\mathbf{r}_1$. The inhomogeneous electric field produced by $\mathbf{d}_1$ scales as R^{-3} and polarizes the second atom, inducing a dipole moment $\mathbf{d}_2 \sim \alpha\mathbf{d}_1/R^{-3}$. The interaction between the instantaneous dipole and the correlated induced dipole yields an attractive potential scaling as

$$V_{\text{vdW}} \sim -\frac{\langle \mathbf{d}_1 \cdot \mathbf{d}_2 \rangle}{R^3} \sim -\frac{1}{R^6}. \quad (11.77)$$

At small internuclear separations where the electron charge clouds overlap, the Pauli exclusion principle forces electrons of identical spin into higher-energy orthogonalized molecular orbitals. This exchange repulsion produces a steep short-range repulsive barrier, commonly modeled as R^{-12} . The combination of long-range dispersion attraction and short-range exchange repulsion is parameterized phenomenologically by the Lennard-Jones 6–12 potential:

$$V_{\text{LJ}}(R) = 4\epsilon \left[\left(\frac{\sigma}{R} \right)^{12} - \left(\frac{\sigma}{R} \right)^6 \right]. \quad (11.78)$$

At asymptotically large distances ($R \gtrsim 137a_0$), retardation effects associated with the finite speed of light soften the R^{-6} van der Waals potential into the R^{-7} Casimir–Polder potential.

11.5 Almost-Degenerate Perturbation Theory

Standard Rayleigh–Schrödinger perturbation theory expands observables in powers of the dimensionless parameter

$$\eta_{nm} \equiv \frac{\langle m|V|n \rangle}{\varepsilon_n - \varepsilon_m}. \quad (11.79)$$

Even if two unperturbed energy levels are not strictly degenerate ($\varepsilon_n \neq \varepsilon_m$), the expansion converges exceedingly slowly—or fails entirely—whenever the energy separation $|\varepsilon_n - \varepsilon_m|$ becomes comparable to or smaller than the coupling matrix element $|\langle m|V|n \rangle|$. In such *quasidegenerate* (or almost-degenerate) systems, one must adopt a hybrid approach: diagonalize the interaction exactly within the quasidegenerate subspace, and subsequently treat the residual coupling to all other distant states via standard perturbation theory.

11.5.1 Two-Level Subspace Diagonalization

Suppose that two unperturbed eigenstates, $|n\rangle$ and $|m\rangle$, are energetically close to each other, $|\varepsilon_n - \varepsilon_m| \lesssim |\langle m|V|n \rangle|$, while well separated from all other levels $|i\rangle$. We decompose the total interaction Hamiltonian into two parts:

$$V = V_1 + V_2, \quad (11.80)$$

where V_1 acts strictly within the two-dimensional subspace $\mathcal{H}_2 = \text{span}\{|n\rangle, |m\rangle\}$,

$$V_1 = |n\rangle\langle n|V|n\rangle\langle n| + |m\rangle\langle m|V|m\rangle\langle m| + |n\rangle\langle n|V|m\rangle\langle m| + |m\rangle\langle m|V|n\rangle\langle n|, \quad (11.81)$$

and $V_2 = V - V_1$ contains the couplings involving the remaining states.

We first solve the intermediate Hamiltonian

$$H_1 = H_0 + V_1 \quad (11.82)$$

exactly. For any state $|i\rangle \notin \mathcal{H}_2$, $V_1|i\rangle = 0$, so $|i\rangle$ remains an exact eigenstate of H_1 with unchanged eigenvalue ε_i . Within $\mathcal{H}_2$, we seek stationary states of the form

$$|\psi'\rangle = \alpha|n\rangle + \beta|m\rangle, \quad H_1|\psi'\rangle = E'|\psi'\rangle. \quad (11.83)$$

Projecting onto $\langle n|$ and $\langle m|$, we obtain the secular matrix equation:

$$\begin{pmatrix} E_n^{(1)} & V_{nm} \\ V_{mn} & E_m^{(1)} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E' \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad (11.84)$$

where $E_n^{(1)} \equiv \varepsilon_n + \langle n|V|n\rangle$, $E_m^{(1)} \equiv \varepsilon_m + \langle m|V|m\rangle$, and $V_{nm} \equiv \langle n|V|m\rangle = V_{mn}^*$.

The exact eigenvalues of the secular matrix are

$$E'_\pm = \frac{E_n^{(1)} + E_m^{(1)}}{2} \pm \sqrt{\left(\frac{E_n^{(1)} - E_m^{(1)}}{2}\right)^2 + |V_{nm}|^2}. \quad (11.85)$$

Equation (11.85) clearly manifests the phenomenon of *level repulsion* (or avoided crossing): the energy difference between the hybridized states is always strictly greater than the unperturbed difference:

$$\Delta E' = \sqrt{(E_n^{(1)} - E_m^{(1)})^2 + 4|V_{nm}|^2} \geq 2|V_{nm}|. \quad (11.86)$$

The corresponding eigenvectors $|\psi'_\pm\rangle = \alpha_\pm|n\rangle + \beta_\pm|m\rangle$ are determined by

$$\frac{\beta_\pm}{\alpha_\pm} = \frac{E'_\pm - E_n^{(1)}}{V_{nm}} = \frac{E_m^{(1)} - E_n^{(1)}}{2V_{nm}} \pm \sqrt{\left(\frac{E_m^{(1)} - E_n^{(1)}}{2V_{nm}}\right)^2 + 1}. \quad (11.87)$$

Having non-perturbatively diagonalized $\mathcal{H}_2$, the residual operator V_2 has vanished matrix elements within $\mathcal{H}_2$ ($\langle\psi'_\pm|V_2|\psi'_\mp\rangle = 0$), so ordinary perturbation theory in V_2 converges rapidly without small denominators.

11.5.2 Application: Nearly Free Electrons in a Periodic Lattice

A classic physical application of almost-degenerate perturbation theory is the nearly-free electron model (the *weak-binding approximation*) for an electron in a one-dimensional crystalline lattice of period d . The periodic potential satisfies $v(x) = v(x+d)$. Imposing Born–von Kármán periodic boundary conditions over a crystal of length $L = Nd$, the unperturbed eigenstates are plane waves:

$$\psi_k(x) = \frac{1}{\sqrt{L}} e^{ikx}, \quad \varepsilon_k = \frac{\hbar^2 k^2}{2m}, \quad (11.88)$$

with allowed wave vectors $k = 2\pi j/L$ ($j \in \mathbb{Z}$).

Expanding the periodic potential in a discrete Fourier series over reciprocal lattice vectors $G_n = nK = n(2\pi/d)$,

$$v(x) = \sum_{n=-\infty}^{\infty} v_n e^{inKx}, \quad v_n = \frac{1}{d} \int_0^d dx v(x) e^{-inKx}, \quad (11.89)$$

the matrix elements of the potential in the plane-wave basis take the selection-rule form:

$$\langle k'|v|k\rangle = \frac{1}{L} \int_0^L dx e^{-i(k'-k)x} \sum_n v_n e^{inKx} = \sum_n v_n \delta_{k',k+nK}. \quad (11.90)$$

The crystal potential only couples plane waves whose wave vectors differ by a reciprocal lattice vector nK .

Perturbation Away from the Brillouin Zone Boundaries

When the electron wave vector k is far from any zone boundary, $|\varepsilon_k - \varepsilon_{k+nK}| \gg |v_n|$ for all $n \neq 0$. Applying standard non-degenerate perturbation theory, the wave function to first order is

$$\Psi_k(x) = \frac{e^{ikx}}{\sqrt{L}} \left[1 + \sum_{n \neq 0} \frac{v_n}{\varepsilon_k - \varepsilon_{k+nK}} e^{inKx} \right] = e^{ikx} u_k(x), \quad (11.91)$$

which explicitly has the form of a Bloch wave with lattice periodicity $u_k(x+d) = u_k(x)$. The second-order energy is

$$E_k = \varepsilon_k + v_0 + \sum_{n \neq 0} \frac{|v_n|^2}{\varepsilon_k - \varepsilon_{k+nK}}. \quad (11.92)$$

Band Gap Opening at the Zone Boundary

However, as k approaches a Brillouin zone boundary,

$$k \approx -\frac{nK}{2}, \quad (11.93)$$

the two unperturbed states $|k\rangle$ and $|k+nK\rangle$ have nearly equal unperturbed kinetic energies:

$$\varepsilon_k = \frac{\hbar^2 k^2}{2m} \approx \frac{\hbar^2 (k+nK)^2}{2m} = \varepsilon_{k+nK}. \quad (11.94)$$

Here the denominator in Eq. (11.92) vanishes and non-degenerate perturbation theory collapses.

Employing almost-degenerate perturbation theory within the two-state subspace spanned by $\{|k\rangle, |k+nK\rangle\}$, we diagonalize the 2×2 effective Hamiltonian:

$$\begin{pmatrix} \varepsilon_k + v_0 & v_n^* \\ v_n & \varepsilon_{k+nK} + v_0 \end{pmatrix}. \quad (11.95)$$

From Eq. (11.85), the exact energy eigenvalues are

$$E_{k\pm} = v_0 + \frac{\varepsilon_k + \varepsilon_{k+nK}}{2} \pm \sqrt{\left(\frac{\varepsilon_k - \varepsilon_{k+nK}}{2}\right)^2 + |v_n|^2}. \quad (11.96)$$

Far from the boundary ($|\varepsilon_k - \varepsilon_{k+nK}| \gg |v_n|$), Eq. (11.96) smoothly recovers the unperturbed parabolic dispersion shifts $E_{k+} \rightarrow \varepsilon_k + v_0$ and $E_{k-} \rightarrow \varepsilon_{k+nK} + v_0$.

Precisely at the Bragg reflection plane $k = -nK/2$, where $\varepsilon_k = \varepsilon_{k+nK}$, the degeneracy is lifted:

$$E_{\pm} \left(-\frac{nK}{2} \right) = \varepsilon_{nK/2} + v_0 \pm |v_n|. \quad (11.97)$$

The continuous free-particle parabolic dispersion splits, opening a forbidden *energy band gap* of width

$$E_g = 2|v_n| \quad (11.98)$$

at every Brillouin zone boundary $k = \pm n\pi/d$. At these points, the propagating waves form standing waves whose probability densities either maximize electron density at the attractive ionic potentials (lower energy branch) or in the interatomic interstitial regions (upper energy branch), providing the physical mechanism of electronic band structure in solids.

11.6 Brillouin–Wigner Perturbation Theory

While Rayleigh–Schrödinger perturbation theory provides an explicit power series in powers of the perturbation parameter λ , the expressions for higher-order state corrections become increasingly intricate due to nested energy denominators and renormalization terms. An alternative formulation, due to Brillouin and Wigner, reorganizes the perturbation expansion into a remarkably transparent, iterative structure by anchoring the energy denominators to the exact perturbed eigenvalue E_n rather than the unperturbed level ε_n .

11.6.1 Iterative Integral Equation for the State Vector

We begin with the stationary Schrödinger equation written in the form

$$(E_n - H_0)|N\rangle = \lambda V|N\rangle. \quad (11.99)$$

Projecting onto an arbitrary unperturbed eigenstate $\langle m|$, we have

$$(E_n - \varepsilon_m)\langle m|N\rangle = \lambda\langle m|V|N\rangle. \quad (11.100)$$

Adopting intermediate normalization, $\langle n|N\rangle = 1$, the exact state vector can be decomposed into its parallel and orthogonal components relative to $|n\rangle$:

$$|N\rangle = |n\rangle\langle n|N\rangle + \sum_{m \neq n} |m\rangle\langle m|N\rangle = |n\rangle + \sum_{m \neq n} |m\rangle\langle m|N\rangle. \quad (11.101)$$

Substituting $\langle m|N\rangle$ from Eq. (11.100) for all $m \neq n$, we obtain the fundamental Brillouin–Wigner recurrence relation:

$$|N\rangle = |n\rangle + \sum_{m \neq n} |m\rangle \frac{1}{E_n - \varepsilon_m} \lambda \langle m|V|N\rangle. \quad (11.102)$$

In operator notation, introducing the projection operator onto the orthogonal complement of $|n\rangle$, $Q_n = 1 - |n\rangle\langle n| = \sum_{m \neq n} |m\rangle\langle m|$, Eq. (11.102) can be written compactly as

$$|N\rangle = |n\rangle + \frac{Q_n}{E_n - H_0} \lambda V|N\rangle. \quad (11.103)$$

11.6.2 Brillouin–Wigner Series and Self-Consistent Energy Equation

Iterating Eq. (11.102) recursively generates the Brillouin–Wigner series for the perturbed state vector:

$$\begin{aligned} |N\rangle = & |n\rangle + \lambda \sum_{m \neq n} \frac{\langle m|V|n\rangle}{E_n - \varepsilon_m} |m\rangle \\ & + \lambda^2 \sum_{j \neq n} \sum_{m \neq n} \frac{\langle j|V|m\rangle \langle m|V|n\rangle}{(E_n - \varepsilon_j)(E_n - \varepsilon_m)} |j\rangle \\ & + \lambda^3 \sum_{k \neq n} \sum_{j \neq n} \sum_{m \neq n} \frac{\langle k|V|j\rangle \langle j|V|m\rangle \langle m|V|n\rangle}{(E_n - \varepsilon_k)(E_n - \varepsilon_j)(E_n - \varepsilon_m)} |k\rangle + \dots \end{aligned} \quad (11.104)$$

Unlike the Rayleigh–Schrödinger expansion, Eq. (11.104) does not represent a strict Taylor expansion in λ , because the unknown energy eigenvalue E_n itself depends implicitly on λ .

Substituting the expansion (11.104) into the exact energy relation $E_n = \varepsilon_n + \lambda\langle n|V|N\rangle$, we obtain a self-consistent, non-linear algebraic equation for the perturbed energy:

$$E_n = \varepsilon_n + \lambda\langle n|V|n\rangle + \lambda^2 \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{E_n - \varepsilon_m} + \lambda^3 \sum_{j \neq n} \sum_{m \neq n} \frac{\langle n|V|j\rangle \langle j|V|m\rangle \langle m|V|n\rangle}{(E_n - \varepsilon_j)(E_n - \varepsilon_m)} + \dots \quad (11.105)$$

11.6.3 Relationship to Rayleigh–Schrödinger Theory

The connection between the two formulations becomes apparent upon performing a Taylor expansion of the Brillouin–Wigner energy denominators around the unperturbed level ε_n :

$$\frac{1}{E_n - \varepsilon_m} = \frac{1}{(\varepsilon_n - \varepsilon_m) + (E_n - \varepsilon_n)} = \frac{1}{\varepsilon_n - \varepsilon_m} \sum_{p=0}^{\infty} \left(-\frac{E_n - \varepsilon_n}{\varepsilon_n - \varepsilon_m} \right)^p. \quad (11.106)$$

Inserting this geometric expansion order-by-order reproduces the Rayleigh–Schrödinger series term by term. For instance, expanding the state vector to second order yields

$$\begin{aligned} |N\rangle &= |n\rangle + \lambda \sum_{m \neq n} \left(\frac{1}{\varepsilon_n - \varepsilon_m} - \frac{\lambda \langle n|V|n\rangle}{(\varepsilon_n - \varepsilon_m)^2} \right) \langle m|V|n\rangle |m\rangle \\ &\quad + \lambda^2 \sum_{j \neq n} \sum_{m \neq n} \frac{\langle j|V|m\rangle \langle m|V|n\rangle}{(\varepsilon_n - \varepsilon_j)(\varepsilon_n - \varepsilon_m)} |j\rangle + \mathcal{O}(\lambda^3), \end{aligned} \quad (11.107)$$

which is identically the Rayleigh–Schrödinger expression.

The Brillouin–Wigner approach offers significant practical advantages when solved iteratively. By retaining E_n in the denominators, the truncated Brillouin–Wigner equation (such as the second-order truncation $E_n = \varepsilon_n + \langle n|V|n\rangle + \sum_{m \neq n} \frac{|\langle m|V|n\rangle|^2}{E_n - \varepsilon_m}$) sums an infinite subset of geometric corrections from all orders of Rayleigh–Schrödinger perturbation theory. This non-perturbative resummation substantially broadens the radius of convergence, especially in the presence of strong level mixing or near-degeneracies.

11.7 Non-Perturbative Methods: The Variational Principle

When a Hamiltonian cannot be separated into a solvable part plus a small perturbation, or when the interaction parameter is large, perturbative series fail. A powerful, non-perturbative technique for estimating ground-state energies and wave functions is provided by the *Rayleigh–Ritz variational principle*.

11.7.1 The Rayleigh–Ritz Variational Theorem

The variational principle states that for any arbitrary normalizable trial state $|\psi\rangle$, the expectation value of the Hamiltonian H provides a strict upper bound to the exact ground-state energy E_0 :

$$\langle H \rangle_\psi \equiv \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0. \quad (11.108)$$

Equality holds if and only if $|\psi\rangle$ is an exact eigenstate belonging to the ground-state eigenvalue E_0 .

To establish this result, we express H in terms of its complete set of orthonormal eigenstates $\{|N\rangle\}$ and eigenvalues $\{E_N\}$ (with $E_0 \leq E_1 \leq E_2 \leq \dots$):

$$H = \sum_N E_N |N\rangle \langle N|. \quad (11.109)$$

Evaluating the expectation value in the state $|\psi\rangle$, we find

$$\langle \psi | H | \psi \rangle = \sum_N E_N |\langle N | \psi \rangle|^2 \geq E_0 \sum_N |\langle N | \psi \rangle|^2 = E_0 \langle \psi | \psi \rangle, \quad (11.110)$$

which immediately yields Eq. (11.108). If $|\psi\rangle$ contains any non-zero admixture of excited eigenstates $|N\rangle$ ($N \geq 1$), the inequality is strict. Furthermore, if $|\psi\rangle$ deviates from the true ground state by a first-order error $\delta|\psi\rangle = |\psi\rangle - |0\rangle \sim \mathcal{O}(\epsilon)$, the energy expectation value $\langle H \rangle_\psi$ deviates from E_0 only at second order, $\mathcal{O}(\epsilon^2)$, ensuring that variational energy estimates are highly robust.

11.7.2 Variational Optimization and Excited States

In practical calculations, one constructs a family of parameterized trial wave functions $\psi(\mathbf{r}; \alpha_1, \alpha_2, \dots, \alpha_p)$ designed to capture the relevant physical features (asymptotics, cusps, symmetry). Minimizing the functional

$$E(\alpha_1, \dots, \alpha_p) = \frac{\langle \psi(\boldsymbol{\alpha}) | H | \psi(\boldsymbol{\alpha}) \rangle}{\langle \psi(\boldsymbol{\alpha}) | \psi(\boldsymbol{\alpha}) \rangle} \quad (11.111)$$

with respect to the variational parameters $\{\alpha_j\}$ selects the optimal approximation to the ground-state wave function and yields the tightest upper bound to E_0 .

The variational theorem also generalizes to excited states:

1. **Symmetry-restricted subspaces:** If the Hamiltonian commutes with a symmetry operator (e.g., parity or orbital angular momentum $\mathbf{L}^2$), the Hilbert space decomposes into orthogonal symmetry sectors. Restricting trial wave functions to a specific irreducible representation (such as p -wave states with $l = 1$) provides a strict upper bound to the lowest-energy eigenstate within that symmetry sector.
2. **Orthogonality constraints:** If the exact ground state $|0\rangle$ (or an accurate approximation thereof) is known, choosing trial states explicitly orthogonal to $|0\rangle$ ($\langle 0 | \psi \rangle = 0$) provides a variational upper bound to the first excited state energy E_1 .

11.7.3 Lower Bounds from Positive Operators

While the Rayleigh–Ritz principle provides upper bounds, establishing rigorous lower bounds is generally more difficult. However, when an interaction potential V is positive semidefinite,

$$\langle \psi | V | \psi \rangle \geq 0 \quad \text{for all } |\psi\rangle, \quad (11.112)$$

a useful lower bound immediately follows. Let $|0\rangle$ denote the normalized ground state of the full Hamiltonian $H = H_0 + V$ with eigenvalue E_0 , and let ε_0 denote the ground-state eigenvalue of H_0 . Then

$$E_0 = \langle 0 | (H_0 + V) | 0 \rangle = \langle 0 | H_0 | 0 \rangle + \langle 0 | V | 0 \rangle \geq \langle 0 | H_0 | 0 \rangle. \quad (11.113)$$

Applying the variational principle to H_0 , we have $\langle 0 | H_0 | 0 \rangle \geq \varepsilon_0$, which proves that

$$E_0 \geq \varepsilon_0. \quad (11.114)$$

Removing a positive-semidefinite operator from a Hamiltonian strictly lowers (or leaves unchanged) its ground-state energy.

11.8 Problems

1. **Second-order energy as a variational expectation value.** Show that in Rayleigh–Schrödinger perturbation theory, the energy calculated up to second order is identically equal to the expectation value of the total Hamiltonian $H = H_0 + V$ evaluated in the state normalized to first order:

$$|\bar{N}^{(1)}\rangle = \frac{|n\rangle + |N^{(1)}\rangle}{\sqrt{1 + \langle N^{(1)} | N^{(1)} \rangle}}. \quad (11.115)$$

Demonstrate that the difference $\langle \bar{N}^{(1)} | H | \bar{N}^{(1)} \rangle - (E_n^{(0)} + E_n^{(1)} + E_n^{(2)})$ is of order $\mathcal{O}(V^3)$.

2. **Resonant van der Waals interaction between excited atoms.** Calculate the leading-order interaction energy between two neutral hydrogen atoms both prepared in excited $n = 2$ states at large internuclear separation R . Determine the degenerate secular matrix, evaluate the energy eigenvalues, and interpret the physical origin of the resulting R^{-3} dependence in contrast to the R^{-6} dispersion force of the ground state.

3. **Upper and lower bounds on the van der Waals dispersion coefficient.** Consider the second-order dispersion coefficient ζ for two ground-state hydrogen atoms, defined in Eq. (11.76).
- (a) Obtain a rigorous lower bound on ζ by truncating the sum over intermediate states to include only the lowest accessible excited states ($n = 2$). Prove that this truncation provides a strict lower bound.
 - (b) Obtain an upper bound on ζ (the Unsöld approximation) by replacing the excitation energy denominators $\varepsilon_{m_1} + \varepsilon_{m_2} - 2\varepsilon_{1s}$ with a suitable minimum energy threshold and evaluating the resulting sum using the quantum completeness relation.
4. **Electric polarizability of atomic hydrogen.** A hydrogen atom in its $1s$ ground state is subjected to a weak uniform static electric field $\mathcal{E} = \mathcal{E}\hat{\mathbf{z}}$.
- (a) Write down the first-order perturbed wave function and derive the induced electric dipole moment $\langle d_z \rangle = -e\langle z \rangle$ to lowest order in $\mathcal{E}$.
 - (b) Using the Hellmann–Feynman theorem, demonstrate that the induced dipole moment can be equivalently computed from the parametric derivative of the second-order Stark energy shift:

$$\langle d_z \rangle = -\frac{\partial E^{(2)}}{\partial \mathcal{E}}. \quad (11.116)$$

- (c) Utilizing the closure and truncation techniques developed in Problem 3, establish rigorous upper and lower bounds on the static electric polarizability $\alpha = \langle d_z \rangle / \mathcal{E}$ of the ground state.
5. **Wave functions near Brillouin zone boundaries.** Within the weak-binding (nearly-free electron) approximation in one dimension, determine the explicit form of the hybridized wave functions $\Psi_{k\pm}(x)$ in the immediate vicinity of the Brillouin zone boundary $k \approx n\pi/d$. Sketch the spatial probability densities $|\Psi_{k\pm}(x)|^2$ at the zone boundary and give a physical interpretation of the origin of the band gap.
6. **Exactness of the renormalization relation in Brillouin–Wigner theory.**
- (a) Utilizing Brillouin–Wigner perturbation theory, prove that the relation

$$\frac{\partial E_n}{\partial \varepsilon_n} = Z \quad (11.117)$$

holds identically to all orders in the perturbation, where $Z = |\langle n | \bar{N} \rangle|^2$ is the wave function renormalization constant, and the partial derivative is taken with fixed potential matrix elements and constant background energies ε_m ($m \neq n$).

- (b) For a one-dimensional harmonic oscillator subjected to a constant force F , evaluate Z explicitly up to second order using the exact translated state vector Eq. (11.32), and verify that it matches $\partial E_n / \partial \varepsilon_n$ obtained from the second-order energy shift.
7. **Variational calculation for hydrogen using a Gaussian trial function.** Estimate the ground-state energy of the hydrogen atom using a three-dimensional isotropic Gaussian trial wave function,

$$\psi(r; \alpha) = \exp(-\alpha r^2), \quad (11.118)$$

where $\alpha > 0$ is a variational parameter. Minimize the energy expectation value and compare the result with the exact ground-state energy $-e^2/(2a_0)$.

8. **Ground-state bounds for the Yukawa potential.** Consider an attractive Yukawa potential in three dimensions:

$$V(r) = -A \frac{e^{-qr}}{r}, \quad (A > 0, q > 0). \quad (11.119)$$

Using suitable trial wave functions and operator inequalities, determine analytic upper and lower bounds for the ground-state energy.

9. **Electron in a three-dimensional separable periodic potential.** Estimate the ground-state energy of an electron moving in a three-dimensional cubic lattice of size $L = Na$ ($N \gg 1$) subject to the periodic potential

$$V(\mathbf{r}) = V_0 \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{2\pi y}{a}\right) \sin\left(\frac{2\pi z}{a}\right), \quad (11.120)$$

imposing periodic boundary conditions. Determine whether a non-zero energy shift appears at first or second order in V_0 .

10. **Half-harmonic oscillator with a hard wall.** A particle of mass m moves in the one-dimensional half-oscillator potential

$$V(x) = \begin{cases} \frac{1}{2}m\omega_0^2(x-a)^2, & x > 0, \\ \infty, & x \leq 0, \end{cases} \quad (11.121)$$

where $a > 0$. Using the variational method, estimate the ground-state energy. Determine whether the true ground-state energy is strictly greater or less than $\frac{1}{2}\hbar\omega_0$, and justify your answer physically.

11. **Diamagnetic susceptibility of atomic hydrogen.** A hydrogen atom in its $1s$ ground state is placed in a uniform static magnetic field $\mathbf{B} = \mathcal{B}\hat{\mathbf{z}}$. Working in the symmetric gauge $\mathbf{A} = -\frac{1}{2}\mathbf{r} \times \mathbf{B}$, calculate the induced magnetization

$$\mathbf{M} = -\frac{\partial E}{\partial \mathbf{B}} \quad (11.122)$$

to first order in $\mathcal{B}$, neglecting electron spin. Evaluate the diamagnetic susceptibility of the ground state in terms of the Bohr radius a_0 .

Chapter 12

Time Dependent Perturbation Theory

12.1 Time-Dependent Perturbation Theory

In stationary-state perturbation theory, the Hamiltonian was taken to be time-independent, and the objective was to determine perturbed stationary energy levels and eigenstates. We now consider a dynamical situation: a quantum system initially governed by an unperturbed, solvable Hamiltonian H_0 is subjected to an explicit, time-dependent external interaction $V(t)$ starting at some initial time t_0 . Typical physical examples include the photoionization of an atom irradiated by an electromagnetic pulse, or electrical transport induced by an applied time-dependent voltage.

12.1.1 The Schrödinger Equation with Time-Dependent Interactions

Prior to the activation of the perturbation ($t \leq t_0$), the system evolves under the unperturbed Hamiltonian H_0 :

$$i\hbar \frac{\partial}{\partial t} |\psi_S(t)\rangle = H_0 |\psi_S(t)\rangle, \quad (t \leq t_0). \quad (12.1)$$

For $t > t_0$, the perturbation $V(t)$ is switched on, and the state vector evolves in the Schrödinger picture according to

$$i\hbar \frac{\partial}{\partial t} |\psi_S(t)\rangle = [H_0 + V(t)] |\psi_S(t)\rangle, \quad (t > t_0), \quad (12.2)$$

subject to the initial boundary condition that $|\psi_S(t_0)\rangle$ matches the unperturbed state.

Exact closed-form solutions of Eq. (12.2) are rarely accessible. However, when $V(t)$ is weak, the resulting dynamical response can be solved systematically by expanding the state vector in powers of the interaction.

12.1.2 The Interaction Picture

Because the rapid dynamical phase evolution generated by H_0 accounts for the dominant time dependence, it is advantageous to isolate this unperturbed motion by transforming to the *interaction picture* (or Dirac picture). We define the interaction-picture state vector $|\psi_I(t)\rangle$ by

$$|\psi_I(t)\rangle \equiv e^{iH_0 t/\hbar} |\psi_S(t)\rangle \iff |\psi_S(t)\rangle = e^{-iH_0 t/\hbar} |\psi_I(t)\rangle. \quad (12.3)$$

Differentiating Eq. (12.3) with respect to time and using Eq. (12.2), we find

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle &= -H_0 e^{iH_0 t/\hbar} |\psi_S(t)\rangle + e^{iH_0 t/\hbar} \left(i\hbar \frac{\partial}{\partial t} |\psi_S(t)\rangle \right) \\ &= -e^{iH_0 t/\hbar} H_0 |\psi_S(t)\rangle + e^{iH_0 t/\hbar} [H_0 + V(t)] |\psi_S(t)\rangle \\ &= e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar} |\psi_I(t)\rangle. \end{aligned} \quad (12.4)$$

Thus, the unperturbed Hamiltonian cancels out completely, and the interaction-picture state evolves according to the Tomonaga–Schwinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle, \quad (12.5)$$

where the interaction-picture potential operator is defined by

$$V_I(t) \equiv e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar}. \quad (12.6)$$

In the interaction picture, state vectors evolve solely due to the coupling $V_I(t)$, while operators evolve under H_0 via the standard Heisenberg equations of motion. If $V(t) = 0$, the interaction picture coincides with the Heisenberg picture.

12.1.3 The Dyson Series and Time Ordering

Integrating the differential equation (12.5) from the initial time t_0 to t gives the Volterra integral equation:

$$|\psi_I(t)\rangle = |\psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt' V_I(t') |\psi_I(t')\rangle. \quad (12.7)$$

This integral equation lends itself to recursive Picard iteration. Substituting the zeroth-order approximation $|\psi_I(t')\rangle \approx |\psi_I(t_0)\rangle$ into the integrand yields the first-order state correction:

$$|\psi_I^{(1)}(t)\rangle = |\psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt' V_I(t') |\psi_I(t_0)\rangle. \quad (12.8)$$

Re-inserting this expression back into Eq. (12.7) generates the second-order correction:

$$|\psi_I^{(2)}(t)\rangle = |\psi_I^{(1)}(t)\rangle + \frac{1}{(i\hbar)^2} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' V_I(t') V_I(t'') |\psi_I(t_0)\rangle. \quad (12.9)$$

In Eq. (12.9), the operators are chronologically ordered such that $t' \geq t''$. In general, the n -th order contribution involves n nested time integrals with chronologically ordered operators $V_I(t_1) V_I(t_2) \cdots V_I(t_n)$ where $t \geq t_1 \geq t_2 \geq \cdots \geq t_n \geq t_0$.

Introducing the chronological time-ordering operator $\mathcal{T}$, which rearranges products of time-dependent operators in descending chronological order from left to right, the entire infinite Dyson series can be expressed in compact exponential form:

$$|\psi_I(t)\rangle = U_I(t, t_0) |\psi_I(t_0)\rangle, \quad (12.10)$$

where the unitary time-evolution operator in the interaction picture is given by the time-ordered Dyson exponential:

$$\begin{aligned} U_I(t, t_0) &= \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t dt' V_I(t') \right] \\ &= \sum_{n=0}^{\infty} \left(-\frac{i}{\hbar} \right)^n \frac{1}{n!} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n \mathcal{T} [V_I(t_1) V_I(t_2) \cdots V_I(t_n)]. \end{aligned} \quad (12.11)$$

12.2 First-Order Transitions and Fermi's Golden Rule

A central problem in time-dependent quantum mechanics is determining the probability that an external perturbation $V(t)$ induces a transition from a known initial eigenstate $|0\rangle$ of H_0 (with energy ε_0) to a distinct orthogonal eigenstate $|n\rangle$ (with energy ε_n).

12.2.1 First-Order Transition Amplitude

In the Schrödinger picture, the projection of the evolved state $|\psi_S(t)\rangle$ onto the target eigenstate $|n\rangle$ gives the transition amplitude:

$$\mathcal{A}_{0 \rightarrow n}(t) \equiv \langle n | \psi_S(t) \rangle = e^{-i\varepsilon_n t/\hbar} \langle n | \psi_I(t) \rangle. \quad (12.12)$$

Assuming the system is initially prepared in state $|0\rangle$ at time t_0 , so that $|\psi_I(t_0)\rangle = |0\rangle$, the first-order interaction-picture expansion yields

$$\langle n | \psi_I(t) \rangle = \frac{1}{i\hbar} \int_{t_0}^t dt' \langle n | V_I(t') | 0 \rangle = \frac{1}{i\hbar} \int_{t_0}^t dt' e^{i(\varepsilon_n - \varepsilon_0)t'/\hbar} \langle n | V(t') | 0 \rangle. \quad (12.13)$$

The transition probability from state $|0\rangle$ to state $|n\rangle$ ($n \neq 0$) is therefore given in first order by

$$P_{0 \rightarrow n}(t) = |\mathcal{A}_{0 \rightarrow n}(t)|^2 = \frac{1}{\hbar^2} \left| \int_{t_0}^t dt' e^{i\omega_{n0}t'} \langle n | V(t') | 0 \rangle \right|^2, \quad (12.14)$$

where $\omega_{n0} \equiv (\varepsilon_n - \varepsilon_0)/\hbar$ is the Bohr transition frequency.

12.2.2 Sudden Switching of a Constant Perturbation

Consider first a constant perturbation switched on abruptly at $t = 0$:

$$V(t) = \begin{cases} 0, & t < 0, \\ V, & t \geq 0. \end{cases} \quad (12.15)$$

Substituting this profile into Eq. (12.14) with $t_0 = 0$ yields

$$\begin{aligned} P_{0 \rightarrow n}(t) &= \frac{|\langle n | V | 0 \rangle|^2}{\hbar^2} \left| \frac{e^{i(\varepsilon_n - \varepsilon_0)t/\hbar} - 1}{i(\varepsilon_n - \varepsilon_0)/\hbar} \right|^2 \\ &= |\langle n | V | 0 \rangle|^2 \left[\frac{\sin\left(\frac{(\varepsilon_n - \varepsilon_0)t}{2\hbar}\right)}{\frac{\varepsilon_n - \varepsilon_0}{2}} \right]^2. \end{aligned} \quad (12.16)$$

The distribution in Eq. (12.16) exhibits a central diffraction maximum centered at $\varepsilon_n = \varepsilon_0$:

1. At very short times, a Taylor expansion gives $P_{0 \rightarrow n}(t) \approx \frac{1}{\hbar^2} |\langle n | V | 0 \rangle|^2 t^2$.
2. The central peak has a height scaling as $t^2/\hbar^2$ and a half-width determined by the first zero of the sinc function:

$$|\varepsilon_n - \varepsilon_0| \leq \frac{2\pi\hbar}{t}. \quad (12.17)$$

This reflects the fundamental energy–time uncertainty relation:

$$\Delta E \cdot \Delta t \gtrsim 2\pi\hbar. \quad (12.18)$$

Transitions to non-energy-conserving states are allowed for short durations, but as $t \rightarrow \infty$, the probability distribution sharpens into an energy-conserving Dirac delta function.

12.2.3 Transitions to a Continuum: Derivation of Fermi's Golden Rule

When the final state $|n\rangle$ belongs to a dense quasi-continuum (such as a free-particle continuum or ionization states), we evaluate the total transition probability into a group of continuum states by integrating over the final-state density $\rho(\varepsilon_n)$:

$$P_{\text{total}}(t) = \sum_{n \in \text{group}} P_{0 \rightarrow n}(t) \approx \int d\varepsilon_n \rho(\varepsilon_n) |\langle n|V|0\rangle|^2 \left[\frac{\sin\left(\frac{(\varepsilon_n - \varepsilon_0)t}{2\hbar}\right)}{\frac{\varepsilon_n - \varepsilon_0}{2}} \right]^2. \quad (12.19)$$

Assuming that the matrix element $\langle n|V|0\rangle$ and the density of states $\rho(\varepsilon_n)$ vary smoothly across the central resonance peak, they may be evaluated at $\varepsilon_n = \varepsilon_0$ and factored out of the integral. Using the definite integral

$$\int_{-\infty}^{\infty} dx \frac{\sin^2(ax)}{x^2} = \pi a \implies \int_{-\infty}^{\infty} d\varepsilon \left[\frac{\sin\left(\frac{\varepsilon t}{2\hbar}\right)}{\frac{\varepsilon}{2}} \right]^2 = \frac{2\pi t}{\hbar}, \quad (12.20)$$

we find that the total transition probability grows linearly with time:

$$P_{\text{total}}(t) = \Gamma t, \quad (12.21)$$

where the constant transition rate $\Gamma \equiv dP_{\text{total}}/dt$ is given by *Fermi's Golden Rule*:

$$\Gamma = \frac{2\pi}{\hbar} |\langle n|V|0\rangle|^2 \rho(\varepsilon_0). \quad (12.22)$$

Equivalently, taking the distributional limit as $t \rightarrow \infty$,

$$\lim_{t \rightarrow \infty} \frac{1}{\pi t} \frac{\sin^2(\Delta\varepsilon t/2\hbar)}{(\Delta\varepsilon/2)^2} = \frac{2}{\hbar} \delta(\Delta\varepsilon), \quad (12.23)$$

the differential transition rate to an individual continuum mode $|n\rangle$ is expressed as

$$\Gamma_{0 \rightarrow n} = \frac{2\pi}{\hbar} |\langle n|V|0\rangle|^2 \delta(\varepsilon_n - \varepsilon_0). \quad (12.24)$$

Regimes of Validity

Fermi's Golden Rule is valid within an intermediate time window:

$$\frac{2\pi\hbar}{\Delta\varepsilon} \ll t \ll \frac{2\pi\hbar}{\delta\varepsilon}, \quad (12.25)$$

where $\Delta\varepsilon$ is the width of the continuum band and $\delta\varepsilon$ is the discrete level spacing of the system. Furthermore, first-order perturbation theory requires that the depletion of the initial state remains negligible, $t \ll 1/\Gamma$. Over macroscopic times, initial-state depletion leads to the standard exponential decay law:

$$N(t) = N_0 e^{-\Gamma t}. \quad (12.26)$$

12.2.4 Physical Applications

Elastic Scattering and the Born Approximation

Consider an incident particle of mass m in an initial momentum state $|\mathbf{k}\rangle$ normalized in a box of volume L^3 , scattered by a localized potential $V(\mathbf{r})$ into a final state $|\mathbf{k}'\rangle$. The wave functions are plane waves $\psi_{\mathbf{k}}(\mathbf{r}) = L^{-3/2} e^{i\mathbf{k}\cdot\mathbf{r}}$, giving the matrix element

$$\langle \mathbf{k}'|V|\mathbf{k}\rangle = \frac{1}{L^3} \int d^3r e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} V(\mathbf{r}) = \frac{\tilde{V}(\mathbf{k}'-\mathbf{k})}{L^3}, \quad (12.27)$$

where $\tilde{V}(\mathbf{q})$ is the spatial Fourier transform of the interaction potential.

The number of final momentum states in volume element $d^3k' = k'^2 dk' d\Omega' = \frac{mk'}{\hbar^2} d\varepsilon_{k'} d\Omega'$ is

$$dN = L^3 \frac{d^3k'}{(2\pi)^3} = \frac{L^3 mk'}{(2\pi)^3 \hbar^2} d\varepsilon_{k'} d\Omega'. \quad (12.28)$$

Applying Fermi's Golden Rule to calculate the scattering rate $d\Gamma$ into solid angle $d\Omega'$, energy conservation fixes $k' = k$:

$$d\Gamma = \frac{2\pi}{\hbar} \left| \frac{\tilde{V}(\mathbf{k}' - \mathbf{k})}{L^3} \right|^2 \left[\frac{L^3 mk}{(2\pi)^3 \hbar^2} \right] d\Omega' = \frac{mk}{4\pi^2 \hbar^3 L^3} |\tilde{V}(\mathbf{k}' - \mathbf{k})|^2 d\Omega'. \quad (12.29)$$

The incident probability current density is $j_{\text{inc}} = v/L^3 = \hbar k/(mL^3)$. Dividing $d\Gamma$ by $j_{\text{inc}} d\Omega'$, the box volume cancels out, and we recover the first Born approximation for the differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{d\Gamma}{j_{\text{inc}} d\Omega'} = \frac{m^2}{4\pi^2 \hbar^4} |\tilde{V}(\mathbf{k}' - \mathbf{k})|^2. \quad (12.30)$$

Radioactive Decay

Consider the decay of an unstable nucleus from an initial state $|i\rangle$ of energy E_i to a daughter nucleus $|f\rangle$ of energy E_f with the emission of a light particle of mass m and momentum $\hbar\mathbf{k}$. Neglecting nuclear recoil, energy conservation requires $\varepsilon_k = \frac{\hbar^2 k^2}{2m} = E_i - E_f$. Integrating over all emission directions, the total decay constant is

$$\Gamma = \frac{mkL^3}{(2\pi)^2 \hbar^3} \int d\Omega |\langle f, \hbar\mathbf{k} | V | i \rangle|^2. \quad (12.31)$$

Because the continuum normalization scales as $L^{-3/2}$, the volume L^3 cancels, yielding a well-defined intrinsic radioactive decay rate.

12.2.5 Adiabatic Switching

In physical systems, perturbations are often turned on gradually rather than instantaneously. A mathematically clean model for adiabatic switching is

$$V(t) = e^{\eta t} V, \quad (\eta \rightarrow 0^+), \quad (12.32)$$

with the system originating at $t_0 \rightarrow -\infty$. Integrating Eq. (12.13), the first-order state amplitude is

$$\langle n | \psi_I(t) \rangle = \frac{1}{i\hbar} \int_{-\infty}^t dt' e^{\eta t'} e^{i(\varepsilon_n - \varepsilon_0)t'/\hbar} \langle n | V | 0 \rangle = \frac{e^{\eta t + i(\varepsilon_n - \varepsilon_0)t/\hbar}}{\varepsilon_0 - \varepsilon_n + i\eta\hbar} \langle n | V | 0 \rangle. \quad (12.33)$$

The transition probability evaluates to a Lorentzian profile:

$$P_{0 \rightarrow n}(t) = \frac{e^{2\eta t}}{(\varepsilon_0 - \varepsilon_n)^2 + (\eta\hbar)^2} |\langle n | V | 0 \rangle|^2. \quad (12.34)$$

Differentiating with respect to time and taking the adiabatic limit $\eta \rightarrow 0^+$, the Lorentzian representation of the delta function,

$$\lim_{\eta \rightarrow 0^+} \frac{\eta\hbar}{\pi[(\varepsilon_0 - \varepsilon_n)^2 + (\eta\hbar)^2]} = \delta(\varepsilon_0 - \varepsilon_n), \quad (12.35)$$

reproduces Fermi's Golden Rule identically:

$$\frac{dP_{0 \rightarrow n}}{dt} = \frac{2\pi}{\hbar} |\langle n | V | 0 \rangle|^2 \delta(\varepsilon_n - \varepsilon_0). \quad (12.36)$$

The transition rate into a continuum is completely independent of the switching protocol.

12.3 Harmonic Perturbations

Many physical interactions in quantum optics and spectroscopy involve oscillating fields, such as laser radiation interacting with an atomic system. We now generalize the perturbation formalism to a harmonic perturbation:

$$V(t) = V \cos(\omega t) e^{\eta t} = \frac{V}{2} (e^{-i\omega t} + e^{i\omega t}) e^{\eta t}, \quad (\eta \rightarrow 0^+), \quad (12.37)$$

where the positive infinitesimal parameter η implements adiabatic switching from the remote past ($t_0 \rightarrow -\infty$).

12.3.1 First-Order Transition Amplitude and Probability

Assuming the system is prepared in an unperturbed eigenstate $|0\rangle$ at $t \rightarrow -\infty$, the first-order transition amplitude in the interaction picture is

$$\begin{aligned} \langle n | \psi_I(t) \rangle &= \frac{1}{i\hbar} \int_{-\infty}^t dt' e^{\eta t'} \langle n | V | 0 \rangle \frac{e^{-i\omega t'} + e^{i\omega t'}}{2} e^{i(\varepsilon_n - \varepsilon_0)t'/\hbar} \\ &= \frac{e^{\eta t}}{2} \langle n | V | 0 \rangle \left[\frac{e^{i(\varepsilon_n - \varepsilon_0 - \hbar\omega)t/\hbar}}{\varepsilon_0 - \varepsilon_n + \hbar\omega + i\eta\hbar} + \frac{e^{i(\varepsilon_n - \varepsilon_0 + \hbar\omega)t/\hbar}}{\varepsilon_0 - \varepsilon_n - \hbar\omega + i\eta\hbar} \right]. \end{aligned} \quad (12.38)$$

Evaluating the transition probability $P_{0 \rightarrow n}(t) = |\langle n | \psi_I(t) \rangle|^2$, we obtain

$$\begin{aligned} P_{0 \rightarrow n}(t) &= \frac{e^{2\eta t}}{4} |\langle n | V | 0 \rangle|^2 \left\{ \frac{1}{(\varepsilon_0 - \varepsilon_n + \hbar\omega)^2 + (\eta\hbar)^2} + \frac{1}{(\varepsilon_0 - \varepsilon_n - \hbar\omega)^2 + (\eta\hbar)^2} \right. \\ &\quad \left. + 2 \operatorname{Re} \left[\frac{e^{-2i\omega t}}{(\varepsilon_0 - \varepsilon_n + \hbar\omega + i\eta\hbar)(\varepsilon_0 - \varepsilon_n - \hbar\omega - i\eta\hbar)} \right] \right\}. \end{aligned} \quad (12.39)$$

The three terms in Eq. (12.39) have clear physical origins:

1. The first term originates from the positive-frequency component $e^{-i\omega t}$ and is sharply resonant when $\varepsilon_n \approx \varepsilon_0 + \hbar\omega$ (absorption).
2. The second term originates from the negative-frequency component $e^{+i\omega t}$ and is sharply resonant when $\varepsilon_n \approx \varepsilon_0 - \hbar\omega$ (stimulated emission).
3. The third term represents quantum interference between the absorption and emission pathways, oscillating rapidly at the second-harmonic frequency 2ω .

12.3.2 Cycle Averaging and Transition Rates

Differentiating Eq. (12.39) with respect to time yields the instantaneous rate of transition:

$$\begin{aligned} \frac{dP_{0 \rightarrow n}}{dt} &= \frac{e^{2\eta t}}{4} |\langle n | V | 0 \rangle|^2 \left\{ \left[\frac{2\eta}{(\varepsilon_0 - \varepsilon_n + \hbar\omega)^2 + (\eta\hbar)^2} + \frac{2\eta}{(\varepsilon_0 - \varepsilon_n - \hbar\omega)^2 + (\eta\hbar)^2} \right] (1 - \cos 2\omega t) \right. \\ &\quad \left. + 2 \sin(2\omega t) \left[\frac{\varepsilon_0 - \varepsilon_n + \hbar\omega}{(\varepsilon_0 - \varepsilon_n + \hbar\omega)^2 + (\eta\hbar)^2} - \frac{\varepsilon_0 - \varepsilon_n - \hbar\omega}{(\varepsilon_0 - \varepsilon_n - \hbar\omega)^2 + (\eta\hbar)^2} \right] \right\}. \end{aligned} \quad (12.40)$$

When measuring physical transition rates over experimental observation times Δt spanning many optical periods ($\Delta t \gg 2\pi/\omega$), the rapidly oscillating interference terms proportional to $\cos(2\omega t)$ and $\sin(2\omega t)$ average to zero. Taking the adiabatic limit $\eta \rightarrow 0^+$, the Lorentzians reduce to Dirac delta functions:

$$\lim_{\eta \rightarrow 0^+} \frac{\eta\hbar}{\pi[(\varepsilon_0 - \varepsilon_n \pm \hbar\omega)^2 + (\eta\hbar)^2]} = \delta(\varepsilon_n - \varepsilon_0 \mp \hbar\omega). \quad (12.41)$$

The cycle-averaged transition rate $\Gamma_{0 \rightarrow n} \equiv \overline{dP_{0 \rightarrow n}/dt}$ therefore separates into two mutually independent, incoherent channels:

$$\Gamma_{0 \rightarrow n} = \frac{2\pi}{\hbar} \frac{|\langle n|V|0\rangle|^2}{4} [\delta(\varepsilon_n - \varepsilon_0 - \hbar\omega) + \delta(\varepsilon_n - \varepsilon_0 + \hbar\omega)]. \quad (12.42)$$

The physical interpretation of Eq. (12.42) is fundamental:

- **Resonant Absorption:** The $\delta(\varepsilon_n - \varepsilon_0 - \hbar\omega)$ component enforces $\varepsilon_n = \varepsilon_0 + \hbar\omega$. The quantum system absorbs an energy quantum $\hbar\omega$ from the classical field, promoting the system to a higher energy level.
- **Stimulated Emission:** The $\delta(\varepsilon_n - \varepsilon_0 + \hbar\omega)$ component enforces $\varepsilon_n = \varepsilon_0 - \hbar\omega$. If the initial state $|0\rangle$ is an excited state, the oscillating perturbation stimulates the emission of an energy quantum $\hbar\omega$, de-exciting the system to a lower level.

The matrix elements governing absorption and stimulated emission between states $|0\rangle$ and $|n\rangle$ have identical magnitudes ($|\langle n|V|0\rangle| = |\langle 0|V|n\rangle|$), establishing the microscopic quantum mechanical basis for the equality of Einstein's absorption and stimulated emission coefficients.

12.4 Second-Order Transitions

In many physical processes, symmetry selection rules dictate that the direct first-order matrix element vanishes identically:

$$\langle n|V(t)|0\rangle = 0. \quad (12.43)$$

Although direct first-order transitions between $|0\rangle$ and $|n\rangle$ are strictly forbidden, transitions can nonetheless proceed through second- or higher-order pathways mediated by virtual intermediate states.

12.4.1 Derivation of the Second-Order Transition Amplitude

Consider a static interaction V turned on adiabatically from the distant past ($t_0 \rightarrow -\infty$) via the exponential factor $e^{\eta t}$ ($\eta \rightarrow 0^+$). The second-order term in the Dyson expansion of the interaction-picture state vector is

$$|\psi_I^{(2)}(t)\rangle = \frac{1}{(i\hbar)^2} \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' e^{\eta t'} e^{\eta t''} V_I(t') V_I(t'') |0\rangle. \quad (12.44)$$

Inserting a resolution of the identity $\sum_m |m\rangle\langle m| = 1$ over the complete eigenbasis of H_0 , the transition amplitude to state $|n\rangle$ becomes

$$\langle n|\psi_I^{(2)}(t)\rangle = \frac{1}{(i\hbar)^2} \sum_m \langle n|V|m\rangle \langle m|V|0\rangle \int_{-\infty}^t dt' e^{[i(\varepsilon_n - \varepsilon_m)/\hbar + \eta]t'} \int_{-\infty}^{t'} dt'' e^{[i(\varepsilon_m - \varepsilon_0)/\hbar + \eta]t''}. \quad (12.45)$$

Performing the inner integration over t'' , we have

$$\int_{-\infty}^{t'} dt'' e^{[i(\varepsilon_m - \varepsilon_0)/\hbar + \eta]t''} = \frac{e^{[i(\varepsilon_m - \varepsilon_0)/\hbar + \eta]t'}}{i(\varepsilon_m - \varepsilon_0)/\hbar + \eta} = \frac{i\hbar e^{[i(\varepsilon_m - \varepsilon_0)/\hbar + \eta]t'}}{\varepsilon_0 - \varepsilon_m + i\eta\hbar}. \quad (12.46)$$

Substituting this into the t' integral yields

$$\begin{aligned} \langle n|\psi_I^{(2)}(t)\rangle &= \frac{1}{i\hbar} \sum_m \frac{\langle n|V|m\rangle \langle m|V|0\rangle}{\varepsilon_0 - \varepsilon_m + i\eta\hbar} \int_{-\infty}^t dt' e^{[i(\varepsilon_n - \varepsilon_0)/\hbar + 2\eta]t'} \\ &= \frac{e^{2\eta t + i(\varepsilon_n - \varepsilon_0)t/\hbar}}{\varepsilon_0 - \varepsilon_n + 2i\eta\hbar} \sum_m \frac{\langle n|V|m\rangle \langle m|V|0\rangle}{\varepsilon_0 - \varepsilon_m + i\eta\hbar}. \end{aligned} \quad (12.47)$$

12.4.2 Second-Order Golden Rule and Virtual Intermediate States

When the direct coupling vanishes ($\langle n|V|0\rangle = 0$), the second-order transition probability into a continuum of states is obtained by taking the time derivative of $|\langle n|\psi_I^{(2)}(t)\rangle|^2$ and taking the adiabatic limit $\eta \rightarrow 0^+$:

$$\Gamma_{0 \rightarrow n}^{(2)} = \frac{2\pi}{\hbar} \left| \sum_m \frac{\langle n|V|m\rangle \langle m|V|0\rangle}{\varepsilon_0 - \varepsilon_m + i\eta\hbar} \right|^2 \delta(\varepsilon_n - \varepsilon_0). \quad (12.48)$$

Equation (12.48) admits an intuitive physical interpretation:

1. **Virtual Transitions:** The system makes the transition from the initial state $|0\rangle$ to the final state $|n\rangle$ via a two-step sequence through an intermediate state $|m\rangle$:

$$|0\rangle \xrightarrow{V} |m\rangle \xrightarrow{V} |n\rangle. \quad (12.49)$$

Energy need *not* be conserved in the virtual intermediate state: ε_m may differ arbitrarily from ε_0 . However, global energy conservation between the initial and final states is strictly enforced by $\delta(\varepsilon_n - \varepsilon_0)$.

2. **Quantum Interference:** One must sum the transition amplitudes over all possible intermediate states $|m\rangle$ *coherently* before taking the absolute square:

$$V_{n0}^{(2)} \equiv \sum_m \frac{\langle n|V|m\rangle \langle m|V|0\rangle}{\varepsilon_0 - \varepsilon_m + i\eta\hbar}. \quad (12.50)$$

Different virtual intermediate paths can interfere constructively or destructively.

3. **Causal Resolvent and Boundary Conditions:** The infinitesimal $+i\eta\hbar$ in the denominator is essential when the intermediate states $|m\rangle$ belong to a continuous energy spectrum overlapping ε_0 . Via the Sokhotski–Plemelj theorem,

$$\frac{1}{\varepsilon_0 - \varepsilon_m + i\eta} = \mathcal{P} \left(\frac{1}{\varepsilon_0 - \varepsilon_m} \right) - i\pi\delta(\varepsilon_0 - \varepsilon_m), \quad (12.51)$$

it selects outgoing-wave boundary conditions and defines the analytic continuation of the Green's resolvent operator.

12.4.3 Generalization: The Transition Matrix (T -Matrix)

When both first- and second-order matrix elements contribute, quantum mechanics requires adding the amplitudes coherently. The total transition rate into a continuum mode is given by

$$\Gamma_{0 \rightarrow n} = \frac{2\pi}{\hbar} |T_{n0}|^2 \delta(\varepsilon_n - \varepsilon_0), \quad (12.52)$$

where $T_{n0} = \langle n|T|0\rangle$ is the on-shell matrix element of the transition operator T , defined to second order by

$$T_{n0} = \langle n|V|0\rangle + \sum_m \frac{\langle n|V|m\rangle \langle m|V|0\rangle}{\varepsilon_0 - \varepsilon_m + i\eta\hbar} + \mathcal{O}(V^3). \quad (12.53)$$

To all orders, this operator satisfies the Lippmann–Schwinger operator equation $T = V + V \frac{1}{\varepsilon_0 - H_0 + i\eta} T$.

12.5 Forward Scattering Amplitude and the Optical Theorem

Having explored transitions from an initial state $|0\rangle$ into orthogonal states $|n\rangle$, we now turn our attention to the *survival amplitude* (or persistence amplitude) $\langle 0|\psi_S(t)\rangle$ —the probability amplitude that the system remains in its initial unperturbed state $|0\rangle$ at time t .

12.5.1 Equations of Motion for the Survival Amplitude

In the interaction picture, the survival amplitude is related to the Schrödinger amplitude by

$$\langle 0|\psi_S(t)\rangle = e^{-i\varepsilon_0 t/\hbar} \langle 0|\psi_I(t)\rangle. \quad (12.54)$$

Projecting the Tomonaga–Schwinger equation $i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle$ onto the initial state bra $\langle 0|$ and inserting a complete set of unperturbed states, we have

$$i\hbar \frac{\partial}{\partial t} \langle 0|\psi_I(t)\rangle = \langle 0|V_I(t)|\psi_I(t)\rangle = \langle 0|V_I(t)|0\rangle \langle 0|\psi_I(t)\rangle + \sum_{n \neq 0} \langle 0|V_I(t)|n\rangle \langle n|\psi_I(t)\rangle. \quad (12.55)$$

Dividing both sides by $\langle 0|\psi_I(t)\rangle$, we obtain an exact equation of motion for the logarithmic derivative:

$$i\hbar \frac{\partial}{\partial t} \ln \langle 0|\psi_I(t)\rangle = \langle 0|V_I(t)|0\rangle + \sum_{n \neq 0} \langle 0|V_I(t)|n\rangle \frac{\langle n|\psi_I(t)\rangle}{\langle 0|\psi_I(t)\rangle}. \quad (12.56)$$

To determine the evolution to second order in V , we approximate the ratio on the right-hand side using the first-order transition amplitude $\langle n|\psi_I(t)\rangle \approx \frac{1}{i\hbar} \int_{t_0}^t dt' \langle n|V_I(t')|0\rangle$ and setting the denominator to $\langle 0|\psi_I(t)\rangle \approx 1$:

$$i\hbar \frac{\partial}{\partial t} \ln \langle 0|\psi_I(t)\rangle = \langle 0|V_I(t)|0\rangle + \frac{1}{i\hbar} \sum_{n \neq 0} \int_{t_0}^t dt' \langle 0|V_I(t)|n\rangle \langle n|V_I(t')|0\rangle + \mathcal{O}(V^3). \quad (12.57)$$

12.5.2 Adiabatic Evolution and Discrete Spectra

Consider a static perturbation V switched on adiabatically with the factor $e^{\eta t}$ ($\eta \rightarrow 0^+$, $t_0 \rightarrow -\infty$). Substituting the explicit time dependence of the interaction-picture operators, Eq. (12.57) becomes

$$i\hbar \frac{\partial}{\partial t} \ln \langle 0|\psi_I(t)\rangle = \langle 0|V|0\rangle + \sum_{n \neq 0} \frac{|\langle 0|V|n\rangle|^2}{\varepsilon_0 - \varepsilon_n + i\eta\hbar}. \quad (12.58)$$

If the energy spectrum of H_0 is discrete and the level ε_0 is non-degenerate, the denominator $\varepsilon_0 - \varepsilon_n$ never vanishes. In the adiabatic limit $\eta \rightarrow 0^+$, the $+i\eta\hbar$ term drops out, leaving a strictly real quantity. Integrating over time, the survival amplitude evolves as

$$\langle 0|\psi_S(t)\rangle \sim \exp\left(-\frac{iE_0 t}{\hbar}\right), \quad (12.59)$$

where

$$E_0 = \varepsilon_0 + \langle 0|V|0\rangle + \sum_{n \neq 0} \frac{|\langle 0|V|n\rangle|^2}{\varepsilon_0 - \varepsilon_n}. \quad (12.60)$$

This is identically the second-order Rayleigh–Schrödinger energy shift. This demonstrates the Gell-Mann and Low adiabatic theorem: turning on a perturbation infinitesimally slowly transforms an unperturbed discrete eigenstate smoothly into the corresponding stationary eigenstate of the interacting Hamiltonian $H_0 + V$. Stationary-state perturbation theory emerges naturally as an adiabatic limit of time-dependent perturbation theory.

12.5.3 Continuum Spectra: Complex Energy and the Optical Theorem

When the unperturbed state $|0\rangle$ is coupled to a continuum of states spanning energies near ε_0 , the denominator $\varepsilon_0 - \varepsilon_n + i\eta\hbar$ acquires a singularity on the real axis. Applying the Sokhotski–Plemelj distribution identity,

$$\lim_{\eta \rightarrow 0^+} \frac{1}{\varepsilon_0 - \varepsilon_n + i\eta\hbar} = \mathcal{P} \left(\frac{1}{\varepsilon_0 - \varepsilon_n} \right) - i\pi\delta(\varepsilon_0 - \varepsilon_n), \quad (12.61)$$

where $\mathcal{P}$ denotes the Cauchy principal value. Substituting this into Eq. (12.58), we find

$$\sum_{n \neq 0} \frac{|\langle 0|V|n\rangle|^2}{\varepsilon_0 - \varepsilon_n + i\eta\hbar} = \mathcal{P} \sum_{n \neq 0} \frac{|\langle 0|V|n\rangle|^2}{\varepsilon_0 - \varepsilon_n} - i\pi \sum_{n \neq 0} |\langle 0|V|n\rangle|^2 \delta(\varepsilon_0 - \varepsilon_n). \quad (12.62)$$

Recognizing the imaginary term as being proportional to the total decay rate Γ from Fermi's Golden Rule,

$$\Gamma = \frac{2\pi}{\hbar} \sum_{n \neq 0} |\langle 0|V|n\rangle|^2 \delta(\varepsilon_0 - \varepsilon_n), \quad (12.63)$$

Eq. (12.62) can be written as

$$i\hbar \frac{\partial}{\partial t} \ln \langle 0|\psi_I(t)\rangle = \Delta E_0 - \frac{i\hbar}{2}\Gamma. \quad (12.64)$$

Integrating this differential equation, the complete time dependence of the initial-state amplitude is

$$\langle 0|\psi_S(t)\rangle \sim \exp \left[-\frac{i}{\hbar} \left(E_0 - \frac{i\hbar\Gamma}{2} \right) t \right] = e^{-iE_0 t/\hbar} e^{-\Gamma t/2}, \quad (12.65)$$

where the renormalized energy level is given by the principal-value integral:

$$E_0 = \varepsilon_0 + \langle 0|V|0\rangle + \mathcal{P} \int d\varepsilon_n \rho(\varepsilon_n) \frac{|\langle 0|V|n\rangle|^2}{\varepsilon_0 - \varepsilon_n}. \quad (12.66)$$

The physical consequences are profound:

1. **Decay Law:** The survival probability of the initial state decays exponentially:

$$P_0(t) = |\langle 0|\psi_S(t)\rangle|^2 = e^{-\Gamma t}. \quad (12.67)$$

The state $|0\rangle$ ceases to be a true stationary state; it behaves as a metastable resonance characterized by a *complex energy*:

$$\mathcal{E}_0 = E_0 - \frac{i\hbar\Gamma}{2}. \quad (12.68)$$

2. **Energy Shift (Lamb Shift Analogue):** The real part E_0 shifts the unperturbed resonance position due to virtual polarization of the continuum (analogous to the self-energy Lamb shift in quantum electrodynamics).
3. **The Optical Theorem:** The forward scattering self-energy matrix element $\langle 0|T|0\rangle$ has an imaginary part that is directly proportional to the total transition rate into all accessible continuum states:

$$\text{Im} \langle 0|T|0\rangle = -\frac{\hbar}{2}\Gamma = -\frac{\hbar}{2} \sum_{n \neq 0} \Gamma_{0 \rightarrow n}. \quad (12.69)$$

This is the exact dynamical counterpart of the optical theorem in scattering theory: the depletion of the incident (forward) channel is governed by the total cross section summed over all open reaction channels.

12.6 Problems

1. **Photoionization of an electron in a spherical square well.** An electron moves in a three-dimensional spherical square-well potential of radius a and depth V_0 , prepared initially in its spherically symmetric ground state with binding energy $E_B > 0$. At $t = 0$, a uniform oscillating electric field $\mathcal{E}(t) = \mathcal{E} \cos(\omega t) \hat{\mathbf{z}}$ is turned on.
 - (a) Using Fermi's Golden Rule for harmonic perturbations, calculate the ionization transition rate $\Gamma(\omega)$ as a function of the driving frequency ω .
 - (b) Approximate the ejected continuum state by a plane wave, and determine the threshold frequency ω_{th} required for photoemission.
 - (c) Find the angular distribution $d\Gamma/d\Omega$ of the ejected photoelectrons, and verify that it exhibits the characteristic p -wave dipole pattern $\cos^2 \theta$.
2. **Photoionization of atomic hydrogen.** A neutral hydrogen atom initially in its $1s$ ground state is subjected to an oscillating electric field $\mathcal{E}(t) = \mathcal{E} \cos(\omega t) \hat{\mathbf{z}}$.
 - (a) Taking into account proton recoil by using the electron-proton reduced mass $\mu = m_e m_p / (m_e + m_p)$, calculate the total photoionization cross section $\sigma(\omega)$ as a function of frequency for $\hbar\omega > \mathcal{R}_y$.
 - (b) Represent the ionized continuum electron by an outgoing plane wave, and evaluate the spatial dipole matrix element $\langle \mathbf{k}|z|1s \rangle$ explicitly by Fourier transforming the ground-state wave function $\psi_{1s}(\mathbf{r}) = (\pi a_0^3)^{-1/2} e^{-r/a_0}$.
3. **Optical theorem from the forward self-energy.** Using the complex energy shift identity Eq. (12.62) alongside the Born approximation cross section Eq. (12.30), rederive the optical theorem of scattering theory:

$$\text{Im } f(0) = \frac{k}{4\pi} \sigma_{\text{tot}}, \quad (12.70)$$

demonstrating that the imaginary part of the forward scattering amplitude directly measures the depletion rate of the incident beam into all scattered channels.

4. Linear response theory and the Kubo formula.

- (a) Consider a quantum system initially prepared in an unperturbed stationary state $|0\rangle$ at $t = 0$, subjected to a time-dependent perturbation $H'(t)$ for $t > 0$. Let A be a physical observable. Prove that to first order in H' , the non-equilibrium expectation value $\langle A(t) \rangle$ is given by the Kubo response formula:

$$\langle A(t) \rangle = \langle 0|A_I(t)|0 \rangle + \frac{1}{i\hbar} \int_0^t dt' \langle 0|[A_I(t), H'_I(t')]|0 \rangle, \quad (12.71)$$

where $A_I(t)$ and $H'_I(t')$ are interaction-picture operators.

- (b) Apply this linear-response formalism to compute the induced electric dipole moment $d(t) = q\langle x(t) \rangle$ of a one-dimensional harmonic oscillator of frequency ω_0 driven by an oscillating electric field $H'(t) = -q\mathcal{E}x \cos(\omega t)$ switched on abruptly at $t = 0$.
- (c) Recompute the induced dipole moment assuming the field is turned on adiabatically with factor $e^{\eta t}$ ($\eta \rightarrow 0^+$), and compare the resonant dynamic polarizability $\alpha(\omega)$ obtained in the two cases.

Chapter 13

Interaction Of Radiation With Matter

13.1 Interaction of Radiation with Matter: Classical Field Foundations

We begin our treatment of radiation processes in quantum mechanics by reviewing the classical electrodynamics of the free and source-driven radiation field. Throughout this discussion, we employ Gaussian cgs units.

13.1.1 The Transverse Gauge and Maxwell's Equations

In the absence of static charge densities, it is most convenient to describe the radiation field in the *transverse gauge* (also termed the Coulomb or radiation gauge), in which the scalar potential vanishes identically,

$$\varphi(\mathbf{r}, t) = 0, \quad (13.1)$$

and the vector potential is strictly solenoidal (divergence-free):

$$\nabla \cdot \mathbf{A}(\mathbf{r}, t) = 0. \quad (13.2)$$

Under these gauge conditions, the physical electric and magnetic fields are determined entirely by $\mathbf{A}(\mathbf{r}, t)$:

$$\mathcal{E}(\mathbf{r}, t) = -\frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}, t)}{\partial t}, \quad (13.3)$$

$$\mathcal{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (13.4)$$

The total electromagnetic field energy contained in space is given by

$$E_{\text{EM}} = \int d^3r \frac{\mathcal{E}^2(\mathbf{r}, t) + \mathcal{B}^2(\mathbf{r}, t)}{8\pi}, \quad (13.5)$$

and the directional energy flux density is governed by the Poynting vector:

$$\mathbf{S}(\mathbf{r}, t) = \frac{c}{4\pi} \mathcal{E}(\mathbf{r}, t) \times \mathcal{B}(\mathbf{r}, t). \quad (13.6)$$

Coupled to a classical matter current density $\mathbf{j}(\mathbf{r}, t)$, Maxwell's equations reduce to the inhomogeneous wave equation:

$$\square \mathbf{A}(\mathbf{r}, t) \equiv \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{A}(\mathbf{r}, t) = -\frac{4\pi}{c} \mathbf{j}_{\perp}(\mathbf{r}, t), \quad (13.7)$$

where $\mathbf{j}_{\perp}(\mathbf{r}, t)$ denotes the transverse (divergence-free) component of the current density:

$$\mathbf{j}_{\perp}(\mathbf{r}, t) = \mathbf{j}(\mathbf{r}, t) - \nabla \int d^3r' \frac{\nabla' \cdot \mathbf{j}(\mathbf{r}', t)}{4\pi|\mathbf{r} - \mathbf{r}'|}. \quad (13.8)$$

13.1.2 Free Plane Waves and Energy Flux

In source-free regions ($\mathbf{j}_\perp = 0$), the vector potential satisfies the homogeneous wave equation $\square \mathbf{A} = 0$. A monochromatic, linearly polarized plane wave solution is written as

$$\mathbf{A}(\mathbf{r}, t) = \alpha \boldsymbol{\varepsilon}_\lambda e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + \alpha^* \boldsymbol{\varepsilon}_\lambda^* e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)}, \quad (13.9)$$

where the complex conjugate ensures that $\mathbf{A}(\mathbf{r}, t)$ is real. Substituting Eq. (13.9) into the wave equation yields the linear dispersion relation:

$$\omega = ck = c|\mathbf{k}|. \quad (13.10)$$

The transverse gauge constraint $\nabla \cdot \mathbf{A} = 0$ requires the unit polarization vector $\boldsymbol{\varepsilon}_\lambda$ to be strictly perpendicular to the propagation vector $\mathbf{k}$:

$$\boldsymbol{\varepsilon}_\lambda \cdot \mathbf{k} = 0, \quad |\boldsymbol{\varepsilon}_\lambda|^2 = 1. \quad (13.11)$$

For each wave vector $\mathbf{k}$, there exist two mutually orthogonal transverse polarization states $\lambda \in \{1, 2\}$, matching the photon spin-1 helicity states.

Evaluating the electromagnetic energy density $u = (\mathcal{E}^2 + \mathcal{B}^2)/(8\pi)$ for the plane wave (13.9), we obtain

$$u(\mathbf{r}, t) = \frac{\omega^2}{2\pi c^2} \left[|\alpha|^2 - \text{Re} \left(\alpha^2 \boldsymbol{\varepsilon}_\lambda^2 e^{2i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right) \right]. \quad (13.12)$$

Averaging over one optical cycle $T = 2\pi/\omega$, the second harmonic term vanishes, leaving the time-averaged energy density:

$$\bar{u} = \frac{\omega^2}{2\pi c^2} |\alpha|^2. \quad (13.13)$$

Because the wave energy propagates at the speed of light along $\hat{\mathbf{k}}$, the cycle-averaged Poynting vector is

$$\langle \mathbf{S} \rangle = c\bar{u} \hat{\mathbf{k}} = \frac{\omega^2}{2\pi c} |\alpha|^2 \hat{\mathbf{k}}. \quad (13.14)$$

13.1.3 Normal Mode Expansion in a Quantization Box

To facilitate quantization, we enclose the radiation field in a large normalization volume $V = L^3$ and impose periodic boundary conditions on the spatial boundaries. An arbitrary divergence-free vector potential can be expanded in a discrete Fourier series over the orthogonal spatial mode functions:

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k}, \lambda} \frac{1}{\sqrt{V}} \left[A_{\mathbf{k}\lambda} \boldsymbol{\varepsilon}_{\mathbf{k}\lambda} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + A_{\mathbf{k}\lambda}^* \boldsymbol{\varepsilon}_{\mathbf{k}\lambda}^* e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right], \quad (13.15)$$

where $\mathbf{k} = \frac{2\pi}{L}(n_x, n_y, n_z)$ ($n_i \in \mathbb{Z}$), $\omega = ck$, and the sum over λ runs over two orthonormal polarization vectors satisfying $\boldsymbol{\varepsilon}_{\mathbf{k}\lambda} \cdot \mathbf{k} = 0$ and $\boldsymbol{\varepsilon}_{\mathbf{k}\lambda} \cdot \boldsymbol{\varepsilon}_{\mathbf{k}\lambda'}^* = \delta_{\lambda\lambda'}$.

Integrating the total electromagnetic field energy (13.5) over the volume V and averaging over rapid temporal oscillations, the spatial cross-terms integrate to zero due to orthogonality, yielding

$$E_{\text{EM}} = \sum_{\mathbf{k}, \lambda} \frac{\omega^2}{2\pi c^2} |A_{\mathbf{k}\lambda}|^2. \quad (13.16)$$

The total energy of the classical radiation field decomposes into an infinite collection of uncoupled, independent harmonic oscillators labeled by mode wave vector $\mathbf{k}$ and polarization λ .

13.2 The Interaction Hamiltonian

To understand how electromagnetic fields drive transitions and quantum dynamics in atomic and condensed-matter systems, we construct the quantum mechanical interaction Hamiltonian coupling charged particles to radiation.

13.2.1 Minimal Coupling and Gauge Invariance

In classical mechanics, the interaction of a particle of mass m and charge e with an electromagnetic field described by vector and scalar potentials $(\mathbf{A}, \varphi)$ is governed by the principle of minimal substitution:

$$H = \frac{1}{2m} \left[\mathbf{p} - \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + e\varphi(\mathbf{r}, t) + V(\mathbf{r}, t), \quad (13.17)$$

where $V(\mathbf{r}, t)$ accounts for non-electromagnetic or background static potentials. In quantum mechanics, $\mathbf{r}$ and $\mathbf{p} = -i\hbar\nabla$ are promoted to non-commuting operators satisfying canonical commutation relations $[r_j, p_k] = i\hbar\delta_{jk}$. The corresponding time-dependent Schrödinger equation reads

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \left\{ \frac{1}{2m} \left[\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + e\varphi(\mathbf{r}, t) + V(\mathbf{r}, t) \right\} \psi(\mathbf{r}, t). \quad (13.18)$$

Under a local gauge transformation parameterized by an arbitrary scalar field $\chi(\mathbf{r}, t)$,

$$\begin{aligned} \mathbf{A}'(\mathbf{r}, t) &= \mathbf{A}(\mathbf{r}, t) + \nabla \chi(\mathbf{r}, t), \\ \varphi'(\mathbf{r}, t) &= \varphi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \chi(\mathbf{r}, t)}{\partial t}, \end{aligned} \quad (13.19)$$

the physical fields $\mathcal{E}$ and $\mathcal{B}$ remain invariant. The Schrödinger equation retains its form in the new gauge,

$$i\hbar \frac{\partial \psi'(\mathbf{r}, t)}{\partial t} = \left\{ \frac{1}{2m} \left[\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A}'(\mathbf{r}, t) \right]^2 + e\varphi'(\mathbf{r}, t) + V(\mathbf{r}, t) \right\} \psi'(\mathbf{r}, t), \quad (13.20)$$

provided the state wave function acquires a local $U(1)$ phase factor:

$$\psi'(\mathbf{r}, t) = \exp \left[\frac{ie}{\hbar c} \chi(\mathbf{r}, t) \right] \psi(\mathbf{r}, t). \quad (13.21)$$

13.2.2 Mechanical Velocity vs. Canonical Momentum

While canonical momentum $\mathbf{p} = -i\hbar\nabla$ generates spatial translations, it is not gauge invariant. Matrix elements of $\mathbf{p}$ transform under a gauge change as

$$\langle \psi'_1 | \mathbf{p} | \psi'_2 \rangle = \langle \psi_1 | \mathbf{p} | \psi_2 \rangle + \frac{e}{c} \int d^3r \psi_1^*(\mathbf{r}, t) \nabla \chi(\mathbf{r}, t) \psi_2(\mathbf{r}, t). \quad (13.22)$$

In contrast, the kinematic (or mechanical) momentum operator,

$$\boldsymbol{\pi} \equiv \mathbf{p} - \frac{e}{c} \mathbf{A}(\mathbf{r}, t), \quad (13.23)$$

has gauge-invariant matrix elements because the gauge shift of $\mathbf{A}$ precisely cancels the phase gradient of the wave functions. In the Heisenberg representation, the Heisenberg equation of motion for the position operator yields the physical velocity:

$$\mathbf{v}(t) = \frac{d\mathbf{r}(t)}{dt} = \frac{1}{i\hbar} [\mathbf{r}(t), H] = \frac{1}{m} \left[\mathbf{p}(t) - \frac{e}{c} \mathbf{A}(\mathbf{r}(t), t) \right] = \frac{\boldsymbol{\pi}(t)}{m}. \quad (13.24)$$

All physically measurable expectation values and transition probabilities are strictly gauge invariant.

13.2.3 Separation of the Interaction Hamiltonian

Expanding the squared kinetic term in Eq. (13.17), we partition the total Hamiltonian into an unperturbed part H_0 and an interaction term H_{int} :

$$H = H_0 + H_{\text{int}}, \quad (13.25)$$

where $H_0 = \frac{\mathbf{p}^2}{2m} + V(\mathbf{r})$ and

$$H_{\text{int}} = -\frac{e}{2mc} [\mathbf{p} \cdot \mathbf{A}(\mathbf{r}, t) + \mathbf{A}(\mathbf{r}, t) \cdot \mathbf{p}] + \frac{e^2}{2mc^2} \mathbf{A}^2(\mathbf{r}, t) + e\varphi(\mathbf{r}, t). \quad (13.26)$$

Because the canonical commutator is $[\mathbf{p}, \mathbf{A}] = -i\hbar \nabla \cdot \mathbf{A}$, the ordering of $\mathbf{p}$ and $\mathbf{A}$ matters in general:

$$\mathbf{p} \cdot \mathbf{A} - \mathbf{A} \cdot \mathbf{p} = -i\hbar \nabla \cdot \mathbf{A}. \quad (13.27)$$

In the transverse gauge ($\nabla \cdot \mathbf{A} = 0$), the two operators commute, $\mathbf{p} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{p}$, and with $\varphi = 0$, the interaction reduces to

$$H_{\text{int}} = -\frac{e}{mc} \mathbf{A} \cdot \mathbf{p} + \frac{e^2}{2mc^2} \mathbf{A}^2. \quad (13.28)$$

13.2.4 Many-Particle Systems: Paramagnetic and Diamagnetic Currents

For a collection of N particles with charges e_i and masses m_i , the interaction Hamiltonian generalizes to

$$H_{\text{int}} = \sum_{i=1}^N \left\{ -\frac{e_i}{2m_i c} [\mathbf{p}_i \cdot \mathbf{A}(\mathbf{r}_i, t) + \mathbf{A}(\mathbf{r}_i, t) \cdot \mathbf{p}_i] + \frac{e_i^2}{2m_i c^2} \mathbf{A}^2(\mathbf{r}_i, t) + e_i \varphi(\mathbf{r}_i, t) \right\}. \quad (13.29)$$

For identical particles of charge e and mass m , we introduce the microscopic particle density operator:

$$\rho(\mathbf{r}) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i), \quad \int d^3r \rho(\mathbf{r}) = N. \quad (13.30)$$

Similarly, we define the Hermitian *paramagnetic current density* operator:

$$\mathbf{j}(\mathbf{r}) = \frac{1}{2} \sum_{i=1}^N \left[\frac{\mathbf{p}_i}{m} \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \frac{\mathbf{p}_i}{m} \right]. \quad (13.31)$$

The true physical, gauge-invariant current density $\mathbf{J}(\mathbf{r})$ is obtained from the mechanical velocity:

$$\mathbf{J}(\mathbf{r}) = \mathbf{j}(\mathbf{r}) - \frac{e}{mc} \mathbf{A}(\mathbf{r}, t) \rho(\mathbf{r}) \equiv \mathbf{j}_{\text{para}}(\mathbf{r}) + \mathbf{j}_{\text{dia}}(\mathbf{r}), \quad (13.32)$$

where $\mathbf{j}(\mathbf{r})$ is the paramagnetic current and $-\frac{e}{mc} \mathbf{A} \rho$ is the diamagnetic current. In terms of these continuous field operators, the interaction Hamiltonian takes the compact and elegant field-theoretic form:

$$H_{\text{int}} = \int d^3r \left[-\frac{e}{c} \mathbf{j}(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}, t) + \frac{e^2}{2mc^2} \rho(\mathbf{r}) \mathbf{A}^2(\mathbf{r}, t) + e \rho(\mathbf{r}) \varphi(\mathbf{r}, t) \right]. \quad (13.33)$$

The linear $\mathbf{j} \cdot \mathbf{A}$ coupling governs single-photon absorption and emission processes in first order, while the quadratic $\mathbf{A}^2$ term mediates two-photon processes and Thomson/Raman scattering.

13.3 Absorption of Light and Induced Emission

Having established the electromagnetic interaction Hamiltonian, we calculate the rates at which an atomic or molecular system absorbs and emits radiation. In typical optical and spectroscopic experiments, the incident radiation field is weak compared to characteristic atomic field strengths ($\mathcal{E}_{\text{at}} = e/a_0^2 \sim 10^9$ V/cm). Consequently, the diamagnetic $\mathbf{A}^2$ term is negligible for single-photon transitions compared to the paramagnetic $\mathbf{j} \cdot \mathbf{A}$ coupling.

Working in the transverse gauge ($\varphi = 0$, $\nabla \cdot \mathbf{A} = 0$), the interaction Hamiltonian governing single-photon processes is

$$H_{\text{int}}(t) = -\frac{e}{c} \int d^3r \mathbf{j}(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}, t). \quad (13.34)$$

13.3.1 Mode Decomposition of the Interaction

Substituting the box-normalized plane-wave expansion of the vector potential into Eq. (13.34), we obtain

$$H_{\text{int}}(t) = -\frac{e}{c} \sum_{\mathbf{k}, \lambda} \frac{1}{\sqrt{V}} [A_{\mathbf{k}\lambda}(\mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda) e^{-i\omega t} + A_{\mathbf{k}\lambda}^*(\mathbf{j}_{\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda^*) e^{i\omega t}], \quad (13.35)$$

where $\omega = ck$, and $\mathbf{j}_{\mathbf{k}}$ is the spatial Fourier transform of the paramagnetic current operator:

$$\mathbf{j}_{\mathbf{k}} \equiv \int d^3r e^{-i\mathbf{k} \cdot \mathbf{r}} \mathbf{j}(\mathbf{r}) = \frac{1}{2m} \sum_{i=1}^N (\mathbf{p}_i e^{-i\mathbf{k} \cdot \mathbf{r}_i} + e^{-i\mathbf{k} \cdot \mathbf{r}_i} \mathbf{p}_i). \quad (13.36)$$

Hermiticity of the current density operator implies $\mathbf{j}_{\mathbf{k}}^\dagger = \mathbf{j}_{-\mathbf{k}}$.

13.3.2 Transition Rates for Broadband Incoherent Radiation

Consider an atom initially in a stationary state $|0\rangle$ illuminated by an incoherent radiation beam. In thermal or spontaneous emission sources (such as a discharge lamp), individual radiative wave packets possess random, uncorrelated phases. Under such phase incoherence, interference cross-terms between distinct Fourier modes average to zero, and the total transition rate is obtained by incoherently summing the transition rates induced by each Fourier component.

By Fermi's Golden Rule for harmonic perturbations, the positive-frequency component $\propto e^{-i\omega t}$ drives resonant upward transitions $|0\rangle \rightarrow |n\rangle$ with an energy gain $\varepsilon_n - \varepsilon_0 = \hbar\omega$. The absorption rate induced by a single mode $(\mathbf{k}, \lambda)$ is

$$\Gamma_{0 \rightarrow n; \mathbf{k}\lambda}^{\text{abs}} = \frac{2\pi}{\hbar} \frac{e^2}{Vc^2} |A_{\mathbf{k}\lambda}|^2 |\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle|^2 \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega). \quad (13.37)$$

Summing over all wave vectors $\mathbf{k}$ and transverse polarizations λ , the total absorption transition rate is

$$\Gamma_{0 \rightarrow n}^{\text{abs}} = \frac{2\pi e^2}{\hbar c^2} \frac{1}{V} \sum_{\mathbf{k}, \lambda} |A_{\mathbf{k}\lambda}|^2 |\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle|^2 \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega). \quad (13.38)$$

Passing to the continuum limit in a large box, $\frac{1}{V} \sum_{\mathbf{k}} \rightarrow \int \frac{d^3k}{(2\pi)^3} = \int \frac{\omega^2 d\omega d\Omega}{(2\pi c)^3}$, we have

$$\Gamma_{0 \rightarrow n}^{\text{abs}} = \frac{2\pi e^2}{\hbar^2 c^2} \frac{\omega^2}{(2\pi c)^3} \int d\Omega \sum_{\lambda} |\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle|^2 |A_{\mathbf{k}\lambda}|^2, \quad (13.39)$$

evaluated at the Bohr resonance frequency $\omega = (\varepsilon_n - \varepsilon_0)/\hbar$.

13.3.3 Connection to Spectral Intensity and Induced Emission

For a collimated, polarized beam spanning solid angle $d\Omega$, the total cycle-averaged energy flux density (Poynting intensity) is

$$I_{\text{tot}} = \frac{1}{V} \sum_{\mathbf{k}} \frac{\omega^2}{2\pi c} |A_{\mathbf{k}\lambda}|^2 = \int d\omega I(\omega), \quad (13.40)$$

where the spectral intensity per unit angular frequency $I(\omega)$ is defined by

$$I(\omega) = \frac{\omega^4 d\Omega}{(2\pi c)^4} |A_{\mathbf{k}\lambda}|^2. \quad (13.41)$$

Expressing the absorption rate in terms of $I(\omega)$, we arrive at

$$\Gamma_{0 \rightarrow n}^{\text{abs}} = \frac{4\pi^2 e^2}{\hbar^2 c \omega^2} I(\omega) |\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle|^2. \quad (13.42)$$

Conversely, if the system is initially in the upper state $|n\rangle$, the negative-frequency component $\propto e^{i\omega t}$ in Eq. (13.35) stimulates resonant downward transitions $|n\rangle \rightarrow |0\rangle$ with energy loss $\varepsilon_n - \varepsilon_0 = \hbar\omega$. The rate of *induced* (or stimulated) emission is

$$\Gamma_{n \rightarrow 0}^{\text{ind. em.}} = \frac{4\pi^2 e^2}{\hbar^2 c \omega^2} I(\omega) |\langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda^* | n \rangle|^2. \quad (13.43)$$

Because $\mathbf{j}_{\mathbf{k}}^\dagger = \mathbf{j}_{-\mathbf{k}}$, the matrix elements satisfy the microscopic reversibility relation:

$$\langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda^* | n \rangle = (\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle)^*, \quad (13.44)$$

which immediately establishes the fundamental microscopic symmetry:

$$\Gamma_{0 \rightarrow n}^{\text{abs}} = \Gamma_{n \rightarrow 0}^{\text{ind. em.}}. \quad (13.45)$$

13.3.4 Photon Interpretation and Absorption Cross Section

Expressing the total field energy in terms of the mode occupation number $N_{\mathbf{k}\lambda}$,

$$E_{\text{EM}} = \sum_{\mathbf{k}, \lambda} \hbar\omega N_{\mathbf{k}\lambda} = \sum_{\mathbf{k}, \lambda} \frac{\omega^2}{2\pi c^2} |A_{\mathbf{k}\lambda}|^2, \quad (13.46)$$

we identify the classical field amplitude with the photon mode density:

$$|A_{\mathbf{k}\lambda}|^2 = \frac{2\pi\hbar c^2}{\omega} N_{\mathbf{k}\lambda}. \quad (13.47)$$

In terms of photon numbers, the absorption and stimulated emission rates become

$$\Gamma_{0 \rightarrow n}^{\text{abs}} = \sum_{\mathbf{k}, \lambda} \frac{4\pi^2 e^2}{\omega V} |\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle|^2 N_{\mathbf{k}\lambda} \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega). \quad (13.48)$$

For an incident beam containing $N_{\mathbf{k}\lambda}$ photons in mode $(\mathbf{k}, \lambda)$, the photon number density is $n_{\text{ph}} = N_{\mathbf{k}\lambda}/V$, and the incident photon flux is $\Phi_{\text{inc}} = cn_{\text{ph}} = N_{\mathbf{k}\lambda}c/V$. The *total absorption cross section* $\sigma_{\text{abs}}(\omega)$ is defined as the ratio of the total absorption transition rate to the incident photon flux:

$$\sigma_{\text{abs}}(\omega) \equiv \frac{\sum_n \Gamma_{0 \rightarrow n}^{\text{abs}}}{\Phi_{\text{inc}}} = \frac{4\pi^2 e^2}{\omega c} \sum_n |\langle n | \mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda | 0 \rangle|^2 \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega). \quad (13.49)$$

The cross section possesses dimensions of area (cm^2) and characterizes the effective geometric target presented by the atom to incident photons of frequency ω , polarization $\boldsymbol{\varepsilon}_\lambda$, and propagation direction $\hat{\mathbf{k}}$.

13.4 The Quantized Radiation Field

The semiclassical treatment of radiation, where matter is treated quantum mechanically while the electromagnetic field is treated as a classical wave, accurately predicts absorption and stimulated emission. However, it cannot fundamentally account for *spontaneous emission*—the process whereby an isolated excited atom drops to a lower energy state in total darkness (the vacuum). To resolve this, we must promote the electromagnetic field itself to a quantum mechanical system described by state vectors and operators.

13.4.1 Photon Fock States and Field Quantization

In a fully quantized description, the state of the electromagnetic field is specified by giving the occupation number of photons in each normal mode $(\mathbf{k}, \epsilon_\lambda)$. We represent these states in the orthonormal photon number (Fock) basis:

$$|\{N_{\mathbf{k}\lambda}\}\rangle = |N_{\mathbf{k}_1\lambda_1}, N_{\mathbf{k}_2\lambda_2}, \dots, N_{\mathbf{k}\lambda}, \dots\rangle, \quad (13.50)$$

where $\langle\{N'_{\mathbf{k}\lambda}\}|\{N_{\mathbf{k}\lambda}\}\rangle = \prod_{\mathbf{k}\lambda} \delta_{N'_{\mathbf{k}\lambda}, N_{\mathbf{k}\lambda}}$. The total energy of the unperturbed free radiation field is given by the Hamiltonian

$$H_{\text{rad}} = \sum_{\mathbf{k}, \lambda} \hbar\omega \left(a_{\mathbf{k}\lambda}^\dagger a_{\mathbf{k}\lambda} + \frac{1}{2} \right), \quad \omega = ck. \quad (13.51)$$

When an atom absorbs a photon from mode $(\mathbf{k}, \lambda)$, the radiation state transitions from $|N_{\mathbf{k}\lambda}\rangle$ to $|N_{\mathbf{k}\lambda} - 1\rangle$. In order for the transition rate calculated from the quantized interaction Hamiltonian $\hat{H}_{\text{int}}$ to reproduce the semiclassical absorption rate Eq. (13.37),

$$\Gamma_{0 \rightarrow n; \mathbf{k}\lambda}^{\text{abs}} = \frac{2\pi}{\hbar} \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega) \left| \langle n; N_{\mathbf{k}\lambda} - 1 | \hat{H}_{\text{int}} | 0; N_{\mathbf{k}\lambda} \rangle \right|^2, \quad (13.52)$$

the field operator must annihilate a photon from mode $(\mathbf{k}, \lambda)$. Defining the photon annihilation and creation operators $a_{\mathbf{k}\lambda}$ and $a_{\mathbf{k}\lambda}^\dagger$,

$$a_{\mathbf{k}\lambda} |N_{\mathbf{k}\lambda}\rangle = \sqrt{N_{\mathbf{k}\lambda}} |N_{\mathbf{k}\lambda} - 1\rangle, \quad (13.53)$$

$$a_{\mathbf{k}\lambda}^\dagger |N_{\mathbf{k}\lambda}\rangle = \sqrt{N_{\mathbf{k}\lambda} + 1} |N_{\mathbf{k}\lambda} + 1\rangle, \quad (13.54)$$

which satisfy canonical bosonic commutation relations:

$$[a_{\mathbf{k}\lambda}, a_{\mathbf{k}'\lambda'}^\dagger] = \delta_{\mathbf{k}\mathbf{k}'} \delta_{\lambda\lambda'}, \quad [a_{\mathbf{k}\lambda}, a_{\mathbf{k}'\lambda'}] = [a_{\mathbf{k}\lambda}^\dagger, a_{\mathbf{k}'\lambda'}^\dagger] = 0. \quad (13.55)$$

13.4.2 The Quantized Vector Potential Operator

Matching the semiclassical mode amplitude with the operator matrix element identifies the quantized vector potential operator:

$$\hat{\mathbf{A}}(\mathbf{r}) = \sum_{\mathbf{k}, \lambda} \sqrt{\frac{2\pi\hbar c^2}{\omega V}} \left(a_{\mathbf{k}\lambda} \epsilon_\lambda \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} + a_{\mathbf{k}\lambda}^\dagger \epsilon_\lambda^* \frac{e^{-i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} \right). \quad (13.56)$$

In the interaction picture, time evolution under H_{rad} imparts the free phase factor $e^{\pm i\omega t}$ to the mode operators ($a_{\mathbf{k}\lambda}(t) = a_{\mathbf{k}\lambda} e^{-i\omega t}$), giving

$$\hat{\mathbf{A}}(\mathbf{r}, t) = \sum_{\mathbf{k}, \lambda} \sqrt{\frac{2\pi\hbar c^2}{\omega V}} \left(a_{\mathbf{k}\lambda} \epsilon_\lambda e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)} + a_{\mathbf{k}\lambda}^\dagger \epsilon_\lambda^* e^{-i(\mathbf{k}\cdot\mathbf{r} - \omega t)} \right). \quad (13.57)$$

The full matter-radiation interaction Hamiltonian is obtained by replacing the classical potential in Eq. (13.33) with the field operator:

$$\hat{H}_{\text{int}} = \int d^3r \left[-\frac{e}{c} \mathbf{j}(\mathbf{r}) \cdot \hat{\mathbf{A}}(\mathbf{r}) + \frac{e^2}{2mc^2} \rho(\mathbf{r}) \hat{\mathbf{A}}^2(\mathbf{r}) \right]. \quad (13.58)$$

13.4.3 Photon Emission: Spontaneous vs. Induced Emission

Now consider the emission of a photon into mode $(\mathbf{k}, \lambda)$ as an atomic electron de-excites from state $|n\rangle$ to $|0\rangle$. The initial and final uncoupled product states are

$$|i\rangle = |n\rangle \otimes |N_{\mathbf{k}\lambda}\rangle, \quad E_i = \varepsilon_n + \hbar\omega N_{\mathbf{k}\lambda}, \quad (13.59)$$

$$|f\rangle = |0\rangle \otimes |N_{\mathbf{k}\lambda} + 1\rangle, \quad E_f = \varepsilon_0 + \hbar\omega(N_{\mathbf{k}\lambda} + 1). \quad (13.60)$$

The transition matrix element of the linear coupling term $-\frac{e}{c} \int d^3r \mathbf{j}(\mathbf{r}) \cdot \hat{\mathbf{A}}(\mathbf{r})$ is mediated by the creation operator $a_{\mathbf{k}\lambda}^\dagger$:

$$\begin{aligned} \langle f | \hat{H}_{\text{int}} | i \rangle &= -\frac{e}{c} \sqrt{\frac{2\pi\hbar c^2}{\omega V}} \langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \boldsymbol{\varepsilon}_{\lambda}^* | n \rangle \langle N_{\mathbf{k}\lambda} + 1 | a_{\mathbf{k}\lambda}^\dagger | N_{\mathbf{k}\lambda} \rangle \\ &= -\frac{e}{c} \sqrt{\frac{2\pi\hbar c^2}{\omega V}} \langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \boldsymbol{\varepsilon}_{\lambda}^* | n \rangle \sqrt{N_{\mathbf{k}\lambda} + 1}. \end{aligned} \quad (13.61)$$

Applying Fermi's Golden Rule, the transition rate for emission into mode $(\mathbf{k}, \lambda)$ is

$$\begin{aligned} \Gamma_{n \rightarrow 0; \mathbf{k}\lambda}^{\text{em}} &= \frac{2\pi}{\hbar} |\langle f | \hat{H}_{\text{int}} | i \rangle|^2 \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega) \\ &= \frac{4\pi^2 e^2}{\omega V} |\langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \boldsymbol{\varepsilon}_{\lambda}^* | n \rangle|^2 (N_{\mathbf{k}\lambda} + 1) \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega). \end{aligned} \quad (13.62)$$

The factor $(N_{\mathbf{k}\lambda} + 1)$ reveals the profound unification of radiative processes:

1. **Induced (Stimulated) Emission** ($\propto N_{\mathbf{k}\lambda}$): When $N_{\mathbf{k}\lambda} > 0$ photons are already present in the mode, the emission rate is enhanced by the ambient photon population. This stimulated process produces an additional photon identical in wave vector, frequency, phase, and polarization to the incident photons, providing the physical foundation for the LASER (Light Amplification by Stimulated Emission of Radiation).
2. **Spontaneous Emission** ($\propto 1$): Even in the complete absence of external radiation ($N_{\mathbf{k}\lambda} = 0$), the emission rate remains non-zero due to the $+1$ term. An excited atom in vacuum decays spontaneously by emitting a photon into an empty mode.

13.4.4 Vacuum Fluctuations as the Origin of Spontaneous Emission

The existence of spontaneous emission is an inescapable manifestation of *quantum vacuum fluctuations*. In the electromagnetic vacuum state $|0\rangle$, the expectation value of the field operator vanishes, $\langle 0 | \hat{\mathbf{A}}(\mathbf{r}, t) | 0 \rangle = 0$, but its variance and two-point correlation functions are strictly non-zero:

$$\langle 0 | \hat{\mathbf{A}}(\mathbf{r}, t) \cdot \hat{\mathbf{A}}(\mathbf{r}', t) | 0 \rangle \neq 0, \quad \langle 0 | \hat{\boldsymbol{\mathcal{E}}}(\mathbf{r}, t) \cdot \hat{\boldsymbol{\mathcal{E}}}(\mathbf{r}', t) | 0 \rangle \neq 0. \quad (13.63)$$

Just as a harmonic oscillator possesses zero-point ground-state motion with energy $\frac{1}{2}\hbar\omega$, each electromagnetic radiation mode possesses irreducible zero-point quantum electric and magnetic field fluctuations. Spontaneous emission can be interpreted physically as emission *stimulated* by the fluctuating zero-point vacuum fields.

13.5 Einstein's A and B Coefficients and Detailed Balance

Historically, the necessity of spontaneous emission was first recognized by Albert Einstein in 1917 through a thermodynamic argument based on detailed balance and Planck's blackbody radiation law, well before the formal development of quantum electrodynamics.

13.5.1 Thermal Equilibrium in a Radiation Cavity

Consider an enclosure (cavity) held at thermal equilibrium at absolute temperature T , containing an ideal gas of non-interacting atoms interacting with blackbody radiation. According to the Boltzmann distribution of statistical mechanics, the probability of finding a mode of frequency $\omega = ck$ occupied by N photons is proportional to the Gibbs factor $\exp(-N\hbar\omega/k_B T)$. The thermally averaged photon occupation number per mode is given by the Bose–Einstein distribution:

$$\bar{N}_{\mathbf{k}\lambda} = \frac{\sum_{N=0}^{\infty} N e^{-N\hbar\omega/k_B T}}{\sum_{N=0}^{\infty} e^{-N\hbar\omega/k_B T}} = \frac{1}{e^{\hbar\omega/k_B T} - 1}. \quad (13.64)$$

The corresponding average energy per radiation mode is

$$\bar{E}_{\mathbf{k}\lambda} = \hbar\omega \bar{N}_{\mathbf{k}\lambda} = \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1}. \quad (13.65)$$

Now consider a two-level atomic system with ground-state energy ε_0 and excited-state energy ε_n , separated by the resonance transition frequency $\omega = (\varepsilon_n - \varepsilon_0)/\hbar$. In thermal equilibrium, the atomic population ratio obeys the Boltzmann factor:

$$\frac{P_n}{P_0} = \exp\left(-\frac{\varepsilon_n - \varepsilon_0}{k_B T}\right) = e^{-\hbar\omega/k_B T}. \quad (13.66)$$

13.5.2 The Principle of Detailed Balance

Photons of frequency ω in a given cavity mode are exchanged with the atomic ensemble via three distinct processes:

1. **Absorption:** An atom in the lower state $|0\rangle$ absorbs a photon, transitioning to $|n\rangle$. The rate of photon loss is proportional to the photon number N and the ground-state population P_0 :

$$\left(\frac{dN}{dt}\right)_{\text{abs}} = -BNP_0. \quad (13.67)$$

2. **Induced (Stimulated) Emission:** An ambient photon stimulates an atom in $|n\rangle$ to emit an identical photon. By microscopic reversibility (or the symmetry of the interaction matrix elements established in Eq. (13.45)), the transition probability is symmetric, governed by the identical rate coefficient B :

$$\left(\frac{dN}{dt}\right)_{\text{ind}} = BNP_n. \quad (13.68)$$

3. **Spontaneous Emission:** If absorption and induced emission were the only active processes, steady-state equilibrium would require $BNP_0 = BNP_n \implies P_0 = P_n$, which contradicts the Boltzmann distribution since $P_n < P_0$ for any finite temperature. To preserve thermal equilibrium, there must exist an autonomous decay channel—*spontaneous emission*—occurring at a rate independent of the ambient photon number:

$$\left(\frac{dN}{dt}\right)_{\text{spont}} = AP_n. \quad (13.69)$$

In steady state, the net rate of change of the mode photon number vanishes identically:

$$\frac{dN}{dt} = -B\bar{N}P_0 + B\bar{N}P_n + AP_n = 0. \quad (13.70)$$

Solving for the ratio of the spontaneous and stimulated emission coefficients, we obtain

$$A = B\bar{N} \left(\frac{P_0}{P_n} - 1 \right) = B\bar{N} \left(e^{\hbar\omega/k_B T} - 1 \right). \quad (13.71)$$

Substituting Planck's distribution (13.64) for $\bar{N}$, the temperature dependence cancels out completely, yielding the exact relationship:

$$A = B. \quad (13.72)$$

13.5.3 Universal Form of the Radiative Transition Rate

Equation (13.72) demonstrates that the coefficient governing spontaneous decay into a specific mode is identical to the stimulated emission coefficient for that mode. The total rate of emission into mode $(\mathbf{k}, \lambda)$ is therefore given by

$$\left(\frac{dN}{dt} \right)_{\text{em}} = BP_n(N+1). \quad (13.73)$$

This macroscopic thermodynamic deduction perfectly corroborates the microscopic factor of $(N_{\mathbf{k}\lambda} + 1)$ obtained in Eq. (13.62) via second quantization. Detailed balance requires that every stimulated radiative process in quantum physics is necessarily accompanied by a corresponding spontaneous process with unit relative weight.

13.6 Details of Spontaneous Emission: Angular Distribution and Momentum Conservation

Having identified spontaneous emission from the quantized radiation field, we now analyze its directional power distribution, its correspondence with classical radiation from accelerated charges, and the fundamental role of quantum momentum conservation.

13.6.1 Radiated Power and the Classical Correspondence

From Eq. (13.62), the spontaneous transition rate into a specific radiation mode $(\mathbf{k}, \epsilon_\lambda)$ in vacuum ($N_{\mathbf{k}\lambda} = 0$) is

$$\Gamma_{n \rightarrow 0; \mathbf{k}\lambda}^{\text{spont}} = \frac{4\pi^2 e^2}{\omega V} |\langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \epsilon_\lambda^* | n \rangle|^2 \delta(\epsilon_n - \epsilon_0 - \hbar\omega). \quad (13.74)$$

Multiplying by the energy $\hbar\omega$ carried by each emitted photon and summing over all mode wave vectors directed within a differential solid angle $d\Omega$, we obtain the radiated power dP :

$$dP = \sum_{\mathbf{k} \in d\Omega} \hbar\omega \Gamma_{n \rightarrow 0; \mathbf{k}\lambda}^{\text{spont}} = d\Omega \int \frac{\omega^2 d\omega}{(2\pi c)^3} \hbar\omega \frac{4\pi^2 e^2}{\omega} |\langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \epsilon_\lambda^* | n \rangle|^2 \delta(\epsilon_n - \epsilon_0 - \hbar\omega). \quad (13.75)$$

Integrating over the frequency delta function fixes $\omega = (\epsilon_n - \epsilon_0)/\hbar$, yielding the power radiated per unit solid angle with polarization ϵ_λ :

$$\frac{dP}{d\Omega} = \frac{\omega^2 e^2}{2\pi c^3} |\langle 0 | \mathbf{j}_{\mathbf{k}} \cdot \epsilon_\lambda^* | n \rangle|^2. \quad (13.76)$$

This result corresponds directly to classical electrodynamics. A classical localized current density oscillating at frequency ω ,

$$\mathcal{J}(\mathbf{r}, t) = \mathcal{J}(\mathbf{r})e^{-i\omega t} + \mathcal{J}^*(\mathbf{r})e^{i\omega t}, \quad (13.77)$$

generates an asymptotic outgoing radiation field in the far zone ($r \gg \lambda$):

$$\mathbf{A}_{\text{rad}}(\mathbf{r}, t) = \frac{e^{-i\omega(t-r/c)}}{rc} \mathcal{J}_{\mathbf{k}} + \text{c.c.}, \quad \mathbf{k} = \frac{\omega}{c} \hat{\mathbf{r}}, \quad (13.78)$$

where $\mathcal{J}_{\mathbf{k}} = \int d^3r' \mathcal{J}(\mathbf{r}') e^{-i\mathbf{k} \cdot \mathbf{r}'}$. The classical cycle-averaged power radiated per unit solid angle with polarization ϵ_λ is

$$\left(\frac{dP}{d\Omega} \right)_{\text{classical}} = \frac{\omega^2}{2\pi c^3} |\mathcal{J}_{\mathbf{k}} \cdot \epsilon_\lambda^*|^2. \quad (13.79)$$

Comparing Eq. (13.76) with Eq. (13.79), we observe that spontaneous emission in quantum mechanics is physically isomorphic to the classical radiation emitted by an *effective transition current density*:

$$\mathcal{J}_{\mathbf{k}}^{\text{eff}} = \langle 0 | e \mathbf{j}_{\mathbf{k}} | n \rangle. \quad (13.80)$$

13.6.2 Momentum Conservation and Atomic Recoil

Spontaneous emission involves not only energy exchange but also momentum exchange between matter and radiation. To examine the momentum constraints, we analyze the spatial translation symmetry of the current operator.

Let $\mathbf{P} = \sum_{j=1}^N \mathbf{p}_j$ denote the total canonical momentum operator of the N -particle matter system. The finite unitary spatial translation operator $T(\mathbf{a}) = \exp(i\mathbf{P} \cdot \mathbf{a}/\hbar)$ shifts the position operators according to

$$e^{i\mathbf{P} \cdot \mathbf{a}/\hbar} \mathbf{r}_j e^{-i\mathbf{P} \cdot \mathbf{a}/\hbar} = \mathbf{r}_j + \mathbf{a}. \quad (13.81)$$

Because the current density operator $\mathbf{j}(\mathbf{r})$ is an explicit function of the particle coordinates and momenta, it transforms under spatial translation as

$$e^{i\mathbf{P} \cdot \mathbf{a}/\hbar} \mathbf{j}(\mathbf{r}) e^{-i\mathbf{P} \cdot \mathbf{a}/\hbar} = \mathbf{j}(\mathbf{r} - \mathbf{a}). \quad (13.82)$$

Setting $\mathbf{a} = \mathbf{r}$, we can express the current operator at any arbitrary spatial location $\mathbf{r}$ in terms of the current at the coordinate origin:

$$\mathbf{j}(\mathbf{r}) = e^{-i\mathbf{P} \cdot \mathbf{r}/\hbar} \mathbf{j}(\mathbf{0}) e^{i\mathbf{P} \cdot \mathbf{r}/\hbar}. \quad (13.83)$$

Now consider initial and final atomic states $|n\rangle$ and $|0\rangle$ that are eigenstates of total momentum:

$$\mathbf{P}|n\rangle = \hbar \mathbf{q}_n |n\rangle, \quad \mathbf{P}|0\rangle = \hbar \mathbf{q}_0 |0\rangle. \quad (13.84)$$

Evaluating the transition current matrix element, we find

$$\begin{aligned} \langle 0 | \mathbf{j}_{\mathbf{k}} | n \rangle &= \int d^3r e^{-i\mathbf{k} \cdot \mathbf{r}} \langle 0 | \mathbf{j}(\mathbf{r}) | n \rangle \\ &= \int d^3r e^{-i\mathbf{k} \cdot \mathbf{r}} \langle 0 | e^{-i\mathbf{P} \cdot \mathbf{r}/\hbar} \mathbf{j}(\mathbf{0}) e^{i\mathbf{P} \cdot \mathbf{r}/\hbar} | n \rangle \\ &= \langle 0 | \mathbf{j}(\mathbf{0}) | n \rangle \int d^3r e^{-i(\mathbf{k} + \mathbf{q}_0 - \mathbf{q}_n) \cdot \mathbf{r}} \\ &= V \langle 0 | \mathbf{j}(\mathbf{0}) | n \rangle \delta_{\mathbf{k}, \mathbf{q}_n - \mathbf{q}_0}. \end{aligned} \quad (13.85)$$

Equation (13.85) demonstrates that the transition matrix element vanishes identically unless the photon wave vector satisfies

$$\hbar \mathbf{k} = \hbar \mathbf{q}_n - \hbar \mathbf{q}_0. \quad (13.86)$$

Total linear momentum is strictly conserved: the emitted photon carries away the momentum recoil lost by the radiating atom.

13.6.3 Impossibility of Radiation by a Free Electron

A fundamental kinematic consequence of Eq. (13.85) is that an isolated, free charged particle cannot emit or absorb a real photon in vacuum. For a single particle of mass m , energy and momentum conservation would simultaneously require

$$\hbar \mathbf{k} = \hbar \mathbf{q}_n - \hbar \mathbf{q}_0, \quad (13.87)$$

$$\hbar c k = \frac{\hbar^2 q_n^2}{2m} - \frac{\hbar^2 q_0^2}{2m} = \frac{\hbar^2}{2m} (\mathbf{q}_n + \mathbf{q}_0) \cdot (\mathbf{q}_n - \mathbf{q}_0) = \frac{\hbar^2}{2m} (\mathbf{q}_n + \mathbf{q}_0) \cdot \mathbf{k}. \quad (13.88)$$

Dividing by $\hbar k$, this implies

$$c = \frac{\hbar}{2m} (\mathbf{q}_n + \mathbf{q}_0) \cdot \hat{\mathbf{k}} \leq \frac{v_n + v_0}{2} < c, \quad (13.89)$$

which is impossible for subluminal particle velocities. To emit or absorb real radiation, an electron must accelerate in the presence of an external potential (e.g., atomic nucleus in bremsstrahlung or synchrotron magnetic fields) to absorb recoil momentum.

13.7 Electric Dipole Transitions

In typical optical transitions of atoms and molecules, the wavelength of the emitted radiation ($\lambda \sim 5000 \text{ \AA}$) is several orders of magnitude larger than the spatial dimensions of the electron charge distribution ($a_0 \sim 0.5 \text{ \AA}$). This separation of length scales justifies the *long-wavelength approximation* (or electric dipole approximation), in which spatial phase variations across the atomic system are expanded in a Taylor series.

13.7.1 The Electric Dipole Approximation

Evaluating the spatial Fourier transform of the transition current density,

$$\langle 0 | \mathbf{j}_{\mathbf{k}} | n \rangle = \int d^3r e^{-i\mathbf{k} \cdot \mathbf{r}} \langle 0 | \mathbf{j}(\mathbf{r}) | n \rangle, \quad (13.90)$$

we expand the plane wave about the center of mass (chosen as the coordinate origin, $\mathbf{r}_0 = \mathbf{0}$):

$$e^{-i\mathbf{k} \cdot \mathbf{r}} = 1 - i\mathbf{k} \cdot \mathbf{r} - \frac{1}{2}(\mathbf{k} \cdot \mathbf{r})^2 + \dots \quad (13.91)$$

Because $|\mathbf{k} \cdot \mathbf{r}| \sim 2\pi a_0 / \lambda \sim 10^{-3} \ll 1$, the leading monopole term dominates. The zeroth-order current operator is proportional to the total canonical momentum:

$$\mathbf{j}_0 \equiv \int d^3r \mathbf{j}(\mathbf{r}) = \frac{\mathbf{P}}{m}, \quad \mathbf{P} = \sum_{i=1}^N \mathbf{p}_i. \quad (13.92)$$

In the absence of velocity-dependent forces, Heisenberg's equation of motion relates the total momentum to the total position operator $\mathbf{R} = \sum_{i=1}^N \mathbf{r}_i$:

$$\mathbf{P} = \frac{m}{i\hbar} [\mathbf{R}, H_0]. \quad (13.93)$$

The matrix element between stationary states $|n\rangle$ and $|0\rangle$ reduces to

$$\langle 0 | \mathbf{j}_0 | n \rangle = \frac{1}{i\hbar} \langle 0 | (\mathbf{R} H_0 - H_0 \mathbf{R}) | n \rangle = \frac{\varepsilon_n - \varepsilon_0}{i\hbar} \langle 0 | \mathbf{R} | n \rangle = -i\omega \mathbf{d}_{0n}, \quad (13.94)$$

where $\omega = (\varepsilon_n - \varepsilon_0)/\hbar$ is the transition frequency, and

$$\mathbf{d}_{0n} \equiv \langle 0 | \mathbf{R} | n \rangle = \langle 0 | \sum_{i=1}^N \mathbf{r}_i | n \rangle \quad (13.95)$$

is the off-diagonal transition matrix element of the electric dipole operator $\mathbf{D} = e\mathbf{R}$. Transitions mediated by Eq. (13.94) are termed *electric dipole* (*E1*) transitions.

13.7.2 Radiated Power and Angular Distribution

Substituting the dipole matrix element (13.94) into the spontaneous emission formula Eq. (13.76), the power radiated per unit solid angle with polarization ϵ_λ is

$$\frac{dP_\lambda}{d\Omega} = \frac{\omega^4 e^2}{2\pi c^3} |\mathbf{d}_{0n} \cdot \epsilon_\lambda^*|^2. \quad (13.96)$$

Because physical radiation modes are purely transverse ($\epsilon_\lambda \cdot \mathbf{k} = 0$), the radiated polarization is governed by the transverse projection of the transition dipole:

$$(\mathbf{d}_{0n})_\perp = \mathbf{d}_{0n} - (\mathbf{d}_{0n} \cdot \hat{\mathbf{k}})\hat{\mathbf{k}} = -\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{d}_{0n}). \quad (13.97)$$

Summing over the two orthogonal physical polarizations using $\sum_{\lambda=1,2} |\mathbf{u} \cdot \epsilon_\lambda^*|^2 = |\mathbf{u}_\perp|^2 = |\mathbf{u}|^2 - |\mathbf{u} \cdot \hat{\mathbf{k}}|^2$, the unpolarized differential power is

$$\frac{dP}{d\Omega} = \frac{\omega^4 e^2}{2\pi c^3} [|\mathbf{d}_{0n}|^2 - |\mathbf{d}_{0n} \cdot \hat{\mathbf{k}}|^2]. \quad (13.98)$$

If $\mathbf{d}_{0n}$ is oriented along the z -axis, this reproduces the classic Hertzian dipole antenna radiation pattern:

$$\frac{dP}{d\Omega} = \frac{\omega^4 e^2 |\mathbf{d}_{0n}|^2}{2\pi c^3} \sin^2 \theta, \quad (13.99)$$

with vanishing radiation along the dipole axis ($\theta = 0, \pi$). Integrating over all solid angles ($\int d\Omega \sin^2 \theta = \frac{8\pi}{3}$), we obtain the total spontaneously radiated power:

$$P_{\text{tot}} = \frac{4\omega^4 e^2}{3c^3} |\mathbf{d}_{0n}|^2. \quad (13.100)$$

Dividing by the single-photon quantum $\hbar\omega$ yields the spontaneous emission rate (Einstein A coefficient):

$$A_{n \rightarrow 0} = \frac{4\omega^3 e^2}{3\hbar c^3} |\mathbf{d}_{0n}|^2. \quad (13.101)$$

13.7.3 Selection Rules for Electric Dipole Transitions

Let the initial and final states be eigenstates of orbital angular momentum $\mathbf{L}^2$ and L_z , labeled by $|l, m\rangle$ and $|l', m'\rangle$. The commutation relations between $\mathbf{L}$ and $\mathbf{R}$ impose strict selection rules on the non-vanishing of $\mathbf{d}_{0n}$.

Magnetic Quantum Number Selection Rules (Δm)

The z -component of position commutes with L_z :

$$[L_z, R_z] = 0 \implies \langle l', m' | [L_z, R_z] | l, m \rangle = \hbar(m' - m) \langle l', m' | R_z | l, m \rangle = 0. \quad (13.102)$$

Hence, R_z connects only states with identical magnetic quantum numbers:

$$\langle l', m' | R_z | l, m \rangle = 0 \quad \text{unless } \Delta m \equiv m' - m = 0. \quad (13.103)$$

Such transitions are termed π -transitions; the emitted photon is linearly polarized parallel to the quantization axis z .

For the transverse components, the commutation relations are

$$[L_z, R_x] = i\hbar R_y, \quad [L_z, R_y] = -i\hbar R_x. \quad (13.104)$$

Introducing the spherical ladder components $R_{\pm} \equiv R_x \pm iR_y$, we have

$$[L_z, R_{\pm}] = \pm \hbar R_{\pm}, \quad (13.105)$$

which immediately implies

$$\langle l', m' | R_{\pm} | l, m \rangle = 0 \quad \text{unless } \Delta m = m' - m = \pm 1. \quad (13.106)$$

Such transitions are termed $\sigma^{\pm}$ -transitions. Emitted photons along the $+z$ axis are right- or left-circularly polarized, carrying intrinsic helicity $\pm \hbar$ to ensure conservation of angular momentum along z . Thus, the universal $E1$ magnetic selection rule is

$$\Delta m = 0, \pm 1. \quad (13.107)$$

Orbital Angular Momentum Selection Rules (Δl)

To establish the selection rule for the orbital quantum number l , we utilize Dirac's nested commutator identity:

$$[\mathbf{L}^2, [\mathbf{L}^2, \mathbf{R}]] = 2\hbar^2 (\mathbf{R}\mathbf{L}^2 + \mathbf{L}^2\mathbf{R}). \quad (13.108)$$

Taking matrix elements of Eq. (13.108) between $|l, m\rangle$ and $\langle l', m'|$, we obtain

$$[l'(l' + 1) - l(l + 1)]^2 \langle l', m' | \mathbf{R} | l, m \rangle = 2 [l(l + 1) + l'(l' + 1)] \langle l', m' | \mathbf{R} | l, m \rangle. \quad (13.109)$$

Algebraic factorization yields

$$(l + l')(l + l' + 2) [(l - l')^2 - 1] \langle l', m' | \mathbf{R} | l, m \rangle = 0. \quad (13.110)$$

Because $l, l' \geq 0$, the factor $(l + l' + 2)$ is strictly non-zero. The factor $l + l'$ can vanish only if $l = l' = 0$; however, the matrix element between two spherically symmetric s -states vanishes by parity symmetry ($\langle 0, 0 | \mathbf{r} | 0, 0 \rangle = \mathbf{0}$). Consequently, non-vanishing matrix elements strictly require $(l - l')^2 - 1 = 0$, establishing the selection rule:

$$\Delta l = l' - l = \pm 1. \quad (13.111)$$

Furthermore, because the position operator is odd under spatial inversion ($\Pi^\dagger \mathbf{r} \Pi = -\mathbf{r}$), parity must change between initial and final states: $\Pi_i \Pi_f = -1$ (Laporte's rule). Transitions between states of the same parity (such as $s \rightarrow s$ or $d \rightarrow s$) are strictly forbidden in electric dipole order.

13.7.4 The Thomas–Reiche–Kuhn (TRK) Dipole Sum Rule

The electric dipole absorption cross section for a polarized beam is given by

$$\sigma_{\text{dip}}(\omega) = \frac{4\pi^2 e^2 \omega}{c} \sum_n |\mathbf{d}_{n0} \cdot \boldsymbol{\varepsilon}_\lambda|^2 \delta(\varepsilon_n - \varepsilon_0 - \hbar\omega). \quad (13.112)$$

A remarkable consequence of quantum mechanics is that the frequency-integrated absorption cross section satisfies an exact, state-independent conservation law.

Let $\hat{\mathbf{u}}$ be a real unit vector. The fundamental canonical commutation relations between total momentum and total position yield

$$[\mathbf{P} \cdot \hat{\mathbf{u}}, \mathbf{R} \cdot \hat{\mathbf{u}}] = \sum_{i,j=1}^N [p_{i,u}, r_{j,u}] = -i\hbar \sum_{i=1}^N 1 = -i\hbar N. \quad (13.113)$$

Taking the expectation value in an arbitrary stationary state $|0\rangle$ and inserting a complete set of states $\sum_n |n\rangle\langle n| = 1$, we have

$$\sum_n (\langle 0|\mathbf{P} \cdot \hat{\mathbf{u}}|n\rangle\langle n|\mathbf{R} \cdot \hat{\mathbf{u}}|0\rangle - \langle 0|\mathbf{R} \cdot \hat{\mathbf{u}}|n\rangle\langle n|\mathbf{P} \cdot \hat{\mathbf{u}}|0\rangle) = -i\hbar N. \quad (13.114)$$

Using $\langle 0|\mathbf{P}|n\rangle = \frac{m}{i\hbar}(\varepsilon_n - \varepsilon_0)\langle 0|\mathbf{R}|n\rangle$, this identity becomes the *Thomas-Reiche-Kuhn (TRK) sum rule*:

$$\sum_n \frac{2m}{\hbar^2}(\varepsilon_n - \varepsilon_0)|\langle n|\mathbf{R} \cdot \hat{\mathbf{u}}|0\rangle|^2 = N, \quad (13.115)$$

where the dimensionless quantities $f_{n0} \equiv \frac{2m(\varepsilon_n - \varepsilon_0)}{\hbar^2}|\langle n|\mathbf{R} \cdot \hat{\mathbf{u}}|0\rangle|^2$ are the atomic *oscillator strengths*, satisfying $\sum_n f_{n0} = N$.

Integrating the dipole absorption cross section Eq. (13.112) over all frequencies for an initial ground state ($E_n \geq E_0$), we obtain

$$\int_0^\infty d\omega \sigma_{\text{dip}}(\omega) = \frac{4\pi^2 e^2}{\hbar^2 c} \sum_n (\varepsilon_n - \varepsilon_0) |\langle n|\mathbf{R} \cdot \boldsymbol{\varepsilon}_\lambda|0\rangle|^2 = \frac{2\pi^2 e^2 N}{mc}. \quad (13.116)$$

The integrated spectral absorption is a universal constant proportional solely to the number of optically active electrons N , providing a powerful constraint on atomic transition probabilities and dielectric response functions.

13.8 Magnetic Dipole and Electric Quadrupole Transitions

When atomic selection rules or geometric symmetries cause the electric dipole matrix element to vanish ($\mathbf{d}_{0n} = \mathbf{0}$), the transition is termed *dipole-forbidden*. Such transitions are not entirely suppressed; they can proceed through higher-order spatial multipole terms in the retardation expansion of the plane-wave factor $e^{-i\mathbf{k} \cdot \mathbf{r}}$.

13.8.1 Decomposition of the First Retardation Correction

Retaining the linear retardation term in Eq. (13.91), the current matrix element acquires the first-order correction:

$$\langle 0|\mathbf{j}_\mathbf{k} \cdot \boldsymbol{\varepsilon}_\lambda^*|n\rangle^{(1)} = -i\langle 0|\int d^3r (\mathbf{k} \cdot \mathbf{r})(\mathbf{j}(\mathbf{r}) \cdot \boldsymbol{\varepsilon}_\lambda^*)|n\rangle. \quad (13.117)$$

We decompose the dyadic integrand into its antisymmetric and symmetric tensor components with respect to the spatial vectors $\mathbf{k}$ and $\boldsymbol{\varepsilon}_\lambda^*$:

$$\begin{aligned} (\mathbf{k} \cdot \mathbf{r})(\mathbf{j}(\mathbf{r}) \cdot \boldsymbol{\varepsilon}_\lambda^*) &= \frac{1}{2} [(\mathbf{k} \cdot \mathbf{r})(\mathbf{j}(\mathbf{r}) \cdot \boldsymbol{\varepsilon}_\lambda^*) - (\boldsymbol{\varepsilon}_\lambda^* \cdot \mathbf{r})(\mathbf{j}(\mathbf{r}) \cdot \mathbf{k})] \\ &\quad + \frac{1}{2} [(\mathbf{k} \cdot \mathbf{r})(\mathbf{j}(\mathbf{r}) \cdot \boldsymbol{\varepsilon}_\lambda^*) + (\boldsymbol{\varepsilon}_\lambda^* \cdot \mathbf{r})(\mathbf{j}(\mathbf{r}) \cdot \mathbf{k})]. \end{aligned} \quad (13.118)$$

Using the vector identity $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$, the first (antisymmetric) term transforms into a cross product:

$$\frac{1}{2} [(\mathbf{k} \cdot \mathbf{r})(\mathbf{j} \cdot \boldsymbol{\varepsilon}_\lambda^*) - (\boldsymbol{\varepsilon}_\lambda^* \cdot \mathbf{r})(\mathbf{j} \cdot \mathbf{k})] = \frac{1}{2} (\mathbf{k} \times \boldsymbol{\varepsilon}_\lambda^*) \cdot (\mathbf{r} \times \mathbf{j}). \quad (13.119)$$

The matrix element (13.117) thus separates cleanly into two distinct physical contributions:

$$\begin{aligned} \langle 0|\mathbf{j}_\mathbf{k} \cdot \boldsymbol{\varepsilon}_\lambda^*|n\rangle^{(1)} &= -\frac{i}{2} (\mathbf{k} \times \boldsymbol{\varepsilon}_\lambda^*) \cdot \langle 0|\int d^3r \mathbf{r} \times \mathbf{j}(\mathbf{r})|n\rangle \\ &\quad - \frac{i}{2} \sum_{a,b} k_b \varepsilon_{\lambda,a}^* \langle 0|\int d^3r [j_a(\mathbf{r})r_b + r_a j_b(\mathbf{r})]|n\rangle. \end{aligned} \quad (13.120)$$

13.8.2 Magnetic Dipole Transitions ($M1$)

Evaluating the integral in the antisymmetric term using the definition of the paramagnetic current density Eq. (13.31), we find

$$\int d^3r \mathbf{r} \times \mathbf{j}(\mathbf{r}) = \frac{1}{2m} \sum_{i=1}^N (\mathbf{r}_i \times \mathbf{p}_i - \mathbf{p}_i \times \mathbf{r}_i) = \frac{\mathbf{L}}{m}, \quad (13.121)$$

where $\mathbf{L} = \sum_i \mathbf{r}_i \times \mathbf{p}_i$ is the total orbital angular momentum operator. The interaction couples directly to the orbital magnetic dipole moment:

$$\mathbf{M}_{\text{orb}} = \frac{e}{2mc} \mathbf{L} = \frac{\mu_B}{\hbar} \mathbf{L}, \quad (13.122)$$

where $\mu_B = e\hbar/(2mc)$ is the Bohr magneton. Including the intrinsic electron spin $\mathbf{S}$, the full magnetic moment operator is $\mathbf{M} = \frac{e}{2mc}(\mathbf{L} + 2\mathbf{S})$.

Transitions mediated by this term are *magnetic dipole* ($M1$) transitions. The effective coupling Hamiltonian corresponds to the Zeeman energy:

$$H_{M1} = -\mathbf{M} \cdot \mathcal{B}(\mathbf{r}, t), \quad (13.123)$$

where $\mathcal{B} = \nabla \times \mathbf{A} \propto i\mathbf{k} \times \boldsymbol{\varepsilon}_\lambda$.

Because $\mathbf{L}$ is an axial vector, it is strictly invariant under spatial parity inversion ($\Pi^\dagger \mathbf{L} \Pi = +\mathbf{L}$). Consequently, $M1$ transitions conserve atomic parity:

$$\Pi_i \Pi_f = +1, \quad \Delta l = 0. \quad (13.124)$$

The amplitude of an $M1$ transition relative to an electric dipole $E1$ transition scales as

$$\frac{\mathcal{A}_{M1}}{\mathcal{A}_{E1}} \sim \frac{\hbar k/m}{\omega a_0} = \frac{\hbar}{mca_0} = \frac{e^2}{\hbar c} \equiv \alpha \approx \frac{1}{137}. \quad (13.125)$$

Because radiative rates depend on the amplitude squared, $M1$ transition rates are suppressed by a factor of $\alpha^2 \sim (v/c)^2 \sim 10^{-4}$ to 10^{-5} compared to allowed $E1$ rates.

13.8.3 Electric Quadrupole Transitions ($E2$)

The symmetric tensor term in Eq. (13.120) can be re-expressed using Heisenberg's equation of motion. In the absence of velocity-dependent forces,

$$[r_a r_b, H_0] = \frac{i\hbar}{m} (r_a p_b + p_a r_b). \quad (13.126)$$

Taking matrix elements between stationary states and utilizing the transversality condition $\mathbf{k} \cdot \boldsymbol{\varepsilon}_\lambda^* = 0$, we can subtract the trace to define the traceless *electric quadrupole tensor*:

$$Q_{ab} \equiv e \sum_{i=1}^N \left(r_{i,a} r_{i,b} - \frac{1}{3} \delta_{ab} r_i^2 \right). \quad (13.127)$$

The symmetric matrix element becomes

$$\langle 0 | \mathbf{j}_\mathbf{k} \cdot \boldsymbol{\varepsilon}_\lambda^* | n \rangle_{E2} = -\frac{\omega}{2e} \sum_{a,b} k_b \varepsilon_{\lambda,a}^* \langle 0 | Q_{ab} | n \rangle. \quad (13.128)$$

Transitions mediated by Eq. (13.128) are *electric quadrupole* ($E2$) transitions, coupling the electric field gradient $\nabla \mathcal{E}$ to the quadrupole tensor Q_{ab} .

Because Q_{ab} transforms as a symmetric rank-2 tensor under spatial rotations and is even under inversion ($\Pi^\dagger Q_{ab} \Pi = +Q_{ab}$), $E2$ transitions obey the selection rules:

$$\Pi_i \Pi_f = +1, \quad \Delta l = 0, \pm 2 \quad (0 \not\rightarrow 0). \quad (13.129)$$

The relative amplitude of an $E2$ transition compared to an $E1$ transition scales as

$$\frac{\mathcal{A}_{E2}}{\mathcal{A}_{E1}} \sim k a_0 \sim \frac{a_0}{\lambda} \sim \alpha \sim 10^{-3}, \quad (13.130)$$

so $E2$ transition rates are likewise suppressed by $\sim 10^{-6}$ relative to $E1$ rates, comparable in magnitude to $M1$ transitions.

13.8.4 Strictly Forbidden Transitions and Two-Photon Decay

When a transition is forbidden in all single-photon multipole orders—most notably the spherically symmetric $2s_{1/2} \rightarrow 1s_{1/2}$ transition in hydrogen ($J = 1/2 \rightarrow 1/2$, $\Delta l = 0$, parity unchanged, with no single-photon $J = 0 \rightarrow 0$ or $1/2 \rightarrow 1/2$ channel carrying angular momentum $\hbar$)—the state is metastable. Such strictly forbidden decays proceed at second order in perturbation theory through the simultaneous emission of *two photons*:

$$|2s\rangle \rightarrow |1s\rangle + \gamma_1 + \gamma_2, \quad \hbar\omega_1 + \hbar\omega_2 = \varepsilon_{2s} - \varepsilon_{1s}. \quad (13.131)$$

This two-photon decay yields a continuous energy spectrum and endows the $2s$ state of hydrogen with an exceptionally long radiative lifetime of $\tau \approx 0.12$ s, eight orders of magnitude longer than the allowed $2p \rightarrow 1s$ Lyman- α transition ($\tau_{2p} \approx 1.6$ ns).

13.9 Scattering of Light and Thomson Scattering

We now examine the scattering of electromagnetic radiation by charged systems within the framework of quantum electrodynamics. In this formulation, scattering is described as a transition between multi-photon states: an incident photon in mode $(\mathbf{k}, \varepsilon_\lambda)$ is destroyed while a scattered photon in mode $(\mathbf{k}', \varepsilon_{\lambda'})$ is created, while the target matter transitions from an initial state $|0\rangle$ to a final state $|n\rangle$:

$$|0; N_{\mathbf{k}\lambda}, 0_{\mathbf{k}'\lambda'}\rangle \longrightarrow |n; N_{\mathbf{k}\lambda} - 1, 1_{\mathbf{k}'\lambda'}\rangle. \quad (13.132)$$

In the nonrelativistic interaction Hamiltonian Eq. (13.58), this transition proceeds via two distinct microscopic mechanisms:

1. **Direct contact scattering:** First-order perturbation in the quadratic diamagnetic interaction $\frac{e^2}{2mc^2} \rho \hat{\mathbf{A}}^2$ (the “seagull” vertex).
2. **Resonant two-photon scattering:** Second-order perturbation in the linear paramagnetic interaction $-\frac{e}{c} \mathbf{j} \cdot \hat{\mathbf{A}}$, involving virtual intermediate states.

13.9.1 First-Order Scattering from the $\mathbf{A}^2$ Contact Interaction

The quantized diamagnetic interaction Hamiltonian is

$$H_{\mathbf{A}^2} = \frac{e^2}{2mc^2} \int d^3r \rho(\mathbf{r}) \hat{\mathbf{A}}^2(\mathbf{r}). \quad (13.133)$$

The matrix element connecting the initial and final states is

$$\begin{aligned} \mathcal{M}_{\mathbf{A}^2} &= \langle n; N_{\mathbf{k}\lambda} - 1, 1_{\mathbf{k}'\lambda'} | H_{\mathbf{A}^2} | 0; N_{\mathbf{k}\lambda}, 0_{\mathbf{k}'\lambda'} \rangle \\ &= \frac{e^2}{2mc^2} \int d^3r \langle n | \rho(\mathbf{r}) | 0 \rangle \langle N_{\mathbf{k}\lambda} - 1, 1_{\mathbf{k}'\lambda'} | \hat{\mathbf{A}}(\mathbf{r}) \cdot \hat{\mathbf{A}}(\mathbf{r}) | N_{\mathbf{k}\lambda}, 0_{\mathbf{k}'\lambda'} \rangle. \end{aligned} \quad (13.134)$$

Because $\hat{\mathbf{A}}(\mathbf{r})$ is a linear combination of creation and annihilation operators, a non-zero matrix element requires one field operator to destroy a $(\mathbf{k}, \lambda)$ photon ($a_{\mathbf{k}\lambda} \sim \sqrt{N_{\mathbf{k}\lambda}}$) and the other to create a $(\mathbf{k}', \lambda')$ photon ($a_{\mathbf{k}'\lambda'}^\dagger \sim \sqrt{0+1} = 1$). Taking into account the two identical operator permutations, we obtain

$$\begin{aligned} \langle N_{\mathbf{k}\lambda} - 1, 1_{\mathbf{k}'\lambda'} | \hat{\mathbf{A}}^2(\mathbf{r}) | N_{\mathbf{k}\lambda}, 0_{\mathbf{k}'\lambda'} \rangle &= 2 \left(\sqrt{\frac{2\pi\hbar c^2}{\omega V}} \epsilon_\lambda \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} \sqrt{N_{\mathbf{k}\lambda}} \right) \cdot \left(\sqrt{\frac{2\pi\hbar c^2}{\omega' V}} \epsilon_{\lambda'}^* \frac{e^{-i\mathbf{k}'\cdot\mathbf{r}}}{\sqrt{V}} \right) \\ &= 2 \left(\frac{2\pi\hbar c^2}{V\sqrt{\omega\omega'}} \right) \sqrt{N_{\mathbf{k}\lambda}} (\epsilon_\lambda \cdot \epsilon_{\lambda'}^*) e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}. \end{aligned} \quad (13.135)$$

Substituting this into Eq. (13.134), the matrix element becomes

$$\mathcal{M}_{\mathbf{A}^2} = r_0 \frac{2\pi\hbar c^2}{V\sqrt{\omega\omega'}} \sqrt{N_{\mathbf{k}\lambda}} (\epsilon_\lambda \cdot \epsilon_{\lambda'}^*) \langle n | \rho_{\mathbf{k}'-\mathbf{k}} | 0 \rangle, \quad (13.136)$$

where $\rho_{\mathbf{q}} \equiv \int d^3r e^{-i\mathbf{q}\cdot\mathbf{r}} \rho(\mathbf{r}) = \sum_i e^{-i\mathbf{q}\cdot\mathbf{r}_i}$ is the spatial Fourier transform of the particle density operator, and

$$r_0 \equiv \frac{e^2}{mc^2} \approx 2.818 \times 10^{-13} \text{ cm} \quad (13.137)$$

is the *classical electron radius*.

13.9.2 Differential Cross Section and Thomson Scattering

Applying Fermi's Golden Rule, the scattering rate into mode $(\mathbf{k}', \lambda')$ is

$$\Gamma_{0 \rightarrow n; \mathbf{k}\lambda \rightarrow \mathbf{k}'\lambda'} = \frac{2\pi}{\hbar} |\mathcal{M}_{\mathbf{A}^2}|^2 \delta(\varepsilon_0 + \hbar\omega - \varepsilon_n - \hbar\omega'). \quad (13.138)$$

Summing over all scattered wave vectors $\mathbf{k}'$ within a differential solid angle $d\Omega'$,

$$\sum_{\mathbf{k}' \in d\Omega'} \rightarrow V \int \frac{\omega'^2 d\omega' d\Omega'}{(2\pi c)^3}, \quad (13.139)$$

the differential transition rate is

$$\frac{d\Gamma_{0 \rightarrow n}}{d\Omega'} = \left(\frac{cN_{\mathbf{k}\lambda}}{V} \right) r_0^2 \left(\frac{\omega'}{\omega} \right) |\epsilon_\lambda \cdot \epsilon_{\lambda'}^*|^2 |\langle n | \rho_{\mathbf{k}'-\mathbf{k}} | 0 \rangle|^2. \quad (13.140)$$

Dividing by the incident photon flux $\Phi_{\text{inc}} = cN_{\mathbf{k}\lambda}/V$, the normalization volume cancels, yielding the differential cross section:

$$\frac{d\sigma_{0 \rightarrow n}}{d\Omega'} = r_0^2 \left(\frac{\omega'}{\omega} \right) |\epsilon_\lambda \cdot \epsilon_{\lambda'}^*|^2 |\langle n | \rho_{\mathbf{k}'-\mathbf{k}} | 0 \rangle|^2. \quad (13.141)$$

Scattering from a Single Free Electron

For a single free electron, the density matrix element enforces exact conservation of total linear momentum:

$$\langle \mathbf{q}_n | \rho_{\mathbf{k}'-\mathbf{k}} | \mathbf{q}_0 \rangle = \delta_{\mathbf{q}_0+\mathbf{k}, \mathbf{q}_n+\mathbf{k}'}. \quad (13.142)$$

In the nonrelativistic limit ($\hbar\omega \ll mc^2$), Compton recoil is negligible, so $\omega' \approx \omega$. Summing over the electron final momentum $\mathbf{q}_n$ yields the celebrated *Thomson scattering cross section*:

$$\frac{d\sigma_{\text{Th}}}{d\Omega'} = r_0^2 |\epsilon_\lambda \cdot \epsilon_{\lambda'}^*|^2. \quad (13.143)$$

For unpolarized incident light scattered through angle θ , averaging over initial polarizations and summing over final polarizations gives

$$\frac{d\sigma_{\text{Th}}}{d\Omega'} = \frac{r_0^2}{2}(1 + \cos^2 \theta). \quad (13.144)$$

Integrating over all solid angles yields the total Thomson cross section:

$$\sigma_{\text{Th}} = \frac{8\pi}{3}r_0^2 \approx 0.665 \times 10^{-24} \text{ cm}^2 = 0.665 \text{ barn}. \quad (13.145)$$

Coherent Multi-Electron Scattering

For long-wavelength radiation scattered elastically from an atom or droplet containing N electrons without electronic excitation ($|n\rangle = |0\rangle$ and $ka \ll 1$), all N atomic electrons scatter in phase: $\rho_{\mathbf{k}'-\mathbf{k}} \approx \sum_{i=1}^N 1 = N$. The cross section scales as

$$\frac{d\sigma}{d\Omega'} = N^2 r_0^2 |\boldsymbol{\varepsilon}_\lambda \cdot \boldsymbol{\varepsilon}_{\lambda'}^*|^2. \quad (13.146)$$

The constructive N^2 interference enhancement explains the intense scattering of light by microscopic liquid droplets in clouds and aerosols compared to individual gas molecules.

13.9.3 Second-Order Contributions from $\mathbf{j} \cdot \mathbf{A}$

Scattering can also proceed through second order in the linear coupling $H_{\mathbf{j}\cdot\mathbf{A}} = -\frac{e}{c} \int d^3r \mathbf{j} \cdot \hat{\mathbf{A}}$. Quantum mechanics requires summing over two distinct time-ordered Feynman pathways:

1. **Direct channel (absorption then emission):** The incident photon $(\mathbf{k}, \lambda)$ is absorbed, promoting the system to a virtual intermediate state $|m\rangle$ with energy denominator $\varepsilon_0 + \hbar\omega - \varepsilon_m + i\eta$, followed by emission of the scattered photon $(\mathbf{k}', \lambda')$.
2. **Crossed channel (emission then absorption):** The scattered photon $(\mathbf{k}', \lambda')$ is emitted first, leaving the system in $|m\rangle$ with energy denominator $\varepsilon_0 - \hbar\omega' - \varepsilon_m + i\eta$, followed by absorption of the incident photon $(\mathbf{k}, \lambda)$.

Summing these two coherent amplitudes yields

$$\mathcal{M}_{\mathbf{j}\cdot\mathbf{A}}^{(2)} = \frac{2\pi\hbar c^2}{V\sqrt{\omega\omega'}} \sqrt{N_{\mathbf{k}\lambda}} \sum_m \left[\frac{\langle n|\mathbf{j}_{\mathbf{k}'} \cdot \boldsymbol{\varepsilon}_{\lambda'}^*|m\rangle \langle m|\mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda|0\rangle}{\varepsilon_0 - \varepsilon_m + \hbar\omega + i\eta} + \frac{\langle n|\mathbf{j}_{-\mathbf{k}} \cdot \boldsymbol{\varepsilon}_\lambda|m\rangle \langle m|\mathbf{j}_{\mathbf{k}'} \cdot \boldsymbol{\varepsilon}_{\lambda'}^*|0\rangle}{\varepsilon_0 - \varepsilon_m - \hbar\omega' + i\eta} \right]. \quad (13.147)$$

For a free electron, evaluating Eq. (13.147) reveals that the direct and crossed amplitudes have opposite signs and cancel each other to leading order in v/c . The net second-order amplitude is of order $\mathcal{O}(v/c)$ relative to the $\mathbf{A}^2$ contact term. Thus, in the nonrelativistic regime ($\hbar\omega \ll mc^2$, $v \ll c$), free-electron Thomson scattering is governed entirely by the first-order $\mathbf{A}^2$ contact interaction.

13.10 Raman and Rayleigh Scattering: The Kramers–Heisenberg Formula

When electromagnetic radiation scatters off a bound atomic or molecular system, the interaction can induce transitions between internal stationary states. Inelastic light scattering, accompanied by a discrete shift in photon frequency, is termed *Raman scattering*. In the elastic limit where the atomic internal state is unchanged, the process reduces to *Rayleigh scattering*.

13.10.1 Derivation of the Kramers–Heisenberg Transition Amplitude

Consider an atom localized at $\mathbf{r} = \mathbf{0}$, transitioning from an initial state $|0\rangle$ of energy ε_0 to a final state $|n\rangle$ of energy ε_n , while an incident photon $(\mathbf{k}, \varepsilon_\lambda)$ is scattered into mode $(\mathbf{k}', \varepsilon_{\lambda'})$. Global energy conservation requires

$$\varepsilon_0 + \hbar\omega = \varepsilon_n + \hbar\omega'. \quad (13.148)$$

Depending on the sign of the energy transfer $\Delta\varepsilon = \varepsilon_n - \varepsilon_0$, we classify:

- **Stokes scattering** ($\varepsilon_n > \varepsilon_0$): The atom is excited, and the scattered photon is redshifted ($\omega' < \omega$).
- **Anti-Stokes scattering** ($\varepsilon_n < \varepsilon_0$): The atom de-excites from an initial excited state, and the scattered photon is blueshifted ($\omega' > \omega$).

In the electric dipole approximation, we evaluate the coherent sum of the first-order diamagnetic contact amplitude $\mathcal{M}_{\mathbf{A}^2}$ and the second-order paramagnetic amplitude $\mathcal{M}_{\mathbf{j}\cdot\mathbf{A}}^{(2)}$. Using the dipole momentum relation $\mathbf{P} = \frac{m}{i\hbar}[\mathbf{R}, H_0]$ and inserting the energy conservation condition (13.148), the second-order bracket in Eq. (13.147) can be reorganized algebraically as

$$\begin{aligned} \mathcal{M}_{\mathbf{j}\cdot\mathbf{A}}^{(2)} = & \frac{2\pi\hbar e^2}{V\sqrt{\omega\omega'}} \sqrt{N_{\mathbf{k}\lambda}} \left\{ \frac{1}{i\hbar m} \langle n | [\mathbf{P} \cdot \varepsilon_{\lambda'}^*, \mathbf{R} \cdot \varepsilon_\lambda] | 0 \rangle \right. \\ & \left. + \omega\omega' \sum_m \left[\frac{\langle n | \mathbf{R} \cdot \varepsilon_{\lambda'}^* | m \rangle \langle m | \mathbf{R} \cdot \varepsilon_\lambda | 0 \rangle}{\varepsilon_0 - \varepsilon_m + \hbar\omega + i\eta} + \frac{\langle n | \mathbf{R} \cdot \varepsilon_\lambda | m \rangle \langle m | \mathbf{R} \cdot \varepsilon_{\lambda'}^* | 0 \rangle}{\varepsilon_0 - \varepsilon_m - \hbar\omega' + i\eta} \right] \right\}. \end{aligned} \quad (13.149)$$

Evaluating the canonical commutator between total momentum and total position,

$$[\mathbf{P} \cdot \varepsilon_{\lambda'}^*, \mathbf{R} \cdot \varepsilon_\lambda] = -i\hbar(\varepsilon_\lambda \cdot \varepsilon_{\lambda'}^*)N, \quad (13.150)$$

the first term in Eq. (13.149) evaluates to

$$-r_0 \frac{2\pi\hbar c^2}{V\sqrt{\omega\omega'}} \sqrt{N_{\mathbf{k}\lambda}} (\varepsilon_\lambda \cdot \varepsilon_{\lambda'}^*) N \delta_{n0}. \quad (13.151)$$

This term *identically cancels* the first-order $\mathbf{A}^2$ contact amplitude Eq. (13.136)!

The net scattering amplitude $\mathcal{M}_{\text{net}} = \mathcal{M}_{\mathbf{A}^2} + \mathcal{M}_{\mathbf{j}\cdot\mathbf{A}}^{(2)}$ is therefore given entirely by the regularized dipole sum:

$$\mathcal{M}_{\text{net}} = \frac{2\pi\hbar e^2}{V} \sqrt{\frac{N_{\mathbf{k}\lambda}}{\omega\omega'}} \omega\omega' \sum_m \left[\frac{\langle n | \mathbf{R} \cdot \varepsilon_{\lambda'}^* | m \rangle \langle m | \mathbf{R} \cdot \varepsilon_\lambda | 0 \rangle}{\varepsilon_0 - \varepsilon_m + \hbar\omega + i\eta} + \frac{\langle n | \mathbf{R} \cdot \varepsilon_\lambda | m \rangle \langle m | \mathbf{R} \cdot \varepsilon_{\lambda'}^* | 0 \rangle}{\varepsilon_0 - \varepsilon_m - \hbar\omega' + i\eta} \right]. \quad (13.152)$$

Applying Fermi's Golden Rule and dividing by the incident photon flux $\Phi_{\text{inc}} = cN_{\mathbf{k}\lambda}/V$, we obtain the fundamental *Kramers–Heisenberg formula* for the differential scattering cross section:

$$\frac{d\sigma_{0 \rightarrow n}}{d\Omega'} = \frac{e^4}{c^4} \omega\omega'^3 \left| \sum_m \left[\frac{\langle n | \mathbf{R} \cdot \varepsilon_{\lambda'}^* | m \rangle \langle m | \mathbf{R} \cdot \varepsilon_\lambda | 0 \rangle}{\varepsilon_0 - \varepsilon_m + \hbar\omega + i\eta} + \frac{\langle n | \mathbf{R} \cdot \varepsilon_\lambda | m \rangle \langle m | \mathbf{R} \cdot \varepsilon_{\lambda'}^* | 0 \rangle}{\varepsilon_0 - \varepsilon_m - \hbar\omega' + i\eta} \right] \right|^2. \quad (13.153)$$

13.10.2 Selection Rules and Parity Complementarity

Because each transition operator in Eq. (13.152) is an odd-parity dipole operator $\mathbf{R}$, the intermediate state $|m\rangle$ must have parity opposite to $|0\rangle$ ($\Pi_m = -\Pi_0$), while the final state $|n\rangle$ must have parity opposite to $|m\rangle$ ($\Pi_n = -\Pi_m = +\Pi_0$). Consequently:

$$\Pi_n = \Pi_0. \quad (13.154)$$

Two-photon Raman transitions conserve parity ($\Delta l = 0, \pm 2$). In centrosymmetric molecules and atoms, this leads to the *rule of mutual exclusion*: transitions that are electric-dipole allowed in single-photon absorption ($\Delta \Pi = -1$) are Raman-forbidden, while Raman-active transitions ($\Delta \Pi = 0$) are dipole-forbidden in absorption.

13.10.3 Rayleigh Scattering and the ω^4 Law

When the incident photon frequency is well below the lowest electronic excitation threshold ($\hbar\omega \ll \varepsilon_1 - \varepsilon_0$), the atom remains in its ground state ($|n\rangle = |0\rangle$, $\omega' = \omega$). Expanding the energy denominators in powers of $\hbar\omega/(\varepsilon_m - \varepsilon_0) \ll 1$, the cross section (13.153) simplifies to

$$\frac{d\sigma_{\text{Rayleigh}}}{d\Omega'} = \left(\frac{\omega a_0}{c}\right)^4 a_0^2 |\boldsymbol{\varepsilon}_{\lambda'}^* \cdot \boldsymbol{\alpha}(\omega) \cdot \boldsymbol{\varepsilon}_{\lambda}|^2 \propto \omega^4, \quad (13.155)$$

where $\boldsymbol{\alpha}(\omega)$ is the dynamic atomic polarizability tensor. This steep ω^4 frequency dependence is Rayleigh's law: blue light ($\lambda \approx 400$ nm) scatters nearly an order of magnitude more intensely than red light ($\lambda \approx 700$ nm), explaining the blue color of the daytime sky and the redness of sunsets.

13.10.4 The Dipole Gauge Transformation

The exact cancellation between the $\mathbf{A}^2$ contact term and the commutator part of the $\mathbf{j} \cdot \mathbf{A}$ interaction can be understood elegantly via a gauge transformation. Performing the Göppert-Mayer (or Power-Zienau-Woolley) unitary transformation with the gauge generating function

$$\chi(\mathbf{r}, t) = -\mathbf{r} \cdot \mathbf{A}(\mathbf{0}, t), \quad (13.156)$$

the transformed potentials in the dipole approximation become

$$\begin{aligned} \mathbf{A}'(\mathbf{r}, t) &= \mathbf{A} + \nabla\chi = \mathbf{0}, \\ \varphi'(\mathbf{r}, t) &= -\frac{1}{c} \frac{\partial\chi}{\partial t} = -\mathbf{r} \cdot \boldsymbol{\mathcal{E}}(\mathbf{0}, t). \end{aligned} \quad (13.157)$$

In this electric dipole gauge, the interaction Hamiltonian takes the manifest multipole form:

$$H_{\text{dipole}} = -e\mathbf{R} \cdot \hat{\boldsymbol{\mathcal{E}}}(\mathbf{0}). \quad (13.158)$$

The $\mathbf{A}^2$ contact term is absent from the outset, and the Kramers-Heisenberg formula (13.152) emerges directly from second-order perturbation theory in $-e\mathbf{R} \cdot \hat{\boldsymbol{\mathcal{E}}}$, providing a mathematically transparent and physically unified description of optical scattering.

13.11 Problems

1. **Many-particle gauge transformation.** Consider a system of N identical particles of charge e and mass m in an external, time-dependent electromagnetic field described by potentials $(\mathbf{A}, \varphi)$. Show that under a gauge transformation with generating function $\chi(\mathbf{r}, t)$,

$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \nabla\chi(\mathbf{r}, t), \quad (13.159)$$

$$\varphi'(\mathbf{r}, t) = \varphi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial\chi(\mathbf{r}, t)}{\partial t}, \quad (13.160)$$

the N -particle Schrödinger wave function $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t)$ transforms according to

$$\psi'(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = \exp \left[\frac{ie}{\hbar c} \sum_{i=1}^N \chi(\mathbf{r}_i, t) \right] \psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t). \quad (13.161)$$

2. Gauge invariance and local charge conservation.

- (a) Prove that all matrix elements of the total physical current density operator,

$$\mathbf{J}(\mathbf{r}) = \mathbf{j}(\mathbf{r}) - \frac{e}{mc} \mathbf{A}(\mathbf{r}) \rho(\mathbf{r}), \quad (13.162)$$

evaluated between arbitrary states of the system, are strictly invariant under gauge transformations.

- (b) In the Heisenberg picture, show that the density operator $\rho(\mathbf{r}, t) = \sum_i \delta(\mathbf{r} - \mathbf{r}_i(t))$ and current density operator $\mathbf{J}(\mathbf{r}, t)$ satisfy the operator continuity equation:

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{J}(\mathbf{r}, t) = 0. \quad (13.163)$$

3. Commutation relations and field Hamiltonian.

- (a) Using the definition of the mode operators $A_{\mathbf{k}\lambda}^{(\text{op})} = \sqrt{\frac{2\pi\hbar c^2}{\omega}} a_{\mathbf{k}\lambda}$, verify the commutation relations:

$$[A_{\mathbf{k}\lambda}^{(\text{op})}, A_{\mathbf{k}'\lambda'}^{(\text{op})}] = 0, \quad (13.164)$$

$$[A_{\mathbf{k}\lambda}^{(\text{op})\dagger}, A_{\mathbf{k}'\lambda'}^{(\text{op})\dagger}] = 0, \quad (13.165)$$

$$[A_{\mathbf{k}\lambda}^{(\text{op})}, A_{\mathbf{k}'\lambda'}^{(\text{op})\dagger}] = \frac{2\pi\hbar c}{k} \delta_{\mathbf{k}\mathbf{k}'} (\boldsymbol{\varepsilon}_\lambda \cdot \boldsymbol{\varepsilon}_{\lambda'}^*). \quad (13.166)$$

- (b) Express the total free electromagnetic field Hamiltonian $H_{\text{em}} = \int d^3r \frac{\boldsymbol{\varepsilon}^2 + \mathcal{B}^2}{8\pi}$ directly in terms of $A_{\mathbf{k}\lambda}^{(\text{op})}$ and $A_{\mathbf{k}\lambda}^{(\text{op})\dagger}$.

4. Relativistic field commutators and simultaneous measurability.

Let $\hat{\mathcal{B}}(\mathbf{r}, t) = \nabla \times \hat{\mathbf{A}}(\mathbf{r}, t)$ and $\hat{\mathcal{E}}(\mathbf{r}, t) = -\frac{1}{c} \frac{\partial \hat{\mathbf{A}}(\mathbf{r}, t)}{\partial t}$ denote the quantized magnetic and electric field operators.

- (a) Compute the unequal-time commutator $[\hat{\mathcal{B}}_i(\mathbf{r}, t), \hat{\mathcal{E}}_j(\mathbf{r}', t')]$.
- (b) Discuss the physical consequences of this result for the simultaneous measurability of electric and magnetic fields in space-time (the Bohr–Rosenfeld analysis).

5. Vacuum field fluctuations.

- (a) Evaluate the two-point vacuum correlation function of the electric field operator, $\langle 0 | \hat{\mathcal{E}}(\mathbf{r}) \cdot \hat{\mathcal{E}}(\mathbf{r}') | 0 \rangle$, in the absence of photons.
- (b) Define the volume-averaged electric field $\bar{\mathcal{E}}_\Omega = \frac{1}{\Omega} \int_\Omega d^3r \hat{\mathcal{E}}(\mathbf{r})$ over a spherical domain of volume Ω . Calculate the mean-square vacuum fluctuation $\langle 0 | (\bar{\mathcal{E}}_\Omega)^2 | 0 \rangle$, and examine its singular behavior as the averaging volume vanishes ($\Omega \rightarrow 0$).

6. Polarization in Thomson scattering.

A monochromatic beam of right-circularly polarized light with wavelength $\lambda = 700 \text{ nm}$ and intensity $I = 1 \text{ W/cm}^2$ is incident upon a dilute gas of $N = 10^{23}$ stationary electrons. Calculate the frequency, degree of circular polarization, and scattered intensity for left-circularly polarized light observed at scattering angles $\theta = 0^\circ, 90^\circ$, and 180° .

7. Spontaneous emission with nuclear recoil.

Consider a neutral hydrogen atom prepared at rest ($\mathbf{P} = \mathbf{0}$) in the $2P_1$ excited state inside a large box.

- (a) Formulate the electric dipole approximation taking into account the finite mass and recoil momentum of the proton.

- (b) Compute the angular distribution and polarization of the emitted photon.
 - (c) Compare the radiative lifetime of the $2P_1$ state with the idealized case of a clamped, infinitely heavy nucleus.
8. **Electronic Raman scattering in atomic hydrogen.** A hydrogen atom in its $1s$ ground state at rest is irradiated by a monochromatic laser beam with wave vector $\mathbf{k}$.
- (a) Write down the explicit Kramers–Heisenberg matrix element for an inelastic Raman transition to an excited $3d$ state ($1s \rightarrow 3d$).
 - (b) Estimate the Raman scattering cross section and contrast it with elastic Rayleigh scattering.
9. **Multipole selection rules for single-electron atoms.** For a single-electron atom with central potential $V(r)$, derive the complete parity, orbital angular momentum (Δl), and magnetic quantum number (Δm) selection rules governing:
- (a) Magnetic dipole ($M1$) transitions.
 - (b) Electric quadrupole ($E2$) transitions.
10. **X-ray scattering and the static structure factor.** In X-ray diffraction from crystals and liquids, detectors typically measure the total energy-integrated scattered intensity rather than resolving the secondary photon energy.
- (a) Show that the total differential scattering cross section integrated over scattered frequencies is given to high precision by

$$\frac{d\sigma}{d\Omega} = r_0^2 \left(\frac{1 + \cos^2 \theta}{2} \right) S(\mathbf{q}), \quad (13.167)$$

where $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_f$ is the momentum transfer, and $S(\mathbf{q}) = \sum_n |\langle n | \rho_{-\mathbf{q}} | 0 \rangle|^2$ is the static structure factor of the electron density in the ground state $|0\rangle$.

- (b) Justify why the second-order $\mathbf{j} \cdot \mathbf{A}$ interaction is negligible for X-rays ($\hbar\omega \sim 10 \text{ keV} \gg \text{Ry}$).
 - (c) Demonstrate that the elastic component ($n = 0$) in a periodic crystalline lattice produces constructive interference peaks (Laue condition / Bragg diffraction) from which the atomic arrangement can be determined.
11. **Zeeman interaction from the vector potential.** Starting from the plane-wave mode expansion Eq. (13.15) in the long-wavelength limit, or working in the symmetric gauge $\mathbf{A} = -\frac{1}{2}\mathbf{r} \times \mathbf{B}$, show that the interaction of a uniform static magnetic field $\mathbf{B}$ with a spinless charged particle reduces to the orbital Zeeman Hamiltonian:

$$H_{\text{int}} = -\mathbf{M}_{\text{orb}} \cdot \mathbf{B}, \quad \mathbf{M}_{\text{orb}} = \frac{e}{2mc} \mathbf{L}. \quad (13.168)$$

Chapter 14

Spin 12

14.1 Spin 1/2

Fundamental particles such as the electron, proton, neutron, muon, neutrino, as well as hyperons (Λ , Σ , and Ξ), possess an intrinsic angular momentum entirely independent of their spatial motion. This internal angular momentum behaves as if the particle were spinning about an internal axis, with the characteristic feature that measuring its projection along any arbitrary spatial direction $\hat{\mathbf{m}}$ yields only one of two possible values: $+\hbar/2$ or $-\hbar/2$. Such intrinsic angular momentum, corresponding to the total angular momentum quantum number $s = 1/2$, is designated as *spin* 1/2.

14.1.1 Spin Operators and State Space

The spin angular momentum is represented by a vector operator $\mathbf{S} = (S_x, S_y, S_z)$ whose components satisfy the canonical $SU(2)$ angular momentum commutation relations:

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k. \quad (14.1)$$

The experimental observation that the spin projection along an arbitrary unit vector $\hat{\mathbf{m}}$ takes only the values $\pm\hbar/2$ implies that the projection operator $\mathbf{S} \cdot \hat{\mathbf{m}}$ possesses exactly two eigenvalues, $\pm\hbar/2$. Denoting the corresponding normalized eigenstates by $|\hat{\mathbf{m}} \uparrow\rangle$ and $|\hat{\mathbf{m}} \downarrow\rangle$ (or $|\hat{\mathbf{m}}, \pm\rangle$), we have

$$\begin{aligned} \mathbf{S} \cdot \hat{\mathbf{m}}|\hat{\mathbf{m}} \uparrow\rangle &= +\frac{\hbar}{2}|\hat{\mathbf{m}} \uparrow\rangle, \\ \mathbf{S} \cdot \hat{\mathbf{m}}|\hat{\mathbf{m}} \downarrow\rangle &= -\frac{\hbar}{2}|\hat{\mathbf{m}} \downarrow\rangle. \end{aligned} \quad (14.2)$$

If a spin-1/2 particle occupies the state $|\hat{\mathbf{m}} \downarrow\rangle$, an ideal measurement of the spin component along $\hat{\mathbf{m}}$ will yield $-\hbar/2$ with certainty.

For any fixed direction $\hat{\mathbf{m}}$, the two eigenstates $\{|\hat{\mathbf{m}} \uparrow\rangle, |\hat{\mathbf{m}} \downarrow\rangle\}$ span the two-dimensional spin Hilbert space $\mathcal{H}_s \cong \mathbb{C}^2$. Any general spin state $|\Psi\rangle$ can therefore be uniquely expressed as a linear superposition:

$$|\Psi\rangle = |\hat{\mathbf{m}} \uparrow\rangle\langle\hat{\mathbf{m}} \uparrow|\Psi\rangle + |\hat{\mathbf{m}} \downarrow\rangle\langle\hat{\mathbf{m}} \downarrow|\Psi\rangle. \quad (14.3)$$

Here, the probability amplitudes $\langle\hat{\mathbf{m}} \uparrow|\Psi\rangle$ and $\langle\hat{\mathbf{m}} \downarrow|\Psi\rangle$ correspond to finding the spin oriented parallel and antiparallel to $\hat{\mathbf{m}}$, respectively. Because no other measurement outcomes exist, these two complex coefficients completely specify the internal spin state $|\Psi\rangle$.

14.1.2 The Pauli Representation

It is customary to choose the quantization axis along the z -axis, using the orthonormal eigenstates of S_z , denoted $\{|\hat{\mathbf{z}} \uparrow\rangle, |\hat{\mathbf{z}} \downarrow\rangle\}$ (or simply $\{|\uparrow\rangle, |\downarrow\rangle\}$), as the standard computational basis.

In this basis, an arbitrary state vector $|\Psi\rangle$ is represented by a two-component spinor:

$$\chi = \begin{pmatrix} \langle \hat{\mathbf{z}} \uparrow | \Psi \rangle \\ \langle \hat{\mathbf{z}} \downarrow | \Psi \rangle \end{pmatrix}. \quad (14.4)$$

The spin-up and spin-down basis states are represented by the elementary unit spinors

$$|\hat{\mathbf{z}} \uparrow\rangle \doteq \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\hat{\mathbf{z}} \downarrow\rangle \doteq \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (14.5)$$

and the projection operator S_z takes the diagonal matrix form

$$S_z \doteq \begin{pmatrix} \langle \hat{\mathbf{z}} \uparrow | S_z | \hat{\mathbf{z}} \uparrow \rangle & \langle \hat{\mathbf{z}} \uparrow | S_z | \hat{\mathbf{z}} \downarrow \rangle \\ \langle \hat{\mathbf{z}} \downarrow | S_z | \hat{\mathbf{z}} \uparrow \rangle & \langle \hat{\mathbf{z}} \downarrow | S_z | \hat{\mathbf{z}} \downarrow \rangle \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (14.6)$$

The matrix representations of the transverse components S_x and S_y are constrained by the Hermiticity of the operators and the commutation relations (14.1). They must take the general off-diagonal form

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & \delta \\ \delta^* & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i\delta \\ i\delta^* & 0 \end{pmatrix}, \quad (14.7)$$

with $|\delta| = 1$. Indeed, evaluating the matrix element of the commutator $[S_x, S_z] = -i\hbar S_y$ between the basis states yields

$$\langle \hat{\mathbf{z}} \uparrow | [S_x, S_z] | \hat{\mathbf{z}} \downarrow \rangle = -\hbar \langle \hat{\mathbf{z}} \uparrow | S_x | \hat{\mathbf{z}} \downarrow \rangle = -i\hbar \langle \hat{\mathbf{z}} \uparrow | S_y | \hat{\mathbf{z}} \downarrow \rangle, \quad (14.8)$$

which establishes that $\langle \hat{\mathbf{z}} \uparrow | S_y | \hat{\mathbf{z}} \downarrow \rangle = -i \langle \hat{\mathbf{z}} \uparrow | S_x | \hat{\mathbf{z}} \downarrow \rangle$. The phase factor δ depends entirely on the relative phase convention chosen between the two basis states $|\hat{\mathbf{z}} \uparrow\rangle$ and $|\hat{\mathbf{z}} \downarrow\rangle$. Standard convention fixes this relative phase such that $\delta = 1$. The spin vector operator is then conventionally written as

$$\mathbf{S} = \frac{\hbar}{2} \boldsymbol{\sigma}, \quad (14.9)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ denotes the standard set of *Pauli spin matrices*:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (14.10)$$

14.1.3 Algebraic Properties of the Pauli Matrices

The Pauli matrices possess several fundamental algebraic identities:

1. **Eigenvalues and Trace:** Each Pauli matrix σ_i is Hermitian, traceless ($\text{Tr } \sigma_i = 0$), and has eigenvalues ± 1 with $\det \sigma_i = -1$.
2. **Commutation relations:** Direct substitution of Eq. (14.9) into Eq. (14.1) gives

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k. \quad (14.11)$$

3. **Anticommutation relations:** For orthogonal directions ($i \neq j$), direct matrix multiplication confirms that $\sigma_x \sigma_y = i\sigma_z = -\sigma_y \sigma_x$, so that

$$\{\sigma_i, \sigma_j\} \equiv \sigma_i \sigma_j + \sigma_j \sigma_i = 0 \quad (i \neq j). \quad (14.12)$$

Furthermore, each matrix squares to the identity operator:

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (14.13)$$

Hence, the general anticommutation relation reads

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbb{I}. \quad (14.14)$$

4. **Product identity:** Summing the commutator (14.11) and anticommutator (14.14) yields the unifying Clifford algebra identity:

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{I} + i \varepsilon_{ijk} \sigma_k. \quad (14.15)$$

In terms of the spin angular momentum operators, this identity becomes

$$S_i S_j = \frac{\hbar^2}{4} \delta_{ij} + i \frac{\hbar}{2} \varepsilon_{ijk} S_k. \quad (14.16)$$

Contracting the indices gives the Casimir invariant for spin-1/2:

$$\mathbf{S}^2 = S_x^2 + S_y^2 + S_z^2 = \frac{\hbar^2}{4} (\sigma_x^2 + \sigma_y^2 + \sigma_z^2) = \frac{3}{4} \hbar^2 = \hbar^2 s(s+1), \quad \text{with } s = \frac{1}{2}. \quad (14.17)$$

14.1.4 Useful Identities

For any two vector operators $\mathbf{a}$ and $\mathbf{b}$ that commute with the Pauli spin matrices (for instance, classical vector fields or spatial operators commuting with spin), identity (14.15) provides the celebrated product formula:

$$\begin{aligned} (\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) &= a_i b_j \sigma_i \sigma_j = a_i b_j (\delta_{ij} \mathbb{I} + i \varepsilon_{ijk} \sigma_k) \\ &= (\mathbf{a} \cdot \mathbf{b}) \mathbb{I} + i (\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (14.18)$$

If $\mathbf{a}$ and $\mathbf{b}$ do not commute with each other, the order of components must be preserved, yielding

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = \mathbf{a} \cdot \mathbf{b} + i \boldsymbol{\sigma} \cdot (\mathbf{a} \times \mathbf{b}). \quad (14.19)$$

Finally, because the four matrices $\{\mathbb{I}, \sigma_x, \sigma_y, \sigma_z\}$ form an orthogonal basis for the space of 2×2 complex matrices under the Hilbert-Schmidt inner product $\langle A, B \rangle = \text{Tr}(A^\dagger B)$, any arbitrary 2×2 matrix M can be decomposed as

$$M = \alpha \mathbb{I} + \boldsymbol{\beta} \cdot \boldsymbol{\sigma}, \quad (14.20)$$

with scalar coefficients $\alpha = \frac{1}{2} \text{Tr}(M)$ and $\boldsymbol{\beta} = \frac{1}{2} \text{Tr}(M \boldsymbol{\sigma})$.

14.2 Rotations in Spin Space

To express the eigenstates of an arbitrary spin projection operator $\mathbf{S} \cdot \hat{\mathbf{n}}$ in terms of the standard S_z basis, we investigate how rotations act on the spin degrees of freedom. Consider two unit vectors $\hat{\mathbf{m}}$ and $\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is obtained by rotating $\hat{\mathbf{m}}$ through a right-handed angle $\alpha = |\boldsymbol{\alpha}|$ about the unit axis $\hat{\boldsymbol{\alpha}}$:

$$\boldsymbol{\alpha} = \alpha \hat{\boldsymbol{\alpha}}. \quad (14.21)$$

We seek the relationship between the eigenstates $\{|\hat{\mathbf{m}} \uparrow\rangle, |\hat{\mathbf{m}} \downarrow\rangle\}$ of $\mathbf{S} \cdot \hat{\mathbf{m}}$ and the eigenstates $\{|\hat{\mathbf{n}} \uparrow\rangle, |\hat{\mathbf{n}} \downarrow\rangle\}$ of $\mathbf{S} \cdot \hat{\mathbf{n}}$.

14.2.1 Transformation of Spin Operators

For an infinitesimal rotation parameterized by the vector $\boldsymbol{\alpha}$, the unit vector $\hat{\mathbf{m}}$ transforms according to

$$\hat{\mathbf{n}} = \hat{\mathbf{m}} + \boldsymbol{\alpha} \times \hat{\mathbf{m}}. \quad (14.22)$$

The spin projection operator along $\hat{\mathbf{n}}$ is therefore

$$\mathbf{S} \cdot \hat{\mathbf{n}} = \mathbf{S} \cdot \hat{\mathbf{m}} + \mathbf{S} \cdot (\boldsymbol{\alpha} \times \hat{\mathbf{m}}) = \mathbf{S} \cdot \hat{\mathbf{m}} + \varepsilon_{ijk} \alpha_i \hat{m}_j S_k. \quad (14.23)$$

Using the canonical commutation relations $[S_i, S_j] = i\hbar\varepsilon_{ijk}S_k$, we substitute $\varepsilon_{ijk}S_k = \frac{1}{i\hbar}[S_i, S_j]$ to find

$$\mathbf{S} \cdot \hat{\mathbf{n}} = \mathbf{S} \cdot \hat{\mathbf{m}} + \frac{i}{\hbar}[\mathbf{S} \cdot \hat{\mathbf{m}}, \mathbf{S} \cdot \boldsymbol{\alpha}]. \quad (14.24)$$

To first order in $\boldsymbol{\alpha}$, this commutator relation can be written compactly as a unitary transformation:

$$\mathbf{S} \cdot \hat{\mathbf{n}} = e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}(\mathbf{S} \cdot \hat{\mathbf{m}})e^{i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}. \quad (14.25)$$

By integrating the infinitesimal transformation or applying the Baker–Campbell–Hausdorff lemma, Eq. (14.25) holds identically for finite rotation angles α . Hence, the spin operator $\mathbf{S}$ generates rotations in spin space in the same way that the orbital angular momentum operator $\mathbf{L}$ generates rotations of spatial wave functions.

14.2.2 Transformation of Spin States

Applying both sides of Eq. (14.25) to the transformed ket $e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}|\hat{\mathbf{m}} \uparrow\rangle$ yields

$$(\mathbf{S} \cdot \hat{\mathbf{n}}) \left(e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}|\hat{\mathbf{m}} \uparrow\rangle \right) = e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}(\mathbf{S} \cdot \hat{\mathbf{m}})|\hat{\mathbf{m}} \uparrow\rangle = \frac{\hbar}{2} \left(e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}|\hat{\mathbf{m}} \uparrow\rangle \right). \quad (14.26)$$

Up to an overall phase convention, the eigenstates of $\mathbf{S} \cdot \hat{\mathbf{n}}$ are therefore generated by the unitary rotation operator:

$$|\hat{\mathbf{n}} \uparrow\rangle = e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}|\hat{\mathbf{m}} \uparrow\rangle, \quad (14.27)$$

$$|\hat{\mathbf{n}} \downarrow\rangle = e^{-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar}|\hat{\mathbf{m}} \downarrow\rangle. \quad (14.28)$$

The rotation connecting $\hat{\mathbf{m}}$ to $\hat{\mathbf{n}}$ is not unique; one may compose it with an arbitrary rotation about $\hat{\mathbf{n}}$, which physically corresponds to an overall gauge phase on the resulting spin kets.

14.2.3 Matrix Evaluation of the Spin Rotation Operator

In the standard S_z representation ($\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}$), the spin rotation operator takes the 2×2 matrix form

$$d(\boldsymbol{\alpha}) \equiv e^{-i\boldsymbol{\sigma} \cdot \boldsymbol{\alpha}/2}. \quad (14.29)$$

Because $(\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\alpha}})^2 = \mathbb{I}$, the power-series expansion of the exponential collapses into even and odd powers:

$$\begin{aligned} e^{-i\boldsymbol{\sigma} \cdot \boldsymbol{\alpha}/2} &= \sum_{n=0}^{\infty} \frac{(-i\boldsymbol{\alpha}/2)^n}{n!} (\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\alpha}})^n \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \left(\frac{\alpha}{2}\right)^{2k} \mathbb{I} - i(\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\alpha}}) \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \left(\frac{\alpha}{2}\right)^{2k+1} \\ &= \cos\left(\frac{\alpha}{2}\right) \mathbb{I} - i(\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\alpha}}) \sin\left(\frac{\alpha}{2}\right). \end{aligned} \quad (14.30)$$

To determine the spinor representation of an arbitrary state $|\hat{\mathbf{n}} \uparrow\rangle$ with respect to the $\hat{\mathbf{z}}$ basis, we set $\hat{\mathbf{m}} = \hat{\mathbf{z}}$ and let $\boldsymbol{\alpha}$ parameterize the rotation bringing $\hat{\mathbf{z}}$ into $\hat{\mathbf{n}}$:

$$\begin{aligned} \chi_{\hat{\mathbf{n}}\uparrow} &= e^{-i\boldsymbol{\sigma} \cdot \boldsymbol{\alpha}/2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left[\cos\left(\frac{\alpha}{2}\right) \mathbb{I} - i(\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\alpha}}) \sin\left(\frac{\alpha}{2}\right) \right] \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos\left(\frac{\alpha}{2}\right) - i\hat{\alpha}_z \sin\left(\frac{\alpha}{2}\right) \\ (-i\hat{\alpha}_x + \hat{\alpha}_y) \sin\left(\frac{\alpha}{2}\right) \end{pmatrix}. \end{aligned} \quad (14.31)$$

Conversely, for any normalized state $|\Psi\rangle \in \mathbb{C}^2$, there exists a spatial direction $\hat{\mathbf{n}}$ such that

$$\mathbf{S} \cdot \hat{\mathbf{n}}|\Psi\rangle = \frac{\hbar}{2}|\Psi\rangle. \quad (14.32)$$

Thus, every pure spin-1/2 state corresponds to a state that is 100% polarized along some unit direction in space.

14.2.4 Composition and the Double-Valued Nature of $SU(2)$

Successive rotations compose naturally. If a rotation α maps $\hat{\mathbf{m}}$ to $\hat{\mathbf{n}}$ and a subsequent rotation β maps $\hat{\mathbf{n}}$ to $\hat{\mathbf{o}}$, then

$$\boldsymbol{\sigma} \cdot \hat{\mathbf{n}} = d(\alpha)(\boldsymbol{\sigma} \cdot \hat{\mathbf{m}})d^\dagger(\alpha), \quad \boldsymbol{\sigma} \cdot \hat{\mathbf{o}} = d(\beta)(\boldsymbol{\sigma} \cdot \hat{\mathbf{n}})d^\dagger(\beta), \quad (14.33)$$

where $d^\dagger(\alpha) = d(-\alpha)$. Composing these relations gives

$$\boldsymbol{\sigma} \cdot \hat{\mathbf{o}} = [d(\beta)d(\alpha)](\boldsymbol{\sigma} \cdot \hat{\mathbf{m}})[d(\beta)d(\alpha)]^\dagger. \quad (14.34)$$

Hence, the effective composite rotation matrix $d(\gamma)$ satisfies the group homomorphism

$$d(\gamma) = d(\beta)d(\alpha). \quad (14.35)$$

A fundamental characteristic of this spin representation is that a spatial rotation of 2π ($\alpha = 2\pi$) does not return the spinor to itself:

$$d(2\pi\hat{\boldsymbol{\alpha}}) = e^{-i\pi\boldsymbol{\sigma}\cdot\hat{\boldsymbol{\alpha}}} = \cos(\pi)\mathbb{I} - i(\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\alpha}})\sin(\pi) = -\mathbb{I}. \quad (14.36)$$

A complete 360° rotation reverses the sign of any spin-1/2 state:

$$|\Psi\rangle \xrightarrow{2\pi \text{ rotation}} -|\Psi\rangle. \quad (14.37)$$

A rotation by 4π is required to return the state vector identically to $|\Psi\rangle$. Consequently, the mapping from the proper rotation group $SO(3)$ to the unitary spin transformations in $SU(2)$ is two-to-one (a double-valued representation): the elements $d(\alpha)$ and $-d(\alpha)$ correspond to the exact same physical rotation in three-dimensional space.

14.3 Including Spatial Degrees of Freedom

The spin of a quantum particle constitutes an intrinsic degree of freedom independent of its spatial motion. The total state space is formally the tensor product of the orbital (spatial) Hilbert space and the two-dimensional spin Hilbert space:

$$\mathcal{H} = \mathcal{H}_{\text{space}} \otimes \mathcal{H}_{\text{spin}} \cong L^2(\mathbb{R}^3) \otimes \mathbb{C}^2. \quad (14.38)$$

Because spatial and spin operators act on distinct Hilbert spaces, every spatial operator commutes with every spin operator:

$$[\mathbf{S}, \mathbf{r}] = 0, \quad [\mathbf{S}, \mathbf{p}] = 0, \quad [\mathbf{S}, \mathbf{L}] = 0. \quad (14.39)$$

14.3.1 Two-Component Spinor Wave Functions

To completely specify the state of a spin-1/2 particle, one must provide the spatial probability amplitude corresponding to each spin orientation. Choosing the quantization axis along $\hat{\mathbf{z}}$, the complete state $|\Psi\rangle$ is expanded in terms of the product basis $\{|\mathbf{r}, \hat{\mathbf{z}} \uparrow\rangle, |\mathbf{r}, \hat{\mathbf{z}} \downarrow\rangle\}$:

$$\psi_\uparrow(\mathbf{r}) = \langle \mathbf{r}, \hat{\mathbf{z}} \uparrow | \Psi \rangle, \quad \psi_\downarrow(\mathbf{r}) = \langle \mathbf{r}, \hat{\mathbf{z}} \downarrow | \Psi \rangle. \quad (14.40)$$

Here, $\psi_\uparrow(\mathbf{r})$ represents the probability amplitude for finding the particle at position $\mathbf{r}$ with spin projection $S_z = +\hbar/2$, while $\psi_\downarrow(\mathbf{r})$ corresponds to $S_z = -\hbar/2$.

The state is concisely represented as a two-component spinor wave function:

$$\Psi(\mathbf{r}) = \begin{pmatrix} \psi_\uparrow(\mathbf{r}) \\ \psi_\downarrow(\mathbf{r}) \end{pmatrix}, \quad (14.41)$$

where the spin polarization can generally vary as a function of position. The local probability density of finding the particle at position $\mathbf{r}$, regardless of spin orientation, is given by

$$\rho(\mathbf{r}) = \Psi^\dagger(\mathbf{r})\Psi(\mathbf{r}) = |\psi_\uparrow(\mathbf{r})|^2 + |\psi_\downarrow(\mathbf{r})|^2, \quad (14.42)$$

while the total probability of finding the particle with spin up across all space is

$$P_\uparrow = \int d^3r |\psi_\uparrow(\mathbf{r})|^2. \quad (14.43)$$

The total state normalization requires

$$\langle \Psi | \Psi \rangle = \int d^3r \Psi^\dagger(\mathbf{r})\Psi(\mathbf{r}) = \int d^3r (|\psi_\uparrow(\mathbf{r})|^2 + |\psi_\downarrow(\mathbf{r})|^2) = 1. \quad (14.44)$$

14.3.2 Total Angular Momentum and Joint Rotations

The total angular momentum operator $\mathbf{J}$ of a spin-1/2 particle is the sum of its orbital and spin angular momentum operators:

$$\mathbf{J} = \mathbf{L} + \mathbf{S}. \quad (14.45)$$

Because $[\mathbf{L}, \mathbf{S}] = 0$, both $\mathbf{L}$ and $\mathbf{S}$ separately satisfy the angular momentum commutation algebra, and their sum $\mathbf{J}$ likewise satisfies:

$$[J_i, J_j] = [L_i + S_i, L_j + S_j] = [L_i, L_j] + [S_i, S_j] = i\hbar\epsilon_{ijk}(L_k + S_k) = i\hbar\epsilon_{ijk}J_k. \quad (14.46)$$

The unitary operator $U(\boldsymbol{\alpha}) = e^{-i\boldsymbol{\alpha}\cdot\mathbf{J}/\hbar}$ generates simultaneous rotations of both spatial coordinates and internal spin orientations. Acting on a localized, spin-polarized state $|\mathbf{r}_0, \hat{\mathbf{m}} \uparrow\rangle$, the operator factorizes:

$$e^{-i\boldsymbol{\alpha}\cdot\mathbf{J}/\hbar}|\mathbf{r}_0, \hat{\mathbf{m}} \uparrow\rangle = e^{-i\boldsymbol{\alpha}\cdot\mathbf{L}/\hbar}e^{-i\boldsymbol{\alpha}\cdot\mathbf{S}/\hbar}|\mathbf{r}_0, \hat{\mathbf{m}} \uparrow\rangle. \quad (14.47)$$

The factor $e^{-i\boldsymbol{\alpha}\cdot\mathbf{S}/\hbar}$ rotates the spin direction $\hat{\mathbf{m}}$ into $\hat{\mathbf{n}} = \mathcal{R}_\alpha\hat{\mathbf{m}}$, while the spatial translation generator $e^{-i\boldsymbol{\alpha}\cdot\mathbf{L}/\hbar}$ rotates the position vector $\mathbf{r}_0$ to $\mathbf{r}'_0 = \mathcal{R}_\alpha\mathbf{r}_0$:

$$e^{-i\boldsymbol{\alpha}\cdot\mathbf{J}/\hbar}|\mathbf{r}_0, \hat{\mathbf{m}} \uparrow\rangle = |\mathbf{r}'_0, \hat{\mathbf{n}} \uparrow\rangle. \quad (14.48)$$

Consequently, for a transformed state $|\Psi'\rangle = e^{i\boldsymbol{\alpha}\cdot\mathbf{J}/\hbar}|\Psi\rangle$, the transformed spinor wave function at the original point satisfies

$$\langle \mathbf{r}_0, \hat{\mathbf{m}} \uparrow | \Psi' \rangle = \langle \mathbf{r}_0, \hat{\mathbf{m}} \uparrow | e^{i\boldsymbol{\alpha}\cdot\mathbf{J}/\hbar} | \Psi \rangle = \langle \mathbf{r}'_0, \hat{\mathbf{n}} \uparrow | \Psi \rangle, \quad (14.49)$$

which represents the original wave function evaluated at the rotated spatial position and with rotated spin polarization.

14.4 The Spin Magnetic Moment

While the existence of intrinsic spin is deduced in high-energy collisions via angular momentum conservation (for example, in hyperon decays such as $\Lambda^0 \rightarrow p + \pi^-$ where the initial state has zero orbital angular momentum in its rest frame), nonrelativistic observation relies primarily on electromagnetic interactions. In classical electrodynamics, any localized rotating charge distribution develops an intrinsic magnetic dipole moment. Similarly, quantum particles possess magnetic dipole moments associated with their intrinsic spin angular momentum.

14.4.1 Orbital and Spin Magnetic Moments

For a particle of mass m and electric charge e , the orbital magnetic dipole moment associated with its mechanical orbital angular momentum $\mathbf{L}$ is

$$\mathbf{M}_{\text{orb}} = \frac{e}{2mc} \mathbf{L}. \quad (14.50)$$

The intrinsic magnetic dipole moment associated with the spin $\mathbf{S}$ takes the analogous form

$$\mathbf{M}_{\text{spin}} = g \frac{e}{2mc} \mathbf{S}, \quad (14.51)$$

where g is the dimensionless *Landé g -factor*. For the Dirac electron, tree-level relativistic quantum mechanics predicts $g = 2$. Radiative corrections from quantum electrodynamics (QED) yield the anomalous magnetic moment:

$$\frac{g-2}{2} = \frac{\alpha}{2\pi} + \mathcal{O}(\alpha^2) \approx 0.00116, \quad (14.52)$$

where $\alpha = e^2/(\hbar c) \approx 1/137$. Because the electron carries negative charge ($e < 0$), its magnetic moment $\mathbf{M}_{\text{spin}}$ points antiparallel to its spin vector $\mathbf{S}$. The gyromagnetic ratio for electron spin is twice that of its orbital motion, which implies that the total magnetic moment,

$$\mathbf{M}_{\text{tot}} = \frac{e}{2mc} (\mathbf{L} + 2\mathbf{S}), \quad (14.53)$$

is not strictly collinear with the total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

For composite hadrons, internal quark-gluon structure produces substantial anomalous magnetic moments. For the proton ($q = +|e|$), $g_p \approx 5.586$, whereas for the uncharged neutron ($q = 0$), the magnetic moment does not vanish:

$$\mathbf{M}_n \approx -1.913 \frac{|e|}{m_n c} \mathbf{S} \approx -3.826 \frac{|e|}{2m_n c} \mathbf{S}. \quad (14.54)$$

14.4.2 The Zeeman Effect in Hydrogen

The interaction energy of a magnetic dipole moment $\mathbf{M}$ with an external magnetic field $\mathbf{B}$ (or $\mathcal{H}$ in Gaussian units) is

$$H_{\text{spin}} = -\mathbf{M}_{\text{spin}} \cdot \mathbf{B}(\mathbf{r}, t). \quad (14.55)$$

Consider a hydrogen atom placed in a uniform static magnetic field $\mathbf{B} = B\hat{z}$, described in the symmetric gauge by $\mathbf{A} = -\frac{1}{2}\mathbf{r} \times \mathbf{B}$. Neglecting proton recoil and retaining terms linear in the magnetic field, the Hamiltonian reads

$$H = \frac{1}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2 - \frac{e^2}{r} - \frac{e}{mc} \mathbf{S} \cdot \mathbf{B} = H_0 - \mathbf{M}_{\text{tot}} \cdot \mathbf{B}, \quad (14.56)$$

where $H_0 = \frac{\mathbf{p}^2}{2m} - \frac{e^2}{r}$. Aligning the magnetic field along the z -axis gives the perturbation

$$H' = -\frac{eB}{2mc} (L_z + 2S_z) = \frac{\mu_B B}{\hbar} (L_z + 2S_z), \quad (14.57)$$

where $\mu_B \equiv \frac{|e|\hbar}{2mc} \approx 9.274 \times 10^{-21}$ erg/G = 5.788×10^{-5} eV/T is the Bohr magneton.

Without spin, the orbital energy shift for hydrogenic states $|n, l, m_l\rangle$ would be

$$(\Delta E)_{\text{no spin}} = \mu_B B m_l, \quad (14.58)$$

which lifts the $(2l + 1)$ -fold spatial degeneracy symmetrically into $2l + 1$ equidistant levels (the normal Zeeman effect). With spin included, the unperturbed states $|n, l, m_l, m_s\rangle$ split into

$$\Delta E = \mu_B \mathcal{B}(m_l + 2m_s), \quad m_s = \pm \frac{1}{2}. \quad (14.59)$$

For a P -state ($l = 1$), the threefold spatial degeneracy combines with the two spin projections into six states. Because $m_l + 2m_s \in \{-2, -1, 0, +1, +2\}$ (with the middle level 0 being doubly degenerate: $m_l = 1, m_s = -1/2$ and $m_l = -1, m_s = +1/2$), the level splits into five distinct spectroscopic components.

14.4.3 Relativistic Spin-Orbit Coupling

In weak magnetic fields, Eq. (14.59) must be modified due to internal relativistic corrections. A particle moving with velocity $\mathbf{v}$ through an electrostatic field $\mathcal{E}$ experiences in its rest frame an effective motional magnetic field $\mathcal{B}_{\text{eff}} \approx -\frac{\mathbf{v}}{c} \times \mathcal{E}$ to lowest order in v/c . The coupling of this effective field to the spin magnetic moment gives

$$-\mathbf{M}_{\text{spin}} \cdot \mathcal{B}_{\text{eff}} = -\frac{|e|\hbar}{mc} \mathbf{S} \cdot \left(\mathbf{v} \times \frac{\mathcal{E}}{c} \right). \quad (14.60)$$

In addition, kinematics in an accelerating rest frame produces a relativistic kinematic precession known as the *Thomas precession*, which reduces the effective spin precession rate by exactly a factor of $1/2$. The net spin-orbit interaction energy is thus

$$H_{\text{SO}} = -\frac{|e|\hbar}{2mc^2} \mathbf{S} \cdot (\mathbf{v} \times \mathcal{E}). \quad (14.61)$$

For a spherically symmetric central potential $V(r)$, the electric field is radial, $|e|\mathcal{E} = \nabla V(r) = \frac{\mathbf{r}}{r} \frac{dV(r)}{dr}$. Using $\mathbf{r} \times \mathbf{v} = \frac{1}{m}(\mathbf{r} \times \mathbf{p}) = \frac{\mathbf{L}}{m}$, the interaction reduces to the standard spin-orbit Hamiltonian:

$$H_{\text{SO}} = \frac{1}{2m^2 c^2} \frac{1}{r} \frac{dV(r)}{dr} \mathbf{L} \cdot \mathbf{S}. \quad (14.62)$$

The simple linear Zeeman splitting (14.59) (the Paschen–Back regime) holds only when the external magnetic field energy $\mu_B \mathcal{B}$ far exceeds the spin-orbit splitting.

14.4.4 The Pauli Equation

In the nonrelativistic regime where spin-orbit coupling can be treated perturbatively or neglected, the time-dependent Schrödinger equation for the two-component spinor wave function $\Psi(\mathbf{r}, t) = \begin{pmatrix} \psi_{\uparrow}(\mathbf{r}, t) \\ \psi_{\downarrow}(\mathbf{r}, t) \end{pmatrix}$ in an applied electromagnetic field takes the form

$$i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \psi_{\uparrow}(\mathbf{r}, t) \\ \psi_{\downarrow}(\mathbf{r}, t) \end{pmatrix} = \left[\frac{1}{2m} \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right)^2 \mathbb{I} - \frac{ge\hbar}{4mc} \boldsymbol{\sigma} \cdot \mathcal{B}(\mathbf{r}, t) + V(\mathbf{r}, t) \mathbb{I} \right] \begin{pmatrix} \psi_{\uparrow}(\mathbf{r}, t) \\ \psi_{\downarrow}(\mathbf{r}, t) \end{pmatrix}. \quad (14.63)$$

Here $\mathbb{I}$ is the 2×2 identity matrix, and the magnetic coupling operator is represented explicitly by

$$\boldsymbol{\sigma} \cdot \mathcal{B} = \begin{pmatrix} \mathcal{B}_z & \mathcal{B}_x - i\mathcal{B}_y \\ \mathcal{B}_x + i\mathcal{B}_y & -\mathcal{B}_z \end{pmatrix}. \quad (14.64)$$

Equation (14.63) is the celebrated *Pauli equation*, which forms the foundational wave equation for nonrelativistic spin-1/2 electrodynamics.

14.5 Spin Precession

The interaction between an intrinsic spin magnetic moment and an external magnetic field leads to characteristic precessional dynamics. Neglecting the spatial and orbital degrees of freedom, the Hamiltonian governing a spin-1/2 particle in an applied magnetic field $\mathbf{B}(t)$ is

$$H_{\text{spin}}(t) = -\mathbf{M}_{\text{spin}} \cdot \mathbf{B}(t) = -\frac{ge}{2mc} \mathbf{S} \cdot \mathbf{B}(t) = -\frac{ge\hbar}{4mc} \boldsymbol{\sigma} \cdot \mathbf{B}(t). \quad (14.65)$$

14.5.1 Heisenberg Equation of Motion and Larmor Precession

In the Heisenberg picture, the time evolution of the spin vector operator $\mathbf{S}(t)$ is determined by the Heisenberg equation of motion:

$$\begin{aligned} \frac{dS_i(t)}{dt} &= \frac{1}{i\hbar} [S_i(t), H_{\text{spin}}(t)] = -\frac{ge}{2mci\hbar} [S_i(t), S_j(t)] \mathcal{B}_j(t) \\ &= -\frac{ge}{2mci\hbar} (i\hbar \varepsilon_{ijk} S_k(t)) \mathcal{B}_j(t) = \frac{ge}{2mc} \varepsilon_{ijk} S_j(t) \mathcal{B}_k(t). \end{aligned} \quad (14.66)$$

In vector notation, this takes the transparent torque form:

$$\frac{d\mathbf{S}(t)}{dt} = \frac{ge}{2mc} \mathbf{S}(t) \times \mathbf{B}(t) = \mathbf{M}_{\text{spin}}(t) \times \mathbf{B}(t). \quad (14.67)$$

Equation (14.67) is the quantum operator analog of the classical torque equation $\boldsymbol{\tau} = \mathbf{M} \times \mathbf{B} = \frac{d\mathbf{S}}{dt}$. For an electron ($e < 0$), the torque causes the spin vector to precess in a right-handed sense about the magnetic field direction.

For a static, uniform magnetic field aligned with the z -axis, $\mathbf{B} = \mathcal{B}_0 \hat{\mathbf{z}}$, the components decouple:

$$\frac{dS_z(t)}{dt} = 0, \quad \frac{dS_x(t)}{dt} = \omega_0 S_y(t), \quad \frac{dS_y(t)}{dt} = -\omega_0 S_x(t), \quad (14.68)$$

where the *Larmor precession frequency* ω_0 is defined by

$$\omega_0 \equiv \frac{ge\mathcal{B}_0}{2mc}. \quad (14.69)$$

For an electron ($g \approx 2$, $e = -|e|$), $\omega_0 = -|e|\mathcal{B}_0/(mc) < 0$, giving an electron spin resonance (ESR) frequency $|\omega_0|/(2\pi\mathcal{B}_0) \approx 2.80 \text{ MHz/G}$.

Integrating the linear system (14.68) yields

$$\begin{aligned} S_x(t) &= S_x(0) \cos(\omega_0 t) + S_y(0) \sin(\omega_0 t), \\ S_y(t) &= -S_x(0) \sin(\omega_0 t) + S_y(0) \cos(\omega_0 t), \\ S_z(t) &= S_z(0). \end{aligned} \quad (14.70)$$

For example, if the initial state at $t = 0$ is an eigenstate of S_x with eigenvalue $+\hbar/2$ ($|\Psi(0)\rangle = |\hat{\mathbf{x}} \uparrow\rangle$), the initial expectation values are $\langle S_x(0) \rangle = \hbar/2$ and $\langle S_y(0) \rangle = \langle S_z(0) \rangle = 0$. Taking the expectation value of Eq. (14.70) yields

$$\langle S_x(t) \rangle = \frac{\hbar}{2} \cos(\omega_0 t), \quad \langle S_y(t) \rangle = -\frac{\hbar}{2} \sin(\omega_0 t), \quad \langle S_z(t) \rangle = 0. \quad (14.71)$$

The polarization vector rotates steadily in the xy -plane at the Larmor frequency ω_0 .

14.5.2 Schrödinger Picture Description

In the Schrödinger picture, the spin state evolves according to

$$|\Psi(t)\rangle = e^{-iH_{\text{spin}}t/\hbar} |\Psi(0)\rangle = \exp\left(\frac{i\omega_0 t S_z}{\hbar}\right) |\Psi(0)\rangle. \quad (14.72)$$

Comparing Eq. (14.72) with the spin rotation operator $U(\boldsymbol{\alpha}) = \exp(-i\mathbf{S} \cdot \boldsymbol{\alpha}/\hbar)$ from Eq. (14.27), the time-evolution operator represents an active rotation about the $-\hat{\mathbf{z}}$ axis by an angle $\alpha(t) = \omega_0 t$, in complete agreement with the Heisenberg picture result.

14.5.3 Experimental Application: The Muon ($g - 2$) Measurement

Larmor precession provides an exquisitely precise method to measure particle gyromagnetic factors, notably in muon ($g - 2$) experiments. Muons decay via the weak interaction:

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu. \quad (14.73)$$

Because the weak interaction maximally violates parity, the emission direction of the decay electron in the muon rest frame is strongly correlated with the muon spin vector $\mathbf{S}_\mu$, peaking preferentially opposite to the spin direction.

In a storage ring with a transverse magnetic field $\mathcal{B}$, a polarized muon beam orbits at the cyclotron frequency $\omega_c = |e|\mathcal{B}/(\gamma m_\mu c)$. Simultaneously, the muon spin precesses at the relativistic Larmor frequency ω_s . The difference between the spin precession frequency and the orbital cyclotron frequency is directly proportional to the anomalous magnetic moment $a_\mu \equiv (g - 2)/2$:

$$\omega_a \equiv \omega_s - \omega_c = a_\mu \frac{|e|\mathcal{B}}{m_\mu c}. \quad (14.74)$$

Measuring the modulation of decay electrons detected along the beam orbit determines a_μ to high precision, providing one of the most stringent tests of the Standard Model of particle physics.

14.6 Magnetic Resonance and Ramsey Interferometry

In a uniform static magnetic field $\mathcal{B}_0 = \mathcal{B}_0 \hat{\mathbf{z}}$, a spin-1/2 magnetic moment precesses uniformly at the Larmor frequency $\omega_0 = \frac{ge\mathcal{B}_0}{2mc}$. Observed from a non-inertial reference frame rotating about the z -axis at angular frequency $-\omega$, the apparent precession slows. Formally, transforming to the rotating frame generates a fictitious inertial term that augments the magnetic field, producing an effective longitudinal field:

$$\mathcal{B}_{\text{eff}} = \mathcal{B}_0 - \frac{2mc\omega}{ge}. \quad (14.75)$$

At resonance, $\omega = \omega_0$, the effective longitudinal field vanishes identically. If a weak transverse field oscillating in resonance is applied, the spin precesses exclusively around this transverse field in the rotating frame. Even a vanishingly small perturbation can therefore completely invert the spin orientation (from spin-up to spin-down).

14.6.1 Rotating Wave Approximation (RWA) and Rabi Oscillations

Experimentally, an oscillating radio-frequency (RF) magnetic field is applied linearly along the x -axis:

$$\mathcal{B}_x(t) = \mathcal{B}_1 \cos(\omega t), \quad \mathcal{B}_y = 0. \quad (14.76)$$

This linearly polarized field can be decomposed into two counter-rotating circularly polarized components:

$$\mathcal{B}_{\text{rf}}(t) = \frac{\mathcal{B}_1}{2} \begin{pmatrix} \cos(\omega t) \\ -\sin(\omega t) \\ 0 \end{pmatrix} + \frac{\mathcal{B}_1}{2} \begin{pmatrix} \cos(\omega t) \\ \sin(\omega t) \\ 0 \end{pmatrix}. \quad (14.77)$$

The first component rotates in the same sense as the Larmor precession and remains quasi-stationary in the rotating frame. The second component rotates at frequency $+2\omega$ relative to the precessing spin. For weak drive fields ($\mathcal{B}_1 \ll \mathcal{B}_0$) near resonance ($\omega \approx \omega_0$), this counter-rotating term produces negligible high-frequency micromotion (Bloch–Siegert shift) and is neglected in the *Rotating Wave Approximation* (RWA).

The Schrödinger equation governing the spinor state $|\Psi(t)\rangle$ is

$$i\hbar \frac{d|\Psi(t)\rangle}{dt} = -\frac{ge\hbar}{4mc} [\mathcal{B}_0 \sigma_z + \mathcal{B}_1 \cos(\omega t) \sigma_x] |\Psi(t)\rangle. \quad (14.78)$$

To analyze the dynamics, we transform to the rotating frame via the unitary transformation

$$|\Psi(t)\rangle = e^{i\omega t \sigma_z/2} |\Psi'(t)\rangle. \quad (14.79)$$

Differentiating $|\Psi(t)\rangle$ and projecting with $e^{-i\omega t \sigma_z/2}$, we obtain

$$i \frac{d|\Psi'(t)\rangle}{dt} = \left[\frac{\omega - \omega_0}{2} \sigma_z - \omega_1 \cos(\omega t) e^{-i\omega t \sigma_z/2} \sigma_x e^{i\omega t \sigma_z/2} \right] |\Psi'(t)\rangle, \quad (14.80)$$

where $\omega_1 \equiv \frac{ge\mathcal{B}_1}{4mc}$. Expanding the operator product gives

$$\cos(\omega t) \left(e^{-i\omega t \sigma_z/2} \sigma_x e^{i\omega t \sigma_z/2} \right) = \frac{\sigma_x}{2} + \frac{1}{2} [\sigma_x \cos(2\omega t) + \sigma_y \sin(2\omega t)]. \quad (14.81)$$

Averaging over the fast 2ω oscillations (the RWA), the effective Hamiltonian in the rotating frame becomes strictly time-independent:

$$i \frac{d|\Psi'(t)\rangle}{dt} = \left[\frac{\Delta}{2} \sigma_z - \frac{\omega_1}{2} \sigma_x \right] |\Psi'(t)\rangle = -\frac{\Omega}{2} (\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}) |\Psi'(t)\rangle, \quad (14.82)$$

where $\Delta \equiv \omega - \omega_0$ is the detuning,

$$\Omega \equiv \sqrt{(\omega - \omega_0)^2 + \omega_1^2} \quad (14.83)$$

is the generalized *Rabi frequency*, and the effective rotation axis is defined by the unit vector

$$\hat{\mathbf{n}} = \frac{\omega_1}{\Omega} \hat{\mathbf{x}} - \frac{\omega - \omega_0}{\Omega} \hat{\mathbf{z}}. \quad (14.84)$$

The exact solution in the laboratory frame is given by

$$|\Psi(t)\rangle = e^{i\omega t \sigma_z/2} \left[\cos\left(\frac{\Omega t}{2}\right) \mathbb{I} - i(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}) \sin\left(\frac{\Omega t}{2}\right) \right] |\Psi(0)\rangle. \quad (14.85)$$

For a particle initially prepared in the spin-up state $|\Psi(0)\rangle = |\hat{\mathbf{z}} \uparrow\rangle$, the transition amplitude to the spin-down state $|\hat{\mathbf{z}} \downarrow\rangle$ is

$$\langle \hat{\mathbf{z}} \downarrow | \Psi(t) \rangle = i e^{-i\omega t/2} \frac{\omega_1}{\Omega} \sin\left(\frac{\Omega t}{2}\right). \quad (14.86)$$

The probability of observing a spin flip at time t is governed by the celebrated *Rabi formula*:

$$P_{\downarrow}(t) = |\langle \hat{\mathbf{z}} \downarrow | \Psi(t) \rangle|^2 = \frac{\omega_1^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right) = \frac{\omega_1^2}{(\omega - \omega_0)^2 + \omega_1^2} \sin^2\left(\frac{\Omega t}{2}\right). \quad (14.87)$$

Exactly on resonance ($\omega = \omega_0$), the detuning vanishes, so $\Omega = \omega_1$. The transition probability oscillates between 0 and 1:

$$P_{\downarrow}(t) = \sin^2\left(\frac{\omega_1 t}{2}\right). \quad (14.88)$$

A pulse of duration $t = \pi/\omega_1$ (a “ π -pulse”) transfers the population with unit probability ($P_{\downarrow} = 1$), corresponding to an energy absorption of $\Delta E = \hbar\omega_0$ from the driving field. A pulse of duration $t = \pi/(2\omega_1)$ (a “ $\pi/2$ -pulse”) creates an equal coherent superposition of spin-up and spin-down.

14.6.2 Separated Oscillatory Fields: The Ramsey Method

In high-precision spectroscopy and atomic frequency standards (such as cesium atomic clocks), Norman Ramsey introduced the method of separated oscillatory fields to suppress transit-time broadening.

A collimated beam of atoms prepared in the state $|\hat{\mathbf{z}} \uparrow\rangle$ passes through two identical RF interaction zones (cavities) separated by an intermediate drift region of length L free of RF fields. A uniform longitudinal field $\mathcal{B}_0 \hat{\mathbf{z}}$ permeates the entire trajectory. Let τ be the transit time across each cavity and T be the flight time through the field-free drift region ($T \gg \tau$).

The total time evolution can be computed perturbatively in the interaction representation:

$$|\Psi(t)\rangle = e^{i\omega_0 t \sigma_z/2} |\Psi'(t)\rangle. \quad (14.89)$$

The interaction Hamiltonian is $H'(t) = -\frac{e\hbar}{2mc} \mathcal{B}'(t) \sigma_x$, where $\mathcal{B}'(t) = \mathcal{B}_1 \cos(\omega t)$ within the intervals $[0, \tau]$ and $[T, T + \tau]$, and zero elsewhere. To first order in perturbation theory, the spin-flip amplitude at the exit of the second cavity ($t = T + \tau$) is

$$\langle \hat{\mathbf{z}} \downarrow | \Psi(T + \tau) \rangle = \frac{ie}{2mc} \int_0^{T+\tau} dt \mathcal{B}'(t) e^{i\omega_0 t}. \quad (14.90)$$

Evaluating the two separated integral segments and neglecting the counter-rotating terms yields

$$\langle \hat{\mathbf{z}} \downarrow | \Psi(T + \tau) \rangle = \frac{e\mathcal{B}_1}{4mc} \left[\frac{e^{i(\omega_0 - \omega)\tau} - 1}{\omega_0 - \omega} \right] \left(1 + e^{i(\omega_0 - \omega)T} \right). \quad (14.91)$$

The term 1 corresponds to the transition amplitude in the first cavity, while $e^{i(\omega_0 - \omega)T}$ represents the amplitude from the second cavity. The phase shift $(\omega_0 - \omega)T$ accumulated during the drift interval gives rise to quantum interference.

Taking the absolute square of Eq. (14.91), the transition probability becomes

$$P_{\downarrow}(\omega) = \left(\frac{e\mathcal{B}_1}{2mc} \right)^2 \left[\frac{\sin\left(\frac{(\omega_0 - \omega)\tau}{2}\right)}{\frac{\omega_0 - \omega}{2}} \right]^2 \cos^2\left(\frac{(\omega_0 - \omega)T}{2}\right). \quad (14.92)$$

Equation (14.92) exhibits the mathematical structure of a two-slit Young interference pattern:

1. The broad envelope $\left[\frac{\sin((\omega_0 - \omega)\tau/2)}{(\omega_0 - \omega)/2} \right]^2$ represents the single-cavity Fourier diffraction envelope, with linewidth $\Delta\omega_{\text{cavity}} \sim 1/\tau$.
2. The rapidly oscillating factor $\cos^2((\omega_0 - \omega)T/2)$ produces sharp *Ramsey fringes* with a central resonance fringe width $\Delta\omega_{\text{Ramsey}} \sim 1/T \ll 1/\tau$.

By locking the local oscillator frequency ω to the extremely narrow central Ramsey fringe, atomic clocks achieve frequency stabilities surpassing one part in 10^{15} , serving as the primary standard for the International System of Units (SI) second.

14.7 Motion in Inhomogeneous Magnetic Fields and the Stern–Gerlach Effect

When a spinning particle moves through a spatially inhomogeneous magnetic field $\mathcal{B}(\mathbf{r})$, the position-dependent interaction Hamiltonian

$$H_{\text{spin}}(\mathbf{r}) = -\mathbf{M}_{\text{spin}} \cdot \mathcal{B}(\mathbf{r}) \quad (14.93)$$

exerts a translational force on the center-of-mass motion:

$$\mathbf{F} = -\nabla H_{\text{spin}} = \nabla [\mathbf{M}_{\text{spin}} \cdot \mathcal{B}(\mathbf{r})]. \quad (14.94)$$

Depending on whether the spin magnetic moment is aligned parallel or antiparallel to the local field gradient, the particle is accelerated toward regions of stronger or weaker field.

14.7.1 The Stern–Gerlach Experiment

In the historic 1922 experiment of Otto Stern and Walther Gerlach, a collimated beam of neutral silver atoms propagated along the y -axis through a magnetic field possessing a strong transverse gradient along the z -axis ($\partial\mathcal{B}_z/\partial z \neq 0$). A neutral silver atom ($Z = 47$) has 46 inner electrons in closed shells with net angular momentum $J = 0$, while the single valence electron occupies a spherically symmetric $5s$ state ($L = 0, S = 1/2$). The magnetic dipole moment of the atom is therefore purely that of the unpaired electron spin.

Classically, magnetic dipoles with randomly distributed orientations would experience a continuous spectrum of forces $F_z = M_z(\partial\mathcal{B}_z/\partial z)$, resulting in a continuous vertical smudge on the detector plate. Instead, Stern and Gerlach observed that the beam bifurcated cleanly into two distinct, discrete components:

1. Atoms with spin projection $S_z = +\hbar/2$ (magnetic moment antiparallel to $\mathcal{B}$) are deflected toward regions of weaker field.
2. Atoms with spin projection $S_z = -\hbar/2$ (magnetic moment parallel to $\mathcal{B}$) are deflected toward regions of stronger field.

For an orbital angular momentum state l , spatial quantization predicts an odd number ($2l + 1$) of traces. The appearance of an *even* number of traces ($2s + 1 = 2$) provided unequivocal experimental evidence for half-integer intrinsic spin.

14.7.2 Entanglement of Spatial and Spin Degrees of Freedom

Consider an incident beam prepared in an eigenstate of S_x with eigenvalue $+\hbar/2$, described by the initial state

$$|\Psi(0)\rangle = |\psi_0\rangle \otimes |\hat{\mathbf{x}} \uparrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_0(\mathbf{r}, 0) \\ \psi_0(\mathbf{r}, 0) \end{pmatrix}. \quad (14.95)$$

Inside the inhomogeneous field, the force couples the spatial evolution to the internal spin state. After traversing the magnet for duration T , the wave function evolves into an entangled state:

$$\Psi(\mathbf{r}, T) = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_\uparrow(\mathbf{r}, T) \\ \psi_\downarrow(\mathbf{r}, T) \end{pmatrix}. \quad (14.96)$$

When the spatial separation between the up and down wave packets exceeds their spatial widths, their spatial overlap vanishes:

$$\int d^3r \psi_\uparrow^*(\mathbf{r}, T) \psi_\downarrow(\mathbf{r}, T) \approx 0. \quad (14.97)$$

The spatial spin density is given by

$$\mathbf{s}(\mathbf{r}, T) = \Psi^\dagger(\mathbf{r}, T) \frac{\hbar\boldsymbol{\sigma}}{2} \Psi(\mathbf{r}, T) = \frac{\hbar}{4} \hat{\mathbf{z}} [|\psi_\uparrow(\mathbf{r}, T)|^2 - |\psi_\downarrow(\mathbf{r}, T)|^2]. \quad (14.98)$$

Integrating over the entire beam volume gives zero net polarization:

$$\langle \mathbf{S} \rangle_T = \int d^3r \mathbf{s}(\mathbf{r}, T) = \mathbf{0}, \quad (14.99)$$

even though initially $\langle S_x(0) \rangle = \hbar/2$.

This apparent loss of macroscopic transverse polarization arises from spatial dephasing. Because the local field $\mathcal{B}_z(\mathbf{r})$ varies across the transverse profile of the beam, particles traversing different paths precess at slightly different Larmor frequencies:

$$\Delta\omega_0 = \left| \frac{ge}{2mc} \right| \Delta\mathcal{B}_z \approx \left| \frac{ge}{2mc} \right| \left| \frac{\partial\mathcal{B}_z}{\partial z} \right| \Delta z, \quad (14.100)$$

where Δz is the beam aperture. For the transverse coherences to average to zero, the dephasing across the aperture must satisfy

$$(\Delta\omega_0)T \geq 2\pi. \quad (14.101)$$

The net vertical momentum imparted by the magnetic force $F_z = \pm \frac{\hbar}{2} \left| \frac{ge}{2mc} \frac{\partial \mathcal{B}_z}{\partial z} \right|$ is $p_z \approx F_z T$, resulting in an angular beam deflection

$$\varphi \approx \frac{p_z}{p_y} \approx \left| \frac{ge}{2mc} \right| \frac{\hbar T}{2p_y} \left| \frac{\partial \mathcal{B}_z}{\partial z} \right|. \quad (14.102)$$

According to de Broglie diffraction, the intrinsic angular diffraction spread of the wave packet is

$$\delta\varphi \approx \frac{\lambda_{dB}}{\Delta z} = \frac{2\pi\hbar}{p_y \Delta z}. \quad (14.103)$$

The condition for clean spatial beam resolution, $\delta\varphi \leq 2\varphi$, translates to

$$\Delta z \left| \frac{\partial}{\partial z} (E_{\uparrow} - E_{\downarrow}) \right| T \geq 2\pi\hbar, \quad (14.104)$$

which is an explicit manifestation of the quantum uncertainty principle: a measurement that resolves two distinct internal energy levels within transit time T requires an energy gradient uncertainty satisfying $\delta(\Delta E)T \geq 2\pi\hbar$.

14.7.3 Impossibility of a Stern–Gerlach Experiment for Free Electrons

A natural question—first raised by Niels Bohr and Wolfgang Pauli—is whether the Stern–Gerlach technique can be applied directly to a beam of free electrons. For free electrons, Maxwell’s divergence equation,

$$\nabla \cdot \mathbf{B} = \frac{\partial \mathcal{B}_x}{\partial x} + \frac{\partial \mathcal{B}_y}{\partial y} + \frac{\partial \mathcal{B}_z}{\partial z} = 0, \quad (14.105)$$

demands that a non-vanishing gradient $\partial \mathcal{B}_z / \partial z$ is accompanied by transverse gradients, e.g., $\partial \mathcal{B}_x / \partial x = -\partial \mathcal{B}_z / \partial z$ (assuming $\partial \mathcal{B}_y / \partial y = 0$).

An electron moving along the y -axis with velocity v_y experiences a magnetic Lorentz force in the z -direction:

$$F_L = -\frac{e}{c} v_y \mathcal{B}_x. \quad (14.106)$$

Because the transverse wave packet has finite width Δx , the field $\mathcal{B}_x$ varies by $\Delta \mathcal{B}_x \approx \left| \frac{\partial \mathcal{B}_x}{\partial x} \right| \Delta x = \left| \frac{\partial \mathcal{B}_z}{\partial z} \right| \Delta x$. This variation produces an angular divergence of the beam:

$$\delta\theta_L \approx \frac{\Delta p_{z,L}}{p_y} = \frac{eT}{mc} \left| \frac{\partial \mathcal{B}_z}{\partial z} \right| \Delta x. \quad (14.107)$$

Applying the Heisenberg uncertainty relation $\Delta x \geq \hbar / \Delta p_x$ and comparing $\delta\theta_L$ with the spin deflection angle 2φ (with $g = 2$):

$$\delta\theta_L \geq \left(\frac{eT}{mc} \left| \frac{\partial \mathcal{B}_z}{\partial z} \right| \right) \frac{\hbar}{\Delta p_x} \approx 2\varphi \left(\frac{p_y}{\Delta p_x} \right). \quad (14.108)$$

For a well-collimated beam, the longitudinal momentum must vastly exceed the transverse momentum spread, $p_y \gg \Delta p_x$. Hence,

$$\delta\theta_L \gg 2\varphi. \quad (14.109)$$

The beam broadening induced by the Lorentz force inevitably washes out the spin splitting. For neutral atoms or heavy ions, this limitation is avoided because either the net charge vanishes ($q = 0$) or the large atomic mass ($M \gg m_e$) suppresses the Lorentz deflection while retaining the electronic magnetic moment.

14.8 Problems

1. Eigenstates and Rotations in Spin-1/2 Spaces:

- (a) Determine the normalized eigenvectors and corresponding eigenvalues of the Pauli operators σ_x , σ_y , and σ_z in the standard representation where σ_z is diagonal. Repeat the construction in the representation where σ_x is diagonal.
- (b) Using the spinor representation of rotations $d(\boldsymbol{\alpha}) = \exp(-i\boldsymbol{\sigma} \cdot \boldsymbol{\alpha}/2)$, derive an explicit vector expression for the unit vector $\hat{\mathbf{n}}$ obtained by rotating the unit vector $\hat{\mathbf{m}}$ through a right-handed angle $\alpha = |\boldsymbol{\alpha}|$ about the direction $\hat{\boldsymbol{\alpha}}$.

2. Spin Resonance Dynamics and Pulses:

- (a) In a magnetic resonance configuration with static field $\mathcal{B}_0 \hat{\mathbf{z}}$ and transverse rotating field $\mathcal{B}_1(t) = \mathcal{B}_1(\cos \omega t \hat{\mathbf{x}} - \sin \omega t \hat{\mathbf{y}})$, calculate the time-dependent Heisenberg operator $\mathbf{S}(t) = (S_x(t), S_y(t), S_z(t))$ explicitly.
- (b) For a spin initially prepared along the $+z$ -direction ($|\Psi(0)\rangle = |\hat{\mathbf{z}} \uparrow\rangle$), compute the probabilities $P_{+x}(t)$ and $P_{+y}(t)$ that a measurement at time t yields spin projected along $+\hat{\mathbf{x}}$ and $+\hat{\mathbf{y}}$, respectively.
- (c) Analyze the physical effect of a resonant 90° ($\pi/2$) pulse in both the Schrödinger picture (state evolution) and the Heisenberg picture (operator evolution).

3. Two-Level Systems and Non-Linear Optical Dipole Response:

The algebraic framework of spin-1/2 provides a universal description for two-level quantum systems (qubits). Consider an atom restricted to two energy eigenstates $|+\rangle$ and $|-\rangle$ with unperturbed eigenenergies ε_+ and ε_- ($\varepsilon_+ > \varepsilon_-$).

- (a) Express the unperturbed atomic Hamiltonian H_0 in terms of the identity matrix and Pauli pseudo-spin matrices.
- (b) Suppose the permanent electric dipole moments vanish, $\langle +|ez|+\rangle = \langle -|ez|-\rangle = 0$, while the transition dipole matrix element is real: $\langle +|ez|-\rangle = \mu$. Write down the total Hamiltonian when an oscillating electric field $\mathbf{E}(t) = \mathcal{E}_0 \cos(\omega t) \hat{\mathbf{z}}$ is applied.
- (c) Utilizing the rotating wave approximation (neglecting counter-rotating driving terms), solve for the time-dependent induced electric dipole polarization $\langle P_z(t) \rangle = \langle \Psi(t) | ez | \Psi(t) \rangle$ assuming the atom begins in the ground state $|-\rangle$. Determine the non-linear response as a function of field amplitude $\mathcal{E}_0$, detuning $\omega - \omega_0$, and time t , and extract the linear susceptibility limit to first order in $\mathcal{E}_0$.

Chapter 15

Addition Of Angular Momenta

15.1 Addition of Angular Momenta: The Two Spin-1/2 System

In classical mechanics, the total angular momentum of a composite system comprising two subsystems with angular momenta $\mathbf{J}_1$ and $\mathbf{J}_2$ is simply given by the vector sum $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$. In quantum mechanics, however, the subsystems are prepared in states labeled by their individual angular momentum quantum numbers (j_1, m_1) and (j_2, m_2) . Because angular momentum components along different spatial axes do not mutually commute, constructing the eigenstates of the total angular momentum $\mathbf{J}^2$ and J_z requires a change of basis known as the addition of angular momenta.

To introduce the foundational concepts, we first examine the simplest non-trivial composite system: two distinguishable spin-1/2 particles (such as the proton and neutron forming a deuteron, or the two electrons in an atom).

15.1.1 The Uncoupled Representation

Let $\mathbf{S}_1$ and $\mathbf{S}_2$ denote the spin operators of particle 1 and particle 2, acting on their respective two-dimensional Hilbert spaces $\mathcal{H}_1 \cong \mathbb{C}^2$ and $\mathcal{H}_2 \cong \mathbb{C}^2$. Because the operators act on independent particle degrees of freedom, their components commute:

$$[\mathbf{S}_1, \mathbf{S}_2] = 0. \quad (15.1)$$

The composite Hilbert space $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ is four-dimensional. Choosing the quantization axis along $\hat{\mathbf{z}}$, the *uncoupled basis* consists of the four simultaneous eigenstates of $\{S_1^2, S_{1z}, S_2^2, S_{2z}\}$:

$$|\uparrow\uparrow\rangle \equiv |++\rangle, \quad |\uparrow\downarrow\rangle \equiv |+-\rangle, \quad |\downarrow\uparrow\rangle \equiv |-+\rangle, \quad |\downarrow\downarrow\rangle \equiv |--\rangle, \quad (15.2)$$

where the first index corresponds to the eigenvalue $m_1 = \pm 1/2$ and the second to $m_2 = \pm 1/2$.

15.1.2 The Coupled Representation: Triplet and Singlet States

The total spin operator of the combined system is defined by

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2. \quad (15.3)$$

From $[S_{1i}, S_{1j}] = i\hbar\epsilon_{ijk}S_{1k}$ and $[S_{2i}, S_{2j}] = i\hbar\epsilon_{ijk}S_{2k}$, together with Eq. (15.1), the total spin satisfies the standard angular momentum Lie algebra:

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k. \quad (15.4)$$

(In what follows, we set $\hbar = 1$ for algebraic conciseness; $\hbar$ factors may be restored by dimensional analysis).

The operators $S^2 = \mathbf{S} \cdot \mathbf{S}$ and $S_z = S_{1z} + S_{2z}$ commute with each other and with S_1^2 and S_2^2 . We therefore seek the orthonormal *coupled basis* $|s, m\rangle$ satisfying

$$\begin{aligned} S^2|s, m\rangle &= s(s+1)|s, m\rangle, \\ S_z|s, m\rangle &= m|s, m\rangle. \end{aligned} \quad (15.5)$$

Evaluating the action of S_z on the uncoupled states gives

$$\begin{aligned} S_z|\uparrow\uparrow\rangle &= \left(\frac{1}{2} + \frac{1}{2}\right)|\uparrow\uparrow\rangle = +1|\uparrow\uparrow\rangle, \\ S_z|\uparrow\downarrow\rangle &= \left(\frac{1}{2} - \frac{1}{2}\right)|\uparrow\downarrow\rangle = 0, \\ S_z|\downarrow\uparrow\rangle &= \left(-\frac{1}{2} + \frac{1}{2}\right)|\downarrow\uparrow\rangle = 0, \\ S_z|\downarrow\downarrow\rangle &= \left(-\frac{1}{2} - \frac{1}{2}\right)|\downarrow\downarrow\rangle = -1|\downarrow\downarrow\rangle. \end{aligned} \quad (15.6)$$

The total magnetic quantum number m takes values $+1, 0, -1$, implying that the four-dimensional space decomposes into a triplet representation ($s = 1$, three states) and a singlet representation ($s = 0$, one state):

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{3} \oplus \mathbf{1}. \quad (15.7)$$

The state with maximum projection, $m = +1$, must belong to the $s = 1$ multiplet:

$$|s = 1, m = 1\rangle = |\uparrow\uparrow\rangle. \quad (15.8)$$

To confirm this, we express S^2 in terms of ladder operators $S_{\pm} = S_x \pm iS_y$:

$$S^2 = S_1^2 + S_2^2 + 2\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{3}{2} + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+}, \quad (15.9)$$

where $S_1^2 = S_2^2 = \frac{3}{4}$. Because $S_{1+}|\uparrow\rangle = S_{2+}|\uparrow\rangle = 0$, we find $S^2|\uparrow\uparrow\rangle = \left(\frac{3}{2} + 2 \cdot \frac{1}{4}\right)|\uparrow\uparrow\rangle = 2|\uparrow\uparrow\rangle = 1(1+1)|\uparrow\uparrow\rangle$.

Applying the total lowering operator $S_- = S_{1-} + S_{2-}$ using the general matrix element identity

$$S_-|s, m\rangle = \sqrt{s(s+1) - m(m-1)}|s, m-1\rangle, \quad (15.10)$$

we obtain

$$\begin{aligned} |s = 1, m = 0\rangle &= \frac{1}{\sqrt{2}}S_-|1, 1\rangle = \frac{1}{\sqrt{2}}(S_{1-} + S_{2-})|\uparrow\uparrow\rangle \\ &= \frac{1}{\sqrt{2}}(|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle). \end{aligned} \quad (15.11)$$

Lowering once more yields the state with $m = -1$:

$$|s = 1, m = -1\rangle = \frac{1}{\sqrt{2}}S_-|1, 0\rangle = |\downarrow\downarrow\rangle. \quad (15.12)$$

The $s = 0, m = 0$ singlet state must lie in the subspace spanned by $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$ and be orthogonal to $|1, 0\rangle$:

$$|s = 0, m = 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (15.13)$$

Inverting these relations expresses the uncoupled product states as linear superpositions of the total spin eigenstates:

$$\begin{aligned}
 |\uparrow\uparrow\rangle &= |1, 1\rangle, \\
 |\uparrow\downarrow\rangle &= \frac{1}{\sqrt{2}}(|1, 0\rangle + |0, 0\rangle), \\
 |\downarrow\uparrow\rangle &= \frac{1}{\sqrt{2}}(|1, 0\rangle - |0, 0\rangle), \\
 |\downarrow\downarrow\rangle &= |1, -1\rangle.
 \end{aligned} \tag{15.14}$$

15.1.3 Spin-Spin Coupling and Projection Operators

The scalar product $\mathbf{S}_1 \cdot \mathbf{S}_2$ arises frequently in microscopic interactions (e.g., hyperfine structure, ferromagnetic exchange, nuclear forces). Rewriting the identity

$$\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2} (S^2 - S_1^2 - S_2^2) = \frac{1}{2} S^2 - \frac{3}{4} \mathbb{I}, \tag{15.15}$$

we find that the coupled states $|s, m\rangle$ diagonalize $\mathbf{S}_1 \cdot \mathbf{S}_2$:

$$\mathbf{S}_1 \cdot \mathbf{S}_2 |s = 1, m\rangle = \left(\frac{2}{2} - \frac{3}{4} \right) |1, m\rangle = +\frac{1}{4} |1, m\rangle, \tag{15.16}$$

$$\mathbf{S}_1 \cdot \mathbf{S}_2 |s = 0, m = 0\rangle = \left(0 - \frac{3}{4} \right) |0, 0\rangle = -\frac{3}{4} |0, 0\rangle. \tag{15.17}$$

For an exchange potential $V = J \mathbf{S}_1 \cdot \mathbf{S}_2$, the energy difference between the triplet and singlet configurations is $\Delta E = E_{\text{triplet}} - E_{\text{singlet}} = J$.

Using the spectral decomposition theorem, we can construct the projection operators onto the triplet and singlet subspaces:

$$P_1 = \frac{3}{4} \mathbb{I} + \mathbf{S}_1 \cdot \mathbf{S}_2, \tag{15.18}$$

$$P_0 = \frac{1}{4} \mathbb{I} - \mathbf{S}_1 \cdot \mathbf{S}_2. \tag{15.19}$$

These operators satisfy the standard projector properties: $P_1 + P_0 = \mathbb{I}$, $P_i P_j = \delta_{ij} P_i$, and $P_i^\dagger = P_i$.

15.2 Addition of Two General Angular Momenta

We now formulate the general theory for combining two arbitrary angular momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$. We assume that $\mathbf{J}_1$ and $\mathbf{J}_2$ act in distinct sub-Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ of dimensions $2j_1 + 1$ and $2j_2 + 1$, respectively, so that their components mutually commute:

$$[\mathbf{J}_1, \mathbf{J}_2] = 0. \tag{15.20}$$

15.2.1 Uncoupled and Coupled Representations

In the *uncoupled representation*, the basis vectors are chosen as the simultaneous eigenstates of the complete set of commuting observables (CSCO) $\{J_1^2, J_{1z}, J_2^2, J_{2z}\}$:

$$\begin{aligned}
 J_1^2 |j_1, j_2, m_1, m_2\rangle &= j_1(j_1 + 1) |j_1, j_2, m_1, m_2\rangle, \\
 J_{1z} |j_1, j_2, m_1, m_2\rangle &= m_1 |j_1, j_2, m_1, m_2\rangle, \\
 J_2^2 |j_1, j_2, m_1, m_2\rangle &= j_2(j_2 + 1) |j_1, j_2, m_1, m_2\rangle, \\
 J_{2z} |j_1, j_2, m_1, m_2\rangle &= m_2 |j_1, j_2, m_1, m_2\rangle.
 \end{aligned} \tag{15.21}$$

The total angular momentum operator

$$\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2 \quad (15.22)$$

satisfies the canonical Lie algebra $[J_i, J_j] = i\varepsilon_{ijk}J_k$ (in units where $\hbar = 1$). Since the Casimir operators J_1^2 and J_2^2 are rotation scalars, they commute with all components of $\mathbf{J}$ and hence with $J^2 = \mathbf{J} \cdot \mathbf{J}$ and $J_z = J_{1z} + J_{2z}$.

Consequently, the alternative *coupled representation* is formed by the simultaneous eigenstates of the CSCO $\{J_1^2, J_2^2, J^2, J_z\}$, denoted by $|(j_1 j_2) j, m\rangle$ or simply $|j, m\rangle$:

$$\begin{aligned} J^2 |j_1, j_2, j, m\rangle &= j(j+1) |j_1, j_2, j, m\rangle, \\ J_z |j_1, j_2, j, m\rangle &= m |j_1, j_2, j, m\rangle, \\ J_1^2 |j_1, j_2, j, m\rangle &= j_1(j_1+1) |j_1, j_2, j, m\rangle, \\ J_2^2 |j_1, j_2, j, m\rangle &= j_2(j_2+1) |j_1, j_2, j, m\rangle. \end{aligned} \quad (15.23)$$

Because $[J^2, J_{1z}] \neq 0$ and $[J^2, J_{2z}] \neq 0$, the individual magnetic quantum numbers m_1 and m_2 cannot be simultaneously specified when the total angular momentum magnitude j is well-defined.

15.2.2 Clebsch–Gordan Coefficients and Selection Rules

The unitary transformation connecting the two orthonormal bases is mediated by the *Clebsch–Gordan (CG) coefficients*:

$$|j_1, j_2, j, m\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} |j_1, j_2, m_1, m_2\rangle \langle j_1, j_2, m_1, m_2 | j, m\rangle. \quad (15.24)$$

Operating with the operator identity $J_z = J_{1z} + J_{2z}$ gives

$$\langle j_1, j_2, m_1, m_2 | J_z | j_1, j_2, j, m\rangle = m \langle j_1, j_2, m_1, m_2 | j, m\rangle = (m_1 + m_2) \langle j_1, j_2, m_1, m_2 | j, m\rangle. \quad (15.25)$$

This yields the strict magnetic selection rule:

$$\langle j_1, j_2, m_1, m_2 | j, m\rangle = 0 \quad \text{unless} \quad m = m_1 + m_2. \quad (15.26)$$

The double summation in Eq. (15.24) therefore reduces to a single sum over m_1 (with $m_2 = m - m_1$):

$$|j_1, j_2, j, m\rangle = \sum_{m_1} |j_1, j_2, m_1, m - m_1\rangle \langle j_1, j_2, m_1, m - m_1 | j, m\rangle. \quad (15.27)$$

15.2.3 The Triangle Rule and Multiplet Decomposition

The total dimension of the product space $\mathcal{H}_1 \otimes \mathcal{H}_2$ is $(2j_1 + 1)(2j_2 + 1)$. The maximum possible value of the magnetic projection is $m_{\max} = j_1 + j_2$, attained uniquely by the single state $|j_1, j_2, j_1, j_2\rangle$. Since every multiplet with total angular momentum j contains exactly one state with projection $m = j$, the maximum value of j is uniquely

$$j_{\max} = j_1 + j_2. \quad (15.28)$$

At the next projection level, $m = j_1 + j_2 - 1$, there exist two linearly independent uncoupled states: $|j_1, j_2, j_1 - 1, j_2\rangle$ and $|j_1, j_2, j_1, j_2 - 1\rangle$. One linear combination belongs to the $j = j_1 + j_2$ multiplet (obtained via $J_- = J_{1-} + J_{2-}$ acting on the top state), requiring the existence of a second, orthogonal multiplet initialized with $j = j_1 + j_2 - 1$.

Iterating this counting argument down to the lower boundary establishes the celebrated *triangle inequality*:

$$|j_1 - j_2| \leq j \leq j_1 + j_2, \quad (15.29)$$

where j steps by unit integers: $j \in \{j_1 + j_2, j_1 + j_2 - 1, \dots, |j_1 - j_2|\}$. The Clebsch–Gordan coefficient $\langle j_1, j_2; m_1, m_2 | j, m \rangle$ vanishes unless the triad (j_1, j_2, j) satisfies this triangle condition.

As a consistency check, we sum the dimensions of all irreducible representations appearing in the direct-sum decomposition:

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1). \quad (15.30)$$

Hence, the coupled basis completely and irreducibly spans the full product Hilbert space:

$$\mathcal{H}_{j_1} \otimes \mathcal{H}_{j_2} = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} \mathcal{H}_j. \quad (15.31)$$

15.3 Clebsch–Gordan Coefficients and Recurrence Relations

The Clebsch–Gordan coefficients $\langle j_1, j_2; m_1, m_2 | j, m \rangle$ represent the transformation matrix elements between the uncoupled product basis $\{|j_1, j_2, m_1, m_2\rangle\}$ and the coupled total angular momentum basis $\{|j_1, j_2, j, m\rangle\}$.

15.3.1 Orthogonality and Completeness

Expanding the uncoupled states in the complete orthonormal coupled basis gives

$$|j_1, j_2, m_1, m_2\rangle = \sum_{j=|j_1-j_2|}^{j_1+j_2} |j_1, j_2, j, m_1 + m_2\rangle \langle j_1, j_2; j, m_1 + m_2 | j_1, j_2, m_1, m_2\rangle. \quad (15.32)$$

Under the standard *Condon–Shortley phase convention*, all Clebsch–Gordan coefficients are chosen to be strictly real. Consequently, the adjoint matrix elements satisfy

$$\langle j_1, j_2; j, m | j_1, j_2, m_1, m_2\rangle = \langle j_1, j_2; m_1, m_2 | j, m\rangle. \quad (15.33)$$

The unitarity of the basis transformation yields two fundamental orthogonality relations:

1. Row Orthogonality (Sum over coupled states):

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} \sum_{m=-j}^j \langle j_1, j_2; m'_1, m'_2 | j, m\rangle \langle j_1, j_2; m_1, m_2 | j, m\rangle = \delta_{m_1 m'_1} \delta_{m_2 m'_2}. \quad (15.34)$$

2. Column Orthogonality (Sum over magnetic projections):

$$\sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} \langle j_1, j_2; m_1, m_2 | j, m\rangle \langle j_1, j_2; m_1, m_2 | j', m'\rangle = \delta_{jj'} \delta_{mm'}. \quad (15.35)$$

Specializing to $j = j'$ and $m = m'$ produces the normalization condition:

$$\sum_{\substack{m_1, m_2 \\ m_1+m_2=m}} \langle j_1, j_2; m_1, m_2 | j, m\rangle^2 = 1. \quad (15.36)$$

15.3.2 Recurrence Relations via Ladder Operators

Closed-form algebraic relations for the Clebsch–Gordan coefficients are derived by taking matrix elements of the total raising and lowering operators $J_{\pm} = J_{1\pm} + J_{2\pm}$.

Acting to the right on a coupled state, $J_{\pm}$ shifts m by ± 1 :

$$\langle j_1, j_2; m_1, m_2 | J_{\pm} | j_1, j_2, j, m \rangle = \sqrt{j(j+1) - m(m \pm 1)} \langle j_1, j_2; m_1, m_2 | j, m \pm 1 \rangle. \quad (15.37)$$

Acting to the left on the bra $\langle j_1, j_2; m_1, m_2 |$, the operator $J_{\pm}^{\dagger} = J_{\mp}$ acts as a lowering or raising operator on the uncoupled state:

$$\begin{aligned} \langle j_1, j_2; m_1, m_2 | (J_{1\pm} + J_{2\pm}) | j_1, j_2, j, m \rangle &= \sqrt{j_1(j_1+1) - m_1(m_1 \mp 1)} \langle j_1, j_2; m_1 \mp 1, m_2 | j, m \rangle \\ &\quad + \sqrt{j_2(j_2+1) - m_2(m_2 \mp 1)} \langle j_1, j_2; m_1, m_2 \mp 1 | j, m \rangle. \end{aligned} \quad (15.38)$$

Equating Eqs. (15.37) and (15.38) yields the standard *Clebsch–Gordan recurrence relations*:

$$\begin{aligned} \sqrt{j(j+1) - m(m \pm 1)} \langle j_1, j_2; m_1, m_2 | j, m \pm 1 \rangle \\ = \sqrt{j_1(j_1+1) - m_1(m_1 \mp 1)} \langle j_1, j_2; m_1 \mp 1, m_2 | j, m \rangle \\ + \sqrt{j_2(j_2+1) - m_2(m_2 \mp 1)} \langle j_1, j_2; m_1, m_2 \mp 1 | j, m \rangle. \end{aligned} \quad (15.39)$$

These relations relate triads of coefficients on the (m_1, m_2) grid. Together with the Condon–Shortley convention that $\langle j_1, j_2; j_1, j - j_1 | j, j \rangle > 0$ and the normalization condition (15.36), all Clebsch–Gordan coefficients are uniquely determined.

15.3.3 Coupling Orbital Angular Momentum to Spin 1/2

A central physical application is the addition of orbital angular momentum l and electron spin $s = 1/2$. The allowed values of total angular momentum are $j = l \pm 1/2$ (for $l > 0$; for $l = 0$, $j = 1/2$).

For $j = l + 1/2$, the highest weight state is unique:

$$|l, 1/2; l + 1/2, l + 1/2\rangle = |l, 1/2; l, 1/2\rangle \implies \langle l, 1/2; l, 1/2 | l + 1/2, l + 1/2 \rangle = 1. \quad (15.40)$$

Applying the lowering recursion relation along the boundary $m_2 = \pm 1/2$, we obtain the general analytic expressions:

$$\begin{aligned} \langle l, 1/2; m - 1/2, +1/2 | l + 1/2, m \rangle &= \sqrt{\frac{l + m + 1/2}{2l + 1}}, \\ \langle l, 1/2; m + 1/2, -1/2 | l + 1/2, m \rangle &= \sqrt{\frac{l - m + 1/2}{2l + 1}}. \end{aligned} \quad (15.41)$$

For the lower branch $j = l - 1/2$, orthogonality to the $j = l + 1/2$ state determines:

$$\begin{aligned} \langle l, 1/2; m - 1/2, +1/2 | l - 1/2, m \rangle &= -\sqrt{\frac{l - m + 1/2}{2l + 1}}, \\ \langle l, 1/2; m + 1/2, -1/2 | l - 1/2, m \rangle &= +\sqrt{\frac{l + m + 1/2}{2l + 1}}. \end{aligned} \quad (15.42)$$

15.3.4 Eigenvalues of the Spin-Orbit Operator

The coupled states $|l, s, j, m\rangle$ diagonalize the spin-orbit operator $\mathbf{L} \cdot \mathbf{S}$. From $\mathbf{J} = \mathbf{L} + \mathbf{S}$, we have

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(J^2 - L^2 - S^2). \quad (15.43)$$

Evaluating this on the $|l, s = 1/2, j, m\rangle$ basis yields

$$\mathbf{L} \cdot \mathbf{S} |l, 1/2, j, m\rangle = \frac{1}{2} \left[j(j+1) - l(l+1) - \frac{3}{4} \right] |l, 1/2, j, m\rangle. \quad (15.44)$$

Explicitly:

$$\mathbf{L} \cdot \mathbf{S} |l, 1/2, l+1/2, m\rangle = +\frac{l}{2} |l, 1/2, l+1/2, m\rangle, \quad (15.45)$$

$$\mathbf{L} \cdot \mathbf{S} |l, 1/2, l-1/2, m\rangle = -\frac{l+1}{2} |l, 1/2, l-1/2, m\rangle. \quad (15.46)$$

The energy shift is attractive when $\mathbf{L}$ and $\mathbf{S}$ are antiparallel ($j = l - 1/2$) and repulsive when parallel ($j = l + 1/2$).

15.4 Scattering of Spin-1/2 Particles with Spinless Particles

The quantum mechanical addition of spin and orbital angular momentum plays a foundational role in low-energy hadron scattering, such as pion-nucleon scattering ($\pi + N \rightarrow \pi + N$). In this system, the pseudoscalar pion ($\pi^\pm, \pi^0$) carries spin $s_\pi = 0$, while the nucleon (proton or neutron) has spin $s_N = 1/2$. The orbital angular momentum $\mathbf{L}$ describes the relative motion in the center-of-mass frame, and the spin $\mathbf{S}$ is that of the nucleon.

15.4.1 Symmetries and Conservation Laws

In experimental configurations, incoming beams and polarized targets are naturally prepared in the uncoupled basis $|l, s, m_l, m_s\rangle$ (where $s = 1/2$). However, the strong interaction Hamiltonian H_{int} is strictly rotationally invariant, commuting with the total angular momentum operator $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$[H_{\text{int}}, \mathbf{J}] = 0. \quad (15.47)$$

Consequently, the total angular momentum quantum numbers (j, m) are strictly conserved throughout the scattering process. Individual components L_z and S_z are not separately conserved; spin-flip transitions ($m_s \rightarrow -m_s$) occur freely provided the recoil orbital angular momentum compensates the projection change ($\Delta m_l = -\Delta m_s$).

Furthermore, strong interactions respect spatial inversion symmetry:

$$[H_{\text{int}}, P] = 0, \quad (15.48)$$

where P is the parity operator. Under spatial reflection, the pseudovector spin is invariant ($P^{-1}\mathbf{S}P = \mathbf{S}$), whereas spatial coordinates transform as $\mathbf{r} \rightarrow -\mathbf{r}$. The total spatial parity of a relative orbital state l is $(-1)^l$. Accounting for the intrinsic odd parity of the pseudoscalar pion ($\eta_\pi = -1$) and the positive intrinsic parity of the nucleon, the total parity of the channel is $\eta_{\text{tot}} = (-1) \cdot (-1)^l$. Conservation of parity requires

$$(-1)^{l'} = (-1)^l \implies l' - l = 0, \pm 2, \pm 4, \dots \quad (15.49)$$

Combining this parity constraint with angular momentum conservation ($j = l \pm 1/2 = l' \pm 1/2$) restricts $|l - l'| \leq 1$. The only integer difference compatible with both conditions is

$$l' = l. \quad (15.50)$$

Thus, for elastic scattering of a spin-1/2 particle with a spin-0 particle, orbital angular momentum l is strictly conserved.

15.4.2 Partial Wave Channel Decomposition

Because H_{int} commutes with $\mathbf{J}$ and P , it is diagonal in the coupled representation $|l, s, j, m\rangle$. Moreover, by the Wigner–Eckart theorem, the matrix elements are independent of the magnetic orientation m :

$$\langle l', s, j', m' | H_{\text{int}} | l, s, j, m \rangle = \delta_{ll'} \delta_{jj'} \delta_{mm'} B(l, j). \quad (15.51)$$

For any given partial wave l , the scattering dynamics is completely determined by exactly two scalar channel amplitudes:

$$B_{l,+} \equiv B\left(l, j = l + \frac{1}{2}\right), \quad B_{l,-} \equiv B\left(l, j = l - \frac{1}{2}\right). \quad (15.52)$$

Transforming back to the experimentally observed uncoupled basis yields the transition matrix element:

$$\begin{aligned} \langle l, s, m'_l, m'_s | H_{\text{int}} | l, s, m_l, m_s \rangle &= \sum_{j,m} \langle l, s, m'_l, m'_s | l, s, j, m \rangle \langle l, s, j, m | H_{\text{int}} | l, s, j, m \rangle \langle l, s, j, m | l, s, m_l, m_s \rangle \\ &= \delta_{m'_l+m'_s, m_l+m_s} \sum_{j=l\pm 1/2} \langle l, 1/2; m'_l, m'_s | j, m \rangle \langle l, 1/2; m_l, m_s | j, m \rangle B(l, j). \end{aligned} \quad (15.53)$$

Equation (15.53) demonstrates that all spin-non-flip ($\Delta m_s = 0$) and spin-flip ($\Delta m_s = \pm 1$) transition amplitudes for a given partial wave are linear superpositions of the two fundamental dynamical parameters $B(l, l \pm 1/2)$, weighted by geometric Clebsch–Gordan coefficients.

15.5 Addition of Three Angular Momenta and Recoupling

When describing composite systems involving three independent angular momenta $\mathbf{J}_1$, $\mathbf{J}_2$, and $\mathbf{J}_3$ (such as two nucleons with intrinsic spins coupled to their relative orbital angular momentum, or three-valence-electron atoms), one must construct eigenstates of the total angular momentum:

$$\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2 + \mathbf{J}_3. \quad (15.54)$$

Because the three individual angular momenta mutually commute, the uncoupled basis is spanned by $|j_1, j_2, j_3, m_1, m_2, m_3\rangle$.

15.5.1 Coupling Schemes

To construct the coupled eigenstates $|j, m\rangle$, one pairs two angular momenta first, subsequently coupling the resultant to the third. The pairing choice is not unique, giving rise to distinct intermediate coupling schemes.

First Coupling Scheme: (12)3

We first couple $\mathbf{J}_1$ and $\mathbf{J}_2$ to an intermediate total angular momentum $\mathbf{J}_{12} = \mathbf{J}_1 + \mathbf{J}_2$:

$$|j_1, j_2, j_{12}, j_3; m_{12}, m_3\rangle = \sum_{m_1, m_2} |j_1, j_2, j_3, m_1, m_2, m_3\rangle \langle j_1, j_2; m_1, m_2 | j_{12}, m_{12}\rangle. \quad (15.55)$$

Coupling $\mathbf{J}_{12}$ to $\mathbf{J}_3$ yields the coupled state:

$$\begin{aligned} |(j_1 j_2) j_{12}, j_3; j, m\rangle &= \sum_{m_{12}, m_3} |j_1, j_2, j_{12}, j_3; m_{12}, m_3\rangle \langle j_{12}, j_3; m_{12}, m_3 | j, m\rangle \\ &= \sum_{m_1, m_2, m_3} |j_1, j_2, j_3, m_1, m_2, m_3\rangle \sum_{m_{12}} \langle j_1, j_2; m_1, m_2 | j_{12}, m_{12}\rangle \langle j_{12}, j_3; m_{12}, m_3 | j, m\rangle. \end{aligned} \quad (15.56)$$

Second Coupling Scheme: 1(23)

Alternatively, coupling $\mathbf{J}_2$ and $\mathbf{J}_3$ first to $\mathbf{J}_{23} = \mathbf{J}_2 + \mathbf{J}_3$, and subsequently coupling $\mathbf{J}_1$ to $\mathbf{J}_{23}$, generates an alternative basis:

$$|j_1, (j_2 j_3) j_{23}; j, m\rangle = \sum_{m_1, m_2, m_3} |j_1, j_2, j_3, m_1, m_2, m_3\rangle \sum_{m_{23}} \langle j_2, j_3; m_2, m_3 | j_{23}, m_{23} \rangle \langle j_1, j_{23}; m_1, m_{23} | j, m \rangle. \quad (15.57)$$

15.5.2 Recoupling Coefficients and Wigner 6j-Symbols

The unitary transformation connecting the two coupling schemes is mediated by the *recoupling coefficients* $\langle (j_1 j_2) j_{12}, j_3; j, m | j_1, (j_2 j_3) j_{23}; j', m' \rangle$. By rotational invariance, these coefficients are diagonal in j and m and completely independent of the projection m :

$$\langle (j_1 j_2) j_{12}, j_3; j, m | j_1, (j_2 j_3) j_{23}; j', m' \rangle = \delta_{jj'} \delta_{mm'} \langle (j_1 j_2) j_{12}, j_3; j | j_1, (j_2 j_3) j_{23}; j \rangle. \quad (15.58)$$

Evaluating the overlap involves summing a four-fold product of Clebsch–Gordan coefficients over the magnetic quantum numbers. These transformation brackets are directly proportional to the *Racah W-coefficients* and the symmetric *Wigner 6j-symbols*:

$$\langle (j_1 j_2) j_{12}, j_3; j | j_1, (j_2 j_3) j_{23}; j \rangle = (-1)^{j_1 + j_2 + j_3 + j} \sqrt{(2j_{12} + 1)(2j_{23} + 1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix}. \quad (15.59)$$

The Wigner 6j-symbol exhibits high permutation symmetries across its rows and columns, vanishing unless the four triads (j_1, j_2, j_{12}) , (j_{12}, j_3, j) , (j_2, j_3, j_{23}) , and (j_1, j_{23}, j) each satisfy the triangle condition.

15.5.3 Physical Application: The Two-Nucleon System

A standard application is the proton-neutron (p - n) system in nuclear physics. Let $\mathbf{S}_1$ and $\mathbf{S}_2$ represent the nucleon spins ($s_1 = s_2 = 1/2$), and let $\mathbf{L}$ denote their relative orbital angular momentum.

In Russell–Saunders (LS) coupling, one first couples the intrinsic spins:

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 \implies s = 0 \text{ (singlet)}, \quad s = 1 \text{ (triplet)}. \quad (15.60)$$

Next, the total spin $\mathbf{S}$ couples with the orbital angular momentum $\mathbf{L}$ to produce the total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Using spectroscopic notation $^{2s+1}L_j$:

- For the singlet channel ($s = 0$): $j = l$, yielding states $^1S_0, ^1P_1, ^1D_2, \dots$
- For the triplet channel ($s = 1$): for $l = 0$, $j = 1$ (3S_1 , the deuteron ground state); for $l \geq 1$, $j = l - 1, l, l + 1$ (e.g., $^3P_0, ^3P_1, ^3P_2$).

Nuclear tensor forces ($S_{12} = 3(\boldsymbol{\sigma}_1 \cdot \hat{\mathbf{r}})(\boldsymbol{\sigma}_2 \cdot \hat{\mathbf{r}}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$) conserve $\mathbf{J}$ and parity but mix orbital angular momentum states with $\Delta l = \pm 2$, such as the 3S_1 – 3D_1 admixture in the deuteron.

15.6 Problems

1. **Clebsch–Gordan Coefficients for $j_1 = 1 \otimes j_2 = 1$:** Compute the complete set of Clebsch–Gordan coefficients $\langle 1, 1; m_1, m_2 | j, m \rangle$ for the coupling of two spin-1 (or orbital p -wave) systems ($1 \otimes 1 = 2 \oplus 1 \oplus 0$). Verify the orthonormality relations and list the explicit linear expansions of the quintet ($j = 2$), triplet ($j = 1$), and singlet ($j = 0$) states in terms of the uncoupled basis $|m_1, m_2\rangle$.

2. **Permutation Symmetry of Two P -Wave Bosons:** Consider two identical spinless particles each in an orbital P -state ($l_1 = 1, l_2 = 1$). Show that the total spatial wave function coupled to total orbital angular momentum $L = 1$ is strictly antisymmetric under the exchange of the spatial coordinates of the two particles ($\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$), whereas the states with $L = 2$ and $L = 0$ are symmetric under particle exchange.

Chapter 16

Isotopic Spin

16.1 Isotopic Spin

In nuclear and particle physics, the mathematical apparatus of quantum angular momentum extends to internal symmetry spaces. The proton and neutron share striking physical similarities: their masses differ by less than 0.14% ($m_p \approx 938.27 \text{ MeV}/c^2$, $m_n \approx 939.57 \text{ MeV}/c^2$), both are spin-1/2 Dirac fermions, and under nuclear (strong) forces they interact almost identically. While their electromagnetic interactions differ radically due to electric charge ($q_p = +e$, $q_n = 0$), the strong interaction at nuclear distances is roughly two orders of magnitude stronger than fine-structure electromagnetic corrections. In the limit where electromagnetism and quark mass splittings are neglected, the proton and neutron can be viewed as two orthogonal quantum states of a single entity: the *nucleon* (N).

16.1.1 The Nucleon Isospin Doublet

By analogy with spin-1/2, Heisenberg introduced the internal degree of freedom termed *isotopic spin* (or *isospin*, $\mathbf{I}$). The nucleon is assigned total isospin $I = 1/2$. Choosing the isospin quantization axis along the 3-axis, the proton is identified with the isospin-up state and the neutron with the isospin-down state:

$$I_3^{(N)}|p\rangle = +\frac{1}{2}|p\rangle, \quad (16.1)$$

$$I_3^{(N)}|n\rangle = -\frac{1}{2}|n\rangle. \quad (16.2)$$

The isospin ladder operators $I_{\pm}^{(N)} \equiv I_1^{(N)} \pm iI_2^{(N)}$ transform the nucleon states into one another:

$$I_-^{(N)}|p\rangle = |n\rangle, \quad I_-^{(N)}|n\rangle = 0, \quad (16.3)$$

$$I_+^{(N)}|n\rangle = |p\rangle, \quad I_+^{(N)}|p\rangle = 0. \quad (16.4)$$

The vector operator $\mathbf{I}^{(N)} = \frac{1}{2}\boldsymbol{\tau}$ (where $\boldsymbol{\tau}$ are the Pauli matrices acting in isospin space) satisfies the standard $SU(2)$ Lie algebra:

$$[I_i^{(N)}, I_j^{(N)}] = i\varepsilon_{ijk}I_k^{(N)}. \quad (16.5)$$

The nucleon multiplet forms a fundamental representation of $SU(2)_I$ with Casimir eigenvalue:

$$[\mathbf{I}^{(N)}]^2|N\rangle = \frac{1}{2}\left(\frac{1}{2} + 1\right)|N\rangle = \frac{3}{4}|N\rangle. \quad (16.6)$$

The electric charge operator of the nucleon (in units of elementary charge e) is related to $I_3^{(N)}$ by

$$Q_N = I_3^{(N)} + \frac{1}{2}\mathbb{I}, \quad (16.7)$$

so that $Q_N|p\rangle = +1|p\rangle$ and $Q_N|n\rangle = 0$.

16.1.2 The Pion Isospin Triplet

The three charge states of the pion (π^+ , π^0 , π^-) possess nearly identical masses ($m_{\pi^\pm} \approx 139.57$ MeV/ c^2 , $m_{\pi^0} \approx 134.98$ MeV/ c^2) and zero spin. They form an isospin triplet representation ($I = 1$):

$$\begin{aligned} I_3^{(\pi)}|\pi^+\rangle &= +1|\pi^+\rangle, \\ I_3^{(\pi)}|\pi^0\rangle &= 0|\pi^0\rangle, \\ I_3^{(\pi)}|\pi^-\rangle &= -1|\pi^-\rangle. \end{aligned} \quad (16.8)$$

For pions, the charge operator is simply $Q_\pi = I_3^{(\pi)}$. The ladder operators $I_\pm^{(\pi)} = I_1^{(\pi)} \pm iI_2^{(\pi)}$ act via

$$I_\pm^{(\pi)}|\pi^0\rangle = \sqrt{1(1+1) - 0(0 \pm 1)}|\pi^\pm\rangle = \sqrt{2}|\pi^\pm\rangle, \quad (16.9)$$

and the total pion isospin Casimir operator satisfies $[\mathbf{I}^{(\pi)}]^2 = 1(1+1) = 2$.

16.1.3 The Pion–Nucleon System and Clebsch–Gordan Decomposition

For a combined system of one pion and one nucleon, the total isospin operator is

$$\mathbf{I} = \mathbf{I}^{(\pi)} + \mathbf{I}^{(N)}. \quad (16.10)$$

By the addition of angular momenta ($1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$), the total isospin takes values $I = 3/2$ (a quartet) and $I = 1/2$ (a doublet).

Using standard Clebsch–Gordan coefficients $\langle 1, 1/2; I_3^{(\pi)}, I_3^{(N)} | I, I_3 \rangle$, the coupled states $|I, I_3\rangle$ expand in terms of the physical particle states as follows:

- **Total Isospin $I = 3/2$ Multiplet:**

$$\begin{aligned} \left| \frac{3}{2}, \frac{3}{2} \right\rangle &= |\pi^+ p\rangle, \\ \left| \frac{3}{2}, \frac{1}{2} \right\rangle &= \sqrt{\frac{1}{3}}|\pi^+ n\rangle + \sqrt{\frac{2}{3}}|\pi^0 p\rangle, \\ \left| \frac{3}{2}, -\frac{1}{2} \right\rangle &= \sqrt{\frac{1}{3}}|\pi^- p\rangle + \sqrt{\frac{2}{3}}|\pi^0 n\rangle, \\ \left| \frac{3}{2}, -\frac{3}{2} \right\rangle &= |\pi^- n\rangle. \end{aligned} \quad (16.11)$$

- **Total Isospin $I = 1/2$ Multiplet:**

$$\begin{aligned} \left| \frac{1}{2}, \frac{1}{2} \right\rangle &= \sqrt{\frac{2}{3}}|\pi^+ n\rangle - \sqrt{\frac{1}{3}}|\pi^0 p\rangle, \\ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle &= \sqrt{\frac{1}{3}}|\pi^0 n\rangle - \sqrt{\frac{2}{3}}|\pi^- p\rangle. \end{aligned} \quad (16.12)$$

Inverting these relations yields the decomposition of the experimentally prepared channel states into the total isospin basis:

$$\begin{aligned}
|\pi^+ p\rangle &= \left| \frac{3}{2}, \frac{3}{2} \right\rangle, \\
|\pi^0 p\rangle &= \sqrt{\frac{2}{3}} \left| \frac{3}{2}, \frac{1}{2} \right\rangle - \sqrt{\frac{1}{3}} \left| \frac{1}{2}, \frac{1}{2} \right\rangle, \\
|\pi^- p\rangle &= \sqrt{\frac{1}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \\
|\pi^+ n\rangle &= \sqrt{\frac{1}{3}} \left| \frac{3}{2}, \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| \frac{1}{2}, \frac{1}{2} \right\rangle, \\
|\pi^0 n\rangle &= \sqrt{\frac{2}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle + \sqrt{\frac{1}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle, \\
|\pi^- n\rangle &= \left| \frac{3}{2}, -\frac{3}{2} \right\rangle.
\end{aligned} \tag{16.13}$$

16.1.4 Charge Independence and Scattering Predictions

The physical principle of *charge independence* asserts that strong interactions are invariant under global rotations in isospin space:

$$[H_{\text{int}}, \mathbf{I}] = 0. \tag{16.14}$$

Consequently, H_{int} is an isospin scalar operator. By the Wigner–Eckart theorem applied to $SU(2)_I$, the scattering matrix elements between coupled states are diagonal in I and I_3 , and strictly independent of the third projection I_3 :

$$\langle I', I'_3 | H_{\text{int}} | I'', I''_3 \rangle = \delta_{I' I''} \delta_{I'_3 I''_3} a_{I'}, \tag{16.15}$$

where $a_{3/2}$ and $a_{1/2}$ denote the two independent scalar scattering amplitudes for the $I = 3/2$ and $I = 1/2$ channels.

Consider three fundamental pion-proton scattering reactions at a given center-of-mass energy:

$$\begin{aligned}
\text{(a)} \quad & \pi^+ + p \rightarrow \pi^+ + p \quad (\text{elastic}), \\
\text{(b)} \quad & \pi^- + p \rightarrow \pi^- + p \quad (\text{elastic}), \\
\text{(c)} \quad & \pi^- + p \rightarrow \pi^0 + n \quad (\text{charge exchange}).
\end{aligned} \tag{16.16}$$

Evaluating the transition amplitudes using Eq. (16.13):

1. Reaction (a) is a pure $I = 3/2, I_3 = 3/2$ state:

$$A_a = \langle \pi^+ p | H_{\text{int}} | \pi^+ p \rangle = a_{3/2}. \tag{16.17}$$

2. In reaction (b), the initial state $|\pi^- p\rangle$ projects onto both $I = 3/2$ and $I = 1/2$ with weights $\sqrt{1/3}$ and $-\sqrt{2/3}$. Projecting the scattered state back onto $\langle \pi^- p |$ yields

$$A_b = \left(\sqrt{\frac{1}{3}} \right)^2 a_{3/2} + \left(-\sqrt{\frac{2}{3}} \right)^2 a_{1/2} = \frac{1}{3} a_{3/2} + \frac{2}{3} a_{1/2}. \tag{16.18}$$

3. In reaction (c), projecting the scattered state onto the final state $\langle \pi^0 n |$ yields

$$A_c = \sqrt{\frac{1}{3}} a_{3/2} \sqrt{\frac{2}{3}} + \left(-\sqrt{\frac{2}{3}} \right) a_{1/2} \sqrt{\frac{1}{3}} = \frac{\sqrt{2}}{3} (a_{3/2} - a_{1/2}). \tag{16.19}$$

The differential cross sections $\sigma \propto |A|^2$ therefore satisfy the universal ratio:

$$\sigma_a : \sigma_b : \sigma_c = |a_{3/2}|^2 : \frac{1}{9}|a_{3/2} + 2a_{1/2}|^2 : \frac{2}{9}|a_{3/2} - a_{1/2}|^2. \quad (16.20)$$

Near the $\Delta(1232)$ resonance ($T_\pi \approx 195$ MeV in the lab frame, or $E_{\text{cm}} \approx 1232$ MeV), scattering proceeds almost exclusively through the resonant $I = 3/2, J^P = 3/2^+$ partial wave (P_{33}). In this energy regime, $|a_{3/2}| \gg |a_{1/2}|$, and setting $a_{1/2} \approx 0$ yields the celebrated cross-section ratio:

$$\sigma(\pi^+ p \rightarrow \pi^+ p) : \sigma(\pi^- p \rightarrow \pi^- p) : \sigma(\pi^- p \rightarrow \pi^0 n) = 9 : 1 : 2. \quad (16.21)$$

This striking experimental result confirms the charge independence of nuclear forces and the $SU(2)$ isospin multiplet classification of hadrons.

16.2 Problems

- Polarized Target Scattering at the $\Delta(1232)$ Resonance:** Assume that pion-nucleon scattering at $E_{\text{cm}} = 154$ MeV proceeds entirely through the resonant channel $l = 1, j = 3/2, I = 3/2$ (P_{33} channel). A beam of π^- mesons propagating along the $+y$ -direction in the laboratory impinges on a target of stationary protons polarized completely along the $+z$ -axis ($m_s = +1/2$).
 - Compute the differential cross section $\frac{d\sigma}{d\Omega}(\theta, \phi)$ for elastic scattering $\pi^- + p \rightarrow \pi^- + p$ in the center-of-mass frame, and determine the polarization vector $\mathbf{P} = \langle \boldsymbol{\sigma} \rangle$ of the scattered recoil protons as a function of the scattering angles.
 - Repeat the calculation for the charge-exchange reaction $\pi^- + p \rightarrow \pi^0 + n$, computing both the angular distribution and the polarization of the recoil neutrons.
- Pion Production in Nucleon–Nucleon Collisions:** The deuteron d is an isospin singlet ($I_d = 0$). Assuming strict charge independence for the strong interactions, determine the ratio of the cross sections for the reactions:

$$p + p \rightarrow d + \pi^+ \quad \text{and} \quad p + n \rightarrow d + \pi^0. \quad (16.22)$$

- Parity Violation and Spin Dynamics in Weak Hyperon Decay:** The neutral lambda hyperon Λ^0 has spin $J = 1/2$, isospin $I = 0$, and decays via the weak interaction into a nucleon and a pion: $\Lambda^0 \rightarrow \pi + N$. The decay conserves electric charge and total angular momentum, but violates parity and satisfies the empirical $\Delta I = 1/2$ isospin selection rule (the final πN state has $I = 1/2$).
 - Using the $\Delta I = 1/2$ rule, calculate the branching ratio of decay into protons versus neutrons: $\Gamma(\Lambda^0 \rightarrow \pi^- p) / \Gamma(\Lambda^0 \rightarrow \pi^0 n)$.
 - For the decay $\Lambda^0 \rightarrow \pi^- + p$, what are the allowed values of the relative orbital angular momentum l in the rest frame? (Note that parity is not conserved).
 - Let a_s and a_p denote the decay matrix elements into the s -wave ($l = 0, j = 1/2$) and p -wave ($l = 1, j = 1/2$) channels at relative momentum q . For a Λ^0 hyperon polarized along the z -axis in its rest frame, write down the spin-angular wave function of the outgoing proton and the asymptotic outgoing spherical wave.
 - Derive the angular distribution of the emitted protons, $\frac{dN}{d\Omega} \propto 1 + \alpha P_\Lambda \cos \theta$, expressing the asymmetry parameter α in terms of the decay amplitudes a_s and a_p .
 - Compute the longitudinal polarization of the decay proton as a function of emission angle θ .

- (f) Explain how measurements on the decay of unpolarized Λ^0 hyperons allow one to extract the parity-violating decay parameters.
 - (g) Extend the analysis to the neutral channel $\Lambda^0 \rightarrow \pi^0 + n$.
 - (h) Express the total decay width Γ and lifetime τ in terms of a_s, a_p and the liberated energy $Q = (m_\Lambda - m_N - m_\pi)c^2$.
4. **One-Boson-Exchange Potential and s -Wave Scattering:** At low energies, the pion-nucleon interaction may be approximated by a vector meson (ρ -meson) exchange Yukawa potential:
- $$V(r) = \frac{f^2}{4\pi} (\mathbf{I}^{(\pi)} \cdot \mathbf{I}^{(N)}) \frac{e^{-\mu r}}{r}. \quad (16.23)$$
- (a) Evaluate the operator $\mathbf{I}^{(\pi)} \cdot \mathbf{I}^{(N)}$ in the $I = 3/2$ and $I = 1/2$ channels, and determine the ratio of the effective potential strengths in these two channels.
 - (b) In the first Born approximation, calculate the low-energy s -wave cross sections for the three channels: $\pi^+ p \rightarrow \pi^+ p$, $\pi^- p \rightarrow \pi^- p$, and $\pi^- p \rightarrow \pi^0 n$.
5. **Kaon-Nucleon Cross Section Ratios from $SU(2)_I$:** Antikaons form an isospin doublet ($K^-, \bar{K}^0$) with $I = 1/2$. The Σ hyperons form an $I = 1$ triplet ($\Sigma^+, \Sigma^0, \Sigma^-$), and the Λ^0 is an isospin singlet ($I = 0$). By decomposing the initial and final states into total isospin channels $I = 0$ and $I = 1$, determine the cross-section ratios among the strong reactions:

$$\begin{aligned} K^- + p &\rightarrow \pi^0 + \Lambda^0, \\ K^- + p &\rightarrow \pi^0 + \Sigma^0, \\ K^- + p &\rightarrow \pi^\pm + \Sigma^\mp, \end{aligned} \quad (16.24)$$

in terms of the reduced isospin amplitudes.

Chapter 17

Rotations And Tensor Operators

17.1 Rotations and Tensor Operators: Introduction

Rotational symmetry plays a central organizing role in quantum mechanics. The invariance of an isolated physical system under spatial rotations implies the conservation of total angular momentum and dictates the multiplet structure and degeneracy of energy levels.

In this chapter, we develop the mathematical theory of finite rotations and their representations in quantum mechanics:

1. **Representations of the Rotation Group:** We investigate the unitary operators $U(\mathcal{R}) = \exp(-i\boldsymbol{\alpha} \cdot \mathbf{J}/\hbar)$ representing spatial rotations in the $(2j + 1)$ -dimensional Hilbert spaces of fixed angular momentum j . We express rotations in terms of Euler angles (α, β, γ) and evaluate the matrix elements known as Wigner's rotation matrices $D_{m'm}^{(j)}(\alpha, \beta, \gamma)$.
2. **Spherical Tensor Operators and the Wigner–Eckart Theorem:** We classify quantum mechanical operators according to their transformation properties under rotations, generalizing vector operators to arbitrary irreducible spherical tensor operators $T_q^{(k)}$. We prove the fundamental Wigner–Eckart theorem, which factors any tensor transition matrix element into a geometric Clebsch–Gordan coefficient and a dynamically determined reduced matrix element.
3. **Multipole Transitions in Radiation:** Applying tensor operator methods, we analyze electromagnetic multipole transitions in atomic and nuclear systems, establishing selection rules for electric ($E\ell$) and magnetic ($M\ell$) multipole radiation.
4. **Oscillator Representation of Angular Momentum:** Finally, we present Schwinger's formulation of angular momentum algebra in terms of two uncoupled harmonic oscillators (creation and annihilation operators), providing an elegant method for deriving representation matrices and Clebsch–Gordan coefficients.

17.2 Representations of the Rotation Group

Let $\mathbf{J}$ denote the total angular momentum operator of a physical system. The unitary operator representing a finite rotation through an angle $\alpha = |\boldsymbol{\alpha}|$ about the unit axis $\hat{\boldsymbol{\alpha}}$ is

$$R_{\boldsymbol{\alpha}} = \exp\left(-\frac{i}{\hbar} \mathbf{J} \cdot \boldsymbol{\alpha}\right). \quad (17.1)$$

(In this chapter, we employ natural angular momentum units $\hbar = 1$). Because $\mathbf{J}^2$ is a rotation scalar, it commutes with the rotation operator:

$$[R_{\boldsymbol{\alpha}}, \mathbf{J}^2] = 0. \quad (17.2)$$

Thus, a spatial rotation preserves the total angular momentum quantum number j :

$$\mathbf{J}^2 R_{\alpha} |j, m\rangle = R_{\alpha} \mathbf{J}^2 |j, m\rangle = j(j+1) R_{\alpha} |j, m\rangle. \quad (17.3)$$

Under an arbitrary rotation, the magnetic projection m is generally not conserved. Because the states $\{|j, m'\rangle\}_{m'=-j}^j$ form a complete orthonormal basis for the $(2j+1)$ -dimensional subspace $\mathcal{H}_j$, the rotated state is expanded as

$$R_{\alpha} |j, m\rangle = \sum_{m'=-j}^j |j, m'\rangle D_{m'm}^{(j)}(\alpha). \quad (17.4)$$

Projecting with the bra $\langle j, m'|$, the matrix elements of the rotation operator are defined by

$$D_{m'm}^{(j)}(\alpha) \equiv \langle j, m' | e^{-i\mathbf{J}\cdot\alpha} | j, m \rangle. \quad (17.5)$$

The $(2j+1) \times (2j+1)$ matrix $D^{(j)}(\alpha)$ is known as the *Wigner rotation matrix*. Its matrix elements are purely kinematic, depending only on the abstract Lie algebra of angular momentum.

17.2.1 Group Representation Properties

Successive rotations compose according to the group multiplication law of $SO(3)$ (or $SU(2)$). If a rotation α followed by β is equivalent to a single net rotation γ ,

$$R_{\gamma} = R_{\beta} R_{\alpha}, \quad (17.6)$$

inserting a resolution of the identity $\mathbb{I}_j = \sum_{m''} |j, m''\rangle \langle j, m''|$ yields

$$\langle j, m' | R_{\gamma} | j, m \rangle = \sum_{m''=-j}^j \langle j, m' | R_{\beta} | j, m'' \rangle \langle j, m'' | R_{\alpha} | j, m \rangle. \quad (17.7)$$

In matrix notation, this establishes the group homomorphism:

$$D^{(j)}(\gamma) = D^{(j)}(\beta) D^{(j)}(\alpha). \quad (17.8)$$

The matrices $D^{(j)}$ form an *irreducible unitary representation* of dimension $2j+1$. Unitarity follows from the Hermiticity of $\mathbf{J}$:

$$\left[D^{(j)}(\alpha)^\dagger \right]_{m'm} = \left[D_{mm'}^{(j)}(\alpha) \right]^* = \langle j, m | e^{-i\mathbf{J}\cdot\alpha} | j, m' \rangle^* = \langle j, m' | e^{i\mathbf{J}\cdot\alpha} | j, m \rangle = D_{m'm}^{(j)}(-\alpha). \quad (17.9)$$

Hence,

$$D^{(j)}(\alpha)^\dagger D^{(j)}(\alpha) = \mathbb{I}_{2j+1}. \quad (17.10)$$

For half-integer j , $D^{(j)}(2\pi\hat{n}) = -\mathbb{I}$, making the representation double-valued on the proper rotation group $SO(3)$, but single-valued on the universal covering group $SU(2)$.

17.2.2 Euler Angle Parameterization

Any spatial rotation can be parameterized by three Euler angles (α, β, γ) (or conventionally (ϕ, θ, ψ)):

1. A rotation about the initial z -axis by angle ϕ .
2. A rotation about the intermediate y' -axis by angle θ .
3. A rotation about the new z'' -axis by angle ψ .

Using the transformation law for rotated coordinate axes, $R_{\mathbf{u}}(\theta) = UR_{\mathbf{u}_0}(\theta)U^\dagger$, the composite operator simplifies to successive rotations about the *fixed* laboratory axes:

$$R(\phi, \theta, \psi) = e^{-i\phi J_z} e^{-i\theta J_y} e^{-i\psi J_z}. \quad (17.11)$$

Evaluating the matrix elements in the coupled basis gives

$$D_{m'm}^{(j)}(\phi, \theta, \psi) = \langle j, m' | e^{-i\phi J_z} e^{-i\theta J_y} e^{-i\psi J_z} | j, m \rangle = e^{-im'\phi} d_{m'm}^{(j)}(\theta) e^{-im\psi}, \quad (17.12)$$

where the reduced (or small) Wigner matrix $d^{(j)}(\theta)$ is defined by

$$d_{m'm}^{(j)}(\theta) \equiv \langle j, m' | e^{-i\theta J_y} | j, m \rangle. \quad (17.13)$$

Because J_y has purely imaginary off-diagonal matrix elements in the standard Condon–Shortley basis, the matrix elements $d_{m'm}^{(j)}(\theta)$ are strictly real and satisfy the symmetry relation:

$$d_{m'm}^{(j)}(\theta) = (-1)^{m'-m} d_{mm'}^{(j)}(\theta) = d_{-m, -m'}^{(j)}(\theta). \quad (17.14)$$

17.2.3 Relation to Spherical Harmonics

For integer angular momentum l and $m = 0$, the rotation matrix elements are intimately connected to the spherical harmonics. Setting $\psi = 0$, the rotation $R(\phi, \theta, 0)$ maps the unit vector $\hat{\mathbf{z}}$ into the direction (θ, ϕ) . Projecting onto the position-space basis $|\theta, \phi\rangle$:

$$D_{m0}^{(l)}(\phi, \theta, 0) = \sqrt{\frac{4\pi}{2l+1}} Y_{lm}^*(\theta, \phi). \quad (17.15)$$

Under a spatial rotation $\mathcal{R}$, spherical harmonics transform as basis functions of the irreducible representation $D^{(l)}$:

$$Y_{lm}(\theta', \phi') = \sum_{m'=-l}^l Y_{lm'}(\theta, \phi) D_{m'm}^{(l)}(\mathcal{R}). \quad (17.16)$$

Setting $m = 0$ in Eq. (17.16) and using Eq. (17.15) produces the *spherical harmonic addition theorem*:

$$P_l(\cos \Theta) = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}(\theta, \phi) Y_{lm}^*(\theta_\alpha, \phi_\alpha), \quad (17.17)$$

where $\cos \Theta = \cos \theta \cos \theta_\alpha + \sin \theta \sin \theta_\alpha \cos(\phi - \phi_\alpha)$ is the angle between the two directions.

17.2.4 Direct Products and the Clebsch–Gordan Series

Consider a composite system formed by two angular momenta $\mathbf{J}_1$ and $\mathbf{J}_2$. In the uncoupled basis, the rotation operator factorizes:

$$e^{-i\mathbf{J}\cdot\boldsymbol{\alpha}} = e^{-i\mathbf{J}_1\cdot\boldsymbol{\alpha}} \otimes e^{-i\mathbf{J}_2\cdot\boldsymbol{\alpha}}, \quad (17.18)$$

giving the direct product representation:

$$\left[D^{(j_1)}(\boldsymbol{\alpha}) \otimes D^{(j_2)}(\boldsymbol{\alpha}) \right]_{m'_1 m'_2, m_1 m_2} = D_{m'_1 m_1}^{(j_1)}(\boldsymbol{\alpha}) D_{m'_2 m_2}^{(j_2)}(\boldsymbol{\alpha}). \quad (17.19)$$

Transforming to the coupled total angular momentum basis $|j, m\rangle$ via Clebsch–Gordan coefficients reduces the direct product into a direct sum of irreducible representations (the *Clebsch–Gordan series*):

$$D_{m'_1 m_1}^{(j_1)}(\boldsymbol{\alpha}) D_{m'_2 m_2}^{(j_2)}(\boldsymbol{\alpha}) = \sum_{j=|j_1-j_2|}^{j_1+j_2} \sum_{m', m} \langle j_1, j_2; m'_1, m'_2 | j, m' \rangle \langle j_1, j_2; m_1, m_2 | j, m \rangle D_{m'm}^{(j)}(\boldsymbol{\alpha}). \quad (17.20)$$

For integer $j_1 = l_1, j_2 = l_2$ with $m_1 = m_2 = 0$, Eq. (17.20) yields the product formula for two spherical harmonics:

$$Y_{l_1 m_1}(\theta, \phi) Y_{l_2 m_2}(\theta, \phi) = \sum_{l, m} \sqrt{\frac{(2l_1 + 1)(2l_2 + 1)}{4\pi(2l + 1)}} \langle l_1, l_2; m_1, m_2 | l, m \rangle \langle l_1, l_2; 0, 0 | l, 0 \rangle Y_{lm}(\theta, \phi). \quad (17.21)$$

17.2.5 Orthogonality of Rotation Matrices and Characters

The invariant Haar measure over $SU(2)$ parameterized by Euler angles is

$$d\omega = \frac{1}{8\pi^2} d\phi \sin \theta d\theta d\psi, \quad \int d\omega = 1, \quad (17.22)$$

where $\phi \in [0, 2\pi]$, $\theta \in [0, \pi]$, and $\psi \in [0, 4\pi]$ (for $SU(2)$).

Integrating an irreducible representation over all rotations projects out the scalar ($j = 0$) component:

$$\int d\omega D_{m'm}^{(j)}(\omega) = \delta_{j0} \delta_{m'0} \delta_{m0}. \quad (17.23)$$

Averaging the Clebsch–Gordan series (17.20) and using the identity $\langle j_1, j_2; m_1, m_2 | 0, 0 \rangle = \frac{(-1)^{j_1 - m_1}}{\sqrt{2j_1 + 1}} \delta_{j_1 j_2} \delta_{m_1, -m_2}$ yields the *great orthogonality theorem* for Wigner D -matrices:

$$\int d\omega \left[D_{m'_1 m_1}^{(j_1)}(\omega) \right]^* D_{m'_2 m_2}^{(j_2)}(\omega) = \frac{1}{2j_1 + 1} \delta_{j_1 j_2} \delta_{m'_1 m'_2} \delta_{m_1 m_2}. \quad (17.24)$$

The trace of the representation matrix defines the *character*:

$$\chi^{(j)}(\alpha) \equiv \text{Tr } D^{(j)}(\alpha) = \sum_{m=-j}^j e^{-im\alpha} = \frac{\sin \left[\left(j + \frac{1}{2} \right) \alpha \right]}{\sin \left(\frac{\alpha}{2} \right)}, \quad (17.25)$$

which depends only on the rotation angle α (the conjugacy class). The characters satisfy the orthogonality relation:

$$\int d\omega \chi^{(j_1)}(\omega)^* \chi^{(j_2)}(\omega) = \delta_{j_1 j_2}. \quad (17.26)$$

17.3 Spherical Tensor Operators and the Wigner–Eckart Theorem

Just as states in Hilbert space transform under spatial rotations according to irreducible representations $D^{(j)}$, quantum mechanical operators can be classified by their rotational transformation behavior. An operator A transforms under a finite rotation $R_\alpha = \exp(-i\mathbf{J} \cdot \alpha)$ via the unitary similarity transformation:

$$A \mapsto R_\alpha A R_\alpha^{-1}. \quad (17.27)$$

17.3.1 Definition of Irreducible Spherical Tensor Operators

An *irreducible spherical tensor operator of rank k* (where k is an integer or half-integer) consists of a set of $2k + 1$ operators $T_q^{(k)}$ ($q = -k, -k + 1, \dots, k$) that transform among themselves under spatial rotations according to

$$R_\alpha T_q^{(k)} R_\alpha^{-1} = \sum_{q'=-k}^k T_{q'}^{(k)} D_{q'q}^{(k)}(\alpha). \quad (17.28)$$

A rank-0 tensor $T_0^{(0)}$ is a rotation scalar ($[R_\alpha, T_0^{(0)}] = 0$), while a rank-1 tensor $T_q^{(1)}$ corresponds to a vector operator.

Considering an infinitesimal rotation ϵ , $R_\epsilon \approx \mathbb{I} - i\mathbf{J} \cdot \epsilon$, Eq. (17.28) expands to

$$T_q^{(k)} - i[\mathbf{J} \cdot \epsilon, T_q^{(k)}] = T_q^{(k)} - i\epsilon \cdot \sum_{q'=-k}^k T_{q'}^{(k)} \langle k, q' | \mathbf{J} | k, q \rangle. \quad (17.29)$$

Matching linear powers in ϵ yields the foundational commutator definition:

$$[\mathbf{J}, T_q^{(k)}] = \sum_{q'=-k}^k T_{q'}^{(k)} \langle k, q' | \mathbf{J} | k, q \rangle. \quad (17.30)$$

In spherical components, this system decouples into the canonical ladder commutation relations:

$$[J_z, T_q^{(k)}] = q T_q^{(k)}, \quad (17.31)$$

$$[J_\pm, T_q^{(k)}] = \sqrt{k(k+1) - q(q \pm 1)} T_{q \pm 1}^{(k)}. \quad (17.32)$$

Equations (17.31) and (17.32) provide the standard practical criterion for verifying whether an operator constitutes an irreducible spherical tensor.

17.3.2 Spherical Components of Vector Operators and Solid Harmonics

For a standard Cartesian vector operator $\mathbf{V} = (V_x, V_y, V_z)$, satisfying $[J_i, V_j] = i\epsilon_{ijk} V_k$, the spherical tensor components of rank $k = 1$ are defined by

$$\begin{aligned} V_{+1}^{(1)} &= -\frac{V_x + iV_y}{\sqrt{2}}, \\ V_0^{(1)} &= V_z, \\ V_{-1}^{(1)} &= \frac{V_x - iV_y}{\sqrt{2}}. \end{aligned} \quad (17.33)$$

For the position operator $\mathbf{r}$, writing Cartesian coordinates in terms of spherical angles gives

$$r_q = \sqrt{\frac{4\pi}{3}} r Y_{1q}(\theta, \phi). \quad (17.34)$$

More generally, the regular *solid harmonics*

$$\mathcal{Y}_{lm}(\mathbf{r}) \equiv r^l Y_{lm}(\theta, \phi) \quad (17.35)$$

are homogeneous polynomials of degree l in (x, y, z) . Replacing $\mathbf{r}$ with any commuting vector operator $\mathbf{V}$ (or properly symmetrizing non-commuting components) constructs an irreducible spherical tensor operator of arbitrary integer rank l :

$$T_m^{(l)} = \mathcal{Y}_{lm}(\mathbf{V}). \quad (17.36)$$

For $l = 2$, the five components of the symmetric, traceless quadrupole tensor are

$$\begin{aligned} T_{\pm 2}^{(2)} &= \sqrt{\frac{15}{8\pi}} V_{\pm 1}^2, \\ T_{\pm 1}^{(2)} &= \sqrt{\frac{15}{16\pi}} (V_0 V_{\pm 1} + V_{\pm 1} V_0), \\ T_0^{(2)} &= \sqrt{\frac{5}{16\pi}} (2V_0^2 + V_1 V_{-1} + V_{-1} V_1). \end{aligned} \quad (17.37)$$

17.3.3 The Wigner–Eckart Theorem

Acting with a spherical tensor component $T_q^{(k)}$ on an angular momentum eigenstate $|\alpha, j, m\rangle$ produces a state carrying magnetic projection $m + q$:

$$J_z \left(T_q^{(k)} |\alpha, j, m\rangle \right) = \left([J_z, T_q^{(k)}] + T_q^{(k)} J_z \right) |\alpha, j, m\rangle = (q + m) \left(T_q^{(k)} |\alpha, j, m\rangle \right). \quad (17.38)$$

The action of $T_q^{(k)}$ on $|\alpha, j, m\rangle$ is kinematically isomorphic to coupling two angular momenta k and j . Coupling with Clebsch–Gordan coefficients yields a state of definite total angular momentum (j', m') :

$$|\tilde{\alpha}, j', m'\rangle = \sum_{q, m} T_q^{(k)} |\alpha, j, m\rangle \langle k, j; q, m | j', m'\rangle. \quad (17.39)$$

Rotational invariance ensures that the overlap $\langle \alpha'', j'', m'' | \tilde{\alpha}, j', m'\rangle$ is diagonal in j' and independent of m' .

Inverting the Clebsch–Gordan expansion yields the celebrated *Wigner–Eckart Theorem*:

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = \frac{\langle \alpha', j' || T^{(k)} || \alpha, j \rangle}{\sqrt{2j' + 1}} \langle k, j; q, m | j', m' \rangle. \quad (17.40)$$

Here, $\langle \alpha', j' || T^{(k)} || \alpha, j \rangle$ is the *reduced matrix element*, which contains all physical and dynamical information of the system, while the Clebsch–Gordan coefficient $\langle k, j; q, m | j', m' \rangle$ dictates the entire geometric and magnetic orientation dependence.

Selection Rules

The Wigner–Eckart theorem immediately implies strict selection rules:

1. **Magnetic Selection Rule:** The matrix element vanishes identically unless $m' = m + q$.
2. **Angular Momentum Triangle Rule:** The matrix element vanishes identically unless $|j - k| \leq j' \leq j + k$.

Application to the Angular Momentum Operator

For the angular momentum operator itself ($k = 1$), taking the $q = 0$ component gives

$$\langle \alpha', j', m' | J_0 | \alpha, j, m \rangle = m \delta_{\alpha\alpha'} \delta_{jj'} \delta_{mm'} = \frac{\langle \alpha', j' || \mathbf{J} || \alpha, j \rangle}{\sqrt{2j' + 1}} \langle 1, j; 0, m | j', m' \rangle. \quad (17.41)$$

Using $\langle 1, j; 0, m | j, m \rangle = \frac{m}{\sqrt{j(j+1)}}$, the reduced matrix element is

$$\langle \alpha', j' || \mathbf{J} || \alpha, j \rangle = \delta_{\alpha\alpha'} \delta_{jj'} \sqrt{j(j+1)(2j+1)}. \quad (17.42)$$

17.3.4 Tensor Products of Tensor Operators

Given two irreducible tensor operators $T^{(k_1)}$ and $W^{(k_2)}$, their simple product $T_{q_1}^{(k_1)} W_{q_2}^{(k_2)}$ is generally reducible. An irreducible tensor operator of rank K is constructed by Clebsch–Gordan coupling:

$$Z_Q^{(K)} = [T^{(k_1)} \otimes W^{(k_2)}]_Q^{(K)} = \sum_{q_1, q_2} T_{q_1}^{(k_1)} W_{q_2}^{(k_2)} \langle k_1, k_2; q_1, q_2 | K, Q \rangle. \quad (17.43)$$

For two vector operators $\mathbf{V}$ and $\mathbf{U}$ ($k_1 = k_2 = 1$), the product representation decomposes into $1 \otimes 1 = 0 \oplus 1 \oplus 2$:

$$Z_0^{(0)} = -\frac{1}{\sqrt{3}} \mathbf{V} \cdot \mathbf{U} \quad (\text{Scalar}), \quad (17.44)$$

$$Z_Q^{(1)} = \frac{i}{\sqrt{2}} (\mathbf{V} \times \mathbf{U})_Q \quad (\text{Vector / Cross product}), \quad (17.45)$$

$$Z_Q^{(2)} = [\mathbf{V} \otimes \mathbf{U}]_Q^{(2)} \quad (\text{Symmetric, traceless rank-2 tensor}). \quad (17.46)$$

This decomposition corresponds to partitioning the Cartesian dyadic $V_i U_j$ into its trace, anti-symmetric part, and symmetric traceless tensor:

$$V_i U_j = \frac{\mathbf{V} \cdot \mathbf{U}}{3} \delta_{ij} + \frac{V_i U_j - V_j U_i}{2} + \left(\frac{V_i U_j + V_j U_i}{2} - \frac{\mathbf{V} \cdot \mathbf{U}}{3} \delta_{ij} \right). \quad (17.47)$$

17.4 Multipole Radiation and Selection Rules

In atomic and nuclear transitions, radiative decay occurs between stationary states labeled by total angular momentum (j, m) and spatial parity π . The transition probability for emitting a photon of wavevector $\mathbf{k}$ and transverse polarization $\boldsymbol{\lambda}$ ($\mathbf{k} \cdot \boldsymbol{\lambda} = 0$) during an atomic transition $|\alpha_i, j_i, m_i\rangle \rightarrow |\alpha_f, j_f, m_f\rangle$ is governed by the matrix element of the transverse current operator:

$$\mathcal{M}_{fi} = \langle \alpha_f, j_f, m_f | \mathbf{J}(-\mathbf{k}) \cdot \boldsymbol{\lambda}^* | \alpha_i, j_i, m_i \rangle, \quad (17.48)$$

where $\mathbf{J}(-\mathbf{k}) = \int d^3r e^{i\mathbf{k} \cdot \mathbf{r}} \mathbf{J}(\mathbf{r})$ is the spatial Fourier transform of the matter current density operator.

17.4.1 Multipole Expansion of the Radiation Field

Expanding the spatial plane wave in terms of spherical Bessel functions $j_l(kr)$ and spherical harmonics yields

$$e^{i\mathbf{k} \cdot \mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l j_l(kr) Y_{lm}^*(\Omega_{\mathbf{k}}) Y_{lm}(\Omega_{\mathbf{r}}). \quad (17.49)$$

Contracting the current density with the photon polarization vector in spherical tensor components,

$$\boldsymbol{\lambda}^* \cdot \mathbf{J}(\mathbf{r}) = \sum_{q=-1}^1 (-1)^q (\lambda^*)_{-q} J_q(\mathbf{r}) = \sum_q \lambda_q^* J_q(\mathbf{r}), \quad (17.50)$$

we couple the orbital spherical harmonic $Y_{lm}(\Omega_{\mathbf{r}})$ (rank l) and the vector current $J_q(\mathbf{r})$ (rank 1) into an irreducible spherical tensor operator of rank L :

$$\mathbf{J}(-\mathbf{k}) \cdot \boldsymbol{\lambda}^* = \sum_{L=1}^{\infty} \sum_{M=-L}^L \sum_{l=|L-1|}^{L+1} 4\pi i^l \Phi_M^{(L)*}(l, \Omega_{\mathbf{k}}, \boldsymbol{\lambda}) \mathcal{J}_M^{(L)}(l). \quad (17.51)$$

Here, the transition current multipole operator is defined by

$$\mathcal{J}_M^{(L)}(l) \equiv \sum_{m,q} \langle l, 1; m, q | L, M \rangle \int d^3r j_l(kr) Y_{lm}(\Omega_{\mathbf{r}}) J_q(\mathbf{r}), \quad (17.52)$$

and the photon angular-polarization wave function is

$$\Phi_M^{(L)}(l, \Omega_{\mathbf{k}}, \boldsymbol{\lambda}) \equiv \sum_{m,q} \langle l, 1; m, q | L, M \rangle Y_{lm}(\Omega_{\mathbf{k}}) \lambda_q. \quad (17.53)$$

17.4.2 Electric and Magnetic Multipoles

The triangle condition for $\langle l, 1; m, q | L, M \rangle$ restricts the orbital index l to three values for a given total photon angular momentum L :

1. **Magnetic Multipoles (ML):** For $l = L$, the operator $\mathcal{J}_M^{(L)}(L)$ describes magnetic 2^L -pole radiation. In the long-wavelength limit ($kR \ll 1$), approximating $j_L(kr) \approx \frac{(kr)^L}{(2L+1)!!}$, the lowest order ($L = 1$) reproduces the magnetic dipole operator:

$$\mathcal{J}_M^{(1)}(1) = \frac{k}{i\sqrt{24\pi}} \int d^3r [\mathbf{r} \times \mathbf{J}(\mathbf{r})]_M = \frac{k}{i\sqrt{24\pi}} \frac{2mc}{e} M_{\text{mag},M}. \quad (17.54)$$

2. **Electric Multipoles (EL):** For $l = L \pm 1$, the leading contribution arises from $l = L - 1$, which scales as k^{L-1} and represents electric 2^L -pole radiation. (The $l = L + 1$ term is suppressed by an additional factor of $(kR)^2$).

Because real photons are strictly transverse ($\mathbf{k} \cdot \boldsymbol{\lambda} = 0$), the scalar combination $\Phi_0^{(0)} \propto \mathbf{k} \cdot \boldsymbol{\lambda}$ vanishes identically. Thus, *photons cannot carry zero total angular momentum* ($L = 0$). Monopole photon emission ($E0$ or $M0$) is strictly forbidden; $0 \rightarrow 0$ nuclear transitions proceed via radiationless processes such as internal conversion or electron-positron pair creation.

17.4.3 Selection Rules via the Wigner–Eckart Theorem

Applying the Wigner–Eckart theorem to the matrix element of the tensor operator $\mathcal{J}_M^{(L)}(l)$:

$$\langle \alpha_f, j_f, m_f | \mathcal{J}_M^{(L)}(l) | \alpha_i, j_i, m_i \rangle = \frac{\langle \alpha_f, j_f || \mathcal{J}^{(L)}(l) || \alpha_i, j_i \rangle}{\sqrt{2j_f + 1}} \langle L, j_i; M, m_i | j_f, m_f \rangle. \quad (17.55)$$

This factorization establishes the universal multipole selection rules:

1. **Conservation of Angular Momentum:** The Clebsch–Gordan coefficient enforces

$$|j_i - j_f| \leq L \leq j_i + j_f, \quad \text{with } j_i + j_f \geq L \geq 1, \quad (17.56)$$

and for the projection:

$$m_f - m_i = M. \quad (17.57)$$

In particular, transitions between two $j = 0$ states ($0 \rightarrow 0$) cannot occur via single-photon emission.

2. **Parity Selection Rules:** Under spatial inversion $\mathbf{r} \rightarrow -\mathbf{r}$, the current density is a polar vector: $P\mathbf{J}(\mathbf{r})P^{-1} = -\mathbf{J}(-\mathbf{r})$. Accounting for the parity $(-1)^l$ of the orbital spherical harmonics, the operator $\mathcal{J}_M^{(L)}(l)$ transforms under parity with the sign $(-1)^{l+1}$. Consequently:

- **Electric 2^L -pole (EL):** $l = L \pm 1$, so the operator parity is $\Delta\pi = (-1)^{(L\pm 1)+1} = (-1)^L$. Parity changes if and only if L is odd:

$$\pi_f \pi_i = (-1)^L \quad (EL). \quad (17.58)$$

- **Magnetic 2^L -pole (ML):** $l = L$, so the operator parity is $\Delta\pi = (-1)^{L+1}$:

$$\pi_f \pi_i = (-1)^{L+1} \quad (ML). \quad (17.59)$$

For electric dipole radiation ($E1$, $L = 1$), these rules dictate $\Delta j = 0, \pm 1$ (with $0 \not\rightarrow 0$) and a strict parity inversion $\pi_f = -\pi_i$. For electric quadrupole ($E2$, $L = 2$) and magnetic dipole ($M1$, $L = 1$), parity is conserved ($\pi_f = \pi_i$).

17.5 Angular Momentum and the Harmonic Oscillator: The Jordan–Schwinger Realization

Julian Schwinger developed a formulation connecting quantum angular momentum to the algebra of two uncoupled harmonic oscillators. This realization allows for the algebraic evaluation of representation matrices, Clebsch–Gordan coefficients, and recoupling brackets.

17.5.1 The Two-Oscillator Representation

Consider two independent one-dimensional bosonic harmonic oscillators described by annihilation and creation operators $a_{\pm}$ and $a_{\pm}^{\dagger}$, satisfying canonical commutation relations:

$$[a_{+}, a_{+}^{\dagger}] = [a_{-}, a_{-}^{\dagger}] = \mathbb{I}, \quad [a_{+}, a_{-}] = [a_{+}, a_{-}^{\dagger}] = 0. \quad (17.60)$$

The composite Fock space is spanned by the orthonormal number states

$$|n_{+}, n_{-}\rangle = \frac{(a_{+}^{\dagger})^{n_{+}}}{\sqrt{n_{+}!}} \frac{(a_{-}^{\dagger})^{n_{-}}}{\sqrt{n_{-}!}} |0, 0\rangle, \quad (17.61)$$

where $|0, 0\rangle$ is the vacuum state ($a_{\pm}|0, 0\rangle = 0$), and $N_{\pm} = a_{\pm}^{\dagger}a_{\pm}$ are the number operators.

Grouping the operators into a two-component spinor $a = \begin{pmatrix} a_{+} \\ a_{-} \end{pmatrix}$, we define the vector operator

$$\mathbf{J} \equiv \frac{1}{2} a^{\dagger} \boldsymbol{\sigma} a, \quad (17.62)$$

where $\boldsymbol{\sigma}$ are the Pauli matrices. In components:

$$\begin{aligned} J_x &= \frac{1}{2} (a_{+}^{\dagger} a_{-} + a_{-}^{\dagger} a_{+}), \\ J_y &= \frac{1}{2i} (a_{+}^{\dagger} a_{-} - a_{-}^{\dagger} a_{+}), \\ J_z &= \frac{1}{2} (a_{+}^{\dagger} a_{+} - a_{-}^{\dagger} a_{-}) = \frac{1}{2} (N_{+} - N_{-}). \end{aligned} \quad (17.63)$$

Using $[a_{\mu}^{\dagger} a_{\nu}, a_{\rho}^{\dagger} a_{\sigma}] = \delta_{\nu\rho} a_{\mu}^{\dagger} a_{\sigma} - \delta_{\mu\sigma} a_{\rho}^{\dagger} a_{\nu}$, one verifies that $\mathbf{J}$ satisfies the angular momentum Lie algebra:

$$[J_i, J_j] = i\varepsilon_{ijk} J_k. \quad (17.64)$$

17.5.2 Angular Momentum Multiplets and Raising/Lowering Operators

The Casimir operator evaluates to

$$\begin{aligned} \mathbf{J}^2 &= J_x^2 + J_y^2 + J_z^2 = \frac{1}{2} (J_{+} J_{-} + J_{-} J_{+}) + J_z^2 \\ &= \frac{N}{2} \left(\frac{N}{2} + 1 \right), \end{aligned} \quad (17.65)$$

where $N = N_{+} + N_{-}$ is the total number operator.

Operating on a Fock state $|n_{+}, n_{-}\rangle$:

$$\begin{aligned} \mathbf{J}^2 |n_{+}, n_{-}\rangle &= j(j+1) |n_{+}, n_{-}\rangle, \\ J_z |n_{+}, n_{-}\rangle &= m |n_{+}, n_{-}\rangle, \end{aligned} \quad (17.66)$$

with the quantum numbers identified as

$$j = \frac{n_{+} + n_{-}}{2}, \quad m = \frac{n_{+} - n_{-}}{2}. \quad (17.67)$$

Inverting this map gives the number of quanta in terms of the angular momentum labels:

$$n_+ = j + m, \quad n_- = j - m. \quad (17.68)$$

Each (+) quantum contributes +1/2 unit of angular momentum along z , while each (−) quantum contributes −1/2. The angular momentum eigenstates are expressed in Fock space as

$$|j, m\rangle = \frac{(a_+^\dagger)^{j+m}}{\sqrt{(j+m)!}} \frac{(a_-^\dagger)^{j-m}}{\sqrt{(j-m)!}} |0, 0\rangle. \quad (17.69)$$

The ladder operators take the form of particle-hole excitations between the two oscillator modes:

$$J_+ = a_+^\dagger a_-, \quad J_- = a_-^\dagger a_+. \quad (17.70)$$

Acting with J_+ on $|j, m\rangle$ destroys a (−) quantum and creates a (+) quantum, increasing m by 1 while keeping $j = (n_+ + n_-)/2$ invariant:

$$\begin{aligned} J_+ |j, m\rangle &= a_+^\dagger a_- \frac{(a_+^\dagger)^{j+m}}{\sqrt{(j+m)!}} \frac{(a_-^\dagger)^{j-m}}{\sqrt{(j-m)!}} |0, 0\rangle \\ &= \sqrt{j(j+1) - m(m+1)} |j, m+1\rangle. \end{aligned} \quad (17.71)$$

In addition, one can construct inter-multiplet ladder operators:

$$K_+ = a_+^\dagger a_-^\dagger, \quad K_- = a_+ a_-, \quad (17.72)$$

which shift $j \rightarrow j \pm 1$ while holding the magnetic projection m fixed:

$$K_+ |j, m\rangle = \sqrt{(j+m+1)(j-m+1)} |j+1, m\rangle. \quad (17.73)$$

17.5.3 Evaluation of Wigner d -Matrices via Generating Functions

The Jordan–Schwinger construction facilitates the closed-form evaluation of the reduced Wigner rotation matrix:

$$d_{mm'}^{(j)}(\theta) = \langle j, m | e^{-i\theta J_y} | j, m' \rangle. \quad (17.74)$$

Under a finite rotation about the y -axis, the spinor annihilation/creation operators transform according to the fundamental spin-1/2 representation:

$$\begin{aligned} e^{-i\theta J_y} a_+^\dagger e^{i\theta J_y} &= a_+^\dagger \cos\left(\frac{\theta}{2}\right) + a_-^\dagger \sin\left(\frac{\theta}{2}\right), \\ e^{-i\theta J_y} a_-^\dagger e^{i\theta J_y} &= a_-^\dagger \cos\left(\frac{\theta}{2}\right) - a_+^\dagger \sin\left(\frac{\theta}{2}\right). \end{aligned} \quad (17.75)$$

Introducing auxiliary complex parameters (x_+, x_-) , we define the generating function:

$$\begin{aligned} G(x_+, x_-) &\equiv \sum_{m'=-j}^j \frac{x_+^{j+m'}}{\sqrt{(j+m')!}} \frac{x_-^{j-m'}}{\sqrt{(j-m')!}} d_{mm'}^{(j)}(\theta) \\ &= \frac{1}{(2j)!} \langle j, m | e^{-i\theta J_y} (x_+ a_+^\dagger + x_- a_-^\dagger)^{2j} | 0, 0 \rangle. \end{aligned} \quad (17.76)$$

Applying Eq. (17.75) inside the expectation value and expanding via the binomial theorem yields

$$G(x_+, x_-) = \frac{(x_+ \cos \frac{\theta}{2} - x_- \sin \frac{\theta}{2})^{j+m}}{\sqrt{(j+m)!}} \frac{(x_- \cos \frac{\theta}{2} + x_+ \sin \frac{\theta}{2})^{j-m}}{\sqrt{(j-m)!}}. \quad (17.77)$$

Matching powers of x_+ and x_- leads directly to the explicit representation of $d_{mm'}^{(j)}(\theta)$ in terms of Jacobi polynomials $P_n^{(\alpha, \beta)}(\cos \theta)$:

$$d_{mm'}^{(j)}(\theta) = \sqrt{\frac{(j+m')!(j-m')!}{(j+m)!(j-m)!}} \sum_k \binom{j+m}{k} \binom{j-m}{j-m'-k} (-1)^{j-m'-k} \\ \times \left(\cos \frac{\theta}{2}\right)^{2k+m-m'} \left(\sin \frac{\theta}{2}\right)^{2j-2k-m+m'}. \quad (17.78)$$

17.5.4 Generalization to $SU(3)$ and the Quark Model

Schwinger's two-oscillator model of $SU(2)$ naturally generalizes to three harmonic oscillators a_1, a_2, a_3 satisfying $[a_i, a_j^\dagger] = \delta_{ij}$. The eight bilinear generators

$$F_a = \frac{1}{2} a^\dagger \lambda_a a \quad (a = 1, \dots, 8), \quad (17.79)$$

where λ_a are the Gell-Mann matrices, generate the Lie algebra of $SU(3)_{\text{flavor}}$. The quanta of the three underlying harmonic oscillators correspond physically to the constituent quarks (u, d, s) of Gell-Mann and Zweig, with hadrons emerging as composite multi-quark states.

17.6 Problems

1. **Euler Angles for an Arbitrary Axis Rotation:** Determine the Euler angles (ϕ, θ, ψ) corresponding to a single finite rotation through an angle $\alpha = |\boldsymbol{\alpha}|$ about an arbitrary axis specified by the unit vector $\hat{\mathbf{n}} = (\sin \Theta \cos \Phi, \sin \Theta \sin \Phi, \cos \Theta)$.
2. **Addition Theorem for Spherical Harmonics:** Complete the derivation showing that the transformation law of spherical harmonics under finite rotations, Eq. (17.16), reduces for $m = 0$ to the spherical harmonic addition theorem:

$$P_l(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}') = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}(\theta, \phi) Y_{lm}^*(\theta', \phi'). \quad (17.80)$$

3. **Connection Between Wigner d -Matrices and Spherical Harmonics:** Using the general Jacobi polynomial formula for the small Wigner rotation matrix $d_{mm'}^{(j)}(\theta)$, Eq. (17.78), verify directly that for integer $j = l$ and $m' = 0$:

$$d_{m0}^{(l)}(\theta) = (-1)^m \sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) = \sqrt{\frac{4\pi}{2l+1}} Y_{lm}^*(\theta, 0). \quad (17.81)$$

Chapter 18

Identical Particles

18.1 Identical Particles: Indistinguishability and Symmetric Observables

In classical mechanics, identical particles (such as identical billiard balls) remain fundamentally distinguishable because each particle traces a unique, continuous trajectory through phase space. One can conceptually paint an identifying label on each particle without altering its dynamical evolution. In quantum mechanics, however, identical microscopic particles—such as electrons, photons, or neutrons—are completely and fundamentally *indistinguishable*.

18.1.1 Quantum Indistinguishability

The impossibility of distinguishing identical particles stems directly from the quantum uncertainty principle and the wave nature of matter:

1. Elementary particles possess no internal marker or distinguishing degrees of freedom beyond their intrinsic universal properties (mass, charge, spin).
2. If the spatial wave packets of two identical particles overlap, tracing continuous classical trajectories would require continuous position measurements. By the Heisenberg uncertainty relation $\Delta x \Delta p \geq \hbar/2$, each measurement imparts an uncontrollable momentum disturbance that irrevocably destroys phase coherence and alters the physical state.

Consequently, if one identical particle is localized at position $\mathbf{r}_A$ and another at $\mathbf{r}_B$, the configuration with particle 1 at $\mathbf{r}_A$ and particle 2 at $\mathbf{r}_B$ is physically identical to the configuration with particle 2 at $\mathbf{r}_A$ and particle 1 at $\mathbf{r}_B$.

18.1.2 The N -Particle Wave Function

Consider a system of n identical particles. In the coordinate-spin representation, the state vector $|\Psi\rangle$ is represented by the many-body wave function:

$$\psi(1, 2, \dots, n) = \langle 1, 2, \dots, n | \Psi \rangle, \quad (18.1)$$

where the composite index $i \equiv (\mathbf{r}_i, s_i)$ denotes both the spatial coordinate $\mathbf{r}_i$ and the discrete spin projection s_i (e.g., $s_z = \pm 1/2$ for spin-1/2 fermions) of the i -th coordinate argument.

Here, the label “ i -th particle” refers strictly to the argument occupying the i -th slot in the mathematical wave function ψ . All physical predictions and measurement probabilities must treat every identical particle on a completely equivalent footing.

18.1.3 Symmetric Observables

Because no physical interaction can distinguish between identical particles, every physical observable $A(1, 2, \dots, n)$ must be a completely symmetric operator under the interchange of any pair of particle coordinates:

$$A(\dots, i, \dots, j, \dots) = A(\dots, j, \dots, i, \dots). \quad (18.2)$$

Fundamental examples of symmetric operators include:

- **The n -body Hamiltonian:** For non-interacting particles of mass m in an external potential $V(\mathbf{r})$:

$$H_0 = \sum_{i=1}^n \left[\frac{\mathbf{p}_i^2}{2m} + V(\mathbf{r}_i) \right], \quad (18.3)$$

or with two-body mutual interactions:

$$H = \sum_{i=1}^n \frac{\mathbf{p}_i^2}{2m} + \sum_{i < j}^n v(\mathbf{r}_i - \mathbf{r}_j). \quad (18.4)$$

- **The Total Momentum Operator:**

$$\mathbf{P} = \sum_{i=1}^n \mathbf{p}_i. \quad (18.5)$$

- **The Particle Density Operator:**

$$\rho(\mathbf{r}) = \sum_{i=1}^n \delta(\mathbf{r} - \mathbf{r}_i). \quad (18.6)$$

- **The Paramagnetic Current Density Operator:**

$$\mathbf{J}(\mathbf{r}) = \frac{1}{2m} \sum_{i=1}^n [\mathbf{p}_i \delta(\mathbf{r} - \mathbf{r}_i) + \delta(\mathbf{r} - \mathbf{r}_i) \mathbf{p}_i]. \quad (18.7)$$

Because all physical observables are symmetric, any permutation of identical particles leaves every physical measurement expectation value invariant.

18.2 Permutations and Symmetry

To formulate the quantum mechanics of identical particles quantitatively, it is convenient to introduce operators that carry out permutations among the particles.

18.2.1 Permutation Operators

We define the transposition (or two-particle interchange) operator P_{ij} by its action on an n -particle state $|\Psi\rangle$: it interchanges the states of the i -th and j -th particles. In the coordinate-spin representation, where the state vector $|\Psi\rangle$ has the wave function $\psi(1, 2, \dots, n) = \langle 1, 2, \dots, n | \Psi \rangle$ with $i \equiv (\mathbf{r}_i, s_i)$, this definition reads:

$$\langle 1, 2, \dots, i, \dots, j, \dots, n | P_{ij} | \Psi \rangle = \langle 1, 2, \dots, j, \dots, i, \dots, n | \Psi \rangle. \quad (18.8)$$

Acting directly on the wave function, the representation of P_{ij} in configuration space yields:

$$P_{ij}\psi(1, 2, \dots, i, \dots, j, \dots, n) = \psi(1, 2, \dots, j, \dots, i, \dots, n). \quad (18.9)$$

In other words, P_{ij} evaluates the amplitude for observing the particle in the j -th slot at position $\mathbf{r}_i$ with spin s_i , and the particle in the i -th slot at $\mathbf{r}_j$ with spin s_j .

More general permutations can be denoted by cyclic permutation operators $P_{abc\dots k}$. Under $P_{abc\dots k}$, particle b is replaced by particle a , particle c by particle b , and finally particle a by particle k . In terms of the wave function arguments, the a -th slot is replaced by the b -th, the b -th by the c -th, and so forth:

$$P_{123}\psi(1, 2, 3, \dots, n) = \psi(2, 3, 1, \dots, n). \quad (18.10)$$

For example, for the test function $\psi(1, 2, 3) = r_1^2(r_2^2 - r_3^2)$, applying the transposition P_{12} gives $P_{12}\psi(1, 2, 3) = r_2^2(r_1^2 - r_3^2)$, while applying the cyclic permutation P_{123} yields $P_{123}\psi(1, 2, 3) = r_2^2(r_3^2 - r_1^2)$.

Every permutation can be decomposed into a product of simple transpositions P_{ij} . For instance,

$$P_{12}P_{13}\psi(1, 2, 3) = P_{12}\psi(3, 2, 1) = \psi(2, 3, 1) = P_{123}\psi(1, 2, 3), \quad (18.11)$$

so that $P_{123} = P_{12}P_{13}$. Permutation operators do not, in general, commute with one another:

$$P_{12}P_{13}\psi(1, 2, 3) = \psi(2, 3, 1), \quad \text{whereas} \quad P_{13}P_{12}\psi(1, 2, 3) = \psi(3, 1, 2). \quad (18.12)$$

Hence, the permutation group S_n for $n \geq 3$ is non-Abelian. The inverse of an arbitrary cycle $P_{abc\dots k}$ is the reversed cycle $P_{k\dots cba}$; for any two-particle transposition, the operator is Hermitian, unitary, and an involution:

$$P_{ij} = P_{ji} = P_{ij}^\dagger = P_{ij}^{-1}, \quad P_{ij}^2 = \mathbb{I}. \quad (18.13)$$

18.2.2 Action on Operators and Exchange Degeneracy

Let $A(1, 2, \dots, n)$ be an arbitrary n -particle operator, not necessarily symmetric. Under conjugation by the transposition P_{ij} , the operator transforms as:

$$P_{ij}A(1, 2, \dots, i, \dots, j, \dots, n) = A(1, 2, \dots, j, \dots, i, \dots, n)P_{ij}. \quad (18.14)$$

To illustrate this identity, consider the single-particle momentum operator $\mathbf{p}_1$ in a two-particle system ($n = 2$). For an arbitrary state $|\Psi\rangle$:

$$\begin{aligned} \langle \mathbf{r}_1 s_1, \mathbf{r}_2 s_2 | P_{12} \mathbf{p}_1 | \Psi \rangle &= \langle \mathbf{r}_2 s_2, \mathbf{r}_1 s_1 | \mathbf{p}_1 | \Psi \rangle \\ &= \frac{\hbar}{i} \nabla_2 \langle \mathbf{r}_2 s_2, \mathbf{r}_1 s_1 | \Psi \rangle \\ &= \frac{\hbar}{i} \nabla_2 \langle \mathbf{r}_1 s_1, \mathbf{r}_2 s_2 | P_{12} | \Psi \rangle \\ &= \langle \mathbf{r}_1 s_1, \mathbf{r}_2 s_2 | \mathbf{p}_2 P_{12} | \Psi \rangle. \end{aligned} \quad (18.15)$$

Because this relation holds for all basis states and vectors $|\Psi\rangle$, it follows that $P_{12}\mathbf{p}_1 = \mathbf{p}_2 P_{12}$. More generally, in configuration space:

$$\begin{aligned} P_{ij}A(\dots, i, \dots, j, \dots)\psi(\dots, i, \dots, j, \dots) &= A(\dots, j, \dots, i, \dots)\psi(\dots, j, \dots, i, \dots) \\ &= A(\dots, j, \dots, i, \dots)P_{ij}\psi(\dots, i, \dots, j, \dots), \end{aligned} \quad (18.16)$$

which establishes Eq. (18.14).

If the physical observable A is a *symmetric operator*, $A(\dots, i, \dots, j, \dots) = A(\dots, j, \dots, i, \dots)$, Eq. (18.14) implies that A commutes with every permutation operator P :

$$[P, A] = 0 \iff PAP^{-1} = A. \quad (18.17)$$

Because any physical observable must be symmetric, the expectation value of any observable is strictly invariant under relabeling the particle coordinates:

$$\langle \Psi | A | \Psi \rangle = \sum_{s_1, \dots, s_n} \int d^3r_1 \dots d^3r_n \psi^*(1, \dots, n) A(1, \dots, n) \psi(1, \dots, n) \quad (18.18)$$

is unchanged under any permutation of the coordinates $1, \dots, n$. As long as observables are symmetric, no experimental measurement can differentiate particle permutations, irrespective of the underlying state symmetry.

Furthermore, if $|\Psi\rangle$ is an energy eigenstate of a symmetric Hamiltonian H with eigenvalue E , then:

$$HP|\Psi\rangle = PH|\Psi\rangle = P(E|\Psi\rangle) = EP|\Psi\rangle. \quad (18.19)$$

Thus, $P|\Psi\rangle$ is also an eigenstate of H with the same energy E . If $P|\Psi\rangle$ is linearly independent of $|\Psi\rangle$, the energy level exhibits *exchange degeneracy*. As we now establish, nature eliminates this degeneracy by restricting physical Hilbert spaces to one-dimensional representations of the permutation group.

18.2.3 Two-Particle Systems: Bosons and Fermions

Consider a two-particle system governed by a symmetric Hamiltonian $H(1, 2)$. Since $[P_{12}, H] = 0$, H and P_{12} share a common eigenbasis. Because $P_{12}^2 = \mathbb{I}$, the eigenvalues of P_{12} are restricted to $+1$ and -1 .

For any arbitrary energy eigenfunction $\psi(1, 2)$ with eigenvalue E , we can project out eigenstates of P_{12} :

$$\psi_S(1, 2) = \psi(1, 2) + \psi(2, 1) \quad (\text{symmetric, } P_{12} = +1), \quad (18.20)$$

$$\psi_A(1, 2) = \psi(1, 2) - \psi(2, 1) \quad (\text{antisymmetric, } P_{12} = -1). \quad (18.21)$$

Direct computation verifies:

$$P_{12}\psi_S(1, 2) = \psi(2, 1) + \psi(1, 2) = +\psi_S(1, 2), \quad (18.22)$$

$$P_{12}\psi_A(1, 2) = \psi(2, 1) - \psi(1, 2) = -\psi_A(1, 2). \quad (18.23)$$

Both ψ_S and ψ_A are valid eigenstates of H with eigenvalue E .

Experimentally, the wave function of identical particles is *always* an eigenstate of P_{12} , and the corresponding eigenvalue (± 1) depends exclusively on the intrinsic nature of the particles:

- **Fermions:** Particles whose wave functions are strictly antisymmetric under interchange:

$$P_{12}\psi_A(1, 2) = -\psi_A(1, 2). \quad (18.24)$$

Examples include electrons, protons, and neutrons.

- **Bosons:** Particles whose wave functions are strictly symmetric under interchange:

$$P_{12}\psi_S(1, 2) = +\psi_S(1, 2). \quad (18.25)$$

Examples include photons, π -mesons, and ^4He nuclei.

18.2.4 The Spin-Statistics Connection and n -Particle Systems

The symmetry under particle interchange is fundamentally tied to the intrinsic angular momentum (spin) of the particle via the celebrated *spin-statistics theorem*:

1. Particles with **integer spin** ($s = 0, 1, 2, \dots$) are bosons and possess completely symmetric state vectors.
2. Particles with **half-integer spin** ($s = \frac{1}{2}, \frac{3}{2}, \dots$) are fermions and possess completely antisymmetric state vectors.

In nonrelativistic quantum mechanics, this relationship is introduced as an empirical postulate; its rigorous justification arises from the axioms of relativistic quantum field theory (microcausality and the positive-definiteness of the energy spectrum).

For an assembly of n identical fermions, the total wave function $\psi(1, 2, \dots, n)$ is completely antisymmetric under the transposition of any two coordinate arguments:

$$P_{ij}\psi(1, \dots, i, \dots, j, \dots, n) = -\psi(1, \dots, i, \dots, j, \dots, n), \quad \forall i \neq j, \quad (18.26)$$

or equivalently $P_{ij}|\Psi\rangle = -|\Psi\rangle$. For example, the 47-electron wave function of a silver atom is totally antisymmetric under the interchange of any pair of electron coordinates.

For an assembly of n identical bosons, the wave function is completely symmetric under any transposition:

$$P_{ij}\psi(1, \dots, i, \dots, j, \dots, n) = +\psi(1, \dots, i, \dots, j, \dots, n), \quad \forall i \neq j, \quad (18.27)$$

or $P_{ij}|\Psi\rangle = +|\Psi\rangle$.¹

The permutation symmetry of a quantum state is a constant of motion. If a system starts in a symmetric (or antisymmetric) state, any physical interaction Hamiltonian $V(1, 2, \dots, n)$ is symmetric and therefore commutes with all permutation operators:

$$P_{ij}V|\Psi\rangle = VP_{ij}|\Psi\rangle = \pm V|\Psi\rangle. \quad (18.28)$$

Consequently, transitions between symmetric and antisymmetric states are strictly forbidden.

18.2.5 Spatial Localization and Wave Function Overlap

A fundamental conceptual question arises: if every electron in the universe must belong to a globally antisymmetrized wave function, why can we successfully describe a single electron in a terrestrial hydrogen atom (e.g., in Urbana) using an isolated one-particle Schrödinger equation, without antisymmetrizing against all other electrons (say, in London)?

The answer lies in the spatial overlap of the single-particle wave functions. Suppose electron U is localized in Urbana with normalized spatial wave function $\varphi_U(\mathbf{r})$, while electron L is localized in London with normalized wave function $\varphi_L(\mathbf{r})$. Because the laboratories are macroscopically separated, the wave functions have negligible spatial overlap:

$$\varphi_U(\mathbf{r})\varphi_L(\mathbf{r}) \approx 0 \quad \text{for all } \mathbf{r}. \quad (18.29)$$

The properly antisymmetrized, normalized two-electron wave function (suppressing spin for simplicity) is:

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} [\varphi_U(\mathbf{r}_1)\varphi_L(\mathbf{r}_2) - \varphi_U(\mathbf{r}_2)\varphi_L(\mathbf{r}_1)]. \quad (18.30)$$

¹For $n \geq 3$, the symmetric and antisymmetric representations are the only one-dimensional representations of the permutation group S_n . While multidimensional representations (parastatistics) are mathematically conceivable, in nature all known elementary particles belong strictly to either Bose-Einstein or Fermi-Dirac statistics.

The total probability density $P(\mathbf{r})$ of detecting an electron at position $\mathbf{r}$ is the sum of detecting particle 1 at $\mathbf{r}$ (integrated over $\mathbf{r}_2$) and particle 2 at $\mathbf{r}$ (integrated over $\mathbf{r}_1$):

$$\begin{aligned} P(\mathbf{r}) &= \int d^3r_2 |\psi(\mathbf{r}, \mathbf{r}_2)|^2 + \int d^3r_1 |\psi(\mathbf{r}_1, \mathbf{r})|^2 \\ &= |\varphi_U(\mathbf{r})|^2 \int d^3r_2 |\varphi_L(\mathbf{r}_2)|^2 + |\varphi_L(\mathbf{r})|^2 \int d^3r_1 |\varphi_U(\mathbf{r}_1)|^2 \\ &\quad - 2 \operatorname{Re} \left[\varphi_U(\mathbf{r}) \varphi_L^*(\mathbf{r}) \int d^3r' \varphi_U^*(r') \varphi_L(r') \right]. \end{aligned} \quad (18.31)$$

Because $\varphi_U(\mathbf{r})\varphi_L^*(\mathbf{r}) = 0$ everywhere, the exchange interference term identically vanishes. Due to normalization ($\int |\varphi_{U,L}|^2 d^3r = 1$), Eq. (18.31) reduces exactly to:

$$P(\mathbf{r}) = |\varphi_U(\mathbf{r})|^2 + |\varphi_L(\mathbf{r})|^2. \quad (18.32)$$

If the measurement is conducted in Urbana, where $\varphi_L(\mathbf{r}) = 0$, the probability density is simply:

$$P(\mathbf{r}) = |\varphi_U(\mathbf{r})|^2. \quad (18.33)$$

This is precisely the result obtained by treating the Urbana electron as a single, isolated particle. Thus, (anti)symmetrization is physically relevant only among particles whose wave packets have non-vanishing spatial overlap. In the classical limit, particle trajectories are sharply localized and never overlap, which explains why classical statistical mechanics does not exhibit exchange effects.

18.2.6 Composite Particles

The statistics of composite systems (atoms, molecules, nuclei) follow directly from the statistics of their constituent elementary particles.

Consider two hydrogen atoms, each consisting of an electron and a proton. Let the total wave function be $\psi(\mathbf{r}_{e1}, \mathbf{r}_{p1}; \mathbf{r}_{e2}, \mathbf{r}_{p2})$. Because electrons and protons are fermions, the wave function must be antisymmetric under electron exchange ($\mathbf{r}_{e1} \leftrightarrow \mathbf{r}_{e2}$) and under proton exchange ($\mathbf{r}_{p1} \leftrightarrow \mathbf{r}_{p2}$). Interchanging the two complete hydrogen atoms corresponds to simultaneously interchanging both constituent pairs:

$$\begin{aligned} \psi(\mathbf{r}_{e2}, \mathbf{r}_{p2}; \mathbf{r}_{e1}, \mathbf{r}_{p1}) &= -\psi(\mathbf{r}_{e1}, \mathbf{r}_{p2}; \mathbf{r}_{e2}, \mathbf{r}_{p1}) \\ &= +\psi(\mathbf{r}_{e1}, \mathbf{r}_{p1}; \mathbf{r}_{e2}, \mathbf{r}_{p2}). \end{aligned} \quad (18.34)$$

The overall wave function remains unchanged; thus, atomic hydrogen behaves as a composite boson.

In general:

- A composite particle containing an **even number of fermions** (and any number of bosons) possesses integer total angular momentum and obeys **Bose-Einstein statistics**. For example, ${}^4\text{He}$ (2 protons, 2 neutrons, 2 electrons = 6 fermions) is a boson.
- A composite particle containing an **odd number of fermions** possesses half-integer total angular momentum and obeys **Fermi-Dirac statistics**. For example, ${}^3\text{He}$ (2 protons, 1 neutron, 2 electrons = 5 fermions) is a fermion.

18.2.7 The Pauli Exclusion Principle and Spin Multiplets

For fermions, total antisymmetry immediately dictates the *Pauli exclusion principle*. Since

$$\psi(2, 1, 3, \dots, n) = -\psi(1, 2, 3, \dots, n), \quad (18.35)$$

setting the coordinate arguments of two particles to be identical ($1 = 2$) gives $\psi = -\psi$, whence $\psi \equiv 0$. Consequently, two identical fermions have zero amplitude to be found at the same spatial point with identical spin projections.

For two spin-1/2 fermions, the combined spin space decomposes into a singlet subspace ($S = 0$, dimension 1) and a triplet subspace ($S = 1$, dimension 3). The singlet spin state is:

$$|\chi_{0,0}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad \chi_{0,0}(s_1, s_2) = \frac{1}{\sqrt{2}}[\chi_{\uparrow}(s_1)\chi_{\downarrow}(s_2) - \chi_{\downarrow}(s_1)\chi_{\uparrow}(s_2)], \quad (18.36)$$

which is antisymmetric under spin exchange $s_1 \leftrightarrow s_2$. For the total wave function $\Psi(1, 2) = \psi(\mathbf{r}_1, \mathbf{r}_2)\chi(s_1, s_2)$ to be antisymmetric under simultaneous exchange of space and spin coordinates, a spin singlet requires a **symmetric spatial wave function**:

$$\psi(\mathbf{r}_2, \mathbf{r}_1) = +\psi(\mathbf{r}_1, \mathbf{r}_2). \quad (18.37)$$

Conversely, the three triplet states are:

$$\begin{aligned} |\chi_{1,1}\rangle &= |\uparrow\uparrow\rangle, \\ |\chi_{1,0}\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \\ |\chi_{1,-1}\rangle &= |\downarrow\downarrow\rangle, \end{aligned} \quad (18.38)$$

which are symmetric under spin exchange $s_1 \leftrightarrow s_2$. Therefore, two spin-1/2 fermions in a spin-triplet state require an **antisymmetric spatial wave function**:

$$\psi(\mathbf{r}_2, \mathbf{r}_1) = -\psi(\mathbf{r}_1, \mathbf{r}_2). \quad (18.39)$$

This fundamental coupling between spatial symmetry and spin symmetry is the origin of the exchange interaction in atomic structure, ferromagnetism, and chemical bonding.

18.3 States of Noninteracting Identical Particles

To develop intuition for many-body wave functions and permutation symmetry, we consider an idealized system of N noninteracting identical particles confined within an external potential well $V(\mathbf{r})$.

18.3.1 The Many-Body Hamiltonian and Product States

The single-particle Hamiltonian for particle i is

$$H_0(i) = \frac{\mathbf{p}_i^2}{2m} + V(\mathbf{r}_i). \quad (18.40)$$

Let $\{\varphi_{\alpha}(\mathbf{r})\}$ denote the complete set of orthonormal stationary eigenstates of H_0 , with corresponding single-particle eigenenergies ϵ_{α} :

$$H_0\varphi_{\alpha}(\mathbf{r}) = \epsilon_{\alpha}\varphi_{\alpha}(\mathbf{r}), \quad \int d^3r \varphi_{\alpha}^*(\mathbf{r})\varphi_{\beta}(\mathbf{r}) = \delta_{\alpha\beta}. \quad (18.41)$$

Because the particles do not interact with one another, the total N -body Hamiltonian is simply the sum of individual single-particle Hamiltonians:

$$H = \sum_{i=1}^N H_0(i) = H_0(1) + H_0(2) + \cdots + H_0(N). \quad (18.42)$$

The stationary N -particle Schrödinger equation is

$$H\Psi(1, 2, \dots, N) = E\Psi(1, 2, \dots, N). \quad (18.43)$$

A straightforward separable solution is the simple Hartree product:

$$\Psi_{\text{prod}}(1, 2, \dots, N) = \varphi_a(1)\varphi_b(2) \cdots \varphi_n(N), \quad (18.44)$$

describing particle 1 in single-particle state a , particle 2 in state b , and so forth. Here, the coordinate-spin argument $i \equiv (\mathbf{r}_i, s_i)$ incorporates both spatial coordinates and spin projections. For spin-1/2 fermions, each spatial state $\varphi_\alpha(\mathbf{r})$ splits into two orthonormal spin-orbitals:

$$\varphi_{\alpha,\uparrow}(i) = \varphi_\alpha(\mathbf{r}_i)\chi_\uparrow(s_i), \quad \varphi_{\alpha,\downarrow}(i) = \varphi_\alpha(\mathbf{r}_i)\chi_\downarrow(s_i). \quad (18.45)$$

The total energy of the product state (18.44) is the sum of the individual orbital energies:

$$E = \epsilon_a + \epsilon_b + \cdots + \epsilon_n. \quad (18.46)$$

Although Ψ_{prod} satisfies the differential equation (18.43), it is not an admissible physical state for identical particles because it lacks definite permutation symmetry. Under any permutation $P \in S_N$ of the particle arguments, we obtain another state:

$$\Psi'(1, 2, \dots, N) = P\Psi_{\text{prod}}(1, 2, \dots, N) = \varphi_a(P(1))\varphi_b(P(2)) \cdots \varphi_n(P(N)), \quad (18.47)$$

which possesses the exact same total energy E . This exchange degeneracy must be resolved by constructing linear combinations that transform according to the physical one-dimensional representations of S_N .

18.3.2 Symmetric States for Bosons

For a system of identical bosons, the physical state must be totally symmetric under all permutations. We construct the symmetric state by summing over all $N!$ permutations of the symmetric group S_N :

$$\begin{aligned} \Psi_S(1, 2, \dots, N) &= \mathcal{N}_S \sum_{P \in S_N} P[\varphi_a(1)\varphi_b(2) \cdots \varphi_n(N)] \\ &= \mathcal{N}_S [\varphi_a(1)\varphi_b(2) \cdots \varphi_n(N) + \varphi_a(2)\varphi_b(1) \cdots \varphi_n(N) + \dots], \end{aligned} \quad (18.48)$$

where $\mathcal{N}_S$ is a normalization constant. For any transposition P_{ij} , the sum over all permutations is simply reordered, confirming that $P_{ij}\Psi_S = +\Psi_S$.

More generally, the symmetrization projection operator $\mathcal{S}$ acting on an arbitrary N -body wave function Ψ is defined by:

$$\Psi_S(1, 2, \dots, N) = \mathcal{S}\Psi(1, 2, \dots, N) \equiv \frac{1}{N!} \sum_{P \in S_N} P\Psi(1, 2, \dots, N). \quad (18.49)$$

To determine the normalization constant $\mathcal{N}_S$, let N_α denote the occupation number of the single-particle state φ_α in the configuration $\{a, b, \dots, n\}$, such that $\sum_\alpha N_\alpha = N$. Out of the $N!$ permutations in the sum, there are only $N! / \prod_\alpha N_\alpha!$ distinct products of single-particle wave functions, each appearing with multiplicity $\prod_\alpha N_\alpha!$. Computing the inner product yields:

$$\int d(1) \cdots d(N) \left| \sum_{P \in S_N} P\varphi_a(1) \cdots \varphi_n(N) \right|^2 = N! \prod_\alpha N_\alpha!, \quad (18.50)$$

where $\int d(i) \equiv \sum_{s_i} \int d^3r_i$. Therefore, the properly normalized symmetric N -boson wave function is:

$$\Psi_S(1, 2, \dots, N) = \frac{1}{\sqrt{N! \prod_\alpha N_\alpha!}} \sum_{P \in S_N} P[\varphi_a(1)\varphi_b(2) \cdots \varphi_n(N)]. \quad (18.51)$$

When all occupied single-particle orbitals are distinct ($N_\alpha = 0$ or 1), the prefactor reduces to $1/\sqrt{N!}$.

18.3.3 Antisymmetric States for Fermions: The Slater Determinant

For identical fermions, the physical state must be totally antisymmetric. Every permutation P can be decomposed into a sequence of transpositions; while the decomposition is not unique, the parity of the number of transpositions is an invariant property of P . We define the signature (or sign) of the permutation:

$$(-1)^P = \begin{cases} +1, & \text{if } P \text{ is an even permutation,} \\ -1, & \text{if } P \text{ is an odd permutation.} \end{cases} \quad (18.52)$$

The antisymmetrization projector $\mathcal{A}$ is given by:

$$\Psi_A(1, 2, \dots, N) = \mathcal{A}\Psi(1, 2, \dots, N) \equiv \frac{1}{N!} \sum_{P \in S_N} (-1)^P P\Psi(1, 2, \dots, N). \quad (18.53)$$

Applying $\mathcal{A}$ to the product state (18.44), we obtain the normalized antisymmetric wave function:

$$\begin{aligned} \Psi_A(1, 2, \dots, N) &= \frac{1}{\sqrt{N!}} \sum_{P \in S_N} (-1)^P P[\varphi_a(1)\varphi_b(2)\cdots\varphi_n(N)] \\ &= \frac{1}{\sqrt{N!}} [\varphi_a(1)\varphi_b(2)\cdots\varphi_n(N) - \varphi_a(2)\varphi_b(1)\cdots\varphi_n(N) \pm \dots]. \end{aligned} \quad (18.54)$$

Remarkably, this sum over permutations can be written compactly as a matrix determinant, known as the *Slater determinant*:

$$\Psi_A(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_a(1) & \varphi_a(2) & \cdots & \varphi_a(N) \\ \varphi_b(1) & \varphi_b(2) & \cdots & \varphi_b(N) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_n(1) & \varphi_n(2) & \cdots & \varphi_n(N) \end{vmatrix}. \quad (18.55)$$

The Slater determinant formulation illuminates key properties of fermionic systems:

1. **Pauli Exclusion Principle:** If any two single-particle orbitals are identical (e.g., $a = b$), two rows of the determinant become identical, and the determinant vanishes identically: $\Psi_A \equiv 0$. Thus, no two identical fermions can occupy the same quantum state.
2. **Antisymmetry under Coordinate Interchange:** Interchanging two particle arguments corresponds to swapping two columns of the determinant, which reverses its sign:

$$P_{ij}\Psi_A = -\Psi_A. \quad (18.56)$$

3. **Linear Independence:** If the set of single-particle orbitals $\{\varphi_a, \dots, \varphi_n\}$ is linearly dependent, the determinant vanishes.

To verify the normalization factor $1/\sqrt{N!}$, we compute the norm of the unnormalized sum $\tilde{\Psi}_A = \sum_P (-1)^P P\varphi_a(1)\cdots\varphi_n(N)$:

$$\langle \tilde{\Psi}_A | \tilde{\Psi}_A \rangle = \sum_{P, P' \in S_N} (-1)^P (-1)^{P'} \int d(1) \dots d(N) [P\varphi_a^*(1) \dots \varphi_n^*(N)] [P'\varphi_a(1) \dots \varphi_n(N)]. \quad (18.57)$$

By the orthogonality of the single-particle orbitals ($\int \varphi_\alpha^*(i)\varphi_\beta(i) d(i) = \delta_{\alpha\beta}$), the spatial and spin integrals vanish unless $P = P'$. When $P = P'$, each of the $N!$ terms integrates to unity:

$$\langle \tilde{\Psi}_A | \tilde{\Psi}_A \rangle = \sum_{P \in S_N} 1 = N!, \quad (18.58)$$

confirming that $\Psi_A = \tilde{\Psi}_A/\sqrt{N!}$ is properly normalized to unity.

18.3.4 Two-Particle Illustrative Example

To highlight the dramatic differences between distinguishable particles, bosons, and fermions, consider two particles and two single-particle levels $\varphi_0(\mathbf{r})$ and $\varphi_1(\mathbf{r})$ with energies $\epsilon_0 < \epsilon_1$.

1. **Distinguishable Particles:** Four independent orthogonal states exist:

$$\psi_{00}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_0(\mathbf{r}_1)\varphi_0(\mathbf{r}_2), \quad E = 2\epsilon_0, \quad (18.59)$$

$$\psi_{11}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_1(\mathbf{r}_1)\varphi_1(\mathbf{r}_2), \quad E = 2\epsilon_1,$$

$$\psi_{01}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_0(\mathbf{r}_1)\varphi_1(\mathbf{r}_2), \quad E = \epsilon_0 + \epsilon_1, \quad (18.60)$$

$$\psi_{10}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_1(\mathbf{r}_1)\varphi_0(\mathbf{r}_2), \quad E = \epsilon_0 + \epsilon_1.$$

The intermediate energy level $E = \epsilon_0 + \epsilon_1$ is twofold degenerate.

2. **Identical Bosons (Spin-0):** Only totally symmetric states are physically permissible:

$$\Psi_{00}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_0(\mathbf{r}_1)\varphi_0(\mathbf{r}_2), \quad E = 2\epsilon_0, \quad (18.61)$$

$$\Psi_{11}(\mathbf{r}_1, \mathbf{r}_2) = \varphi_1(\mathbf{r}_1)\varphi_1(\mathbf{r}_2), \quad E = 2\epsilon_1, \quad (18.62)$$

$$\Psi_{01}^S(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} [\varphi_0(\mathbf{r}_1)\varphi_1(\mathbf{r}_2) + \varphi_1(\mathbf{r}_1)\varphi_0(\mathbf{r}_2)], \quad E = \epsilon_0 + \epsilon_1. \quad (18.63)$$

There are only 3 distinct physical states; the twofold exchange degeneracy at $\epsilon_0 + \epsilon_1$ is eliminated.

3. **Identical Fermions (Spin-1/2):** If the two fermions share the same spin projection (e.g., $|\uparrow\uparrow\rangle$), the spin wave function is symmetric, forcing the spatial wave function to be strictly antisymmetric:

$$\Psi_{01}^A(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} [\varphi_0(\mathbf{r}_1)\varphi_1(\mathbf{r}_2) - \varphi_1(\mathbf{r}_1)\varphi_0(\mathbf{r}_2)], \quad E = \epsilon_0 + \epsilon_1. \quad (18.64)$$

States with both particles in φ_0 or both in φ_1 are forbidden by the exclusion principle for parallel spins. However, if the spins form an antisymmetric singlet state $|\chi_{0,0}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, the spatial state can be symmetric, allowing two fermions of opposite spin to doubly occupy the same spatial orbital:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2; s_1, s_2) = \varphi_0(\mathbf{r}_1)\varphi_0(\mathbf{r}_2)\chi_{0,0}(s_1, s_2), \quad E = 2\epsilon_0. \quad (18.65)$$

18.3.5 Ground State Structure: Macroscopic Condensation vs. Fermi Sea

The statistical symmetry fundamentally governs the macroscopic behavior of many-body systems in their ground states:

The Bose-Einstein Condensate

Because bosons are not subject to the exclusion principle, an arbitrary number of bosons can simultaneously occupy the single-particle ground state orbital φ_0 . For an N -boson system, the ground state wave function is simply:

$$\Psi_0(1, 2, \dots, N) = \varphi_0(1)\varphi_0(2) \cdots \varphi_0(N), \quad (18.66)$$

with total ground state energy

$$E_0^{\text{boson}} = N\epsilon_0. \quad (18.67)$$

Even for macroscopic numbers ($N \sim 10^{23}$ particles), all bosons condense into the lowest available single-particle state. This macroscopic occupation of a single quantum state underlies the phenomena of Bose-Einstein condensation and superfluidity in liquid ^4He .

The Fermi Sea and Degeneracy Pressure

For spin-1/2 fermions, the exclusion principle permits at most two particles (one with spin up, one with spin down) to occupy each spatial orbital $\varphi_k(\mathbf{r})$.

Consequently, to construct the N -particle ground state, fermions must successively fill the energy ladder $\epsilon_0 < \epsilon_1 < \epsilon_2 < \dots$ from the bottom up, creating a “Fermi sea”:

- **Even N :** The lowest $N/2$ spatial levels ($k = 0, 1, \dots, \frac{N}{2} - 1$) are doubly occupied. The ground state energy is:

$$E_0^{\text{fermion}} = 2 \sum_{k=0}^{\frac{N}{2}-1} \epsilon_k. \quad (18.68)$$

- **Odd N :** The lowest $(N-1)/2$ levels are doubly occupied, while the highest occupied orbital at level $(N-1)/2$ is singly occupied:

$$E_0^{\text{fermion}} = 2 \sum_{k=0}^{\frac{N-3}{2}} \epsilon_k + \epsilon_{\frac{N-1}{2}}. \quad (18.69)$$

Because particles are forced into progressively higher energy states, the fermionic ground state energy E_0^{fermion} is substantially higher than the bosonic ground state energy $N\epsilon_0$. This vast quantum kinetic energy, termed *degeneracy energy*, gives rise to electron degeneracy pressure, which stabilizes white dwarf stars, supports atomic structure against electrostatic collapse, and governs the transport properties of metals.

18.4 Scattering of Identical Particles

The quantum indistinguishability of identical particles produces profound, observable interference phenomena in scattering experiments.

18.4.1 Scattering of Spinless Bosons

Consider the elastic scattering of two identical spinless bosons, such as two α -particles (^4He nuclei, $s = 0$). In the center-of-mass (CM) frame, let $\mathbf{P}_{\text{tot}} = \hbar\mathbf{k}_1 + \hbar\mathbf{k}_2 = 0$ and define the relative coordinate $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. The total wave function separates as:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i\mathbf{P}_{\text{cm}} \cdot (\mathbf{r}_1 + \mathbf{r}_2)/2\hbar} \psi(\mathbf{r}), \quad (18.70)$$

where $\mathbf{P}_{\text{cm}}$ is the conserved center-of-mass momentum. Under spatial interchange $\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$, the relative vector transforms as $\mathbf{r} \rightarrow -\mathbf{r}$. Because bosons require a symmetric wave function $\Psi(\mathbf{r}_2, \mathbf{r}_1) = +\Psi(\mathbf{r}_1, \mathbf{r}_2)$, the relative wave function must possess *even spatial parity*:

$$\psi(-\mathbf{r}) = \psi(\mathbf{r}). \quad (18.71)$$

Under space inversion, the spherical harmonics transform as $Y_{lm}(-\hat{\mathbf{r}}) = (-1)^l Y_{lm}(\hat{\mathbf{r}})$. Consequently, only **even orbital angular momentum states** ($l = 0, 2, 4, \dots$) are physically permitted. All odd- l partial waves vanish identically.

For distinguishable particles, the standard asymptotic boundary condition for a scattering wave function with incident wavevector $\mathbf{k} = k\hat{\mathbf{z}}$ is

$$\psi_{\text{dist}}(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{r}} + f(\theta) \frac{e^{ikr}}{r}. \quad (18.72)$$

For identical bosons, we must explicitly symmetrize Eq. (18.72) under $\mathbf{r} \rightarrow -\mathbf{r}$. Since under inversion $\theta \rightarrow \pi - \theta$ and $r \rightarrow r$, the physical asymptotic wave function becomes:

$$\psi_S(\mathbf{r}) = \frac{1}{\sqrt{2}} [\psi_{\text{dist}}(\mathbf{r}) + \psi_{\text{dist}}(-\mathbf{r})] \propto e^{i\mathbf{k}\cdot\mathbf{r}} + e^{-i\mathbf{k}\cdot\mathbf{r}} + f_S(\theta) \frac{e^{ikr}}{r}, \quad (18.73)$$

where the symmetrized scattering amplitude is

$$f_S(\theta) = f(\theta) + f(\pi - \theta). \quad (18.74)$$

The two terms in $f_S(\theta)$ have a transparent physical interpretation: a detector positioned at scattering angle θ cannot distinguish whether:

1. The projectile was scattered through an angle θ while the target recoiled at $\pi - \theta$, or
2. The projectile was scattered through $\pi - \theta$ while the target was deflected forward into the detector at angle θ .

Because these two reaction pathways lead to identical final quantum states, their transition amplitudes add coherently.

The resulting differential cross section is:

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= |f_S(\theta)|^2 = |f(\theta) + f(\pi - \theta)|^2 \\ &= |f(\theta)|^2 + |f(\pi - \theta)|^2 + 2 \operatorname{Re} [f^*(\theta) f(\pi - \theta)]. \end{aligned} \quad (18.75)$$

In contrast, if the particles were distinguishable, the detector would register the incoherent classical sum of both possibilities:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{distinguishable}} = |f(\theta)|^2 + |f(\pi - \theta)|^2. \quad (18.76)$$

The cross term $2 \operatorname{Re} [f^*(\theta) f(\pi - \theta)]$ represents the purely quantum-mechanical *exchange interference*.

At a scattering angle of $\theta = 90^\circ$ ($\theta = \pi/2$), we have $f(\theta) = f(\pi - \theta) = f(\pi/2)$. The bosonic cross section is:

$$\left(\frac{d\sigma}{d\Omega} \right)_{90^\circ} = \left| 2f\left(\frac{\pi}{2}\right) \right|^2 = 4 \left| f\left(\frac{\pi}{2}\right) \right|^2, \quad (18.77)$$

which is exactly **twice** the classical/distinguishable result of $2|f(\pi/2)|^2$. This factor-of-two enhancement at 90° is a hallmark of Bose statistics, confirmed experimentally in low-energy α - α scattering.

In terms of the standard partial wave expansion,

$$f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta), \quad (18.78)$$

since $P_l(\cos(\pi - \theta)) = P_l(-\cos \theta) = (-1)^l P_l(\cos \theta)$, the symmetrized amplitude retains only even partial waves:

$$f_S(\theta) = \frac{2}{k} \sum_{l \text{ even}} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta). \quad (18.79)$$

18.4.2 Scattering of Spin-1/2 Fermions

Now consider the elastic scattering of two identical spin-1/2 fermions (such as electron-electron or proton-proton scattering) under a spin-independent central potential. The two-fermion wave function factors into spatial and spin parts:

$$\Psi(1, 2) = \psi(\mathbf{r}) \chi(s_1, s_2). \quad (18.80)$$

Total antisymmetry under simultaneous particle exchange ($1 \leftrightarrow 2$) requires:

Singlet Spin State ($S = 0$)

The singlet spin state $|\chi_{0,0}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ is antisymmetric under spin exchange. Hence, the spatial wave function must be symmetric: $\psi(-\mathbf{r}) = +\psi(\mathbf{r})$. Only **even orbital angular momenta** ($l = 0, 2, 4, \dots$) contribute. The scattering amplitude and cross section are identical to the spinless boson case:

$$f_{\text{singlet}}(\theta) = f(\theta) + f(\pi - \theta), \quad (18.81)$$

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{singlet}} = |f(\theta) + f(\pi - \theta)|^2. \quad (18.82)$$

Triplet Spin State ($S = 1$)

The three triplet spin states $\{|\uparrow\uparrow\rangle, \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), |\downarrow\downarrow\rangle\}$ are symmetric under spin exchange. Consequently, the relative spatial wave function must be *antisymmetric*:

$$\psi(-\mathbf{r}) = -\psi(\mathbf{r}), \quad (18.83)$$

which restricts orbital angular momentum to **odd values** ($l = 1, 3, 5, \dots$).

Antisymmetrizing the asymptotic relative wave function gives:

$$\psi_A(\mathbf{r}) \propto e^{i\mathbf{k}\cdot\mathbf{r}} - e^{-i\mathbf{k}\cdot\mathbf{r}} + f_A(\theta) \frac{e^{ikr}}{r}, \quad (18.84)$$

where the antisymmetric scattering amplitude is:

$$f_A(\theta) = f(\theta) - f(\pi - \theta). \quad (18.85)$$

The triplet differential cross section exhibits *destructive* exchange interference:

$$\begin{aligned} \left(\frac{d\sigma}{d\Omega}\right)_{\text{triplet}} &= |f_A(\theta)|^2 = |f(\theta) - f(\pi - \theta)|^2 \\ &= |f(\theta)|^2 + |f(\pi - \theta)|^2 - 2 \operatorname{Re}[f^*(\theta)f(\pi - \theta)]. \end{aligned} \quad (18.86)$$

At $\theta = 90^\circ$, destructive interference is complete:

$$f_A\left(\frac{\pi}{2}\right) = f\left(\frac{\pi}{2}\right) - f\left(\frac{\pi}{2}\right) = 0 \quad \Rightarrow \quad \left(\frac{d\sigma}{d\Omega}\right)_{\text{triplet}, 90^\circ} = 0. \quad (18.87)$$

Fermions prepared in a triplet spin state *cannot scatter at right angles* in the center-of-mass frame.

In terms of partial waves, the triplet amplitude contains only odd- l components:

$$f_A(\theta) = \frac{2}{k} \sum_{l \text{ odd}} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta). \quad (18.88)$$

Unpolarized Scattering

If the initial beam and target particles are unpolarized, all four spin states are populated with equal statistical probability $1/4$. The observed cross section is the weighted statistical average of 1 part singlet and 3 parts triplet:

$$\begin{aligned} \left(\frac{d\sigma}{d\Omega}\right)_{\text{unpol}} &= \frac{1}{4} \left(\frac{d\sigma}{d\Omega}\right)_{\text{singlet}} + \frac{3}{4} \left(\frac{d\sigma}{d\Omega}\right)_{\text{triplet}} \\ &= |f(\theta)|^2 + |f(\pi - \theta)|^2 - \operatorname{Re}[f^*(\theta)f(\pi - \theta)]. \end{aligned} \quad (18.89)$$

Equation (18.89) is the celebrated *Mott scattering cross section*.

18.4.3 Exchange Effects in Collisions of Composite Particles

Exchange symmetry also governs collisions between composite particles that appear distinguishable at first glance.

Consider the elastic scattering of a ${}^4\text{He}^+$ ion from a neutral ${}^4\text{He}$ atom in their ground states. Although the ion ($+e$) and atom (0) have different net charges, the neutral atom is a bound state of an identical ${}^4\text{He}$ nucleus (${}^4\text{He}^{++}$) and two electrons. Direct scattering of the incident ion by angle θ is quantum-mechanically indistinguishable from resonant *charge-transfer scattering*, in which the incident ion scatters by $\pi - \theta$ while capturing an electron from the target. The coherent total scattering amplitude is:

$$f_{\text{total}}(\theta) = f_{\text{direct}}(\theta) + f_{\text{exchange}}(\pi - \theta). \quad (18.90)$$

The interference between direct and charge-exchange amplitudes creates pronounced oscillatory fringes in the differential cross section.

18.4.4 Homonuclear Diatomic Molecules: Rotational Spectra

In homonuclear diatomic molecules, nuclear permutation symmetry constrains the allowed rotational energy levels:

$$E_{\text{rot}} = \frac{\hbar^2 l(l+1)}{2\mathcal{J}}, \quad (18.91)$$

where $\mathcal{J}$ is the molecular moment of inertia and l is the rotational quantum number.

- **Carbon Dimer (${}^{12}\text{C}$)₂:** The ${}^{12}\text{C}$ nucleus has spin $I = 0$ (boson). Nuclear interchange requires a totally symmetric state. Since the electronic ground state is symmetric (${}^1\Sigma_g^+$), the nuclear spatial wave function must have even parity under coordinate inversion $\mathbf{r} \rightarrow -\mathbf{r}$. Consequently, only **even rotational levels** ($l = 0, 2, 4, \dots$) exist in nature. Odd- l rotational spectral lines are completely missing.
- **Molecular Hydrogen (H_2):** Each proton has spin $I = 1/2$.

- **Parahydrogen:** Nuclear spin singlet ($I_{\text{tot}} = 0$, antisymmetric). Spatial wave function must be symmetric ($l = 0, 2, 4, \dots$). The rotational energies are:

$$E_{\text{rot, para}} = 0, 3\frac{\hbar^2}{\mathcal{J}}, 10\frac{\hbar^2}{\mathcal{J}}, 21\frac{\hbar^2}{\mathcal{J}}, \dots \quad (l = 0, 2, 4, 6, \dots). \quad (18.92)$$

The absolute ground state of the hydrogen molecule ($l = 0$) is parahydrogen.

- **Orthohydrogen:** Nuclear spin triplet ($I_{\text{tot}} = 1$, symmetric). Spatial wave function must be antisymmetric ($l = 1, 3, 5, \dots$). The rotational energies are:

$$E_{\text{rot, ortho}} = \frac{\hbar^2}{\mathcal{J}}, 6\frac{\hbar^2}{\mathcal{J}}, 15\frac{\hbar^2}{\mathcal{J}}, 28\frac{\hbar^2}{\mathcal{J}}, \dots \quad (l = 1, 3, 5, 7, \dots). \quad (18.93)$$

Because nuclear spin flips are mediated only by extremely weak magnetic dipole interactions, spontaneous ortho-para conversions have lifetimes on the order of months in the gas phase. This frozen spin-isomeric ratio explains the anomalous low-temperature specific heat of molecular hydrogen gas.

18.4.5 Three-Fermion Systems and Mixed Symmetry

For two fermions, total antisymmetry always reduces to pairing a symmetric space state with an antisymmetric spin state, or vice versa. For three or more fermions, richer permutational symmetries arise.

For three spin-1/2 fermions, the single-particle spin space has dimension 2 ($\uparrow$ and $\downarrow$). The total spin space has dimension $2^3 = 8$, which decomposes into:

$$\mathbf{2} \otimes \mathbf{2} \otimes \mathbf{2} = \mathbf{4}_{S=3/2} \oplus \mathbf{2}_{S=1/2} \oplus \mathbf{2}_{S=1/2}. \quad (18.94)$$

Because there are only two spin states available, one cannot construct a completely antisymmetric spin state for three particles (by the pigeonhole principle, at least two spins must be identical).

To achieve total antisymmetry of the combined wave function $\Psi(1, 2, 3)$, one may:

1. Combine a totally symmetric spin state ($S = 3/2$, quartet) with a totally antisymmetric spatial wave function $\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3)$, or
2. Form a state of **mixed symmetry** ($S = 1/2$, doublet). For example, let $\chi(s_1, s_2, s_3)$ be antisymmetric in spins 2 and 3:

$$\chi(s_1, s_2, s_3) = \chi_{\uparrow}(s_1) [\chi_{\uparrow}(s_2)\chi_{\downarrow}(s_3) - \chi_{\downarrow}(s_2)\chi_{\uparrow}(s_3)], \quad (18.95)$$

and let $\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3)$ be symmetric in $\mathbf{r}_2$ and $\mathbf{r}_3$. The cyclic linear combination

$$\begin{aligned} \Psi(1, 2, 3) = & \chi(s_1, s_2, s_3)\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) + \chi(s_2, s_3, s_1)\psi(\mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_1) \\ & + \chi(s_3, s_1, s_2)\psi(\mathbf{r}_3, \mathbf{r}_1, \mathbf{r}_2) \end{aligned} \quad (18.96)$$

is completely antisymmetric under the interchange of any two particles.

This mixed-symmetry construction is essential in the constituent quark model of baryons (such as the proton and neutron, which are $S = 1/2$ three-quark bound states).

18.5 Identity and Isotopic Spin

In nuclear and particle physics, the concept of indistinguishability extends naturally through the formalism of *isotopic spin* (isospin).

18.5.1 The Generalized Pauli Principle for Nucleons

Protons and neutrons have nearly identical masses ($m_n \approx 939.6 \text{ MeV}/c^2$, $m_p \approx 938.3 \text{ MeV}/c^2$) and experience identical strong nuclear interactions, a property known as the *charge independence* of nuclear forces. Heisenberg proposed treating the proton and neutron not as fundamentally distinct species, but as two orthogonal states of a single entity: the *nucleon*.

The nucleon is assigned an isospin of $I = 1/2$, with third component I_3 :

$$|p\rangle = |I = \frac{1}{2}, I_3 = +\frac{1}{2}\rangle, \quad |n\rangle = |I = \frac{1}{2}, I_3 = -\frac{1}{2}\rangle. \quad (18.97)$$

If nucleons are treated as identical particles within this larger internal space, how do we impose the Pauli exclusion principle? In physical reality, a state must be antisymmetric under the exchange of two protons or two neutrons, but is not required to be antisymmetric under the exchange of a proton and a neutron.

This requirement is fulfilled by the **Generalized Pauli Principle**:²

²See J. M. Blatt and V. F. Weisskopf, *Theoretical Nuclear Physics* (John Wiley & Sons, New York, 1952), Section III.5.

The complete state vector of an N -nucleon system must be totally antisymmetric under the simultaneous interchange of the spatial, spin, and isospin coordinates of any pair of nucleons:

$$P_{ij}^{(\text{space})} P_{ij}^{(\text{spin})} P_{ij}^{(\text{isospin})} |\Psi\rangle = -|\Psi\rangle. \quad (18.98)$$

18.5.2 Two-Nucleon Systems

Consider a system of two nucleons. The two-particle isospin space decomposes into an isospin singlet ($I = 0$) and an isospin triplet ($I = 1$):

$$\text{Isospin Singlet } (I = 0) : |\phi_{0,0}\rangle = \frac{1}{\sqrt{2}} (|pn\rangle - |np\rangle), \quad (18.99)$$

$$\text{Isospin Triplet } (I = 1) : \begin{cases} |\phi_{1,1}\rangle = |pp\rangle, \\ |\phi_{1,0}\rangle = \frac{1}{\sqrt{2}} (|pn\rangle + |np\rangle), \\ |\phi_{1,-1}\rangle = |nn\rangle. \end{cases} \quad (18.100)$$

Under coordinate interchange $1 \leftrightarrow 2$:

- The isospin singlet $|\phi_{0,0}\rangle$ is **antisymmetric**: $P_{12}^{(\text{isospin})} = -1$.
- The isospin triplet states $|\phi_{1,M_I}\rangle$ are **symmetric**: $P_{12}^{(\text{isospin})} = +1$.

Factoring the total two-nucleon state into spatial, spin, and isospin components:

$$\Psi(1, 2) = \psi_{\text{space}}(\mathbf{r}_1, \mathbf{r}_2) \chi_{\text{spin}}(s_1, s_2) \phi_{\text{isospin}}(\tau_1, \tau_2), \quad (18.101)$$

the permutation eigenvalue under coordinate exchange is:

$$P_{12} = (-1)^L \times (-1)^{S+1} \times (-1)^{I+1} = (-1)^{L+S+I+2} = (-1)^{L+S+I}. \quad (18.102)$$

Imposing total antisymmetry $P_{12} = -1$ yields the selection rule:

$$(-1)^{L+S+I} = -1 \quad \Longleftrightarrow \quad L + S + I \text{ is an odd integer.} \quad (18.103)$$

Implications for Nucleon Pairs

1. **Identical Nucleon Pairs (pp and nn):** Two protons ($I_3 = +1$) or two neutrons ($I_3 = -1$) can only exist in the isospin triplet state ($I = 1$). Therefore, $L + S + 1$ must be odd, which requires:

$$L + S \text{ is even.} \quad (18.104)$$

If the spins are in a singlet ($S = 0$), the orbital angular momentum must be even ($L = 0, 2, \dots$, corresponding to $^1S_0, ^1D_2, \dots$). If the spins are in a triplet ($S = 1$), L must be odd ($L = 1, 3, \dots$, corresponding to $^3P_{0,1,2}, ^3F_{2,3,4}, \dots$). This reproduces the ordinary Pauli principle.

2. **Proton-Neutron Pairs (pn):** A proton and a neutron can have either $I = 1$ (where $L + S$ must be even) or $I = 0$ (where $L + S$ must be odd). Consequently, no symmetry restrictions are placed on their space-spin wave function; both even and odd values of $L + S$ are physically allowed.

18.5.3 The Structure of the Deuteron

The deuteron is the only bound state of two nucleons, consisting of a bound proton and neutron. Experimentally, the deuteron has:

- Total angular momentum $J = 1$.
- Spin triplet configuration ($S = 1$).
- Positive parity, composed primarily of an orbital S -wave ($L = 0$, $\approx 96\%$) with a tensor-force-induced D -wave admixture ($L = 2$, $\approx 4\%$).

Because L is even (0 or 2) and $S = 1$, the space-spin wave function is symmetric under interchange:

$$P_{12}^{(\text{space})} P_{12}^{(\text{spin})} = (-1)^L (-1)^{S+1} = (+1)(+1) = +1. \quad (18.105)$$

By the generalized Pauli principle, the isospin wave function must therefore be **antisymmetric**:

$$I = 0 \quad (\text{isospin singlet}). \quad (18.106)$$

This explains why neither the diproton (pp) nor the dineutron (nn) exists as a bound nuclear system. Both pp and nn necessarily belong to $I = 1$; by the generalized Pauli principle, an $I = 1$ state with $S = 1$ would require odd orbital angular momentum ($L = 1, 3, \dots$). The strong nuclear force is attractive primarily in the S -wave channel ($L = 0$) and is insufficiently attractive in P -waves to form a bound state.

18.5.4 Identical Bosons and Pions

The generalized symmetry principle applies equally to bosonic systems. For identical bosons, the complete wave function must be totally symmetric under the simultaneous interchange of space, spin, and isospin coordinates:

$$P_{ij}^{(\text{space})} P_{ij}^{(\text{spin})} P_{ij}^{(\text{isospin})} |\Psi\rangle = +|\Psi\rangle. \quad (18.107)$$

For a pair of pions (π^+, π^0, π^-), each pion has spin $S = 0$ and isospin $I = 1$. The two-pion isospin states decompose as:

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{5}_{I=2} \oplus \mathbf{3}_{I=1} \oplus \mathbf{1}_{I=0}. \quad (18.108)$$

Under particle exchange:

- The $I = 0$ (singlet) and $I = 2$ (quintet) states are **symmetric** ($P_{12}^{(\text{isospin})} = +1$).
- The $I = 1$ (triplet) states are **antisymmetric** ($P_{12}^{(\text{isospin})} = -1$).

Because pions have zero intrinsic spin ($S = 0$), the symmetry condition requires:

$$(-1)^L \times (+1) \times (-1)^I = +1 \quad \Longleftrightarrow \quad L + I \text{ is an even integer.} \quad (18.109)$$

Hence:

- Two pions in an even relative orbital state ($L = 0, 2, \dots$) can only exist in isospin channels $I = 0$ or $I = 2$.
- Two pions in an odd relative orbital state ($L = 1, 3, \dots$) can only exist in the isospin channel $I = 1$.

This selection rule explains why the vector meson $\rho(770)$ (which has spin $J = 1$ and decays into $\pi\pi$ with orbital angular momentum $L = 1$) has isospin $I = 1$, while the scalar meson $f_0(500)$ ($J = 0$, $L = 0$) has isospin $I = 0$.

18.6 Problems

1. **Coulomb Scattering of Identical Particles:** Calculate the differential cross section for the Coulomb scattering of two identical α particles (spinless bosons with $Z = 2$). Compare this result with the differential cross section for the scattering of two electrons in:
 - (a) A spin singlet state,
 - (b) A spin triplet state, and
 - (c) An unpolarized beam and target configuration.

Express each cross section in terms of the classical Rutherford cross section and discuss the behavior at $\theta = 90^\circ$.

2. **Total Antisymmetry of Three-Fermion States:** Verify explicitly that the mixed-symmetry three-fermion wave function, Eq. (18.96):

$$\Psi(1, 2, 3) = \chi(s_1, s_2, s_3)\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) + \chi(s_2, s_3, s_1)\psi(\mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_1) + \chi(s_3, s_1, s_2)\psi(\mathbf{r}_3, \mathbf{r}_1, \mathbf{r}_2), \quad (18.110)$$

where $\chi(s_1, s_2, s_3)$ is antisymmetric in (s_2, s_3) and $\psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3)$ is symmetric in $(\mathbf{r}_2, \mathbf{r}_3)$, is completely antisymmetric under the transposition of any two particles ($1 \leftrightarrow 2$, $2 \leftrightarrow 3$, and $1 \leftrightarrow 3$).

3. **Low-Energy Nucleon-Nucleon Scattering:** Using the effective range expansion and the low-energy s -wave scattering parameters for singlet and triplet states, calculate explicitly the total cross section for the elastic scattering of:
 - (a) Unpolarized neutrons by unpolarized protons (n - p scattering), and
 - (b) Unpolarized neutrons by unpolarized neutrons (n - n scattering).

Explain how the Generalized Pauli Principle accounts for the differences between these two cross sections.

4. **Isotopic Spin States of Two Pions:** Construct the explicit Clebsch–Gordan decomposition for the isospin states $|I, M_I\rangle$ of a two-pion system (π^+, π^0, π^- with $I_1 = I_2 = 1$). For each total isospin multiplet ($I = 0, 1, 2$):
 - (a) Determine the symmetry of the isospin state under particle interchange.
 - (b) Deduce the required permutation symmetry of the spatial wave function.
 - (c) Identify the allowed values of the relative orbital angular momentum L .

Chapter 19

Second Quantization

19.1 Second Quantization: Introduction

When dealing with quantum systems consisting of only a few identical particles, one can explicitly construct symmetrized or antisymmetrized wave functions in configuration space, as demonstrated in Chapter 18. However, for macroscopic many-body systems—such as the conduction electrons in a metal ($N \sim 10^{23}$ fermions), nucleons in heavy nuclei, or atoms in liquid ^4He and cold atomic Bose-Einstein condensates—the direct manipulation of N -particle coordinate wave functions and $N!$ -term Slater determinants becomes extraordinarily cumbersome and computationally intractable.

To overcome these difficulties, we introduce the formal machinery of *second quantization* (the occupation-number representation). In this approach:

1. Identical particles are described not by labeled spatial coordinates, but by state vectors in *Fock space*, where states are characterized by the occupation numbers of single-particle quantum orbitals.
2. Symmetrization and antisymmetrization postulates are automatically and algebraically enforced by the canonical commutation relations (for bosons) or anticommutation relations (for fermions) of creation and annihilation operators.
3. Physical observables—including single-particle densities, currents, kinetic energy, and mutual two-body interactions—are expressed as compact field operators acting directly on Fock space.

In this chapter, we develop the foundations of second quantization and apply it to many-body quantum phenomena:

- **Creation and Annihilation Operators:** We define Fock space, introduce the vacuum state $|0\rangle$, and formulate the canonical commutation algebra for bosons and the canonical anticommutation algebra for fermions.
- **Second-Quantized Observables:** We represent one-body and two-body operators in terms of field operators $\psi(\mathbf{r})$ and $\psi^\dagger(\mathbf{r})$ in the coordinate and momentum representations.
- **Pair Correlation Functions and Quantum Statistics:** We evaluate the two-particle correlation function $g(\mathbf{r}_1, \mathbf{r}_2)$ for ideal Fermi and Bose gases, demonstrating spatial antibunching (the Pauli exchange hole) for fermions and spatial bunching for bosons.
- **The Hanbury Brown and Twiss Effect:** We explore intensity interferometry and photon correlation measurements, illustrating how quantum interference manifests in intensity fluctuations.

- **Many-Body Hamiltonians:** We formulate the full second-quantized Hamiltonian describing interacting many-particle systems, providing the foundation for modern condensed matter and nuclear theory.

19.2 Creation and Annihilation Operators

In the quantum harmonic oscillator, the ladder operators a and $a^\dagger$ respectively annihilate and create quanta of vibrational excitation. We now generalize this concept to many-body systems by introducing operators that remove and add physical *particles*. These creation and annihilation operators establish the occupation-number representation (or second quantization), where the permutation symmetry of identical particles is encoded directly into canonical operator algebras.

19.2.1 Single-Level Systems

To introduce the formalism, consider first an external potential well $V(\mathbf{r})$ with single-particle stationary eigenstates $\{\varphi_\alpha(\mathbf{r})\}$.

Bosons

Let $|n\rangle$ denote an n -boson state where all n particles occupy the lowest single-particle orbital $\varphi_0(\mathbf{r})$. Since bosons are unconstrained by the exclusion principle, n can be any non-negative integer ($n = 0, 1, 2, \dots$). The state $|0\rangle$ denotes the *vacuum state*, containing no particles.

We define the annihilation operator a_0 and creation operator $a_0^\dagger$ by:

$$a_0|n\rangle = \sqrt{n}|n-1\rangle, \quad (19.1)$$

$$a_0^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (19.2)$$

Acting on $|n\rangle$, the annihilation operator a_0 lowers the particle number by one, with $a_0|0\rangle = 0$. The creation operator $a_0^\dagger$ increments the particle number by one. Evaluating the commutator on an arbitrary state $|n\rangle$:

$$(a_0 a_0^\dagger - a_0^\dagger a_0)|n\rangle = [(n+1) - n]|n\rangle = |n\rangle, \quad (19.3)$$

which establishes the canonical Bose commutation relation:

$$[a_0, a_0^\dagger] \equiv a_0 a_0^\dagger - a_0^\dagger a_0 = \mathbb{I}. \quad (19.4)$$

By repeated application of $a_0^\dagger$ on the vacuum, any normalized state $|n\rangle$ can be constructed as:

$$|n\rangle = \frac{(a_0^\dagger)^n}{\sqrt{n!}}|0\rangle. \quad (19.5)$$

The operator $a_0^\dagger$ is the Hermitian adjoint of a_0 . In the dual bra space, their actions are reversed:

$$\langle n|a_0^\dagger = \sqrt{n}\langle n-1|, \quad \langle n|a_0 = \sqrt{n+1}\langle n+1|. \quad (19.6)$$

The particle number operator is defined by:

$$N_0 \equiv a_0^\dagger a_0, \quad N_0|n\rangle = n|n\rangle. \quad (19.7)$$

Fermions

Now suppose the particles are fermions, and consider a single spin-orbital φ_0 with fixed spin projection. By the Pauli exclusion principle, the orbital can accommodate at most one fermion. The Hilbert space is two-dimensional, spanned by the vacuum $|0\rangle$ and the singly occupied state $|1\rangle$.

The fermionic creation and annihilation operators are defined by:

$$a_0|0\rangle = 0, \quad a_0|1\rangle = |0\rangle, \quad (19.8)$$

$$a_0^\dagger|0\rangle = |1\rangle, \quad a_0^\dagger|1\rangle = 0. \quad (19.9)$$

The condition $a_0^\dagger|1\rangle = 0$ algebraically enforces the Pauli exclusion principle: one cannot create a second fermion in an already occupied state. In the ordered basis $\{|0\rangle, |1\rangle\}$, these operators have the 2×2 matrix representations:

$$a_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad a_0^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (19.10)$$

Evaluating the anticommutator $\{A, B\} \equiv AB + BA$ on both basis states:

$$\{a_0, a_0^\dagger\}|0\rangle = (a_0a_0^\dagger + a_0^\dagger a_0)|0\rangle = |0\rangle, \quad (19.11)$$

$$\{a_0, a_0^\dagger\}|1\rangle = (a_0a_0^\dagger + a_0^\dagger a_0)|1\rangle = |1\rangle. \quad (19.12)$$

Thus, fermionic operators satisfy the canonical anticommutation relations:

$$\{a_0, a_0^\dagger\} = \mathbb{I}, \quad \{a_0, a_0\} = 2a_0^2 = 0, \quad \{a_0^\dagger, a_0^\dagger\} = 2(a_0^\dagger)^2 = 0. \quad (19.13)$$

The nilpotent property $(a_0^\dagger)^2 = 0$ is the operator expression of the Pauli exclusion principle, while $a_0^2 = 0$ reflects the impossibility of removing two fermions from a single state. The number operator $N_0 = a_0^\dagger a_0$ satisfies $N_0|0\rangle = 0$ and $N_0|1\rangle = |1\rangle$, with eigenvalues $\{0, 1\}$.

19.2.2 Two-Level Systems and Permutation Parity

Next, consider particles that can occupy two distinct single-particle states, φ_0 and φ_1 .

Two-Level Bosons

The basis states are denoted by $|n_0, n_1\rangle$, where n_0 and n_1 are the occupation numbers of levels φ_0 and φ_1 , respectively. The ladder operators act independently on their respective levels:

$$\begin{aligned} a_0|n_0, n_1\rangle &= \sqrt{n_0}|n_0 - 1, n_1\rangle, & a_0^\dagger|n_0, n_1\rangle &= \sqrt{n_0 + 1}|n_0 + 1, n_1\rangle, \\ a_1|n_0, n_1\rangle &= \sqrt{n_1}|n_0, n_1 - 1\rangle, & a_1^\dagger|n_0, n_1\rangle &= \sqrt{n_1 + 1}|n_0, n_1 + 1\rangle. \end{aligned} \quad (19.14)$$

Because the order in which bosons are added to or removed from different levels does not affect the physical state, operators corresponding to different levels commute:

$$[a_i, a_j^\dagger] = \delta_{ij}\mathbb{I}, \quad [a_i, a_j] = 0, \quad [a_i^\dagger, a_j^\dagger] = 0 \quad (i, j \in \{0, 1\}). \quad (19.15)$$

Arbitrary Fock states are generated from the vacuum $|0, 0\rangle$ by:

$$|n_0, n_1\rangle = \frac{(a_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(a_0^\dagger)^{n_0}}{\sqrt{n_0!}} |0, 0\rangle. \quad (19.16)$$

The total particle number operator is

$$N = N_0 + N_1 = a_0^\dagger a_0 + a_1^\dagger a_1, \quad N|n_0, n_1\rangle = (n_0 + n_1)|n_0, n_1\rangle. \quad (19.17)$$

Two-Level Fermions

For fermions, the occupation numbers are restricted to $n_0, n_1 \in \{0, 1\}$, giving four basis states:

$$|0, 0\rangle, \quad |1, 0\rangle, \quad |0, 1\rangle, \quad |1, 1\rangle. \quad (19.18)$$

We define the ladder operators for orbital φ_1 by:

$$a_1^\dagger|0, 0\rangle = |0, 1\rangle, \quad a_1^\dagger|1, 0\rangle = |1, 1\rangle, \quad a_1^\dagger|0, 1\rangle = a_1^\dagger|1, 1\rangle = 0, \quad (19.19)$$

$$a_1|0, 1\rangle = |0, 0\rangle, \quad a_1|1, 1\rangle = |1, 0\rangle, \quad a_1|0, 0\rangle = a_1|1, 0\rangle = 0. \quad (19.20)$$

Similarly, when acting on states with $n_1 = 0$, $a_0^\dagger$ and a_0 act normally:

$$a_0^\dagger|0, 0\rangle = |1, 0\rangle, \quad a_0^\dagger|1, 0\rangle = 0, \quad a_0|1, 0\rangle = |0, 0\rangle, \quad a_0|0, 0\rangle = 0. \quad (19.21)$$

To determine the action of $a_0^\dagger$ on $|0, 1\rangle$, we must enforce antisymmetry under particle interchange. In the two-particle state $|1, 1\rangle$, we can physically interchange the particles by:

1. Annihilating the particle in φ_1 : $|1, 1\rangle \rightarrow |1, 0\rangle = a_1|1, 1\rangle$.
2. Moving the particle from φ_0 into φ_1 : $|1, 0\rangle \rightarrow |0, 1\rangle = a_1^\dagger a_0|1, 0\rangle$.
3. Re-inserting the first particle into φ_0 via $a_0^\dagger$: $|0, 1\rangle \rightarrow a_0^\dagger|0, 1\rangle$.

The complete particle exchange is executed by the operator sequence $a_0^\dagger a_1^\dagger a_0 a_1|1, 1\rangle = a_0^\dagger|0, 1\rangle$. For the state to change sign under particle interchange, this resulting state must equal $-|1, 1\rangle$:

$$a_0^\dagger|0, 1\rangle = -|1, 1\rangle. \quad (19.22)$$

Hermitian conjugation then requires:

$$a_0|1, 1\rangle = -|0, 1\rangle, \quad a_0^\dagger|1, 1\rangle = 0, \quad a_0|0, 1\rangle = 0. \quad (19.23)$$

With these sign conventions, creation and annihilation operators of different single-particle states *anticommute*:

$$\{a_i, a_j^\dagger\} = \delta_{ij}\mathbb{I}, \quad \{a_i, a_j\} = 0, \quad \{a_i^\dagger, a_j^\dagger\} = 0 \quad (i, j \in \{0, 1\}). \quad (19.24)$$

The Fock states are constructed by acting on the vacuum in a fixed standard order:

$$|n_0, n_1\rangle = (a_1^\dagger)^{n_1} (a_0^\dagger)^{n_0} |0, 0\rangle \quad (n_0, n_1 \in \{0, 1\}). \quad (19.25)$$

19.2.3 General Discrete Basis in Fock Space

For an arbitrary complete orthonormal single-particle basis $\{\varphi_i\}$, a general many-body state is specified by its set of occupation numbers:

$$|n_0, n_1, n_2, \dots\rangle \equiv |\{n_i\}\rangle. \quad (19.26)$$

Bose–Einstein Statistics

For bosons, the occupation numbers can take any non-negative integer value ($n_i = 0, 1, 2, \dots$). The operators satisfy the canonical commutation relations (CCR):

$$[a_i, a_j^\dagger] = \delta_{ij}\mathbb{I}, \quad [a_i, a_j] = 0, \quad [a_i^\dagger, a_j^\dagger] = 0. \quad (19.27)$$

The normalized basis states of Fock space are given by:

$$|n_0, n_1, n_2, \dots\rangle = \prod_i \frac{(a_i^\dagger)^{n_i}}{\sqrt{n_i!}} |0\rangle. \quad (19.28)$$

Fermi–Dirac Statistics

For fermions, the occupation numbers are restricted to $n_i \in \{0, 1\}$. The operators satisfy the canonical anticommutation relations (CAR):

$$\{a_i, a_j^\dagger\} = \delta_{ij}\mathbb{I}, \quad \{a_i, a_j\} = 0, \quad \{a_i^\dagger, a_j^\dagger\} = 0. \quad (19.29)$$

Choosing an arbitrary but fixed ordering of single-particle states ($0 < 1 < 2 < \dots$), the normalized basis states are:

$$|n_0, n_1, n_2, \dots\rangle = \dots (a_2^\dagger)^{n_2} (a_1^\dagger)^{n_1} (a_0^\dagger)^{n_0} |0\rangle. \quad (19.30)$$

The action of a_i and $a_i^\dagger$ incurs a Jordan–Wigner phase factor depending on the number of fermions occupying states prior to index i :

$$a_i |n_0, n_1, \dots\rangle = (-1)^{\sum_{j<i} n_j} n_i |n_0, \dots, n_i - 1, \dots\rangle. \quad (19.31)$$

For both statistics, the total number operator is

$$N = \sum_i a_i^\dagger a_i, \quad [a_i^\dagger a_i, a_j^\dagger a_j] = 0, \quad (19.32)$$

so that the individual occupation numbers n_i are mutually compatible observables.

19.2.4 Field Operators in Continuous Space

When describing particles moving in continuous space, it is advantageous to transform from a discrete orbital basis to local field operators.

Consider a box of volume V with periodic boundary conditions, where single-particle states are plane waves:

$$\varphi_{\mathbf{p}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}, \quad (19.33)$$

with quantized momenta $\mathbf{p} = \frac{2\pi\hbar}{L}(n_x, n_y, n_z)$. The operator $a_{\mathbf{p}s}^\dagger$ creates a particle with momentum $\mathbf{p}$ and spin projection s .

We define the spatial *field creation operator* $\psi_s^\dagger(\mathbf{r})$ as the Fourier transform:

$$\psi_s^\dagger(\mathbf{r}) \equiv \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} a_{\mathbf{p}s}^\dagger. \quad (19.34)$$

The operator $\psi_s^\dagger(\mathbf{r})$ creates a coherent wave packet localized at position $\mathbf{r}$. Indeed, evaluating the amplitude at point $\mathbf{r}'$ of the state created by $\psi_s^\dagger(\mathbf{r})$ yields:

$$\frac{1}{V} \sum_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} e^{i\mathbf{p}\cdot\mathbf{r}'/\hbar} = \delta(\mathbf{r} - \mathbf{r}'). \quad (19.35)$$

Thus, $\psi_s^\dagger(\mathbf{r})$ creates a particle at point $\mathbf{r}$ with spin s .

Correspondingly, its Hermitian adjoint is the *field annihilation operator*:

$$\psi_s(\mathbf{r}) \equiv \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} a_{\mathbf{p}s}, \quad (19.36)$$

which removes a particle at position $\mathbf{r}$ with spin s .

Canonical Field (Anti)commutation Relations

Using the fundamental relations for momentum modes $[a_{\mathbf{p}s}, a_{\mathbf{p}'s'}^\dagger]_{\mp} = \delta_{\mathbf{p}\mathbf{p}'}\delta_{ss'}$ (where the upper sign $-$ denotes bosons and the lower sign $+$ denotes fermions), we obtain the field algebra:

$$\psi_s(\mathbf{r})\psi_{s'}(\mathbf{r}') \mp \psi_{s'}(\mathbf{r}')\psi_s(\mathbf{r}) = 0, \quad (19.37)$$

$$\psi_s^\dagger(\mathbf{r})\psi_{s'}^\dagger(\mathbf{r}') \mp \psi_{s'}^\dagger(\mathbf{r}')\psi_s^\dagger(\mathbf{r}) = 0, \quad (19.38)$$

and

$$\begin{aligned} \psi_s(\mathbf{r})\psi_{s'}^\dagger(\mathbf{r}') \mp \psi_{s'}^\dagger(\mathbf{r}')\psi_s(\mathbf{r}) &= \frac{1}{V} \sum_{\mathbf{p}, \mathbf{p}'} e^{i(\mathbf{p}\cdot\mathbf{r} - \mathbf{p}'\cdot\mathbf{r}')/\hbar} [a_{\mathbf{p}s}, a_{\mathbf{p}'s'}^\dagger]_{\mp} \\ &= \frac{\delta_{ss'}}{V} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot(\mathbf{r} - \mathbf{r}')/\hbar} = \delta(\mathbf{r} - \mathbf{r}')\delta_{ss'}. \end{aligned} \quad (19.39)$$

In compact notation:

$$[\psi_s(\mathbf{r}), \psi_{s'}^\dagger(\mathbf{r}')]_{\mp} = \delta(\mathbf{r} - \mathbf{r}')\delta_{ss'}, \quad [\psi_s(\mathbf{r}), \psi_{s'}(\mathbf{r}')]_{\mp} = 0. \quad (19.40)$$

19.2.5 Configuration-Space States and Completeness

Using the field creation operators, we construct the n -particle coordinate state (suppressing spin indices for simplicity):

$$|\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n\rangle \equiv \frac{1}{\sqrt{n!}} \psi^\dagger(\mathbf{r}_n) \cdots \psi^\dagger(\mathbf{r}_2) \psi^\dagger(\mathbf{r}_1) |0\rangle. \quad (19.41)$$

Because of the commutation (or anticommutation) of the field operators, these states automatically possess the correct physical permutation symmetry:

$$|\dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots\rangle = \pm |\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots\rangle. \quad (19.42)$$

Adding a particle to $|\mathbf{r}_1, \dots, \mathbf{r}_n\rangle$ yields:

$$\psi^\dagger(\mathbf{r})|\mathbf{r}_1, \dots, \mathbf{r}_n\rangle = \sqrt{n+1} |\mathbf{r}_1, \dots, \mathbf{r}_n, \mathbf{r}\rangle. \quad (19.43)$$

Acting with the annihilation operator $\psi(\mathbf{r})$ on the state (19.41), we successively commute $\psi(\mathbf{r})$ past each $\psi^\dagger(\mathbf{r}_k)$ using Eq. (19.39) until it annihilates the vacuum ($\psi(\mathbf{r})|0\rangle = 0$):

$$\psi(\mathbf{r})|\mathbf{r}_1, \dots, \mathbf{r}_n\rangle = \frac{1}{\sqrt{n}} \sum_{k=1}^n (\pm 1)^{n-k} \delta(\mathbf{r} - \mathbf{r}_k) |\mathbf{r}_1, \dots, \widehat{\mathbf{r}}_k, \dots, \mathbf{r}_n\rangle, \quad (19.44)$$

where $\widehat{\mathbf{r}}_k$ indicates the omission of the k -th coordinate. A particle can be extracted only from one of the occupied locations $\mathbf{r}_k$, leaving behind a properly symmetrized $(n-1)$ -particle state.

The dual bra state is defined by Hermitian conjugation:

$$\langle \mathbf{r}_1, \dots, \mathbf{r}_n | = \frac{1}{\sqrt{n!}} \langle 0 | \psi(\mathbf{r}_1) \psi(\mathbf{r}_2) \cdots \psi(\mathbf{r}_n). \quad (19.45)$$

The overlap between two configuration-space basis states is:

$$\langle \mathbf{r}'_1, \dots, \mathbf{r}'_{n'} | \mathbf{r}_1, \dots, \mathbf{r}_n \rangle = \frac{\delta_{nn'}}{n!} \sum_{P \in S_n} (\pm 1)^P P [\delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) \cdots \delta(\mathbf{r}_n - \mathbf{r}'_n)]. \quad (19.46)$$

An arbitrary n -particle state $|\Phi\rangle$ characterized by a wave function $\varphi(\mathbf{r}_1, \dots, \mathbf{r}_n)$ can be expressed as a continuous superposition:

$$|\Phi\rangle = \int d^3r_1 \dots d^3r_n \varphi(\mathbf{r}_1, \dots, \mathbf{r}_n) |\mathbf{r}_1, \dots, \mathbf{r}_n\rangle. \quad (19.47)$$

Projecting $|\Phi\rangle$ onto a position bra yields:

$$\langle \mathbf{r}'_1, \dots, \mathbf{r}'_n | \Phi \rangle = \frac{1}{n!} \sum_{P \in S_n} (\pm 1)^P P \varphi(\mathbf{r}'_1, \dots, \mathbf{r}'_n). \quad (19.48)$$

Even if the test function φ was not initially symmetrized, the physical amplitude $\langle \mathbf{r}'_1, \dots, \mathbf{r}'_n | \Phi \rangle$ is automatically and perfectly (anti)symmetrized. If φ is already properly symmetrized and normalized, then:

$$\langle \mathbf{r}_1, \dots, \mathbf{r}_n | \Phi \rangle = \varphi(\mathbf{r}_1, \dots, \mathbf{r}_n), \quad \langle \Phi | \Phi \rangle = \int d^3r_1 \dots d^3r_n |\varphi(\mathbf{r}_1, \dots, \mathbf{r}_n)|^2 = 1. \quad (19.49)$$

Finally, the completeness relation in the n -particle subspace is given by:

$$\mathbb{I}_n = \int d^3r_1 \dots d^3r_n |\mathbf{r}_1, \dots, \mathbf{r}_n\rangle \langle \mathbf{r}_1, \dots, \mathbf{r}_n|, \quad (19.50)$$

and the resolution of the identity across the entire Fock space is:

$$\mathbb{I} = |0\rangle \langle 0| + \sum_{n=1}^{\infty} \mathbb{I}_n = \sum_{n=0}^{\infty} \int d^3r_1 \dots d^3r_n |\mathbf{r}_1, \dots, \mathbf{r}_n\rangle \langle \mathbf{r}_1, \dots, \mathbf{r}_n|. \quad (19.51)$$

This establishes the complete equivalence between the configuration-space wave function mechanics and the second-quantized field operator formalism.

19.3 Second-Quantized Operators

Having constructed Fock space and field operators, we now establish how physical observables—such as particle density, current density, kinetic energy, and spin density—are represented in second quantization.

19.3.1 One-Body Operators: Density and Particle Number

Consider the particle density operator $\rho(\mathbf{r})$. In the field operator representation, it is given by:

$$\rho(\mathbf{r}) = \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}). \quad (19.52)$$

To prove that Eq. (19.52) represents the physical density, we evaluate its matrix element between two arbitrary n -particle states $|\Phi\rangle$ and $|\Phi'\rangle$. Inserting a resolution of the identity $\mathbb{I}_{n-1}$ for the $(n-1)$ -particle subspace:

$$\begin{aligned} \langle \Phi' | \rho(\mathbf{r}) | \Phi \rangle &= \langle \Phi' | \psi^\dagger(\mathbf{r}) \mathbb{I}_{n-1} \psi(\mathbf{r}) | \Phi \rangle \\ &= \int d^3\mathbf{r}_1 \dots d^3\mathbf{r}_{n-1} \langle \Phi' | \psi^\dagger(\mathbf{r}) | \mathbf{r}_1 \dots \mathbf{r}_{n-1} \rangle \langle \mathbf{r}_1 \dots \mathbf{r}_{n-1} | \psi(\mathbf{r}) | \Phi \rangle. \end{aligned} \quad (19.53)$$

Using the identity $\psi^\dagger(\mathbf{r}) |\mathbf{r}_1 \dots \mathbf{r}_{n-1}\rangle = \sqrt{n} |\mathbf{r}_1 \dots \mathbf{r}_{n-1}, \mathbf{r}\rangle$, this becomes:

$$\langle \Phi' | \rho(\mathbf{r}) | \Phi \rangle = n \int d^3\mathbf{r}_1 \dots d^3\mathbf{r}_{n-1} \Phi'^*(\mathbf{r}_1, \dots, \mathbf{r}_{n-1}, \mathbf{r}) \Phi(\mathbf{r}_1, \dots, \mathbf{r}_{n-1}, \mathbf{r}). \quad (19.54)$$

Because the physical wave functions are completely (anti)symmetric under the interchange of any coordinates, this expression is identically equal to:

$$\langle \Phi' | \rho(\mathbf{r}) | \Phi \rangle = \int d^3\mathbf{r}_1 \dots d^3\mathbf{r}_n \Phi'^*(\mathbf{r}_1, \dots, \mathbf{r}_n) \left[\sum_{i=1}^n \delta(\mathbf{r} - \mathbf{r}_i) \right] \Phi(\mathbf{r}_1, \dots, \mathbf{r}_n). \quad (19.55)$$

Equation (19.55) is precisely the matrix element of the classical first-quantized density operator $\sum_{i=1}^n \delta(\mathbf{r} - \mathbf{r}_i)$.

Physically, $\psi^\dagger(\mathbf{r})\psi(\mathbf{r})$ measures the local particle density by probing whether a particle can be annihilated at position $\mathbf{r}$ and immediately restored.

For particles with spin projection s , the local spin-resolved density is $\psi_s^\dagger(\mathbf{r})\psi_s(\mathbf{r})$. Summing over spin components yields the total density operator:

$$\rho(\mathbf{r}) = \sum_s \psi_s^\dagger(\mathbf{r})\psi_s(\mathbf{r}). \quad (19.56)$$

Integrating $\rho(\mathbf{r})$ over all space gives the total number operator:

$$N = \int d^3r \rho(\mathbf{r}) = \sum_s \int d^3r \psi_s^\dagger(\mathbf{r})\psi_s(\mathbf{r}). \quad (19.57)$$

Substituting the plane-wave expansions $\psi_s(\mathbf{r}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} a_{\mathbf{p}s}$ into Eq. (19.57):

$$\begin{aligned} N &= \sum_s \sum_{\mathbf{p}, \mathbf{p}'} a_{\mathbf{p}s}^\dagger a_{\mathbf{p}'s} \int \frac{d^3r}{V} e^{i(\mathbf{p}' - \mathbf{p})\cdot\mathbf{r}/\hbar} \\ &= \sum_s \sum_{\mathbf{p}, \mathbf{p}'} a_{\mathbf{p}s}^\dagger a_{\mathbf{p}'s} \delta_{\mathbf{p}\mathbf{p}'} = \sum_{\mathbf{p}, s} a_{\mathbf{p}s}^\dagger a_{\mathbf{p}s}, \end{aligned} \quad (19.58)$$

recovering the sum of momentum-space occupation numbers.

19.3.2 Kinetic Energy and Current Density

In momentum space, the total kinetic energy operator is obtained by weighting each momentum mode by its single-particle kinetic energy $\epsilon_{\mathbf{p}} = p^2/2m$:

$$T = \sum_{\mathbf{p}, s} \frac{p^2}{2m} a_{\mathbf{p}s}^\dagger a_{\mathbf{p}s}. \quad (19.59)$$

To express T in terms of spatial field operators, we invert the Fourier relations:

$$a_{\mathbf{p}s} = \int \frac{d^3r}{\sqrt{V}} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi_s(\mathbf{r}), \quad a_{\mathbf{p}s}^\dagger = \int \frac{d^3r}{\sqrt{V}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi_s^\dagger(\mathbf{r}). \quad (19.60)$$

Noting that $\mathbf{p}e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} = -i\hbar\nabla e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$, substitution yields:

$$T = \frac{\hbar^2}{2m} \frac{1}{V} \sum_{\mathbf{p}, s} \int d^3r d^3r' (\nabla e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}) \cdot (\nabla' e^{-i\mathbf{p}\cdot\mathbf{r}'/\hbar}) \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}'). \quad (19.61)$$

Integrating by parts and executing the sum over $\mathbf{p}$ via $\frac{1}{V} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')/\hbar} = \delta(\mathbf{r} - \mathbf{r}')$, we arrive at:

$$T = \frac{\hbar^2}{2m} \sum_s \int d^3r \nabla \psi_s^\dagger(\mathbf{r}) \cdot \nabla \psi_s(\mathbf{r}) = -\frac{\hbar^2}{2m} \sum_s \int d^3r \psi_s^\dagger(\mathbf{r}) \nabla^2 \psi_s(\mathbf{r}). \quad (19.62)$$

Similarly, the second-quantized particle current density operator is:

$$\mathbf{J}(\mathbf{r}) = \frac{\hbar}{2mi} \sum_s \left[\psi_s^\dagger(\mathbf{r}) \nabla \psi_s(\mathbf{r}) - \left(\nabla \psi_s^\dagger(\mathbf{r}) \right) \psi_s(\mathbf{r}) \right], \quad (19.63)$$

and for spin-1/2 fermions, the local spin density operator is:

$$\mathbf{S}(\mathbf{r}) = \frac{\hbar}{2} \sum_{s,s'} \psi_s^\dagger(\mathbf{r}) \boldsymbol{\sigma}_{ss'} \psi_{s'}(\mathbf{r}), \quad (19.64)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli spin matrices.

The Concept of “Second Quantization”: Notice the striking formal correspondence: the many-body operators $\rho(\mathbf{r})$, T , and $\mathbf{J}(\mathbf{r})$ have exactly the same functional forms as the quantum expectation values $\varphi^*(\mathbf{r})\varphi(\mathbf{r})$, $\int \varphi^* \left(-\frac{\hbar^2}{2m} \nabla^2\right) \varphi d^3r$, and $\frac{\hbar}{2mi}(\varphi^* \nabla \varphi - \varphi \nabla \varphi^*)$ of single-particle wave mechanics, with the classical wave function $\varphi(\mathbf{r})$ replaced by the quantum field operator $\psi(\mathbf{r})$. This historical terminology does not imply a repeated quantization of physics, but rather a powerful operator language for identical-particle systems.

19.3.3 Ground State of the Noninteracting Fermi Gas

To illustrate the utility of the formalism, we examine the ground state $|\Phi_0\rangle$ of an ideal gas of spin-1/2 fermions of mass m in volume V .

In the ground state, all single-particle momentum states up to the Fermi momentum $p_F = \hbar k_F$ are filled:

$$n_{\mathbf{p}s} \equiv \langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{p}s} | \Phi_0 \rangle = \Theta(p_F - p) = \begin{cases} 1, & p \leq p_F, \\ 0, & p > p_F, \end{cases} \quad (19.65)$$

for both spin projections $s = \uparrow, \downarrow$. In the thermodynamic limit ($V \rightarrow \infty$), the total particle number is:

$$N = \sum_{\mathbf{p},s} n_{\mathbf{p}s} = 2V \int \frac{d^3p}{(2\pi\hbar)^3} \Theta(p_F - p) = 2V \frac{4\pi}{3} \left(\frac{p_F}{2\pi\hbar} \right)^3 = \frac{p_F^3}{3\pi^2\hbar^3} V. \quad (19.66)$$

Hence, the average particle density $n = N/V$ fixes the Fermi momentum:

$$p_F = \hbar k_F = \hbar (3\pi^2 n)^{1/3}. \quad (19.67)$$

The expectation value of the density operator is:

$$\langle \rho(\mathbf{r}) \rangle = \sum_s \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) | \Phi_0 \rangle = \frac{1}{V} \sum_{\mathbf{p}, \mathbf{p}', s} e^{i(\mathbf{p}' - \mathbf{p}) \cdot \mathbf{r} / \hbar} \langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{p}'s} | \Phi_0 \rangle. \quad (19.68)$$

Since removing a fermion of momentum $\mathbf{p}'$ from $|\Phi_0\rangle$ and restoring $|\Phi_0\rangle$ requires re-inserting a fermion of identical momentum and spin:

$$\langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{p}'s'} | \Phi_0 \rangle = \delta_{\mathbf{p}\mathbf{p}'} \delta_{ss'} n_{\mathbf{p}s}. \quad (19.69)$$

Consequently, the spatial density is strictly uniform:

$$\langle \rho(\mathbf{r}) \rangle = \frac{1}{V} \sum_{\mathbf{p},s} n_{\mathbf{p}s} = \frac{N}{V} = n. \quad (19.70)$$

19.3.4 The One-Particle Density Matrix

A fundamental quantity characterizing spatial coherence in many-body systems is the *one-particle density matrix*:

$$G_s(\mathbf{r} - \mathbf{r}') \equiv \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}') | \Phi_0 \rangle. \quad (19.71)$$

Physically, $G_s(\mathbf{r} - \mathbf{r}')$ represents the quantum amplitude for annihilating a fermion of spin s at $\mathbf{r}'$ and recreating it at $\mathbf{r}$ while preserving the ground state.

Substituting the Fourier expansion of the field operators and utilizing Eq. (19.69):

$$G_s(\mathbf{r} - \mathbf{r}') = \frac{1}{V} \sum_{\mathbf{p}} e^{-i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')/\hbar} n_{\mathbf{p}s} = \int_{p \leq p_F} \frac{d^3p}{(2\pi\hbar)^3} e^{-i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')/\hbar}. \quad (19.72)$$

Let $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ and $R = |\mathbf{R}|$. In spherical coordinates with polar angle $\mu = \cos \theta$:

$$\begin{aligned} G_s(\mathbf{R}) &= \frac{1}{(2\pi)^2} \int_0^{k_F} k^2 dk \int_{-1}^1 d\mu e^{-ikR\mu} \\ &= \frac{1}{2\pi^2 R} \int_0^{k_F} k \sin(kR) dk \\ &= \frac{k_F^3}{2\pi^2} \frac{\sin(k_F R) - k_F R \cos(k_F R)}{(k_F R)^3}. \end{aligned} \quad (19.73)$$

Introducing the dimensionless variable $x \equiv k_F R = p_F |\mathbf{r} - \mathbf{r}'|/\hbar$ and using $k_F^3/2\pi^2 = \frac{3}{2}n_s = \frac{3n}{4}$ (with $n_s = n/2$ being the density per spin channel):

$$G_s(\mathbf{r} - \mathbf{r}') = \frac{3n}{2} \left[\frac{\sin x - x \cos x}{x^3} \right] = \frac{3n}{2} \frac{j_1(x)}{x}, \quad (19.74)$$

where $j_1(x)$ is the spherical Bessel function of order 1.

Limiting Behaviors

- **Coincident Points** ($R \rightarrow 0$): Since $\lim_{x \rightarrow 0} \frac{j_1(x)}{x} = \frac{1}{3}$, we find:

$$G_s(0) = \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) | \Phi_0 \rangle = \frac{n}{2}, \quad (19.75)$$

which precisely equals the single-spin density n_s .

- **Short Distances** ($x \ll 1$): Expanding the trigonometric functions:

$$G_s(\mathbf{r} - \mathbf{r}') = \frac{n}{2} \left[1 - \frac{(k_F |\mathbf{r} - \mathbf{r}'|)^2}{10} + \mathcal{O}(x^4) \right]. \quad (19.76)$$

- **Long Distances** ($x \gg 1$): The density matrix exhibits Friedel oscillations decaying as:

$$G_s(\mathbf{R}) \sim -\frac{3n}{2} \frac{\cos(k_F R)}{(k_F R)^2}, \quad (19.77)$$

reflecting the sharp discontinuity of the Fermi-Dirac distribution at the Fermi surface.

19.4 Pair Correlation Functions

In a quantum gas, exchange symmetry induces spatial correlations even in the absence of dynamical interparticle forces. Fermions of the same spin tend to avoid one another due to the Pauli exclusion principle, creating an “exchange hole,” whereas bosons exhibit an enhanced probability of being found in close spatial proximity (“Bose bunching”). These phenomena are quantitatively characterized by *pair correlation functions*.

19.4.1 Definition of the Pair Correlation Function

Consider the probability density of detecting a particle with spin s' at position $\mathbf{r}'$, given that a particle with spin s is already detected at position $\mathbf{r}$.

In second quantization, measuring a particle at $\mathbf{r}$ corresponds to applying the annihilation operator $\psi_s(\mathbf{r})$ to the state $|\Phi_0\rangle$, resulting in an unnormalized $(N-1)$ -particle state:

$$|\Phi'(\mathbf{r}, s)\rangle = \psi_s(\mathbf{r})|\Phi_0\rangle. \quad (19.78)$$

The local density of particles with spin s' in this conditioned state is:

$$\begin{aligned} \langle \Phi'(\mathbf{r}, s) | \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') | \Phi'(\mathbf{r}, s) \rangle &= \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle \\ &\equiv n_s n_{s'} g_{ss'}(\mathbf{r} - \mathbf{r}'), \end{aligned} \quad (19.79)$$

where $n_s = \langle \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) \rangle$ is the average density of spin- s particles ($n_s = n/2$ for an unpolarized spin-1/2 gas), and $g_{ss'}(\mathbf{r} - \mathbf{r}')$ is the dimensionless *pair correlation function*.

Equivalently, one can consider removing two particles at positions $\mathbf{r}$ and $\mathbf{r}'$ and projecting onto a complete set of $(N-2)$ -particle eigenstates $\{|\Phi_i\rangle\}$:

$$\begin{aligned} \sum_i |\langle \Phi_i | \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle|^2 &= \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \left[\sum_i |\Phi_i\rangle \langle \Phi_i| \right] \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle \\ &= \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle, \end{aligned} \quad (19.80)$$

confirming Eq. (19.79). In a completely uncorrelated classical system, $g_{ss'}(\mathbf{r} - \mathbf{r}') = 1$ for all separations.

19.4.2 Pair Correlations in the Ideal Fermi Gas

Expanding the field operators in momentum modes:

$$\left(\frac{n}{2}\right)^2 g_{ss'}(\mathbf{r} - \mathbf{r}') = \frac{1}{V^2} \sum_{\mathbf{p}, \mathbf{p}', \mathbf{q}, \mathbf{q}'} e^{-i(\mathbf{p}-\mathbf{p}') \cdot \mathbf{r} / \hbar} e^{-i(\mathbf{q}-\mathbf{q}') \cdot \mathbf{r}' / \hbar} \langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{q}s'}^\dagger a_{\mathbf{q}'s'} a_{\mathbf{p}'s} | \Phi_0 \rangle. \quad (19.81)$$

To preserve the ground state, the annihilated momentum and spin quantum numbers must match those created.

Opposite Spins ($s \neq s'$)

Because the spin labels are distinct, the particles are distinguishable by their spin state. Annihilation of a spin- s fermion with momentum $\mathbf{p}'$ and a spin- s' fermion with momentum $\mathbf{q}'$ requires recreation of identical momenta: $\mathbf{p} = \mathbf{p}'$ and $\mathbf{q} = \mathbf{q}'$. Using the canonical anticommutation relations $[a_{\mathbf{p}s}, a_{\mathbf{q}s'}^\dagger] = 0$:

$$\langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{q}s'}^\dagger a_{\mathbf{q}s'} a_{\mathbf{p}s} | \Phi_0 \rangle = \langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{p}s} a_{\mathbf{q}s'}^\dagger a_{\mathbf{q}s'} | \Phi_0 \rangle = n_{\mathbf{p}s} n_{\mathbf{q}s'}. \quad (19.82)$$

Substituting this into Eq. (19.81):

$$\left(\frac{n}{2}\right)^2 g_{ss'}(\mathbf{r} - \mathbf{r}') = \left(\frac{1}{V} \sum_{\mathbf{p}} n_{\mathbf{p}s}\right) \left(\frac{1}{V} \sum_{\mathbf{q}} n_{\mathbf{q}s'}\right) = \left(\frac{n}{2}\right)^2, \quad (19.83)$$

which yields:

$$g_{ss'}(\mathbf{r} - \mathbf{r}') = 1 \quad (s \neq s'). \quad (19.84)$$

The exclusion principle does not act between particles with opposite spins; their spatial positions remain entirely uncorrelated in an ideal gas.

Parallel Spins ($s = s'$)

When both particles have identical spin projection, there are two distinct pairing possibilities for the momentum indices:

1. **Direct Pairing:** $\mathbf{p} = \mathbf{p}'$ and $\mathbf{q} = \mathbf{q}'$.
2. **Exchange Pairing:** $\mathbf{p} = \mathbf{q}'$ and $\mathbf{q} = \mathbf{p}'$.

If $\mathbf{p} = \mathbf{q}$, the expectation value vanishes identically because $(a_{\mathbf{p}s}^\dagger)^2 = 0$. For $\mathbf{p} \neq \mathbf{q}$, anticommuting the creation and annihilation operators produces a relative minus sign for the exchange contraction:

$$\langle \Phi_0 | a_{\mathbf{p}s}^\dagger a_{\mathbf{q}s}^\dagger a_{\mathbf{q}'s} a_{\mathbf{p}'s} | \Phi_0 \rangle = (\delta_{\mathbf{p}\mathbf{p}'} \delta_{\mathbf{q}\mathbf{q}'} - \delta_{\mathbf{p}\mathbf{q}'} \delta_{\mathbf{q}\mathbf{p}'}) n_{\mathbf{p}s} n_{\mathbf{q}s}. \quad (19.85)$$

Inserting Eq. (19.85) into Eq. (19.81):

$$\begin{aligned} \left(\frac{n}{2}\right)^2 g_{ss}(\mathbf{r} - \mathbf{r}') &= \frac{1}{V^2} \sum_{\mathbf{p}, \mathbf{q}} \left[1 - e^{-i(\mathbf{p}-\mathbf{q}) \cdot (\mathbf{r}-\mathbf{r}')/\hbar} \right] n_{\mathbf{p}s} n_{\mathbf{q}s} \\ &= \left(\frac{n}{2}\right)^2 - [G_s(\mathbf{r} - \mathbf{r}')]^2, \end{aligned} \quad (19.86)$$

where $G_s(\mathbf{r} - \mathbf{r}')$ is the one-particle density matrix derived in Eq. (19.73).

Dividing by $(n/2)^2$ and defining the dimensionless distance $x \equiv k_F |\mathbf{r} - \mathbf{r}'|$, we obtain the celebrated *Fermi exchange correlation function*:

$$g_{ss}(\mathbf{r} - \mathbf{r}') = 1 - \left[\frac{3(\sin x - x \cos x)}{x^3} \right]^2 = 1 - \left[\frac{3j_1(x)}{x} \right]^2. \quad (19.87)$$

Physical Consequences of the Exchange Hole

- **Complete Coincident Suppression:** As $|\mathbf{r} - \mathbf{r}'| \rightarrow 0$ ($x \rightarrow 0$), $\lim_{x \rightarrow 0} \frac{3j_1(x)}{x} = 1$, leading to:

$$g_{ss}(0) = 0. \quad (19.88)$$

The probability of finding two identical fermions with parallel spin at the same spatial point is strictly zero.

- **The Exchange Hole:** For separations $|\mathbf{r} - \mathbf{r}'| \lesssim k_F^{-1} = \lambda_F/2\pi$, g_{ss} is strongly suppressed below unity. Each fermion is surrounded by a depleted region of parallel-spin density known as the *Fermi hole* (or exchange hole).
- **Short-Distance Expansion:** For $x \ll 1$:

$$g_{ss}(\mathbf{r} - \mathbf{r}') \approx \frac{(k_F |\mathbf{r} - \mathbf{r}'|)^2}{5}. \quad (19.89)$$

- **Asymptotic Decoupling:** At macroscopic separations ($x \gg 1$), $g_{ss} \rightarrow 1$, approaching the uncorrelated value.

19.4.3 Hartree–Fock Factorization (Wick’s Theorem)

Combining Eqs. (19.84), (19.86), and (19.71), the two-particle density operator factors identically as:

$$\begin{aligned} \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle &= \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) | \Phi_0 \rangle \langle \Phi_0 | \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') | \Phi_0 \rangle \\ &\quad - \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_{s'}(\mathbf{r}') | \Phi_0 \rangle \langle \Phi_0 | \psi_{s'}^\dagger(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle. \end{aligned} \quad (19.90)$$

This algebraic identity—the direct Hartree term minus the exchange Fock term—is an exact manifestation of Wick’s theorem for any quantum state that can be represented as a single Slater determinant.

19.4.4 Pair Correlations for Bosons: Bose Bunching

Now consider a gas of noninteracting spinless bosons in a Fock state with definite occupation numbers:

$$|\Phi\rangle = |n_{\mathbf{p}_0}, n_{\mathbf{p}_1}, \dots\rangle, \quad n = \frac{1}{V} \sum_{\mathbf{p}} n_{\mathbf{p}}. \quad (19.91)$$

The four-operator expectation value in Eq. (19.81) evaluates differently for bosons because:

1. Bosonic operators satisfy commutation relations rather than anticommutation relations ($[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = \delta_{\mathbf{p}\mathbf{q}}$), so the exchange term enters with a **positive** sign.
2. Multiple bosons can occupy the exact same single-particle mode ($n_{\mathbf{p}} > 1$), yielding $a_{\mathbf{p}}^\dagger a_{\mathbf{p}}^\dagger a_{\mathbf{p}} a_{\mathbf{p}} = n_{\mathbf{p}}(n_{\mathbf{p}} - 1)$.

Explicit calculation yields:

$$\langle \Phi | a_{\mathbf{p}}^\dagger a_{\mathbf{q}}^\dagger a_{\mathbf{q}'} a_{\mathbf{p}'} | \Phi \rangle = (1 - \delta_{\mathbf{p}\mathbf{q}}) (\delta_{\mathbf{p}\mathbf{p}'} \delta_{\mathbf{q}\mathbf{q}'} + \delta_{\mathbf{p}\mathbf{q}'} \delta_{\mathbf{q}\mathbf{p}'}) n_{\mathbf{p}} n_{\mathbf{q}} + \delta_{\mathbf{p}\mathbf{q}} \delta_{\mathbf{p}\mathbf{p}'} \delta_{\mathbf{q}\mathbf{q}'} n_{\mathbf{p}} (n_{\mathbf{p}} - 1). \quad (19.92)$$

Summing over all momenta gives the exact two-body expectation value:

$$\langle \Phi | \psi^\dagger(\mathbf{r}) \psi^\dagger(\mathbf{r}') \psi(\mathbf{r}') \psi(\mathbf{r}) | \Phi \rangle = n^2 + \left| \frac{1}{V} \sum_{\mathbf{p}} n_{\mathbf{p}} e^{-i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')/\hbar} \right|^2 - \frac{1}{V^2} \sum_{\mathbf{p}} n_{\mathbf{p}} (n_{\mathbf{p}} + 1). \quad (19.93)$$

Special Case: Pure Condensate State

If all N bosons occupy a single momentum state $\mathbf{p}_0$ (a pure number-state Bose-Einstein condensate), Eq. (19.93) reduces to:

$$\langle \psi^\dagger(\mathbf{r}) \psi^\dagger(\mathbf{r}') \psi(\mathbf{r}') \psi(\mathbf{r}) \rangle = n^2 + n^2 - \frac{N(N+1)}{V^2} = \frac{N(N-1)}{V^2} = n^2 \left(1 - \frac{1}{N} \right). \quad (19.94)$$

In a pure Fock state of a single mode, there is no spatial bunching because density fluctuations are strictly suppressed by the fixed total particle number.

Thermal or Broad Momentum Distribution

Consider instead an incoherent or smoothly distributed beam of bosons with a Gaussian momentum spread centered at $\mathbf{p}_0$:

$$n_{\mathbf{p}} = c e^{-\alpha(\mathbf{p} - \mathbf{p}_0)^2/2\hbar^2}. \quad (19.95)$$

In the thermodynamic limit ($V \rightarrow \infty$, $n = \text{const}$), the discrete self-interaction term $\frac{1}{V^2} \sum_{\mathbf{p}} n_{\mathbf{p}} (n_{\mathbf{p}} + 1)$ scales as $\mathcal{O}(1/V)$ and is negligible compared to the macroscopic terms. Converting the sum to an integral:

$$\begin{aligned} \langle \Phi | \psi^\dagger(\mathbf{r}) \psi^\dagger(\mathbf{r}') \psi(\mathbf{r}') \psi(\mathbf{r}) | \Phi \rangle &= n^2 \left[1 + \left| \int \frac{d^3 p}{(2\pi\hbar)^3} \frac{n_{\mathbf{p}}}{n} e^{-i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}')/\hbar} \right|^2 \right] \\ &= n^2 \left(1 + e^{-|\mathbf{r} - \mathbf{r}'|^2/\alpha} \right). \end{aligned} \quad (19.96)$$

The bosonic pair correlation function is therefore:

$$g(\mathbf{r} - \mathbf{r}') = 1 + e^{-|\mathbf{r} - \mathbf{r}'|^2/\alpha}. \quad (19.97)$$

At zero spatial separation ($\mathbf{r} = \mathbf{r}'$), we obtain:

$$g(0) = 2. \quad (19.98)$$

The probability of finding two bosons simultaneously at the same location is exactly **twice** the probability of finding them far apart! This constructive quantum interference between identical bosonic pathways is termed *Bose bunching*. It is the quantum-statistical mechanism underlying the Hanbury Brown and Twiss effect.

19.5 The Hanbury Brown and Twiss Experiment

The spatial clustering of bosons predicted by quantum statistics was dramatically demonstrated in optics by R. Hanbury Brown and R. Q. Twiss in 1956.¹ Their experiment established the field of intensity interferometry and photon correlation spectroscopy, measuring the joint probability of detecting two photons simultaneously at different spatial locations in an incoherent beam of light.

19.5.1 Experimental Configuration

To measure two-photon correlations without detector occlusion, a collimated beam of pseudothermal or thermal light is directed onto a 50:50 non-polarizing beam splitter (half-silvered mirror). The split beams illuminate two independent photodetectors, labeled 1 and 2:

- Detector 1 measures the instantaneous intensity $I_1(t)$.
- Detector 2 measures the intensity $I_2(t + \tau)$ after an adjustable electronic delay τ .

The experimental observable is the time-averaged intensity cross-correlation:

$$\overline{I_1(t)I_2(t + \tau)} \equiv \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T I_1(t)I_2(t + \tau) dt. \quad (19.99)$$

Because the longitudinal spatial separation along the beam corresponds to $\Delta r = c\tau$, this measurement probes the spatial pair correlation function $g(\Delta r)$ of the photons.

Experimentally, as $\tau \rightarrow 0$ (coincident detection), the normalized intensity correlation function:

$$g^{(2)}(\tau) \equiv \frac{\overline{I_1(t)I_2(t + \tau)}}{\bar{I}_1\bar{I}_2} \quad (19.100)$$

rises to a peak value of $g^{(2)}(0) = 2$, decaying to $g^{(2)}(\infty) = 1$ for delays larger than the coherence time τ_c . This direct observation of photon bunching confirmed the $g(0) = 2$ behavior derived in Eq. (19.96).

19.5.2 Classical Wave Optics Interpretation

While photon bunching appears at first to be a uniquely quantum-mechanical property of Bose–Einstein statistics, it can be understood completely within classical wave theory.² The bosonic nature of photons is prefigured in the linearity and superposition principle of classical Maxwellian electrodynamics.

Consider an idealized thermal light source containing two independent, point-like monochromatic emitters, A and B , separated in the source:

- Emitter A radiates with complex field amplitude α and wavenumber $k = \omega/c$.
- Emitter B radiates with complex field amplitude β and wavenumber $k' = \omega'/c$.
- The relative phase between α and β fluctuates randomly on time scales longer than the optical period.

Let r_1 and r'_1 denote the optical path lengths from sources A and B to detector 1, and r_2 and r'_2 denote the distances to detector 2. By the classical superposition principle, the scalar electric field amplitudes at detectors 1 and 2 are:

$$a_1 = \alpha e^{ikr_1} + \beta e^{ik'r'_1}, \quad (19.101)$$

$$a_2 = \alpha e^{ikr_2} + \beta e^{ik'r'_2}. \quad (19.102)$$

¹R. Hanbury Brown and R. Q. Twiss, *Nature* **177**, 27 (1956); **178**, 1447 (1956).

²See E. M. Purcell, *Nature* **178**, 1449 (1956).

The instantaneous intensity at detector 1 is:

$$I_1 = |a_1|^2 = |\alpha|^2 + |\beta|^2 + 2 \operatorname{Re} \left[\alpha^* \beta e^{i(k'r'_1 - kr_1)} \right]. \quad (19.103)$$

Averaging over the random relative phases eliminates the interference cross term ($\overline{\alpha^* \beta} = 0$):

$$\bar{I}_1 = |\alpha|^2 + |\beta|^2, \quad \bar{I}_2 = |\alpha|^2 + |\beta|^2. \quad (19.104)$$

The product of individual average intensities $\bar{I}_1 \bar{I}_2 = (|\alpha|^2 + |\beta|^2)^2$ is strictly constant and independent of the detector separation.

Now compute the instantaneous intensity product:

$$\begin{aligned} I_1 I_2 &= |a_1 a_2|^2 \\ &= \left| \alpha^2 e^{ik(r_1+r_2)} + \beta^2 e^{ik'(r'_1+r'_2)} + \alpha\beta \left(e^{ikr_1} e^{ik'r'_2} + e^{ik'r'_1} e^{ikr_2} \right) \right|^2. \end{aligned} \quad (19.105)$$

Expanding this expression and performing the statistical phase average, all cross terms containing unequal powers of α and β vanish identically:

$$\begin{aligned} \overline{I_1 I_2} &= |\alpha|^4 + |\beta|^4 + |\alpha|^2 |\beta|^2 \left| e^{ikr_1} e^{ik'r'_2} + e^{ik'r'_1} e^{ikr_2} \right|^2 \\ &= (|\alpha|^2 + |\beta|^2)^2 + 2|\alpha|^2 |\beta|^2 \cos [k'(r'_1 - r'_2) - k(r_1 - r_2)] \\ &= \bar{I}_1 \bar{I}_2 + 2|\alpha|^2 |\beta|^2 \cos [k'(r'_1 - r'_2) - k(r_1 - r_2)]. \end{aligned} \quad (19.106)$$

For a paraxial, well-collimated beam, the path differences are approximately equal: $r'_1 - r'_2 \approx r_1 - r_2 \equiv \Delta r$. Equation (19.106) reduces to:

$$\overline{I_1 I_2} = \bar{I}_1 \bar{I}_2 + 2|\alpha|^2 |\beta|^2 \cos [(k' - k)\Delta r]. \quad (19.107)$$

Integrating Eq. (19.107) over a continuous Gaussian spectral distribution of wavenumbers k, k' reproduces precisely the Gaussian pair correlation profile:

$$\overline{I_1 I_2} = \bar{I}_1 \bar{I}_2 \left(1 + e^{-\Delta r^2 / \alpha} \right). \quad (19.108)$$

Thus, classical intensity fluctuations of Gaussian random light give an exact classical account of photon bunching.

19.5.3 Quantum Two-Photon Interference Interpretation

From the quantum-mechanical perspective, the terms in Eq. (19.105) represent probability amplitudes for two-photon detection events:

- The α^2 amplitude corresponds to both detected photons originating from atom A , contributing $|\alpha|^4$ to the coincident count rate.
- The β^2 amplitude corresponds to both detected photons originating from atom B , contributing $|\beta|^4$.
- The $\alpha\beta$ term corresponds to one photon originating from atom A and the other from atom B . Crucially, there are **two indistinguishable quantum pathways** for this process:
 1. Photon from A strikes detector 1, while photon from B strikes detector 2 (amplitude $\alpha\beta e^{ikr_1} e^{ik'r'_2}$).
 2. Photon from A strikes detector 2, while photon from B strikes detector 1 (amplitude $\alpha\beta e^{ikr_2} e^{ik'r'_1}$).

Because photons are identical bosons, these two alternative detection histories are fundamentally indistinguishable. Their transition amplitudes add *constructively*:

$$\mathcal{A}_{\text{boson}} = \alpha\beta \left(e^{ikr_1} e^{ik'r'_2} + e^{ik'r'_1} e^{ikr_2} \right), \quad (19.109)$$

producing the interference cosine term $+2|\alpha|^2 |\beta|^2 \cos[\dots]$ in Eq. (19.106).

19.5.4 The Fermionic Analogue: Antibunching

What would happen if the Hanbury Brown and Twiss experiment were conducted with an unpolarized beam of identical fermions, such as electrons or neutrons?

For fermions, exchange antisymmetry demands that the two alternative two-particle detection amplitudes interfere *destructively*:

$$\mathcal{A}_{\text{fermion}} = \alpha\beta \left(e^{ikr_1} e^{ik'r'_2} - e^{ik'r'_1} e^{ikr_2} \right). \quad (19.110)$$

Squaring this amplitude produces a negative interference cross term:

$$\overline{I_1 I_2}_{\text{fermion}} = \bar{I}_1 \bar{I}_2 - 2|\alpha|^2 |\beta|^2 \cos[(k' - k)\Delta r]. \quad (19.111)$$

In the coincident limit $\Delta r \rightarrow 0$, destructive interference is complete:

$$g^{(2)}(0) = 0. \quad (19.112)$$

Fermions exhibit **antibunching**: the probability of detecting two identical fermions at the same time and position is suppressed to zero. Modern experiments with ballistic electrons in semiconductor two-dimensional electron gases and ultracold atomic Fermi gases have verified this fermionic antibunching with remarkable precision.

19.6 The Many-Body Hamiltonian

To complete the second-quantization formalism, we construct the Hamiltonian operator describing particles interacting through mutual pairwise forces, evaluate the ground state energy of the interacting electron gas, and derive the Heisenberg equations of motion for field operators.

19.6.1 The Two-Body Interaction Operator

Suppose the particles interact via a spin-independent two-body potential $v(\mathbf{r} - \mathbf{r}')$. In second quantization, the interaction Hamiltonian is given by:

$$\mathcal{V} = \frac{1}{2} \sum_{s,s'} \int d^3r d^3r' v(\mathbf{r} - \mathbf{r}') \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}). \quad (19.113)$$

The ordering of the field operators in Eq. (19.113) is crucial:

1. The operators are *normally ordered*: all creation operators appear to the left of all annihilation operators.
2. Annihilating two particles before recreating them ensures that a particle cannot interact with itself. Acting on the vacuum or a one-particle state, $\mathcal{V}$ yields zero.
3. The prefactor 1/2 compensates for double-counting each interacting pair of particles in the double spatial integral.

Combining the kinetic energy operator (19.62) and the interaction operator (19.113), the complete many-body Hamiltonian in second quantization is:

$$H = \sum_s \int d^3r \frac{\hbar^2}{2m} \nabla \psi_s^\dagger(\mathbf{r}) \cdot \nabla \psi_s(\mathbf{r}) + \frac{1}{2} \sum_{s,s'} \int d^3r d^3r' v(\mathbf{r} - \mathbf{r}') \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}). \quad (19.114)$$

19.6.2 Ground State Energy of the Interacting Fermi Gas

We treat the two-body potential $v(\mathbf{r} - \mathbf{r}')$ as a perturbation to the ideal Fermi gas ground state $|\Phi_0\rangle$.

Zero-Order Energy: Kinetic Contribution

In the absence of interactions, the unperturbed ground state energy is the integrated kinetic energy of the filled Fermi sphere:

$$E^{(0)} = \sum_{\mathbf{p}, s} \frac{p^2}{2m} n_{\mathbf{p}s} = 2V \int_0^{p_F} \frac{d^3p}{(2\pi\hbar)^3} \frac{p^2}{2m} = \frac{3}{5} \frac{p_F^2}{2m} N = \frac{3}{5} N E_F. \quad (19.115)$$

The mean kinetic energy per fermion is $3/5$ of the Fermi energy $E_F = p_F^2/2m$.

First-Order Energy: Hartree and Fock Contributions

To first order in perturbation theory, the energy shift $E^{(1)}$ is the expectation value of $\mathcal{V}$ in the unperturbed state $|\Phi_0\rangle$:

$$E^{(1)} = \langle \Phi_0 | \mathcal{V} | \Phi_0 \rangle = \frac{1}{2} \int d^3r d^3r' v(\mathbf{r} - \mathbf{r}') \sum_{s, s'} \langle \Phi_0 | \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi_0 \rangle. \quad (19.116)$$

Using the pair correlation function relation (19.79) and Wick's theorem (19.90):

$$E^{(1)} = \frac{1}{2} \int d^3r d^3r' v(\mathbf{r} - \mathbf{r}') \left[n^2 - \sum_s G_s(\mathbf{r} - \mathbf{r}')^2 \right]. \quad (19.117)$$

Equation (19.117) separates into two physically distinct contributions:

1. The Direct (Hartree) Energy:

$$E_{\text{Hartree}} = \frac{1}{2} n^2 \int d^3r d^3r' v(\mathbf{r} - \mathbf{r}') = \frac{1}{2} N n v_0, \quad (19.118)$$

where $v_0 = \int d^3r v(\mathbf{r})$. This represents the classical electrostatic self-energy of a continuous, uniform charge density n , neglecting quantum correlations.

2. The Exchange (Fock) Energy:

$$E_{\text{ex}} = -\frac{1}{2} \int d^3r d^3r' v(\mathbf{r} - \mathbf{r}') \sum_s [G_s(\mathbf{r} - \mathbf{r}')]^2. \quad (19.119)$$

The exchange energy arises from the Pauli exclusion hole between fermions of parallel spin. Because identical fermions avoid each other at short separations, the repulsive interaction energy is lower than the classical Hartree estimate. The exchange energy subtracts this overcounted short-range repulsion.

Substituting the one-particle density matrix $G_s(r)$ from Eq. (19.73) into Eq. (19.119), the exchange energy per particle is:

$$\frac{E_{\text{ex}}}{N} = -\frac{9n}{4} \int d^3r v(\mathbf{r}) \left[\frac{\sin(k_F r) - k_F r \cos(k_F r)}{(k_F r)^3} \right]^2. \quad (19.120)$$

19.6.3 The Electron Gas in the Jellium Model

A cornerstone of modern condensed matter physics is the degenerate electron gas (the *jellium model*), which describes the conduction electrons in simple metals. The electrons interact via the long-range Coulomb potential:

$$v(\mathbf{r} - \mathbf{r}') = \frac{e^2}{|\mathbf{r} - \mathbf{r}'|}. \quad (19.121)$$

To ensure macroscopic electrostatic stability and charge neutrality, the electrons are embedded in a rigid, uniform positive background charge of density $+ne$ provided by the ionized atomic cores.

The positive background contributes:

1. Background-background self-energy: $+\frac{1}{2}e^2n^2 \int \frac{d^3r d^3r'}{|\mathbf{r}-\mathbf{r}'|}$,
2. Electron-background attraction: $-e^2n^2 \int \frac{d^3r d^3r'}{|\mathbf{r}-\mathbf{r}'|}$.

These two background terms precisely cancel the divergent direct Hartree energy of the electron gas! The only surviving first-order interaction energy is the **exchange energy**:

$$\begin{aligned} \frac{E_{\text{ex}}}{N} &= -\frac{9\pi ne^2}{k_F^2} \int_0^\infty \frac{dx}{x^5} (\sin x - x \cos x)^2 \\ &= -\frac{3}{4\pi} \hbar k_F \frac{e^2}{\hbar} = -\frac{3}{4\pi} e^2 k_F. \end{aligned} \quad (19.122)$$

Wigner–Seitz Parameterization

The thermodynamic state of the electron gas is characterized by the dimensionless Wigner–Seitz density parameter r_s , defined as the radius d of a sphere enclosing one electron measured in units of the Bohr radius $a_0 = \hbar^2/me^2$:

$$\frac{4\pi}{3} d^3 n = 1 \quad \implies \quad d = \left(\frac{9\pi}{4} \right)^{1/3} \frac{1}{k_F}. \quad (19.123)$$

Thus,

$$r_s \equiv \frac{d}{a_0} = \left(\frac{9\pi}{4} \right)^{1/3} \frac{me^2}{\hbar^2 k_F} = \frac{1.919}{k_F a_0}. \quad (19.124)$$

Small r_s corresponds to the high-density limit, whereas large r_s corresponds to the low-density, strongly correlated regime.

In units of the Rydberg ($1 \text{ Ry} = e^2/2a_0 = 13.6 \text{ eV}$), the total energy per electron to first order is:

$$\frac{E}{N} = \left(\frac{2.21}{r_s^2} - \frac{0.916}{r_s} \right) \left(\frac{e^2}{2a_0} \right). \quad (19.125)$$

- The kinetic term scales as $r_s^{-2} \propto n^{2/3}$ and dominates at high densities ($r_s \rightarrow 0$).
- The exchange term scales as $-r_s^{-1} \propto -n^{1/3}$ and provides net electrostatic binding.

For $r_s > 2.21/0.916 \approx 2.41$, the energy is negative, demonstrating that the electron gas is bound against dissociation. By the Rayleigh–Ritz variational theorem, Eq. (19.125) is a rigorous upper bound on the true ground state energy.

Correlation Energy

Real metals have densities in the range $1.8 \lesssim r_s \lesssim 5.5$. At these densities, higher-order perturbation terms (the *correlation energy*) are non-negligible. The exact high-density expansion derived by Gell-Mann and Brueckner³ using diagrammatic resummed perturbation theory yields:

$$\frac{E}{N} = \left(\frac{2.21}{r_s^2} - \frac{0.916}{r_s} + 0.0622 \ln r_s - 0.094 + \mathcal{O}(r_s \ln r_s) \right) \left(\frac{e^2}{2a_0} \right). \quad (19.126)$$

The non-analytic $\ln r_s$ behavior originates from dynamic screening of long-wavelength Coulomb interactions (plasmon collective excitations).

19.6.4 Interaction Hamiltonian in Momentum Space

In many-body perturbation theory and Feynman diagrammatics, it is convenient to represent $\mathcal{V}$ in the plane-wave momentum basis. Substituting $\psi_s(\mathbf{r}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} a_{\mathbf{p}s}$ into Eq. (19.113):

$$\mathcal{V} = \frac{1}{2V} \sum_{\mathbf{p}, \mathbf{p}', \mathbf{q}, \mathbf{q}'} \sum_{s, s'} v_{\mathbf{p}' - \mathbf{p}} \delta_{\mathbf{p} + \mathbf{q}, \mathbf{p}' + \mathbf{q}'} a_{\mathbf{p}'s}^\dagger a_{\mathbf{q}'s'}^\dagger a_{\mathbf{q}s'} a_{\mathbf{p}s}, \quad (19.127)$$

where $v_{\mathbf{k}} = \int d^3r e^{-i\mathbf{k}\cdot\mathbf{r}/\hbar} v(\mathbf{r})$ is the Fourier transform of the two-body potential.

Defining the transferred momentum $\hbar\mathbf{q} = \mathbf{p}' - \mathbf{p}$, Eq. (19.127) takes the canonical form:

$$\mathcal{V} = \frac{1}{2V} \sum_{\mathbf{k}, \mathbf{p}, \mathbf{q}} \sum_{s, s'} v(\mathbf{q}) a_{\mathbf{k} + \mathbf{q}, s}^\dagger a_{\mathbf{p} - \mathbf{q}, s'}^\dagger a_{\mathbf{p}s'} a_{\mathbf{k}s}. \quad (19.128)$$

This operator describes a fundamental two-body scattering process: two particles with initial momenta $(\mathbf{k}, \mathbf{p})$ exchange momentum $\hbar\mathbf{q}$ via the interaction $v(\mathbf{q})$ and scatter into final momentum states $(\mathbf{k} + \mathbf{q}, \mathbf{p} - \mathbf{q})$, preserving total momentum.

19.6.5 Heisenberg Equations of Motion for Field Operators

In the Heisenberg picture, the time evolution of a time-independent operator $X(t)$ is governed by:

$$i\hbar \frac{\partial X(t)}{\partial t} = [X(t), H]. \quad (19.129)$$

At equal times, the field operators satisfy the canonical equal-time (anti)commutation relations. Evaluating the commutator of the field annihilation operator $\psi_s(\mathbf{r}, t)$ with the full many-body Hamiltonian (19.114):

$$i\hbar \frac{\partial \psi_s(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2 \nabla^2}{2m} \psi_s(\mathbf{r}, t) + \left[\sum_{s'} \int d^3r' v(\mathbf{r} - \mathbf{r}') \psi_{s'}^\dagger(\mathbf{r}', t) \psi_{s'}(\mathbf{r}', t) \right] \psi_s(\mathbf{r}, t). \quad (19.130)$$

Equation (19.130) is the *nonlinear operator Schrödinger equation*. The term in brackets represents the operator-valued Hartree potential generated dynamically by all other particles in the system:

$$V_{\text{eff}}(\mathbf{r}, t) = \int d^3r' v(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}', t). \quad (19.131)$$

Because the potential is itself an operator dependent on the local density, the motion of each quantum particle is dynamically coupled to the collective quantum fluctuations of the entire many-body medium.

³M. Gell-Mann and K. A. Brueckner, *Phys. Rev.* **106**, 364 (1957).

19.7 Problems

1. **Matrix Representation of Two-Level Fermion Operators:** In the ordered four-dimensional Fock basis $\{|0, 0\rangle, |1, 0\rangle, |0, 1\rangle, |1, 1\rangle\}$ for two single-particle levels φ_0 and φ_1 :

- (a) Construct the explicit 4×4 matrix representations of the annihilation and creation operators $a_0, a_0^\dagger, a_1$, and $a_1^\dagger$, adhering to the Jordan–Wigner phase convention defined in Eqs. (19.19)–(19.23).
 (b) Explicitly verify the canonical anticommutation relations:

$$\{a_i, a_j^\dagger\} = \delta_{ij} \mathbb{I}_4, \quad \{a_i, a_j\} = 0, \quad \{a_i^\dagger, a_j^\dagger\} = 0 \quad (i, j \in \{0, 1\}). \quad (19.132)$$

2. **Quasiparticles and Holes in the Interacting Fermi Gas:** Consider an N -electron Fermi gas in its ground state $|\Phi_0\rangle$ with Fermi momentum p_F .

- (a) *Quasiparticle State:* Calculate to first order in the two-body potential $v(\mathbf{r} - \mathbf{r}')$ the excitation energy of an $(N + 1)$ -particle state created by adding a fermion with momentum $\mathbf{p}$ ($p > p_F$) and spin s :

$$|\mathbf{p}s\rangle = a_{\mathbf{p}s}^\dagger |\Phi_0\rangle. \quad (19.133)$$

Measure the excitation energy relative to the N -particle ground state energy: $\varepsilon_p = E_{N+1}(\mathbf{p}) - E_N^{(0)}$.

- (b) *Hole State:* Repeat the first-order calculation for an $(N - 1)$ -particle state created by removing a fermion with momentum $\mathbf{p}$ ($p \leq p_F$) and spin s :

$$|\mathbf{p}s\rangle^{-1} = a_{\mathbf{p}s} |\Phi_0\rangle, \quad (19.134)$$

measuring the excitation energy as $\varepsilon_h(\mathbf{p}) = E_{N-1}(\mathbf{p}) - E_N^{(0)}$.

- (c) *Coulomb Interaction (Jellium):* For electrons interacting via the Coulomb force in a neutralizing positive background, evaluate the quasiparticle dispersion relation $\varepsilon(p)$ explicitly and show that the effective velocity $v(p) = \partial\varepsilon/\partial p$ diverges logarithmically at the Fermi surface $p = p_F$ when screening is neglected.

3. **Hartree–Fock Factorization for Arbitrary Inhomogeneous Systems:** Let $|\Phi\rangle$ be an N -fermion state represented by a single Slater determinant constructed from an arbitrary complete set of orthonormal single-particle orbitals $\{\varphi_i(\mathbf{r})\}_{i=1}^N$:

$$|\Phi\rangle = a_N^\dagger a_{N-1}^\dagger \cdots a_1^\dagger |0\rangle, \quad \text{where} \quad a_i^\dagger = \int d^3r \varphi_i(\mathbf{r}) \psi^\dagger(\mathbf{r}). \quad (19.135)$$

Using the canonical anticommutation relations of the field operators:

- (a) Express $\psi(\mathbf{r})$ in terms of the orbital annihilation operators a_i .
 (b) Prove that the two-particle correlation function factorizes generally as:

$$\begin{aligned} \langle \Phi | \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi \rangle &= \langle \Phi | \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) | \Phi \rangle \langle \Phi | \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') | \Phi \rangle \\ &\quad - \langle \Phi | \psi_s^\dagger(\mathbf{r}) \psi_{s'}(\mathbf{r}') | \Phi \rangle \langle \Phi | \psi_{s'}^\dagger(\mathbf{r}') \psi_s(\mathbf{r}) | \Phi \rangle, \end{aligned} \quad (19.136)$$

thereby generalizing Eq. (19.90) to arbitrary external potentials (e.g., atoms, molecules, and quantum dots).

4. **Exchange Energy and Spin Alignment:** Consider two electrons confined to a box of volume V occupying orthogonal plane-wave spatial orbitals $\varphi_{\mathbf{p}_1}(\mathbf{r})$ and $\varphi_{\mathbf{p}_2}(\mathbf{r})$ with $\mathbf{p}_1 \neq \mathbf{p}_2$.

- (a) Write down the properly symmetrized state vectors for the spin singlet ($S = 0$) and spin triplet ($S = 1$) configurations.
- (b) Calculate the first-order energy shift for both states under the mutual Coulomb interaction $v(\mathbf{r}_1 - \mathbf{r}_2) = e^2/|\mathbf{r}_1 - \mathbf{r}_2|$.
- (c) Show that the energy difference between the triplet and singlet configurations is given by:

$$E_{\text{triplet}} - E_{\text{singlet}} = -2K_{\text{ex}}, \quad (19.137)$$

where $K_{\text{ex}} = \frac{1}{V^2} \int d^3r_1 d^3r_2 \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \cos \left[\frac{(\mathbf{p}_1 - \mathbf{p}_2) \cdot (\mathbf{r}_1 - \mathbf{r}_2)}{\hbar} \right] > 0$ is the positive-definite exchange integral.

- (d) Conclude why the Coulomb interaction favors parallel spin alignment (ferromagnetism / Hund's first rule).

Chapter 20

Atoms

20.1 Atoms

Accurate calculations of atomic properties are notoriously challenging. This complexity stems fundamentally from two primary sources:

1. The mutual Coulomb repulsion among the electrons, which constitutes an interacting many-body problem.
2. The relativistic spin-orbit interaction coupling the electron spins to the internal electric fields within the atom.

Consequently, in this chapter we outline the overarching physical principles governing atomic structure and discuss the primary theoretical and computational frameworks used to approximate atomic states.¹ Throughout this chapter, we adopt natural units with $\hbar = 1$ unless stated otherwise.

20.2 Two-Electron Atoms

The simplest atomic systems beyond hydrogen are those containing two electrons, such as the helium atom (He) and helium-like ions (H^- , Li^+ , Be^{++} , etc.). Positively and negatively charged ions share the same underlying structure as neutral two-electron systems and can be treated within a unified theoretical framework. In lighter atoms, relativistic spin-orbit coupling is relatively weak and may be neglected to a first approximation. Under the assumption of an infinitely massive, fixed nucleus of charge Ze , the nonrelativistic Hamiltonian is given by

$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} - \frac{Ze^2}{r_1} - \frac{Ze^2}{r_2} + \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (20.1)$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ denote the electron coordinates measured with respect to the nucleus. (For highly precise calculations, small corrections arising from finite nuclear motion must also be incorporated.)

Because of the electron-electron Coulomb repulsion term $e^2/|\mathbf{r}_1 - \mathbf{r}_2|$, the Schrödinger equation cannot be solved in closed analytical form. Rigorous mathematical analysis confirms that well-behaved bound states exist, yet in practice one must rely on approximation methods to determine the ground-state energy and wave function.

¹For comprehensive treatments, see E. U. Condon and G. H. Shortley, *The Theory of Atomic Spectra* (Cambridge University Press, Cambridge, 1959); H. Bethe and R. Jackiw, *Intermediate Quantum Mechanics*, 2nd ed. (W. A. Benjamin, New York, 1968); and J. Slater, *Quantum Theory of Atomic Structure* (McGraw-Hill, New York, 1960).

20.2.1 First-Order Perturbation Theory

The simplest approach treats the mutual Coulomb repulsion as a perturbation to the non-interacting system. In the unperturbed system, each electron moves independently in the nuclear Coulomb potential $-Ze^2/r$. The spatial ground state is formed by placing both electrons in the hydrogenic 1s orbital:

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2; s_1, s_2) = \psi_{1s}(\mathbf{r}_1)\psi_{1s}(\mathbf{r}_2)\chi_{0,0}(s_1, s_2), \quad (20.2)$$

where the normalized hydrogenic 1s orbital is

$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0}, \quad (20.3)$$

with $a_0 = \hbar^2/(me^2)$ denoting the Bohr radius. Because the spatial wave function is symmetric under electron exchange ($\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$), the Pauli exclusion principle mandates that the total wave function be antisymmetric, requiring the spin state to be the antisymmetric singlet state:

$$|\chi_{0,0}\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (20.4)$$

Since the ground-state energy of an electron in an isolated hydrogenic atom of nuclear charge Z is $-Z^2e^2/(2a_0)$, the unperturbed ground-state energy is

$$E_0^{(0)} = -2 \left(\frac{Z^2e^2}{2a_0} \right) = -\frac{Z^2e^2}{a_0}. \quad (20.5)$$

The first-order energy correction ΔE is the expectation value of the mutual Coulomb repulsion in the unperturbed state:

$$\Delta E = \int d^3r_1 d^3r_2 \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} |\psi_{1s}(\mathbf{r}_1)|^2 |\psi_{1s}(\mathbf{r}_2)|^2 = \frac{e^2}{\pi^2} \left(\frac{Z}{a_0} \right)^6 \int d^3r_1 d^3r_2 \frac{e^{-2Z(r_1+r_2)/a_0}}{|\mathbf{r}_1 - \mathbf{r}_2|}. \quad (20.6)$$

This two-center integral can be evaluated by expressing the Coulomb kernel in Fourier representation:

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \int \frac{d^3k}{(2\pi)^3} \frac{4\pi}{k^2} e^{i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}. \quad (20.7)$$

Substituting this into Eq. (20.6) decouples the spatial integrals:

$$\Delta E = \frac{e^2}{2\pi^4} \left(\frac{Z}{a_0} \right)^6 \int \frac{d^3k}{k^2} \left| \int d^3r e^{i\mathbf{k}\cdot\mathbf{r} - 2Zr/a_0} \right|^2. \quad (20.8)$$

The single-particle Fourier integral evaluates directly via spherical coordinates:

$$\int d^3r e^{i\mathbf{k}\cdot\mathbf{r} - 2Zr/a_0} = \frac{16\pi(Z/a_0)}{[k^2 + (2Z/a_0)^2]^2}. \quad (20.9)$$

Inserting Eq. (20.9) into Eq. (20.8) yields

$$\Delta E = \frac{4Ze^2}{\pi a_0} \int_0^\infty \frac{x^2 dx}{(x^2 + 1)^4} = \frac{5}{8} \frac{Ze^2}{a_0}. \quad (20.10)$$

Combining the zeroth-order and first-order contributions gives the total ground-state energy to first order:

$$E_0 \approx E_0^{(0)} + \Delta E = -\left(Z - \frac{5}{8} \right) \frac{Ze^2}{a_0}. \quad (20.11)$$

By the Rayleigh-Ritz variational principle, this first-order energy represents a strict upper bound to the true ground-state energy, as it equals the expectation value $\langle \psi_0 | H | \psi_0 \rangle$ evaluated with an unperturbed trial wave function.

Experimentally, the ground-state energy of a two-electron atom can be determined by measuring the first ionization energy E_i (the minimum energy required to remove one electron to infinity):

$$E_i = E'_0 - E_0, \quad (20.12)$$

where E'_0 is the exact nonrelativistic ground-state energy of the resulting hydrogenic one-electron ion:

$$E'_0 = -\frac{Z^2 e^2}{2a_0}. \quad (20.13)$$

For neutral helium ($Z = 2$), the experimental ionization energy is $E_i = 1.807$ Ry (where $1 \text{ Ry} = e^2/(2a_0) \approx 13.6 \text{ eV}$), implying an experimental ground-state energy of

$$E_0 = -E'_0 - E_i = -4 \text{ Ry} - 1.807 \text{ Ry} = -5.807 \text{ Ry}. \quad (20.14)$$

In contrast, Eq. (20.11) predicts $E_0 = -2(2 - 5/8)(2 \text{ Ry}) = -5.500 \text{ Ry}$. While reasonably close, the discrepancy of 0.307 Ry reveals the limitations of first-order perturbation theory.

20.2.2 Variational Method and Screening

The principal physical defect of the unperturbed state (20.2) is that it neglects mutual screening: each electron perceives not only the bare nuclear charge $+Ze$, but also the partially screening cloud of negative charge distributed by the other electron. This suggests using a variational trial wave function with an effective nuclear charge $Z' < Z$:

$$\psi_{1s}(r; Z') = \frac{1}{\sqrt{\pi}} \left(\frac{Z'}{a_0} \right)^{3/2} e^{-Z'r/a_0}. \quad (20.15)$$

To compute the expectation value $E_0(Z') = \langle H \rangle_{Z'}$, we evaluate each term using known virial and Coulomb properties:

- The mutual electron repulsion scales linearly with Z' , giving $\frac{5}{8} \frac{Z' e^2}{a_0}$ by virtue of Eq. (20.10).
- The kinetic energy per electron in a hydrogenic state with parameter Z' is $\frac{Z'^2 e^2}{2a_0}$.
- The nuclear attraction per electron is

$$\left\langle -\frac{Ze^2}{r} \right\rangle = \frac{Z}{Z'} \left\langle -\frac{Z'e^2}{r} \right\rangle = \frac{Z}{Z'} \left(-\frac{Z'^2 e^2}{a_0} \right) = -\frac{ZZ'e^2}{a_0}. \quad (20.16)$$

Summing these contributions yields the expectation value of the total energy as a function of Z' :

$$E_0(Z') = 2 \left(\frac{Z'^2 e^2}{2a_0} - \frac{ZZ'e^2}{a_0} \right) + \frac{5}{8} \frac{Z' e^2}{a_0} = \left(Z'^2 - 2ZZ' + \frac{5}{8} Z' \right) \frac{e^2}{a_0}. \quad (20.17)$$

Minimizing $E_0(Z')$ with respect to Z' ,

$$\frac{dE_0(Z')}{dZ'} = \left(2Z' - 2Z + \frac{5}{8} \right) \frac{e^2}{a_0} = 0, \quad (20.18)$$

gives the optimal effective nuclear charge:

$$Z' = Z - \frac{5}{16}. \quad (20.19)$$

As expected, screening reduces the effective nuclear charge by $5/16$. Substituting this optimal value into Eq. (20.17) gives the variational ground-state energy:

$$E_0 = - \left(Z - \frac{5}{16} \right)^2 \frac{e^2}{a_0}. \quad (20.20)$$

For helium ($Z = 2$), this yields $Z' = 27/16 = 1.6875$ and

$$E_0 = -2 \left(\frac{27}{16} \right)^2 \text{Ry} = -\frac{729}{128} \text{Ry} \approx -5.695 \text{Ry}, \quad (20.21)$$

which is an improvement of nearly 0.2 Ry over first-order perturbation theory.

However, for the hydride ion H^- ($Z = 1$), Eq. (20.20) yields $E_0 = -(11/16)^2(2 \text{Ry}) \approx -0.945 \text{Ry}$. Because this energy is higher than that of an isolated hydrogen atom plus an electron at rest (-1.0Ry), the single-parameter trial function fails to predict that H^- is stable against autoionization. Accurate theoretical binding energies (which correctly demonstrate the stability of H^-) require multiparameter trial wave functions that explicitly include the interelectronic coordinate $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$ (such as the Hylleraas trial functions), as well as relativistic corrections and nuclear recoil.

20.3 The Hartree Approximation

In multi-electron atoms, solving the full N -body Schrödinger equation directly is computationally intractable. However, to a good first approximation, each electron can be envisioned as moving independently in an effective, spherically averaged potential created by the positively charged nucleus and the charge distribution of the remaining $N - 1$ electrons. The central theoretical challenge is thus to simultaneously determine both this average potential and the single-particle wave functions (orbitals) that generate it. This self-consistent interplay arises because the effective field experienced by any one electron depends on the quantum states occupied by all the others.

20.3.1 The Hartree Product and Self-Consistent Field

As a starting point, we represent the total N -electron wave function as an unsymmetrized Hartree product of single-particle orbitals:

$$\Psi(\mathbf{r}_1, s_1; \dots; \mathbf{r}_N, s_N) = \varphi_1(1)\varphi_2(2) \cdots \varphi_N(N), \quad (20.22)$$

where the shorthand notation $i \equiv (\mathbf{r}_i, s_i)$ denotes the spatial and spin coordinates of the i th electron. While the simple product (20.22) lacks the antisymmetry under particle exchange required by the Pauli principle, the exclusion principle is partially enforced by requiring the single-particle spin-orbitals $\{\varphi_i\}$ to be mutually orthonormal:

$$\langle \varphi_i | \varphi_j \rangle = \delta_{ij}. \quad (20.23)$$

The average charge density at position $\mathbf{r}'$ produced by the j th electron in orbital φ_j is $-e|\varphi_j(\mathbf{r}')|^2$. By Coulomb's law, another electron located at $\mathbf{r}$ experiences an electrostatic repulsion potential energy from this electron given by

$$\int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\varphi_j(\mathbf{r}')|^2. \quad (20.24)$$

Summing over all other electrons $j \neq i$ and including the nuclear attraction $-Ze^2/r$, the net effective potential experienced by the i th electron is

$$V_i(\mathbf{r}) = -\frac{Ze^2}{r} + \sum_{j \neq i} \int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\varphi_j(\mathbf{r}')|^2. \quad (20.25)$$

The stationary orbital $\varphi_i(\mathbf{r})$ is therefore governed by the single-particle Schrödinger equation:

$$\left[-\frac{\nabla^2}{2m} + V_i(\mathbf{r}) \right] \varphi_i(\mathbf{r}) = \varepsilon_i \varphi_i(\mathbf{r}), \quad (20.26)$$

where ε_i represents the single-particle orbital energy. Equations (20.25) and (20.26) constitute the *Hartree equations*. Because $V_i(\mathbf{r})$ depends on the unknown wave functions of all other electrons, this coupled system is solved iteratively via the *self-consistent field* (SCF) method: one guesses an initial set of orbitals, constructs the potentials $V_i(\mathbf{r})$, solves Eq. (20.26) to obtain a new set of orbitals, and repeats the cycle until numerical convergence is achieved.

20.3.2 Central-Field Approximation and Total Energy

Because the potential $V_i(\mathbf{r})$ generally depends on the state index i , the resulting orbitals $\varphi_i(\mathbf{r})$ for different i are not automatically orthogonal in space. In atomic physics, it is customary and physically justified to perform a spherical average over angles in $V_i(\mathbf{r})$, replacing it with an effective central potential:

$$V_i(r) = \frac{1}{4\pi} \int V_i(\mathbf{r}) d\Omega. \quad (20.27)$$

Under this central-field approximation, the spatial orbitals separate into radial functions and standard spherical harmonics $Y_{lm_l}(\theta, \phi)$, allowing the atomic states to be classified using the familiar quantum numbers (n, l, m_l, m_s) .

The Hartree equations can also be derived variationally by minimizing the expectation value of the total Hamiltonian with respect to independent variations of the single-particle orbitals φ_i^* , subject to the normalization constraints $\int |\varphi_i|^2 d^3r = 1$. The resulting total ground-state energy is not simply the sum of the orbital energies $\sum_i \varepsilon_i$, because summing the eigenvalues counts the mutual Coulomb repulsion energy twice:

$$\begin{aligned} \sum_i \varepsilon_i &= \sum_i \int d^3r \varphi_i^*(\mathbf{r}) \left[-\frac{\nabla^2}{2m} - \frac{Ze^2}{r} + \sum_{j \neq i} \int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\varphi_j(\mathbf{r}')|^2 \right] \varphi_i(\mathbf{r}) \\ &= \sum_i \langle T_i + V_{\text{nuc},i} \rangle + \sum_{i \neq j} \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\varphi_i(\mathbf{r})|^2 |\varphi_j(\mathbf{r}')|^2. \end{aligned} \quad (20.28)$$

To obtain the true expectation value of the Hamiltonian, one must subtract the double-counted electronic repulsion:

$$E = \sum_i \varepsilon_i - \frac{1}{2} \sum_{i \neq j} \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\varphi_i(\mathbf{r})|^2 |\varphi_j(\mathbf{r}')|^2. \quad (20.29)$$

20.4 The Thomas-Fermi Approximation

For heavy atoms containing a large number of electrons ($Z \gg 1$), numerical solution of the Hartree equations becomes increasingly involved. In this regime, the global features of the self-consistent potential can be accurately captured by a statistical, semiclassical model developed independently by Thomas and Fermi.

In a large atom, the interaction of a given electron with itself is negligible compared to the collective interaction with the remaining $N - 1 \approx Z$ electrons. Therefore, the average potential $V(\mathbf{r})$ experienced by any atomic electron closely approximates the total electrostatic potential generated by the nucleus and the continuous electronic charge cloud:

$$V(\mathbf{r}) = -\frac{Ze^2}{r} + \int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} n(\mathbf{r}'), \quad (20.30)$$

where the total electron number density is

$$n(\mathbf{r}) = \sum_{j=1}^N |\varphi_j(\mathbf{r})|^2. \quad (20.31)$$

20.4.1 Semiclassical Local Fermi Gas Formulation

The Thomas-Fermi method is grounded in the semiclassical condition that, throughout the bulk of a heavy atom, the kinetic energy of the electrons is sufficiently high that the potential $V(\mathbf{r})$ varies slowly over the local de Broglie wavelength. Under this condition, the single-particle wave functions behave locally as plane waves:

$$\varphi_i(\mathbf{r}) \sim e^{i\mathbf{p}_i(\mathbf{r}) \cdot \mathbf{r}}, \quad (20.32)$$

with the local momentum $\mathbf{p}_i(\mathbf{r})$ governed by the classical energy relation

$$\varepsilon_i = \frac{p_i^2(\mathbf{r})}{2m} + V(\mathbf{r}). \quad (20.33)$$

In the atomic ground state, the electrons fill available single-particle states up to a common Fermi energy ε_F . At any point $\mathbf{r}$, the local momentum varies from zero up to a maximum Fermi momentum:

$$p_F(\mathbf{r}) = \sqrt{2m[\varepsilon_F - V(\mathbf{r})]}. \quad (20.34)$$

Only regions where $\varepsilon_F > V(\mathbf{r})$ are classically accessible. Treating the electron cloud locally as a degenerate, non-interacting Fermi gas, the two spin states per spatial phase-space volume $h^3 = (2\pi)^3$ yield the standard relation between local density and Fermi momentum:

$$n(\mathbf{r}) = 2 \int_{p \leq p_F(\mathbf{r})} \frac{d^3p}{(2\pi)^3} = \frac{p_F^3(\mathbf{r})}{3\pi^2}. \quad (20.35)$$

Substituting Eq. (20.34) gives the local density directly in terms of the potential:

$$n(\mathbf{r}) = \frac{\{2m[\varepsilon_F - V(\mathbf{r})]\}^{3/2}}{3\pi^2}. \quad (20.36)$$

20.4.2 The Thomas-Fermi Equation

Applying the Laplacian to the electrostatic potential (20.30) and using $\nabla^2(1/|\mathbf{r}-\mathbf{r}'|) = -4\pi\delta(\mathbf{r}-\mathbf{r}')$, we recover Poisson's equation for $r > 0$:

$$\nabla^2 V(\mathbf{r}) = -4\pi e^2 n(\mathbf{r}). \quad (20.37)$$

Assuming spherical symmetry and substituting the density from Eq. (20.36) leads to a closed, nonlinear differential equation for $V(r)$:

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d[-V(r)]}{dr} \right) = \frac{4e^2}{3\pi} \{2m[\varepsilon_F - V(r)]\}^{3/2}. \quad (20.38)$$

The physical boundary conditions are:

1. As $r \rightarrow 0$, the potential is dominated by the bare nuclear attraction:

$$V(r) \rightarrow -\frac{Ze^2}{r}. \quad (20.39)$$

2. At the boundary of a neutral atom ($N = Z$), the electron density vanishes at infinity, requiring $V(\infty) = 0$ and consequently setting the Fermi level to zero:

$$\varepsilon_F = 0. \quad (20.40)$$

To convert Eq. (20.38) into universal dimensionless form, we introduce the dimensionless variable x and screening function $\Phi(x)$ defined by

$$V(r) = -\frac{Ze^2}{r}\Phi(x), \quad r = \frac{bx}{Z^{1/3}}, \quad (20.41)$$

where the characteristic length scale is

$$b = \frac{1}{2} \left(\frac{3\pi}{4} \right)^{2/3} a_0 \approx 0.8853 a_0, \quad (20.42)$$

with $a_0 = \hbar^2/(me^2)$ being the Bohr radius. Under this transformation, Eq. (20.38) reduces to the universal *Thomas-Fermi equation*:

$$x^{1/2} \frac{d^2\Phi(x)}{dx^2} = \Phi^{3/2}(x), \quad (20.43)$$

subject to the universal boundary conditions $\Phi(0) = 1$ and $\Phi(\infty) = 0$.

Numerical integration of Eq. (20.43) yields the asymptotic behaviors

$$\Phi(x) \approx \begin{cases} 1 - 1.588x, & x \rightarrow 0, \\ \frac{144}{x^3}, & x \rightarrow \infty. \end{cases} \quad (20.44)$$

Near the nucleus ($r \rightarrow 0$), the potential expands as

$$V(r) = -\frac{Ze^2}{r} + 1.80 Z^{4/3} \frac{e^2}{a_0} + \mathcal{O}(r), \quad (20.45)$$

where the second term represents the finite electrostatic potential at the nucleus produced by the surrounding electron cloud.

20.4.3 Physical Implications and Limitations

The Thomas-Fermi formulation provides powerful scaling laws for heavy atoms:

- **Spatial scale:** The characteristic atomic radius contracts with nuclear charge as $r \propto Z^{-1/3}$.
- **Potential and density:** At a fixed scaled radius x , the potential scales as $V \propto Z^{4/3}$, the electron density scales as $n \propto Z^2$, and the average electronic momentum scales as $\bar{p} \propto Z^{2/3}$.

The model also offers insight into the sequential appearance of angular momentum subshells across the periodic table. For an electron with orbital angular momentum l , the effective radial potential is

$$V_{\text{eff}}(r) = V(r) + \frac{l(l+1)}{2mr^2}. \quad (20.46)$$

A bound state of angular momentum l can only form if $V_{\text{eff}}(r)$ possesses a negative minimum. Analysis of the Thomas-Fermi potential reveals that:

- For $Z < 4$, a negative minimum exists only for $l = 0$ (s orbitals).
- For $4 \leq Z < 19$, minima exist for $l = 0$ and $l = 1$ (s and p orbitals).
- For $19 \leq Z < 53$, minima exist for $l = 0, 1, 2$ (s, p, and d orbitals).

These thresholds capture the onset of subshell filling in the periodic table, where experimentally p electrons appear at $Z = 5$ (B), d electrons at $Z = 21$ (Sc), and f electrons at $Z = 58$ (Ce).

Despite its utility, the Thomas-Fermi approximation has notable limitations:

1. **Nuclear vicinity** ($r \lesssim a_0/Z$): The potential varies too rapidly, violating the semiclassical validity condition.
2. **Atomic perimeter** ($r \gtrsim a_0$): The local kinetic energy becomes small, and the model predicts an unphysical power-law decay ($\Phi \sim x^{-3}$) instead of the true exponential decay of electronic bound states, leading to an infinite atomic radius and the absence of chemical binding (Teller's theorem).
3. **Exchange neglect**: The standard Thomas-Fermi model treats electrons as a classical charge distribution without quantum exchange. Incorporating exchange leads to the Thomas-Fermi-Dirac model.

20.5 The Hartree-Fock Approximation

To properly incorporate the Pauli exclusion principle beyond the simple product ansatz of the Hartree approximation, the trial wave function must be fully antisymmetric under particle exchange. For an N -electron system, this is achieved by representing the state as a Slater determinant formed from a set of orthonormal single-particle spin-orbitals $\{\varphi_i\}$:

$$\Psi(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_1(1) & \varphi_1(2) & \cdots & \varphi_1(N) \\ \varphi_2(1) & \varphi_2(2) & \cdots & \varphi_2(N) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_N(1) & \varphi_N(2) & \cdots & \varphi_N(N) \end{vmatrix}, \quad (20.47)$$

where each variable $i \equiv (\mathbf{r}_i, s_i)$ encompasses spatial coordinates and spin projections. This formulation constitutes the *Hartree-Fock approximation*.

20.5.1 Total Energy and the Exchange Interaction

We now evaluate the expectation value of the atomic Hamiltonian in the determinantal state (20.47). The total electronic kinetic energy is given by

$$\langle T \rangle = \sum_{i=1}^N \int d^3r \frac{|\nabla \varphi_i(\mathbf{r})|^2}{2m}, \quad (20.48)$$

and the electron-nucleus Coulomb attraction is

$$\langle V_{\text{nuc}} \rangle = - \sum_{i=1}^N \int d^3r \frac{Ze^2}{r} |\varphi_i(\mathbf{r})|^2. \quad (20.49)$$

In second quantization, the electron-electron Coulomb interaction operator is written as

$$V_{ee} = \frac{1}{2} \sum_{s,s'} \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}), \quad (20.50)$$

where $\psi_s(\mathbf{r})$ is the field operator destroying an electron of spin s at position $\mathbf{r}$. For a single determinantal state, Wick's theorem yields the two-particle density matrix (pair correlation function):

$$\begin{aligned} \langle \psi_s^\dagger(\mathbf{r}) \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \psi_s(\mathbf{r}) \rangle &= \langle \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) \rangle \langle \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}') \rangle \\ &\quad - \delta_{s,s'} \langle \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}') \rangle \langle \psi_{s'}^\dagger(\mathbf{r}') \psi_{s'}(\mathbf{r}) \rangle. \end{aligned} \quad (20.51)$$

The local spin density and the single-particle density matrix are expressible in terms of the occupied orbitals as

$$\sum_s \langle \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}) \rangle = \sum_{i=1}^N |\varphi_i(\mathbf{r})|^2 = n(\mathbf{r}), \quad (20.52)$$

$$\langle \psi_s^\dagger(\mathbf{r}) \psi_s(\mathbf{r}') \rangle = \sum_{\substack{i=1 \\ (s_i=s)}}^N \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}'). \quad (20.53)$$

Combining Eqs. (20.48)–(20.53), the total energy expectation value becomes

$$\begin{aligned} \langle H \rangle &= \sum_{i=1}^N \int d^3r \left[\frac{|\nabla \varphi_i(\mathbf{r})|^2}{2m} - \frac{Ze^2}{r} |\varphi_i(\mathbf{r})|^2 \right] \\ &\quad + \frac{1}{2} \sum_{i,j=1}^N \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} |\varphi_i(\mathbf{r})|^2 |\varphi_j(\mathbf{r}')|^2 \\ &\quad - \frac{1}{2} \sum_{i,j=1}^N \delta_{s_i, s_j} \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \varphi_i^*(\mathbf{r}) \varphi_j^*(\mathbf{r}') \varphi_j(\mathbf{r}) \varphi_i(\mathbf{r}'). \end{aligned} \quad (20.54)$$

The second sum is the direct classical Hartree repulsion, whereas the final sum is the quantum *exchange energy*. Notice that for $i = j$, the direct and exchange integrals are identical and cancel identically, automatically eliminating the spurious self-interaction present in the original Hartree method.

20.5.2 The Hartree-Fock Equations and Koopmans' Theorem

Minimizing the energy functional (20.54) with respect to orbital variations $\varphi_i^*(\mathbf{r})$, subject to the orthonormalization constraints $\int d^3r \varphi_i^*(\mathbf{r}) \varphi_j(\mathbf{r}) = \delta_{ij}$, yields the self-consistent *Hartree-Fock equations*:

$$\begin{aligned} \left(-\frac{\nabla^2}{2m} - \frac{Ze^2}{r} \right) \varphi_i(\mathbf{r}) + \int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \sum_{j=1}^N |\varphi_j(\mathbf{r}')|^2 \varphi_i(\mathbf{r}) \\ - \sum_{j=1}^N \delta_{s_i, s_j} \left[\int d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \varphi_j^*(\mathbf{r}') \varphi_i(\mathbf{r}') \right] \varphi_j(\mathbf{r}) = \varepsilon_i \varphi_i(\mathbf{r}). \end{aligned} \quad (20.55)$$

The nonlocal exchange operator acts on φ_i through an integral kernel involving $\varphi_j^*(\mathbf{r}') \varphi_i(\mathbf{r}')$. Because it contains the antisymmetrized pair amplitude $\varphi_j(\mathbf{r}') \varphi_i(\mathbf{r}) - \varphi_j(\mathbf{r}) \varphi_i(\mathbf{r}')$, the exclusion principle creates a depletion of electrons of parallel spin around each electron (the *exchange hole* or Fermi hole).

Multiplying Eq. (20.55) by $\varphi_i^*(\mathbf{r})$ and integrating over all space yields the orbital eigenvalue ε_i :

$$\begin{aligned} \varepsilon_i = & \int d^3r \left(\frac{|\nabla \varphi_i(\mathbf{r})|^2}{2m} - \frac{Ze^2}{r} |\varphi_i(\mathbf{r})|^2 \right) \\ & + \sum_{j=1}^N \int d^3r d^3r' \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \varphi_i^*(\mathbf{r}) \varphi_j^*(\mathbf{r}') [\varphi_j(\mathbf{r}') \varphi_i(\mathbf{r}) - \delta_{s_i, s_j} \varphi_j(\mathbf{r}) \varphi_i(\mathbf{r}')]. \end{aligned} \quad (20.56)$$

Comparing Eq. (20.56) with the total energy (20.54), one observes that $-\varepsilon_i$ corresponds to the energy required to remove an electron from orbital φ_i to infinity, under the approximation that the remaining orbitals are “frozen” (unrelaxed). This physical interpretation is known as *Koopmans’ theorem*. In reality, the remaining $N - 1$ electrons relax into a slightly lower energy configuration once an electron is removed, so $-\varepsilon_i$ provides a slight overestimate of the ionization potential.

The ground-state energy computed via Hartree-Fock is lower and more accurate than that obtained from the Hartree product. However, it still exceeds the exact nonrelativistic ground-state energy by roughly 1 eV per electron pair. This remaining energy discrepancy is defined as the *correlation energy*:

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}. \quad (20.57)$$

It stems from dynamical correlations: electrons avoid one another due to Coulomb repulsion irrespective of spin, whereas Hartree-Fock only correlates electrons with parallel spins via the exchange hole.

20.5.3 Shell Structure and Electron Configurations

In the central-field approximation, the effective potential is spherically averaged, allowing the Hartree-Fock orbitals to be labeled by the quantum numbers (n, l, m_l, m_s) :

$$\varphi_i(\mathbf{r}, s) = R_{nl}(r) Y_{lm_l}(\theta, \phi) \chi_{m_s}(s). \quad (20.58)$$

Here $n \geq l + 1$ is the principal quantum number, and the radial wave function $R_{nl}(r)$ has $n - l - 1$ radial nodes. Unlike the pure Coulomb potential where states with different l are degenerate, the screening in multi-electron atoms lifts this degeneracy: states of lower orbital angular momentum l penetrate closer to the nucleus, experiencing less screening and therefore having lower energies ($\varepsilon_{ns} < \varepsilon_{np} < \varepsilon_{nd} < \varepsilon_{nf}$).

For a given n and l , there are $(2s + 1)(2l + 1) = 2(2l + 1) = 4l + 2$ degenerate states, forming a subshell:

- s subshell ($l = 0$): 2 electrons,
- p subshell ($l = 1$): 6 electrons,
- d subshell ($l = 2$): 10 electrons,
- f subshell ($l = 3$): 14 electrons.

An atomic state is characterized by its *electron configuration*, which specifies the occupation number of each subshell. For example, ground-state nitrogen ($Z = 7$) has the configuration $(1s)^2(2s)^2(2p)^3$.

20.5.4 The Aufbau Principle Across the Rows

The systematic filling of subshells with increasing atomic number Z (the *Aufbau principle*) governs the rows of the periodic table:

- **Row 1** ($Z = 1, 2$): Filling the 1s shell (H : $1s^1$, He : $1s^2$).
- **Row 2** ($Z = 3\text{--}10$): Filling 2s (Li, Be) followed by 2p (B through Ne).
- **Row 3** ($Z = 11\text{--}18$): Core is neon; filling 3s (Na, Mg) followed by 3p (Al through Ar).
- **Row 4** ($Z = 19\text{--}36$): Energetic competition occurs between 4s and 3d. The 4s shell fills first in K and Ca. Across the 3d transition metals (Sc to Zn), the 3d orbitals fill, with occasional transfer to achieve half-filled or filled subshells (Cr : $3d^5 4s^1$, Cu : $3d^{10} 4s^1$). The row terminates with the filling of the 4p subshell (Ga to Kr). Because 3d orbitals are more localized than the outer 4s/4p electrons, they contribute little to primary valence chemistry but provide localized magnetic moments in elements such as Fe, Co, and Ni.
- **Row 5** ($Z = 37\text{--}54$): An analogous competition occurs between 5s and 4d, terminating with the filled 4d shell in Pd ($4d^{10}$) and Ag ($4d^{10} 5s^1$), and completing with the 5p shell in Xe.
- **Row 6** ($Z = 55\text{--}86$): Filling begins with 6s (Cs, Ba), passes through La ($5d^1$), and enters the lanthanide series (4f filling from Ce to Lu). Because 4f electrons are strongly screened by outer shells, lanthanides exhibit nearly identical chemical properties. The 5d and 6p shells then fill out to radon (Rn).
- **Row 7** ($Z = 87\text{--}118$): The 7s, 6d, and 5f (actinide) shells compete in heavy radioactive elements.

20.6 The Periodic Table and Chemical Periodicity

The chemical properties of an element are largely dictated by its valence electrons—those occupying the outermost, least tightly bound shells. Consequently, elements that share identical outer-shell configurations occupy the same vertical group (or column) in the periodic table and display strikingly similar chemical behavior.

20.6.1 Noble Gases (Inert Gases)

The rightmost column of the periodic table consists of the noble gases (He, Ne, Ar, Kr, Xe, Rn). Excluding helium (which has a completed $1s^2$ shell), each noble gas is characterized by a filled valence p subshell ($ns^2 np^6$), corresponding to an octet of electrons.

To promote an electron from a closed shell to an unoccupied orbital requires increasing the principal quantum number n to $n + 1$, which involves a substantial excitation energy. For example, promoting an electron in neon from its ground-state configuration $(1s)^2(2s)^2(2p)^6$ to the excited configuration $(1s)^2(2s)^2(2p)^5(3s)^1$ requires approximately 16 eV. This large energy gap, combined with the spherical symmetry and zero total angular momentum of completed shells, renders noble gases exceptionally stable and chemically inert under standard conditions.

Closed-shell stability is fundamentally different from a simple large excitation gap: in atomic hydrogen, the excitation energy to the 2p state is 10.2 eV, yet hydrogen is chemically reactive because an open shell can form covalent or ionic bonds that lower the overall system energy.

20.6.2 Alkali Metals

At the opposite extreme, the first column on the left contains the alkali metals (Li, Na, K, Rb, Cs, Fr). These elements possess a single outer s electron orbiting outside a stable, spherically symmetric noble-gas core:

$$[\text{core}] ns^1. \quad (20.59)$$

Because the inner core electrons screen all but approximately one unit of nuclear charge, this outermost valence electron is weakly bound. For instance, in lithium, exciting the valence electron from the $(1s)^2(2s)^1$ ground state to $(1s)^2(2p)^1$ requires only about 2 eV, and the first ionization energy is merely 5.39 eV. As a result, the alkali metals readily lose their outer electron to achieve a closed-shell electronic configuration, forming ionic compounds with oxidation state +1.

20.6.3 Halogens

In the column immediately preceding the noble gases are the halogens (F, Cl, Br, I, At), which possess the valence configuration

$$ns^2 np^5. \quad (20.60)$$

Each halogen lacks just one electron to complete its outer p shell. Electrostatically, this configuration can be described as a filled shell containing a single positively charged “hole.” Because the effective nuclear charge perceived by the valence electrons is large ($Z_{\text{eff}} \approx Z - \text{inner shielding}$), the halogens have large electron affinities and high electronegativities. They avidly capture an additional electron to attain a stable octet, forming univalent anions (e.g., F^- , Cl^-) or engaging in single covalent bonds.

20.7 Splitting of Configurations and LS Coupling

An electronic configuration (such as $(1s)^2(2s)^2(2p)^2$ for carbon) specifies only the distribution of electrons among subshells (nl), but does not uniquely specify an individual quantum state. For example, the excited configuration $(1s)^2(2s)^1(2p)^1$ of beryllium represents $2 \times 6 = 12$ distinct quantum states, since the 2s electron possesses two spin projections ($m_s = \pm 1/2$) and the 2p electron has six accessible spin-orbital states ($m_l = 0, \pm 1$ and $m_s = \pm 1/2$).

States belonging to the same configuration are split in energy by two physical mechanisms:

1. Residual electrostatic Coulomb interactions between valence electrons, which depend on their spatial and spin symmetry and split the configuration into energy levels separated on the order of electron volts ($\sim 0.1\text{--}1$ eV).
2. Relativistic spin-orbit coupling, which further splits each electrostatic multiplet into fine-structure components with typical splittings on the order of hundredths of an electron volt ($\sim 10^{-2}$ eV in light atoms).

20.7.1 Russell-Saunders (LS) Coupling

In lighter atoms ($Z \lesssim 30$), the residual electrostatic interaction significantly exceeds the spin-orbit interaction. This regime is governed by *Russell-Saunders* (or *LS*) coupling.

In the absence of external fields, the total angular momentum of the atom,

$$\mathbf{J} = \mathbf{L} + \mathbf{S}, \quad (20.61)$$

is a rigorous constant of motion. When spin-orbit coupling is neglected to zero order, the total orbital angular momentum $\mathbf{L} = \sum_i \mathbf{l}_i$ and total spin angular momentum $\mathbf{S} = \sum_i \mathbf{s}_i$ commute

separately with the nonrelativistic Hamiltonian:

$$[H_0, \mathbf{L}] = 0, \quad [H_0, \mathbf{S}] = 0. \quad (20.62)$$

Consequently, atomic energy eigenstates can be labeled by the total orbital and spin angular momentum quantum numbers L , S , and their magnetic projections M_L , M_S . Although the Hamiltonian commutes with the individual electron spins $\mathbf{s}_i$, the physical states are generally not eigenstates of individual spin projections because the antisymmetrization required by the Pauli principle entangles the electron spins. For example, two electrons in an antisymmetric spin-singlet state,

$$|\chi_{0,0}\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad (20.63)$$

is an eigenstate of total spin $S = 0$ and $M_S = 0$, but neither electron has a definite s_{iz} .

In the absence of spin-orbit coupling, all $(2L + 1)(2S + 1)$ states of a given L and S are degenerate. Spectroscopists designate these terms using the standard notation

$$^{2S+1}L_J, \quad (20.64)$$

where the multiplicity $2S + 1$ is written as a left superscript, L is represented by a capital letter (S, P, D, F, G, ... for $L = 0, 1, 2, 3, 4, \dots$), and the total angular momentum quantum number J appears as a right subscript. When spin-orbit coupling is introduced, each term splits into J -levels ranging over

$$|L - S| \leq J \leq L + S. \quad (20.65)$$

20.7.2 Classification of Terms for Equivalent Electrons

Completed (closed) subshells have zero net orbital, spin, and total angular momenta:

$$L_{\text{core}} = 0, \quad S_{\text{core}} = 0, \quad J_{\text{core}} = 0. \quad (20.66)$$

Hence, atomic terms are entirely determined by open-shell valence electrons:

- **Hydrogen** ($1s^1$): $L = 0$, $S = 1/2 \Rightarrow {}^2S_{1/2}$.
- **Helium** ($1s^2$): Closed shell $\Rightarrow {}^1S_0$.
- **Boron** ($2p^1$): A single p electron has $L = 1$, $S = 1/2$. The allowed J values are $J = 1/2$ and $J = 3/2$, yielding the terms ${}^2P_{1/2}$ (degeneracy $2J + 1 = 2$) and ${}^2P_{3/2}$ (degeneracy $2J + 1 = 4$), accounting for all $2(2l + 1) = 6$ states.
- **Carbon** ($2p^2$): Two equivalent electrons in the 2p subshell yield $\binom{6}{2} = 15$ anti-symmetrized states:
 - $L = 2$: The spatial wave function is symmetric under particle interchange; the spin state must therefore be antisymmetric (singlet, $S = 0$). This yields the 1D_2 term ($2J + 1 = 5$ states).
 - $L = 1$: The spatial wave function is antisymmetric under interchange; the spin state must be symmetric (triplet, $S = 1$). This yields ${}^3P_0, {}^3P_1, {}^3P_2$ ($1 + 3 + 5 = 9$ states).
 - $L = 0$: The spatial wave function is symmetric; the spin state must be antisymmetric ($S = 0$), yielding the 1S_0 term (1 state).

Altogether, $5 + 9 + 1 = 15$ states.

20.7.3 Hund's Rules

The energetic ordering of multiplets arising from a given electron configuration is governed by *Hund's rules*:

1. **Maximum Spin Multiplicity:** The term with the maximum total spin S has the lowest energy.
2. **Maximum Orbital Angular Momentum:** For a given value of S , the term with the maximum total orbital angular momentum L has the lowest energy.
3. **Spin-Orbit Splitting (J -Value):** For an unfilled subshell:

- If the subshell is *not more than half-filled*, the lowest energy level has the minimum total angular momentum:

$$J = |L - S|. \quad (20.67)$$

- If the subshell is *more than half-filled*, the lowest energy level has the maximum total angular momentum:

$$J = L + S. \quad (20.68)$$

Physical Origin of Hund's Rules.

- *First Rule:* Maximizing S produces a symmetric total spin state. By the Pauli principle, the spatial wave function must be antisymmetric under electron interchange. An antisymmetric spatial wave function vanishes whenever two electrons approach each other ($\psi(\mathbf{r}_1, \mathbf{r}_1) = 0$), which minimizes interelectronic Coulomb repulsion and lowers the electrostatic energy.
- *Second Rule:* For electrons with parallel spins, maximizing L corresponds classically to electrons orbiting in the same spatial sense. This reduces close encounters between electrons compared to counter-rotating configurations, thereby further lowering repulsive Coulomb energy.
- *Third Rule:* As will be derived in subsequent sections, the spin-orbit interaction takes the form $\xi(r)\mathbf{L} \cdot \mathbf{S}$ for less than half-filled shells (with $\xi > 0$, favoring parallel antiparallel alignment $\mathbf{J} = \mathbf{L} - \mathbf{S}$), but reverses sign ($-\xi(r)\mathbf{L} \cdot \mathbf{S}$) for more than half-filled shells due to hole conjugation, favoring parallel alignment $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

20.7.4 Ground-State Determinations Across the 2p Shell

- **Boron ($2p^1$):** Unfilled shell (1/6 full). $L = 1$, $S = 1/2$. By Rule 3, the ground state has $J = |1 - 1/2| = 1/2$, namely $^2P_{1/2}$.
- **Carbon ($2p^2$):** Terms are 1D , 3P , 1S . Rule 1 selects $S = 1$ (3P). Because the subshell is less than half-full, Rule 3 gives $J = |1 - 1| = 0$, identifying the ground state as 3P_0 .
- **Nitrogen ($2p^3$):** $\binom{6}{3} = 20$ states. Maximizing spin gives $S = 3/2$ (all three spins parallel). Pauli exclusion forces the electrons into distinct spatial orbitals with $m_l = +1, 0, -1$, so $M_L = \sum m_l = 0$, giving $L = 0$. Hence the term is $^4S_{3/2}$.
- **Oxygen ($2p^4$):** Four electrons are equivalent to two holes in a filled $2p^6$ shell, producing the same terms as $2p^2$: 1D , 3P , 1S . Hund's first rule selects 3P . Because the shell is more than half-filled, Rule 3 yields $J = 1 + 1 = 2$, identifying the ground state as 3P_2 .
- **Fluorine ($2p^5$):** A single hole in the 2p subshell yields $^2P_{3/2}$ as the ground state.
- **Neon ($2p^6$):** Completed subshell, ground state 1S_0 .

20.7.5 Multideterminant Representation of Terms

While single-determinant states often suffice for closed shells or maximum-spin configurations, many angular momentum eigenstates cannot be written as a single Slater determinant. For example, the 1S_0 state of a p^2 configuration is an entangled state given by a linear combination of three Slater determinants:

$$|^1S_0\rangle = \frac{1}{\sqrt{3}} \left[\det(p_1 \uparrow, p_{-1} \downarrow) - \det(p_0 \uparrow, p_0 \downarrow) + \det(p_{-1} \uparrow, p_1 \downarrow) \right]. \quad (20.69)$$

In such cases, the total energy of each spectroscopic term is calculated by first obtaining the self-consistent Hartree-Fock orbitals, constructing the appropriate linear combinations of determinants using Clebsch-Gordan coefficients, and evaluating the expectation value of the full Hamiltonian.

20.8 Spin-Orbit Interaction and Landé's Interval Rule

We now examine the fine-structure splitting of an LS multiplet caused by relativistic spin-orbit coupling. For an N -electron atom, the spin-orbit Hamiltonian is given by

$$H_{so} = \sum_{i=1}^N \mathbf{X}(\mathbf{r}_i) \cdot \mathbf{s}_i, \quad (20.70)$$

where $\mathbf{s}_i$ is the spin operator of the i th electron, and the effective magnetic field operator $\mathbf{X}(\mathbf{r}_i)$ experienced in its rest frame (including the Thomas precession factor of $1/2$) is

$$\mathbf{X}(\mathbf{r}_i) = -\frac{|e|\hbar}{2m^2c^2} \mathbf{p}_i \times \mathbf{E}_i(\mathbf{r}_i). \quad (20.71)$$

Here $\mathbf{p}_i$ is the momentum operator and $\mathbf{E}_i(\mathbf{r}_i)$ is the total microscopic electric field acting on the i th electron due to the nucleus and all other electrons.

20.8.1 Effective Hamiltonian via the Wigner-Eckart Theorem

In light atoms, where electronic velocities are nonrelativistic ($v/c \ll 1$), H_{so} can be treated as a weak perturbation on the degenerate $(2L+1)(2S+1)$ states belonging to an unperturbed LS term. We denote the unperturbed basis states by $|NLSM_L M_S\rangle$, where N encompasses all additional quantum numbers (such as the electron configuration).

Because the unperturbed nonrelativistic Hamiltonian does not depend on spin, each eigenstate is a linear combination of spatial states of definite (L, M_L) tensored with spin states of definite (S, M_S) :

$$|NLSM_L M_S\rangle = \sum_{\alpha} c_{\alpha} |\psi_{NLM_L}^{(\alpha)}\rangle |\chi_{SM_S}^{(\alpha)}\rangle. \quad (20.72)$$

The matrix elements of H_{so} within the degenerate multiplet take the form

$$\langle NLSM_L M_S | \mathbf{X}_i \cdot \mathbf{s}_i | NLSM'_L M'_S \rangle = \sum_{\alpha, \alpha'} c_{\alpha}^* c_{\alpha'} \langle \alpha NLM_L | \mathbf{X}_i | \alpha' NLM'_L \rangle \cdot \langle \alpha SM_S | \mathbf{s}_i | \alpha' SM'_S \rangle. \quad (20.73)$$

By the Wigner-Eckart theorem (specifically the projection theorem for vector operators), the matrix elements of any vector operator within a subspace of constant angular momentum are strictly proportional to the matrix elements of the total angular momentum operator itself:

$$\langle \alpha SM_S | \mathbf{s}_i | \alpha' SM'_S \rangle \propto \langle SM_S | \mathbf{S} | SM'_S \rangle, \quad (20.74)$$

$$\langle \alpha NLM_L | \mathbf{X}_i | \alpha' NLM'_L \rangle \propto \langle LM_L | \mathbf{L} | LM'_L \rangle. \quad (20.75)$$

Crucially, the constants of proportionality depend on the internal structure (N, L, S) but are entirely independent of the magnetic quantum numbers M_L, M'_L and M_S, M'_S .

Summing over all electrons $i = 1, \dots, N$, the matrix elements of the microscopic spin-orbit Hamiltonian within the LS multiplet reduce to

$$\langle NLSM_L M_S | H_{\text{so}} | NLSM'_L M'_S \rangle = \xi(NLS) \langle NLSM_L M_S | \mathbf{L} \cdot \mathbf{S} | NLSM'_L M'_S \rangle, \quad (20.76)$$

where $\xi(NLS)$ is the *spin-orbit coupling constant* for the multiplet. Thus, within the subspace of a given LS term, the perturbation is rigorously represented by the effective operator

$$H_{\text{so}}^{\text{eff}} = \xi(NLS) \mathbf{L} \cdot \mathbf{S}. \quad (20.77)$$

20.8.2 Fine-Structure Splitting and Landé's Interval Rule

To diagonalize $H_{\text{so}}^{\text{eff}}$, we transform from the uncoupled basis $\{|M_L, M_S\rangle\}$ to the coupled total angular momentum basis $\{|J, M_J\rangle\}$, where $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Squaring both sides,

$$\mathbf{J}^2 = (\mathbf{L} + \mathbf{S})^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L} \cdot \mathbf{S}, \quad (20.78)$$

which yields the identity

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2} (\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2). \quad (20.79)$$

Since the coupled states $|NLSJM_J\rangle$ are simultaneous eigenstates of $\mathbf{J}^2, \mathbf{L}^2, \mathbf{S}^2$, and J_z , the perturbation matrix is diagonal in this basis:

$$\begin{aligned} \langle NLSJM_J | H_{\text{so}} | NLSJ'M'_J \rangle &= \delta_{JJ'} \delta_{M_J M'_J} \Delta E_{\text{so}}(J), \\ \Delta E_{\text{so}}(J) &= \frac{1}{2} \xi(NLS) [J(J+1) - L(L+1) - S(S+1)]. \end{aligned} \quad (20.80)$$

Each J -level retains a residual $(2J+1)$ -fold magnetic degeneracy with respect to M_J .

The energy separation between two adjacent fine-structure levels J and $J-1$ within the same multiplet is

$$\Delta E_{\text{so}}(J) - \Delta E_{\text{so}}(J-1) = \frac{1}{2} \xi(NLS) [J(J+1) - (J-1)J] = \xi(NLS) J. \quad (20.81)$$

This striking linear dependence on J is *Landé's interval rule*.

The sign of the spin-orbit constant $\xi(NLS)$ determines the ordering of fine-structure levels:

- **Regular Multiplet** ($\xi > 0$): The lowest energy level corresponds to the minimum total angular momentum, $J = |L - S|$. This occurs for subshells that are less than half-filled.
- **Inverted Multiplet** ($\xi < 0$): The lowest energy level corresponds to the maximum total angular momentum, $J = L + S$. This occurs for subshells that are more than half-filled.

Landé's interval rule therefore provides the theoretical foundation for Hund's third rule.

20.9 Spin-Orbit Splitting in the Hartree-Fock Approximation

We now calculate the spin-orbit parameter $\xi(NLS)$ explicitly within the central-field Hartree-Fock framework and establish Hund's third rule from first principles.

20.9.1 Central-Field Spin-Orbit Operator

In a spherically symmetric central potential $V(r)$, the electric field is purely radial:

$$\mathbf{E}(\mathbf{r}) = -\nabla V(r) = -\frac{\mathbf{r}}{r} \frac{dV(r)}{dr}. \quad (20.82)$$

Consequently, the rest-frame magnetic field operator (20.71) becomes

$$\mathbf{X}_i = -\frac{|e|\hbar}{2m^2c^2} \mathbf{p}_i \times \left(-\frac{\mathbf{r}_i}{r_i} \frac{dV}{dr_i} \right) = \frac{1}{2m^2c^2r_i} \frac{dV}{dr_i} (\mathbf{r}_i \times \mathbf{p}_i) = \frac{1}{2m^2c^2r_i} \frac{dV}{dr_i} \mathbf{l}_i, \quad (20.83)$$

where $\mathbf{l}_i = \mathbf{r}_i \times \mathbf{p}_i$ is the orbital angular momentum operator of the i th electron. The spin-orbit Hamiltonian for the atom is therefore

$$H_{\text{so}} = \sum_{i=1}^N \frac{1}{2m^2c^2r_i} \frac{dV}{dr_i} \mathbf{l}_i \cdot \mathbf{s}_i. \quad (20.84)$$

20.9.2 Matrix Elements within an LS Multiplet

To determine the constant $\xi(NLS)$, it is sufficient to evaluate the diagonal matrix element of H_{so} in the state with maximum magnetic quantum numbers, $M_L = L$ and $M_S = S$:

$$\langle NLSLS | H_{\text{so}} | NLSLS \rangle = \xi(NLS) LS. \quad (20.85)$$

We express the state $|NLSLS\rangle$ as a linear combination of Slater determinants $|N\beta\rangle$:

$$|NLSLS\rangle = \sum_{\beta} a_{\beta} |N\beta\rangle, \quad \sum_{\beta} |a_{\beta}|^2 = 1. \quad (20.86)$$

Each determinant $|N\beta\rangle$ is constructed from central-field spin-orbitals $\varphi_{nlm_l m_s}(\mathbf{r}, s)$, sharing the same radial functions $R_{nl}(r)$ and having eigenvalues $\sum m_l = L$ and $\sum m_s = S$.

Since H_{so} is a sum of single-particle operators, matrix elements between different determinants $\langle N\beta | H_{\text{so}} | N\beta' \rangle$ with $\beta \neq \beta'$ vanish if they differ by more than one orbital. Moreover, because $l_{ix}s_{ix} + l_{iy}s_{iy}$ changes m_l and m_s while preserving $m_l + m_s$, only the diagonal term $l_{iz}s_{iz}$ contributes to the expectation value in an eigenstate of L_z and S_z :

$$\langle N\beta | H_{\text{so}} | N\beta \rangle = \sum_{nlm_l m_s \in \beta} \zeta_{nl} m_l m_s, \quad (20.87)$$

where the single-electron spin-orbit radial integral is defined by

$$\zeta_{nl} = \frac{1}{2m^2c^2} \int_0^{\infty} dr r^2 |R_{nl}(r)|^2 \frac{1}{r} \frac{dV(r)}{dr}. \quad (20.88)$$

Because the potential $V(r)$ is spherically symmetric, ζ_{nl} is independent of m_l and m_s .

For any completely filled subshell, the sum over all magnetic sub-states vanishes identically:

$$\sum_{m_l=-l}^l m_l \sum_{m_s=\pm 1/2} m_s = \left(\sum_{m_l=-l}^l m_l \right) \left(\sum_{m_s=\pm 1/2} m_s \right) = 0 \times 0 = 0. \quad (20.89)$$

Thus, closed shells contribute nothing to the spin-orbit splitting, and the sum in Eq. (20.87) restricts entirely to the partially filled valence subshell (nl).

20.9.3 Regular vs. Inverted Multiplets

Assuming a single open subshell (nl) containing k electrons:

Case 1: Subshell less than half-filled ($k \leq 2l + 1$). According to Hund's first rule, the ground-state term possesses the maximum total spin $S = k/2$. In the state $M_S = S$, every valence electron has its spin aligned upward ($m_s = +1/2$). Therefore,

$$\sum_{\text{occ}} m_l m_s = \frac{1}{2} \sum_{\text{occ}} m_l = \frac{1}{2} M_L. \quad (20.90)$$

Since $S = k/2$, we can write $\frac{1}{2} = \frac{1}{2S}$ (or equivalently for $M_L = L$, $\frac{1}{2}L = \frac{1}{2S}LS$). Consequently,

$$\langle N\beta | H_{\text{so}} | N\beta \rangle = \frac{\zeta_{nl}}{2S} M_L S. \quad (20.91)$$

Because this value is identical for every determinant $|N\beta\rangle$ occurring in the expansion of $|NLSLS\rangle$, we find

$$\langle NLSLS | H_{\text{so}} | NLSLS \rangle = \frac{\zeta_{nl}}{2S} LS. \quad (20.92)$$

Comparing this with Eq. (20.85) yields

$$\xi(NLS) = +\frac{\zeta_{nl}}{2S} \quad (\text{less than half-filled}). \quad (20.93)$$

Since the attractive central potential satisfies $dV/dr > 0$ everywhere, the radial integral ζ_{nl} is strictly positive. Hence $\xi(NLS) > 0$: the multiplet is *regular*, and the level of lowest energy has minimum $J = |L - S|$.

Case 2: Subshell more than half-filled ($k > 2l + 1$). Here we employ the particle-hole equivalence. In the maximum-spin state $M_S = S$, all empty orbitals (holes) have spin down ($m_s = -1/2$). Since the sum over the entire filled shell is zero,

$$\sum_{\text{occupied}} m_l m_s = - \sum_{\text{unoccupied}} m_l m_s = - \left(\sum_{\text{holes}} m_l \right) \left(-\frac{1}{2} \right) = -\frac{1}{2} M_L^{\text{holes}} = -\frac{1}{2S} M_L S, \quad (20.94)$$

where we have used the fact that the total orbital projection of the occupied electrons is $M_L = -M_L^{\text{holes}}$. This leads directly to

$$\xi(NLS) = -\frac{\zeta_{nl}}{2S} \quad (\text{more than half-filled}). \quad (20.95)$$

Thus $\xi(NLS) < 0$: the multiplet is *inverted*, and the level of lowest energy has maximum $J = L + S$.

Case 3: Exactly half-filled subshell ($k = 2l + 1$). All $2l + 1$ orbital states are singly occupied with spin up ($m_s = +1/2$). Hence $M_L = \sum_{m_l=-l}^l m_l = 0$, implying $L = 0$. Since $L = 0$, the operator $\mathbf{L} \cdot \mathbf{S}$ vanishes identically, and there is no first-order spin-orbit splitting.

20.9.4 Magnitude and the Transition to *jj*-Coupling

In order of magnitude, the electric field near the core is $E \sim Ze/r^2$, giving

$$\zeta_{nl} \sim \frac{1}{m^2 c^2} \left\langle \frac{1}{r} \frac{dV}{dr} \right\rangle \sim \frac{Ze^2}{m^2 c^2} \left\langle \frac{1}{r^3} \right\rangle. \quad (20.96)$$

Because core-penetrating electrons experience an effective nuclear charge $Z_{\text{eff}} \sim Z$, the expectation value scales as $\langle 1/r^3 \rangle \sim Z^3/a_0^3$, leading to the parametric scaling

$$\zeta_{nl} \sim Z^4 \alpha^2 \text{ Ry}, \quad (20.97)$$

where $\alpha = e^2/(\hbar c) \approx 1/137$ is the fine-structure constant. Even when screening reduces the effective charge to $Z_{\text{eff}} \sim Z - \sigma$, the spin-orbit splitting grows rapidly with Z (typically as Z^2 to Z^4).

For light atoms ($Z \lesssim 30$), the fine-structure splitting ($\sim 10^{-4}$ to 10^{-2} eV) is much smaller than the electrostatic multiplet splitting ($\sim 1\text{--}5$ eV), justifying the *LS* (Russell-Saunders) coupling scheme. In heavy atoms ($Z \gg 30$), however, ζ_{nl} becomes comparable to or exceeds the Coulomb exchange splitting. In this regime, the spin-orbit interaction must be diagonalized first for each individual electron, coupling $\mathbf{l}_i + \mathbf{s}_i = \mathbf{j}_i$ to form single-particle states characterized by (n, l, j, m_j) . The total atomic states are then formed by coupling individual $\mathbf{j}_i$ vectors to total $\mathbf{J}$ —a framework known as the *jj-coupling scheme*.

20.10 The Zeeman Effect and the Paschen-Back Effect

When an atom is placed in an external uniform magnetic field $\mathcal{H} = \mathcal{H}\hat{\mathbf{z}}$, the interaction lifts the magnetic degeneracy of the fine-structure levels. To first order in the field, the magnetic interaction Hamiltonian is given by

$$H_{\text{mag}} = -\boldsymbol{\mu} \cdot \mathcal{H} = -\frac{e}{2mc}(\mathbf{L} + 2\mathbf{S}) \cdot \mathcal{H}, \quad (20.98)$$

where $e = -|e|$ is the electron charge and the factor of 2 represents the electron spin *g*-factor ($g_s \approx 2$). Introducing the Bohr magneton $\mu_B = |e|\hbar/(2mc)$, we may write Eq. (20.98) for a field along the *z*-axis as

$$H_{\text{mag}} = \frac{\mu_B}{\hbar}(L_z + 2S_z)\mathcal{H}. \quad (20.99)$$

20.10.1 Weak-Field Zeeman Effect and the Landé *g*-Factor

When the external field is weak compared to the internal atomic spin-orbit field ($\mathcal{H} \ll 10^5$ G), the spin-orbit coupling remains the dominant perturbation, and the atomic states are well-described by the coupled basis $|NLSJM_J\rangle$. The energy shift is then computed via first-order perturbation theory:

$$\Delta E(J, M_J) = \langle NLSJM_J | H_{\text{mag}} | NLSJM_J \rangle = \frac{\mu_B \mathcal{H}}{\hbar} \langle NLSJM_J | (J_z + S_z) | NLSJM_J \rangle, \quad (20.100)$$

where we have rewritten $L_z + 2S_z = J_z + S_z$. Since $|NLSJM_J\rangle$ is an eigenstate of J_z with eigenvalue $M_J\hbar$,

$$\langle NLSJM_J | J_z | NLSJM_J \rangle = M_J\hbar. \quad (20.101)$$

To evaluate the matrix element of S_z , we invoke the projection theorem for vector operators (a direct consequence of the Wigner-Eckart theorem). For any vector operator $\mathbf{V}$, its expectation value within a manifold of fixed total angular momentum J is proportional to that of $\mathbf{J}$:

$$\langle NLSJM_J | \mathbf{V} | NLSJM_J \rangle = \frac{\langle NLSJM_J | \mathbf{J} \cdot \mathbf{V} | NLSJM_J \rangle}{J(J+1)\hbar^2} \langle NLSJM_J | \mathbf{J} | NLSJM_J \rangle. \quad (20.102)$$

Setting $\mathbf{V} = \mathbf{S}$, we express the operator $\mathbf{J} \cdot \mathbf{S}$ in terms of standard angular momentum Casimir invariants:

$$\mathbf{J} \cdot \mathbf{S} = (\mathbf{L} + \mathbf{S}) \cdot \mathbf{S} = \mathbf{S}^2 + \mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2). \quad (20.103)$$

Taking the expectation value yields

$$\langle NLSJM_J | S_z | NLSJM_J \rangle = M_J\hbar \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \quad (20.104)$$

Substituting Eq. (20.104) back into Eq. (20.100), the Zeeman energy shift takes the compact form

$$\Delta E(J, M_J) = g \mu_B \mathcal{H} M_J, \quad (20.105)$$

where the dimensionless *Landé g-factor* for *LS* coupling is

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}. \quad (20.106)$$

Under the weak-field Zeeman effect, each fine-structure level of angular momentum J splits symmetrically into $2J + 1$ equally spaced magnetic sublevels labeled by $M_J = -J, -J + 1, \dots, +J$, with energy separation $\Delta E = g \mu_B \mathcal{H}$.

- If $S = 0$, the magnetic moment arises entirely from orbital motion, giving $g = 1$ (the *normal Zeeman effect*).
- If $L = 0$, the magnetic moment is purely spin-derived, giving $g = 2$.
- For general open-shell configurations, g assumes rational fractional values (the *anomalous Zeeman effect*).

20.10.2 Strong-Field Regime: The Paschen-Back Effect

When the external magnetic field is strong enough that the magnetic energy shift substantially exceeds the fine-structure splitting ($\mu_B \mathcal{H} \gg \zeta_{nl}$), the external field decouples $\mathbf{L}$ and $\mathbf{S}$. In this limit, J ceases to be a good quantum number, and the appropriate basis is the uncoupled representation $|NLM_L SM_S\rangle$.

The unperturbed dominant splitting is generated by H_{mag} :

$$\langle NLM_L SM_S | H_{\text{mag}} | NLM_L SM_S \rangle = \mu_B \mathcal{H} (M_L + 2M_S). \quad (20.107)$$

Levels sharing the same value of $M_L + 2M_S$ remain degenerate under the magnetic interaction alone. The spin-orbit interaction $H_{\text{so}}^{\text{eff}} = \xi(NLS) \mathbf{L} \cdot \mathbf{S}$ now acts as a secondary perturbation. Since only $L_z S_z$ possesses non-vanishing diagonal matrix elements in the uncoupled basis, the spin-orbit correction is

$$\langle NLM_L SM_S | H_{\text{so}} | NLM_L SM_S \rangle = \xi(NLS) M_L M_S. \quad (20.108)$$

Combining both contributions yields the total energy shift in the *Paschen-Back regime*:

$$\Delta E(M_L, M_S) = \mu_B \mathcal{H} (M_L + 2M_S) + \xi(NLS) M_L M_S. \quad (20.109)$$

20.10.3 Intermediate and Ultra-Strong Fields

In the intermediate field regime, where $\mu_B \mathcal{H} \sim \zeta_{nl}$, neither the coupled $\{|J, M_J\rangle\}$ nor the uncoupled $\{|M_L, M_S\rangle\}$ basis diagonalizes the Hamiltonian. One must construct and numerically diagonalize the full $(2L+1)(2S+1)$ -dimensional perturbation matrix of $H_{\text{mag}} + H_{\text{so}}$ within the *LS* multiplet. Because $J_z = L_z + S_z$ commutes with both H_{mag} and H_{so} , matrix elements between states with different total $M = M_L + M_S$ vanish, allowing the secular equation to factorize into decoupled blocks of constant M .

Finally, in ultrastrong magnetic fields ($\mathcal{H} \gtrsim 10^8$ – 10^9 G, such as those found on white dwarfs and neutron stars), the diamagnetic quadratic vector-potential term

$$H_{\text{dia}} = \frac{e^2}{8mc^2} \mathcal{H}^2 (x^2 + y^2) \quad (20.110)$$

becomes significant, severely distorting the spherical symmetry of the atomic orbitals and mixing different electronic configurations.

20.11 Problems

1. Ground State and Excited States of Two-Electron Atoms:

- (a) Using the hydrogenic trial wave function $\psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{Z_{\text{eff}}^3}{\pi a_0^3} \exp[-Z_{\text{eff}}(r_1 + r_2)/a_0]$ for the helium isoelectronic sequence with nuclear charge Z , determine the optimal effective nuclear charge $Z_{\text{eff}} = Z - \frac{5}{16}$ via the variational method. Compute the corresponding ground-state energy estimate and compare it with the unperturbed value (neglecting electron-electron repulsion).
- (b) Consider the excited configuration $(1s)(2s)$ of helium. Construct the properly symmetrized spatial and spin wave functions for the orthohelium (spin triplet, $S = 1$) and parahelium (spin singlet, $S = 0$) states. Express the energy splitting between the singlet and triplet levels in terms of the direct Coulomb integral J and the exchange integral K :

$$K = \int d^3r_1 \int d^3r_2 \phi_{1s}^*(\mathbf{r}_1) \phi_{2s}^*(\mathbf{r}_2) \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \phi_{1s}(\mathbf{r}_2) \phi_{2s}(\mathbf{r}_1), \quad (20.111)$$

and explain why the triplet state lies lower in energy (Hund's first rule).

2. **The Thomas-Fermi Model of Heavy Atoms:** In the Thomas-Fermi statistical approximation, the electrostatic potential $\Phi(r)$ satisfies the nonlinear differential equation

$$\frac{1}{r} \frac{d^2}{dr^2}(r\Phi) = \frac{4e}{3\pi} \left(\frac{2m}{\hbar^2} \right)^{3/2} (e\Phi)^{3/2}. \quad (20.112)$$

- (a) Introducing the dimensionless variable $x = r/b$ and the screening function $\chi(x) = r\Phi(r)/(Ze)$, where $b = \frac{1}{2} \left(\frac{3\pi}{4} \right)^{2/3} \frac{\hbar^2}{me^2} Z^{-1/3} \approx 0.8853 a_0 Z^{-1/3}$, derive the dimensionless Thomas-Fermi equation

$$\frac{d^2\chi}{dx^2} = \frac{\chi^{3/2}}{x^{1/2}}. \quad (20.113)$$

State the boundary conditions satisfied by $\chi(x)$ for a neutral isolated atom.

- (b) Show that the total kinetic energy T and the total electrostatic potential energy V of the electron cloud satisfy the virial relation $2T + V = 0$.
- (c) Demonstrate that the total binding energy of a neutral atom scales with atomic number Z as $E_b \propto -Z^{7/3}$.

3. **Hartree-Fock Theory and Koopmans' Theorem:** Consider a closed-shell atomic system described by a single Slater determinant of orthonormal single-particle spatial orbitals $\{\phi_i(\mathbf{r})\}$.

- (a) Show that the Hartree-Fock single-particle eigenvalue ε_k for an occupied orbital ϕ_k is given by

$$\varepsilon_k = \langle \phi_k | h_0 | \phi_k \rangle + \sum_j (2J_{kj} - K_{kj}), \quad (20.114)$$

where $h_0 = -\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{r}$, and J_{kj} and K_{kj} are the direct and exchange Coulomb integrals, respectively.

- (b) Prove *Koopmans' theorem*: assuming that the remaining orbitals do not relax (the "frozen orbital" approximation), show that $-\varepsilon_k$ equals the energy required to remove an electron from orbital ϕ_k to infinity (the unrelaxed ionization potential).

4. **Zeeman and Paschen-Back Transitions in Multi-Electron Atoms:** An atom with total orbital angular momentum $\mathbf{L}$ and total spin $\mathbf{S}$ is placed in an external uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$. The perturbation Hamiltonian is

$$H' = \frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B} + \zeta(NLS)\mathbf{L} \cdot \mathbf{S}. \quad (20.115)$$

- (a) In the weak-field limit ($\mu_B B \ll \zeta \hbar^2$), compute the first-order energy shifts $\Delta E(J, M_J) = g\mu_B B M_J$ for all fine-structure levels of the carbon atom in its ground state configuration $1s^2 2s^2 2p^2$ (3P_0 , 3P_1 , 3P_2).
- (b) In the strong-field limit ($\mu_B B \gg \zeta \hbar^2$), determine the energy shifts $\Delta E(M_L, M_S)$ in the uncoupled basis $|M_L, M_S\rangle$.
- (c) For the $J = 1$ and $J = 0, 2$ manifold with $M = M_L + M_S = 0$, construct the 3×3 secular matrix of H' for arbitrary magnetic field strength B , and find the eigenvalues interpolating between the Zeeman and Paschen-Back regimes.

Chapter 21

Molecules

21.1 Molecules: Energy Hierarchy and Separation of Scales

The problem of determining the electronic states in a molecule is considerably more intricate than in isolated atoms, because the effective potential experienced by the electrons lacks spherical symmetry.¹ In a molecule, one envisions the atomic nuclei as having well-defined classical equilibrium positions—points of minimum potential energy—about which they undergo small, slow oscillations, while the electrons move rapidly in the multi-center Coulomb field produced by the nuclei and each other.

This intuitive physical picture is valid because the nuclei are vastly heavier than the electrons. The ratio of the electron mass m to a typical nuclear mass M satisfies

$$\frac{m}{M} \sim 10^{-4} \text{ to } 10^{-5}. \quad (21.1)$$

As a direct consequence, the quantum zero-point fluctuations of the nuclei are substantially smaller than those of the electrons. Not only do the nuclei move far more sluggishly than the electrons, but they can be tightly localized in space without violating the Heisenberg uncertainty principle.

21.1.1 Hierarchy of Molecular Energy Scales

From the perspective of the fast-moving electrons, the heavy nuclei are essentially stationary. The electrons adapt almost instantaneously to the instantaneous nuclear configuration, occupying eigenstates of the static multi-center Coulomb potential. The slow vibrational motion of the nuclei acts merely as an adiabatic perturbation that deforms these electronic states.

In a molecule with characteristic linear dimension a , the uncertainty principle dictates that typical electronic momenta are $p_e \sim \hbar/a$. The typical spacing between electronic energy levels is therefore

$$E_{\text{elect}} \sim \frac{\hbar^2}{ma^2}. \quad (21.2)$$

These electronic excitations are typically of order a few electron volts ($\sim 1\text{--}10\text{ eV}$), precisely as in isolated atoms.

Conversely, to the heavy nuclei, the rapid electronic cloud appears as a time-averaged, smeared distribution. As the nuclei displace from their equilibrium positions, the electronic ground-state energy shifts, providing an effective restoring force. The nuclei thus oscillate

¹For standard references on molecular quantum mechanics, see L. Pauling, *The Nature of the Chemical Bond*, 3rd ed. (Cornell University Press, Ithaca, NY, 1960); J. C. Slater, *Quantum Theory of Molecules and Solids* (McGraw-Hill, New York, 1963); and C. J. Ballhausen and H. B. Gray, *Molecular Orbital Theory* (W. A. Benjamin, New York, 1963).

about positions that minimize the total energy—the sum of the electronic eigenenergy and the mutual Coulomb repulsion between nuclei.

We can estimate the characteristic frequency ω of these nuclear vibrations by modeling the nuclear potential near equilibrium as a harmonic well, $V(R) \approx \frac{1}{2}M\omega^2 R^2$, where R denotes the displacement from equilibrium. If a nucleus were displaced by an atomic distance $\sim a$, the electronic energy would change by an amount comparable to an electronic excitation energy, $\hbar^2/(2ma^2)$. Balancing these scales,

$$\frac{1}{2}M\omega^2 a^2 \sim \frac{\hbar^2}{2ma^2}, \quad (21.3)$$

which yields the characteristic vibrational frequency

$$\omega \sim \left(\frac{m}{M}\right)^{1/2} \frac{\hbar}{ma^2}. \quad (21.4)$$

The nuclear vibrational excitation quantum $\hbar\omega$ is suppressed relative to electronic excitation energies by a factor of $(m/M)^{1/2}$:

$$E_{\text{vib}} \sim \hbar\omega \sim \left(\frac{m}{M}\right)^{1/2} E_{\text{elect}}. \quad (21.5)$$

Because $(m/M)^{1/2} \sim 10^{-2}$, vibrational quanta typically range from 10^{-2} to 10^{-1} eV (infrared regime).

The zero-point momentum of a nucleus in this harmonic well is estimated by $P^2/(2M) \sim \frac{1}{2}\hbar\omega$, giving

$$P \sim \left(\frac{M}{m}\right)^{1/4} \left(\frac{\hbar}{a}\right), \quad (21.6)$$

which is roughly an order of magnitude larger than electronic momenta. However, because of the nuclear mass, the typical nuclear velocity $v_N = P/M$ is much smaller:

$$v_N \sim \left(\frac{m}{M}\right)^{3/4} \left(\frac{\hbar}{ma}\right) \sim \left(\frac{m}{M}\right)^{3/4} v_e. \quad (21.7)$$

The root-mean-square displacement δ of a nucleus from its equilibrium site satisfies $\frac{1}{2}M\omega^2\delta^2 \sim \frac{1}{2}\hbar\omega$, leading to

$$\left(\frac{\delta}{a}\right)^2 \sim \frac{\hbar\omega}{M\omega^2 a^2} \sim \frac{E_{\text{vib}}}{E_{\text{elect}}} \sim \left(\frac{m}{M}\right)^{1/2}, \quad (21.8)$$

or

$$\frac{\delta}{a} \sim \left(\frac{m}{M}\right)^{1/4} \sim \frac{1}{10}. \quad (21.9)$$

Hence, nuclear excursions remain small fractions of the internuclear separation, confirming the validity of localized nuclear coordinates.

21.1.2 Rotational Degrees of Freedom

In addition to electronic and vibrational excitations, a molecule can undergo rigid rotation as a whole about its center of mass. Such collective rotations involve virtually no internal structural deformation. For a rotational state with angular momentum quantum number l , the energy is

$$E_{\text{rot}} = \frac{\hbar^2 l(l+1)}{2\mathcal{I}}, \quad (21.10)$$

where $\mathcal{I} \sim Ma^2$ is the molecular moment of inertia. Comparing this with electronic energies,

$$E_{\text{rot}} \sim \frac{\hbar^2}{Ma^2} \sim \left(\frac{m}{M}\right) E_{\text{elect}}. \quad (21.11)$$

Rotational excitations are thus smaller than electronic excitations by a factor of $m/M \sim 10^{-4}$ to 10^{-3} , placing them in the microwave region (10^{-4} – 10^{-3} eV).

21.1.3 Summary of Molecular Scales

To lowest order, a general molecular energy eigenstate decomposes into a sum of electronic, vibrational, and rotational energies:

$$E = E_{\text{elect}} + E_{\text{vib}} + E_{\text{rot}}. \quad (21.12)$$

The hierarchy of molecular energy scales is concisely summarized by the relative ratios:

$$E_{\text{elect}} : E_{\text{vib}} : E_{\text{rot}} \sim 1 : \left(\frac{m}{M}\right)^{1/2} : \left(\frac{m}{M}\right). \quad (21.13)$$

This striking separation of timescales and energies provides the physical foundation for the Born-Oppenheimer adiabatic approximation.

21.2 The Born-Oppenheimer Method and the H_2^+ Molecular Ion

We now formulate the mathematical description of molecular structure directly from the nonrelativistic Schrödinger equation. A molecule containing N_e electrons and N_N nuclei is governed by the total Hamiltonian

$$H = T_e + T_N + V_{ee} + V_{eN} + V_{NN}, \quad (21.14)$$

where

$$T_e = \sum_{i=1}^{N_e} \frac{\mathbf{p}_i^2}{2m} \quad (21.15)$$

is the electronic kinetic energy,

$$T_N = \sum_{\alpha=1}^{N_N} \frac{\mathbf{P}_\alpha^2}{2M_\alpha} \quad (21.16)$$

is the nuclear kinetic energy, and the Coulomb potential energies are

$$V_{ee} = \sum_{i < j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad V_{eN} = - \sum_{i,\alpha} \frac{Z_\alpha e^2}{|\mathbf{r}_i - \mathbf{R}_\alpha|}, \quad V_{NN} = \sum_{\alpha < \beta} \frac{Z_\alpha Z_\beta e^2}{|\mathbf{R}_\alpha - \mathbf{R}_\beta|}. \quad (21.17)$$

Here, $\mathbf{r} \equiv \{\mathbf{r}_i\}$ denotes the coordinates of all electrons, and $\mathbf{R} \equiv \{\mathbf{R}_\alpha\}$ denotes the positions of all nuclei.

Because $m/M_\alpha \ll 1$, the nuclear kinetic energy $T_N \sim \mathcal{O}(1/M)$ is diminutive compared to the electronic energy scales. The *Born-Oppenheimer* (or *adiabatic*) approximation treats T_N as a small perturbation, expanding the molecular dynamics in powers of the small parameter $(m/M)^{1/4}$.

21.2.1 The Adiabatic Decomposition

To zeroth order in m/M , we neglect T_N entirely. In this clamped-nucleus limit, the nuclear coordinates $\mathbf{R}$ enter the electronic Hamiltonian merely as fixed external parameters rather than dynamical operators. The electronic Schrödinger equation reads

$$[T_e + V_{ee}(\mathbf{r}) + V_{eN}(\mathbf{r}; \mathbf{R})] \varphi_n(\mathbf{r}; \mathbf{R}) = [\varepsilon_n(\mathbf{R}) - V_{NN}(\mathbf{R})] \varphi_n(\mathbf{r}; \mathbf{R}), \quad (21.18)$$

or equivalently,

$$H_e(\mathbf{r}; \mathbf{R}) \varphi_n(\mathbf{r}; \mathbf{R}) \equiv [T_e + V_{ee} + V_{eN} + V_{NN}] \varphi_n(\mathbf{r}; \mathbf{R}) = \varepsilon_n(\mathbf{R}) \varphi_n(\mathbf{r}; \mathbf{R}). \quad (21.19)$$

The eigenvalues $\varepsilon_n(\mathbf{R})$ define the adiabatic potential energy surfaces (PES) of the molecule as functions of nuclear configuration $\mathbf{R}$.

For each fixed $\mathbf{R}$, the electronic eigenfunctions $\{\varphi_n(\mathbf{r}; \mathbf{R})\}$ constitute a complete orthonormal basis in the electronic Hilbert space:

$$\int d\mathbf{r} \varphi_n^*(\mathbf{r}; \mathbf{R}) \varphi_m(\mathbf{r}; \mathbf{R}) = \delta_{nm}. \quad (21.20)$$

We can therefore expand the exact total molecular wave function $\Psi(\mathbf{r}, \mathbf{R})$, which satisfies the full stationary Schrödinger equation

$$(T_e + T_N + V_{ee} + V_{eN} + V_{NN})\Psi(\mathbf{r}, \mathbf{R}) = E\Psi(\mathbf{r}, \mathbf{R}), \quad (21.21)$$

as an exact spectral expansion:

$$\Psi(\mathbf{r}, \mathbf{R}) = \sum_{m=1}^{\infty} \Phi_m(\mathbf{R}) \varphi_m(\mathbf{r}; \mathbf{R}). \quad (21.22)$$

Here, the expansion coefficients $\Phi_m(\mathbf{R})$ represent the nuclear wave functions associated with each electronic surface m .

Substituting Eq. (21.22) into Eq. (21.21) and using Eq. (21.18), we obtain

$$\sum_m [\varepsilon_m(\mathbf{R}) + T_N] \Phi_m(\mathbf{R}) \varphi_m(\mathbf{r}; \mathbf{R}) = E \sum_m \Phi_m(\mathbf{R}) \varphi_m(\mathbf{r}; \mathbf{R}). \quad (21.23)$$

Projecting Eq. (21.23) onto the electronic eigenstate $\varphi_n^*(\mathbf{r}; \mathbf{R})$ and integrating over all electronic coordinates yields

$$\sum_m \int d\mathbf{r} \varphi_n^*(\mathbf{r}; \mathbf{R}) T_N [\Phi_m(\mathbf{R}) \varphi_m(\mathbf{r}; \mathbf{R})] + \varepsilon_n(\mathbf{R}) \Phi_n(\mathbf{R}) = E \Phi_n(\mathbf{R}). \quad (21.24)$$

Since $T_N = -\sum_{\alpha} \frac{\hbar^2}{2M_{\alpha}} \nabla_{\mathbf{R}_{\alpha}}^2$, the Laplacian acts on both $\Phi_m(\mathbf{R})$ and the parametric $\mathbf{R}$ -dependence of $\varphi_m(\mathbf{r}; \mathbf{R})$ via the product rule:

$$\nabla_{\mathbf{R}_{\alpha}}^2 (\Phi_m \varphi_m) = (\nabla_{\mathbf{R}_{\alpha}}^2 \Phi_m) \varphi_m + 2(\nabla_{\mathbf{R}_{\alpha}} \Phi_m) \cdot (\nabla_{\mathbf{R}_{\alpha}} \varphi_m) + \Phi_m (\nabla_{\mathbf{R}_{\alpha}}^2 \varphi_m). \quad (21.25)$$

Consequently, Eq. (21.24) takes the form of a coupled system of equations for the nuclear wave functions:

$$[T_N + \varepsilon_n(\mathbf{R})] \Phi_n(\mathbf{R}) = E \Phi_n(\mathbf{R}) - \sum_m A_{nm} \Phi_m(\mathbf{R}), \quad (21.26)$$

where the non-adiabatic coupling operator A_{nm} is defined by

$$A_{nm} \Phi_m(\mathbf{R}) \equiv -\sum_{\alpha} \frac{\hbar^2}{2M_{\alpha}} \int d\mathbf{r} \varphi_n^*(\mathbf{r}; \mathbf{R}) [2\nabla_{\mathbf{R}_{\alpha}} \Phi_m(\mathbf{R}) \cdot \nabla_{\mathbf{R}_{\alpha}} \varphi_m(\mathbf{r}; \mathbf{R}) + \Phi_m(\mathbf{R}) \nabla_{\mathbf{R}_{\alpha}}^2 \varphi_m(\mathbf{r}; \mathbf{R})]. \quad (21.27)$$

21.2.2 Validity of the Born-Oppenheimer Approximation

The off-diagonal terms A_{nm} ($n \neq m$) induce non-adiabatic transitions between distinct electronic manifolds. To estimate their magnitude, consider each piece of Eq. (21.27):

1. The term involving $\frac{\hbar^2}{2M_{\alpha}} \nabla_{\mathbf{R}_{\alpha}}^2 \varphi_m$ is of order $(m/M)(\hbar^2/2ma^2) \sim (m/M)E_{\text{elect}}$. Since the gap between electronic energy levels is $\Delta\varepsilon \sim E_{\text{elect}}$, inter-level mixing from this term is suppressed by $\mathcal{O}(m/M) \sim 10^{-4}$.

2. For the gradient-cross term, the nuclear harmonic oscillator wave function has spatial derivative $\nabla_{\mathbf{R}}\Phi_m \sim (\delta M\omega/\hbar)\Phi_m$, where δ is the vibrational amplitude and ω is the vibrational frequency. Because $\nabla_{\mathbf{R}}\varphi_m \sim \varphi_m/a$, the product satisfies

$$\frac{\hbar^2}{M}(\nabla_{\mathbf{R}}\Phi_m) \cdot (\nabla_{\mathbf{R}}\varphi_m) \sim \frac{\hbar^2}{M} \frac{\delta M\omega}{\hbar a} \Phi_m \varphi_m = \hbar\omega \left(\frac{\delta}{a}\right) \Phi_m \varphi_m \sim \left(\frac{m}{M}\right)^{3/4} E_{\text{elect}} \Phi_m \varphi_m. \quad (21.28)$$

The matrix elements of this term are of order $\hbar\omega \sim (m/M)^{1/2} E_{\text{elect}}$, which is small compared to the electronic level spacing E_{elect} .

Thus, provided the electronic potential curves remain well-separated ($\Delta\varepsilon_n \gg \hbar\omega$, i.e., in the absence of conical intersections or avoided crossings), A_{nm} can be neglected to leading order.

Setting $A_{nm} \rightarrow 0$ decouples the electronic states completely, reducing Eq. (21.26) to the uncoupled nuclear Schrödinger equation:

$$[T_N + \varepsilon_n(\mathbf{R})] \Phi_{n\nu}(\mathbf{R}) = E_{n\nu} \Phi_{n\nu}(\mathbf{R}), \quad (21.29)$$

where ν indexes the vibrational-rotational quantum states on the n -th electronic surface. In this adiabatic limit, the molecular stationary states factorize:

$$\Psi_{n\nu}(\mathbf{r}, \mathbf{R}) = \Phi_{n\nu}(\mathbf{R}) \varphi_n(\mathbf{r}; \mathbf{R}). \quad (21.30)$$

The total electronic energy $\varepsilon_n(\mathbf{R})$ acts as an effective potential that binds the nuclei together like an elastic spring, whose stiffness depends on the electronic state n .

21.2.3 The H_2^+ Ion: Molecular Orbitals and LCAO Approximation

The simplest molecular system is the singly ionized molecular hydrogen ion H_2^+ , consisting of two protons fixed at $\mathbf{R}_A$ and $\mathbf{R}_B$ (internuclear distance $R = |\mathbf{R}_A - \mathbf{R}_B|$) and a single electron at $\mathbf{r}$. Clamping the nuclei, the electronic Hamiltonian is

$$H = -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{|\mathbf{r} - \mathbf{R}_A|} - \frac{e^2}{|\mathbf{r} - \mathbf{R}_B|} + \frac{e^2}{R}. \quad (21.31)$$

Although Eq. (21.31) can be solved by separation of variables in prolate spheroidal coordinates, its physics is transparently captured by the *linear combination of atomic orbitals* (LCAO) variational ansatz:

$$\psi(\mathbf{r}) = \alpha\psi_A(\mathbf{r}) + \beta\psi_B(\mathbf{r}), \quad (21.32)$$

where ψ_A and ψ_B are normalized ground-state hydrogen 1s orbitals centered on nuclei A and B , respectively:

$$\psi_A(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}} e^{-|\mathbf{r} - \mathbf{R}_A|/a_0}, \quad \psi_B(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}} e^{-|\mathbf{r} - \mathbf{R}_B|/a_0}. \quad (21.33)$$

Because the Hamiltonian is invariant under spatial reflection across the perpendicular bisector plane of the internuclear axis, parity is a good quantum number. Parity conservation requires $\beta = \pm\alpha$, yielding two symmetry-adapted molecular orbitals:

$$\psi_{\pm}(\mathbf{r}) = C_{\pm} [\psi_A(\mathbf{r}) \pm \psi_B(\mathbf{r})]. \quad (21.34)$$

The normalization condition $\int d^3r |\psi_{\pm}|^2 = 1$ gives

$$C_{\pm} = \frac{1}{\sqrt{2 \pm 2S(R)}}, \quad (21.35)$$

where $S(R)$ is the *overlap integral*:

$$S(R) \equiv \int d^3r \psi_A(\mathbf{r})\psi_B(\mathbf{r}) = \left(1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2}\right) e^{-R/a_0}. \quad (21.36)$$

Integrals of this two-center form are evaluated systematically using confocal elliptical (prolate spheroidal) coordinates:

$$\mu = \frac{|\mathbf{r} - \mathbf{R}_A| + |\mathbf{r} - \mathbf{R}_B|}{R}, \quad \nu = \frac{|\mathbf{r} - \mathbf{R}_A| - |\mathbf{r} - \mathbf{R}_B|}{R}, \quad d^3r = \frac{R^3}{8}(\mu^2 - \nu^2)d\mu d\nu d\phi, \quad (21.37)$$

with $1 \leq \mu < \infty$, $-1 \leq \nu \leq 1$, and $0 \leq \phi < 2\pi$.

The expectation value of the Hamiltonian in states $\psi_{\pm}$ yields the adiabatic energy curves:

$$\varepsilon_{\pm}(R) = \langle H \rangle_{\pm} = \frac{\langle A|H|A \rangle \pm \langle A|H|B \rangle}{1 \pm S(R)}, \quad (21.38)$$

where $\langle A|H|A \rangle = \langle B|H|B \rangle$ by symmetry. Evaluating the individual matrix elements,

$$\begin{aligned} \langle A|H|A \rangle &= \varepsilon_1 + \frac{e^2}{R} - \int d^3r \psi_A^2(\mathbf{r}) \frac{e^2}{|\mathbf{r} - \mathbf{R}_B|} \\ &= \varepsilon_1 + \frac{e^2}{R} \left(1 + \frac{R}{a_0}\right) e^{-2R/a_0}, \end{aligned} \quad (21.39)$$

and the exchange/resonance integral is

$$\begin{aligned} \langle A|H|B \rangle &= \left(\varepsilon_1 + \frac{e^2}{R}\right) S(R) - \int d^3r \psi_A(\mathbf{r})\psi_B(\mathbf{r}) \frac{e^2}{|\mathbf{r} - \mathbf{R}_B|} \\ &= \left(\varepsilon_1 + \frac{e^2}{R}\right) S(R) - \frac{e^2}{a_0} \left(1 + \frac{R}{a_0}\right) e^{-R/a_0}, \end{aligned} \quad (21.40)$$

where $\varepsilon_1 = -e^2/(2a_0) = -1 \text{ Ry} = -13.6 \text{ eV}$ is the ground-state energy of an isolated hydrogen atom.

21.2.4 Bonding Versus Antibonding Orbitals

Substituting Eqs. (21.39) and (21.40) into Eq. (21.38), we obtain the two molecular energy curves:

- **Bonding Orbital $\psi_+(\mathbf{r})$ (Gerade symmetry, $\sigma_g 1s$):** The spatial wave function is symmetric and constructively interferes in the region between the two nuclei. The electron density builds up between the protons, screening their mutual Coulomb repulsion and simultaneously maximizing the attractive electrostatic interaction with both nuclei. The resulting potential curve $\varepsilon_+(R)$ possesses an energy minimum at $R_{\text{eq}} \approx 1.3 \text{ \AA}$ with a binding energy $D_e = -(\varepsilon_+ - \varepsilon_1) \approx 1.76 \text{ eV}$ (experimental values: $R_{\text{eq}} = 1.06 \text{ \AA}$, $D_e = 2.79 \text{ eV}$).
- **Antibonding Orbital $\psi_-(\mathbf{r})$ (Ungerade symmetry, $\sigma_u^* 1s$):** The wave function is antisymmetric and vanishes identically on the nodal midplane ($\nu = 0$). Electron density is depleted between the nuclei, leaving the proton charges unscreened. As a consequence, $\varepsilon_-(R)$ is monotonically repulsive for all $R > 0$, and no bound state forms.

21.2.5 Asymptotic Limits and United-Atom Behavior

At large internuclear separations ($R \gg a_0$), the system smoothly dissociates into a neutral hydrogen atom and an isolated proton: $\varepsilon_{\pm}(R) \rightarrow -1 \text{ Ry}$.

In the united-atom limit ($R \rightarrow 0$), the two protons coalesce into a single helium nucleus ($Z = 2$):

- For the bonding state ψ_+ , L'Hôpital's rule reveals that $\lim_{R \rightarrow 0} \psi_+(\mathbf{r})$ simply reduces to a hydrogen $1s$ orbital with effective charge $Z_{\text{eff}} = 1$, with electronic energy $\varepsilon_+ - e^2/R \rightarrow -3 \text{ Ry}$. However, the true ground state of a He^+ ion with $Z = 2$ has energy $-Z^2 \text{ Ry} = -4 \text{ Ry}$. Because the trial orbital does not allow the radial wave function to shrink as Z increases, the variational LCAO energy underestimates the binding and overestimates the equilibrium bond length.
- For the antibonding state ψ_- , expanding as $R \rightarrow 0$ yields

$$\lim_{R \rightarrow 0} \psi_-(\mathbf{r}) \propto \hat{\mathbf{R}} \cdot \nabla_{\mathbf{r}} e^{-r/a_0} \propto \cos \theta e^{-r/a_0}, \quad (21.41)$$

where $\hat{\mathbf{R}}$ is the molecular axis and θ is the polar angle. This is an atomic p -wave ($l = 1$) state. The centrifugal barrier adds $+8 \text{ Ry}$ of kinetic energy, ensuring that $\varepsilon_-(R) - e^2/R > 0$ near the origin, preventing any bonding minimum.

21.2.6 Muon-Catalyzed Fusion

An interesting manifestation of molecular physics arises in muonic molecules. A negative muon μ^- ($m_{\mu} \approx 207 m_e$) can bind a proton and a deuteron into a $(pd\mu)^+$ ion. Because the Bohr radius scales inversely with mass, $a_{\mu} = a_0(m/m_{\mu}) \approx a_0/207$, the equilibrium internuclear distance shrinks by a factor of 200 down to $\sim 2.5 \times 10^{-11} \text{ cm}$.

At this minute separation, the nuclear Coulomb barrier penetration probability increases by tens of orders of magnitude, enabling rapid nuclear fusion at room temperature:



The muon is liberated to catalyze subsequent fusion events. Because $m_{\mu}/M \sim 1/10$, the Born-Oppenheimer approximation breaks down for muonic molecules, and the coupled non-adiabatic three-body dynamics must be treated rigorously.

21.3 The Neutral Hydrogen Molecule H_2

The neutral hydrogen molecule H_2 is a two-electron, two-proton system. In the absence of spin-orbit coupling, the total electron spin $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$ is a conserved quantity, and the stationary states classify into spin singlets ($S = 0$, antisymmetric spin) and spin triplets ($S = 1$, symmetric spin). By the Pauli exclusion principle, the spatial wave function must be symmetric for singlet states and antisymmetric for triplet states.

The electronic Hamiltonian in the clamped-nuclei frame (protons at $\mathbf{R}_A$ and $\mathbf{R}_B$, separation $R = |\mathbf{R}_A - \mathbf{R}_B|$) is

$$H = -\frac{\hbar^2}{2m} \nabla_1^2 - \frac{\hbar^2}{2m} \nabla_2^2 - \frac{e^2}{|\mathbf{r}_1 - \mathbf{R}_A|} - \frac{e^2}{|\mathbf{r}_2 - \mathbf{R}_B|} - \frac{e^2}{|\mathbf{r}_1 - \mathbf{R}_B|} - \frac{e^2}{|\mathbf{r}_2 - \mathbf{R}_A|} + \frac{e^2}{R} + \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|}. \quad (21.43)$$

21.3.1 Molecular Orbital Versus Valence Bond Formulations

Two complementary variational approaches are traditionally applied to calculate the ground and low-lying electronic states of H_2 :

1. The Molecular Orbital (MO) Approach

Analogous to the Hartree-Fock method in atomic physics, each electron is placed independently into the bonding molecular orbital $\psi_+(\mathbf{r}) \propto \psi_A(\mathbf{r}) + \psi_B(\mathbf{r})$ of the H_2^+ ion [Eq. (21.34)]. The resulting two-electron singlet state is

$$\Psi_{\text{MO}}(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{2[1 + S(R)]} [\psi_A(\mathbf{r}_1) + \psi_B(\mathbf{r}_1)] [\psi_A(\mathbf{r}_2) + \psi_B(\mathbf{r}_2)] \chi_{\text{singlet}}. \quad (21.44)$$

Expanding the spatial factor reveals

$$\Psi_{\text{MO}} \propto \underbrace{[\psi_A(\mathbf{r}_1)\psi_A(\mathbf{r}_2) + \psi_B(\mathbf{r}_1)\psi_B(\mathbf{r}_2)]}_{\text{ionic configurations: } \text{H}^+\text{H}^- \text{ and } \text{H}^-\text{H}^+} + \underbrace{[\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2) + \psi_B(\mathbf{r}_1)\psi_A(\mathbf{r}_2)]}_{\text{covalent configurations: } \text{H-H}}. \quad (21.45)$$

The major deficiency of the simple MO wave function is that it assigns an equal 50% probability weight to the polar (ionic) configurations H^+H^- at all internuclear distances. At large separations $R \gg a_0$, electrostatic repulsion suppresses ionic states because $\text{H}^+ + \text{H}^-$ lies roughly 12.8 eV higher in energy than two isolated neutral H atoms. Thus, Eq. (21.44) fails to dissociate properly to neutral atoms.

2. The Valence Bond (Heitler-London) Approach

To guarantee correct dissociation at large separations, Heitler and London eliminated the ionic components, taking a linear combination of product atomic states where each electron is localized on a distinct nucleus:

$$\psi_s(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2(1 + S^2)}} [\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2) + \psi_B(\mathbf{r}_1)\psi_A(\mathbf{r}_2)] \chi_{\text{singlet}}, \quad (21.46)$$

$$\psi_t(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2(1 - S^2)}} [\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2) - \psi_B(\mathbf{r}_1)\psi_A(\mathbf{r}_2)] \chi_{\text{triplet}}, \quad (21.47)$$

where $S \equiv S(R) = \int d^3r \psi_A(\mathbf{r})\psi_B(\mathbf{r})$ is the atomic overlap integral [Eq. (21.36)].

21.3.2 Evaluation of the Heitler-London Energy Curves

The expectation value of the Hamiltonian (21.43) in the singlet (+) and triplet (−) states evaluates to

$$\varepsilon_{\pm}(R) = \langle H \rangle_{\pm} = \frac{\langle AB|H|AB \rangle \pm \langle BA|H|AB \rangle}{1 \pm S^2}, \quad (21.48)$$

where $|AB \rangle \equiv \psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2)$. Using the unperturbed hydrogenic relations $(-\frac{\hbar^2}{2m}\nabla_1^2 - \frac{e^2}{r_{1A}})\psi_A(1) = \varepsilon_1\psi_A(1)$, the direct matrix element is

$$\langle AB|H|AB \rangle = 2\varepsilon_1 + \frac{e^2}{R} + V_C(R), \quad (21.49)$$

where $V_C(R)$ represents the direct electrostatic Coulomb interaction:

$$V_C(R) \equiv \int d^3r_1 d^3r_2 |\psi_A(\mathbf{r}_1)|^2 |\psi_B(\mathbf{r}_2)|^2 \left[\frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} - \frac{e^2}{|\mathbf{r}_2 - \mathbf{R}_A|} - \frac{e^2}{|\mathbf{r}_1 - \mathbf{R}_B|} \right]. \quad (21.50)$$

Similarly, the exchange matrix element is

$$\langle BA|H|AB \rangle = S^2 \left(2\varepsilon_1 + \frac{e^2}{R} \right) + V_{\text{ex}}(R), \quad (21.51)$$

with the exchange interaction integral given by

$$V_{\text{ex}}(R) \equiv \int d^3r_1 d^3r_2 \psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_2)\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_1) \left[\frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} - \frac{e^2}{|\mathbf{r}_2 - \mathbf{R}_A|} - \frac{e^2}{|\mathbf{r}_1 - \mathbf{R}_B|} \right]. \quad (21.52)$$

$V_{\text{ex}}(R)$ arises from quantum exchange and is roughly proportional to the square of the overlap integral, S^2 .

Substituting Eqs. (21.49) and (21.51) into Eq. (21.48) yields

$$\varepsilon_{\pm}(R) = 2\varepsilon_1 + \frac{e^2}{R} + \frac{V_C(R) \pm V_{\text{ex}}(R)}{1 \pm S^2}, \quad (21.53)$$

which may be reorganized into

$$\varepsilon_{\pm}(R) = 2\varepsilon_1 + \frac{\left[V_C(R) + \frac{e^2}{R} \right] \pm \left[V_{\text{ex}}(R) + S^2 \frac{e^2}{R} \right]}{1 \pm S^2}. \quad (21.54)$$

21.3.3 Physical Mechanism of Chemical Binding

Over the physically relevant range of internuclear distances, the direct Coulomb term $V_C(R) + e^2/R$ is positive, whereas the exchange combination $V_{\text{ex}}(R) + S^2 e^2/R$ is negative. Consequently, the singlet curve $\varepsilon_+(R)$ lies substantially below the triplet curve $\varepsilon_-(R)$:

- **Singlet State** $\varepsilon_+(R)$ ($^1\Sigma_g^+$): The symmetric spatial wave function concentrates electron density directly between the two protons. The negative exchange contribution overcomes the nuclear repulsion, producing a deep potential well with a minimum at $R_{\text{eq}} \approx 1.5 a_0 \approx 0.80 \text{ \AA}$ and a binding energy $D_e \approx 3.14 \text{ eV}$ (experimental: $R_{\text{eq}} = 0.74 \text{ \AA}$, $D_e = 4.75 \text{ eV}$).
- **Triplet State** $\varepsilon_-(R)$ ($^3\Sigma_u^+$): The antisymmetric spatial wave function forces a nodal plane between the nuclei, depleting electron density in the bonding region. The energy $\varepsilon_-(R) - 2\varepsilon_1$ is strictly positive and repulsive for all R , precluding bound state formation.

The origin of chemical binding is often contrasted with atomic Hund's rules. In free atoms, parallel spins (triplet) lower the energy by keeping electrons apart and reducing electron-electron Coulomb repulsion. In molecules, however, the spatial antisymmetry of the triplet forces the electrons away from the internuclear region where the electron-nuclear attraction is strongest. The severe loss of attractive nuclear potential energy easily outweighs the reduction in electron-electron repulsion, causing the singlet state to form the stable chemical bond.

21.3.4 Limiting Behaviors: United Atom and Van der Waals Dispersion

1. **United-Atom Limit** ($R \rightarrow 0$): As $R \rightarrow 0$, the two protons coalesce into a helium nucleus ($Z = 2$). The ground state of He is the spin singlet $1s^2$ (1S_0 , $E \approx -5.8 \text{ Ry}$), while the lowest triplet state $(1s)(2s)$ (3S_1) lies approximately $1.45 \text{ Ry} \approx 19.8 \text{ eV}$ higher. Thus, the large singlet-triplet splitting persists continuously down to $R = 0$, reflecting the energetic cost of placing an electron into an excited atomic orbital.
2. **Asymptotic Long-Range Dispersion** ($R \gg a_0$): At large separations, the Heitler-London wave functions fall off exponentially as e^{-2R/a_0} . At second order in perturbation theory, mutual quantum dipole-dipole fluctuations generate the attractive van der Waals potential:

$$V_{\text{vdW}}(R) = -\frac{C_6}{R^6}, \quad C_6 \approx 6.5 \text{ a.u.} \approx 6.5 e^2 a_0^5. \quad (21.55)$$

Although the van der Waals attraction is roughly three orders of magnitude weaker than the covalent chemical bond and cannot bind light diatomic molecules against zero-point vibrations, it is of paramount importance in the condensation of noble gases and the tertiary structure of macromolecules.

21.4 Pairing of Electrons, Saturation of Covalent Bonds, and Closed-Shell Repulsion

As established in the analysis of the hydrogen molecule, large spatial overlap between the atomic wave functions of two electrons in a spin-singlet configuration generates a strong attractive covalent bond. Conversely, spatial overlap in a spin-triplet configuration gives rise to quantum exchange repulsion. This fundamental dichotomy explains why a hydrogen atom and a helium atom cannot bind to form a stable HHe molecule, and more generally, governs the rules of chemical valence and bond saturation.

21.4.1 Why Helium and Hydrogen Do Not Form a Bound Molecule

The helium atom in its ground state has two electrons occupying the $1s$ spatial orbital with opposing spins ($\uparrow\downarrow$). Now consider bringing a neutral hydrogen atom, whose single electron has (say) spin up ($\uparrow$), in proximity to the helium atom:

- The hydrogen electron cannot undergo exchange with the spin-down ($\downarrow$) electron of helium into the helium $1s$ level, because the Pauli exclusion principle forbids two electrons with identical spin from occupying the same spatial orbital.
- Exchange can only occur between the hydrogen electron and the spin-up ($\uparrow$) electron of helium. Because both electrons possess parallel spins, their spatial wave function must be antisymmetric upon exchange. Exactly as in the triplet state of H_2 , this exchange antisymmetry depletes the electron density in the internuclear region and induces a strong repulsive interaction at short distances.

This absence of binding is equally apparent from the united-atom perspective. As the internuclear separation $R \rightarrow 0$, the H-He system approaches a united lithium atom ($Z = 3$). In lithium, the $1s$ shell can accommodate only two electrons; the third electron is promoted to the $2s$ shell, where it is weakly bound with an ionization potential of only $0.4 \text{ Ry} \approx 5.4 \text{ eV}$. Consequently, the electronic curve $\varepsilon(R) - 2e^2/R$ terminates at $R = 0$ at too high an energy for $\varepsilon(R)$ to possess a potential well capable of supporting a bound state.

21.4.2 Slater Determinant Analysis of Closed-Shell Repulsion

The mathematical mechanism behind this repulsion is demonstrated by constructing a three-electron Slater determinant:

$$\Psi(1, 2, 3) = C \det \begin{pmatrix} \psi_A(\mathbf{r}_1)\chi_{\uparrow}(1) & \psi_A(\mathbf{r}_1)\chi_{\downarrow}(1) & \psi_B(\mathbf{r}_1)\chi_{\uparrow}(1) \\ \psi_A(\mathbf{r}_2)\chi_{\uparrow}(2) & \psi_A(\mathbf{r}_2)\chi_{\downarrow}(2) & \psi_B(\mathbf{r}_2)\chi_{\uparrow}(2) \\ \psi_A(\mathbf{r}_3)\chi_{\uparrow}(3) & \psi_A(\mathbf{r}_3)\chi_{\downarrow}(3) & \psi_B(\mathbf{r}_3)\chi_{\uparrow}(3) \end{pmatrix}, \quad (21.56)$$

where ψ_A is the helium $1s$ orbital, ψ_B is the hydrogen $1s$ orbital, and C is the normalization constant.

Evaluating the normalization integral yields

$$\begin{aligned} C^{-2} &= 6 \int d^3r_1 d^3r_2 d^3r_3 \psi_A(\mathbf{r}_1)\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_3) [\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_3) - \psi_B(\mathbf{r}_1)\psi_A(\mathbf{r}_3)] \psi_A(\mathbf{r}_2) \\ &= 6 [1 - S^2(R)], \end{aligned} \quad (21.57)$$

where $S(R) \equiv \int d^3r \psi_A(\mathbf{r})\psi_B(\mathbf{r})$ is the overlap integral. Because the spin-up and spin-down states are orthogonal, only exchange between electrons with parallel spins contributes to Eq. (21.57).

The expectation value of the total electronic Hamiltonian is then

$$\varepsilon(R) = \frac{1}{1 - S^2(R)} \int d^3r_1 d^3r_2 d^3r_3 \psi_A(\mathbf{r}_1)\psi_A(\mathbf{r}_2)\psi_B(\mathbf{r}_3) H [\psi_A(\mathbf{r}_1)\psi_B(\mathbf{r}_3) - \psi_B(\mathbf{r}_1)\psi_A(\mathbf{r}_3)] \psi_A(\mathbf{r}_2). \quad (21.58)$$

As in the triplet state of H_2 , the exchange integral associated with parallel spins enters with a net positive (repulsive) sign, driving the total energy upward as the overlap $S(R)$ increases.

21.4.3 Electron Pairing, Chemical Inertness, and Valence

Two electrons possessing opposite spins that occupy the same spatial orbital (such as the $1s^2$ pair in helium) are defined as *paired*.

- When an electron from another atom attempts to overlap with a paired set of electrons, the interaction is inevitably repulsive, because parallel-spin exchange with one member of the pair is unavoidable.
- This explains the chemical inertness of the noble gases (He, Ne, Ar, Kr, Xe). Because all electrons reside in completely filled shells, all electrons are paired, presenting an effective repulsive core to any approaching atom.
- Closed-shell core electrons in heavier atoms likewise remain chemically inert: because core orbitals are tightly bound and have small spatial radii, their overlap with neighboring atomic orbitals is negligible at typical equilibrium bonding distances. Chemical binding is mediated almost entirely by the outer, loosely bound *valence electrons*.

The *valency* of an atom in a given configuration is the number of unpaired valence electrons available to form chemical bonds. For non-transition elements, the active valence electrons belong to the outermost s and p subshells, and the maximum valence is given by $2S_{\text{max}}$, where S_{max} is the maximum total spin of the valence shell. Unpaired d and f electrons in transition metals and lanthanides are often radially localized inside the outer s shell and participate in more complex bonding phenomena.

21.4.4 Saturation of Covalent Bonds

Once two unpaired electrons belonging to different atoms combine into a spatial bonding orbital with singlet spin ($S = 0$), they become paired. Any third electron brought into contact with this newly formed bonding pair experiences exchange repulsion.

Consequently, covalent bonding forces *saturate*:

1. Each covalent electron-pair bond utilizes exactly one electron from each partner atom, forming a localized two-center two-electron ($2c-2e$) bond.
2. An individual valence electron participates in at most one localized covalent bond.
3. In stable molecules composed of main-group elements, every valence electron is accommodated either in a localized bonding pair or a non-bonding lone pair. The total molecular spin is therefore predominantly singlet ($S = 0$), rendering most organic and inorganic molecules diamagnetic. Molecules with open shells and permanent paramagnetism (such as O_2 and NO) represent notable exceptions governed by molecular orbital degeneracies.

For instance, alkali metal atoms (Li, Na, K) have a single s valence electron outside a closed inert gas core. They readily form single covalent bonds or ionic salts, exhibiting a valency of one.

21.5 Spatially Directed Orbitals: σ and π Bonds

Unlike spherically symmetric atomic s -orbitals, atomic p -orbitals exhibit distinct directional characteristics, concentrating electron density along mutually orthogonal spatial axes. Because

the strength of a covalent chemical bond depends directly on the spatial overlap between interacting electron wave functions, p -electrons bond most effectively when their spatial lobes align along or across the internuclear axis.

21.5.1 Cartesian p -Orbitals

In an isolated central field, atomic p -states are eigenstates of orbital angular momentum $\mathbf{L}^2 = 2\hbar^2$ ($l = 1$) and its projection $L_z = m_l\hbar$ ($m_l = 0, \pm 1$), described by spherical harmonics $Y_{1,m_l}(\theta, \phi)$. To describe directional bonding in molecules, it is advantageous to form real, orthonormal Cartesian linear combinations:

$$\begin{aligned} |p_z\rangle &= Y_{10}(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta = \sqrt{\frac{3}{4\pi}} \frac{z}{r}, \\ |p_x\rangle &= \frac{1}{\sqrt{2}} [-Y_{11}(\theta, \phi) + Y_{1,-1}(\theta, \phi)] = \sqrt{\frac{3}{4\pi}} \sin \theta \cos \phi = \sqrt{\frac{3}{4\pi}} \frac{x}{r}, \\ |p_y\rangle &= \frac{i}{\sqrt{2}} [Y_{11}(\theta, \phi) + Y_{1,-1}(\theta, \phi)] = \sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi = \sqrt{\frac{3}{4\pi}} \frac{y}{r}. \end{aligned} \quad (21.59)$$

The orbital $|p_z\rangle$ exhibits rotational symmetry about the z -axis with positive and negative lobes along $+z$ and $-z$, while $|p_x\rangle$ and $|p_y\rangle$ have analogous geometries oriented along the x - and y -axes, respectively.

21.5.2 Formation of σ and π Molecular Bonds

Let the internuclear line connecting two bonded atoms define the molecular z -axis. When two atoms approach each other, their atomic orbitals overlap and form covalent bonds whose symmetry and strength depend on their orientation relative to $\hat{\mathbf{z}}$:

1. **The σ Bond (Axial Overlap):** Two $|p_z\rangle$ orbitals directed toward each other experience maximal head-on overlap along the internuclear axis. Because $L_z|p_z\rangle = 0$, the resulting molecular orbital has zero angular momentum projection along the bond axis ($m_l = 0$) and is cylindrically symmetric under rotations about $\hat{\mathbf{z}}$. Such an axial bond is designated a σ bond (by analogy with atomic s -states). Covalent bonds formed by s -electrons (such as in H_2) are also σ bonds.
2. **The π Bond (Lateral Overlap):** Orbitals that point perpendicular to the bond axis, such as $|p_x\rangle$ and $|p_y\rangle$, overlap sideways. In terms of the angular momentum along the internuclear axis,

$$L_z^2|p_x\rangle = \hbar^2|p_x\rangle, \quad L_z^2|p_y\rangle = \hbar^2|p_y\rangle, \quad (21.60)$$

so that each orbital possesses projection $|m_l| = 1$. The resulting molecular orbital has a nodal plane containing the internuclear axis and is termed a π bond (analogous to atomic p -states).

Because lateral overlap is significantly smaller than head-on axial overlap, π bonds are substantially weaker than σ bonds. To a good approximation, the total binding energy of multiple bonds is additive:

$$E_{\text{bond}} \approx E_{\sigma} + n_{\pi}E_{\pi}. \quad (21.61)$$

21.5.3 Multiple Bonding in Homonuclear Diatomics

The interplay between σ and π bonds is illustrated by the series C_2 , N_2 , and O_2 :

Carbon (C₂)

The ground configuration of carbon is $1s^2 2s^2 2p^2$, having two unpaired $2p$ electrons in its valence shell. When two carbon atoms form C₂:

- The first pair of $2p$ electrons occupies the $|p_z\rangle$ orbitals, forming a strong σ bond with singlet spin ($S = 0$).
- Once the σ bonding orbital is fully occupied by two paired electrons, additional electrons cannot enter the $|p_z\rangle$ channel without experiencing severe exchange repulsion.
- The second pair of valence electrons occupies the orthogonal $|p_x\rangle$ orbitals, forming a weaker π bond. Thus, C₂ is held together by a double bond ($\sigma + \pi$).

Nitrogen (N₂)

Each nitrogen atom has the valence configuration $2s^2 2p^3$, possessing three unpaired $2p$ electrons. In the N₂ molecule:

- One pair forms a central σ bond ($|p_z\rangle$).
- The remaining two pairs form two mutually orthogonal π bonds ($|p_x\rangle$ and $|p_y\rangle$).

This combination of one σ bond and two π bonds constitutes a triple bond ($\sigma + 2\pi$). The exceptionally large dissociation energy of N₂ (9.76 eV) reflects the cumulative strength of this saturated triple-bond system.

21.5.4 Paramagnetism of Molecular Oxygen (O₂)

The oxygen atom has eight electrons, with ground configuration $1s^2 2s^2 2p^4$ (four $2p$ electrons per atom, for a total of eight valence p -electrons in O₂).

Constructing the molecular orbital energy levels:

1. Six of the eight electrons completely fill the bonding orbitals: one $\sigma_g(2p_z)$ and two degenerate $\pi_u(2p_{x,y})$ orbitals, forming an underlying triple bond.
2. The remaining two electrons must enter the next available molecular orbitals, which are the degenerate antibonding $\pi_g^*(2p_x)$ and $\pi_g^*(2p_y)$ orbitals.
3. In accordance with Hund's first rule, the energy is minimized by placing each electron into a separate spatial orbital to reduce electron-electron Coulomb repulsion, while aligning their spins in a symmetric spin triplet ($S = 1$):

$$\Psi_{\text{valence}}(\text{O}_2) \propto [\pi_g^*(x; 1)\pi_g^*(y; 2) - \pi_g^*(y; 1)\pi_g^*(x; 2)] \chi_{\text{triplet}}. \quad (21.62)$$

Consequently, the ground state of molecular oxygen is a spin triplet ($^3\Sigma_g^-$) possessing a permanent magnetic dipole moment ($S = 1$). Liquid and gaseous oxygen are distinctly paramagnetic, providing a direct experimental demonstration of quantum exchange interactions in molecules.

21.6 Orbital Hybridization, Molecular Geometry, and Hydrogen Bonding

To maximize the overlap between bonding electrons and thereby maximize covalent binding energy, an atomic electron orbital should ideally concentrate its probability density in a single

direction toward the bonded partner. However, an isolated atomic s -orbital is spherically symmetric (even parity), while a pure p -orbital possesses two symmetrical lobes of opposite signs pointing in antiparallel directions (odd parity).

By forming quantum mechanical linear combinations of states with different orbital angular momenta, one can construct highly asymmetric, directional wave functions that protrude prominently in the bonding direction while being suppressed in the reverse direction. This phenomenon is known as *orbital hybridization*.

21.6.1 Principles of Hybridization: The Li_2 Molecule

The energetic rationale for hybridization involves an optimization between single-particle promotion energy and molecular covalent binding:

1. Exciting an electron from an atomic s -state to a higher-energy p -state costs a promotion energy ΔE_{sp} .
2. However, a hybrid orbital formed by coherent superposition exhibits greater directional overlap with a neighboring orbital, yielding an enhanced covalent binding energy ΔE_{bond} .
3. If the gain in bonding energy exceeds the promotion penalty, the hybridized state represents the true variational ground state.

A clear example occurs in dilithium Li_2 . The valence electron of atomic lithium occupies the $2s$ orbital. A trial molecular orbital can be constructed by mixing the $2s$ state with the $2p_z$ state:

$$|\psi_{\text{hyb}}\rangle \propto |s\rangle + \lambda|p_z\rangle, \quad (\lambda > 0). \quad (21.63)$$

Because the s -amplitude is positive everywhere while the p_z -amplitude is positive along $+z$ and negative along $-z$, the amplitudes interfere constructively along $+z$ and destructively along $-z$. A variational minimization yields an optimal mixing parameter $\lambda \approx 0.3$, corresponding to approximately 10% p -character. The resulting directional lobe significantly enhances the overlap with the second lithium atom.

21.6.2 Tetrahedral sp^3 Hybridization

When an atom engages in multiple covalent bonds, the valence s -orbital can mix simultaneously with all three Cartesian p -orbitals. If the $2s$ and $2p$ atomic levels were strictly degenerate, the configuration that maximizes directional binding across four equivalent bonds is the symmetric sp^3 set:

$$\begin{aligned} |1\rangle &= \frac{1}{2} (|s\rangle + |p_x\rangle + |p_y\rangle + |p_z\rangle), \\ |2\rangle &= \frac{1}{2} (|s\rangle + |p_x\rangle - |p_y\rangle - |p_z\rangle), \\ |3\rangle &= \frac{1}{2} (|s\rangle - |p_x\rangle + |p_y\rangle - |p_z\rangle), \\ |4\rangle &= \frac{1}{2} (|s\rangle - |p_x\rangle - |p_y\rangle + |p_z\rangle). \end{aligned} \quad (21.64)$$

These four hybrid orbitals form a complete, orthonormal basis within the sp^3 valence subspace:

$$\langle i|j\rangle = \delta_{ij}, \quad (i, j = 1, 2, 3, 4). \quad (21.65)$$

The spatial geometry of these orbitals is readily deduced from the Cartesian forms of the p -orbitals [Eq. (21.59)]:

$$|p_x\rangle + |p_y\rangle + |p_z\rangle \propto \frac{x + y + z}{r}. \quad (21.66)$$

This combination reaches its maximum along the body diagonal of the positive octant, $(1, 1, 1)$, and its minimum along $(-1, -1, -1)$. Adding the spherically symmetric $|s\rangle$ orbital produces a strongly directed lobe oriented along the vector $(1, 1, 1)$.

Similarly, orbitals $|2\rangle$, $|3\rangle$, and $|4\rangle$ point along $(1, -1, -1)$, $(-1, 1, -1)$, and $(-1, -1, 1)$, respectively. The four hybrid lobes point toward the vertices of a regular tetrahedron centered at the nucleus. The angle θ between any two lobes satisfies

$$\cos \theta = \frac{(1)(1) + (1)(-1) + (1)(-1)}{\sqrt{3}\sqrt{3}} = -\frac{1}{3}, \quad \theta = \arccos\left(-\frac{1}{3}\right) \approx 109.47^\circ. \quad (21.67)$$

21.6.3 Molecular Geometries: H_2O and NH_3

The Water Molecule (H_2O)

The oxygen atom possesses six valence electrons ($2s^2 2p^4$):

- In the unhybridized picture, two unpaired p -electrons (e.g., $|p_x\rangle$ and $|p_y\rangle$) bond with the $1s$ electrons of the two hydrogen atoms, while $|p_z\rangle$ and $2s$ remain non-bonding lone pairs. Since $|p_x\rangle$ and $|p_y\rangle$ are mutually perpendicular, this naive model predicts a bond angle of 90° .
- In the idealized sp^3 fully hybridized picture, two hybrid orbitals form covalent σ bonds with the hydrogen atoms, while the remaining two orbitals house the two non-bonding lone pairs ($S = 0$). This predicts the tetrahedral angle of 109.5° .

The experimentally observed bond angle is 104.5° . This intermediate value reflects two competing physical effects:

1. *Partial Hybridization:* The atomic $2s$ orbital in oxygen lies approximately 16 eV below the $2p$ subshell. Complete hybridization is energetically too costly, so the bonding orbitals retain more p -character ($sp^{3.2}$), pulling the angle toward 90° .
2. *Lone-Pair Repulsion:* The non-bonding lone pairs are spatially more diffuse and closer to the oxygen nucleus than the shared bonding pairs. Their electrostatic and Pauli repulsion exerts pressure on the two O–H bonds, compressing the angle from 109.5° down to 104.5° .

Ammonia (NH_3) and Inversion Tunneling

The nitrogen atom has five valence electrons ($2s^2 2p^3$). Intermediate hybridization produces three bonding orbitals that combine with three hydrogen $1s$ orbitals, leaving one non-bonding lone pair pointing along the molecular symmetry axis. The resulting trigonal pyramid has an experimental H–N–H bond angle of 107.3° .

The ammonia molecule exhibits a classic quantum double-well potential: the nitrogen atom can reside on either side of the plane defined by the three protons. Quantum mechanical tunneling between these two equivalent configurations splits the ground vibrational state into a symmetric state $|+\rangle$ and an antisymmetric state $|-\rangle$. The energy splitting is

$$\Delta E_{\text{tunnel}} \approx 10^{-4} \text{ eV} \implies \nu = \frac{\Delta E}{h} \approx 23.8 \text{ GHz} \quad (\lambda \approx 1.26 \text{ cm}). \quad (21.68)$$

This inversion transition in ammonia served as the operating frequency for the first quantum maser developed by Townes and coworkers in 1954.

21.6.4 Hydrogen Bonding and the Physics of Liquid Water

The electronic structure of H_2O imparts a large electric dipole moment ($\mu \approx 1.85 \text{ D}$) and explains the macroscopic behavior of water:

- The tetrahedral geometry endows the water molecule with four functional "arms": two positively polarized protons stripped of electron density, and two negatively polarized lone-pair lobes.
- When two water molecules approach, a proton on one molecule experiences a strong electrostatic attraction toward a lone pair on an adjacent oxygen. This directional intermolecular interaction is the *hydrogen bond*.
- A typical hydrogen bond in water has a binding energy of $\sim 0.2 \text{ eV}$ ($\approx 20 \text{ kJ/mol}$), intermediate between weak van der Waals dispersion ($\sim 0.01 \text{ eV}$) and strong covalent bonds ($\sim 3\text{--}5 \text{ eV}$).
- In the hydrogen bond, the intramolecular O–H distance is $\approx 1.0 \text{ \AA}$, while the intermolecular $\text{H} \cdots \text{O}$ distance is $\approx 1.8 \text{ \AA}$.

Because each water molecule can form up to four hydrogen bonds simultaneously (two as donor, two as acceptor), liquid water self-assembles into extensive, fluctuating tetrahedral networks of $(\text{H}_2\text{O})_n$ clusters.

Thermal agitation continuously breaks and reforms these bonds, leading to the characteristic thermodynamic anomalies of water: its exceptionally high boiling point, heat capacity, surface tension, and the density maximum at 4°C . At low temperatures, large clusters increase the viscosity through geometric entanglement; heating ruptures the hydrogen-bonded network, rapidly decreasing the viscosity. Furthermore, the capacity of water to form hydrogen bonds with polar solutes underpins its effectiveness as a universal biological solvent.

21.7 Hydrocarbons: Hybridization Schemes and Delocalized π -Systems

The diverse structural chemistry of carbon compounds provides a striking illustration of the intimate connection between quantum orbital hybridization and molecular geometry.

In its atomic ground state, carbon has the electron configuration $1s^2 2s^2 2p^2$ with two unpaired $2p$ electrons, which would suggest a chemical valency of two. However, the promotional energy required to excite one $2s$ electron into the vacant $2p$ orbital to produce the tetravalent configuration $1s^2 2s^1 2p^3$ is approximately 4 eV . This modest energetic penalty is far outweighed by the energy released in forming four covalent bonds rather than two (each covalent bond yielding $\sim 3\text{--}4 \text{ eV}$ of stabilization). Carbon compounds therefore predominantly feature tetravalent carbon atoms, adopting three distinct hybridization geometries.

21.7.1 Tetrahedral sp^3 Carbon: Alkanes (CH_4)

In methane (CH_4), the $2s$ orbital mixes equally with all three $2p$ orbitals to produce four equivalent sp^3 hybrid orbitals [Eq. (21.64)]. Each orbital contains one electron and forms a localized two-electron σ bond with the $1s$ orbital of a hydrogen atom.

Because the four hybrid orbitals point toward the vertices of a regular tetrahedron, the molecule assumes a rigid tetrahedral geometry with an exact H–C–H bond angle of

$$\theta = \arccos\left(-\frac{1}{3}\right) \approx 109.47^\circ. \quad (21.69)$$

All four C–H bonds are axially symmetric σ bonds.

21.7.2 Trigonal sp^2 Carbon: Alkenes (C_2H_4)

In ethylene (C_2H_4), the carbon atoms are coplanar with the four hydrogen atoms. This geometry is accounted for by sp^2 hybridization, in which the $2s$ orbital mixes with only two of the p -orbitals (say, p_x and p_y):

$$\begin{aligned} |1\rangle &= |p_z\rangle, \\ |2\rangle &= \sqrt{\frac{1}{3}}|s\rangle + \sqrt{\frac{2}{3}}|p_x\rangle, \\ |3\rangle &= \sqrt{\frac{1}{3}}|s\rangle - \sqrt{\frac{1}{6}}|p_x\rangle + \sqrt{\frac{1}{2}}|p_y\rangle, \\ |4\rangle &= \sqrt{\frac{1}{3}}|s\rangle - \sqrt{\frac{1}{6}}|p_x\rangle - \sqrt{\frac{1}{2}}|p_y\rangle. \end{aligned} \quad (21.70)$$

The three hybrid orbitals $|2\rangle$, $|3\rangle$, and $|4\rangle$ lie in the xy -plane, mutually oriented at 120° angles. On each carbon atom:

- Two of the sp^2 hybrids form strong σ bonds with hydrogen $1s$ orbitals.
- The third sp^2 hybrid overlaps head-on with the corresponding sp^2 hybrid of the second carbon atom along the internuclear x -axis, forming a robust C–C σ bond.
- The unhybridized $|p_z\rangle$ orbital $|1\rangle$ is oriented perpendicular to the molecular plane. The $|p_z\rangle$ orbitals of the two carbon atoms overlap sideways to form a π bond.

Because the π bond requires mutual parallelism of the two p_z lobes, rotating one CH_2 group relative to the other about the C=C axis destroys the lateral π -overlap. This creates a large torsional barrier ($\sim 2.7\text{ eV} \approx 65\text{ kcal/mol}$), enforcing strict molecular planarity and making cis-trans isomerism possible.

21.7.3 Linear sp Carbon: Alkynes (C_2H_2)

In acetylene (C_2H_2), the molecule forms a linear chain $H-C \equiv C-H$ along the x -axis. Here, the $2s$ orbital mixes with only a single p -orbital (p_x) to form two collinear digonal sp hybrids:

$$|\sigma_{\pm}\rangle = \frac{1}{\sqrt{2}}(|s\rangle \pm |p_x\rangle). \quad (21.71)$$

These two orbitals point in opposite directions along $\pm\hat{x}$ at an angle of 180° :

- One sp orbital forms a σ bond with a hydrogen $1s$ orbital, while the other forms a C–C σ bond.
- The two remaining unhybridized orbitals, $|p_y\rangle$ and $|p_z\rangle$, are mutually perpendicular and perpendicular to the molecular axis. They overlap laterally with their counterparts on the adjacent carbon, forming two orthogonal π bonds (π_y and π_z).

The carbon-carbon triple bond in acetylene thus comprises one central σ bond and two perpendicular π bonds ($\sigma + 2\pi$), yielding a cylindrically symmetric electron sheath around the internuclear axis.

21.7.4 Delocalized π -Systems and Aromaticity: Benzene (C_6H_6)

When conjugate multiple bonds occur in cyclic networks, the localized bond picture gives way to quantum delocalization. The prototypical example is benzene (C_6H_6), which consists of a planar, regular hexagonal ring of six carbon atoms:

1. Each carbon atom utilizes sp^2 hybridization with 120° angles, matching the internal angles of a regular hexagon. The sp^2 orbitals form a rigid, localized planar σ -skeleton containing six C–C bonds and six C–H bonds.
2. Each carbon retains one unhybridized p_z orbital directed perpendicular to the ring plane, containing one valence electron.

Classically, the six p_z electrons could pair into three localized π bonds in two equivalent ways (the Kekulé structures). In quantum mechanics, the true ground state is not a localized static alternation of single and double bonds, but a coherent linear combination of both configurations.

The six p_z atomic orbitals combine to form six delocalized molecular orbitals extending across the entire ring:

$$\psi_k(\mathbf{r}) = \frac{1}{\sqrt{6}} \sum_{j=1}^6 e^{i2\pi kj/6} \phi_{p_z}(\mathbf{r} - \mathbf{R}_j), \quad (k = 0, \pm 1, \pm 2, 3). \quad (21.72)$$

The three lowest-energy delocalized molecular orbitals ($k = 0, \pm 1$) are completely filled by the six π -electrons, forming a closed shell in momentum space (Hückel's $4n + 2$ rule with $n = 1$).

Delocalization lowers the kinetic energy of the π -electrons by increasing their effective de Broglie wavelength, conferring an extra *resonance stabilization energy* of ≈ 1.5 eV (≈ 36 kcal/mol) relative to localized cyclohexatriene. Furthermore, when an external magnetic field is applied perpendicular to the ring, the delocalized electrons circulate freely around the hexagonal perimeter. These macroscopic *ring currents* induce an anomalously large diamagnetic susceptibility, a hallmark of aromatic systems.

Chapter 22

Relativistic Spin Zero Particles Klein Gordon Equation

22.1 Relativistic Spin-Zero Particles: Klein-Gordon Equation

Generalizing quantum mechanics to describe relativistic particles is a nontrivial problem that introduces several profound conceptual departures from the nonrelativistic framework. We begin by examining spinless (scalar) particles before addressing particles with intrinsic spin. The properties of spin, as an intrinsic angular momentum, are intimately tied to the spacetime structure of special relativity; consequently, the relativistic wave equation depends sensitively on the particle's spin. Prototypical physical examples of relativistic spin-zero particles include the π and K mesons.

22.1.1 Heuristic Motivation and Nonlocality

A free nonrelativistic particle satisfies the familiar energy-momentum dispersion relation

$$E = \frac{p^2}{2m}, \quad (22.1)$$

which is invariant under a Galilean boost to a frame moving with velocity $-\mathbf{v}$. In the boosted frame, the energy E' and momentum $\mathbf{p}'$ transform according to

$$\begin{aligned} E' &= E + \mathbf{p} \cdot \mathbf{v} + \frac{1}{2}mv^2, \\ \mathbf{p}' &= \mathbf{p} + m\mathbf{v}, \end{aligned} \quad (22.2)$$

which preserves the form $E' = p'^2/(2m)$. For any closed nonrelativistic system, Eq. (22.2) holds with E representing the total energy, $\mathbf{p}$ the total momentum, and m the total mass. Up to an additive constant independent of $\mathbf{v}$, Eq. (22.1) is the unique energy-momentum relation invariant under Galilean transformations.

Quantum mechanically, energy and momentum govern the temporal and spatial phase evolution of the wave function:

$$\psi(\mathbf{r}, t) = e^{i(\mathbf{p} \cdot \mathbf{r} - Et)/\hbar}. \quad (22.3)$$

To enforce the dispersion relation (22.1), the state must satisfy the free Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{1}{2m} \left(\frac{\hbar}{i} \nabla \right)^2 \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi. \quad (22.4)$$

Equation (22.4) directly follows from the classical relation (22.1) via the operator replacements

$$E \rightarrow i\hbar \frac{\partial}{\partial t}, \quad \mathbf{p} \rightarrow \frac{\hbar}{i} \nabla. \quad (22.5)$$

Minimal electromagnetic coupling is incorporated by introducing the vector potential $\mathbf{A}(\mathbf{r}, t)$ and scalar potential $\Phi(\mathbf{r}, t)$ through the gauge-invariant substitutions

$$E \rightarrow E - e\Phi, \quad \mathbf{p} \rightarrow \mathbf{p} - \frac{e}{c}\mathbf{A}, \quad (22.6)$$

or equivalently in operator form,

$$i\hbar \frac{\partial}{\partial t} \rightarrow i\hbar \frac{\partial}{\partial t} - e\Phi, \quad \frac{\hbar}{i} \nabla \rightarrow \frac{\hbar}{i} \nabla - \frac{e}{c}\mathbf{A}, \quad (22.7)$$

where e denotes the particle's electric charge.

In seeking a relativistic generalization, the wave equation must reproduce the Einstein energy-momentum relation:

$$E = c\sqrt{p^2 + m^2c^2}. \quad (22.8)$$

The most direct attempt is to define the momentum-space amplitude

$$\psi_{\mathbf{p}}(t) = \int d^3r e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi(\mathbf{r}, t), \quad (22.9)$$

and postulate the first-order evolution equation

$$i\hbar \frac{\partial}{\partial t} \psi_{\mathbf{p}}(t) = c\sqrt{p^2 + m^2c^2} \psi_{\mathbf{p}}(t). \quad (22.10)$$

While this equation yields the desired positive energy eigenvalues, transforming back to coordinate space reveals a critical defect:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = \int d^3r' K(\mathbf{r} - \mathbf{r}') \psi(\mathbf{r}', t), \quad (22.11)$$

where the integral kernel is

$$K(\mathbf{r} - \mathbf{r}') = \int \frac{d^3p}{(2\pi\hbar)^3} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')/\hbar} c\sqrt{p^2 + m^2c^2}. \quad (22.12)$$

Equation (22.11) is explicitly nonlocal: the time derivative $\partial\psi/\partial t$ at position $\mathbf{r}$ depends on the values of $\psi(\mathbf{r}', t)$ at distant points. The kernel $K(\mathbf{r} - \mathbf{r}')$ remains appreciable over spatial separations on the order of the reduced Compton wavelength $\lambda_c = \hbar/mc$, which is the characteristic length scale of the particle. Consequently, an initially localized wave packet confined within a region smaller than λ_c immediately develops non-vanishing amplitudes at points outside the light cone. Thus, Eq. (22.10) violates microscopic causality when applied to states localized within a Compton wavelength.

22.1.2 The Klein-Gordon Equation

To obtain a strictly local differential equation, one begins by squaring the relativistic energy-momentum relation:

$$E^2 = p^2c^2 + m^2c^4. \quad (22.13)$$

Applying the operator substitutions (22.5) then yields

$$\left(\frac{i\hbar}{c} \frac{\partial}{\partial t} \right)^2 \psi(\mathbf{r}, t) = \left[\left(\frac{\hbar}{i} \nabla \right)^2 + m^2c^2 \right] \psi(\mathbf{r}, t), \quad (22.14)$$

or in standard covariant form,

$$\left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \left(\frac{mc}{\hbar} \right)^2 \right] \psi(\mathbf{r}, t) = 0. \quad (22.15)$$

Equation (22.15) is the *Klein-Gordon equation*. In four-vector notation, with $\partial_\mu \partial^\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2$, it reads $\left(\partial_\mu \partial^\mu + \frac{m^2 c^2}{\hbar^2}\right) \psi = 0$.

Under a Lorentz transformation $x \rightarrow x'$, the d'Alembertian operator $\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2$ is an invariant scalar operator. For a spinless particle, the wave function transforms as a scalar field:

$$\psi'(\mathbf{r}', t') = \psi(\mathbf{r}, t). \quad (22.16)$$

Coupling to an external electromagnetic field is achieved via minimal substitution (22.7):

$$\frac{1}{c^2} \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right)^2 \psi(\mathbf{r}, t) = \left[\left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right)^2 + m^2 c^2 \right] \psi(\mathbf{r}, t). \quad (22.17)$$

22.1.3 Physical Peculiarities: Negative Energies and Conserved Current

The Klein-Gordon equation exhibits several distinctive features:

1. **Second Order in Time:** Because the equation contains a second time derivative, specifying the physical state at $t = 0$ requires knowledge of both $\psi(\mathbf{r}, 0)$ and $\partial\psi(\mathbf{r}, 0)/\partial t$. This doubling of the degrees of freedom indicates that the theory contains additional internal states, ultimately corresponding to particle and antiparticle degrees of freedom.
2. **Negative Energy Solutions:** Plane-wave solutions $\psi \propto e^{i(\mathbf{p} \cdot \mathbf{r} - Et)/\hbar}$ satisfy Eq. (22.14) for both signs of energy:

$$E = \pm c\sqrt{p^2 + m^2 c^2}. \quad (22.18)$$

The negative-energy states possess the unusual property that increasing the momentum magnitude $|\mathbf{p}|$ drives the energy toward more negative values. As we shall see, these states describe antiparticles.

3. **Indefinite Probability Density:** The standard nonrelativistic density $|\psi|^2$ is not conserved: $\frac{d}{dt} \int d^3r |\psi|^2 \neq 0$. Instead, using the reality of the differential operator in Eq. (22.15), we take ψ^* times the equation for ψ minus ψ times the equation for ψ^* :

$$\psi^* \left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \left(\frac{mc}{\hbar} \right)^2 \right] \psi - \psi \left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \left(\frac{mc}{\hbar} \right)^2 \right] \psi^* = 0. \quad (22.19)$$

This directly produces the local continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0, \quad (22.20)$$

where the probability current density is

$$\mathbf{j}(\mathbf{r}, t) = \frac{\hbar}{2im} (\psi^* \nabla \psi - \psi \nabla \psi^*), \quad (22.21)$$

and the associated density is

$$\rho(\mathbf{r}, t) = \frac{i\hbar}{2mc^2} \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right). \quad (22.22)$$

Although the spatial integral $\int d^3r \rho(\mathbf{r}, t)$ is strictly conserved in time, $\rho(\mathbf{r}, t)$ is not positive definite. For a negative-energy plane wave with $E = -c\sqrt{p^2 + m^2 c^2}$, we find $\rho = E/(mc^2) < 0$. Therefore, ρ cannot be interpreted as a probability density. Instead, $e\rho(\mathbf{r}, t)$ is interpreted as a *charge density*, and $e\mathbf{j}(\mathbf{r}, t)$ as an *electric current density*.

In the presence of an electromagnetic field, the gauge-invariant current and density satisfying the continuity equation (22.20) become

$$\mathbf{j}(\mathbf{r}, t) = \frac{1}{2m} \left[\psi^* \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right) \psi + \psi \left(-\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right) \psi^* \right], \quad (22.23)$$

$$\rho(\mathbf{r}, t) = \frac{1}{2mc^2} \left[\psi^* \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \psi + \psi \left(-i\hbar \frac{\partial}{\partial t} - e\Phi \right) \psi^* \right]. \quad (22.24)$$

Chapter 23

Relativistic Spin Zero Particles

Chapter 22: Relativistic Spin-Zero Particles — Overview

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23.1 Negative Energy States and Antiparticles

To construct a consistent physical interpretation of the Klein-Gordon equation, we examine the transformation properties of its plane-wave solutions under Lorentz boosts and analyze the emergence of antiparticle states.

23.1.1 Positive-Energy Solutions and Lorentz Covariance

Consider first a free particle at rest ($\mathbf{p} = \mathbf{0}$). A positive-energy solution has rest energy $E = mc^2$, so its wave function takes the form

$$\psi(\mathbf{r}, t) = e^{-imc^2 t/\hbar}. \quad (23.1)$$

Under a Lorentz boost to a reference frame moving with velocity $-\mathbf{v}$ relative to the particle, the particle appears with velocity $\mathbf{v}$, relativistic momentum $\mathbf{p} = \gamma m\mathbf{v}$, and energy $E_p = \gamma mc^2$, where $\gamma = (1 - v^2/c^2)^{-1/2}$. Since ψ transforms as a scalar field according to Eq. (22.16), the wave function in the primed frame is

$$\psi'(\mathbf{r}', t') = e^{-imc^2 t'/\hbar} = e^{i(\mathbf{p}' \cdot \mathbf{r}' - E_p t')/\hbar}, \quad (23.2)$$

where we have used the Lorentz invariance of the four-scalar product $p_\mu x^\mu = E_p t' - \mathbf{p}' \cdot \mathbf{r}' = mc^2 t$.

Evaluating the conserved density and current from Eqs. (22.21) and (22.22) for the boosted positive-energy state yields

$$\rho(\mathbf{r}', t') = \frac{E_p}{mc^2} = \gamma, \quad (23.3)$$

and

$$\mathbf{j}(\mathbf{r}', t') = \frac{\mathbf{p}'}{m} = \frac{\mathbf{p}'c^2}{E_p} \rho(\mathbf{r}', t') = \mathbf{v} \rho(\mathbf{r}', t'). \quad (23.4)$$

The current density is simply the density multiplied by the relativistic group velocity

$$\mathbf{v} = \frac{\mathbf{p}'c^2}{E_p}. \quad (23.5)$$

The quantity $(\rho, \mathbf{j}/c)$ transforms as a relativistic four-vector. This behavior has an intuitive geometric origin: under a Lorentz boost, a volume element undergoes Lorentz contraction along the direction of motion by a factor of $\gamma^{-1} = \sqrt{1 - v^2/c^2}$, causing the particle density to scale by $\gamma = E_p/(mc^2)$.

23.1.2 Negative-Energy States and Antiparticle Interpretation

Next, consider the negative-energy solutions. A state at rest with energy $E = -mc^2$ is represented by

$$\psi(\mathbf{r}, t) = e^{+imc^2 t/\hbar}. \quad (23.6)$$

Substituting Eq. (23.6) into Eq. (22.22) gives

$$\rho(\mathbf{r}, t) = -1. \quad (23.7)$$

A negative probability density is physically untenable. However, if ρ is interpreted as proportional to an electric charge density, a state with negative density ρ corresponds to a state carrying the opposite charge. We therefore interpret a negative-energy particle state of rest energy $-mc^2$ as representing an *antiparticle* at rest with positive physical energy $+mc^2$.

Applying a boost of velocity $-\mathbf{v}$ to this rest frame state gives

$$\psi'(\mathbf{r}', t') = e^{imc^2 t/\hbar} = e^{-i(\mathbf{p}' \cdot \mathbf{r}' - E_p t')/\hbar}. \quad (23.8)$$

The phase of ψ' corresponds to negative energy $-E_p$ and momentum $-\mathbf{p}'$. The associated conserved density and current are

$$\rho(\mathbf{r}', t') = -\frac{E_p}{mc^2} = -\gamma, \quad (23.9)$$

$$\mathbf{j}(\mathbf{r}', t') = -\frac{\mathbf{p}'}{m} = \frac{\mathbf{p}'c^2}{E_p} \rho(\mathbf{r}', t') = -\mathbf{v} |\rho(\mathbf{r}', t')|. \quad (23.10)$$

The negative-energy solution with momentum $-\mathbf{p}'$ describes a physical antiparticle propagating with positive physical energy E_p and momentum $+\mathbf{p}'$, moving at velocity $\mathbf{v}$. The flux of an antiparticle of charge $-e$ moving with velocity $\mathbf{v}$ yields an electric current density

$$\mathbf{j}_{\text{em}} = (-e)\mathbf{v}|\rho| = e\mathbf{j}, \quad (23.11)$$

which is completely consistent with the convention that $e\rho$ and $e\mathbf{j}$ represent the physical electromagnetic charge density and current density, respectively.

23.1.3 Charge Conjugation and Normalization

To establish that the negative-energy solutions rigorously describe antiparticles coupling with the opposite sign of the charge, consider the Klein-Gordon equation in an external electromagnetic field (22.17). Taking the complex conjugate and noting that $\Phi(\mathbf{r}, t)$ and $\mathbf{A}(\mathbf{r}, t)$ are real, we obtain

$$\frac{1}{c^2} \left[i\hbar \frac{\partial}{\partial t} + e\Phi(\mathbf{r}, t) \right]^2 \psi^*(\mathbf{r}, t) = \left\{ \left[\frac{\hbar}{i} \nabla + \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + m^2 c^2 \right\} \psi^*(\mathbf{r}, t). \quad (23.12)$$

Equation (23.12) shows that if $\psi(\mathbf{r}, t)$ satisfies the Klein-Gordon equation for a particle of mass m and charge e , then the *charge-conjugate* wave function

$$\psi_c(\mathbf{r}, t) \equiv \psi^*(\mathbf{r}, t) \quad (23.13)$$

satisfies the identical wave equation for a particle of identical mass m but with opposite electric charge $-e$.

Because complex conjugation inverts the sign of the temporal phase ($e^{-iEt/\hbar} \rightarrow e^{+iEt/\hbar}$), the complex conjugate of a negative-energy state ψ with charge e represents a physical positive-energy state ψ_c with charge $-e$. For instance, the negative-energy solution (23.8) conjugates to

$$\psi^*(\mathbf{r}', t') = e^{i(\mathbf{p}' \cdot \mathbf{r}' - E_p t')/\hbar}, \quad (23.14)$$

describing an antiparticle with physical energy E_p , momentum $\mathbf{p}'$, and charge $-e$.

Evaluating the density and current using ψ^* and the conjugated charge $-e$, we define

$$\rho_c(\mathbf{r}, t) = \frac{1}{2mc^2} \left[\psi \left(i\hbar \frac{\partial}{\partial t} + e\Phi \right) \psi^* + \psi^* \left(-i\hbar \frac{\partial}{\partial t} + e\Phi \right) \psi \right] = -\rho(\mathbf{r}, t), \quad (23.15)$$

and similarly for the current density:

$$\mathbf{j}_c(\mathbf{r}, t) = -\mathbf{j}(\mathbf{r}, t). \quad (23.16)$$

States can therefore be normalized according to total charge:

$$\int d^3r \rho(\mathbf{r}, t) = \pm 1, \quad (23.17)$$

where $+1$ corresponds to particle states and -1 to antiparticle states. The normalization (23.17) is a Lorentz scalar and is conserved in time by virtue of the continuity equation.

23.2 First-Order Klein-Gordon Equation

To clarify the physical interpretation of the Klein-Gordon equation and facilitate comparison with the first-order Dirac and Schrödinger equations, it is advantageous to recast the second-order equation into an equivalent system of two coupled first-order equations in time.¹

23.2.1 The Feshbach-Villars Representation

We introduce the auxiliary field

$$\psi^0(\mathbf{r}, t) \equiv \left(\frac{\partial}{\partial t} + \frac{ie}{\hbar} \Phi(\mathbf{r}, t) \right) \psi(\mathbf{r}, t). \quad (23.18)$$

From the minimally coupled Klein-Gordon equation (22.17), the time evolution of ψ^0 satisfies

$$\left(\frac{\partial}{\partial t} + \frac{ie}{\hbar} \Phi(\mathbf{r}, t) \right) \psi^0(\mathbf{r}, t) = \left[c^2 \left(\nabla - \frac{ie}{\hbar c} \mathbf{A}(\mathbf{r}, t) \right)^2 - \frac{m^2 c^4}{\hbar^2} \right] \psi(\mathbf{r}, t). \quad (23.19)$$

Equations (23.18) and (23.19) form a coupled first-order system. To decouple the rest-mass contributions and symmetrize the equations, we define the linear combinations

$$\begin{aligned} \varphi &= \frac{1}{2} \left(\psi + \frac{i\hbar}{mc^2} \psi^0 \right), \\ \chi &= \frac{1}{2} \left(\psi - \frac{i\hbar}{mc^2} \psi^0 \right). \end{aligned} \quad (23.20)$$

¹See the classic review by H. Feshbach and F. Villars, *Rev. Mod. Phys.* **30**, 24 (1958).

In terms of φ and χ , the original fields are recovered as $\psi = \varphi + \chi$ and $\psi^0 = \frac{mc^2}{i\hbar}(\varphi - \chi)$. A straightforward algebraic rearrangement of Eqs. (23.18) and (23.19) yields

$$\begin{aligned} \left(i\hbar\frac{\partial}{\partial t} - e\Phi\right)\varphi &= \frac{1}{2m} \left(\frac{\hbar}{i}\nabla - \frac{e}{c}\mathbf{A}\right)^2 (\varphi + \chi) + mc^2\varphi, \\ \left(i\hbar\frac{\partial}{\partial t} - e\Phi\right)\chi &= -\frac{1}{2m} \left(\frac{\hbar}{i}\nabla - \frac{e}{c}\mathbf{A}\right)^2 (\varphi + \chi) - mc^2\chi. \end{aligned} \quad (23.21)$$

We assemble φ and χ into a two-component wave function:

$$\Psi(\mathbf{r}, t) = \begin{pmatrix} \varphi(\mathbf{r}, t) \\ \chi(\mathbf{r}, t) \end{pmatrix}. \quad (23.22)$$

Introducing the Pauli matrices acting on this two-dimensional charge space,

$$\tau_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \tau_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (23.23)$$

(reserving the symbol $\boldsymbol{\sigma}$ exclusively for genuine physical spin), Eqs. (23.21) condense into a single Schrödinger-like equation:

$$i\hbar\frac{\partial\Psi}{\partial t} = H\Psi, \quad (23.24)$$

with the Feshbach-Villars Hamiltonian

$$H = \frac{1}{2m} \left(\frac{\hbar}{i}\nabla - \frac{e}{c}\mathbf{A}\right)^2 (\tau_3 + i\tau_2) + mc^2\tau_3 + e\Phi. \quad (23.25)$$

Crucially, the two components of Ψ do not describe spin; the particle remains strictly spinless ($s = 0$). Instead, the two-component structure reflects the internal charge degree of freedom (particle versus antiparticle).

23.2.2 Indefinite Metric and Generalized Hermiticity

In the two-component formalism, the conserved charge density (22.24) acquires an exceptionally clean representation:

$$\rho(\mathbf{r}, t) = |\varphi|^2 - |\chi|^2 = \Psi^\dagger \tau_3 \Psi, \quad (23.26)$$

where $\Psi^\dagger = (\varphi^*, \chi^*)$. The associated current density is

$$\mathbf{j}(\mathbf{r}, t) = \frac{\hbar}{2im} \left[\Psi^\dagger \tau_3 (\tau_3 + i\tau_2) \nabla \Psi - (\nabla \Psi^\dagger) \tau_3 (\tau_3 + i\tau_2) \Psi \right] - \frac{e\mathbf{A}}{mc} \Psi^\dagger \tau_3 (\tau_3 + i\tau_2) \Psi. \quad (23.27)$$

Because ρ is not positive definite, the natural Hilbert space inner product must be modified by the indefinite metric τ_3 :

$$\langle \Psi | \Psi' \rangle \equiv \int d^3r \Psi^\dagger(\mathbf{r}, t) \tau_3 \Psi'(\mathbf{r}, t), \quad (23.28)$$

with the normalization condition given by

$$\langle \Psi | \Psi \rangle = \int d^3r \Psi^\dagger \tau_3 \Psi = \pm 1. \quad (23.29)$$

Although H is not Hermitian in the conventional sense—since $(\tau_3 + i\tau_2)^\dagger = \tau_3 - i\tau_2 \neq \tau_3 + i\tau_2$ —it is *pseudo-Hermitian* with respect to the metric τ_3 :

$$H = \tau_3 H^\dagger \tau_3. \quad (23.30)$$

This condition ensures that the Hamiltonian is self-adjoint under the inner product (23.28):

$$\langle \Psi | H \Psi' \rangle = \langle H \Psi | \Psi' \rangle = [\langle \Psi' | H \Psi \rangle]^*. \quad (23.31)$$

Consequently, energy eigenvalues are strictly real, and eigenstates corresponding to distinct eigenvalues are orthogonal in the generalized sense:

$$\langle \Psi_E | \Psi_{E'} \rangle = \int d^3r \Psi_E^\dagger(\mathbf{r}) \tau_3 \Psi_{E'}(\mathbf{r}) = 0 \quad \text{for } E \neq E'. \quad (23.32)$$

23.2.3 Charge Conjugation

Under charge conjugation, $\psi \rightarrow \psi^*$, which maps $\varphi \rightarrow \chi^*$ and $\chi \rightarrow \varphi^*$. In the two-component language, the charge-conjugate wave function is

$$\Psi_c = \tau_1 \Psi^*. \quad (23.33)$$

Taking the complex conjugate of $H(e)$ in Eq. (23.25) and noting that $\mathbf{p}^* = -\mathbf{p}$ and $\tau_3 + i\tau_2$ is real, we have

$$H^*(e) = \frac{1}{2m} \left(-\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2 (\tau_3 + i\tau_2) + e\Phi + mc^2 \tau_3. \quad (23.34)$$

Since τ_1 anticommutes with both τ_2 and τ_3 , similarity transformation by τ_1 yields

$$\tau_1 H^*(e) \tau_1 = - \left[\frac{1}{2m} \left(\mathbf{p} + \frac{e}{c} \mathbf{A} \right)^2 (\tau_3 + i\tau_2) - e\Phi + mc^2 \tau_3 \right] = -H(-e). \quad (23.35)$$

It follows directly from $i\hbar \frac{\partial \Psi}{\partial t} = H(e)\Psi$ that

$$i\hbar \frac{\partial \Psi_c}{\partial t} = H(-e)\Psi_c, \quad (23.36)$$

demonstrating that Ψ_c evolves according to the Hamiltonian of an identical particle carrying the opposite electric charge $-e$.

23.2.4 Free-Particle Plane Waves and Completeness

For a free particle with momentum $\mathbf{p}$ and positive energy $E_p = \sqrt{p^2 c^2 + m^2 c^4}$, the normalized scalar wave function is $\psi_{\mathbf{p}}^{(+)}(\mathbf{r}, t) = \sqrt{\frac{mc^2}{E_p}} e^{i(\mathbf{p} \cdot \mathbf{r} - E_p t)/\hbar}$. Applying Eqs. (23.18) and (23.20) gives the two-component spinor

$$\Psi_{\mathbf{p}}^{(+)}(\mathbf{r}, t) = \Psi_{\mathbf{p}}^{(+)} e^{i(\mathbf{p} \cdot \mathbf{r} - E_p t)/\hbar}, \quad (23.37)$$

where

$$\Psi_{\mathbf{p}}^{(+)} = \frac{1}{2\sqrt{E_p mc^2}} \begin{pmatrix} mc^2 + E_p \\ mc^2 - E_p \end{pmatrix}. \quad (23.38)$$

Similarly, the negative-energy solution corresponding to an antiparticle of momentum $\mathbf{p}$ is

$$\Psi_{\mathbf{p}}^{(-)}(\mathbf{r}, t) = \Psi_{\mathbf{p}}^{(-)} e^{-i(\mathbf{p} \cdot \mathbf{r} - E_p t)/\hbar}, \quad (23.39)$$

where

$$\Psi_{\mathbf{p}}^{(-)} = \frac{1}{2\sqrt{E_p mc^2}} \begin{pmatrix} mc^2 - E_p \\ mc^2 + E_p \end{pmatrix} = \tau_1 \Psi_{\mathbf{p}}^{(+)}. \quad (23.40)$$

In the nonrelativistic limit ($p \ll mc$), these spinors reduce to

$$\Psi_{\mathbf{p}}^{(+)} \approx \begin{pmatrix} 1 \\ -\frac{v^2}{4c^2} \end{pmatrix}, \quad \Psi_{\mathbf{p}}^{(-)} \approx \begin{pmatrix} -\frac{v^2}{4c^2} \\ 1 \end{pmatrix}. \quad (23.41)$$

The lower component χ is suppressed by $\mathcal{O}(v^2/c^2)$ relative to the upper component φ for positive-energy states. Neglecting χ in Eq. (23.21) reduces the system to the standard nonrelativistic Schrödinger equation for φ (with rest energy mc^2 included).

The free solutions satisfy the generalized orthonormality relations:

$$\begin{aligned}\Psi_{\mathbf{p}}^{(+)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(+)} &= +1, \\ \Psi_{\mathbf{p}}^{(-)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(-)} &= -1, \\ \Psi_{\mathbf{p}}^{(+)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(-)} &= 0.\end{aligned}\tag{23.42}$$

Any arbitrary state $\Psi(\mathbf{r}, t)$ can be expanded in terms of this complete set:

$$\Psi(\mathbf{r}, t) = \int \frac{d^3p}{(2\pi\hbar)^3} \left[u_{\mathbf{p}}(t) \Psi_{\mathbf{p}}^{(+)} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} + v_{\mathbf{p}}^*(t) \Psi_{\mathbf{p}}^{(-)} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \right], \tag{23.43}$$

where $u_{\mathbf{p}}(t)$ is the probability amplitude for a particle of momentum $\mathbf{p}$ and charge $+e$, and $v_{\mathbf{p}}(t)$ is the amplitude for an antiparticle of momentum $\mathbf{p}$ and charge $-e$. By orthogonality, the amplitudes are projected out via

$$\begin{aligned}u_{\mathbf{p}}(t) &= \int d^3r \Psi_{\mathbf{p}}^{(+)\dagger} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \tau_3 \Psi(\mathbf{r}, t), \\ v_{\mathbf{p}}^*(t) &= - \int d^3r \Psi_{\mathbf{p}}^{(-)\dagger} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \tau_3 \Psi(\mathbf{r}, t).\end{aligned}\tag{23.44}$$

The overall normalization condition becomes

$$\int \frac{d^3p}{(2\pi\hbar)^3} (|u_{\mathbf{p}}|^2 - |v_{\mathbf{p}}|^2) = \pm 1. \tag{23.45}$$

Because only the difference of squares is constrained, $u_{\mathbf{p}}$ and $v_{\mathbf{p}}$ can individually be arbitrarily large, anticipating the Bose-Einstein statistics of integer-spin fields.

Finally, the expectation value of the kinetic energy is

$$\langle E \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} (E_p |u_{\mathbf{p}}|^2 + E_p |v_{\mathbf{p}}|^2) = \int d^3r \Psi^\dagger(\mathbf{r}) \tau_3 H_0 \Psi(\mathbf{r}), \tag{23.46}$$

where $H_0 = \frac{p^2}{2m}(\tau_3 + i\tau_2) + mc^2\tau_3$. Similarly, the expectation value of momentum is

$$\langle \mathbf{p} \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} (\mathbf{p} |u_{\mathbf{p}}|^2 + \mathbf{p} |v_{\mathbf{p}}|^2) = \int d^3r \Psi^\dagger(\mathbf{r}) \tau_3 \left(\frac{\hbar}{i} \nabla \right) \Psi(\mathbf{r}). \tag{23.47}$$

Both physical quantities are strictly positive and correctly weighted sums over particle and antiparticle contributions.

23.3 Free-Particle Wave Packets and Localization

We now examine the dynamics of wave packets in the Klein-Gordon theory. In particular, we analyze the propagation of localized packets and uncover the fundamental limits on single-particle spatial localization imposed by relativity.

23.3.1 Wave Packet Propagation and Group Velocity

A pure positive-energy wave packet is constructed by superposing positive-frequency solutions:

$$\Psi^{(+)}(\mathbf{r}, t) = \int \frac{d^3p}{(2\pi\hbar)^3} u_{\mathbf{p}} \Psi_{\mathbf{p}}^{(+)} e^{i(\mathbf{p}\cdot\mathbf{r} - E_p t)/\hbar}. \tag{23.48}$$

If the amplitude distribution $u_{\mathbf{p}}$ is sharply peaked about a central momentum $\mathbf{p}_0$, the method of stationary phase shows that the center of the packet propagates with the relativistic group velocity

$$\mathbf{v}_g = \nabla_{\mathbf{p}} E_p|_{\mathbf{p}=\mathbf{p}_0} = \frac{\mathbf{p}_0 c^2}{E_{p_0}} = \frac{\mathbf{p}_0}{\sqrt{m^2 + p_0^2/c^2}}. \quad (23.49)$$

Similarly, an antiparticle wave packet formed entirely of negative-energy solutions,

$$\Psi^{(-)}(\mathbf{r}, t) = \int \frac{d^3 p}{(2\pi\hbar)^3} v_{-\mathbf{p}}^* \Psi_{-\mathbf{p}}^{(-)} e^{i(\mathbf{p}\cdot\mathbf{r} + E_p t)/\hbar}, \quad (23.50)$$

with $v_{\mathbf{p}}$ peaked around $\mathbf{p}_0$, also moves with the physical group velocity $\mathbf{v}_g = \mathbf{p}_0 c^2 / E_{p_0}$.

23.3.2 Impossibility of Exact Single-Particle Localization

Consider an attempt to prepare a state strictly localized at the origin at $t = 0$:

$$\Psi(\mathbf{r}, 0) = \delta(\mathbf{r}) \begin{pmatrix} \varphi \\ \chi \end{pmatrix}, \quad (23.51)$$

where φ and χ are constant spinors. Projecting onto the positive- and negative-energy basis via Eq. (23.44) gives

$$\begin{aligned} u_{\mathbf{p}} &= \Psi_{\mathbf{p}}^{(+)\dagger} \tau_3 \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \frac{E_p(\varphi + \chi) + mc^2(\varphi - \chi)}{2\sqrt{E_p mc^2}}, \\ v_{\mathbf{p}}^* &= -\Psi_{\mathbf{p}}^{(-)\dagger} \tau_3 \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \frac{E_p(\varphi + \chi) - mc^2(\varphi - \chi)}{2\sqrt{E_p mc^2}}. \end{aligned} \quad (23.52)$$

Because $E_p = \sqrt{p^2 c^2 + m^2 c^4}$ depends nonlinearly on momentum, there exists no choice of constants φ and χ that causes $v_{\mathbf{p}}^*$ to vanish for all $\mathbf{p}$. Consequently, a state localized at a sharp point unavoidably contains both particle ($u_{\mathbf{p}} \neq 0$) and antiparticle ($v_{\mathbf{p}} \neq 0$) components. A strictly localized wave packet formed purely from positive-energy states does not exist in relativistic quantum mechanics.

Attempting to localize a particle within its Compton wavelength $\lambda_c = \hbar/mc$ necessarily supplies sufficient energy uncertainty ($\Delta E \sim c\Delta p \gtrsim mc^2$) to create particle-antiparticle pairs from the vacuum.

23.3.3 The Position Operator: Even and Odd Decompositions

The origin of this mixing becomes explicit upon examining the action of the coordinate operator $\mathbf{r}$ on a positive-energy wave packet:

$$\begin{aligned} \mathbf{r}\Psi^{(+)}(\mathbf{r}, t) &= \int \frac{d^3 p}{(2\pi\hbar)^3} u_{\mathbf{p}}(t) \Psi_{\mathbf{p}}^{(+)} \left(\frac{\hbar}{i} \nabla_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \right) \\ &= \int \frac{d^3 p}{(2\pi\hbar)^3} [i\hbar \nabla_{\mathbf{p}} u_{\mathbf{p}}(t)] \Psi_{\mathbf{p}}^{(+)} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} + \int \frac{d^3 p}{(2\pi\hbar)^3} u_{\mathbf{p}}(t) [i\hbar \nabla_{\mathbf{p}} \Psi_{\mathbf{p}}^{(+)}] e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \end{aligned} \quad (23.53)$$

Differentiating the explicit spinor basis vectors (23.38) and (23.40) with respect to $\mathbf{p}$ yields the transition identity

$$\nabla_{\mathbf{p}} \Psi_{\mathbf{p}}^{(\pm)} = -\frac{\mathbf{p}c^2}{2E_p^2} \Psi_{\mathbf{p}}^{(\mp)}. \quad (23.54)$$

The gradient $\nabla_{\mathbf{p}}$ connects positive- and negative-energy spinors. The position operator therefore decomposes into an *even* part $\mathbf{r}_+$ and an *odd* part $\mathbf{r}_-$:

$$\mathbf{r} = \mathbf{r}_+ + \mathbf{r}_-, \quad (23.55)$$

whose actions on $\Psi^{(+)}(\mathbf{r}, t)$ are

$$\mathbf{r}_+ \Psi^{(+)}(\mathbf{r}, t) = \int \frac{d^3 p}{(2\pi\hbar)^3} [i\hbar \nabla_{\mathbf{p}} u_{\mathbf{p}}(t)] \Psi_{\mathbf{p}}^{(+)} e^{i\mathbf{p} \cdot \mathbf{r}/\hbar}, \quad (23.56)$$

$$\mathbf{r}_- \Psi^{(+)}(\mathbf{r}, t) = - \int \frac{d^3 p}{(2\pi\hbar)^3} u_{\mathbf{p}}(t) \frac{i\hbar \mathbf{p} c^2}{2E_p^2} \Psi_{\mathbf{p}}^{(-)} e^{i\mathbf{p} \cdot \mathbf{r}/\hbar}. \quad (23.57)$$

The even operator $\mathbf{r}_+$ (closely related to the Newton-Wigner position operator) preserves the sign of the energy subspace: it maps positive-energy states into positive-energy states and negative-energy states into negative-energy states. The odd operator $\mathbf{r}_-$, by contrast, induces transitions between the positive- and negative-energy subspaces, transforming particles into antiparticles and vice versa.

The most localized state constructed exclusively from positive-energy states is an eigenstate of the even position operator $\mathbf{r}_+$. The eigenvalue problem $\mathbf{r}_+ \Psi^{(+)}(\mathbf{r}) = \mathbf{r}_0 \Psi^{(+)}(\mathbf{r})$ requires

$$i\hbar \nabla_{\mathbf{p}} u_{\mathbf{p}} = \mathbf{r}_0 u_{\mathbf{p}} \implies u_{\mathbf{p}} \propto e^{-i\mathbf{p} \cdot \mathbf{r}_0/\hbar}, \quad (23.58)$$

yielding the Newton-Wigner localized state

$$\Psi_{\mathbf{r}_0}^{(+)}(\mathbf{r}) = \int \frac{d^3 p}{(2\pi\hbar)^3} e^{i\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}_0)/\hbar} \Psi_{\mathbf{p}}^{(+)}. \quad (23.59)$$

In coordinate space, this state is not a spatial Dirac delta function; rather, it possesses an intrinsic spatial extension characterized by exponential tails decaying over the Compton wavelength $\lambda_c \sim \hbar/mc$. Thus, single-particle relativistic quantum mechanics cannot resolve positions sharper than λ_c .

Furthermore, because external potentials $V(\mathbf{r})$ are functions of the position operator $\mathbf{r}$, any spatially varying potential necessarily couples to the odd operator $\mathbf{r}_-$. Consequently, exposing a positive-energy particle to an external potential generally induces transitions to negative-energy states, manifesting physically as the spontaneous creation of particle-antiparticle pairs.

23.4 Klein's Paradox and Pair Production

The coupling between positive- and negative-energy states leads to remarkable physical consequences when a relativistic particle interacts with an intense external potential. The most striking manifestation is *Klein's paradox*, which highlights the breakdown of a strict single-particle interpretation and signals the onset of particle-antiparticle pair creation.

23.4.1 Scattering by an Electrostatic Step Barrier

Consider a positively charged spinless particle of mass m and charge $e > 0$, incident from the left ($x < 0$) with momentum p on a one-dimensional electrostatic potential step of height $V_0 \equiv e\Phi$:

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & x > 0. \end{cases} \quad (23.60)$$

In the asymptotic region $x < 0$, the particle has energy $E_p = \sqrt{p^2 c^2 + m^2 c^4}$, and the stationary wave function consists of incident and reflected plane waves:

$$\psi(x) = a e^{ipx/\hbar} + b e^{-ipx/\hbar}, \quad x < 0. \quad (23.61)$$

In the barrier region $x > 0$, the stationary Klein-Gordon equation (22.17) reads

$$(E_p - V_0)^2 \psi(x) = -\hbar^2 c^2 \frac{d^2 \psi(x)}{dx^2} + m^2 c^4 \psi(x), \quad (23.62)$$

admitting transmitted solutions of the form

$$\psi(x) = d e^{ikx}, \quad x > 0, \quad (23.63)$$

where the wave number k satisfies the dispersion relation

$$\hbar^2 c^2 k^2 + m^2 c^4 = (E_p - V_0)^2. \quad (23.64)$$

Matching $\psi(x)$ and its derivative $d\psi/dx$ continuously across $x = 0$ yields the standard transmission and reflection amplitudes:

$$b = \frac{p - \hbar k}{p + \hbar k} a, \quad (23.65)$$

$$d = \frac{2p}{p + \hbar k} a. \quad (23.66)$$

23.4.2 The Three Potential Regimes

The physical nature of the solution depends crucially on the barrier height V_0 :

1. **Low Barrier** ($E_p > V_0 + mc^2$): The kinetic energy exceeds the barrier height plus rest mass. The transmitted wave number

$$k = \frac{1}{\hbar c} \sqrt{(E_p - V_0)^2 - m^2 c^4} \quad (23.67)$$

is real and positive. The wave is partially reflected and partially transmitted, in direct analogy with nonrelativistic tunneling.

2. **Intermediate Barrier** ($E_p - mc^2 < V_0 < E_p + mc^2$): The effective kinetic energy inside the barrier is negative, rendering $k = i\kappa$ purely imaginary with

$$\kappa = \frac{1}{\hbar c} \sqrt{m^2 c^4 - (E_p - V_0)^2}. \quad (23.68)$$

The wave function decays exponentially for $x > 0$ ($\psi \propto e^{-\kappa x}$), resulting in total reflection ($|b/a| = 1$). However, the conserved charge density inside the barrier,

$$\rho(x) = \frac{E_p - V_0}{mc^2} |d|^2 e^{-2\kappa x}, \quad (23.69)$$

exhibits an intriguing feature: while positive for $V_0 < E_p$, it becomes *negative* when $V_0 > E_p$, despite the incident flux carrying positive charge.

3. **Supercritical Barrier** ($V_0 > E_p + mc^2$): When the barrier height exceeds the threshold $E_p + mc^2$, the term $(E_p - V_0)^2 - m^2 c^4$ becomes positive once again, so k becomes real! Propagating wave solutions re-emerge inside the classically forbidden barrier.

23.4.3 Resolution of Klein's Paradox: Pair Creation

To ascertain the physical direction of propagation in the supercritical regime $V_0 > E_p + mc^2$, we calculate the relativistic group velocity $v_g = \frac{\partial E_p}{\partial(\hbar k)}$ inside the barrier. Differentiating Eq. (23.64) gives

$$v_g = \frac{\hbar c^2 k}{E_p - V_0}. \quad (23.70)$$

Because $E_p - V_0 < -mc^2 < 0$, the denominator is negative. Requiring an outgoing wave carrying flux away from the interface ($v_g > 0$) demands choosing a *negative* wave number:

$$k = -\frac{1}{\hbar c} \sqrt{(E_p - V_0)^2 - m^2 c^4} < 0. \quad (23.71)$$

Substituting $k < 0$ into Eq. (23.65) yields a striking result:

$$\left| \frac{b}{a} \right| = \left| \frac{p + \hbar|k|}{p - \hbar|k|} \right| > 1. \quad (23.72)$$

The reflected flux strictly exceeds the incident flux! Furthermore, the transmitted charge density and current,

$$\rho = \frac{E_p - V_0}{mc^2} |d|^2 < 0, \quad j = \frac{\hbar k}{m} |d|^2 < 0, \quad (23.73)$$

are both negative.

This apparent paradox is resolved by recognizing that the supercritical potential induces spontaneous *particle-antiparticle pair production* at the barrier. Negatively charged antiparticles feel an attractive potential in the region $x > 0$ and propagate to the right, producing the negative charge density and negative current. Concurrently, the created positively charged particles are repelled to the left, adding constructively to the fully reflected incident wave to produce a reflected particle current greater than the incident beam.

Energy conservation is rigorously satisfied: the energy required to create a free particle on the left is E_p , while the energy of the antiparticle on the right is $\sqrt{\hbar^2 c^2 k^2 + m^2 c^4} - V_0$. From Eq. (23.64),

$$E_p + \sqrt{\hbar^2 c^2 k^2 + m^2 c^4} - V_0 = 0. \quad (23.74)$$

The potential drop across the barrier is so steep that it compensates for the rest mass of the pair ($V_0 > 2mc^2$), enabling spontaneous pair creation from the vacuum with net zero energy cost. Single-particle quantum mechanics therefore breaks down in supercritical external potentials, mandating a transition to relativistic quantum field theory.

23.4.4 Wave Packet Perturbations and Zitterbewegung

Even weak external perturbations mix positive- and negative-energy states. Consider perturbing a pure positive-energy wave packet at $t = 0$ by an infinitesimal spatial phase factor $e^{i\mathbf{k}\cdot\mathbf{r}/\hbar}$:

$$\Psi'(\mathbf{r}, 0) \equiv e^{i\mathbf{k}\cdot\mathbf{r}/\hbar} \Psi^{(+)}(\mathbf{r}, 0) = \int \frac{d^3 p}{(2\pi\hbar)^3} u_{\mathbf{p}-\mathbf{k}} \Psi_{\mathbf{p}-\mathbf{k}}^{(+)} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \quad (23.75)$$

Expanding to first order in $\mathbf{k}$ using the spinor relation (23.54),

$$\Psi_{\mathbf{p}-\mathbf{k}}^{(+)} \approx \Psi_{\mathbf{p}}^{(+)} + \frac{\mathbf{k} \cdot \mathbf{p} c^2}{2E_p^2} \Psi_{\mathbf{p}}^{(-)}, \quad (23.76)$$

we find that the phase factor inevitably excites negative-energy components. The subsequent time evolution is

$$\Psi'(\mathbf{r}, t) = \int \frac{d^3 p}{(2\pi\hbar)^3} \left[u_{\mathbf{p}-\mathbf{k}} \Psi_{\mathbf{p}-\mathbf{k}}^{(+)} e^{i(\mathbf{p}\cdot\mathbf{r} - E_p t)/\hbar} + u_{\mathbf{p}} \frac{\mathbf{k} \cdot \mathbf{p} c^2}{2E_p^2} \Psi_{\mathbf{p}}^{(-)} e^{i(\mathbf{p}\cdot\mathbf{r} + E_p t)/\hbar} \right]. \quad (23.77)$$

The cross-term between positive- and negative-energy amplitudes produces rapid interference oscillations at frequencies

$$\omega \geq \frac{2mc^2}{\hbar}. \quad (23.78)$$

This rapid internal microscopic trembling is known as *zitterbewegung*. For subcritical perturbations, these virtual oscillations dephase within a characteristic timescale $\Delta t \sim \hbar/(2mc^2)$ once the wave packet leaves the interaction zone. Permanent, real particle-antiparticle pairs emerge only when the external potential supplies energy exceeding the pair-production threshold $\Delta E \geq 2mc^2$.

23.5 Scattering by a Potential and Stimulated Pair Production

We investigate potential scattering for relativistic spinless particles within the first-order Feshbach-Villars formalism, utilizing the time-dependent Born approximation. This framework illustrates elastic and inelastic scattering and elucidates the mechanism of dynamic pair production in time-dependent external fields.

23.5.1 Born Approximation for Positive-Energy Scattering

Consider a particle incident with momentum $\mathbf{p}$ and positive energy $E_p = \sqrt{p^2 c^2 + m^2 c^4}$, interacting with a time-dependent electrostatic potential $e\Phi(\mathbf{r}, t)$. The wave equation is

$$i\hbar \frac{\partial \Psi}{\partial t} = (H_0 + e\Phi) \Psi, \quad (23.79)$$

where $H_0 = \frac{p^2}{2m}(\tau_3 + i\tau_2) + mc^2\tau_3$ is the free Hamiltonian. Decomposing the full state into the incident wave and a scattered correction,

$$\Psi(\mathbf{r}, t) = \Psi_{\mathbf{p}}^{(+)}(\mathbf{r}, t) + \Psi_{\text{sc}}(\mathbf{r}, t), \quad (23.80)$$

first-order perturbation theory gives

$$\left(i\hbar \frac{\partial}{\partial t} - H_0 \right) \Psi_{\text{sc}}(\mathbf{r}, t) = e\Phi(\mathbf{r}, t) \Psi_{\mathbf{p}}^{(+)}(\mathbf{r}, t). \quad (23.81)$$

Integrating with the retarded boundary condition $\Psi_{\text{sc}} \rightarrow 0$ as $t \rightarrow -\infty$ yields

$$\Psi_{\text{sc}}(t) = -\frac{i}{\hbar} \int_{-\infty}^t dt' e^{-iH_0(t-t')/\hbar} e\Phi(t') \Psi_{\mathbf{p}}^{(+)}(t'). \quad (23.82)$$

The amplitude $u_{\mathbf{p}'}(t)$ to detect the scattered particle in a positive-energy state $\Psi_{\mathbf{p}'}^{(+)}$ of momentum $\mathbf{p}'$ is obtained via the projection (23.44):

$$u_{\mathbf{p}'}(t) = \int d^3r \Psi_{\mathbf{p}'}^{(+)\dagger} \tau_3 e^{-i\mathbf{p}' \cdot \mathbf{r}/\hbar} \Psi_{\text{sc}}(\mathbf{r}, t). \quad (23.83)$$

Allowing H_0 to act to the left on the eigenbra $\Psi_{\mathbf{p}'}^{(+)\dagger} \tau_3$, which has eigenvalue $E_{p'}$, we obtain

$$u_{\mathbf{p}'}(t) = -\frac{i}{\hbar} \int_{-\infty}^t dt' e^{iE_{p'}(t-t')/\hbar} e^{-iE_p t'/\hbar} e\Phi_{\mathbf{p}'-\mathbf{p}}(t') \left[\Psi_{\mathbf{p}'}^{(+)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(+)} \right], \quad (23.84)$$

where $\Phi_{\mathbf{q}}(t) = \int d^3r e^{-i\mathbf{q} \cdot \mathbf{r}/\hbar} \Phi(\mathbf{r}, t)$ is the spatial Fourier transform of the potential. Using the explicit spinor definitions (23.38), the matrix element between positive-energy states evaluates to

$$\Psi_{\mathbf{p}'}^{(+)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(+)} = \frac{E_p + E_{p'}}{2\sqrt{E_p E_{p'}}}. \quad (23.85)$$

For a monochromatic potential oscillating with frequency ω ,

$$\Phi_{\mathbf{p}'-\mathbf{p}}(t) = \Phi_{\mathbf{p}'-\mathbf{p}} e^{-i\omega t}, \quad (23.86)$$

Fermi's golden rule yields the transition rate

$$\Gamma_{\mathbf{p}(+) \rightarrow \mathbf{p}'(+)} = \frac{2\pi}{\hbar} |e\Phi_{\mathbf{p}'-\mathbf{p}}|^2 \frac{(2E_p + \hbar\omega)^2}{4E_p(E_p + \hbar\omega)} \delta(E_p + \hbar\omega - E_{p'}). \quad (23.87)$$

Dividing by the incident flux $v = dE_p/dp$ and multiplying by the final-state density $\frac{dn'}{dE_{p'}} = \frac{p'^2 d\Omega}{(2\pi\hbar)^3 (dE_{p'}/dp')}$, we obtain the differential cross section:

$$\frac{d\sigma_{\mathbf{p} \rightarrow \mathbf{p}'}}{d\Omega} = \frac{p'}{p} \frac{|e\Phi_{\mathbf{p}'-\mathbf{p}}|^2}{4\pi^2\hbar^4} \frac{(2E_p + \hbar\omega)^2}{4E_p(E_p + \hbar\omega)}. \quad (23.88)$$

In the static limit ($\omega = 0$, $E_{p'} = E_p$, $p' = p$), the kinematic factor reduces to unity, recovering the standard Born cross section:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{static}} = \frac{|e\Phi_{\mathbf{p}'-\mathbf{p}}|^2}{4\pi^2\hbar^4}. \quad (23.89)$$

23.5.2 Transitions to Negative-Energy States and Stimulated Pair Production

We now examine the amplitude for scattering into a negative-energy antiparticle state $\Psi_{\mathbf{p}'}^{(-)} e^{-i\mathbf{p}' \cdot \mathbf{r}/\hbar}$. From Eq. (23.44), the complex conjugate amplitude is

$$v_{\mathbf{p}'}^*(t) = - \int d^3r \Psi_{\mathbf{p}'}^{(-)\dagger} e^{i\mathbf{p}' \cdot \mathbf{r}/\hbar} \tau_3 \Psi_{\text{sc}}(\mathbf{r}, t), \quad (23.90)$$

which evaluates to

$$v_{\mathbf{p}'}^*(t) = \frac{i}{\hbar} \int_{-\infty}^t dt' e^{-iE_{p'}(t-t')/\hbar} e^{-iE_p t'/\hbar} e\Phi_{-\mathbf{p}'-\mathbf{p}}(t') \left[\Psi_{\mathbf{p}'}^{(-)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(+)} \right]. \quad (23.91)$$

The cross-overlap matrix element between opposite-energy branches is

$$\Psi_{\mathbf{p}'}^{(-)\dagger} \tau_3 \Psi_{\mathbf{p}}^{(+)} = \frac{E_p - E_{p'}}{2\sqrt{E_p E_{p'}}}. \quad (23.92)$$

For a harmonic potential with frequency ω , the transition rate to the antiparticle mode $\mathbf{p}'$ is

$$\Gamma_{\mathbf{p}(+) \rightarrow \mathbf{p}'(-)} = \frac{2\pi}{\hbar} |e\Phi_{-\mathbf{p}'-\mathbf{p}}|^2 \frac{(E_p - E_{p'})^2}{4E_p E_{p'}} \delta(\hbar\omega + E_p + E_{p'}). \quad (23.93)$$

Because the total normalization $\int d^3r \Psi^\dagger \tau_3 \Psi = +1$ is an exact constant of motion, the incident particle does not convert into an antiparticle (which would violate electric charge conservation). Rather, Eq. (23.93) describes the creation of an antiparticle of momentum $\mathbf{p}'$ accompanied by an additional particle.

The energy conservation delta function enforces the pair-production threshold condition:

$$-\hbar\omega = E_p + E_{p'} \geq 2mc^2. \quad (23.94)$$

Spontaneous pair creation requires an external field oscillating at high negative frequency, providing the necessary energy $\geq 2mc^2$ and transferring spatial momentum $\mathbf{p} + \mathbf{p}'$ via the Fourier component $\Phi_{-\mathbf{p}'-\mathbf{p}}$.

The appearance of the initial particle momentum $\mathbf{p}$ in the threshold formula (23.94) has an elegant physical explanation: even in the absence of an incident beam, a sufficiently strong, high-frequency external potential produces pairs spontaneously from the vacuum. When an incident particle of momentum $\mathbf{p}$ is already present, Bose-Einstein statistics enhances the transition rate into that occupied mode by an additional factor of $(n_{\mathbf{p}} + 1)$. Equation (23.93) precisely computes the *stimulated emission* rate of particle-antiparticle pairs due to the occupation of the incoming mode $\mathbf{p}$.

23.6 Bound-State Problems: The Relativistic Coulomb Potential

We now investigate stationary bound states of relativistic spin-zero particles in a static electrostatic potential $\Phi(\mathbf{r})$. A prominent experimental application is the pionic atom (such as a π^- meson bound in the Coulomb field of an atomic nucleus), which admits an exact analytical solution in the Klein-Gordon framework.

23.6.1 Stationary States and Vacuum Polarization

For a stationary state of energy E , the wave function separates as

$$\psi(\mathbf{r}, t) = \psi(\mathbf{r}) e^{-iEt/\hbar}. \quad (23.95)$$

The corresponding physical charge density given by Eq. (22.24) is

$$e\rho(\mathbf{r}) = \frac{e[E - e\Phi(\mathbf{r})]}{mc^2} |\psi(\mathbf{r})|^2. \quad (23.96)$$

In classically allowed regions where $E > e\Phi(\mathbf{r})$, the charge density has the sign of the particle charge e . Conversely, in classically forbidden domains where $e\Phi(\mathbf{r}) > E$, the local charge density changes sign.

This sign reversal is a direct manifestation of virtual antiparticle admixture: according to Eq. (23.43), the bound-state wave packet decomposes into free particle and antiparticle components. Positively charged components are attracted toward regions of low electrostatic potential $e\Phi$, while negatively charged antiparticle components are displaced toward higher potential. The external potential thus polarizes the virtual particle-antiparticle degrees of freedom in the wave function, producing an effective *vacuum polarization* that screens the nuclear charge.

23.6.2 Exact Solution for the Pionic Hydrogen-Like Atom

Consider an attractive central Coulomb potential produced by a nucleus of atomic number Z :

$$e\Phi(r) = -\frac{Ze^2}{r}. \quad (23.97)$$

The stationary Klein-Gordon equation (22.17) takes the form

$$\left(E + \frac{Ze^2}{r}\right)^2 \psi(\mathbf{r}) = (-\hbar^2 c^2 \nabla^2 + m^2 c^4) \psi(\mathbf{r}). \quad (23.98)$$

Expanding into spherical harmonics $\psi(\mathbf{r}) = R_l(r) Y_{lm}(\theta, \phi)$, the radial wave equation becomes

$$\left[-\frac{\hbar^2}{2m} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{\hbar^2 l(l+1) - (Ze^2)^2/c^2}{2mr^2} - \frac{EZe^2}{mc^2 r} \right] R_l(r) = \frac{E^2 - m^2 c^4}{2mc^2} R_l(r). \quad (23.99)$$

Equation (23.99) can be mapped directly onto the nonrelativistic radial Coulomb Schrödinger equation by defining effective parameters:

$$m' \equiv \frac{E}{c^2}, \quad (23.100)$$

$$l'(l'+1) \equiv l(l+1) - (Z\alpha)^2, \quad (23.101)$$

$$E' \equiv \frac{E^2 - m^2 c^4}{2E}, \quad (23.102)$$

where $\alpha = e^2/(\hbar c) \approx 1/137$ is the fine-structure constant. In terms of these primed variables, Eq. (23.99) becomes

$$\left[-\frac{\hbar^2}{2m'} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{\hbar^2 l'(l'+1)}{2m'r^2} - \frac{Ze^2}{r} \right] R(r) = E' R(r). \quad (23.103)$$

This is isomorphic to the nonrelativistic Kepler problem with effective mass m' , effective energy E' , and non-integer orbital angular momentum l' .

23.6.3 Energy Eigenvalues and Fine Structure

The bound states of the nonrelativistic Coulomb problem with arbitrary complex angular momentum correspond to the poles of the partial-wave S -matrix (Regge poles):

$$S_l \propto \frac{\Gamma(l+1+i\gamma)}{\Gamma(l+1-i\gamma)}, \quad \gamma = -\frac{m'Ze^2}{\hbar\sqrt{2m'E'}}. \quad (23.104)$$

Poles occur when the gamma function argument in the numerator equals a non-positive integer:

$$l' + 1 + i\gamma = -\nu, \quad \nu = 0, 1, 2, \dots \quad (23.105)$$

Defining the effective principal quantum number $n' \equiv l' + \nu + 1$, the eigenvalue condition reads

$$E' = -\frac{m'Z^2e^4}{2\hbar^2n'^2}. \quad (23.106)$$

Multiplying by $2m' = 2E/c^2$ and substituting definition (23.102) yields

$$E^2 - m^2c^4 = -\left(\frac{Z\alpha}{n'}\right)^2 E^2, \quad (23.107)$$

which immediately solves for the exact relativistic energy:

$$E = \frac{mc^2}{\sqrt{1 + \left(\frac{Z\alpha}{n'}\right)^2}}. \quad (23.108)$$

To determine l' , we complete the square in Eq. (23.101):

$$\left(l' + \frac{1}{2}\right)^2 = \left(l + \frac{1}{2}\right)^2 - (Z\alpha)^2, \quad (23.109)$$

yielding the two roots

$$l'_\pm = -\frac{1}{2} \pm \sqrt{\left(l + \frac{1}{2}\right)^2 - (Z\alpha)^2}. \quad (23.110)$$

As $r \rightarrow 0$, the regular solution behaves as $R(r) \sim r^{l'}$. For $l \geq 1$, the lower branch gives $l'_- < -1$, rendering the wave function non-square-integrable at the origin. For $l = 0$, both roots would yield square-integrable functions; however, a linear combination $\psi = \psi_{l'_+} + i\psi_{l'_-}$ produces a divergent radial current density:

$$j_r \propto \frac{l'_+ - l'_-}{r} r^{l'_+ + l'_-} = \frac{2\sqrt{(l + 1/2)^2 - (Z\alpha)^2}}{r^2}. \quad (23.111)$$

The flux integral $\oint j_r r^2 d\Omega$ would then be a non-vanishing constant independent of the sphere radius, representing an unphysical point source or sink of probability at the origin. Physical boundary conditions therefore uniquely select the upper branch l'_+ :

$$l' = -\frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - (Z\alpha)^2}. \quad (23.112)$$

Introducing the standard principal quantum number $n = \nu + l + 1$, the effective quantum number is

$$n' = n - \left(l + \frac{1}{2}\right) + \sqrt{\left(l + \frac{1}{2}\right)^2 - (Z\alpha)^2}. \quad (23.113)$$

Expanding Eq. (23.108) in powers of the fine-structure parameter $(Z\alpha)$ yields

$$E = mc^2 - \frac{mZ^2e^4}{2\hbar^2n^2} \left[1 + \left(\frac{Z\alpha}{n}\right)^2 \left(\frac{n}{l+1/2} - \frac{3}{4}\right) \right] + \mathcal{O}((Z\alpha)^6). \quad (23.114)$$

The first term is the particle's rest energy, the second reproduces the Bohr spectrum, and the third represents the relativistic fine-structure shift. Notice that the accidental nonrelativistic l -degeneracy is lifted: states with lower orbital angular momentum l penetrate closer to the nucleus, feeling a stronger relativistic attraction and thus acquiring larger binding energies.

23.6.4 Fall to the Center and Critical Coupling

The exact energy eigenvalue (23.108) remains real provided that the radical in Eq. (23.112) is non-negative:

$$Z\alpha \leq l + \frac{1}{2}. \quad (23.115)$$

For s -states ($l = 0$), this imposes the critical threshold

$$Z\alpha < \frac{1}{2} \implies Z < 68. \quad (23.116)$$

When $Z\alpha > l + 1/2$, the effective attractive $1/r^2$ potential overcomes the centrifugal barrier, causing the classical orbit and the quantum wave packet to collapse into the center (the "fall to the center" catastrophe). For physical pionic atoms with $Z > 68$, this singular behavior is regularized by the finite spatial distribution of the nuclear charge and quantum electrodynamic vacuum polarization corrections.²

23.7 Nonrelativistic Limit and Relativistic Corrections

To connect the Klein-Gordon theory with familiar nonrelativistic quantum mechanics, we systematically eliminate the small component χ in powers of $1/c$ and extract the leading relativistic corrections to the Schrödinger equation.

23.7.1 Systematic Elimination of the Small Component

For a positive-energy particle state in the nonrelativistic regime, the rapid phase oscillation associated with the rest mass can be factored out:

$$\Psi(\mathbf{r}, t) = e^{-imc^2t/\hbar} \begin{pmatrix} \tilde{\varphi}(\mathbf{r}, t) \\ \tilde{\chi}(\mathbf{r}, t) \end{pmatrix}, \quad (23.117)$$

²Experimental spectroscopy of π^- mesic atoms is discussed by D. A. Jenkins and R. Kunselman, *Phys. Rev. Lett.* **17**, 1148 (1966).

where $\tilde{\varphi}$ and $\tilde{\chi}$ vary slowly compared to the rest-mass frequency $\omega_0 = mc^2/\hbar$. Consequently, the time derivative acting on the lower component satisfies

$$i\hbar \frac{\partial \chi}{\partial t} \approx mc^2 \chi. \quad (23.118)$$

Substituting this estimate into the lower component equation of the Feshbach-Villars system (23.21), we obtain

$$(2mc^2 - e\Phi + \mathcal{O}(v^2/c^2)) \chi = -\frac{1}{2m} \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right)^2 (\varphi + \chi). \quad (23.119)$$

To leading order in v/c , neglecting terms of relative order v^2/c^2 and assuming weak electrostatic potentials $|e\Phi| \ll mc^2$, the lower component is enslaved to the upper component:

$$\chi \approx -\frac{1}{4m^2 c^2} \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right)^2 \varphi \sim \mathcal{O}\left(\frac{v^2}{c^2}\right) \varphi. \quad (23.120)$$

The lower component is suppressed by $(v/c)^2$ relative to φ , consistent with the free-particle limit (23.41).

23.7.2 Leading Relativistic Hamiltonian

Setting $\chi = 0$ in the upper component equation of (23.21) immediately recovers the standard nonrelativistic Schrödinger equation with rest mass included:

$$i\hbar \frac{\partial \varphi}{\partial t} = \left[mc^2 + \frac{1}{2m} \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right)^2 + e\Phi \right] \varphi. \quad (23.121)$$

To determine the leading relativistic corrections, we substitute the approximation (23.120) back into the upper equation of (23.21):

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \varphi = \left[mc^2 + \frac{1}{2m} \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right)^2 - \frac{1}{8m^3 c^2} \left(\frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} \right)^4 \right] \varphi. \quad (23.122)$$

The differential operator appearing on the right-hand side is simply the binomial Taylor expansion of the relativistic kinetic energy operator:

$$c\sqrt{m^2 c^2 + \mathbf{\Pi}^2} = mc^2 + \frac{\mathbf{\Pi}^2}{2m} - \frac{\mathbf{\Pi}^4}{8m^3 c^2} + \mathcal{O}\left(\frac{\mathbf{\Pi}^6}{m^5 c^4}\right), \quad (23.123)$$

where $\mathbf{\Pi} = \mathbf{p} - \frac{e}{c} \mathbf{A} = \frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A}$ is the gauge-covariant mechanical momentum. Crucially, because the Klein-Gordon field has spin zero, there are no spin-orbit interactions or spin-related Darwin terms; the leading relativistic correction is purely kinematic in origin.

23.7.3 Orbital Magnetic Moment in a Uniform Field

In the presence of a weak uniform magnetic field $\mathcal{H} = \nabla \times \mathbf{A}$ described by the symmetric gauge $\mathbf{A} = \frac{1}{2} \mathcal{H} \times \mathbf{r}$, the mechanical momentum operator satisfies

$$\mathbf{\Pi}^2 = p^2 - \frac{e}{c} \mathcal{H} \cdot \mathbf{L} + \mathcal{O}(\mathcal{H}^2), \quad (23.124)$$

where $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is the canonical orbital angular momentum. Retaining terms up to order $(v/c)^3$ and linear in $\mathcal{H}$, Eq. (23.122) expands to

$$i\hbar \frac{\partial \varphi}{\partial t} = -\frac{\hbar^2 \nabla^2}{2m} \left(1 + \frac{\hbar^2 \nabla^2}{4m^2 c^2} \right) \varphi + (mc^2 + e\Phi) \varphi - \frac{e\mathcal{H}}{2mc} \cdot \mathbf{L} \left(1 + \frac{\hbar^2 \nabla^2}{2m^2 c^2} \right) \varphi. \quad (23.125)$$

From Eq. (23.125), the effective orbital magnetic moment operator $\boldsymbol{\mu}_L$ is identified as

$$\boldsymbol{\mu}_L = \frac{e\mathbf{L}}{2mc} \left(1 - \frac{p^2}{2m^2c^2} \right). \quad (23.126)$$

Relativistic kinematics decreases the effective orbital gyromagnetic ratio by a fractional factor $\sim v^2/(2c^2)$, reflecting the relativistic mass increase $m_{\text{eff}} = \gamma m \approx m(1 + v^2/2c^2)$ of the moving particle.

23.8 Scalar Interactions

In preceding sections, we considered particles coupled to the electromagnetic field via minimal gauge coupling $(\Phi, \mathbf{A})$, which forms a spacetime four-vector. Gauge interactions couple with opposite signs to particles and antiparticles ($\pm e$). In hadron and nuclear physics, however, particles also experience *scalar interactions*, mediated by scalar fields (such as the σ -meson in effective nuclear models or the Higgs field), which couple with identical strength and sign to both particles and antiparticles.

23.8.1 Covariant Formulation of Scalar Potentials

In the nonrelativistic domain, a spinless particle in a spin- and charge-independent potential $S(\mathbf{r}, t)$ obeys the Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2 \nabla^2}{2m} + S(\mathbf{r}, t) \right] \psi(\mathbf{r}, t). \quad (23.127)$$

To generalize this interaction relativistically while preserving Lorentz covariance, we treat $S(\mathbf{r}, t)$ as a Lorentz scalar field. Rather than shifting the time derivative (which produces vector coupling), the scalar potential couples directly to the mass term. Shifting $m^2c^2 \rightarrow m^2c^2 + 2mS(\mathbf{r}, t)$ in the Klein-Gordon equation (22.15) gives

$$-\frac{\hbar^2}{c^2} \frac{\partial^2 \psi}{\partial t^2} = [-\hbar^2 \nabla^2 + m^2c^2 + 2mS(\mathbf{r}, t)] \psi(\mathbf{r}, t). \quad (23.128)$$

In four-vector notation, this is $(\partial_\mu \partial^\mu + \frac{m^2c^2 + 2mS}{\hbar^2}) \psi = 0$, ensuring that ψ transforms as a genuine Lorentz scalar.

Equation (23.128) corresponds to the classical relativistic dispersion relation

$$E^2 = c^2p^2 + m^2c^4 + 2mc^2S(\mathbf{r}, t). \quad (23.129)$$

Expanding Eq. (23.129) in the nonrelativistic limit ($p \ll mc$, $|S| \ll mc^2$) yields

$$E = \sqrt{m^2c^4 + c^2p^2 + 2mc^2S} \approx mc^2 + \frac{p^2}{2m} + S(\mathbf{r}, t), \quad (23.130)$$

reproducing Eq. (23.127). Because $S(\mathbf{r}, t)$ is real, taking the complex conjugate of Eq. (23.128) leaves the equation invariant: ψ and ψ^* satisfy the identical wave equation with no sign inversion of S . Particles and antiparticles therefore experience identical forces in a scalar potential.

23.8.2 Stationary Bound States in a Static Scalar Potential

For a static scalar potential $S(\mathbf{r})$, stationary states $\psi(\mathbf{r}, t) = \psi(\mathbf{r})e^{-iEt/\hbar}$ satisfy

$$\frac{E^2 - m^2c^4}{2mc^2} \psi(\mathbf{r}) = \left[-\frac{\hbar^2 \nabla^2}{2m} + S(\mathbf{r}) \right] \psi(\mathbf{r}). \quad (23.131)$$

In the nonrelativistic limit, the left-hand side reduces directly to the nonrelativistic binding energy:

$$\frac{E^2 - m^2 c^4}{2mc^2} = \frac{(E - mc^2)(E + mc^2)}{2mc^2} \approx E - mc^2. \quad (23.132)$$

Equation (23.131) is exactly isomorphic to an equivalent nonrelativistic Schrödinger eigenvalue problem:

$$\left[-\frac{\hbar^2 \nabla^2}{2m} + S(\mathbf{r}) \right] \psi(\mathbf{r}) = E' \psi(\mathbf{r}). \quad (23.133)$$

Once the nonrelativistic energy eigenvalue E' is determined, the exact relativistic energy eigenvalues are obtained from

$$E = \pm \sqrt{m^2 c^4 + 2mc^2 E'}. \quad (23.134)$$

Because the wave equation depends solely on E^2 , the spectrum is strictly symmetric under $E \leftrightarrow -E$. This exact reflection symmetry reflects the fact that scalar interactions treat particles and antiparticles on completely equal footings.

To prevent imaginary frequencies and tachyon-like spatial instabilities ($E^2 < 0$), the nonrelativistic eigenvalues must satisfy the physical lower bound:

$$E' > -\frac{1}{2}mc^2. \quad (23.135)$$

When the attractive scalar potential is sufficiently deep that $E' \leq -mc^2/2$, the effective mass vanishes or becomes imaginary, indicating vacuum instability and spontaneous scalar field condensation.

23.9 Neutral Particles and Real Scalar Fields

Throughout our discussion of the Klein-Gordon theory, the conserved density $\rho(\mathbf{r}, t)$ has been interpreted as an electric charge density. We now clarify how neutral spinless particles ($q = 0$) are incorporated into the relativistic framework. Two distinct physical categories arise:

23.9.1 Non-Self-Conjugate Neutral Particles

The first category consists of neutral particles that are distinct from their antiparticles. A canonical example is the neutral kaon system (K^0 and $\bar{K}^0$). Although both carry zero electric charge, they are distinguished by an internal global quantum number: strangeness ($S = +1$ for K^0 and $S = -1$ for $\bar{K}^0$, or hypercharge Y).

Because fundamental interactions (such as the strong interaction) conserve this internal flavor charge, we can define a generalized conserved charge density. The complex Klein-Gordon field $\psi(\mathbf{r}, t)$ describes both K^0 and $\bar{K}^0$, and $\rho(\mathbf{r}, t)$ represents the local density of strangeness rather than electric charge. Positive-frequency modes correspond to strangeness +1 particles, while negative-frequency modes describe strangeness -1 antiparticles.

23.9.2 Truly Neutral (Self-Conjugate) Particles

The second category comprises particles that are strictly identical to their antiparticles, having no distinguishing quantum numbers whatsoever. A prime example is the neutral pion π^0 (in contrast to the charged pair $\pi^\pm$).

For self-conjugate particles, nature exhibits an exact symmetry between positive- and negative-energy states. If $\psi(\mathbf{r}, t)$ satisfies the free Klein-Gordon equation, its charge-conjugate wave function $\psi^*(\mathbf{r}, t)$ satisfies the identical equation. Consequently, the field can be chosen to be strictly *real*:

$$\psi(\mathbf{r}, t) = \psi^*(\mathbf{r}, t). \quad (23.136)$$

For a real scalar field, the conserved current and density vanish identically:

$$\begin{aligned}\rho(\mathbf{r}, t) &= \frac{i\hbar}{2mc^2} \left(\psi \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi}{\partial t} \right) = 0, \\ \mathbf{j}(\mathbf{r}, t) &= \frac{\hbar}{2im} (\psi \nabla \psi - \psi \nabla \psi) = \mathbf{0}.\end{aligned}\quad (23.137)$$

A real scalar field carries no conserved additive charge. This situation directly parallels classical electromagnetism, where the photon (a neutral spin-1 boson that is its own antiparticle) is described by a real vector potential $A^\mu(\mathbf{r}, t)$. Upon canonical second quantization, the real Klein-Gordon field operator creates and destroys quanta (π^0 mesons) that are their own antiparticles.

23.10 Problems

1. **Galilean Transformation and Wave Function Phase:** Show generally (without assuming free particles) that under an active Galilean boost $\mathbf{r}' = \mathbf{r} - \mathbf{v}t$, $t' = t$, the solution $\psi'(\mathbf{r}', t')$ to the nonrelativistic Schrödinger equation in the moving frame is related to the solution $\psi(\mathbf{r}, t)$ in the laboratory frame by

$$\psi'(\mathbf{r}', t) = \exp \left[-\frac{i}{\hbar} \left(m\mathbf{v} \cdot \mathbf{r} - \frac{1}{2}mv^2t \right) \right] \psi(\mathbf{r}, t). \quad (23.138)$$

Explain why, in sharp contrast to the relativistic Klein-Gordon wave function where ψ transforms as a spacetime scalar even when $v \ll c$, the nonrelativistic Schrödinger wave function requires a velocity-dependent gauge phase factor (forming a ray representation of the Galilei group).

2. **Magnetic Quench and Antiparticle Production:** A π^+ meson occupies a stationary state in a uniform magnetic field $\mathcal{H} = \mathcal{H}\hat{\mathbf{z}}$. Its longitudinal momentum along the field vanishes ($p_z = 0$), and its orbital angular momentum projection along the field is $L_z = +\hbar$. At $t = 0$, the magnetic field is suddenly switched off.

Calculate the angular and energy distributions of π^- mesons (antiparticles) produced via sudden non-adiabatic perturbation of the vacuum.

Hint: In the symmetric gauge $\mathbf{A} = \frac{1}{2}\mathcal{H} \times \mathbf{r}$, the stationary Klein-Gordon equation maps onto a two-dimensional harmonic oscillator.

3. **Gyromagnetic Ratios in the Pionic Atom:** Using the relativistic corrections derived for the magnetic moment operator in Eq. (23.126), calculate the gyromagnetic ratios (effective Landé g -factors) for the bound eigenstates $|n, l, m\rangle$ of the relativistic spin-zero Coulomb problem.
4. **Scalar Coulomb Potential:** Suppose an attractive central Coulomb interaction were mediated by a Lorentz scalar potential, $S(r) = -Ze^2/r$, rather than the time component of a four-vector gauge field.
 - (a) Determine the bound-state energy spectrum and compare the resulting fine-structure splitting with the vector Coulomb result in Eq. (23.114).
 - (b) Investigate whether any critical restrictions arise on the coupling strength $Z\alpha$ in this scalar case.

Chapter 24

Relativistic Spin 1/2 Particles Dirac Equation

24.1 Relativistic Spin-1/2 Particles: Dirac Equation

We now turn to the quantum-mechanical description of particles with intrinsic spin $1/2$ moving at relativistic velocities (such as electrons, positrons, muons, and quarks). Because spin constitutes an intrinsic angular momentum tensor intimately connected with spacetime rotations and boosts, special relativity imposes stringent symmetry requirements on the structure of the wave equation.¹ Before constructing the Dirac equation, we first analyze how spin states transform under general proper orthochronous Lorentz transformations.

¹See the comprehensive review by H. Feshbach and F. Villars, *Rev. Mod. Phys.* **30**, 24 (1958).

Chapter 25

Relativistic Spin Vz Particles Dirac Equation 534

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25.2 Lorentz Transformation of Spin and the Dirac Algebra

Spin represents the intrinsic angular momentum of a particle. In nonrelativistic quantum mechanics, angular momentum is conventionally treated as a three-dimensional axial vector. In relativistic spacetime, however, angular momentum does not transform as a four-vector; rather, it forms the spatial components of an antisymmetric second-rank tensor. In this section, we deduce the transformation properties of relativistic spin-1/2 states, uncover the necessity of a four-dimensional internal state space, and construct the Dirac Clifford algebra.

25.2.1 Spin as a Second-Rank Spacetime Tensor

Consider the canonical orbital angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, with components $L_i = \varepsilon_{ijk} r_j p_k$. Under a Lorentz boost along the z -axis with velocity v , the transverse spatial coordinates and

momenta remain invariant:

$$x' = x, \quad y' = y, \quad p'_x = p_x, \quad p'_y = p_y. \quad (25.1)$$

Consequently, the longitudinal angular momentum component is unaffected:

$$L'_z = x'p'_y - y'p'_x = xp_y - yp_x = L_z. \quad (25.2)$$

If $\mathbf{L}$ transformed as the spatial part of a four-vector A^μ , its longitudinal component would undergo a Lorentz transformation: $L'_z = \gamma(L_z - \frac{v}{c}L^0) \neq L_z$. Instead, $\mathbf{L}$ transforms as the space-space components of an antisymmetric second-rank tensor $L^{\mu\nu} = x^\mu p^\nu - x^\nu p^\mu$.

An illuminating analogy is provided by the electromagnetic field strength tensor $F^{\mu\nu}$, whose components combine the electric field $\mathcal{E}$ and magnetic field $\mathcal{H}$:

$$F^{\mu\nu} = \begin{pmatrix} 0 & \mathcal{E}_1 & \mathcal{E}_2 & \mathcal{E}_3 \\ -\mathcal{E}_1 & 0 & \mathcal{H}_3 & -\mathcal{H}_2 \\ -\mathcal{E}_2 & -\mathcal{H}_3 & 0 & \mathcal{H}_1 \\ -\mathcal{E}_3 & \mathcal{H}_2 & -\mathcal{H}_1 & 0 \end{pmatrix}. \quad (25.3)$$

Under a boost with velocity $\mathbf{v}$, the electromagnetic fields transform as

$$\begin{aligned} \mathcal{H}'_{\parallel} &= \mathcal{H}_{\parallel}, & \mathcal{E}'_{\parallel} &= \mathcal{E}_{\parallel}, \\ \mathcal{H}'_{\perp} &= \frac{\mathcal{H}_{\perp} - (\mathbf{v}/c) \times \mathcal{E}}{\sqrt{1 - v^2/c^2}}, & \mathcal{E}'_{\perp} &= \frac{\mathcal{E}_{\perp} + (\mathbf{v}/c) \times \mathcal{H}}{\sqrt{1 - v^2/c^2}}. \end{aligned} \quad (25.4)$$

Under spatial inversion (parity $\mathcal{P}$), the electric field is a true polar vector ($\mathcal{E} \rightarrow -\mathcal{E}$), whereas the magnetic field is an axial pseudovector ($\mathcal{H} \rightarrow \mathcal{H}$). Orbital angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is likewise parity-even ($\mathbf{L} \rightarrow \mathbf{L}$).

Because the intrinsic spin $\mathbf{S} = \frac{1}{2}\hbar\boldsymbol{\sigma}$ is parity-even, it must transform like $\mathcal{H}$. Lorentz covariance therefore requires the existence of a conjugate polar vector operator acting on the internal spin space, which we denote by $i\boldsymbol{\alpha}/2$, analogous to $\mathcal{E}$. Together, $\boldsymbol{\sigma}$ and $i\boldsymbol{\alpha}$ form an antisymmetric second-rank spin tensor:

$$\sigma^{\mu\nu} = \begin{pmatrix} 0 & i\alpha_1 & i\alpha_2 & i\alpha_3 \\ -i\alpha_1 & 0 & \sigma_3 & -\sigma_2 \\ -i\alpha_2 & -\sigma_3 & 0 & \sigma_1 \\ -i\alpha_3 & \sigma_2 & -\sigma_1 & 0 \end{pmatrix}, \quad (25.5)$$

transforming under boosts as

$$\begin{aligned} \boldsymbol{\sigma}'_{\parallel} &= \boldsymbol{\sigma}_{\parallel}, & i\boldsymbol{\alpha}'_{\parallel} &= i\boldsymbol{\alpha}_{\parallel}, \\ \boldsymbol{\sigma}'_{\perp} &= \frac{\boldsymbol{\sigma}_{\perp} - (\mathbf{v}/c) \times i\boldsymbol{\alpha}}{\sqrt{1 - v^2/c^2}}, & i\boldsymbol{\alpha}'_{\perp} &= \frac{i\boldsymbol{\alpha}_{\perp} + (\mathbf{v}/c) \times \boldsymbol{\sigma}}{\sqrt{1 - v^2/c^2}}. \end{aligned} \quad (25.6)$$

25.2.2 Algebraic Properties of the Internal Operators

Since $\boldsymbol{\alpha}$ transforms as a vector under spatial rotations generated by $\mathbf{S} = \frac{1}{2}\boldsymbol{\sigma}$, it satisfies the standard rotational commutation relation:

$$[\alpha_i, \sigma_j] = 2i\varepsilon_{ijk}\alpha_k. \quad (25.7)$$

For a spin-1/2 particle, the Pauli relation in the rest frame is

$$\sigma_i\sigma_j = \delta_{ij} + i\varepsilon_{ijk}\sigma_k. \quad (25.8)$$

Because the fundamental quantum algebra of physical observables must be Lorentz invariant, Eq. (25.8) must hold in every Lorentz frame ($\sigma'_i \sigma'_j = \delta_{ij} + i\varepsilon_{ijk} \sigma'_k$).

Considering a boost along the z -axis with velocity v :

$$\sigma'_x = \frac{\sigma_x + i(v/c)\alpha_y}{\sqrt{1 - v^2/c^2}}, \quad \sigma'_y = \frac{\sigma_y - i(v/c)\alpha_x}{\sqrt{1 - v^2/c^2}}, \quad \sigma'_z = \sigma_z. \quad (25.9)$$

Requiring $\sigma_x'^2 = \mathbf{1}$ for all $v < c$ yields:

$$\frac{\sigma_x^2 + i(v/c)\{\sigma_x, \alpha_y\} - (v^2/c^2)\alpha_y^2}{1 - v^2/c^2} = \mathbf{1}, \quad (25.10)$$

which demands

$$\{\sigma_x, \alpha_y\} = 0, \quad \alpha_y^2 = \mathbf{1}. \quad (25.11)$$

Similarly, evaluating the cross-products $\sigma'_y \sigma'_z = i\sigma'_x$ and $\sigma'_x \sigma'_y = i\sigma'_z$ fixes the full commutation and anticommutation structure:

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij}\mathbf{1}, \quad (25.12)$$

$$[\alpha_i, \alpha_j] = 2i\varepsilon_{ijk}\sigma_k, \quad (25.13)$$

$$[\alpha_i, \sigma_j] = 2i\varepsilon_{ijk}\alpha_k, \quad (25.14)$$

$$\{\alpha_i, \sigma_j\} = 0 \quad (i \neq j). \quad (25.15)$$

Notice that α satisfies an identical algebra to σ . However, $\alpha \neq \sigma$ because they possess opposite spatial parities: σ is parity-even ($\mathcal{P}\sigma\mathcal{P}^{-1} = \sigma$), whereas α is parity-odd ($\mathcal{P}\alpha\mathcal{P}^{-1} = -\alpha$).

25.2.3 Parity and the Four-Dimensional Representation

Let β denote the unitary parity operator acting on the internal spin space. Choosing $\beta^2 = \mathbf{1}$ (so that $\beta^{-1} = \beta^\dagger = \beta$), the transformation rules require

$$\beta\sigma = \sigma\beta, \quad \beta\alpha = -\alpha\beta. \quad (25.16)$$

Taking the determinant of $\beta\alpha_i\beta = -\alpha_i$ in an N -dimensional matrix representation gives

$$\det(\alpha_i) = \det(-\alpha_i) = (-1)^N \det(\alpha_i). \quad (25.17)$$

Since $\alpha_i^2 = \mathbf{1}$, $\det(\alpha_i) \neq 0$, implying that N must be even ($N = 2, 4, 6, \dots$).

For $N = 2$, the set of Pauli matrices $\{\mathbf{1}, \sigma\}$ forms a complete basis for all 2×2 matrices. Any matrix commuting with σ would necessarily be proportional to the identity, making it impossible to satisfy $\{\beta, \alpha\} = 0$. Therefore, the minimal dimension is $N = 4$.

Relativistic invariance thus dictates that a spin-1/2 particle must possess at least *four* internal degrees of freedom: two spin orientations ($\pm\hbar/2$) for the particle, and two for the corresponding antiparticle.

In the *standard Dirac-Pauli representation*, the 4×4 matrices are expressed in terms of 2×2 blocks:

$$\sigma = \begin{pmatrix} \boldsymbol{\tau} & 0 \\ 0 & \boldsymbol{\tau} \end{pmatrix}, \quad \alpha = \begin{pmatrix} 0 & \boldsymbol{\tau} \\ \boldsymbol{\tau} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad (25.18)$$

where $\boldsymbol{\tau} = (\tau_1, \tau_2, \tau_3)$ denotes the conventional 2×2 Pauli spin matrices. All these matrices are Hermitian and traceless.

25.2.4 Finite Lorentz Transformations

Spatial rotations through an angle ϕ are generated by the unitary operator

$$R_\phi = \exp\left(-\frac{i}{2}\boldsymbol{\sigma} \cdot \boldsymbol{\phi}\right), \quad (25.19)$$

under which internal operators transform as $O' = R_\phi O R_\phi^{-1}$.

Similarly, Lorentz boosts along direction $\hat{\mathbf{v}}$ are generated by the boost parameter (rapidity) $\boldsymbol{\omega} = \omega \hat{\mathbf{v}}$, where $\tanh \omega = v/c$:

$$\mathcal{L}_{\mathbf{v}} = \exp\left(\frac{1}{2}\boldsymbol{\alpha} \cdot \boldsymbol{\omega}\right), \quad \mathcal{L}_{\mathbf{v}}^{-1} = \exp\left(-\frac{1}{2}\boldsymbol{\alpha} \cdot \boldsymbol{\omega}\right). \quad (25.20)$$

Using the hyperbolic identity

$$\exp\left(\frac{1}{2}\boldsymbol{\alpha} \cdot \boldsymbol{\omega}\right) = \cosh\left(\frac{\omega}{2}\right) + (\boldsymbol{\alpha} \cdot \hat{\boldsymbol{\omega}}) \sinh\left(\frac{\omega}{2}\right), \quad (25.21)$$

one readily verifies that $\mathcal{L}_{\mathbf{v}}$ generates the exact transformation law (25.6).

25.2.5 The Dirac Gamma Matrices and Clifford Algebra

Under a Lorentz boost, the parity operator β transforms as

$$\beta' = \mathcal{L}_{\mathbf{v}} \beta \mathcal{L}_{\mathbf{v}}^{-1} = \frac{\beta - (\mathbf{v}/c) \cdot \beta \boldsymbol{\alpha}}{\sqrt{1 - v^2/c^2}}, \quad (25.22)$$

while the longitudinal and transverse components of $\beta \boldsymbol{\alpha}$ transform according to

$$\beta' \boldsymbol{\alpha}'_{\parallel} = \frac{\beta \boldsymbol{\alpha}_{\parallel} - (\mathbf{v}/c) \beta}{\sqrt{1 - v^2/c^2}}, \quad \beta' \boldsymbol{\alpha}'_{\perp} = \beta \boldsymbol{\alpha}_{\perp}. \quad (25.23)$$

Equations (25.22) and (25.23) establish that β and $\beta \boldsymbol{\alpha}$ constitute the time and space components of a genuine spacetime four-vector!

We define the *Dirac gamma matrices* γ^μ ($\mu = 0, 1, 2, 3$) in the standard Bjorken-Drell convention:¹

$$\gamma^0 \equiv \beta = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad \gamma^i \equiv \beta \alpha_i = \begin{pmatrix} 0 & \tau_i \\ -\tau_i & 0 \end{pmatrix}. \quad (25.24)$$

Here γ^0 is Hermitian ($\gamma^{0\dagger} = \gamma^0$), while γ^i are anti-Hermitian ($\gamma^{i\dagger} = -\gamma^i$), which is compactly expressed as $\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0$.

The gamma matrices satisfy the fundamental *Dirac Clifford algebra*:

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu} \mathbf{1}_{4 \times 4}, \quad (25.25)$$

with the Minkowski metric $g^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$. The spin tensor (25.5) is expressed via the commutator of gamma matrices:

$$\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]. \quad (25.26)$$

¹See J. D. Bjorken and S. D. Drell, *Relativistic Quantum Mechanics*, McGraw-Hill (1964).

25.2.6 Basis of the Clifford Algebra

The space of all 4×4 complex matrices is 16-dimensional. A complete, linearly independent, orthogonal basis under the Frobenius trace inner product is given by the 16 Dirac bilinear matrices:

$$\Gamma_A \in \{\mathbf{1}, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu \gamma_5, \gamma_5\}, \quad (25.27)$$

where the chirality operator γ_5 is defined by

$$\gamma_5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}. \quad (25.28)$$

The operator γ_5 satisfies $\gamma_5^2 = \mathbf{1}$, $\gamma_5^\dagger = \gamma_5$, and anticommutes with all four Dirac matrices: $\{\gamma_5, \gamma^\mu\} = 0$. Under parity, $\beta\gamma_5\beta = -\gamma_5$, identifying γ_5 as a *pseudoscalar*. Correspondingly, $\gamma^\mu\gamma_5$ is an *axial vector* (pseudovector). The 16 basis matrices thus transform under Lorentz transformations as:

Operator	Spacetime Transformation Character	Count
$\mathbf{1}$	Scalar	1
γ^μ	Vector	4
$\sigma^{\mu\nu}$	Rank-2 Antisymmetric Tensor	6
$\gamma^\mu\gamma_5$	Axial Vector (Pseudovector)	4
γ_5	Pseudoscalar	1
Total		16

All 15 non-identity basis matrices are traceless, confirming their linear independence and completeness.

25.3 The Dirac Equation and the Electron g -Factor

With the transformation laws and algebraic properties of relativistic spin-1/2 operators established, we now formulate the first-order relativistic wave equation for spin-1/2 particles.

25.3.1 Derivation of the Free Dirac Equation

Because $\gamma^\mu = (\beta, \beta\boldsymbol{\alpha})$ forms a spacetime four-vector and the four-momentum $p^\mu = (E/c, \mathbf{p})$ is likewise a four-vector, the Lorentz scalar product

$$\gamma^\mu p_\mu = \frac{\beta E}{c} - \beta\boldsymbol{\alpha} \cdot \mathbf{p} \quad (25.29)$$

is a Lorentz invariant operator. Squaring Eq. (25.29) using the operator algebra $\{\beta, \boldsymbol{\alpha}\} = 0$, $\beta^2 = \mathbf{1}$, and $(\boldsymbol{\alpha} \cdot \mathbf{p})^2 = p^2\mathbf{1}$, we find

$$\left(\frac{\beta E}{c} - \beta\boldsymbol{\alpha} \cdot \mathbf{p}\right)^2 = \beta^2 \left(\frac{E}{c}\right)^2 + (\beta\boldsymbol{\alpha} \cdot \mathbf{p})^2 - (\beta^2\boldsymbol{\alpha} + \beta\boldsymbol{\alpha}\beta) \cdot \frac{\mathbf{p}E}{c} = \frac{E^2}{c^2} - p^2 = m^2c^2. \quad (25.30)$$

The scalar operator (25.29) must therefore equal $\pm mc$. Selecting the conventional sign $+mc$, we obtain

$$\beta E - \beta c\boldsymbol{\alpha} \cdot \mathbf{p} = mc^2. \quad (25.31)$$

Promoting energy and momentum to differential operators via $E \rightarrow i\hbar\frac{\partial}{\partial t}$ and $\mathbf{p} \rightarrow \frac{\hbar}{i}\boldsymbol{\nabla}$ and letting them act on a four-component spinor wave function $\psi(\mathbf{r}, t)$, we have

$$\beta i\hbar\frac{\partial\psi}{\partial t} = \left(\beta c\boldsymbol{\alpha} \cdot \frac{\hbar}{i}\boldsymbol{\nabla} + mc^2\right)\psi. \quad (25.32)$$

Multiplying on the left by β yields the canonical Hamiltonian form of the *Dirac equation*:

$$i\hbar \frac{\partial \psi}{\partial t} = H_D \psi, \quad H_D = c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta mc^2 = c\boldsymbol{\alpha} \cdot \frac{\hbar}{i} \boldsymbol{\nabla} + \beta mc^2. \quad (25.33)$$

In covariant four-vector notation, multiplying Eq. (25.32) by $\beta = \gamma^0$ gives

$$(i\hbar \gamma^\mu \partial_\mu - mc) \psi = 0, \quad (25.34)$$

or in Feynman slash notation, $(i\hbar \not{\partial} - mc)\psi = 0$, where $\not{\partial} \equiv \gamma^\mu a_\mu$.

25.3.2 Electromagnetic Coupling and the Nonrelativistic Limit: Origin of $g = 2$

Coupling to an external electromagnetic four-potential $A^\mu = (\Phi, \mathbf{A})$ is achieved via minimal substitution:

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \psi = \left[c\boldsymbol{\alpha} \cdot \left(\frac{\hbar}{i} \boldsymbol{\nabla} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 \right] \psi. \quad (25.35)$$

To evaluate the nonrelativistic limit, we express the four-component spinor in terms of two two-component spinors:

$$\psi(\mathbf{r}, t) = \begin{pmatrix} \varphi(\mathbf{r}, t) \\ \chi(\mathbf{r}, t) \end{pmatrix}. \quad (25.36)$$

Using the standard representation (25.18), the Dirac equation splits into two coupled equations:

$$\begin{aligned} \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \varphi &= c\boldsymbol{\Pi} \cdot \boldsymbol{\tau} \chi + mc^2 \varphi, \\ \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \chi &= c\boldsymbol{\Pi} \cdot \boldsymbol{\tau} \varphi - mc^2 \chi, \end{aligned} \quad (25.37)$$

where $\boldsymbol{\Pi} = \frac{\hbar}{i} \boldsymbol{\nabla} - \frac{e}{c} \mathbf{A}$ is the kinetic momentum operator and $\boldsymbol{\tau}$ denotes the Pauli matrices.

For a positive-energy state in the nonrelativistic limit, factoring out the rest-mass phase $\psi \sim e^{-imc^2 t/\hbar} \tilde{\psi}$ gives $i\hbar \frac{\partial \chi}{\partial t} \approx mc^2 \chi$. In the weak-field regime ($|e\Phi| \ll mc^2$), the second equation of (25.37) yields

$$2mc^2 \chi \approx c(\boldsymbol{\Pi} \cdot \boldsymbol{\tau}) \varphi \implies \chi \approx \frac{1}{2mc} (\boldsymbol{\Pi} \cdot \boldsymbol{\tau}) \varphi \sim \mathcal{O}\left(\frac{v}{c}\right) \varphi. \quad (25.38)$$

Here φ is the "large" component and χ is the "small" component.

Substituting χ from Eq. (25.38) into the upper equation of (25.37) yields

$$i\hbar \frac{\partial \varphi}{\partial t} = \left[\frac{1}{2m} (\boldsymbol{\Pi} \cdot \boldsymbol{\tau})^2 + e\Phi + mc^2 \right] \varphi. \quad (25.39)$$

Using the fundamental Pauli identity $(\boldsymbol{\tau} \cdot \mathbf{a})(\boldsymbol{\tau} \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\boldsymbol{\tau} \cdot (\mathbf{a} \times \mathbf{b})$, the operator evaluates to

$$(\boldsymbol{\Pi} \cdot \boldsymbol{\tau})^2 = \boldsymbol{\Pi}^2 + i\boldsymbol{\tau} \cdot (\boldsymbol{\Pi} \times \boldsymbol{\Pi}). \quad (25.40)$$

Evaluating the cross-product of the non-commuting kinetic momenta:

$$\boldsymbol{\Pi} \times \boldsymbol{\Pi} = -\frac{e\hbar}{ic} (\boldsymbol{\nabla} \times \mathbf{A} + \mathbf{A} \times \boldsymbol{\nabla}) = i\frac{e\hbar}{c} \boldsymbol{\mathcal{H}}, \quad (25.41)$$

where $\boldsymbol{\mathcal{H}} = \boldsymbol{\nabla} \times \mathbf{A}$ is the magnetic field. Equation (25.39) becomes

$$i\hbar \frac{\partial \varphi}{\partial t} = \left[\frac{1}{2m} \left(\frac{\hbar}{i} \boldsymbol{\nabla} - \frac{e}{c} \mathbf{A} \right)^2 - \frac{e\hbar}{2mc} \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{H}} + e\Phi + mc^2 \right] \varphi. \quad (25.42)$$

Equation (25.42) is precisely the nonrelativistic *Pauli equation*. Crucially, the spin magnetic dipole interaction term

$$H_{\text{mag}} = -\boldsymbol{\mu}_S \cdot \boldsymbol{\mathcal{H}} = -g \left(\frac{e}{2mc} \mathbf{S} \right) \cdot \boldsymbol{\mathcal{H}}, \quad \mathbf{S} = \frac{\hbar}{2} \boldsymbol{\sigma}, \quad (25.43)$$

emerges automatically with the Landé gyromagnetic factor

$$g = 2. \quad (25.44)$$

The Dirac equation does not require an *ad hoc* introduction of the anomalous electron gyromagnetic ratio: $g = 2$ is an inescapable consequence of relativistic invariance for spin-1/2 fields.²

25.3.3 Conserved Probability Current

Because $\boldsymbol{\alpha}$ and β are Hermitian in the standard representation, taking the Hermitian conjugate of Eq. (25.35) gives

$$\left(-i\hbar \frac{\partial}{\partial t} - e\Phi \right) \psi^\dagger = \psi^\dagger \left[c\boldsymbol{\alpha} \cdot \left(-\frac{\hbar}{i} \boldsymbol{\nabla} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 \right]. \quad (25.45)$$

Multiplying Eq. (25.35) on the left by $\psi^\dagger$ and subtracting Eq. (25.45) multiplied on the right by ψ yields the local conservation law:

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0, \quad (25.46)$$

where

$$\rho(\mathbf{r}, t) = \psi^\dagger(\mathbf{r}, t) \psi(\mathbf{r}, t) = \sum_{a=1}^4 |\psi_a(\mathbf{r}, t)|^2 \geq 0, \quad (25.47)$$

$$\mathbf{j}(\mathbf{r}, t) = c \psi^\dagger(\mathbf{r}, t) \boldsymbol{\alpha} \psi(\mathbf{r}, t). \quad (25.48)$$

Unlike the Klein-Gordon density (22.22), the Dirac density ρ is *strictly positive definite*. It admits a consistent interpretation as a local spatial probability density, with $\mathbf{j}$ representing the probability flux. The velocity operator in the Dirac theory is identified as $\mathbf{v}_{\text{op}} = \frac{1}{i\hbar} [\mathbf{r}, H_D] = c\boldsymbol{\alpha}$.

In covariant notation, defining the Dirac adjoint spinor $\bar{\psi} \equiv \psi^\dagger \gamma^0$, the four-current density is

$$j^\mu \equiv c \bar{\psi} \gamma^\mu \psi = (c\rho, \mathbf{j}), \quad \partial_\mu j^\mu = 0. \quad (25.49)$$

25.3.4 Lorentz Transformation of the Wave Function

Under a Lorentz boost $\mathbf{x} \rightarrow \mathbf{x}' = \Lambda \mathbf{x}$ characterized by velocity $\mathbf{v}$, the matrices transform as $\boldsymbol{\alpha}' = \mathcal{L}_{\mathbf{v}} \boldsymbol{\alpha} \mathcal{L}_{\mathbf{v}}^{-1}$ and $\beta' = \mathcal{L}_{\mathbf{v}} \beta \mathcal{L}_{\mathbf{v}}^{-1}$. To preserve the numerical form of $\boldsymbol{\alpha}$ and β across all inertial reference frames, the spinor wave function must transform according to

$$\psi'(\mathbf{r}', t') = S(\Lambda) \psi(\mathbf{r}, t), \quad S(\Lambda) = \mathcal{L}_{\mathbf{v}}^{-1} = \exp \left(-\frac{1}{2} \boldsymbol{\alpha} \cdot \boldsymbol{\omega} \right). \quad (25.50)$$

With this spinor transformation law, the Dirac equation retains its identical structural form:

$$i\hbar \gamma^\mu \frac{\partial \psi'}{\partial x'^\mu} - mc \psi' = 0 \quad (25.51)$$

in every inertial reference frame.

²Quantum electrodynamic radiative loop corrections modify this tree-level prediction by the Schwinger anomalous magnetic moment $\Delta g/2 = \frac{\alpha}{2\pi} + \mathcal{O}(\alpha^2)$. For composite hadrons such as the proton and neutron, strong vacuum fluctuations induce large non-perturbative anomalous moments.

25.4 Free-Particle Solutions of the Dirac Equation

We construct the explicit plane-wave solutions of the free Dirac equation. By first solving the equation in the particle rest frame and subsequently applying active Lorentz boosts, we obtain the four linearly independent eigenspinors corresponding to the two spin states of the particle and antiparticle.

25.4.1 Solutions in the Rest Frame and Intrinsic Parity

For a particle at rest ($\mathbf{p} = \mathbf{0}$), stationary states take the form

$$\psi(\mathbf{r}, t) = e^{-iEt/\hbar} u, \quad (25.52)$$

where u is a position- and time-independent four-component spinor. Substituting Eq. (25.52) into the Dirac equation (25.33) yields the rest-frame eigenvalue equation

$$Eu = \beta mc^2 u. \quad (25.53)$$

Because the eigenvalues of $\beta = \gamma^0$ are ± 1 , the spectrum splits into two degenerate branches:

- Positive-energy solutions: $E = +mc^2$, corresponding to $\beta u = +u$.
- Negative-energy solutions: $E = -mc^2$, corresponding to $\beta u = -u$.

In the standard Dirac-Pauli representation (25.18), four orthonormal rest-frame spinors are chosen as simultaneous eigenstates of β and the rest-frame spin operator σ_z :

$$u_{0\uparrow}^{(+)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u_{0\downarrow}^{(+)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_{0\uparrow}^{(-)} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_{0\downarrow}^{(-)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (25.54)$$

The superscript $(\pm)$ labels the sign of the energy eigenvalue $b = \pm 1$. The subscript arrows denote the spin projection: $u_{0\uparrow}^{(+)}$ and $u_{0\downarrow}^{(-)}$ satisfy $\sigma_z u = +u$, while $u_{0\downarrow}^{(+)}$ and $u_{0\uparrow}^{(-)}$ satisfy $\sigma_z u = -u$. As shown below, the negative-energy state $u_{0\uparrow}^{(-)}$ physically represents an antiparticle with spin *up*.

Under spatial inversion, $\psi(\mathbf{r}, t) \rightarrow \beta \psi(-\mathbf{r}, t)$. In the rest frame, the positive-energy states possess intrinsic parity $\beta = +1$, whereas the negative-energy states possess intrinsic parity $\beta = -1$. A spin-1/2 fermion and its antifermion therefore have *opposite intrinsic parity*.

25.4.2 Boosted Spinors for Arbitrary Momentum

To obtain the eigenspinors for a particle of arbitrary momentum $\mathbf{p}$ and positive energy $E_p = \sqrt{p^2 c^2 + m^2 c^4}$, we apply an active Lorentz boost from the rest frame with velocity $\mathbf{v} = \mathbf{p}c^2/E_p$. According to Eq. (25.50), the boosted wave function is

$$\psi(\mathbf{r}, t) = e^{\pm i(\mathbf{p} \cdot \mathbf{r} - E_p t)/\hbar} u_{\mathbf{p}\sigma}^{(\pm)}, \quad (25.55)$$

where the boosted spinor is generated by

$$u_{\mathbf{p}\sigma}^{(\pm)} = \exp\left(-\frac{1}{2} \boldsymbol{\alpha} \cdot \boldsymbol{\omega}\right) u_{0\sigma}^{(\pm)} = \left[\cosh\left(\frac{\omega}{2}\right) - (\boldsymbol{\alpha} \cdot \hat{\mathbf{v}}) \sinh\left(\frac{\omega}{2}\right)\right] u_{0\sigma}^{(\pm)}. \quad (25.56)$$

Using the relativistic kinematic relations $\tanh \omega = v/c = pc/E_p$ and $\cosh \omega = \gamma = E_p/(mc^2)$, the half-angle hyperbolic functions evaluate to

$$\cosh\left(\frac{\omega}{2}\right) = \sqrt{\frac{E_p + mc^2}{2mc^2}}, \quad \sinh\left(\frac{\omega}{2}\right) = \frac{pc}{\sqrt{2mc^2(E_p + mc^2)}}. \quad (25.57)$$

The boost operator therefore takes the compact form

$$u_{\mathbf{p}\sigma}^{(\pm)} = \sqrt{\frac{E_p + mc^2}{2mc^2}} \left(\mathbf{1} + \frac{c\boldsymbol{\alpha} \cdot \mathbf{p}}{E_p + mc^2} \right) u_{0\sigma}^{(\pm)}. \quad (25.58)$$

Acting on the rest spinors (25.54) in the standard representation yields the four explicit free Dirac spinors:

$$\begin{aligned} u_{\mathbf{p}\uparrow}^{(+)} &= \sqrt{\frac{E_p + mc^2}{2mc^2}} \begin{pmatrix} 1 \\ 0 \\ \frac{p_z c}{E_p + mc^2} \\ \frac{(p_x + ip_y)c}{E_p + mc^2} \end{pmatrix}, & u_{\mathbf{p}\downarrow}^{(+)} &= \sqrt{\frac{E_p + mc^2}{2mc^2}} \begin{pmatrix} 0 \\ 1 \\ \frac{(p_x - ip_y)c}{E_p + mc^2} \\ \frac{-p_z c}{E_p + mc^2} \end{pmatrix}, \\ u_{\mathbf{p}\uparrow}^{(-)} &= \sqrt{\frac{E_p + mc^2}{2mc^2}} \begin{pmatrix} \frac{p_z c}{E_p + mc^2} \\ \frac{(p_x + ip_y)c}{E_p + mc^2} \\ 1 \\ 0 \end{pmatrix}, & u_{\mathbf{p}\downarrow}^{(-)} &= \sqrt{\frac{E_p + mc^2}{2mc^2}} \begin{pmatrix} \frac{(p_x - ip_y)c}{E_p + mc^2} \\ \frac{-p_z c}{E_p + mc^2} \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \quad (25.59)$$

The positive-energy plane waves

$$\psi_{\mathbf{p}\sigma}^{(+)}(\mathbf{r}, t) = u_{\mathbf{p}\sigma}^{(+)} e^{i(\mathbf{p} \cdot \mathbf{r} - E_p t)/\hbar} \quad (25.60)$$

describe physical particles of energy E_p , momentum $\mathbf{p}$, and rest-frame spin σ . The negative-energy solutions

$$\psi_{\mathbf{p}\sigma}^{(-)}(\mathbf{r}, t) = u_{\mathbf{p}\sigma}^{(-)} e^{-i(\mathbf{p} \cdot \mathbf{r} - E_p t)/\hbar} \quad (25.61)$$

carry formal eigenvalues $-\mathbf{p}$ and $-E_p$; under the Feynman-Stückelberg interpretation, they correspond to physical antiparticles of positive energy E_p , physical momentum $+\mathbf{p}$, and physical spin σ .

25.4.3 Orthogonality, Dirac Adjoint, and Completeness

The spinors (25.59) satisfy the standard Hermitian inner-product relations:

$$u_{\mathbf{p}\sigma}^{(b)\dagger} u_{\mathbf{p}\sigma'}^{(b')} = \frac{E_p}{mc^2} \delta_{bb'} \delta_{\sigma\sigma'} = \gamma \delta_{bb'} \delta_{\sigma\sigma'}. \quad (25.62)$$

The inner product $u^\dagger u$ scales with the Lorentz factor $\gamma = E_p/(mc^2)$, precisely as expected for a probability density undergoing Lorentz length contraction.

To construct a manifestly Lorentz-invariant bilinear, we introduce the *Dirac adjoint spinor*:

$$\bar{u} \equiv u^\dagger \gamma^0 = u^\dagger \beta. \quad (25.63)$$

Under a boost generated by $\mathcal{L}$, the Dirac adjoint transforms as $\bar{u}' = \bar{u} \mathcal{L}$, ensuring that the Dirac scalar product $\bar{u}_1 u_2$ is strictly Lorentz invariant. Evaluating this invariant product gives

$$\bar{u}_{\mathbf{p}\sigma}^{(b)} u_{\mathbf{p}\sigma'}^{(b')} = b \delta_{bb'} \delta_{\sigma\sigma'} = \pm \delta_{bb'} \delta_{\sigma\sigma'}. \quad (25.64)$$

Specifically, $\bar{u}_{\mathbf{p}\sigma}^{(+)} u_{\mathbf{p}\sigma}^{(+)} = +1$ and $\bar{u}_{\mathbf{p}\sigma}^{(-)} u_{\mathbf{p}\sigma}^{(-)} = -1$.

The set of four eigenspinors forms a complete basis in the four-dimensional spinor space:

$$\sum_{b=\pm} \sum_{\sigma=\uparrow, \downarrow} b u_{\mathbf{p}\sigma}^{(b)} \bar{u}_{\mathbf{p}\sigma}^{(b)} = \mathbf{1}_{4 \times 4}. \quad (25.65)$$

In covariant four-momentum notation ($\not{p} = \gamma^\mu p_\mu = \gamma^0 E_p/c - \boldsymbol{\gamma} \cdot \mathbf{p}$), the spinors satisfy the algebraic Dirac equations:

$$(\not{p} - mc)u_{\mathbf{p}\sigma}^{(+)} = 0, \quad \bar{u}_{\mathbf{p}\sigma}^{(+)}(\not{p} - mc) = 0, \quad (25.66)$$

$$(\not{p} + mc)u_{\mathbf{p}\sigma}^{(-)} = 0, \quad \bar{u}_{\mathbf{p}\sigma}^{(-)}(\not{p} + mc) = 0. \quad (25.67)$$

The corresponding energy projection operators are

$$\Lambda_+(\mathbf{p}) \equiv \sum_{\sigma} u_{\mathbf{p}\sigma}^{(+)} \bar{u}_{\mathbf{p}\sigma}^{(+)} = \frac{\not{p} + mc}{2mc}, \quad (25.68)$$

$$\Lambda_-(\mathbf{p}) \equiv -\sum_{\sigma} u_{\mathbf{p}\sigma}^{(-)} \bar{u}_{\mathbf{p}\sigma}^{(-)} = \frac{-\not{p} + mc}{2mc}. \quad (25.69)$$

25.5 Dirac Currents and the Gordon Decomposition

We now analyze the conserved currents in the Dirac theory, examine the Lorentz transformation properties of spinor bilinears, and investigate the Gordon decomposition into orbital (convective) and intrinsic spin-dipole components.

25.5.1 Covariant Bilinears and Probability Conservation

Let $\psi(\mathbf{r}, t)$ be an arbitrary solution of the Dirac equation. Under a Lorentz boost characterized by $S(\Lambda) = \mathcal{L}^{-1}$, the Dirac adjoint spinor $\bar{\psi} \equiv \psi^\dagger \gamma^0$ transforms as

$$\bar{\psi}'(\mathbf{r}', t') = \psi'^\dagger \gamma^0 = (\mathcal{L}^{-1} \psi)^\dagger \gamma^0 = \psi^\dagger (\mathcal{L}^{-1})^\dagger \gamma^0 = \psi^\dagger \gamma^0 \mathcal{L} = \bar{\psi}(\mathbf{r}, t) \mathcal{L}. \quad (25.70)$$

From the 16 Clifford algebra basis elements (25.27), we construct the five standard relativistic covariant bilinears:

Bilinear	Lorentz Transformation Property	Count
$\bar{\psi}\psi$	Scalar	1
$\bar{\psi}\gamma^\mu\psi$	Vector	4
$\bar{\psi}\sigma^{\mu\nu}\psi$	Rank-2 Antisymmetric Tensor	6
$\bar{\psi}\gamma^\mu\gamma_5\psi$	Axial Vector (Pseudovector)	4
$\bar{\psi}\gamma_5\psi$	Pseudoscalar	1

The vector current four-vector $j^\mu \equiv c\bar{\psi}\gamma^\mu\psi = (c\rho, \mathbf{j})$ consists of the local probability density

$$\rho(\mathbf{r}, t) = \psi^\dagger(\mathbf{r}, t)\psi(\mathbf{r}, t) = \sum_{a=1}^4 |\psi_a(\mathbf{r}, t)|^2 \geq 0, \quad (25.71)$$

and the spatial current density

$$\mathbf{j}(\mathbf{r}, t) = c\psi^\dagger(\mathbf{r}, t)\boldsymbol{\alpha}\psi(\mathbf{r}, t), \quad (25.72)$$

which satisfy the continuity equation $\partial_\mu j^\mu = \frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0$.

Unlike the Klein-Gordon scalar density (22.22), the Dirac density ρ is *strictly positive definite*, preserving the probability normalization $\int d^3r \psi^\dagger\psi = 1$ for both positive- and negative-energy states. However, this feature introduces a fundamental instability: because negative-energy states are normalized positively, nothing in single-particle quantum mechanics forbids a positive-energy electron from cascading down to negative-energy states via radiative photon emission. This catastrophe was ultimately resolved by Dirac's hole theory and second quantization.

25.5.2 The Velocity Operator and Zitterbewegung

In the Heisenberg picture, the time evolution of the position operator is governed by

$$\frac{d\mathbf{r}}{dt} = \frac{1}{i\hbar}[\mathbf{r}, H_D] = c\boldsymbol{\alpha}. \quad (25.73)$$

The instantaneous velocity operator is $\mathbf{v}_{\text{op}} = c\boldsymbol{\alpha}$. Because $\alpha_i^2 = \mathbf{1}$, the eigenvalues of each velocity component are strictly

$$\text{spec}(c\alpha_i) = \pm c. \quad (25.74)$$

An instantaneous measurement of any velocity component can only yield the speed of light c .

This apparent paradox arises because the operator $c\boldsymbol{\alpha}$ is odd: it couples positive- and negative-energy subspaces. The position operator decomposes into even and odd components:

$$\mathbf{r} = \mathbf{r}_{(+)} + \mathbf{r}_{(-)}. \quad (25.75)$$

Under free time evolution, the instantaneous position contains a rapidly oscillating cross-term (zitterbewegung):

$$\mathbf{r}(t) = \mathbf{r}(0) + \frac{c^2\mathbf{p}}{H_D}t + \frac{i\hbar c}{2H_D} \left(\boldsymbol{\alpha}(0) - \frac{c\mathbf{p}}{H_D} \right) e^{-2iH_D t/\hbar}. \quad (25.76)$$

The high-frequency trembling at $\omega \approx 2mc^2/\hbar$ has an amplitude on the order of the Compton wavelength $\lambda_c = \hbar/mc$. When wave packets are averaged over macroscopic times, the zitterbewegung dephases, leaving the classical group velocity $\langle \mathbf{v}_{\text{op}} \rangle = \frac{\mathbf{p}c^2}{E_p}$.

25.5.3 The Gordon Decomposition

To isolate the distinct physical origins of the probability current, we use the minimally coupled Dirac equation (25.35) and its adjoint (25.45). Writing

$$j_i = c\psi^\dagger \alpha_i \psi = c\bar{\psi} \gamma^i \psi, \quad (25.77)$$

and using the Dirac equation to replace $\bar{\psi}$ and ψ , we separate the current into a convective (orbital) part and an internal (spin-polarization) part:

$$\mathbf{j}(\mathbf{r}, t) = \mathbf{j}_{\text{conv}}(\mathbf{r}, t) + \mathbf{j}_{\text{int}}(\mathbf{r}, t), \quad \rho(\mathbf{r}, t) = \rho_{\text{conv}}(\mathbf{r}, t) + \rho_{\text{int}}(\mathbf{r}, t). \quad (25.78)$$

The convective components take the familiar Schrödinger-like form:

$$\mathbf{j}_{\text{conv}} = \frac{\hbar}{2im} (\bar{\psi} \nabla \psi - (\nabla \bar{\psi}) \psi) - \frac{e\mathbf{A}}{mc} \bar{\psi} \psi, \quad (25.79)$$

$$\rho_{\text{conv}} = \frac{i\hbar}{2mc^2} \left(\bar{\psi} \frac{\partial \psi}{\partial t} - \frac{\partial \bar{\psi}}{\partial t} \psi \right) - \frac{e\Phi}{mc^2} \bar{\psi} \psi. \quad (25.80)$$

The internal components are governed by spatial gradients of the spin magnetization $\mathbf{M}$ and electric polarization $\mathbf{P}$:

$$\mathbf{j}_{\text{int}} = c\nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}, \quad (25.81)$$

$$\rho_{\text{int}} = -\nabla \cdot \mathbf{P}, \quad (25.82)$$

where

$$\mathbf{M} = \frac{\hbar}{2mc} \bar{\psi} \boldsymbol{\sigma} \psi, \quad \mathbf{P} = -\frac{i\hbar}{2mc} \bar{\psi} \boldsymbol{\alpha} \psi. \quad (25.83)$$

For an electron of charge e , $e\mathbf{M}$ represents the intrinsic magnetic dipole moment density generated by spin, with gyromagnetic ratio $g = 2$. Concurrently, $e\mathbf{P}$ represents the electric

dipole moment induced when a magnetic dipole moves relativistically relative to the laboratory frame. In covariant notation, the Gordon decomposition condenses to

$$j^\mu = \frac{i\hbar}{2m} (\bar{\psi} \partial^\mu \psi - (\partial^\mu \bar{\psi}) \psi) - \frac{e}{mc} A^\mu \bar{\psi} \psi + \frac{\hbar}{2m} \partial_\nu (\bar{\psi} \sigma^{\mu\nu} \psi). \quad (25.84)$$

Both convective and internal currents are conserved separately:

$$\partial_\mu j_{\text{conv}}^\mu = 0, \quad \partial_\mu j_{\text{int}}^\mu = 0. \quad (25.85)$$

For an unconfined plane wave, $\mathbf{M}$ and $\mathbf{P}$ are spatially homogeneous, so $\mathbf{j}_{\text{int}} = \mathbf{0}$ and $\rho_{\text{int}} = 0$. For a localized wave packet, integrating the total current over all space causes the internal magnetization boundary term $\int d^3r \nabla \times \mathbf{M} = \mathbf{0}$ to vanish identically, leaving the macroscopic transport current

$$\mathbf{J} \equiv \int d^3r \mathbf{j}(\mathbf{r}, t) = \int d^3r \mathbf{j}_{\text{conv}}(\mathbf{r}, t) = \sum_\sigma \int \frac{d^3p}{(2\pi\hbar)^3} |a'_{\mathbf{p}\sigma}|^2 \frac{\mathbf{p}c^2}{E_p} = \langle \mathbf{v}_g \rangle, \quad (25.86)$$

in exact concordance with relativistic classical particle kinematics.

25.6 Nonrelativistic Limit: Fine-Structure Hamiltonian, Darwin Term, and Thomas Precession

We now perform a systematic expansion of the coupled two-component Dirac equations (25.37) to order $\mathcal{O}(v^2/c^2)$ (or order $1/c^2$). This reveals the complete relativistic fine-structure Hamiltonian, resolves the non-unitary norm leakage via a Foldy-Wouthuysen canonical transformation, and elucidates the physical origins of the Darwin contact term and the Thomas precession in the spin-orbit interaction.

25.6.1 Iterative Solution for the Small Component

From the lower component of the Dirac equation (25.37), the exact relation between χ and φ is

$$\chi = \frac{1}{2mc} (\mathbf{\Pi} \cdot \boldsymbol{\tau}) \varphi - \frac{1}{2mc^2} \left(i\hbar \frac{\partial}{\partial t} - mc^2 - e\Phi \right) \chi, \quad (25.87)$$

where $\mathbf{\Pi} = \frac{\hbar}{i} \nabla - \frac{e}{c} \mathbf{A} = \mathbf{p} - \frac{e}{c} \mathbf{A}$. Iterating Eq. (25.87) once to include the next order in $1/c$ yields

$$\chi = \frac{1}{2mc} (\mathbf{\Pi} \cdot \boldsymbol{\tau}) \varphi - \frac{1}{4m^2 c^3} \left(i\hbar \frac{\partial}{\partial t} - mc^2 - e\Phi \right) (\mathbf{\Pi} \cdot \boldsymbol{\tau}) \varphi + \mathcal{O}\left(\frac{1}{c^4}\right). \quad (25.88)$$

Substituting Eq. (25.88) into the upper equation of (25.37) gives

$$\left(i\hbar \frac{\partial}{\partial t} - mc^2 - e\Phi \right) \varphi = \frac{1}{2m} (\mathbf{\Pi} \cdot \boldsymbol{\tau})^2 \varphi - \frac{1}{4m^2 c^2} (\mathbf{\Pi} \cdot \boldsymbol{\tau}) \left(i\hbar \frac{\partial}{\partial t} - mc^2 - e\Phi \right) (\mathbf{\Pi} \cdot \boldsymbol{\tau}) \varphi. \quad (25.89)$$

To leading order, $(i\hbar \partial_t - mc^2 - e\Phi) \varphi \approx \frac{p^2}{2m} \varphi$. Commuting the operators past one another with $\boldsymbol{\mathcal{E}} = -\nabla \Phi - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t}$, Eq. (25.89) evaluates to

$$\left(i\hbar \frac{\partial}{\partial t} - mc^2 - e\Phi \right) \varphi = \left[\frac{\mathbf{\Pi}^2}{2m} - \frac{e\hbar}{2mc} \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{H}} - \frac{p^4}{8m^3 c^2} - \frac{e\hbar}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\boldsymbol{\mathcal{E}} \times \mathbf{p}) - \frac{ie\hbar}{4m^2 c^2} \mathbf{p} \cdot \boldsymbol{\mathcal{E}} \right] \varphi. \quad (25.90)$$

The final term in Eq. (25.90), $-\frac{ie\hbar}{4m^2 c^2} \mathbf{p} \cdot \boldsymbol{\mathcal{E}}$, is manifestly *non-Hermitian*.

25.6.2 Norm Conservation and the Foldy-Wouthuysen Transformation

A non-Hermitian operator violates probability conservation. The physical origin of this non-Hermiticity lies in the definition of the two-component spinor φ : the total Dirac probability is normalized over both components:

$$\int d^3r \psi^\dagger \psi = \int d^3r (\varphi^\dagger \varphi + \chi^\dagger \chi) = 1. \quad (25.91)$$

To lowest order, $\chi \approx \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{2mc} \varphi$, so that $\chi^\dagger \chi \approx \varphi^\dagger \frac{p^2}{4m^2 c^2} \varphi$. Consequently,

$$\int d^3r \varphi^\dagger \left(\mathbf{1} + \frac{p^2}{4m^2 c^2} \right) \varphi = \int d^3r \left[\left(\mathbf{1} + \frac{p^2}{8m^2 c^2} \right) \varphi \right]^\dagger \left[\left(\mathbf{1} + \frac{p^2}{8m^2 c^2} \right) \varphi \right] = 1 + \mathcal{O}\left(\frac{1}{c^4}\right). \quad (25.92)$$

To preserve unitarity in the upper two-component subspace, we perform a canonical transformation (the $\mathcal{O}(c^{-2})$ Foldy-Wouthuysen transformation) to the normalized physical wave function:

$$\Psi(\mathbf{r}, t) \equiv \left(\mathbf{1} + \frac{p^2}{8m^2 c^2} \right) \varphi(\mathbf{r}, t), \quad \int d^3r \Psi^\dagger \Psi = 1. \quad (25.93)$$

Multiplying Eq. (25.90) on the left by $(\mathbf{1} + \frac{p^2}{8m^2 c^2})$ and expressing the result in terms of Ψ , the commutator with the electrostatic potential produces

$$\frac{1}{8m^2 c^2} [p^2, e\Phi] - \frac{ie\hbar}{4m^2 c^2} \mathbf{p} \cdot \boldsymbol{\mathcal{E}} = \frac{\hbar^2}{8m^2 c^2} \nabla^2 (e\Phi). \quad (25.94)$$

The non-Hermitian term is canceled exactly, yielding the hermitian Pauli-Schrödinger equation:

$$\begin{aligned} i\hbar \frac{\partial \Psi}{\partial t} &= H_{\text{eff}} \Psi, \\ H_{\text{eff}} &= mc^2 + \frac{1}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2 - \frac{p^4}{8m^3 c^2} - \frac{e\hbar}{2mc} \boldsymbol{\sigma} \cdot \boldsymbol{\mathcal{H}} \\ &\quad - \frac{e\hbar}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\boldsymbol{\mathcal{E}} \times \mathbf{p}) + e\Phi + \frac{\hbar^2}{8m^2 c^2} \nabla^2 (e\Phi). \end{aligned} \quad (25.95)$$

25.6.3 Physical Identification of Relativistic Corrections

Equation (25.95) contains three distinct relativistic corrections of order $\mathcal{O}(v^2/c^2)$:

1. **Relativistic Mass Correction:** The term $-\frac{p^4}{8m^3 c^2}$ represents the first relativistic correction to the Newtonian kinetic energy, stemming from the expansion of $\sqrt{p^2 c^2 + m^2 c^4} - mc^2$.
2. **Spin-Orbit Interaction:** The term $-\frac{e\hbar}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\boldsymbol{\mathcal{E}} \times \mathbf{p})$ describes the coupling of the electron spin to the electric field. In a spherically symmetric potential $e\Phi(r) = V(r)$, $\boldsymbol{\mathcal{E}} = -\frac{\mathbf{r}}{r} \frac{dV}{dr}$, which gives

$$H_{\text{SO}} = \frac{1}{2m^2 c^2} \frac{1}{r} \frac{dV}{dr} \mathbf{L} \cdot \mathbf{S}, \quad \mathbf{S} = \frac{\hbar}{2} \boldsymbol{\sigma}, \quad \mathbf{L} = \mathbf{r} \times \mathbf{p}. \quad (25.96)$$

3. **The Darwin Term:** The term $H_{\text{Darwin}} = \frac{\hbar^2}{8m^2 c^2} \nabla^2 (e\Phi)$ is a purely quantum-relativistic contact interaction. By Poisson's equation $\nabla^2 \Phi = -4\pi \rho_{\text{charge}}$, the Darwin term vanishes in vacuum and acts exclusively at the source. For a nuclear point charge $e\Phi = -Ze^2/r$,

$$H_{\text{Darwin}} = \frac{\pi \hbar^2 Z e^2}{2m^2 c^2} \delta(\mathbf{r}). \quad (25.97)$$

Because only s -wave ($l = 0$) wave functions are non-vanishing at the origin ($\psi(0) \neq 0$), the Darwin term shifts the energies of s -states upward, playing a decisive role in lifting the degeneracy between $2S_{1/2}$ and $2P_{1/2}$ states in the nonrelativistic atom.

The Darwin term admits an intuitive physical interpretation in terms of *zitterbewegung*: the electron's rapid microscopic trembling over its Compton wavelength $\delta r \sim \lambda_c = \hbar/mc$ samples the potential averaged over a sphere of radius λ_c :

$$\overline{V(\mathbf{r} + \delta \mathbf{r})} \approx V(\mathbf{r}) + \frac{1}{6} \langle (\delta \mathbf{r})^2 \rangle \nabla^2 V(\mathbf{r}) \approx V(\mathbf{r}) + \frac{\hbar^2}{6m^2 c^2} \nabla^2 V(\mathbf{r}), \quad (25.98)$$

reproducing the functional form and order of magnitude of the Darwin term.

25.6.4 Thomas Precession and the Spin-Orbit Coupling

To understand the factor of 1/2 in the spin-orbit interaction, we decompose the interaction into its laboratory and rest-frame parts:

$$-\frac{e\hbar}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\boldsymbol{\mathcal{E}} \times \mathbf{p}) = -\frac{e\hbar}{2mc} \boldsymbol{\sigma} \cdot \left[\frac{\boldsymbol{\mathcal{E}} \times \mathbf{p}}{mc} \right] + \frac{e\hbar}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\boldsymbol{\mathcal{E}} \times \mathbf{p}). \quad (25.99)$$

To order v/c , a particle moving with velocity $\mathbf{v} = \mathbf{p}/m$ through an electric field $\boldsymbol{\mathcal{E}}$ experiences an effective magnetic field in its rest frame:

$$\boldsymbol{\mathcal{H}}_{\text{rest}} = -\frac{\mathbf{v}}{c} \times \boldsymbol{\mathcal{E}} = \frac{\boldsymbol{\mathcal{E}} \times \mathbf{p}}{mc}. \quad (25.100)$$

The interaction energy of the intrinsic magnetic dipole moment $\boldsymbol{\mu}_S = \frac{e\hbar}{2mc} \boldsymbol{\sigma}$ ($g = 2$) with this motional magnetic field is $-\boldsymbol{\mu}_S \cdot \boldsymbol{\mathcal{H}}_{\text{rest}}$, which would predict twice the correct spin-orbit splitting.

The second term in Eq. (25.99) represents the kinematic correction known as *Thomas precession*. Because successive non-collinear Lorentz boosts do not commute, a particle accelerating along a curved orbit experiences a continuous Wigner rotation of its rest frame relative to the laboratory frame. For an acceleration $\dot{\mathbf{v}}$ and velocity $\mathbf{v} \ll c$, the Thomas precession frequency is

$$\boldsymbol{\omega}_T = \frac{1}{2c^2} \dot{\mathbf{v}} \times \mathbf{v}. \quad (25.101)$$

This spatial precession contributes an additional kinematic energy $\Delta E = \mathbf{S} \cdot \boldsymbol{\omega}_T$. Using Newton's second law $m\dot{\mathbf{v}} = e\boldsymbol{\mathcal{E}}$,

$$\Delta E = \mathbf{S} \cdot \left(\frac{e\boldsymbol{\mathcal{E}} \times \mathbf{v}}{2mc^2} \right) = \frac{e\hbar}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\boldsymbol{\mathcal{E}} \times \mathbf{p}). \quad (25.102)$$

The Thomas precession subtracts exactly half of the motional magnetic interaction, providing a complete physical explanation of the Landé factor in the atomic fine structure.

25.7 The Second-Order Dirac Equation and Pauli Moment Coupling

In many relativistic calculations—particularly in central Coulomb scattering, semiclassical approximations, and the study of electromagnetic interactions—it is advantageous to cast the Dirac equation in a form that is second order in spacetime derivatives, making its connection to the Klein-Gordon equation explicit.

25.7.1 Derivation of the Second-Order Equation

We begin with the first-order Dirac equation in Hamiltonian form:

$$\beta \left(i\hbar \frac{\partial}{\partial t} - H \right) \psi(\mathbf{r}, t) = 0, \quad (25.103)$$

where the Hamiltonian in an external electromagnetic field $(\Phi, \mathbf{A})$ is

$$H = c\boldsymbol{\alpha} \cdot \boldsymbol{\Pi} + \beta mc^2 + e\Phi, \quad \boldsymbol{\Pi} \equiv \frac{\hbar}{i}\boldsymbol{\nabla} - \frac{e}{c}\mathbf{A}. \quad (25.104)$$

Equation (25.103) can be written in terms of the first-order differential operator

$$\mathcal{D}_1 \equiv \beta \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) - c\beta\boldsymbol{\alpha} \cdot \boldsymbol{\Pi} - mc^2. \quad (25.105)$$

We now define the conjugate operator P :

$$P \equiv \frac{\beta \left(i\hbar \frac{\partial}{\partial t} - H \right) + 2mc^2}{2mc^2} = \frac{1}{2mc^2} \left[\beta \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) - c\beta\boldsymbol{\alpha} \cdot \boldsymbol{\Pi} + mc^2 \right]. \quad (25.106)$$

Acting on Eq. (25.103) from the left with $2mc^2 P$ gives

$$\left[\beta \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) - c\beta\boldsymbol{\alpha} \cdot \boldsymbol{\Pi} + mc^2 \right] \left[\beta \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) - c\beta\boldsymbol{\alpha} \cdot \boldsymbol{\Pi} - mc^2 \right] \psi = 0. \quad (25.107)$$

Expanding this product and using the Dirac matrix anticommutation relations $\{\beta, \boldsymbol{\alpha}\} = 0$, $\beta^2 = \mathbf{1}$, we find

$$\left[\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right)^2 - c^2(\boldsymbol{\alpha} \cdot \boldsymbol{\Pi})^2 + c\beta \left[\boldsymbol{\alpha} \cdot \boldsymbol{\Pi}, \beta \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \right] - m^2 c^4 \right] \psi = 0. \quad (25.108)$$

Using the operator identities

$$(\boldsymbol{\alpha} \cdot \boldsymbol{\Pi})^2 = \boldsymbol{\Pi}^2 - \frac{e\hbar}{c}\boldsymbol{\Sigma} \cdot \boldsymbol{\mathcal{H}}, \quad (25.109)$$

where $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}$, and

$$\left[\boldsymbol{\Pi}, i\hbar \frac{\partial}{\partial t} - e\Phi \right] = -i\hbar e\boldsymbol{\mathcal{E}}, \quad (25.110)$$

where $\boldsymbol{\mathcal{E}} = -\boldsymbol{\nabla}\Phi - \frac{1}{c}\frac{\partial\mathbf{A}}{\partial t}$ is the electric field and $\boldsymbol{\mathcal{H}} = \boldsymbol{\nabla} \times \mathbf{A}$ is the magnetic field, the cross-commutator becomes

$$c\beta \left[\boldsymbol{\alpha} \cdot \boldsymbol{\Pi}, \beta \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \right] = -c\boldsymbol{\alpha} \cdot \left[\boldsymbol{\Pi}, i\hbar \frac{\partial}{\partial t} - e\Phi \right] = i\hbar e c \boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{E}}. \quad (25.111)$$

Dividing by c^2 , we arrive at the *second-order Dirac equation*:

$$\left[\frac{1}{c^2} \left(i\hbar \frac{\partial}{\partial t} - e\Phi \right)^2 - \left(\frac{\hbar}{i}\boldsymbol{\nabla} - \frac{e}{c}\mathbf{A} \right)^2 - m^2 c^2 + \frac{e\hbar}{c} (\boldsymbol{\Sigma} \cdot \boldsymbol{\mathcal{H}} - i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{E}}) \right] \psi = 0. \quad (25.112)$$

25.7.2 Relativistic Interpretation and Pauli Coupling

Equation (25.112) is formally identical to the minimally coupled Klein-Gordon equation for a charged scalar particle, supplemented by the non-minimal Pauli coupling term:

$$H_{\text{Pauli}} = \frac{e\hbar}{c} (\boldsymbol{\Sigma} \cdot \boldsymbol{\mathcal{H}} - i\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{E}}). \quad (25.113)$$

In relativistic covariant notation, this term is simply the contraction of the electromagnetic field strength tensor $F_{\mu\nu}$ with the Dirac spin tensor $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$:

$$\left[D_\mu D^\mu + \left(\frac{mc}{\hbar} \right)^2 + \frac{e}{2\hbar c} \sigma^{\mu\nu} F_{\mu\nu} \right] \psi = 0, \quad (25.114)$$

where $D_\mu = \partial_\mu + \frac{ie}{\hbar c} A_\mu$.

- The magnetic term, $\frac{e\hbar}{c}\boldsymbol{\Sigma} \cdot \boldsymbol{\mathcal{H}}$, represents the intrinsic magnetic dipole interaction $-\boldsymbol{\mu} \cdot \boldsymbol{\mathcal{H}}$, with gyromagnetic ratio $g = 2$.
- The anti-Hermitian electric term, $-i\frac{e\hbar}{c}\boldsymbol{\alpha} \cdot \boldsymbol{\mathcal{E}}$, reflects the relativistic transformation of the magnetic dipole moment in a moving frame, generating spin-orbit and Thomas precession effects.

25.7.3 Projection onto the First-Order Dirac Space

Because Eq. (25.112) is of second order in time, it admits twice as many linearly independent solutions as the first-order Dirac equation. While every solution of the first-order Dirac equation satisfies Eq. (25.112), the converse does not hold.

However, the operator P defined in Eq. (25.106) serves as an exact projector:

1. Because P is a polynomial in $\beta(i\hbar\partial_t - H)$, it commutes with the Dirac operator:

$$\left[P, \beta \left(i\hbar \frac{\partial}{\partial t} - H \right) \right] = 0. \quad (25.115)$$

The second-order equation can therefore be written as

$$P\beta \left(i\hbar \frac{\partial}{\partial t} - H \right) \psi = \beta \left(i\hbar \frac{\partial}{\partial t} - H \right) (P\psi) = 0. \quad (25.116)$$

2. Consequently, if ψ is any solution to the second-order equation (25.112), the state

$$\varphi \equiv P\psi \quad (25.117)$$

is guaranteed to be a genuine solution to the physical first-order Dirac equation:

$$i\hbar \frac{\partial \varphi}{\partial t} = H\varphi. \quad (25.118)$$

3. If ψ already satisfies the first-order Dirac equation, then $\beta(i\hbar\partial_t - H)\psi = 0$, which immediately yields

$$P\psi = \frac{0 + 2mc^2}{2mc^2} \psi = \psi. \quad (25.119)$$

Thus, $P^2 = P$ on the solution space, confirming that P projects the broader solution space of the second-order equation onto the physical subspace of the Dirac equation.

25.8 The Dirac Hydrogen Atom: Exact Spectrum, Fine Structure, and Hidden Symmetry

A crowning triumph of relativistic quantum mechanics was Dirac's exact theoretical explanation of the fine-structure spectrum of the hydrogen atom, removing the ad hoc spin-orbit and Thomas precession corrections of early quantum theory.

Consider an electron of charge $e = -|e|$ moving in the static attractive central Coulomb potential of a nucleus of charge Ze :

$$e\Phi(r) = -\frac{Ze^2}{r}. \quad (25.120)$$

The stationary first-order Dirac Hamiltonian is

$$H = c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta mc^2 - \frac{Ze^2}{r}. \quad (25.121)$$

Rather than solving the coupled first-order radial differential equations directly, we employ the Martin-Glauber operator method³ based on the second-order Dirac equation (25.112), which reveals the underlying algebraic structure with exceptional elegance.

25.8.1 Reduction to a Radial Klein-Gordon Problem

For a stationary eigenstate of energy E , setting $\mathbf{A} = \mathbf{0}$ and $\mathcal{E} = -\nabla\Phi = -\frac{Ze}{r^2}\hat{\mathbf{r}}$ in the second-order Dirac equation (25.112) yields

$$\left[\frac{1}{c^2} \left(E + \frac{Ze^2}{r} \right)^2 - \left(\frac{\hbar}{i} \nabla \right)^2 - m^2 c^2 + \frac{i\hbar Ze^2}{r^2 c} \alpha_r \right] \psi = 0, \quad (25.122)$$

where $\alpha_r \equiv \boldsymbol{\alpha} \cdot \hat{\mathbf{r}}$. Decomposing the Laplacian into radial and orbital angular momentum operators,

$$-\hbar^2 \nabla^2 = -\frac{\hbar^2}{r} \frac{\partial^2}{\partial r^2} r + \frac{\mathbf{L}^2}{r^2}, \quad (25.123)$$

Eq. (25.122) becomes

$$\left[\frac{E^2 - m^2 c^4}{c^2} + \frac{2EZe^2}{rc^2} + \frac{\hbar^2}{r} \frac{\partial^2}{\partial r^2} r - \frac{\mathbf{L}^2 - (Ze^2/c)^2 - i\hbar(Ze^2/c)\alpha_r}{r^2} \right] \psi = 0. \quad (25.124)$$

25.8.2 The Dirac Angular Momentum Operator K

In a relativistic central field, neither orbital angular momentum $\mathbf{L}$ nor electron spin $\mathbf{S} = \frac{1}{2}\hbar\boldsymbol{\sigma}$ is separately conserved. Only the total angular momentum

$$\mathbf{J} = \mathbf{L} + \frac{\hbar}{2}\boldsymbol{\Sigma}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix} \quad (25.125)$$

commutes with the Dirac Hamiltonian: $[\mathbf{J}, H] = 0$.

To completely classify the states, Dirac introduced the invariant operator

$$K \equiv \beta \left(\mathbf{1} + \frac{\boldsymbol{\Sigma} \cdot \mathbf{L}}{\hbar} \right) = \beta \left(\frac{\boldsymbol{\Sigma} \cdot \mathbf{J}}{\hbar} + \frac{1}{2} \right). \quad (25.126)$$

Using the fundamental operator relations $\boldsymbol{\alpha} = \gamma_5 \boldsymbol{\Sigma}$ and $\{\beta, \boldsymbol{\alpha}\} = 0$, one readily verifies that K commutes with the basic central-field scalars:

$$[K, \boldsymbol{\alpha} \cdot \mathbf{p}] = 0, \quad [K, \boldsymbol{\alpha} \cdot \mathbf{r}] = 0, \quad [K, r] = 0, \quad [K, \mathbf{J}] = 0. \quad (25.127)$$

Consequently, K is a rigorous constant of the motion:

$$[K, H] = 0. \quad (25.128)$$

The square of K is intimately tied to total angular momentum:

$$K^2 = \mathbf{1} + 2 \frac{\boldsymbol{\Sigma} \cdot \mathbf{L}}{\hbar} + \left(\frac{\boldsymbol{\Sigma} \cdot \mathbf{L}}{\hbar} \right)^2 = \frac{\mathbf{J}^2}{\hbar^2} + \frac{1}{4}. \quad (25.129)$$

Because $\mathbf{J}^2$ has eigenvalues $j(j+1)\hbar^2$, the eigenvalues k of K satisfy

$$k^2 = j(j+1) + \frac{1}{4} = \left(j + \frac{1}{2} \right)^2 \implies j = |k| - \frac{1}{2}. \quad (25.130)$$

Moreover, since K anticommutes with the chirality matrix γ_5 :

$$\{K, \gamma_5\} = 0, \quad (25.131)$$

if $|k\rangle$ is an eigenstate of K with eigenvalue k , then $\gamma_5|k\rangle$ is an eigenstate with eigenvalue $-k$. Thus, the spectrum of K consists of all non-zero integers:

$$k = \pm 1, \pm 2, \pm 3, \dots \quad (j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots). \quad (25.132)$$

³P. C. Martin and R. J. Glauber, *Phys. Rev.* **109**, 1307 (1958).

25.8.3 The Martin-Glauber Auxiliary Operator Λ

To diagonalize the effective centrifugal term in Eq. (25.124), we introduce the dimensionless operator

$$\Lambda \equiv -\beta K - i \frac{Ze^2}{\hbar c} \alpha_r. \quad (25.133)$$

Using $\{\beta, \alpha_r\} = 0$ and $[K, \alpha_r] = 0$, we find

$$[\Lambda, K] = 0, \quad [\Lambda, \mathbf{J}] = 0, \quad (25.134)$$

and its square is

$$\Lambda^2 = K^2 - \left(\frac{Ze^2}{\hbar c} \right)^2 = \left(j + \frac{1}{2} \right)^2 - (Z\alpha)^2, \quad (25.135)$$

where $\alpha \equiv e^2/(\hbar c) \approx 1/137.036$ is the fine-structure constant.

Direct multiplication demonstrates that Λ generates the effective angular barrier in Eq. (25.124):

$$\hbar^2 \Lambda(\Lambda + 1) = \mathbf{L}^2 - \left(\frac{Ze^2}{c} \right)^2 - i\hbar \left(\frac{Ze^2}{c} \right) \alpha_r. \quad (25.136)$$

Equation (25.124) can therefore be rewritten in the extraordinarily compact form:

$$\left[\frac{E^2 - m^2 c^4}{\hbar^2 c^2} + \frac{2EZ\alpha}{\hbar c r} + \frac{1}{r} \frac{\partial^2}{\partial r^2} r - \frac{\Lambda(\Lambda + 1)}{r^2} \right] \psi(\mathbf{r}) = 0. \quad (25.137)$$

25.8.4 Energy Eigenvalues: The Sommerfeld-Dirac Formula

Within a simultaneous eigenspace of K and Λ , the operator Λ assumes the scalar eigenvalue $\pm\lambda$, where

$$\lambda \equiv +\sqrt{k^2 - (Z\alpha)^2} = +\sqrt{\left(j + \frac{1}{2} \right)^2 - (Z\alpha)^2}. \quad (25.138)$$

Setting $\Lambda(\Lambda + 1) = l'(l' + 1)$, regularity of the wave function at $r \rightarrow 0$ selects the positive roots:

$$\begin{aligned} \text{For } \Lambda = +\lambda: \quad l' &= \lambda, \\ \text{For } \Lambda = -\lambda: \quad l' &= \lambda - 1. \end{aligned} \quad (25.139)$$

Mapping Eq. (25.137) directly onto the radial Klein-Gordon Coulomb equation, the bound-state energy spectrum is given by

$$E = \frac{mc^2}{\left[1 + \left(\frac{Z\alpha}{n'} \right)^2 \right]^{1/2}}, \quad n' = l' + 1 + \nu, \quad (\nu = 0, 1, 2, \dots). \quad (25.140)$$

Introducing the nonrelativistic principal quantum number $n \equiv \nu + |k| = \nu + j + 1/2$, the effective quantum number n' becomes

$$n' = n - |k| + \lambda = n - \left(j + \frac{1}{2} \right) + \sqrt{\left(j + \frac{1}{2} \right)^2 - (Z\alpha)^2}. \quad (25.141)$$

This yields the celebrated *Sommerfeld-Dirac fine-structure formula*:

$$E_{nj} = mc^2 \left\{ 1 + \frac{(Z\alpha)^2}{\left[n - \left(j + \frac{1}{2} \right) + \sqrt{\left(j + \frac{1}{2} \right)^2 - (Z\alpha)^2} \right]^2} \right\}^{-1/2}. \quad (25.142)$$

To ensure that the ground state ($j = 1/2$) possesses a real-valued energy and a regular radial wave function, the nuclear charge must satisfy

$$Z\alpha < 1 \implies Z < 137. \quad (25.143)$$

25.8.5 Physical Classification and Fine-Structure Splitting

While the orbital angular momentum $\mathbf{L}^2$ does not commute with H , the states can be labeled by the nonrelativistic orbital angular momentum quantum number l . Using the operator identity $K(K - \beta) = \mathbf{L}^2/\hbar^2$ and taking $\beta \rightarrow +1$ in the nonrelativistic limit, l is determined by $k(k - 1) = l(l + 1)$:

$$\begin{aligned} l &= k - 1 = j - \frac{1}{2} \quad \text{for } k > 0 \quad (\text{spin and orbit parallel, } j = l + \tfrac{1}{2}), \\ l &= |k| = j + \frac{1}{2} \quad \text{for } k < 0 \quad (\text{spin and orbit antiparallel, } j = l - \tfrac{1}{2}). \end{aligned} \quad (25.144)$$

Expanding Eq. (25.142) in powers of $Z\alpha \ll 1$:

$$E_{nj} \approx mc^2 - \frac{m(Z\alpha)^2 c^2}{2n^2} \left[1 + \frac{(Z\alpha)^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] + \mathcal{O}((Z\alpha)^6). \quad (25.145)$$

Remarkably, the energy levels depend exclusively on n and j , but are independent of l .

- States with differing j for the same n are split: for example, the $2P_{3/2}$ ($j = 3/2$) and $2P_{1/2}$ ($j = 1/2$) levels are separated by $\Delta E_{\text{FS}} = \frac{mc^2(Z\alpha)^4}{32} \approx 4.53 \times 10^{-5} \text{ eV}$ (10.9 GHz).
- States sharing the same n and j but opposite parities (e.g., $2S_{1/2}$ and $2P_{1/2}$) are strictly degenerate in the single-particle Dirac theory.

25.8.6 Accidental Degeneracy and the Johnson-Lippmann Symmetry Operator

The unexpected twofold degeneracy between states of opposite $k = \pm(j + 1/2)$ is governed by a hidden dynamical symmetry. Johnson and Lippmann⁴ identified the invariant operator:

$$B \equiv \gamma_5 \left(\Lambda + \frac{KH}{mc^2} \right). \quad (25.146)$$

B is Hermitian, commutes with the Hamiltonian, and anticommutes with K :

$$[B, H] = 0, \quad \{B, K\} = 0. \quad (25.147)$$

Consequently, if $|k, E\rangle$ is an energy eigenstate with K -eigenvalue k , the transformed state $B|k, E\rangle$ is a distinct state with the identical energy E , but with inverted eigenvalue $-k$:

$$K(B|k, E\rangle) = -BK|k, E\rangle = -k(B|k, E\rangle). \quad (25.148)$$

The operator B explicitly maps the $2S_{1/2}$ state into the $2P_{1/2}$ state, and the $3P_{3/2}$ state into the $3D_{3/2}$ state, establishing the algebraic origin of their degeneracy.

For the state of maximum $j = n - 1/2$ (such as $1S_{1/2}, 2P_{3/2}, 3D_{5/2}$), $B|k, E\rangle \equiv 0$ identically, which explains why these extreme states remain uniquely non-degenerate.

25.9 Hyperfine Structure and the 21-cm Hydrogen Line

The exact energy levels of the Dirac hydrogen atom undergo two critical physical corrections:

1. *Quantum Electrodynamical (QED) Corrections:* The interaction of the bound electron with quantum vacuum fluctuations of the electromagnetic radiation field lifts the accidental degeneracy between states of opposite parity (such as $2S_{1/2}$ and $2P_{1/2}$), producing the *Lamb shift*.

⁴M. H. Johnson and B. A. Lippmann, *Phys. Rev.* **78**, 329(A) (1950).

2. *Hyperfine Structure:* The magnetic dipole interaction between the electron spin and the intrinsic magnetic dipole moment of the proton splits each fine-structure level into hyperfine components labeled by total angular momentum $\mathbf{F} = \mathbf{J} + \mathbf{I}$.

25.9.1 The Nuclear Magnetic Dipole Interaction

The proton possesses intrinsic spin $\mathbf{I} = \frac{1}{2}\hbar\boldsymbol{\sigma}_p$ and a magnetic dipole moment

$$\mathbf{m}_p = g_p \frac{|e|\hbar}{4Mc} \boldsymbol{\sigma}_p = \frac{1}{2} g_p \mu_N \boldsymbol{\sigma}_p, \quad (25.149)$$

where M is the proton mass, $\mu_N \equiv \frac{|e|\hbar}{2Mc}$ is the nuclear magneton, and $g_p \approx 5.58569$ is the proton Landé g -factor.

Assuming the proton is localized at the coordinate origin, its vector potential is

$$\mathbf{A}(\mathbf{r}) = \frac{\mathbf{m}_p \times \mathbf{r}}{r^3} = -\mathbf{m}_p \times \nabla \left(\frac{1}{r} \right) = -\frac{1}{2} g_p \mu_N (\boldsymbol{\sigma}_p \times \nabla) \frac{1}{r}. \quad (25.150)$$

The corresponding magnetic field $\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r})$ is

$$\mathbf{B}(\mathbf{r}) = -\frac{1}{2} g_p \mu_N [\nabla \times (\boldsymbol{\sigma}_p \times \nabla)] \frac{1}{r} = -\frac{1}{2} g_p \mu_N [\boldsymbol{\sigma}_p \nabla^2 - (\boldsymbol{\sigma}_p \cdot \nabla) \nabla] \frac{1}{r}. \quad (25.151)$$

The interaction Hamiltonian coupling the electron's spin magnetic moment $\boldsymbol{\mu}_e = -\frac{|e|\hbar}{2mc} \boldsymbol{\sigma} = -\mu_B \boldsymbol{\sigma}$ to the nuclear field $\mathbf{B}(\mathbf{r})$ is

$$H_{\text{hfs}} = -\boldsymbol{\mu}_e \cdot \mathbf{B}(\mathbf{r}) = \mu_B \boldsymbol{\sigma} \cdot \mathbf{B}(\mathbf{r}) = -\frac{1}{2} g_p \mu_B \mu_N [(\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p) \nabla^2 - (\boldsymbol{\sigma} \cdot \nabla)(\boldsymbol{\sigma}_p \cdot \nabla)] \frac{1}{r}, \quad (25.152)$$

where $\mu_B = \frac{|e|\hbar}{2mc}$ is the Bohr magneton.

25.9.2 Fermi Contact Interaction in s -States

To first order in perturbation theory, the energy shift of an atomic state with spatial wave function $\psi(\mathbf{r})$ is

$$\Delta E_{\text{hfs}} = \langle H_{\text{hfs}} \rangle = -\frac{1}{2} g_p \mu_B \mu_N \int d^3r |\psi(\mathbf{r})|^2 [(\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p) \nabla^2 - \langle (\boldsymbol{\sigma} \cdot \nabla)(\boldsymbol{\sigma}_p \cdot \nabla) \rangle] \frac{1}{r}. \quad (25.153)$$

For any spherically symmetric s -state ($l = 0$), the spatial distribution is isotropic, so the directional derivative tensor contracts isotropically:

$$\langle (\boldsymbol{\sigma} \cdot \nabla)(\boldsymbol{\sigma}_p \cdot \nabla) \rangle_{l=0} = \frac{1}{3} \langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle \nabla^2. \quad (25.154)$$

Using the Poisson identity $\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^3(\mathbf{r})$, the bracketed differential operator reduces to the celebrated *Fermi contact term*:

$$\begin{aligned} \Delta E_{\text{hfs}} &= -\frac{1}{3} g_p \mu_B \mu_N \langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle \int d^3r |\psi(\mathbf{r})|^2 \nabla^2 \left(\frac{1}{r} \right) \\ &= \frac{4\pi}{3} g_p \mu_B \mu_N |\psi(0)|^2 \langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle. \end{aligned} \quad (25.155)$$

Because only s -electrons have non-vanishing probability density at the nucleus ($r = 0$), the Fermi contact term is the sole contribution to hyperfine structure for $l = 0$.

For a nonrelativistic hydrogenic ns -orbital,

$$|\psi_{ns}(0)|^2 = \frac{1}{\pi(na_0)^3}, \quad (25.156)$$

where $a_0 = \hbar^2/(me^2)$ is the Bohr radius. Substituting Eq. (25.156) into Eq. (25.155) and expressing the magnetons in terms of the Rydberg energy $R_\infty = \frac{e^2}{2a_0}$ and the fine-structure constant $\alpha = \frac{e^2}{\hbar c}$, we obtain

$$\Delta E_{\text{hfs}} = \frac{2}{3} \left(\frac{e^2}{2a_0} \right) g_p \left(\frac{m}{M} \right) \frac{\alpha^2}{n^3} \langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle. \quad (25.157)$$

25.9.3 Hyperfine Multiplets and the 21-Centimeter Line

The total angular momentum of the atom is $\mathbf{F} = \mathbf{S}_e + \mathbf{I}$, where $S_e = 1/2$ and $I = 1/2$. The operator $\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p$ evaluates according to the coupled spin states:

$$\mathbf{F}^2 = \left(\frac{\hbar}{2} \boldsymbol{\sigma} + \frac{\hbar}{2} \boldsymbol{\sigma}_p \right)^2 = \frac{3}{2} \hbar^2 \mathbf{1} + \frac{\hbar^2}{2} \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \implies \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p = 2F(F+1) - 3. \quad (25.158)$$

Hence:

- **Triplet State** ($F = 1$, **spins parallel**): $\langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle = +1$ (statistical weight $2F+1 = 3$).
- **Singlet State** ($F = 0$, **spins antiparallel**): $\langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle = -3$ (statistical weight $2F+1 = 1$).

Because $\langle \boldsymbol{\sigma} \cdot \boldsymbol{\sigma}_p \rangle < 0$ for $F = 0$, the singlet ground state lies lower in energy than the triplet state.

The relative energy shifts satisfy the center-of-gravity theorem:

$$\sum_F (2F+1) \Delta E(F) = 3 \cdot (+1) + 1 \cdot (-3) = 0, \quad (25.159)$$

meaning that the unweighted centroid of the hyperfine multiplet is unshifted to first order.

For the $1S_{1/2}$ ground state ($n = 1$), the total hyperfine splitting between the $F = 1$ and $F = 0$ levels is

$$\Delta E = \frac{8}{3} \left(\frac{e^2}{2a_0} \right) g_p \left(\frac{m}{M} \right) \alpha^2 \approx 5.874 \times 10^{-6} \text{ eV}. \quad (25.160)$$

The transition between these two hyperfine levels corresponds to a radiation frequency and vacuum wavelength of

$$\nu = \frac{\Delta E}{h} \approx 1420.40575 \text{ MHz}, \quad \lambda = \frac{c}{\nu} \approx 21.106 \text{ cm}. \quad (25.161)$$

This 21-cm hydrogen line is among the most famous spectroscopic transitions in physics, serving as the primary radio-astronomical probe for mapping the neutral hydrogen distribution and spiral arm dynamics across the Milky Way and the observable Universe.

For excited states with principal quantum number $n = 2$:

- The $2S_{1/2}$ state has $|\psi_{2s}(0)|^2 = \frac{1}{8} |\psi_{1s}(0)|^2$, giving a splitting exactly $\frac{1}{8}$ that of the ground state ($\Delta\nu \approx 177.55 \text{ MHz}$).
- For the $2P_{1/2}$ state, although $|\psi(0)|^2 = 0$, the non-contact dipolar and orbital magnetic interactions yield a non-zero splitting equal to $\frac{1}{24}$ of the ground-state splitting ($\Delta\nu \approx 59.19 \text{ MHz}$).

25.10 The Lamb Shift, Mass Renormalization, and Radiative Corrections

In Dirac's single-particle theory of the hydrogen atom, states sharing the same principal quantum number n and total angular momentum j but differing in orbital angular momentum $l = j \pm 1/2$ (such as $2S_{1/2}$ and $2P_{1/2}$) are strictly degenerate. However, in 1947, Willis Lamb and Robert Retherford⁵ established experimentally that the $2S_{1/2}$ level lies approximately 1057 MHz above the $2P_{1/2}$ level.

This splitting—the *Lamb shift*—originates from the interaction of the bound electron with quantum vacuum fluctuations of the quantized electromagnetic radiation field. The physical explanation of this shift by Hans Bethe⁶ marked the birth of modern quantum electrodynamics (QED) and the systematic program of mass renormalization.

25.10.1 Second-Order Radiative Perturbation Theory

The minimal coupling between an electron with current density $\mathbf{j}(\mathbf{r})$ and the quantized radiation field $\mathbf{A}(\mathbf{r})$ is

$$H_{\text{int}} = -\frac{e}{c} \int d^3r \mathbf{j}(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}). \quad (25.162)$$

In second-order perturbation theory, the electron can spontaneously emit a virtual transverse photon of wavevector $\mathbf{k}$, angular frequency $\omega_k = ck$, and polarization $\boldsymbol{\lambda}$, transitioning from state $|n\rangle$ to an intermediate atomic state $|n'\rangle$, before reabsorbing it:

$$\Delta E_n = \sum_{n'} \sum_{\mathbf{k}\boldsymbol{\lambda}} \frac{|\langle n', \mathbf{k}\boldsymbol{\lambda} | H_{\text{int}} | n, 0 \rangle|^2}{\varepsilon_n - \varepsilon_{n'} - \hbar ck}. \quad (25.163)$$

In the electric dipole approximation, the interaction matrix element is

$$\langle n', \mathbf{k}\boldsymbol{\lambda} | H_{\text{int}} | n, 0 \rangle = -\frac{e}{m} \sqrt{\frac{2\pi\hbar^2}{\omega_k V}} \langle n' | \mathbf{p} \cdot \boldsymbol{\lambda}^* | n \rangle, \quad (25.164)$$

where $\mathbf{p}$ is the electron momentum operator. Replacing the sum over photon modes by a phase-space integral, $\sum_{\mathbf{k}} \rightarrow \frac{V}{(2\pi)^3} \int d^3k$, and performing the angular integration over the two transverse photon polarizations,

$$\int d\Omega_{\mathbf{k}} \sum_{\lambda=1}^2 |\langle n' | \mathbf{p} \cdot \boldsymbol{\lambda}^* | n \rangle|^2 = \frac{8\pi}{3} |\langle n' | \mathbf{p} | n \rangle|^2. \quad (25.165)$$

The unrenormalized second-order energy shift evaluates to

$$\Delta E_n = \frac{2e^2}{3\pi\hbar c^3 m^2} \int_0^\infty \omega d\omega \sum_{n'} \frac{|\langle n' | \mathbf{p} | n \rangle|^2}{\varepsilon_n - \varepsilon_{n'} - \omega}, \quad (25.166)$$

where $\omega \equiv \hbar ck$. At large frequencies ($\omega \gg |\varepsilon_n - \varepsilon_{n'}|$), the denominator approaches $-\omega$, causing the integral over ω to diverge *linearly*:

$$\Delta E_n \sim -\frac{2e^2}{3\pi\hbar c^3 m^2} \langle n | p^2 | n \rangle \int_0^\infty d\omega \rightarrow -\infty. \quad (25.167)$$

⁵W. E. Lamb and R. C. Retherford, *Phys. Rev.* **72**, 241 (1947).

⁶H. A. Bethe, *Phys. Rev.* **72**, 339 (1947).

25.10.2 Free-Electron Self-Energy and Mass Renormalization

The resolution of this divergence lies in the realization that a free electron also interacts continuously with the vacuum radiation field. For an unbound electron in a plane-wave momentum eigenstate $|\mathbf{p}\rangle$, Eq. (25.166) yields the free self-energy:

$$\Delta E_p = -\frac{2e^2}{3\pi\hbar c^3 m^2} \langle \mathbf{p} | p^2 | \mathbf{p} \rangle \int_0^\infty d\omega = -\left(\frac{4e^2}{3\pi\hbar c^3 m^2} \int_0^\infty d\omega \right) \frac{p^2}{2}. \quad (25.168)$$

Remarkably, ΔE_p is strictly proportional to p^2 , which represents a shift in the electron's effective kinetic energy.

If m_0 denotes the theoretical "bare" mass of the electron in the absence of electromagnetic coupling, the physical kinetic energy of the free electron is

$$E_{\text{kin}} = \frac{p^2}{2m_0} + \Delta E_p = \left[\frac{1}{m_0} - \frac{4e^2}{3\pi\hbar c^3 m^2} \int_0^\infty d\omega \right] \frac{p^2}{2} \equiv \frac{p^2}{2m}, \quad (25.169)$$

where m is the experimentally observed, physical mass. The relation

$$\frac{1}{m} = \frac{1}{m_0} - \frac{4e^2}{3\pi\hbar c^3 m^2} \int_0^\infty d\omega \quad (25.170)$$

defines the *electromagnetic mass renormalization*.

Because all physical experiments inevitably measure the dressed electron with mass m , writing $p^2/(2m)$ in the atomic Hamiltonian already implicitly includes the free electromagnetic self-energy. To avoid double-counting, the free-particle self-energy must be subtracted from the bound-state shift. The physically observable radiative level shift is therefore

$$\Delta E'_n = \frac{2e^2}{3\pi\hbar c^3 m^2} \int_0^\infty \omega d\omega \left(\sum_{n'} \frac{|\langle n' | \mathbf{p} | n \rangle|^2}{\varepsilon_n - \varepsilon_{n'} - \omega} + \frac{\langle n | p^2 | n \rangle}{\omega} \right). \quad (25.171)$$

Using the completeness relation $\langle n | p^2 | n \rangle = \sum_{n'} |\langle n' | \mathbf{p} | n \rangle|^2$, the integrand combines into

$$\Delta E'_n = \frac{2e^2}{3\pi\hbar c^3 m^2} \sum_{n'} |\langle n' | \mathbf{p} | n \rangle|^2 (\varepsilon_{n'} - \varepsilon_n) \int_0^\infty \frac{d\omega}{\varepsilon_{n'} - \varepsilon_n + \omega}. \quad (25.172)$$

The linear ultraviolet divergence is canceled identically, leaving only a gentle logarithmic divergence.

25.10.3 Bethe's Evaluation and the Oscillator Sum Rule

In a fully relativistic field-theoretic calculation, electron-positron pair creation and relativistic kinematics provide an ultraviolet cutoff at frequencies of order the electron rest energy, $\omega_{\text{max}} \approx mc^2$. Imposing this physical cutoff,

$$\Delta E'_n \approx \frac{2e^2}{3\pi\hbar c^3 m^2} \sum_{n'} |\langle n' | \mathbf{p} | n \rangle|^2 (\varepsilon_{n'} - \varepsilon_n) \ln \left| \frac{mc^2}{\varepsilon_{n'} - \varepsilon_n} \right|. \quad (25.173)$$

Following Bethe, we define the logarithmically weighted average excitation energy K_0 :

$$\ln K_0 \equiv \frac{\sum_{n'} |\langle n' | \mathbf{p} | n \rangle|^2 (\varepsilon_{n'} - \varepsilon_n) \ln |\varepsilon_{n'} - \varepsilon_n|}{\sum_{n'} |\langle n' | \mathbf{p} | n \rangle|^2 (\varepsilon_{n'} - \varepsilon_n)}. \quad (25.174)$$

The remaining sum over dipole oscillator strengths is evaluated using Heisenberg commutators:

$$\begin{aligned} \sum_{n'} |\langle n' | \mathbf{p} | n \rangle|^2 (\varepsilon_{n'} - \varepsilon_n) &= \langle n | \mathbf{p} \cdot [H_0, \mathbf{p}] | n \rangle = -\frac{1}{2} \langle n | [\mathbf{p}, \cdot [\mathbf{p}, H_0]] | n \rangle \\ &= \frac{\hbar^2}{2} \langle n | \nabla^2 V(r) | n \rangle = \frac{\hbar^2}{2} \langle n | 4\pi e^2 \delta^3(\mathbf{r}) | n \rangle \\ &= 2\pi e^2 \hbar^2 |\psi(0)|^2. \end{aligned} \quad (25.175)$$

Because $|\psi(0)|^2$ vanishes for all states with $l > 0$, the dominant Lamb shift occurs exclusively in s -states!

Substituting Eq. (25.175) into Eq. (25.173) for an ns -state ($|\psi_{ns}(0)|^2 = \frac{1}{\pi n^3 a_0^3}$) gives

$$\Delta E'_{ns} = \frac{4\alpha^3}{3\pi n^3} \left(\frac{e^2}{2a_0} \right) \ln \left(\frac{mc^2}{K_0} \right) = \frac{8\alpha^3}{3\pi n^3} R_\infty \ln \left(\frac{mc^2}{K_0} \right). \quad (25.176)$$

For the $2S$ level of hydrogen, Bethe numerically integrated over the hydrogenic discrete spectrum and continuum to find

$$K_0(2S) \approx 17.8 R_\infty \approx 242 \text{ eV} \implies \ln \left(\frac{mc^2}{K_0} \right) \approx 7.63. \quad (25.177)$$

This nonrelativistic estimate yields

$$\Delta \nu(2S_{1/2}) = \frac{\Delta E'_{2s}}{h} \approx +1040 \text{ MHz}. \quad (25.178)$$

A small negative shift of the $2P_{1/2}$ level ($\approx -17 \text{ MHz}$) and higher-order relativistic and vacuum polarization corrections bring the theoretical value to 1057.86 MHz , in breathtaking agreement with the Lamb-Retherford measurement ($1057.77 \pm 0.10 \text{ MHz}$).

25.10.4 Physical Picture and the Lamb-Retherford Experiment

Welton⁷ provided an intuitive semiclassical interpretation of the Lamb shift: the zero-point electromagnetic fluctuations cause the point electron to undergo rapid spatial jitter (Zitterbewegung) over a mean-square radius $\langle (\delta \mathbf{r})^2 \rangle \sim \frac{2\alpha}{\pi} \left(\frac{\hbar}{mc} \right)^2 \ln \left(\frac{mc^2}{K_0} \right)$. The electron samples a smeared potential $\langle V(\mathbf{r} + \delta \mathbf{r}) \rangle \approx V(\mathbf{r}) + \frac{1}{6} \langle (\delta \mathbf{r})^2 \rangle \nabla^2 V$. For a Coulomb potential, $\nabla^2 V = 4\pi e^2 \delta^3(\mathbf{r})$, which only shifts states with probability density at the origin ($l = 0$), pushing the $2S_{1/2}$ state upward.

Level	Energy Shift Relative to $2P_{1/2}$	Physical Origin
$2P_{3/2}$	+10,969 MHz	Fine-structure spin-orbit splitting
$2S_{1/2}$	+1057 MHz	QED radiative vacuum fluctuations (Lamb shift)
$2P_{1/2}$	0 MHz (reference)	Unperturbed Dirac energy

Table 25.1: Fine-structure and Lamb-shift energy level separations in the $n = 2$ manifold of atomic hydrogen.

The Lamb-Retherford experimental technique ingeniously exploited the disparate lifetimes of the $2S_{1/2}$ and $2P_{1/2}$ states:

- Parity and selection rules forbid single-photon electric dipole transitions $2S_{1/2} \rightarrow 1S_{1/2}$. The $2S_{1/2}$ state is metastable, decaying solely via two-photon emission with a long lifetime $\tau_{2S} \approx 0.14 \text{ s}$.

⁷T. A. Welton, *Phys. Rev.* **74**, 1157 (1948).

- In contrast, the $2P_{1/2}$ state decays rapidly via electric dipole emission with lifetime $\tau_{2P} \approx 1.6 \times 10^{-9}$ s.
- An atomic beam of excited hydrogen atoms traveled through a high-vacuum chamber until all short-lived states decayed, leaving a pure beam of metastable $2S_{1/2}$ atoms.
- Applying a tunable microwave electromagnetic field at the resonant frequency $\nu \approx 1057$ MHz stimulated transitions $2S_{1/2} \rightarrow 2P_{1/2}$, followed by immediate radiative decay to the $1S_{1/2}$ ground state. By monitoring the depletion of metastable atoms as a function of microwave frequency, Lamb and Retherford mapped the resonance curve with extraordinary precision.

25.11 Dirac Hole Theory, Charge Conjugation, and the Spin-Statistics Connection

The existence of negative-energy continuum solutions ($E \leq -mc^2$) presents a profound conceptual challenge for relativistic single-particle wave equations. In the presence of electromagnetic coupling, an electron in a positive-energy state ($E \geq +mc^2$) would inevitably undergo radiative cascade into unbounded negative-energy states, destabilizing all atomic matter.

To resolve this crisis, Paul Dirac proposed an ingenious physical hypothesis that laid the conceptual groundwork for modern relativistic quantum field theory and antimatter.

25.11.1 The Dirac Sea and Positrons as Holes

Because spin-1/2 particles obey the Pauli exclusion principle, Dirac postulated that the physical *vacuum* is not empty space, but rather a completely filled, uniform "sea" of electrons occupying all negative-energy states from $-\infty$ up to $-mc^2$.

- By the exclusion principle, an electron occupying a positive-energy state cannot make radiative transitions into the negative-energy sea because every negative-energy orbital is already fully occupied. Positive-energy states are thus rendered stable against radiative collapse.
- While the negative-energy sea possesses an infinite negative charge and mass density, in the absence of external fields this sea is spatially homogeneous and isotropic. Physical detectors register only observable *deviations* from this unperturbed vacuum background.

A remarkable consequence emerges when an external perturbation (such as an energetic photon with $\hbar\omega > 2mc^2$) imparts energy to an electron in the negative-energy sea, ejecting it into an empty positive-energy state. What remains in the sea is an unoccupied orbital—a *hole*.

- If the removed electron had energy $-E_p$, momentum $-\mathbf{p}$, spin $-\sigma$, and electric charge $e = -|e|$, the absence of this electron alters the total state of the system relative to the vacuum by:

$$\begin{aligned}
 E &= -(-E_p) = +E_p > 0, \\
 \mathbf{P} &= -(-\mathbf{p}) = +\mathbf{p}, \\
 S_z &= -(-\sigma) = +\sigma, \\
 Q &= -(-|e|) = +|e| > 0.
 \end{aligned} \tag{25.179}$$

- The hole behaves in all physical respects as an independent particle having positive energy, ordinary mass m , and positive electric charge $+|e|$. This antiparticle is the *positron*, discovered experimentally by Carl Anderson in 1932.

- *Pair production* ($\gamma \rightarrow e^- + e^+$) is simply the excitation of an electron from the negative-energy sea to the positive-energy continuum, leaving a hole behind.

25.11.2 Vacuum Subtraction and Observable Currents

Consider a state containing $n_{\mathbf{p}\sigma}^{(+)} \in \{0, 1\}$ electrons in positive-energy states and $n_{\mathbf{p}\sigma}^{(-)} \in \{0, 1\}$ electrons in negative-energy states. The total energy is

$$E_{\text{tot}} = \sum_{\mathbf{p}\sigma} \left[E_p n_{\mathbf{p}\sigma}^{(+)} + (-E_p) n_{\mathbf{p}\sigma}^{(-)} \right]. \quad (25.180)$$

Introducing the positron occupation number

$$\bar{n}_{\mathbf{p}\sigma} \equiv 1 - n_{\mathbf{p}\sigma}^{(-)} \in \{0, 1\}, \quad (25.181)$$

the total energy decomposes into

$$E_{\text{tot}} = \sum_{\mathbf{p}\sigma} E_p \left(n_{\mathbf{p}\sigma}^{(+)} + \bar{n}_{\mathbf{p}\sigma} \right) + \sum_{\mathbf{p}\sigma} (-E_p). \quad (25.182)$$

The divergent term $E_{\text{vac}} = \sum_{\mathbf{p}\sigma} (-E_p)$ represents the zero-point energy of the unperturbed Dirac sea. Subtracting this constant vacuum reference energy yields the physically observed energy:

$$E'_{\text{tot}} \equiv E_{\text{tot}} - E_{\text{vac}} = \sum_{\mathbf{p}\sigma} E_p \left(n_{\mathbf{p}\sigma}^{(+)} + \bar{n}_{\mathbf{p}\sigma} \right) \geq 0, \quad (25.183)$$

which is manifestly positive-definite.

In identical fashion, the observable total momentum is

$$\mathbf{P}'_{\text{tot}} = \sum_{\mathbf{p}\sigma} \mathbf{p} \left(n_{\mathbf{p}\sigma}^{(+)} + \bar{n}_{\mathbf{p}\sigma} \right), \quad (25.184)$$

and the physical electromagnetic four-current is

$$\langle J^\mu \rangle_{\text{phys}} = \sum_{\mathbf{p}\sigma} \left[(e j_{\mathbf{p}\sigma}^\mu) n_{\mathbf{p}\sigma}^{(+)} + (-e j_{\mathbf{p}\sigma}^\mu) \bar{n}_{\mathbf{p}\sigma} \right], \quad (25.185)$$

where $j_{\mathbf{p}\sigma}^\mu$ is the probability current density. The factor $-e = +|e|$ in the second term demonstrates that positrons carry positive electric charge.

25.11.3 Charge Conjugation

The physical equivalence between negative-energy electron holes and positive-energy positrons is formulated mathematically via the operation of *charge conjugation* ($\mathcal{C}$).

Taking the complex conjugate of the Dirac equation in an external electromagnetic field ($\Phi, \mathbf{A}$):

$$\left(-i\hbar \frac{\partial}{\partial t} - e\Phi \right) \psi^* = \left[c\boldsymbol{\alpha}^* \cdot \left(-\frac{\hbar}{i} \boldsymbol{\nabla} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 \right] \psi^*. \quad (25.186)$$

In the Dirac-Pauli representation, α_x, α_z , and β are real matrices, while α_y is purely imaginary:

$$\alpha_x^* = \alpha_x, \quad \alpha_z^* = \alpha_z, \quad \beta^* = \beta, \quad \alpha_y^* = -\alpha_y. \quad (25.187)$$

Because the Dirac matrices anticommute, we can express $\boldsymbol{\alpha}^*$ as

$$\boldsymbol{\alpha}^* = -\alpha_y \boldsymbol{\alpha} \alpha_y. \quad (25.188)$$

Substituting Eq. (25.188) into Eq. (25.186) and multiplying from the left by $\beta\alpha_y = \gamma^2 = \gamma_y$ gives

$$\left(i\hbar\frac{\partial}{\partial t} + e\Phi\right)(\beta\alpha_y\psi^*) = \left[c\boldsymbol{\alpha} \cdot \left(\frac{\hbar}{i}\boldsymbol{\nabla} + \frac{e}{c}\mathbf{A}\right) + \beta mc^2\right](\beta\alpha_y\psi^*). \quad (25.189)$$

Thus, the *charge-conjugate wave function* defined by

$$\psi_C(\mathbf{r}, t) \equiv i\beta\alpha_y\psi^*(\mathbf{r}, t) = i\gamma^2\psi^*(\mathbf{r}, t) = \mathcal{C}\bar{\psi}^T(\mathbf{r}, t) \quad (25.190)$$

satisfies the Dirac equation with the sign of the charge reversed ($e \rightarrow -e$).

In explicit 4×4 matrix form, the charge-conjugation operator is

$$\psi_C(\mathbf{r}, t) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \psi^*(\mathbf{r}, t). \quad (25.191)$$

Complex conjugation inverts the spacetime plane-wave phase:

$$\left[e^{-i(-E_pt - (-\mathbf{p})\cdot\mathbf{r})/\hbar}\right]^* = e^{-i(E_pt - \mathbf{p}\cdot\mathbf{r})/\hbar}. \quad (25.192)$$

Hence, if ψ is a negative-energy electron solution with momentum $-\mathbf{p}$ and spin projection $-\sigma$, the charge-conjugate state ψ_C is an authentic positive-energy positron solution with physical momentum $+\mathbf{p}$ and spin $+\sigma$:

$$\psi(\mathbf{r}, t) = u_{\mathbf{p}\sigma}^{(-)} e^{i(\mathbf{p}\cdot\mathbf{r} - E_pt)/\hbar} \iff \psi_C(\mathbf{r}, t) = u_{\mathbf{p}\sigma}^{(+)} e^{-i(\mathbf{p}\cdot\mathbf{r} - E_pt)/\hbar}. \quad (25.193)$$

Furthermore, the probability four-current and spin polarization are strictly invariant under charge conjugation:

$$j_C^\mu = c\bar{\psi}_C\gamma^\mu\psi_C = j^\mu, \quad \langle\boldsymbol{\sigma}\rangle_C = \langle\boldsymbol{\sigma}\rangle. \quad (25.194)$$

25.11.4 Vacuum Polarization and Relativistic Many-Body Effects

In an inhomogeneous electrostatic potential $\Phi(\mathbf{r})$, the negative-energy eigenstates ψ_E are spatially distorted relative to free plane waves.

- The negative-energy sea redistributes: negative charge is displaced away from regions of repulsive potential ($\Phi < 0$) and pulled toward regions of positive potential ($\Phi > 0$).
- This induced charge displacement produces *vacuum polarization*, acting as a virtual dielectric medium that partially screens the bare nuclear charge at large distances.
- In the hydrogen atom, vacuum polarization (the Uehling potential) shifts the $2S_{1/2}$ level downward by -27.13 MHz, contributing approximately 2.5% of the observed net Lamb shift.

25.11.5 The Spin-Statistics Connection

The structural difference between the relativistic Klein-Gordon and Dirac equations reveals the deep relativistic connection between spin and quantum statistics:

1. **Spin-0 (Klein-Gordon):** Expanding an arbitrary scalar state in terms of positive- and negative-energy modes yields the normalization condition [Eq. (23.45)]:

$$\int \frac{d^3p}{(2\pi\hbar)^3} (|u_{\mathbf{p}}|^2 - |v_{\mathbf{p}}|^2) = \pm 1. \quad (25.195)$$

Because of the relative minus sign in the conserved current metric, $|u_{\mathbf{p}}|^2$ and $|v_{\mathbf{p}}|^2$ can be arbitrarily large without violating normalization. This unbounded occupation capacity is the defining hallmark of *bosons*.

2. **Spin-1/2 (Dirac):** Expanding an arbitrary Dirac state in terms of positive- and negative-energy plane waves yields the positive-definite probability normalization:

$$\sum_{\sigma} \int \frac{d^3p}{(2\pi\hbar)^3} (|a_{\mathbf{p}\sigma}|^2 + |b_{\mathbf{p}\sigma}|^2) = 1. \quad (25.196)$$

The plus sign strictly bounds individual mode amplitudes: $|a_{\mathbf{p}\sigma}|^2 \leq 1$ and $|b_{\mathbf{p}\sigma}|^2 \leq 1$. This structural bound reflects the Pauli exclusion principle, demonstrating that spin-1/2 particles must be *fermions*.

Had spin-1/2 particles been quantized as bosons, the negative-energy states could not be stabilized by filling a sea, and the system would be fundamentally unstable. Fully relativistic consistency requires the spin-statistics theorem: integer-spin particles are bosons, and half-integer-spin particles are fermions.

25.12 Problems

1. **The Relativistic Klein Paradox and Vacuum Polarization at a Potential Step:**

Consider an electron beam with incident energy $E > mc^2$ and momentum $p = \sqrt{E^2/c^2 - m^2c^2}$ incident from $x < 0$ onto an electrostatic step potential:

$$e\Phi(x) = \begin{cases} 0, & x < 0, \\ V, & x > 0. \end{cases} \quad (25.197)$$

- (a) By matching the four-component Dirac plane-wave spinors at the interface $x = 0$, calculate the reflection coefficient R and transmission coefficient T for:
- The subcritical regime, $V < E + mc^2$.
 - The supercritical regime, $V > E + mc^2$ (the *Klein paradox*).

Provide a complete physical interpretation of the supercritical reflection $R > 1$ within the framework of Dirac hole theory, identifying the physical mechanism in terms of field-induced electron-positron pair creation at the barrier interface.

- (b) Consider the subcritical limit with no incident particle in the presence of a weak step $0 < V \ll 2mc^2$. Formulate the spatial profile of the induced vacuum polarization charge density $\rho_{\text{vac}}(x)$ across the boundary.

2. **Algebraic Symmetry Operators of the Dirac-Coulomb Problem:** For the relativistic Dirac hydrogen Hamiltonian $H = c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta mc^2 - Ze^2/r$:

- (a) Prove the fundamental commutation relations:

$$[K, \boldsymbol{\alpha} \cdot \mathbf{p}] = 0, \quad [K, \boldsymbol{\alpha} \cdot \mathbf{r}] = 0, \quad [K, H] = 0, \quad (25.198)$$

where $K = \beta \left(\mathbf{1} + \frac{\boldsymbol{\Sigma} \cdot \mathbf{L}}{\hbar} \right)$.

- (b) Define the Johnson-Lippmann operator:

$$B = \gamma_5 \left(\Lambda + \frac{KH}{mc^2} \right), \quad \Lambda = -\beta K - i \frac{Ze^2}{\hbar c} \alpha_r, \quad (25.199)$$

with $\alpha_r = \boldsymbol{\alpha} \cdot \hat{\mathbf{r}}$. Prove that B commutes with the Hamiltonian, $[B, H] = 0$, and anticommutes with K , $\{B, K\} = 0$. Evaluate the commutator $[B, \Lambda]$.

- (c) Demonstrate that B is strictly Hermitian: $B^\dagger = B$.

3. **Radiative Lifetime of the 21-Centimeter Hyperfine Transition:** The hyperfine interaction splits the $1S_{1/2}$ ground state of atomic hydrogen into a triplet state ($F = 1$) and a singlet state ($F = 0$).

- (a) Show that electric dipole (E1) transitions between these states are strictly forbidden by parity conservation.
- (b) Using the magnetic dipole (M1) transition probability per unit time:

$$A(F = 1 \rightarrow F = 0) = \frac{4\omega^3}{3\hbar c^3} |\langle F = 0, M_F = 0 | \boldsymbol{\mu} | F = 1, M_F \rangle|^2, \quad (25.200)$$

where $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + 2\mathbf{S})/\hbar + g_p\mu_N\mathbf{I}/\hbar$, calculate the radiative transition rate and show that the spontaneous lifetime of the upper hyperfine state is $\tau \approx 1.1 \times 10^7$ years $\approx 3.5 \times 10^{14}$ s.

4. **Spin-Orbit and Darwin Contributions to Fine Structure:** From the nonrelativistic Foldy-Wouthuysen expansion of the Dirac equation to order $\mathcal{O}(v^2/c^2)$, isolate the three relativistic corrections to the Coulomb Hamiltonian:

$$H' = H_{\text{kin}} + H_{\text{so}} + H_{\text{Darwin}} = -\frac{p^4}{8m^3c^2} + \frac{Ze^2}{2m^2c^2r^3} \mathbf{L} \cdot \mathbf{S} + \frac{\pi Ze^2\hbar^2}{2m^2c^2} \delta^3(\mathbf{r}). \quad (25.201)$$

Compute the expectation value of each separate term in the hydrogenic state $|n, l, j, m_j\rangle$, and demonstrate that their sum reproduces the full fine-structure shift $\Delta E_{nj} = -\frac{m(Z\alpha)^4c^2}{2n^3} \left(\frac{1}{j+1/2} - \frac{3}{4n} \right)$, confirming that l -dependence cancels identically for all states.

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